Preface

In this book, we will learn justifications of certain basic properties of the number systems $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ that are often assumed without details in undergraduate mathematics programmes. The book is written with a typical second year undergraduate student of Mathematics in mind, but it should be accessible for a younger mathematical audience as well.

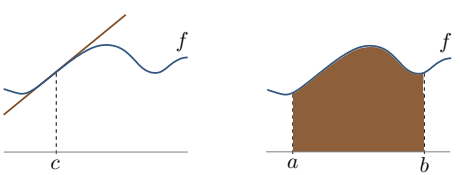
Our starting point is the existence of a set and a (successor) function satisfying the Peano axioms for the natural numbers, from which we systematically construct and prove properties of the integers, rationals, reals, and complex numbers.

With the exception of the first introductory chapter, until the construction of the reals is completed, we stick purely to the objects hitherto defined, but in later chapters, especially in the exercises, we also use freely some of the standard objects studied in undergraduate mathematics, for example, the elementary functions. The reader interested in a rigorous treatment of such things (starting just from the least upper bound property of $\mathbb{R}$) is referred e.g. to [Sasane (2015)].

Below we motivate the need in analysis for a rigorous treatment of the number systems, in particular the real numbers.

Roughly speaking, analysis is calculus made rigorous. In calculus, given a real-valued function f of a real variable, the focus is on two things:

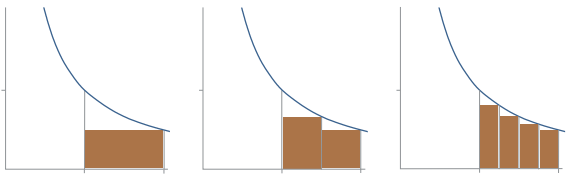
- (Differentiation) What is the rate of change of f at a point?
- (Integration) What is the area under the graph of f over an interval?



In calculus, while there are these two quite different topics of study, the fundamental theorem of calculus is a bridge between these different worlds,

saying that the processes of differentiation and integration are inverses of each other: $\int_a^b f'(x)dx = f(b) - f(a)$ and $\frac{d}{dx}\int_a^x f(\xi)d\xi = f(x)$. This interaction between differentiation and integration provides a powerful body of understanding and calculational technique, familiar to us as 'calculus'.

A thorough treatment of calculus, i.e., of the subject of analysis, must start with a careful study of the number system in which the action takes place, namely the set $\mathbb{R}$ of real numbers. To see why, let's consider an example. Suppose we want to find the area A under the graph of the function $f(x) = \frac{1}{x}$ from $x = 1$ to $x = 2$.



We wish to obtain the area A by approximating it with the sum of rectangular areas, each time doubling the number of rectangles, hence 'exhausting' more and more of the required area. The area a_n at the nth stage is

$$a_n = \frac{1}{2^{n-1}}\left(\frac{1}{1+\frac{1}{2^{n-1}}} + \frac{1}{1+\frac{2}{2^{n-1}}} + \cdots + \frac{1}{1+\frac{2^{n-1}-1}{2^{n-1}}} + \frac{1}{1+\frac{2^{n-1}}{2^{n-1}}}\right).$$

The idea is that for large n, a_n approximates A. Clearly $a_1 \leqslant a_2 \leqslant a_3 \leqslant \cdots$, and they are all less than some big number[1]. Since a_n misses A by smaller and smaller amounts as n increases, we expect that A should be the 'smallest' number exceeding the numbers $a_1, a_2, a_3, \cdots$. Does such a number exist? We seem to need the following:

(F) For an increasing sequence $a_1, a_2, a_3, \cdots$, of numbers all of which are less than a certain number, there is a smallest number which is bigger than each of $a_1, a_2, a_3, \cdots$.

Note that each $a_n \in \mathbb{Q}$ (set of rationals). Does (F) hold for rational numbers?

This question might seem frivolous to a scientist who is just interested in 'real world applications'. But such a sloppy attitude can lead to trouble. Indeed, results in calculus during the 16th to the 18th century relying on a mixture of deductive reasoning and intuition, involving vaguely defined terms, were later shown to be *incorrect*. To give a quick example of how things might easily go wrong, one might naively, but incorrectly[2], guess that the answer to the question above is 'yes'. This prompts the question

[1] Each a_n is less than (the area of the square of height and width 1, which is) 1.

[2] In fact, for our sequence $a_1, a_2, a_3, \cdots$ of areas above, there is no smallest rational number which is bigger than each of the a_n. Considering it as a real sequence, $A = \log_e 2$, which can be shown to be irrational.

of whether there is a bigger set of numbers than the rational numbers for which the property happens to be true? The answer is 'yes', and this is the *real number system* $\mathbb{R}$.

In the first part, we will spend some time learning about the foundations of the number systems, starting with the natural number system $\mathbb{N}$ of the 'counting numbers', and progressively enlarging the number system set whenever we meet an arithmetic hurdle, until we meet an 'analytical' hurdle with the rationals, which is finally remedied by the real number system: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$. We will first learn about the manner in which these number systems are constructed. Subsequently, we shall also delve deeper into the 'algebraic structure' of the integers obtained from its arithmetic, since it plays a fundamental role in all of mathematics. In particular, we will learn about

- the division algorithm,
- divisibility, the greatest common divisor, and Euclid's algorithm,
- prime numbers, and the fundamental theorem of arithmetic, and
- modular arithmetic.

We will also learn about countable and uncountable sets, the decimal representation as well as continued fraction representations of real numbers, and end our study with an introduction to the complex numbers, along with a proof of the fundamental theorem of algebra.

This book is based on lecture notes I wrote for the course MA103 *Introduction to Abstract Mathematics*, given to first year undergraduates in the BSc Mathematics and Economics programme at the London School of Economics. I recorded corresponding lectures during 2021-22, and can be found on YouTube [Sasane (2021)].

There are ample exercises throughout the text, with detailed solutions included at the end of the book. The ends of proofs, examples, and remarks are indicated with '□', '◇', and '∗', respectively. Page number references are indicated by 'p.number': so p.vii refers to this page, p.1 to the first page of Chapter 1, etc. Sources that were used while writing the book are included in the bibliography, but no claim to originality is made in case a reference is missing. I am grateful to Sara Maad Sasane (Lund University), Raymond Mortini (Université de Lorraine), Adam Ostaszewski and Konrad Swanepoel (London School of Economics) for useful comments. It is a pleasure to thank the World Scientific publishing team, in particular, Rochelle Kronzek and Rok Ting Tan.

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2025.

Contents

Preface v

1. Sets, relations, functions 1
   - 1.1 The number line . . . 1
   - 1.2 Sets and functions . . . 4
   - 1.3 Equivalence relations . . . 8
2. Natural numbers 13
   - 2.1 Peano axioms and the natural numbers . . . 13
   - 2.2 Recursive definitions . . . 14
   - 2.3 Addition . . . 16
   - 2.4 Multiplication . . . 17
   - 2.5 Peano playing . . . 18
   - 2.6 Order . . . 20
   - 2.7 Finite nonempty sets . . . 22
3. Integers 25
   - 3.1 Addition . . . 27
   - 3.2 Multiplication . . . 29
   - 3.3 Order . . . 32
4. Rational numbers 35
   - 4.1 Addition and multiplication . . . 37
   - 4.2 Order . . . 38

5. Real numbers 41
5.1 Addition and multiplication . . . 43
5.2 Order . . . 46
5.3 Least upper bound property . . . 49
5.4 Completeness of $\mathbb{R}$ . . . 55

6. The ring of integers 63
6.1 The division algorithm . . . 63
6.2 Divisibility, gcd, and the Euclid algorithm . . . 64
6.3 Fundamental theorem of arithmetic . . . 71

7. Modular arithmetic 75

8. Cardinality 81

9. Real numbers revisited 91
9.1 Irrational numbers . . . 91
9.2 Decimal representation . . . 93
9.3 Continued fraction representation . . . 97
9.4 Rational zeroes theorem and algebraic numbers . . . 102
9.5 Extended real number system $\overline{\mathbb{R}}$ . . . 105

10. Complex numbers 109
10.1 Addition and multiplication . . . 109
10.2 Geometric representation of complex numbers . . . 112
10.3 Topology of $\mathbb{C}$ . . . 117
10.4 Fundamental theorem of algebra . . . 121

Solutions 127

Notes 221

Bibliography 223

Index 225

Chapter 1

Sets, relations, functions

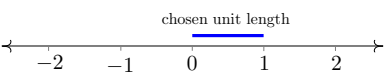
In this chapter, we review some preliminaries that will be needed in the book. It is assumed that the reader has had, or is going through, a first course in (one variable) calculus, and hence is already familiar with most of the topics of this chapter.

1.1 The number line

Now and then in this chapter (e.g. in the exercises) we refer to the integers, rationals, and reals, and these will be constructed carefully in the later chapters, but here we rely on the reader's prior knowledge of these number systems (at whatever intuitive/rigour level). We are familiar with

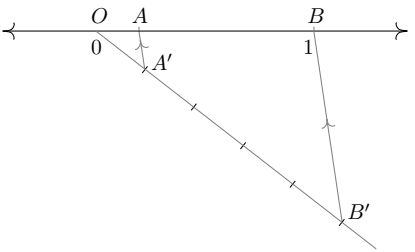
the natural numbers $\mathbb{N} = \{1, 2, 3, \cdots\}$,
the integers $\mathbb{Z} = \{\cdots, -3, -2, -1, 0, 1, 2, 3, \cdots\}$,
the rational numbers $\mathbb{Q} = \{\frac{n}{d} : n, d \in \mathbb{Z}, d \neq 0\}$,

and we are accustomed to visualising these numbers on the 'number line'. The *number line* is any line in the plane, on which we have chosen a point O as the 'origin', representing the number 0, and chosen a unit length by marking off a point on the right of O, where the number 1 is placed. In this way, we get all the positive integers, $1, 2, 3, 4, \cdots$ by repeatedly marking off successively the unit length towards the right, and all the negative integers $-1, -2, -3, \cdots$ by repeatedly marking off successively the unit length towards the left.



We can also depict all rational numbers. First, there is a procedure for dividing a unit length on the number line into d ($\in \mathbb{N}$) equal parts, allowing us to construct the rational number $\frac{1}{d}$ on the number line. We use ℓ to denote lengths, and Δ to denote triangles. Take an arbitrary length $\ell(OA')$ along a ray starting at O in any direction other than that of the number line itself. Let B' be a point on the ray such that $\ell(OB') = d \cdot \ell(OA')$. Draw

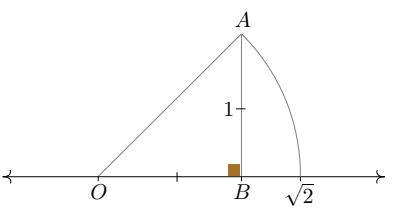
AA' parallel to BB' to meet the number line at A. We conclude that the $\Delta OAA'$ is similar to $\Delta OBB'$, and so $\ell(OA) = \frac{1}{d}$. Having obtained $\frac{1}{d}$, we can now construct $\frac{n}{d}$ on the number line for *any* $n \in \mathbb{Z}$, by repeating the length $\frac{1}{d}$ n times towards the right of 0 if $n > 0$, and towards the left $-n$ times from 0 if n is negative.



Construction of rationals: Given the length 1 $(=\ell(OB))$, we can construct the length $\frac{1}{5}$, and so A corresponds to the rational number $\frac{1}{5}$.

Hence we can depict all the rational numbers on the number line. Does this exhaust the number line? That is, suppose that we start with all the points on the number line being coloured black, and suppose that at a later time, we colour all the rational ones by red: are there any black points left over? The answer is 'yes', and we demonstrate this below. We will show that there does 'exist', based on geometric reasoning, a point on the number line, whose square is 2, but we will also argue that this number, denoted by $\sqrt{2}$, is not a rational number.

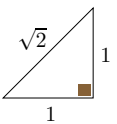
Firstly, the following picture shows that $\sqrt{2}$ exists as a point on the number line. Indeed, in the right angled triangle ΔOBA, by Pythagoras's theorem, we have $(\ell(OA))^2 = (\ell(OB))^2 + (\ell(AB))^2 = 1^2 + 1^2 = 2$, and so $\ell(OA)$ is a number, denoted say by $\sqrt{2}$, whose square is 2. Taking O as the center and radius $\ell(OA)$, we draw a circle intersecting the number line at a point C, corresponding to the number $\sqrt{2}$. Is $\sqrt{2}$ a rational number? We will show below that it is not.



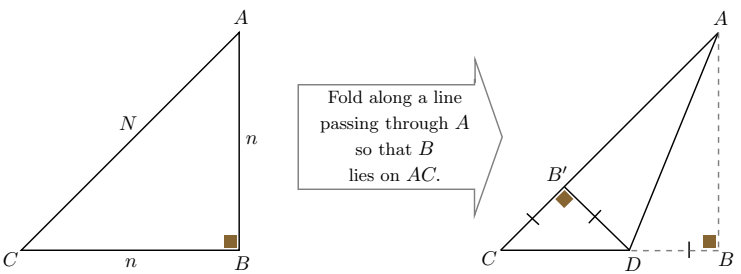
Exercise 1.1. Depict $-\frac{11}{6}$ and $\sqrt{3}$ on the number line.

Theorem 1.1. (An 'origami' proof of the irrationality of $\sqrt{2}$).
There is no rational number $q \in \mathbb{Q}$ such that $q^2 = 2$.

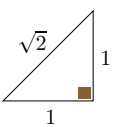
Proof. Let $\sqrt{2}$ be a rational number. Then some scaling of the triangle



by an integer will produce a similar triangle, all of whose sides are integers. Consider the smallest such triangle, say ΔABC, with integer lengths $\ell(BC) = \ell(AB) = n$, and $\ell(AC) = N$, where $n, N \in \mathbb{N}$. Now perform the following origami: fold along a line passing through A so that B lies on AC, giving rise to the point B' on AC. The 'crease' in the paper is actually the angle bisector AD of the angle $\angle BAC$.



In $\Delta CB'D$, $\angle CB'D = 90°$, and $\angle B'CD = \angle ACB = 45°$. Thus $\Delta CB'D$ is an isosceles right triangle. So $\ell(B'D) = \ell(CB') = \ell(AC) - \ell(AB') = N - n \in \mathbb{N}$, while $\ell(CD) = \ell(CB) - \ell(DB) = n - \ell(B'D) = n - (N - n) = 2n - N \in \mathbb{N}$. So $\Delta CB'D$ is similar to the triangle

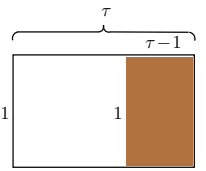


has integer side lengths, and is smaller than ΔABC, contradicting the choice of ΔABC. So there is no rational number q such that $q^2 = 2$. □

We will meet a different proof yet again, after the rigorous construction of the numbers sytems, based on the 'rational zeroes theorem' (see p.103).

Exercise 1.2. The real number $\tau := \frac{1+\sqrt{5}}{2} \approx 1.618\cdots$ is called the *golden ratio*. It is believed to create geometrical figures of particularly pleasing proportions, for example, the *golden rectangle*, with sides in the ratio $1 : \tau$. Using the smaller

side, if a square is separated from the golden rectangle, we obtain yet another golden rectangle, thanks to the relation $\tau^2 - \tau - 1 = 0$, that is, $\tau - 1 = \frac{1}{\tau}$. See the following picture.



Give a geometric proof that τ cannot be rational. Conclude that also $\sqrt{5} \notin \mathbb{Q}$.

Thus we have seen that the elements of $\mathbb{Q}$ can be depicted on the number line, and that not all the points on the number line belong to $\mathbb{Q}$. We think of $\mathbb{R}$ as *all* the points on the number line. As mentioned before, if we take out everything on the number line (the black points) except for the rational numbers $\mathbb{Q}$ (the red points), then there will be holes amongst the rational numbers (for example there will be a missing black point where $\sqrt{2}$ lies on the number line). We can think of the real numbers as 'filling in' these holes between the rational numbers.

1.2 Sets and functions

The notion of a *set* involves mentally putting together or collecting 'objects', and viewing them as whole. For example, one may talk about the set of all human beings, or the set of all integers, or the set of all straight lines in a plane. The objects in the set are referred to as its *elements* or *members*. Sets will usually be denoted by capital letters, and their elements by a small letter. Given an element a in S, we say 'a *belongs to* S' or 'a *is an element of* S', or 'a *is in* S', and write '$a \in S$'. For example, $0 \in \mathbb{Z}$ means the integer 0 belongs to the set $\mathbb{Z}$ of integers. If it is not the case[1] that a is an element of a set S, we write $a \notin S$. The set devoid of any elements is denoted by $\varnothing$, and is called the *empty set*. A set containing precisely one element is called a *singleton*.

Although the notion of sets of concrete objects is familiar to us from an early age, things start getting subtle when one tries to talk about sets of sets. Indeed, in the above informal discussion, we have agreed that the set has an identity separate from that of each of its constituents. Now if we try to consider a set consisting of all sets, then such a set is a member of

[1]The phrase 'it is not the case that $\cdots$' in front of a mathematical statement T will be denoted by $\neg$, and $\neg T$ is called the 'negation of T'. Thus, $\neg\, T$ is true means that the mathematical statement 'T does not hold' is true.

itself. We could instead try to exclude such strange sets by saying we only consider 'ordinary' sets, with the property that such a set is not its own member. Now consider the set S of all ordinary sets. A natural question is whether S is ordinary. If it is ordinary, then by the defining property it is not its own member, and hence is not ordinary, a contradiction. If it is not ordinary, then by the defining property, it is its own member, and hence is ordinary, a contradiction. This situation[2] shows that we cannot rely on the simplistic definition of 'a collection of objects' for a set, but must concede that not all collections of objects can be deemed to be sets.

Axiomatic set theory is the subject in which an attempt is made to define a set and postulate several basic axioms about sets so as to avoid paradoxes as above. A set of axioms known as the Zermelo-Frankel axioms (abbreviated ZFC, with 'C' standing for the 'axiom of choice') are widely accepted as being quite natural. We shall not delve into this[3], because our interest in sets is primarily as building blocks of the number systems, and in proving their arithmetic properties. Based on this attitude towards sets as a means rather than as an end, we adopt the 'naive set theory' approach, where the paradoxes are avoided by confining ourselves only to certain sets, imagining they are all subsets of a 'universal set' (which we will not specify, but assume is large enough to include all the symbols, expressions, and concepts that we are dealing with, and its existence, taken for granted by us, is a matter of axiomatic set theory). Below, we shall proceed by recalling some preliminaries from naive set theory (operations on sets, the notions of functions and relations, etc.).

Two sets are equal when they have exactly the same elements. There are two ways of specifying a set directly in terms of its members. One is to list them all within curly brackets, e.g. the set $\{1, 2, 3\}$ is the set whose elements are the numbers $1, 2$ and 3. Another way is to specify a set by giving a 'characteristic property of the set', that is, a property which is satisfied by the elements of the set and by no other objects. If $P(x)$ denotes the statement that the property holds for the object x, then $\{x : P(x)\}$ or $\{x \,|\, P(x)\}$ is the notation for the set of all elements x for which this property holds. For example, the set $\{n \in \mathbb{N} : n < 4\}$ is equal to the set $\{1, 2, 3\}$.

Let S, T be sets. Then S *is a subset of* T, written $S \subset T$, if for each $x \in S$, we have that $x \in T$. Sometimes we also write $T \supset S$ in this case. Also, we define $T \backslash S = \{x \in T : x \notin S\}$. If S is a set, then the set of all subsets of S is called the *power set of* S, and is denoted by $\mathcal{P}(S)$.

[2] Called Russell's paradox, see [Ciesielski (1997), Chapter 1].

[3] The interested reader is referred for example to [Ciesielski (1997)].

Given a family $\mathcal{F}$ of sets, the *union* of the elements of $\mathcal{F}$ is

$$\bigcup_{A\in\mathcal{F}} A = \{x : x \in A \text{ for at least one } A \in \mathcal{F}\},$$

and the *intersection* of the elements of $\mathcal{F}$ is

$$\bigcap_{A\in\mathcal{F}} A = \{x : x \in A \text{ for all } A \in \mathcal{F}\}.$$

Sets A and B are *disjoint* if $A \cap B = \varnothing$.

In order to talk about relations and functions, we need the notion of an ordered pair. If X and Y are sets, and we take elements x, y from X, Y, respectively, then we would like to talk about the ordered pair (x, y), where x is the *first* member of the pair, and y is the *second* member of the pair. It is tempting to define the ordered pair as the set $\{x, y\}$, but this is the same as the set $\{y, x\}$, and so it does not distinguish its members as a first element and a second element. So one may define the ordered pair (x, y) in set-theoretic terms as the set $S = \{\{x\}, \{x, y\}\}$. Then this definition singles out the 'first' member of the ordered pair as the element x having the property that for all $A \in S$, $x \in A$. Of course, if $x = y$, then $S = \{\{x\}, \{x, x\}\} = \{\{x\}, \{x\}\} = \{\{x\}\}$. The crucial fact manifested by this definition is the following result.

Theorem 1.2.
Two ordered pairs (x, y) and $(\widetilde{x}, \widetilde{y})$ are equal if and only if $x = \widetilde{x}$ and $y = \widetilde{y}$.

Proof. (If part:) Let $x = \widetilde{x}$ and $y = \widetilde{y}$. Then $\{x\} = \{\widetilde{x}\}$ and $\{x, y\} = \{\widetilde{x}, \widetilde{y}\}$, showing that $(x, y) = \{\{x\}, \{x, y\}\} = \{\{\widetilde{x}\}, \{\widetilde{x}, \widetilde{y}\}\} = (\widetilde{x}, \widetilde{y})$.

(Only if part:) Let $(x, y) = (\widetilde{x}, \widetilde{y})$. If $x = y$, then $(x, y) = \{\{x\}\} = \{\{\widetilde{x}\}, \{\widetilde{x}, \widetilde{y}\}\}$. Thus $\{\widetilde{x}\} \in \{\{x\}\}$, giving $\{\widetilde{x}\} = \{x\}$ and so $\widetilde{x} = x$. Also, $\{\widetilde{x}, \widetilde{y}\} \in \{\{x\}\}$, implying $\{\widetilde{x}, \widetilde{y}\} = \{x\}$. So $\widetilde{y} \in \{\widetilde{x}, \widetilde{y}\} = \{x\}$, giving $\widetilde{y} = x$. So $\widetilde{y} = x = y$.

If $x \neq y$, then $\{x, y\} \neq \{\widetilde{x}\}$. So $(x, y) = (\widetilde{x}, \widetilde{y})$ implies that $\{x, y\} = \{\widetilde{x}, \widetilde{y}\}$ and $\{x\} = \{\widetilde{x}\}$. The latter gives $x = \widetilde{x}$, and then $y \in \{x, y\} = \{\widetilde{x}, \widetilde{y}\} = \{x, \widetilde{y}\}$ implies that $y = \widetilde{y}$ (since $y \neq x$). □

If X, Y are sets, then the *Cartesian product $X \times Y$ of X and Y* is defined by $X \times Y = \{(x, y) : x \in X \text{ and } y \in Y\}$. If X, Y, Z are sets, $x \in X, y \in Y, z \in Z$, then one can define a triple (x, y, z) as the ordered pair $((x, y), z)$. More generally, if $X_1, \cdots, X_n$ are n sets, where $n \in \mathbb{N}$, then an *n-tuple* $(x_1, \cdots, x_n)$ can be defined analogously, where $x_k \in X_k$ for each $k \in \{1, \cdots, n\}$. The Cartesian product of $X_1, \cdots, X_n$ is denoted by $X_1 \times \cdots \times X_n$, and if all the sets are equal to the same set X, then we simply use the notation X^n. Thus $\mathbb{R}^3 = \{(x, y, z) : x, y, z \in \mathbb{R}\}$.

A *relation* R is a subset of $X \times Y$ for some sets X and Y. If $(x, y) \in R$,

then we think of $x \in X$ and $y \in Y$ as 'related by the relation R', and write xRy. A *function* f is a relation with the following two properties:

- For every $x \in X$, there exists a $y \in Y$ such that $(x, y) \in R$.
- If $(x_1, y_1), (x_2, y_2) \in R$ and $x_1 = x_2$, then $y_1 = y_2$.

Functions are also called *maps* or *mappings*. We write $f : X \to Y$. If $x \in X$ and $(x, y) \in R$, then we write '$f(x) = y$', or '$x \mapsto y$', and say 'f maps x to $f(x) = y$'. Sometimes we write $X \ni x \mapsto f(x) \in Y$, which is read as 'element x belonging to X is mapped to the element $f(x)$ belonging to Y'.

The set X is the *domain of* f, and Y is the *codomain of* f. The *range of* f is the set $\operatorname{ran} f = \{y \in Y : \text{there exists an } x \in X \text{ such that } f(x) = y\}$. The range of f is also referred to as the *image of* f. The *identity map* $\mathrm{id}_X : X \to X$ is defined by $\mathrm{id}_X(x) = x$ for all $x \in X$. If X is clear from context, then we simply write id for the identity function.

Let $f : X \to Y$ be a function. Given $U \subset X$, we define the subset $f(U)$ of Y by $f(U) = \{y \in Y : y = f(x) \text{ for some } x \in U\}$. Thus $\operatorname{ran} f = f(X)$. Given a subset V of Y, we define the *inverse image* $f^{-1}(V)$ *of* V *under* f by $f^{-1}(V) = \{x \in X : f(x) \in V\}$. So $f^{-1}(V) \subset X$. We have $f^{-1}(\varnothing) = \varnothing$ and $f^{-1}(Y) = X$.

If $f : X \to Y$ is a function and S is a subset of X, then the *restriction of* f *to* S is the function $f|_S : S \to Y$ given by $f|_S(x) = f(x)$ for all $x \in S$.

Given sets X, Y, Z and maps $f : X \to Y$ and $g : Y \to Z$, the *composition* $g \circ f : X \to Z$ is defined by $(g \circ f)(x) = g(f(x))$ for all $x \in X$. A function $f : X \to Y$ is *injective* if for all $x_1, x_2 \in X$ such that $x_1 \neq x_2$, we have $f(x_1) \neq f(x_2)$. We also refer to injective functions as *one-to-one* functions. We say f is *surjective* if $Y = \operatorname{ran} f$. Surjective functions are also called *onto* functions. The composition of injective maps is injective, and the composition of surjective maps is surjective. An injective and surjective function is called a *bijective function* or a *bijection*. If $f : X \to Y$ is bijective, then there is a unique function $g : Y \to X$ such that $f \circ g = \mathrm{id}_Y$ and $g \circ f = \mathrm{id}_X$, and g is called the *inverse of* f. Vice versa, if $f : X \to Y$ and $g : Y \to X$ are maps such that $f \circ g = \mathrm{id}_Y$ and $g \circ f = \mathrm{id}_X$, then f (and g) is bijective, and g is the inverse of f (and f is the inverse of g).

Let I be a set which is used as an 'index set' for a family $\mathcal{F}$ of sets, i.e., for each element $i \in I$, we have a set $A_i \in \mathcal{F}$. More formally, there is a map $I \ni i \mapsto A_i \in \mathcal{F}$. Then the union $\bigcup_{A \in \mathcal{F}} A$ and the intersection $\bigcap_{A \in \mathcal{F}} A$ are denoted by $\bigcup_{i \in I} A_i$ and $\bigcap_{i \in I} A_i$, respectively.

1.3 Equivalence relations

In this section, we recall the important notion of equivalence relations, because it will play a significant role in the rest of the book. For example, to construct the integers from natural numbers, we will need to identify pairs of natural numbers using an equivalence relation.

Definition 1.1. (Relation).
A *relation R on a set S* is a subset of the $S \times S := \{(a, b) : a, b \in S\}$.
If $(a, b) \in R$, then we write aRb.

For example, if we take S to be the set of all human beings, then the set $R_{\text{sibling}} = \{(a, b) \in S \times S : a, b \text{ have the same biological parents}\}$ is a relation on the set of all human beings. Here we mention that a person is one's own sibling since pairs of the form (a, a) also belong to R_{sibling}. As another example, we can take the set $S = \mathbb{Z}$, the set of all integers, and $R_{\text{mod }2} = \{(m, n) \in \mathbb{Z} \times \mathbb{Z} : m - n \text{ is divisible by } 2\}$. Then $R_{\text{mod }2}$ is a relation on the set of all integers. As a third example, let S be the set of all straight lines in the plane. If ℓ_1 is parallel to ℓ_2, then we write $\ell_1 \parallel \ell_2$. A line is considered to be parallel to itself. Then $R_{\text{parallel}} = \{(\ell_1, \ell_2) \in S \times S : \ell_1 \parallel \ell_2\}$ is a relation on the set of all straight lines.

Sometimes we use the symbol $\sim$ to denote a relation, and then instead of aRb, we will write $a \sim b$.

Definition 1.2. (Equivalence relation).
A relation R on a set S is an *equivalence relation* if it satisfies the following:
(ER1) R is *reflexive*, that is, for all $a \in S$, aRa.
(ER2) R is *symmetric*, that is, if $a, b \in S$ and aRb, then bRa.
(ER3) R is *transitive*, that is, if $a, b, c \in S$, aRb and bRc, then aRc.

In our example above, where $S = \{\text{all human beings}\}$, R_{sibling} can easily be checked to be an equivalence relation. Similarly $R_{\text{mod }2}$ is an equivalence relation on $\mathbb{Z}$, and R_{parallel} is an equivalence class on the set of all straight lines in the plane.

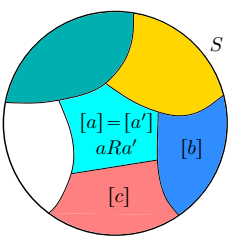
Why are equivalence relations useful? They help 'partition' the set into 'equivalence classes', and help to break down the big set into smaller subsets, such that all the elements in each subset are related to each other, and hence 'equivalent' in some way. For example, R_{sibling} enables one to partition the set of human beings into equivalence classes consisting of groups of brothers/sisters. On the other hand, $R_{\text{mod }2}$ partitions $\mathbb{Z}$ into the sets {even integers} and {odd integers}. Finally, R_{parallel} partitions the set of all straight lines in the plane into collections of mutually parallel lines.

Definition 1.3. (Equivalence class).
If R is an equivalence relation of a set S, then the *equivalence class of* a, denoted by $[a]$, is defined to be the set $[a] = \{b \in S : aRb\}$.

Given any $a, b \in S$, either $[a] = [b]$ or $[a] \cap [b] = \varnothing$. Indeed, let $[a] \cap [b] \neq \varnothing$. Suppose $c \in [a] \cap [b]$, that is, aRc and bRc. By symmetry, cRb. As aRc and cRb, by transitivity, we obtain aRb, and again by symmetry, bRa. If $d \in [a]$, then aRd. As bRa and aRd, by transitivity, bRd. So $d \in [b]$ too. So we have shown that $[a] \subset [b]$. In the same way, one can show $[b] \subset [a]$ as well. So $[a] = [b]$.

Clearly, $\bigcup_{a \in S} [a] \subset S$. For $a \in S$, we have aRa (reflexivity), and so $a \in [a]$. Thus $S \subset \bigcup_{a \in S} [a]$. So $S = \bigcup_{a \in S} [a]$.

As any two distinct (i.e., unequal) equivalence classes do not overlap at all, it follows that S is partitioned into equivalence classes by R.



The idea is that an equivalence relation is really an 'attention focusing device', where we have chosen to ignore other distinguishing features of objects which are related, and have put them together in an equivalence class. So an equivalence relation gives one a 'pair of glasses' through which we 'clump together' things which are 'essentially the same' (equivalent under the relation) and see them as one object. For example, if our set is the collection of children in a school bus and we consider the equivalence relation R_1 of 'having the same sex', then through these glasses, we see only two equivalence classes: boys and girls. On the other hand, if we consider the equivalence relation R_2 of 'having the same age', then through these glasses, we see groups of children sorted by age.

Exercise 1.3. Find out which of the properties of reflexivity, symmetry or transitivity are valid for each of the following relations R on the given set S, and hence determine which amongst them are equivalence relations.
(1) $S = \mathbb{R}$, xRy if $x < y + 1$.
(2) $S = \mathbb{Z}$, mRn if $m \mid n$, that is, if n is divisible by m.
(3) $S = \mathbb{R} \backslash \{0\}$, xRy if $xy^{-1} \in \mathbb{Q}$.
(4) $S = \mathbb{N}$, mRn if there exists an $k \in \mathbb{Z}$ such that $m = 2^k n$.

Exercise 1.4. Let R be a relation on the set $\mathbb{R}$ of real numbers.
Viewing R as a subset of the (x,y)-plane, explain the geometric meaning of the reflexive, symmetric and transitive properties.

Exercise 1.5. Which of the following $R \subset \mathbb{R}\times\mathbb{R}$ is an equivalence relation on $\mathbb{R}$?
(1) $R = \{(x,x) \in \mathbb{R}^2 : x \in \mathbb{R}\}$.
(2) $R = \varnothing$.
(3) $R = \{(x,y) \in \mathbb{R}^2 : y = 0\}$.
(4) $R = \{(x,y) \in \mathbb{R}^2 : xy + 1 = 0\}$.
(5) $R = \{(x,y) \in \mathbb{R}^2 : x^2y - xy^2 - x + y = 0\}$.

Exercise 1.6. (Equivalence relation induced by a map $f : X \to Y$).
Let $f : X \to Y$ be a map. Define the relation $\sim$ on X by $x \sim x'$ if $f(x) = f(x')$.
(1) Show that $\sim$ is an equivalence relation.
(2) Recall that the range of f is the set $\operatorname{ran} f = \{f(x) : x \in X\}$.
For $y \in \operatorname{ran} f$, consider the inverse image $f^{-1}(\{y\}) = \{x \in X : f(x) = y\}$.
For all $x \in X$, show that $[x] = f^{-1}(\{f(x)\})$.
(3) Denote the set of all equivalence classes of X by $\overline{X}$. Define $\overline{f} : \overline{X} \to \operatorname{ran} f$ as follows. For any equivalence class $S \in \overline{X}$, take any $x \in S$, and set $\overline{f}(S) = f(x)$. Show that $\overline{f}$ is *well-defined*, i.e., it does not depend on the choice of the representative selected from the equivalence class S.
(4) Prove that $\overline{f} : \overline{X} \to \operatorname{ran} f$ is bijective.
(5) For the following maps, consider the corresponding induced equivalence relation on their respective domains.
(a) The complex absolute value $|\cdot| : \mathbb{C} \to \mathbb{R}$.
(b) $f : \mathbb{R}^2 \to \mathbb{R}$ given by $f(x,y) = xy$.
In each case, find the equivalence classes, and sketch these in the domain.

Exercise 1.7. Define a relation $\sim$ on $\mathbb{R}$ as follows: For $a, b \in \mathbb{R}$, $a \sim b$ if $a - b \in \mathbb{Z}$.
Show that $\sim$ is an equivalence relation, and $[a] = \{a + n : n \in \mathbb{Z}\}$ for $a \in \mathbb{R}$.

Exercise 1.8. A *sequence in* $\mathbb{Q}$ is a function $f : \mathbb{N} \to \mathbb{Q}$.
Instead of writing $f(n)$ for the value of f at a natural number n, we write a_n.
The entire sequence is then referred to with the notation $(a_n)_{n\in\mathbb{N}}$.
A *Cauchy sequence in* $\mathbb{Q}$ is a sequence $(a_n)_{n\in\mathbb{N}}$ in $\mathbb{Q}$ such that for every rational $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that whenever $m, n \in \mathbb{N}$ satisfy $m, n > N$, we have $|a_n - a_m| < \epsilon$.
A sequence $(a_n)_{n\in\mathbb{N}}$ in $\mathbb{Q}$ *converges in* $\mathbb{Q}$ *to* $r \in \mathbb{Q}$ if for every rational $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $n \in \mathbb{N}$ satisfying $n > N$, we have $|a_n - r| < \epsilon$.
(1) Show that $(\frac{1}{n})_{n\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{Q}$.
(2) Show that every sequence $(a_n)_{n\in\mathbb{N}}$ in $\mathbb{Q}$ that is convergent in $\mathbb{Q}$ is a Cauchy sequence in $\mathbb{Q}$.
(3) Show that if the integer $m \geqslant 2$, then $(1 + \frac{1}{m} + \cdots + \frac{1}{m^{n-1}})_{n\in\mathbb{N}}$ converges in $\mathbb{Q}$ to $\frac{1}{1-\frac{1}{m}}$.

(4) If $(a_n)_{n\in\mathbb{N}}$ is a sequence in $\mathbb{Q}$ which is convergent in $\mathbb{Q}$ to $r \in \mathbb{Q}$, and there exist $N \in \mathbb{N}$ and $c, d \in \mathbb{Q}$ such that $d \geqslant a_n \geqslant c$ for all natural numbers $n > N$, then show that $d \geqslant r \geqslant c$.

(5) Show that $(\frac{1}{1!} + \cdots + \frac{1}{n!})_{n\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{Q}$.

(6) Show that $(\frac{1}{1!} + \cdots + \frac{1}{n!})_{n\in\mathbb{N}}$ is not convergent in $\mathbb{Q}$.

(7) Let $S \subset \mathbb{Q}$. An element $u \in \mathbb{Q}$ is an *upper bound of* S if for all $x \in S$, $x \leqslant u$. If S possesses an upper bound, then S is *bounded above*. An element $u_* \in \mathbb{Q}$ is a *least upper bound of* S if u_* is an upper bound of S, and for any upper bound u of S, we have $u_* \leqslant u$. The aim of this part of the exercise is to show that not every nonempty subset S of $\mathbb{Q}$ possesses a least upper bound (in $\mathbb{Q}$). Show, for example by using the result from the previous part, that the set $S = \{\frac{1}{1!} + \cdots + \frac{1}{n!} : n \in \mathbb{N}\} \subset \mathbb{Q}$ does not possess a least upper bound in $\mathbb{Q}$.

(8) Let $\mathscr{C}$ denote the set of all Cauchy sequences in $\mathbb{Q}$. Define the relation $\sim$ on $\mathscr{C}$ as follows: for $(a_n)_{n\in\mathbb{N}}, (b_n)_{n\in\mathbb{N}} \in \mathscr{C}$, $(a_n)_{n\in\mathbb{N}} \sim (b_n)_{n\in\mathbb{N}}$ if $(a_n - b_n)_{n\in\mathbb{N}}$ converges to 0 in $\mathbb{Q}$. Show that $\sim$ is an equivalence relation. The set of all equivalence classes of $\mathscr{C}$ under $\sim$ is denoted by $\mathscr{C}/\sim$.

(9) Given a $q \in \mathbb{Q}$, the constant sequence $(q)_{n\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{Q}$. Show that the map $\mathbb{Q} \ni q \mapsto [(q)_{n\in\mathbb{N}}] \in \mathscr{C}/\sim$ is injective.
We will return to this equivalence relation when we construct $\mathbb{R}$ from $\mathbb{Q}$: The set of real numbers will be $\mathscr{C}/\sim$. This part of the exercise illustrates that $\mathbb{Q}$ can be considered to be contained in $\mathbb{R}$. We will clarify later how the arithmetic operations are defined in $\mathscr{C}/\sim$.

Chapter 2

Natural numbers

In this chapter, beginning with an axiomatic framework, we will construct the natural numbers. We are familiar with natural numbers from an early age when we learn to count. The basic idea is that for each natural number, there is 'next one' or a 'successor' (which is used to label the next object in the set we are counting). Thus the natural numbers are built, intuitively speaking, as a sequence of objects, starting with 1 and then taking 'successive successors'. The Italian mathematician Giuseppe Peano (1858-1932) formulated axioms for the natural numbers in 1889 which reflect the above intuition. The familiar properties of the natural numbers can then be proved as theorems.

2.1 Peano axioms and the natural numbers

The *Peano axioms* for a set $\mathbb{N}$ and a (successor) function $\sigma : \mathbb{N} \to \mathbb{N}$ state the following:

(N1) The set $\mathbb{N}$ contains an element 1 such that for every $n \in \mathbb{N}$, $\sigma(n) \neq 1$.

(N2) The map σ is injective.

(N3) (*Induction axiom*). Suppose a subset $S \subset \mathbb{N}$ has the properties that

- $1 \in S$, and
- if $n \in S$, then $\sigma(n) \in S$.

Then $S = \mathbb{N}$.

The element $\sigma(n)$ is called the *successor of* $n \in \mathbb{N}$. The property (N3) is the induction property, and forms the basis for why induction works.

The existence/construction of such a set $\mathbb{N}$ and a successor function σ that satisfies the Peano axioms relies on the axioms of set theory. For example, a possible definition is

$$1 := \{\varnothing\},$$
$$2 := \{\varnothing, \{\varnothing\}\},$$
$$3 := \{\varnothing, \{\varnothing\}, \{\varnothing, \{\varnothing\}\}\},$$

and so on. The successor of $n \in \mathbb{N}$ is $\sigma(n) = n \cup \{n\}$. To talk about the set of *all* natural numbers (the 'and so on' part above), one requires an axiom

called the 'axiom of infinity' from set theory (which is a disguised version of the induction axiom). We will not get into these matters here[1]. Instead, we simply take for granted the existence[2] of a set $\mathbb{N}$ and a successor function σ that satisfies the Peano axioms. From this starting point, in the rest of the section, we will prove the familiar arithmetic properties of the natural numbers, which will in turn be used in order to construct the integers, and also to derive similar properties for the integers.

Let us show that every natural number not equal to 1 has a 'predecessor'.

Theorem 2.1. *If $n \in \mathbb{N}\backslash\{1\}$, then there is an $m \in \mathbb{N}$ such that $\sigma(m) = n$.*

Proof. Let $S = \{1\} \cup \{k \in \mathbb{N} : \text{there exists}^3 \text{ an } m \in \mathbb{N} \text{ such that } \sigma(m) = k\}$. We have $1 \notin \{k \in \mathbb{N} : \exists m \in \mathbb{N} \text{ such that } \sigma(m) = k\}$, but $1 \in S$. If $n \in S \subset \mathbb{N}$, then $\sigma(n) \in \{k \in \mathbb{N} : \exists m \in \mathbb{N} \text{ such that } \sigma(m) = k\} \subset S$. By the induction axiom (N3), $S = \mathbb{N}$. Hence if $n \in \mathbb{N}\backslash\{1\} = S\backslash\{1\}$, then there exists an $m \in \mathbb{N}$ such that $\sigma(m) = k$. □

Henceforth, we'll denote the successor $\sigma(1)$ of 1 by 2, $\sigma(2)$ by 3, and so on.

Exercise 2.1. Consider the set $S = \{n \in \mathbb{N} : \sigma(n) \neq n\}$.
Using Peano's axioms, show that $S = \mathbb{N}$. Conclusion: for all $n \in \mathbb{N}$, $\sigma(n) \neq n$.

2.2 Recursive definitions

The Peano axioms allow us to make 'recursive/inductive definitions'. This means that we can define a sequence of objects O_n indexed by the natural numbers, via

$$(*) \begin{cases} \bullet \text{ specifying the initial 'value' } O_1, \\ \bullet \text{ giving a rule for determining } O_{\sigma(n)} \text{ from } O_n \\ \quad \text{(so that each object is defined using the preceding one).} \end{cases}$$

Intuitively $(*)$ defines the sequence $O_1, O_2, O_3, \cdots$ uniquely, though proving this using the Peano axioms is cumbersome. A natural approach to show this would be to define S to be the set of n such that $(*)$ determines O_k for $k \leqslant n$. Then $1 \in S$. Also whenever $n \in S$, also $\sigma(n) \in S$. So (N3) would yield $S = \mathbb{N}$. The problem is the usage of $\leqslant$, which has not been defined yet, and so is inadmissible in the above argument.

We now prove the 'recursion theorem', which justifies this. The unique mapping $\mathbb{N} \ni n \mapsto O_n$ we seek will be denoted by $\boldsymbol{O}$, and we let $\mathcal{O}$ denote the nonempty collection of objects from which $O_1, O_2, O_3, \cdots$ are chosen. The rule determining $O_{\sigma(n)}$ from O_n is denoted by $\boldsymbol{\sigma}$.

[1] The interested reader is referred to [Ciesielski (1997)].
[2] The uniqueness 'up to isomorphisms' is shown in Proposition 2.1, p.16.
[3] We often use '$\exists$' for the phrase 'there exists'.

Theorem 2.2. (Recursion theorem).
Let $\mathcal{O}$ be a nonempty set, $O_1 \in \mathcal{O}$, and $\boldsymbol{\sigma} : \mathcal{O} \to \mathcal{O}$. Then there is exactly one map $\boldsymbol{O} : \mathbb{N} \to \mathcal{O}$ such that $\boldsymbol{O}(1) = O_1$, and for all $n \in \mathbb{N}$, $\boldsymbol{O}(\sigma(n)) = \boldsymbol{\sigma}(\boldsymbol{O}(n))$.

Proof. *Existence:* Let $\mathcal{C}$ be the set of all subsets T of $\mathbb{N} \times \mathcal{O}$ such that
(1) $(1, O_1) \in T$, and
(2) whenever $(n, O) \in T$, we have $(\sigma(n), \boldsymbol{\sigma}(O)) \in T$.
Clearly, the full set $T := \mathbb{N} \times \mathcal{O} \in \mathcal{C}$. Thus $\mathcal{C}$ is not empty. The set

$$T_* := \bigcap_{T \in \mathcal{C}} T$$

also satisfies (1) and (2). Thus $T_* \in \mathcal{C}$ as well. Clearly $T_* \subset T$ for all $T \in \mathcal{C}$. Let $S := \{n \in \mathbb{N} : \text{there exists exactly one } O \in \mathcal{O} \text{ such that } (n, O) \in T_*\}$.

We claim $1 \in S$. We have $(1, O_1) \in T_*$. Let $(1, O_*) \in T_*$ for an $O_* \neq O_1$. Then $(1, O_1) \in T_* \backslash \{(1, O_*)\} =: T$. Then T satisfies (1). We claim T also satisfies (2). If $(n, O) \in T$, then $(n, O) \in T_*$, and as T_* satisfies (2), $(\sigma(n), \boldsymbol{\sigma}(O)) \in T_*$. But $(\sigma(n), \boldsymbol{\sigma}(O)) \neq (1, O_*)$ because $\sigma(n) \neq 1$, and so $(\sigma(n), \boldsymbol{\sigma}(O)) \in T$. Thus $T \in \mathcal{C}$, and so $T_* \subset T = T_* \backslash \{(1, O_*)\}$, a contradiction. Hence $1 \in S$.

Let $n \in S$. We claim $\sigma(n) \in S$. Let O_n be the unique element in O such that $(n, O_n) \in T_*$. Since $T_* \in \mathcal{C}$, $(\sigma(n), \boldsymbol{\sigma}(O_n)) \in T_*$. Let $(\sigma(n), O_*) \in T_*$ for an $O_* \neq \boldsymbol{\sigma}(O_n)$. Then $(\sigma(n), \boldsymbol{\sigma}(O_n)) \in T_* \backslash \{(\sigma(n), O_*)\} =: \widetilde{T}$. As 1 is not a successor, $(1, O_1) \neq (\sigma(n), O_*)$. Thus $(1, O_1) \in \widetilde{T}$, i.e., $\widetilde{T}$ satisfies (1) above. We show that $\widetilde{T}$ also satisfies (2). Let $(m, O) \in \widetilde{T}$. Then $(m, O) \in T_* \in \mathcal{C}$, and so $(\sigma(m), \boldsymbol{\sigma}(O)) \in T_*$. If $(\sigma(m), \boldsymbol{\sigma}(O)) = (\sigma(n), O_*)$, then $\sigma(n) = \sigma(m)$ (implying $m = n$ by the injectivity of σ) and $\boldsymbol{\sigma}(O) = O_* \neq \boldsymbol{\sigma}(O_n)$ (implying $O \neq O_n$), showing that $(n, O) = (m, O) \in \widetilde{T} \subset T_*$, as well as $(n, O_n) \in T_*$, but $O \neq O_n$, a contradiction to the fact that $n \in S$. So $(\sigma(m), \boldsymbol{\sigma}(O)) \neq (\sigma(n), O_*)$. Thus we have shown that $(\sigma(m), \boldsymbol{\sigma}(O)) \in \widetilde{T}$. Consequently, $\widetilde{T}$ satisfies (2). So $\widetilde{T} \in \mathcal{C}$. But then $T_* \subset \widetilde{T} = T_* \backslash \{(\sigma(n), O_*)\}$, a contradiction.

By (N3), it follows that $S = \mathbb{N}$. For $n \in \mathbb{N} = S$, define $\boldsymbol{O}(n)$ to be the unique element O_n in $\mathcal{O}$ such that $(n, O_n) \in T_*$. As $(1, O_1) \in T_*$, it follows that $\boldsymbol{O}(1) = O_1$. Also, for all $n \in \mathbb{N}$, if $(n, O_n) \in T_* \in \mathcal{C}$, then by (2), $(\sigma(n), \boldsymbol{\sigma}(O_n)) \in T_*$, and so $\boldsymbol{O}(\sigma(n)) = \boldsymbol{\sigma}(O_n) = \boldsymbol{\sigma}(\boldsymbol{O}(n))$.

Uniqueness: Let $\widetilde{\boldsymbol{O}} : \mathbb{N} \to \mathcal{O}$ be such that $\widetilde{\boldsymbol{O}}(1) = O_1$ and for all $n \in \mathbb{N}$, $\widetilde{\boldsymbol{O}}(\sigma(n)) = \boldsymbol{\sigma}(\widetilde{\boldsymbol{O}}(n))$. Let $S = \{n \in \mathbb{N} : \widetilde{\boldsymbol{O}}(n) = \boldsymbol{O}(n)\}$, where $\boldsymbol{O}$ is the map from the existence part. Then $1 \in S$ because $\widetilde{\boldsymbol{O}}(1) = O_1 = \boldsymbol{O}(1)$. Moreover, if $n \in S$, then $\widetilde{\boldsymbol{O}}(\sigma(n)) = \boldsymbol{\sigma}(\widetilde{\boldsymbol{O}}(n)) = \boldsymbol{\sigma}(\boldsymbol{O}(n)) = \boldsymbol{O}(\sigma(n))$, and so $\sigma(n) \in S$ as well. By (N3), $S = \mathbb{N}$, that is, $\widetilde{\boldsymbol{O}} = \boldsymbol{O}$, as wanted. □

The following result shows the promised uniqueness of $\mathbb{N}$ up to isomorphisms.

Proposition 2.1. *Let* $(\mathbb{N}, \sigma)$ *and* $(\mathbb{N}', \sigma')$ *satisfy the Peano axioms, with* $1' \in \mathbb{N}'$ *such that for all* $n \in \mathbb{N}'$, $\sigma'(n) \neq 1'$. *Then there is a unique bijective map* $f : \mathbb{N} \to \mathbb{N}'$ *such that* $f(1) = 1'$, *and* $f \circ \sigma = \sigma' \circ f$.

The latter means in particular that the following diagram commutes:

$$\begin{array}{ccc} \mathbb{N} & \xrightarrow{f} & \mathbb{N}' \\ {\scriptstyle\sigma}\downarrow & & \downarrow{\scriptstyle\sigma'} \\ \mathbb{N} & \xrightarrow{f} & \mathbb{N}' \end{array}$$

Proof. In the recursion theorem, take $\mathcal{O} = \mathbb{N}'$, $O_1 = 1'$, and $\boldsymbol{\sigma} = \sigma'$. Then there is a unique map $f : \mathbb{N} \to \mathbb{N}'$ such that $f(1) = 1'$, and for all $n \in \mathbb{N}$, $f(\sigma(n)) = \sigma'(f(n))$. Swapping the roles of $\mathbb{N}$ and $\mathbb{N}'$, we obtain a map $g : \mathbb{N}' \to \mathbb{N}$ such that $g(1') = 1$ and for all $n' \in \mathbb{N}'$, $g(\sigma'(n')) = \sigma(g(n'))$.

We claim $g \circ f = \mathrm{id}_{\mathbb{N}}$. The map $g \circ f : \mathbb{N} \to \mathbb{N}$, satisfies $(g \circ f)(1) = g(1') = 1$, and for all $n \in \mathbb{N}$, $(g \circ f)(\sigma(n)) = g(\sigma'(f(n))) = \sigma(g(f(n))) = \sigma((g \circ f)(n))$. Also, $\mathrm{id}_{\mathbb{N}} : \mathbb{N} \to \mathbb{N}$ satisfies $\mathrm{id}_{\mathbb{N}}(1) = 1$, and $\mathrm{id}_{\mathbb{N}}(\sigma(n)) = \sigma(n) = \sigma(\mathrm{id}_{\mathbb{N}}(n))$ for all $\mathbb{N} \in \mathbb{N}$. By the uniqueness part of the recursion theorem (applied to $\mathcal{O} = \mathbb{N}$, $O_1 = 1$, and $\boldsymbol{\sigma} = \sigma$), we conclude that $g \circ f = \mathrm{id}_{\mathbb{N}}$.

The proof that $f \circ g = \mathrm{id}_{\mathbb{N}'}$ is obtained by repeating, mutatis mutandis, the argument from the previous paragraph.

Thus f, g are inverses of each other. □

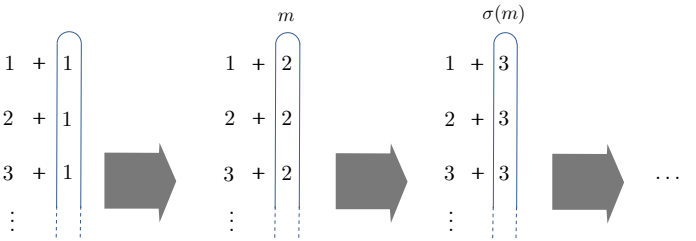
Next we will define addition and multiplication of natural numbers recursively.

2.3 Addition

Theorem 2.3. *There is a unique map* $+ : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ *such that*
(1) *for all* $n \in \mathbb{N}$, $n + 1 = \sigma(n)$
(2) *for all* $m, n \in \mathbb{N}$, $n + \sigma(m) = \sigma(n + m)$.

Proof. *Existence*: Let $n \in \mathbb{N}$. With $\mathcal{O} = \mathbb{N}$, $O_1 = \sigma(n)$, $\boldsymbol{\sigma} = \sigma$ in the recursion theorem, we get a unique map $\boldsymbol{O}_n : \mathbb{N} \to \mathbb{N}$ such that $\boldsymbol{O}_n(1) = \sigma(n)$ and $\boldsymbol{O}_n(\sigma(m)) = \sigma(\boldsymbol{O}_n(m))$ for all $m \in \mathbb{N}$. Define the map $+ : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ by $n + m = \boldsymbol{O}_n(m)$ for all $m, n \in \mathbb{N}$. Then $n + 1 = \boldsymbol{O}_n(1) = \sigma(n)$, and $n + \sigma(m) = \boldsymbol{O}_n(\sigma(m)) = \sigma(\boldsymbol{O}_n(m)) = \sigma(n + m)$ for all $m, n \in \mathbb{N}$.

Uniqueness: Let $\widetilde{+} : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ be any map satisfying $n \mathbin{\widetilde{+}} 1 = \sigma(n)$ and $n \mathbin{\widetilde{+}} m = \sigma(n + m)$ for all $n, m \in \mathbb{N}$. We want to show $\widetilde{+}$ coincides with $+$ from the existence part. Let $n \in \mathbb{N}$. Let $S_n = \{m \in \mathbb{N} : n \mathbin{\widetilde{+}} m = n + m\}$. Then $1 \in S_n$ because $n \mathbin{\widetilde{+}} 1 = \sigma(n) = n + 1$. If $m \in S_n$, then $n \mathbin{\widetilde{+}} m = n + m$, and $n \mathbin{\widetilde{+}} \sigma(m) = \sigma(n \mathbin{\widetilde{+}} m) = \sigma(n + m) = n + \sigma(m)$. Hence $S_n = \mathbb{N}$ by (N3). So for all $n, m \in \mathbb{N}$, $n \mathbin{\widetilde{+}} m = n + m$, as wanted. □



Example 2.1. (What is $2+2$?)
We have $2+2=2+\sigma(1)=\sigma(2+1)=\sigma(\sigma(2))=\sigma(3)=4$. ◇

Exercise 2.2. Using the definition of addition in $\mathbb{N}$, show that $2+3=1+4=5$.

Theorem 2.4. (Principle of induction).
Let $P(1), P(2), P(3), \cdots$ be a sequence of statements, one for each $n \in \mathbb{N}$. Suppose $P(1)$ is true, and whenever $P(n)$ is true, also $P(n+1)$ is true. Then $P(n)$ is true for all $n \in \mathbb{N}$.

Proof. Let $S := \{n \in \mathbb{N} : P(n) \text{ is true}\}$. Thus $1 \in S$, and whenever $n \in S$, also $\sigma(n) \in S$. By (N3), $S = \mathbb{N}$. So $P(n)$ is true for all $n \in \mathbb{N}$. □

We remark that once we have the notion of order (and 'at most'), we will also learn about a variant of the principle of induction, called the principle of *strong* induction, in Exercise 2.20, p.22.

2.4 Multiplication

Theorem 2.5. *There is a unique map $\times : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ such that*
(1) *for all $n \in \mathbb{N}$, $n \times 1 = n$*
(2) *for all $m, n \in \mathbb{N}$, $n \times \sigma(m) = (n \times m) + n$.*

Proof. *Existence:* Let $n \in \mathbb{N}$. In the recursion theorem, taking $\mathcal{O} = \mathbb{N}$, $O_1 = n$, $\boldsymbol{\sigma}(m) = m + n$ for all $m \in \mathbb{N}$, we obtain a unique map $\boldsymbol{O}_n : \mathbb{N} \to \mathbb{N}$ such that $\boldsymbol{O}_n(1) = n$ and $\boldsymbol{O}_n(\sigma(m)) = \boldsymbol{\sigma}(\boldsymbol{O}_n(m)) = \boldsymbol{O}_n(m) + n$ for all $m \in \mathbb{N}$. Define the map $\times : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ by $n \times m = \boldsymbol{O}_n(m)$ for all $m, n \in \mathbb{N}$. Then $n \times 1 = \boldsymbol{O}_n(1) = n$, and $n \times \sigma(m) = \boldsymbol{O}_n(\sigma(m)) = \boldsymbol{O}_n(m) + n = (n \times m) + n$ for all $m, n \in \mathbb{N}$.
Uniqueness: Let $\tilde{\times} : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ be any map satisfying $n \mathbin{\tilde{\times}} 1 = n$ and $n \mathbin{\tilde{\times}} \sigma(m) = (n \mathbin{\tilde{\times}} m) + n$ for all $n, m \in \mathbb{N}$. We want to show $\tilde{\times}$ coincides with $\times$ from the existence part. Let $n \in \mathbb{N}$. Let $S_n = \{m \in \mathbb{N} : n \mathbin{\tilde{\times}} m = n \times m\}$. Then $1 \in S_n$ because $n \mathbin{\tilde{\times}} 1 = n = n \times 1$. If $m \in S_n$, then $n \mathbin{\tilde{\times}} m = n \times m$, and $n \mathbin{\tilde{\times}} \sigma(m) = (n \mathbin{\tilde{\times}} m) + n = (n \times m) + n = n \times \sigma(m)$. Thus $S_n = \mathbb{N}$ by (N3). So for all $n, m \in \mathbb{N}$, $n \mathbin{\tilde{\times}} m = n \times m$, as wanted. □

Example 2.2. (What is 2×2?)
We have $2 \times 2 = 2 \times \sigma(1) = (2 \times 1) + 2 = 2 + 2 = 4$. ◇

Exercise 2.3. Using the definition of multiplication in $\mathbb{N}$, show that $2 \times 3 = 6$.

We often skip writing $\times$, and so we use the notation mn instead of $m \times n$ for natural numbers $m, n \in \mathbb{N}$.

2.5 Peano playing

Using the Peano axioms and the definitions of addition and multiplication, the usual rules of arithmetic, namely the commutativity and associativity of addition and of multiplication, and the distributive law can be proved.

Theorem 2.6. (Associativity of $+$).
For all $n, m, \ell \in \mathbb{N}$, $(n+m)+\ell = n+(m+\ell)$.

Proof. Let $S = \{\ell \in \mathbb{N} : (n+m)+\ell = n+(m+\ell) \text{ for all } n, m \in \mathbb{N}\}$. Then $1 \in S$ because $(n+m)+1 = \sigma(n+m) = n+\sigma(m) = n+(m+1)$. If $\ell \in S$, then

$$\begin{aligned}(n+m)+(\ell+1) &= ((n+m)+\ell)+1 && \text{(definition of addition)}\\ &= (n+(m+\ell))+1 && (\ell \in S)\\ &= n+((m+\ell)+1) && \text{(definition of addition)}\\ &= n+(m+(\ell+1)) && \text{(definition of addition)}\end{aligned}$$

for all $m, n \in \mathbb{N}$. Thus $\sigma(\ell) \in S$. By (N3), $S = \mathbb{N}$, that is, for all $\ell \in \mathbb{N}$, $(n+m)+\ell = n+(m+\ell)$ for all $n, m \in \mathbb{N}$. □

Exercise 2.4. (Commutativity of $+$).
(1) Considering $S = \{n \in \mathbb{N} : n+1 = 1+n\}$ and using (N3), show that $S = \mathbb{N}$.
(2) Let $S' = \{m \in \mathbb{N} : \text{for all } n \in \mathbb{N},\ n+m = m+n\}$. Show that $S' = \mathbb{N}$.

Exercise 2.5. In Exercise 2.1, p.14, we showed that $\sigma(n) \neq n$ for all $n \in \mathbb{N}$, that is, $n+1 \neq n$ for all $n \in \mathbb{N}$. Let $n, m \in \mathbb{N}$. Show that $n+m \neq n$ by considering the set $S = \{n \in \mathbb{N} : n+m \neq n \text{ for all } m \in \mathbb{N}\}$. (We remark that by the commutativity of addition, we also have $m+n \neq n$ for all $m, n \in \mathbb{N}$.)

Exercise 2.6. Show that for all $n \in \mathbb{N}$, $1 \times n = n$.

Theorem 2.7. (Commutativity of $\times$). *For all* $n, m \in \mathbb{N}$, $n \times m = m \times n$.

Proof.
1° We first show $(m+1) \times n = (m \times n) + n$ for all $m, n \in \mathbb{N}$. To this end, we define $S = \{n \in \mathbb{N} : (m+1) \times n = (m \times n) + n \text{ for all } m \in \mathbb{N}\}$. Then $1 \in S$ because $(m+1) \times 1 = m+1 = (m \times 1) + 1$ for all $m \in \mathbb{N}$.

If $n \in S$, then for all $m \in \mathbb{N}$,

$$\begin{aligned}
(m+1)\times\sigma(n) &= ((m+1)\times n) + (m+1) && \text{(multiplication definition)}\\
&= ((m\times n) + n) + (m+1) && (n \in S)\\
&= (m\times n) + (n + (m+1)) && \text{(addition associativity)}\\
&= (m\times n) + ((n + m)+1) && \text{(addition associativity)}\\
&= (m\times n) + ((m + n)+1) && \text{(addition commutativity)}\\
&= (m\times n) + (m + (n+1)) && \text{(addition associativity)}\\
&= ((m\times n) + m) + (n+1) && \text{(addition associativity)}\\
&= (m\times\sigma(n)) + \sigma(n) && \text{(multiplication definition).}
\end{aligned}$$

By (N3), $S = \mathbb{N}$, i.e., for all $n \in \mathbb{N}$, $(m+1)\times n = (m\times n) + n$ for all $m \in \mathbb{N}$.

2° By Exercise 2.6, $1 \times n = n$ for all $n \in \mathbb{N}$. So $1 \times n = n = n \times 1$ for all $n \in \mathbb{N}$.

3° Let $S' = \{m \in \mathbb{N} : m\times n = n\times m \text{ for all } n \in \mathbb{N}\}$. By 2°, $1 \in S'$. If $m \in S'$, then

$$\begin{aligned}
n \times \sigma(m) &= (n \times m) + n && \text{(multiplication definition)}\\
&= (m \times n) + n && (m \in S')\\
&= (m + 1) \times n && \text{(by 1°)}\\
&= \sigma(m) \times n && \text{(addition definition)}
\end{aligned}$$

for all $n \in \mathbb{N}$. By (N3), $S' = \mathbb{N}$. So for all $m, n \in \mathbb{N}$, $m \times n = n \times m$. □

Exercise 2.7. (Distributive law). Show that $m\times(n+\ell) = (m\times n)+(m\times\ell)$ for all $\ell, m, n \in \mathbb{N}$. *Hint*: Consider $\{\ell \in \mathbb{N} : m\times(n+\ell) = (m\times n)+(m\times\ell)$ for all $m, n \in \mathbb{N}\}$. We remark that by the commutativity of $\times$, then we also have that

$$(n + \ell) \times m = (n \times m) + (\ell \times m) \text{ for all } \ell, m, n \in \mathbb{N}.$$

Exercise 2.8. (Associativity of $\times$). Show that $(m \times n) \times \ell = m \times (n \times \ell)$ for all $\ell, m, n \in \mathbb{N}$. *Hint*: Consider $S = \{\ell \in \mathbb{N} : (m \times n) \times \ell = m \times (n \times \ell)$ for all $m, n \in \mathbb{N}\}$.

Exercise 2.9. (Generalised associativity from associativity). Let S be a set and $\cdot : S \times S \to S$ be a map which enjoys the associativity property, i.e., for all $a, b, c \in S$, we have $a \cdot (b \cdot c) = (a \cdot b) \cdot c$. Prove that irrespective of bracketing, every expression involving $\cdot$ in which $n \in \mathbb{N}$ elements $a_1, \cdots, a_n$ appear in this order once, gives a unique element; e.g., $(a_1 \cdot a_2) \cdot ((a_3 \cdot a_4) \cdot a_5) = (((a_1 \cdot a_2) \cdot a_3) \cdot a_4) \cdot a_5$.

Exercise 2.10. (Postfix notation). In the usual arithmetic notation, the arithmetic operations $+, -, \times, /$ come between the operands, but in *postfix notation*, the operator succeeds the operands. For example, $mn+$ is the postfix notation for $m + n$. Postfix notation[4] does not require brackets to indicate which operator is to be executed first. For evaluating an expression in postfix form, one proceeds as follows. Scan the expression from left to right till an operator is encountered. Apply the operator to the operands immediately preceding the operator. This gives a reduced expression. Repeat the process until no operator is left . The final number is the evaluation of the expression so obtained in the usual arithmetic notation. E.g., $a \times (b + c)$ in the usual notation is $abc + \times$ in postfix notation.
Find the postfix notation for $(a + b) \times (c + d + e/f - g/h)$.
Find the usual notation for $ab \times cd/ + efgh - / \times \times$.

[4] Used in 'stack-oriented programming languages', see, e.g., [Sestoft (2017)].

Example 2.3. (No 'additive identity' in $\mathbb{N}$, that is, no 'zero' in $\mathbb{N}$).
We show that there is no $m \in \mathbb{N}$ such that for all $n \in \mathbb{N}$, we have $n = n + m$. Suppose, on the contrary, that there exists such an m. Taking $n = 1$, we get $1 = 1 + m = m + 1 = \sigma(m)$, a contradiction to (N1). ◇

2.6 Order

Definition 2.1. (Order $<$ on $\mathbb{N}$).
For natural numbers $k, K \in \mathbb{N}$, we say that k *is less than* K, written as $k < K$, if there exists an $m \in \mathbb{N}$ such that $K = k + m$. The notation $K > k$ means $k < K$.

Example 2.4. ($1 < 2$).

(a) $1 < 2$, because we can write $2 = 1 + m$, with $m = 1 \in \mathbb{N}$.
In fact, for all $n \in \mathbb{N}$, $n < \sigma(n)$ since $\sigma(n) = n + 1$.

(b) The equation $1 = 2 + m$ does not have a solution $m \in \mathbb{N}$. Otherwise, we get $1 = (1 + 1) + m = 1 + (1 + m) = 1 + (m + 1) = (m + 1) + 1 = \sigma(m + 1)$, a contradiction to (N1). ◇

Examples 2.3 and 2.4(b), show arithmetic 'flaws' with the natural numbers, and these will remedied by the integers.

Exercise 2.11. (Transitivity of $<$).
Suppose that $m, n, k \in \mathbb{N}$ are such that $m < n$ and $n < k$. Prove that $m < k$.

Exercise 2.12. Let $m, n \in \mathbb{N}$ be such that $m < n$.
Show that for all $k \in \mathbb{N}$, $m + k < n + k$ and $m \times k < n \times k$.

Theorem 2.8. (Trichotomy law).
For all $m, n \in \mathbb{N}$, *exactly one of the following three statements holds*:

1° $m = n$.

2° $m < n$.

3° $m > n$.

Proof. To show this we will use the result from Exercise 2.5, p.18, saying that $m + k \neq m$ for all $m, k \in \mathbb{N}$. Thus, the cases 1° and 2° are mutually exclusive. Moreover, 1° and 3° are mutually exclusive. Finally, 2° and 3° are also mutually exclusive, since otherwise we have $n = m + k$ (by $m < n$) and $m = n + \ell$ (by $m > n$), giving $n = m + k = (n + \ell) + k = n + (\ell + k)$, contradicting the result from Exercise 2.5.

Now fix $m \in \mathbb{N}$. Define

$$S_m = \{n \in \mathbb{N} : \text{exactly one of the cases } 1^\circ, 2^\circ, 3^\circ \text{ holds}\}.$$

We claim that $1 \in S_m$. Indeed:

- If $m = 1$, then 1° holds.
- If $m \neq 1$, then it has a predecessor $\ell \in \mathbb{N}$ by Theorem 2.1, p.14, and so $m = \sigma(\ell) = \ell + 1 = 1 + \ell$. So $m > 1$, i.e., 3° holds for $n = 1$.

Now suppose $n \in S_m$, that is, exactly one of the cases 1°, 2°, 3° holds. We want to show that $\sigma(n) \in S_m$. We consider the three cases separately.

1° $m = n$. Then $\sigma(n) = \sigma(m) = m + 1$. So $m < \sigma(n)$, i.e., 2° holds for $\sigma(n)$.

2° $m < n$. Then $n = m + k$ for some $k \in \mathbb{N}$.
So $\sigma(n) = \sigma(m + k) = m + \sigma(k)$. So $m < \sigma(n)$, i.e., 2° holds for $\sigma(n)$.

3° $m > n$. Then $m = n + k$.
If $k = 1$, then $\sigma(n) = n + 1 = m$, and so we have case 1° for $\sigma(n)$.
If $k \neq 1$, then it has a predecessor: $k = \sigma(\ell)$ for an $\ell \in \mathbb{N}$. Then we have $m = n + k = n + \sigma(\ell) = \sigma(n + \ell) = \sigma(\ell + n) = \ell + \sigma(n) = \sigma(n) + \ell$. Thus we have $m > \sigma(n)$, i.e., case 3° holds for $\sigma(n)$.

Hence whenever $n \in S_m$, also $\sigma(n) \in S_m$. By (N3), $S_m = \mathbb{N}$. As $m \in \mathbb{N}$ was arbitrary, we have shown the trichotomy law for all $m, n \in \mathbb{N}$. □

The following exercise justifies calling $\sigma(n)$ the successor of $n \in \mathbb{N}$.

Exercise 2.13. For any $n \in \mathbb{N}$, show that there is no $m \in \mathbb{N}$ such that $n < m < n + 1$.
Hint: Argue by contradiction. Write $m = n + k$. Consider the cases $k = 1$ and $k \neq 1$.

Exercise 2.14. (Additive and multiplicative cancellation rules). Let $m, n, k \in \mathbb{N}$.
(1) Show that if $m + k = n + k$, then $m = n$.
(2) Show that if $m \times k = n \times k$, then $m = n$.
Hints: If $m \neq n$, what does the trichotomy law allow? Use Exercise 2.12, p.20.

If $m, n \in \mathbb{N}$, and either $m = n$ or $m < n$, then we write $m \leqslant n$. The symbol $\geqslant$ is defined similarly: $m \geqslant n$ if $m = n$ or $m > n$.

Exercise 2.15. Show that for all $n \in \mathbb{N}$, we have $1 \leqslant n$.

Exercise 2.16. Let $m, n \in \mathbb{N}$ be such that $m < n$. Prove that $m + 1 \leqslant n$.

Exercise 2.17. Let $m, n \in \mathbb{N}$. Show that the following are equivalent:
(1) $m < n$.
(2) For all $k \in \mathbb{N}$, $m + k < n + k$.
(3) There exists a $k \in \mathbb{N}$ such that $m + k < n + k$.

Exercise 2.18. Let $m, n \in \mathbb{N}$. Show that $m \times n = 1$ if and only if $m = n = 1$.

Exercise 2.19. Let $m, n \in \mathbb{N}$. Show that $n \times m \geqslant n$.
Also, show that if $m \neq 1$, then $n \times m > 1$.

Exercise 2.20. (Strong induction). Suppose $P(1), P(2), P(3), \cdots$ is a sequence of statements, one for each $n \in \mathbb{N}$. Let $P(1)$ be true, and whenever for some $n \in \mathbb{N}$ it so happens that $P(k)$ is true for all $k \in \{m \in \mathbb{N} : m \leqslant n\}$, we have that also $P(n+1)$ is true. Show that $P(n)$ is true for all $n \in \mathbb{N}$.

Definition 2.2. Let S be a nonempty subset of $\mathbb{N}$.
An element $\ell \in S$ is called a *least element of* S if for all $n \in S$, $\ell \leqslant n$.
An element $u \in S$ is called a *greatest element of* S if for all $n \in \mathbb{N}$, $n \leqslant u$.

Exercise 2.21. Show that $\mathbb{N}$ has no greatest element.
Hint: If u is a greatest element, then consider $\sigma(u)$.

Theorem 2.9. (Well-ordering principle).
Any nonempty subset of $\mathbb{N}$ *has a least element.*

Proof. Let S be a nonempty subset of $\mathbb{N}$. Define the set of 'lower bounds of S', $\Lambda = \{\ell \in \mathbb{N} : \text{ for all } n \in S,\ \ell \leqslant n\}$. Then $1 \in \Lambda$ by Exercise 2.15, p.21. Also, $\Lambda \neq \mathbb{N}$: indeed if $n \in S$, then $\sigma(n) > n$, and so by the trichotomy law, $\neg(\sigma(n) \leqslant n)$, showing $\sigma(n) \notin \Lambda$.

Thus there exists an $\ell_* \in \Lambda$ with $\sigma(\ell_*) \notin \Lambda$ (otherwise by (N3), $\Lambda = \mathbb{N}$). We claim that $\ell_* \in S$. Let $\ell_* \notin S$. We know that $\ell_* \leqslant n$ for all $n \in S$. But as ℓ_* cannot equal any $n \in S$, the trichotomy law shows $\ell_* < n$ for all $n \in S$. By Exercise 2.16, p.21, $\ell_* + 1 \leqslant n$ for all $n \in S$. But then $\sigma(\ell_*) = \ell_* + 1 \in \Lambda$, a contradiction.

Now $\ell_* \in \Lambda$ and $\ell_* \in S$ means that ℓ_* is a least element of S. □

Exercise 2.22. Let S be a nonempty subset of $\mathbb{N}$. Show that the least element of S (which exists, by the well-ordering principle) is unique. The least element of S is called its *minimum*, denoted by $\min S$. (Similarly a greatest element, *if* it exists, is unique, and it is called the *maximum of* S, and is denoted by $\max S$.)

2.7 Finite nonempty sets

If $n \in \mathbb{N}$, then we denote the set $\{m \in \mathbb{N} : m \leqslant n\}$ by $\{1, \cdots, n\}$.

Definition 2.3. (Finite and infinite sets).
A set S is *finite* if it is empty or there exists an $n \in \mathbb{N}$ and a bijective map $f : S \to \{1, \cdots, n\}$. A set S is *infinite* if it is not finite.

Theorem 2.10. *Let S be a nonempty finite set. Then there is a unique $n \in \mathbb{N}$ for which there is a bijective map from S to $\{1, \cdots, n\}$.*
(This n is then called the *cardinal number of* S).

To prove this theorem, we will first show the following two lemmas.

Lemma 2.1. *Let $n \in \mathbb{N}$, S be a nonempty set, $a \in S$, and $S\backslash\{a\} \neq \varnothing$. Then the following are equivalent:*
(1) *There is an injective map $f : S \to \{1, \cdots, n+1\}$.*
(2) *There is an injective map $g : S\backslash\{a\} \to \{1, \cdots, n\}$.*

Proof. (1)⇒(2): Let $f : S \to \{1, \cdots, n+1\}$ be an injective map.

First, suppose for all $x \in S$, $f(x) \neq n+1$. Define $g : S\backslash\{a\} \to \{1, \cdots, n\}$ by $g(x) = f(x)$ for all $x \in S\backslash\{a\}$. Then g is injective since f is injective, and we are done. The remaining case is when $n+1 \in \operatorname{ran} f$.

If $f(a) = n+1$, then we define $g : S\backslash\{a\} \to \{1, \cdots, n\}$ by $g(x) = f(x)$ for $x \in S\backslash\{a\}$. Then g is well-defined, since the range of values assumed by g does not include $n+1$: by the injectivity of f, we have, for all $x \in S$ with $x \neq a$, that $f(x) \neq f(a) = n+1$. Also, g is injective, since if $x, x' \in S\backslash\{a\}$ and $g(x) = g(x')$, then $f(x) = f(x')$, and by the injectivity of f, $x = x'$.

Now let $f(a) \in \{1, \cdots, n\}$, and $a' \in S$ be such that $f(a') = n+1$. Then $a' \neq a$ (otherwise $f(a) = f(a') = n+1 \notin \{1, \cdots, n\}$, a contradiction), and by the injectivity of f, also a' is the only element in S whose image is $n+1$. Define $g : S\backslash\{a\} \to \{1, \cdots, n\}$ by $g(x) = f(x)$ if $x \in S\backslash\{a, a'\}$, and $g(a') = f(a)$. Then g is well-defined as the range of values taken by g is contained in $\{1, \cdots, n\}$. Moreover, g is injective, since if $x, x' \in S\backslash\{a, a'\}$ are such that $g(x) = g(x')$, then $f(x) = f(x')$ and so $x = x'$ by the injectivity of f, and also, there is no element $x'' \in S\backslash\{a, a'\}$ such that $g(x'') = g(a')$ (since otherwise $f(x'') = f(a)$, which implies $x'' = a$, a contradiction).

(2)⇒(1): Let $g : S\backslash\{a\} \to \{1, \cdots, n\}$ be injective. Define $f : S \to \{1, \cdots, n+1\}$ by $f(x) = g(x)$ if $x \in S\backslash\{a\}$, and $f(a) = n+1$. If $x, x' \in S\backslash\{a\}$ are such that $f(x) = f(x')$, then $g(x) = g(x')$ and so $x = x'$. Moreover, if $x \in S\backslash\{a\}$, then $f(x) = g(x) \in \{1, \cdots, n\}$, and as $n+1 \notin \{1, \cdots, n\}$, we get $f(x) \neq n+1 = f(a)$. Hence f is injective. □

Lemma 2.2. *Let $m, n \in \mathbb{N}$. If there exists an injective map from $\{1, \cdots, m\}$ to $\{1, \cdots, n\}$, then $m \leqslant n$.*

Proof. Let T be the set of all $m \in \mathbb{N}$ such that for all $n \in \mathbb{N}$, if there exists an injective map from $\{1, \cdots, m\}$ to $\{1, \cdots, n\}$, then $m \leqslant n$. We want to show that $T = \mathbb{N}$. Firstly, $1 \in T$ because if $n \in \mathbb{N}$ and there exists an injective map from $\{1\}$ to $\{1, \cdots, n\}$, then $1 \leqslant n$ trivially (Exercise 2.15, p.21). Now suppose that $m \in T$. We want to show that $m+1 \in T$ too. Let $n \in \mathbb{N}$, and $f : \{1, \cdots, m+1\} \to \{1, \cdots, n\}$ be an injective map. As $m \in \mathbb{N}$, we have $m \geqslant 1$, and so $m+1 \geqslant 1+1 = 2$. If $n = 1$, then we get $f(1) = 1 = f(2)$, contradicting the injectivity of f. So $n \neq 1$. Hence n has a predecessor, say

$k \in \mathbb{N}$, that is, $n = \sigma(k) = k + 1$. Thus $\{1, \cdots, n\} = \{1, \cdots, k + 1\}$. Let $S = \{1, \cdots, m + 1\}$ and $a = m + 1 \in S$. Then $S \backslash \{a\} = \{1, \cdots, m\} \neq \varnothing$. Applying Lemma 2.1, p.23, to $S = \{1, \cdots, m + 1\}$ with $a = m + 1 \in S$, we get the existence of an injective map $g : \{1, \cdots, m\} \to \{1, \cdots, k\}$. By the induction hypothesis, $m \leqslant k$. Adding 1 to both sides, we get $m+1 \leqslant k+1 = n$, completing the induction step. Thus $T = \mathbb{N}$, as wanted. □

We are now ready to prove Theorem 2.10, p.22.

Proof. Let S be a nonempty finite set. Suppose that $n, N \in \mathbb{N}$ are such that $n < N$. Let $f_n : S \to \{1, \cdots, n\}$ and $f_N : S \to \{1, \cdots, N\}$ be bijections. Then $f_n \circ f_N^{-1} : \{1, \cdots, N\} \to \{1, \cdots, n\}$ is a bijection, and in particular, an injective map. By Lemma 2.2 above, we get $N \leqslant n$, a contradiction. □

Exercise 2.23. Show that $\mathbb{N}$ is not finite.

Exercise 2.24. In this exercise, we will show a useful variant of Lemma 2.1, p.23. Let $n \in \mathbb{N}$, S be a nonempty set, and $a \in S$. Show that the following are equivalent:
(1) There is a *bijective* map $f : S \to \{1, \cdots, n + 1\}$.
(2) $S \backslash \{a\} \neq \varnothing$ and there is a *bijective* map $g : S \backslash \{a\} \to \{1, \cdots, n\}$.

Exercise 2.25. Show that if A is a finite set, then every subset of A is finite. Moreover, show that if the subset of the finite set is nonempty, then the cardinal number of the subset is less than or equal to the cardinal number of A.
(It follows that if A has an infinite subset, then A is infinite.)

Exercise 2.26. Show that if A, B are finite, then $A \cup B$ is finite.
Hint: Consider the disjoint case first.

Exercise 2.27. Let A be a nonempty set and $n \in \mathbb{N}$.
Show that the following are equivalent:
(1) There is a surjective function $f : \{1, \cdots, n\} \to A$.
(2) There is a injective function $g : A \to \{1, \cdots, n\}$.
(3) A is finite and has cardinal number at most n.

Exercise 2.28. Suppose that A is an infinite set and B, C are nonempty sets. Let $f : B \to A$ be a surjective map and $g : A \to C$ be an injective map.
Prove that B, $\operatorname{ran} g$ and C are infinite sets.

Chapter 3

Integers

By the definition of $<$, given $m, n \in \mathbb{N}$, the equation $m + x = n$ is solvable in the unknown $x \in \mathbb{N}$ if and only if $m < n$. By the trichotomy law, it follows that that if $m \geqslant n$, then the equation $m + x = n$ is *not* solvable for $x \in \mathbb{N}$. Now the aim is to 'extend' the natural number system to a 'bigger' system, the integers $\mathbb{Z}$, with an extension of the operation $+$, which allows a unique solution to such an equation for *arbitrary* $m, n \in \mathbb{N}$. Intuitively, we just think of the symbol $n - m$ as being the integer solution x. As the ordered pair (m, n) determines this x, we could think of x as a pair (m, n), where $m, n \in \mathbb{N}$. But then we realise that the equation $m + x = n$ is equivalent to $(m + 1) + x = n + 1$, and so the integer x ought to be also the pair $(m + 1, n + 1)$. So we must identify the pairs (m, n) and $(m + 1, n + 1)$. In fact, we need to identify all pairs $(m, n), (m + 1, n + 1), (m + 2, n + 2), \cdots$, and think of it as being the integer x. This suggests that maybe there is an equivalence relation $\sim$ on $\mathbb{N} \times \mathbb{N}$ such that the equivalence class $[(m, n)]$ under $\sim$ containing (m, n) is exactly this list of ordered pairs. Note that we should deem the pairs $(k, \ell), (m, n)$ as equivalent, that is, $(k, \ell) \sim (m, n)$, if they yield the same solution x, and so we should have $\ell - k = n - m$, or rearranging, $k + n = \ell + m$. This prompts defining the relation $\sim$ by $(k, \ell) \sim (m, n)$ if $k + n = \ell + m$. This is a relation, and we check that it is an *equivalence* relation below.

Theorem 3.1. *Let $\sim$ be the relation on $\mathbb{N} \times \mathbb{N}$ defined as follows:*

$$\text{for } k, \ell, m, n \in \mathbb{N}, \quad (k, \ell) \sim (m, n) \text{ if } k + n = \ell + m.$$

Then $\sim$ is an equivalence relation.

Proof. We check that $\sim$ is reflexive, symmetric, and transitive.
Reflexivity: Let $(m, n) \in \mathbb{N} \times \mathbb{N}$. Then $(m, n) \sim (m, n)$ since $m + n = n + m$.
Symmetry: Suppose that $(k, \ell), (m, n) \in \mathbb{N} \times \mathbb{N}$, and we have $(k, \ell) \sim (m, n)$. Then $k + n = \ell + m$. Thus also $m + \ell = n + k$, showing that $(m, n) \sim (k, \ell)$.
Transitivity: Suppose that $(k, \ell), (m, n), (p, q) \in \mathbb{N} \times \mathbb{N}$ satisfy $(k, \ell) \sim (m, n)$ and $(m, n) \sim (p, q)$. Then we have $k + n = \ell + m$ and $m + q = n + p$. Hence

$(k+n)+(m+q)=(\ell+m)+(n+p)$, and so $(k+q)+(m+n)=(\ell+p)+(m+n)$. By the additive cancellation rule (Exercise 2.14, p.21), we conclude that $k+q=\ell+p$, and so $(k,\ell)\sim(p,q)$. □

The equivalence relation $\sim$ partitions $\mathbb{N}\times\mathbb{N}$ into equivalence classes. The equivalence class of $(m,n)\in\mathbb{N}\times\mathbb{N}$ is

$$\begin{aligned}[(m,n)] &= \{(k,\ell)\in\mathbb{N}\times\mathbb{N} : (k,\ell)\sim(m,n)\}\\ &= \{(k,\ell)\in\mathbb{N}\times\mathbb{N} : k+n=\ell+m\}.\end{aligned}$$

Definition 3.1. ($\mathbb{Z}$, the set of integers).
The *set* $\mathbb{Z}$ *of integers* is the set of equivalence classes of $\mathbb{N}\times\mathbb{N}$ under the equivalence relation $\sim$ described in Theorem 3.1, p.25.

The symbol $\mathbb{Z}$ comes from 'Zahlen', meaning 'numbers' in German.

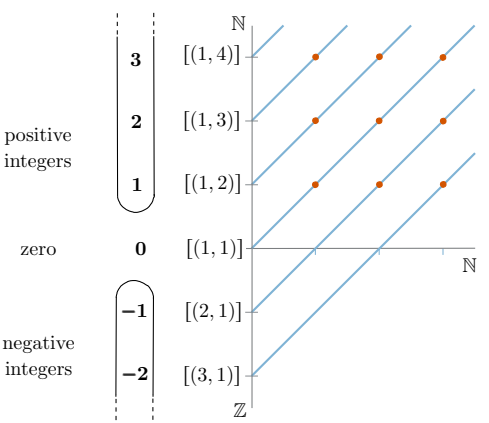
Example 3.1. Intuitively $[(1,1)]$ represents the integer '$(1-1=)\ 0$'. We have

$$\begin{aligned}[(1,1)] &= \{(m,n)\in\mathbb{N}\times\mathbb{N} : m+1=n+1\}=\{(m,n)\in\mathbb{N}\times\mathbb{N} : m=n\}\\ &= \{(m,m) : m\in\mathbb{N}\}=\{(1,1),\ (2,2),\ (3,3),\cdots\}.\end{aligned}$$

Similarly, $[(2,1)]$ represents the integer '$(1-2=)\ -1$'. We have

$$\begin{aligned}[(2,1)] &= \{(m,n)\in\mathbb{N}\times\mathbb{N} : m+1=n+2\}\\ &= \{(m,n)\in\mathbb{N}\times\mathbb{N} : m=n+1\}\\ &= \{(n+1,n) : n\in\mathbb{N}\}=\{(2,1),\ (3,2),\ (4,3),\cdots\}.\end{aligned}$$

The following picture depicts the equivalence classes visually. ◇



The following exercise allows us to view each natural number as an integer.

Exercise 3.1. ($\mathbb{N}\subset\mathbb{Z}$). Show that the map $\mathbb{N}\ni n\mapsto[(1,\sigma(n))]\in\mathbb{Z}$ is injective.

Exercise 3.2. Show that if $n,m,k\in\mathbb{N}$, then $[(m+k,n+k)]=[(m,n)]$.

3.1 Addition

We know that we can think of the integer $[(m,n)]$ as $n-m$ in the notation from elementary school. So to add $[(k,\ell)]$ and $[(m,n)]$, we ought to get $(\ell-k)+(n-m)=(\ell+n)-(k+m)$, which is the integer $[(k+m,\ell+n)]$. This motivates the following definition.

Definition 3.2. (Addition in $\mathbb{Z}$).
The *sum* $\boldsymbol{a}+\boldsymbol{b}$ of $\boldsymbol{a}=[(k,\ell)]$ and $\boldsymbol{b}=[(m,n)]\in\mathbb{Z}$, is defined to be the integer $\boldsymbol{a}+\boldsymbol{b}=[(k,\ell)]+[(m,n)]:=[(k+m,\ell+n)]$.

We ought to check that this notion is *well-defined*. What does this mean? Firstly, an integer $\boldsymbol{a}\in\mathbb{Z}$ is an equivalence class, and so it is a set with many members, and when we write $\boldsymbol{a}=[(k,\ell)]$, we have just picked *one* member (k,ℓ) belonging to the equivalence class $\boldsymbol{a}$. Then we know that the equivalence class $[(k,\ell)]$ corresponding to this (k,ℓ) is equal to $\boldsymbol{a}$. Secondly, after picking such representatives (k,ℓ) and (m,n) for $\boldsymbol{a}$ and $\boldsymbol{b}$, we are defining the sum $\boldsymbol{a}+\boldsymbol{b}$ by taking the equivalence class of $(k+m,\ell+n)$. But if we had chosen different representatives, say $(k',\ell')\in\boldsymbol{a}$, and $(m',n')\in\mathbf{b}$, do we get the same integer? Thus we ask: Is $[(k'+m',\ell'+n')]=[(k+m,\ell+n)]$? This is the question of well-defined-ness. We now check that the answer is 'yes'.

Theorem 3.2. (Addition is well-defined). *If* $k,\ell,m,n,k',\ell',m',n'\in\mathbb{N}$ *are such that* $(k,\ell)\sim(k',\ell')$ *and* $(m,n)\sim(m',n')$, *then*

$$(k+m,\ell+n)\sim(k'+m',\ell'+n').$$

(*And so* $[(k+m,\ell+n)]=[(k'+m',\ell'+n')]$.)

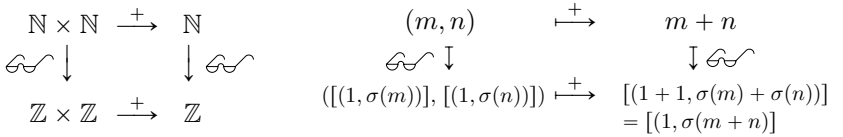
Proof. As $(k,\ell)\sim(k',\ell')$ and $(m,n)\sim(m',n')$, we know that

$$\begin{aligned} k+\ell' &= \ell+k', \\ m+n' &= n+m'. \end{aligned}$$

Adding, $(k+\ell')+(m+n')=(\ell+k')+(n+m')$. By the associativity and commutativity of addition in $\mathbb{N}$, $(k+m)+(\ell'+n')=(\ell+n)+(k'+m')$. Hence $(k+m,\ell+n)\sim(k'+m',\ell'+n')$. □

Ideally, we should use a different symbol for addition, since we already used $+$ for the addition of natural numbers, but the following result shows that the new addition is in fact an extension of the old addition. Recall that by putting on our 'integer glasses', the natural number $n\in\mathbb{N}$ is the integer $[(1,\sigma(n))]\in\mathbb{Z}$.

Theorem 3.3. *If* $m,n\in\mathbb{N}$, *then* $[(1,\sigma(m))]+[(1,\sigma(n))]=[(1,\sigma(m+n))]$.



Proof. To show that $[(1+1,\sigma(m)+\sigma(n))]=[(1,\sigma(m+n))]$, we need to check that $(1+1,\sigma(m)+\sigma(n))\sim(1,\sigma(m+n))$. But this is clear using the commutativity and associativity of natural number addition:

$$\begin{aligned}(1+1)+\sigma(m+n) &= (1+1)+((m+n)+1)=((m+1)+(n+1))+1\\ &= (\sigma(m)+\sigma(n))+1.\end{aligned}$$

□

Example 3.2. (**0** and additive inverses).
We define *zero* to be the integer $\mathbf{0}=[(1,1)]\in\mathbb{Z}$. For any $\boldsymbol{a}=[(m,n)]\in\mathbb{Z}$,

$$\boldsymbol{a}+\mathbf{0}=[(m,n)]+[(1,1)]=[(m+1,n+1)]\overset{(*)}{=}[(m,n)]=\boldsymbol{a}.$$

Justification of $(*)$: $(m+1,n+1)\sim(m,n)$ since $(m+1)+n=(n+1)+m$. In Exercise 3.3 below, we will show that addition in $\mathbb{Z}$ is commutative, using which it follows that $\mathbf{0}+\boldsymbol{a}=\boldsymbol{a}+\mathbf{0}=\boldsymbol{a}$.

For an integer $\boldsymbol{a}=[(m,n)]\in\mathbb{Z}$, we define its *additive inverse* to be the[1] integer $-\boldsymbol{a}:=[(n,m)]$. Then we have

$$\begin{aligned}\boldsymbol{a}+(-\boldsymbol{a})&=[(m,n)]+[(n,m)]=[(m+n,n+m)]=[(1,1)]=\mathbf{0},\\ -\boldsymbol{a}+\boldsymbol{a}&=[(n,m)]+[(m,n)]=[(n+m,m+n)]=[(1,1)]=\mathbf{0}.\end{aligned}$$

We note that $-(-\boldsymbol{a})=-[(n,m)]=[(m,n)]=\boldsymbol{a}$.

In particular, if $n\in\mathbb{N}$, and we view this as the integer $[(1,\sigma(n))]$, then its additive inverse in $\mathbb{Z}$ is the integer $[(\sigma(n),1)]$. ◇

Exercise 3.3. Prove that addition in $\mathbb{Z}$ is associative and commutative.

Exercise 3.4. (Additive cancellation rule).
Let $\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\in\mathbb{Z}$ be such that $\boldsymbol{a}+\boldsymbol{c}=\boldsymbol{b}+\boldsymbol{c}$. Show that $\boldsymbol{a}=\boldsymbol{b}$.

We now show that result that prompted the construction of $\mathbb{Z}$.

Theorem 3.4.
Let $\boldsymbol{a},\boldsymbol{b}\in\mathbb{Z}$. Then there exists a unique solution $\boldsymbol{x}\in\mathbb{Z}$ to $\boldsymbol{a}+\boldsymbol{x}=\boldsymbol{b}$.

Proof. Set $\boldsymbol{x}=\boldsymbol{b}+(-\boldsymbol{a})\in\mathbb{Z}$. Using the commutativity and associativity of addition in $\mathbb{Z}$, we get $\boldsymbol{a}+\boldsymbol{x}=\boldsymbol{a}+(\boldsymbol{b}+(-\boldsymbol{a}))=\boldsymbol{b}+(\boldsymbol{a}+(-\boldsymbol{a}))=\boldsymbol{b}+\mathbf{0}=\boldsymbol{b}$. To show uniqueness, suppose that $\boldsymbol{x},\boldsymbol{x}'\in\mathbb{Z}$ are such that $\boldsymbol{a}+\boldsymbol{x}=\boldsymbol{b}=\boldsymbol{a}+\boldsymbol{x}'$. Adding $-\boldsymbol{a}$ to both sides yields $-\boldsymbol{a}+(\boldsymbol{a}+\boldsymbol{x})=-\boldsymbol{a}+(\boldsymbol{a}+\boldsymbol{x}')$, and thanks to associativity, $(-\boldsymbol{a}+\boldsymbol{a})+\boldsymbol{x}=(-\boldsymbol{a}+\boldsymbol{a})+\boldsymbol{x}'$, i.e., $\mathbf{0}+\boldsymbol{x}=\mathbf{0}+\boldsymbol{x}'$. Thus $\boldsymbol{x}=\boldsymbol{x}'$. □

[1] This is well-defined, since if $(m,n)\sim(m',n')$, then $(n,m)\sim(n',m')$.

Theorem 3.5. *For any integer* $\boldsymbol{a} \in \mathbb{Z}$, *exactly one of the following hold*:
1° $\boldsymbol{a} = [(1, \sigma(k))]$ *for a* $k \in \mathbb{N}$.
(*Integer corresponding to the natural number* $k \in \mathbb{N}$.)
2° $\boldsymbol{a} = \boldsymbol{0}$.
3° $\boldsymbol{a} = [(\sigma(k), 1)] = -[(1, \sigma(k))]$ *for a* $k \in \mathbb{N}$.
(*Additive inverse of the integer corresponding to* $k \in \mathbb{N}$.)

Proof. 1° and 2° are mutually exclusive, since otherwise $(1, \sigma(k)) \sim (1, 1)$, giving $1 + 1 = \sigma(k) + 1$, so that $\sigma(k) = 1$, contradicting (N1). Similarly 2° and 3° are mutually exclusive. Finally 1° and 3° are mutually exclusive: otherwise $(\sigma(k'), 1) \sim (1, \sigma(k))$ for some $k, k' \in \mathbb{N}$, i.e., $k' + 1 + k + 1 = 1 + 1$, so that $\sigma(k + k') = 1$, which is again a contradiction to (N1).
Let $\boldsymbol{a} = [(m, n)] \in \mathbb{Z}$. By the trichotomy law for $\mathbb{N}$, one of the following hold:
1° $m < n$. Then $n = m + k$ for some $k \in \mathbb{N}$.
If $m = 1$, then we have $\boldsymbol{a} = [(m, n)] = [(1, 1 + k)] = [(1, \sigma(k))]$.
If $m \neq 1$, then m has a predecessor, say $\ell \in \mathbb{N}$, i.e., $m = \sigma(\ell) = \ell + 1$, and so $\boldsymbol{a} = [(m, n)] = [(m, m + k)] = [(\ell + 1, \ell + 1 + k)] = [(1, \sigma(k))]$.
2° $m = n$. Then $\boldsymbol{a} = [(m, m)] = \boldsymbol{0}$.
3° $m > n$. Just as in 1°, if $m = n + k$, then we have $\boldsymbol{a} = [(\sigma(k), 1)]$. □

So the set $\mathbb{Z}$ is partitioned into three mutually disjoint subsets:
- the 'positive integers' (integers corresponding to the natural numbers),
- zero, and
- the 'negative integers' (additive inverses of the natural numbers).

Review the picture on p.26. The terminology (positive/negative) used above will become clear when we learn about the extension of the order relation from $\mathbb{N}$ to the integers in §3.3.

3.2 Multiplication

For simplicity, we denote multiplication in $\mathbb{N}$ with a $\cdot$ instead of the symbol $\times$ used previously, or simply by placing the factors next to each other: for example we will write $m \cdot n$ or mn instead of $m \times n$, where $m, n \in \mathbb{N}$.

To define multiplication formally, again we note that we expect the product of $[(m, n)]$ and $[(k, \ell)]$ to be the integer (in elementary school notation) $(\ell - k)(n - m) = \ell n + km - kn - \ell m$, that is, $[(kn + \ell m, km + \ell n)]$. This motivates the following:

Definition 3.3. (Multiplication in $\mathbb{Z}$).
The *product* $\boldsymbol{a} \cdot \boldsymbol{b}$ of $\boldsymbol{a} = [(k, \ell)] \in \mathbb{Z}$ and $\boldsymbol{b} = [(m, n)] \in \mathbb{Z}$ is defined to be the integer $\boldsymbol{a} \cdot \boldsymbol{b} = [(k, \ell)] \cdot [(m, n)] := [(kn + \ell m, km + \ell n)]$.

Again we need to check well-definedness.

Theorem 3.6. (Multiplication is well-defined).
If $k,\ell,m,n,k',\ell',m',n'\in\mathbb{N}$ *are such that* $(k,\ell)\sim(k',\ell')$ *and* $(m,n)\sim(m',n')$, *then* $(kn+\ell m, km+\ell n)\sim(k'n'+\ell'm', k'm'+\ell'n')$.
(*Thus* $[(k'n'+\ell'm', k'm'+\ell'n')]=[(kn+\ell m, km+\ell n)]$.)

Proof. As $(k,\ell)\sim(k',\ell')$ and $(m,n)\sim(m',n')$,

$$k+\ell' = \ell+k' \qquad (\star)$$
$$m+n' = n+m'. \qquad (\star\star)$$

We want to show $(kn+\ell m, km+\ell n)\sim(k'n'+\ell'm', k'm'+\ell'n')$, that is,

$$kn+\ell m+k'm'+\ell'n' = km+\ell n+k'n'+\ell'm'. \qquad (*)$$

The terms on the left-hand side suggest considering $n\cdot(\star)$, $m\cdot(\star)$, $k'\cdot(\star\star)$, $\ell'\cdot(\star\star)$, which are, respectively:

$$kn+\ell'n = \ell n+k'n$$
$$\ell m+k'm = km+\ell'm$$
$$k'm'+k'n = k'n'+k'm$$
$$\ell'n'+\ell'm = \ell'm'+\ell'n.$$

Adding these, we obtain

$$\left.\begin{array}{r} kn+\ell m+\ell'n'+k'm' \\ +(k'n+\ell'm+k'm+\ell'n)\end{array}\right\} = \left\{\begin{array}{l}\ell n+km+k'n'+\ell'm' \\ +(k'n+\ell'm+k'm+\ell'n).\end{array}\right.$$

By the additive cancellation rule in $\mathbb{N}$, we obtain $(*)$. □

Next we show that multiplication in $\mathbb{Z}$ is an extension of the multiplication in $\mathbb{N}$.

Theorem 3.7. *If* $m,n\in\mathbb{N}$, *then* $[(1,\sigma(m))]\cdot[(1,\sigma(n))]=[(1,\sigma(mn))]$.

Proof. We must show $[(1\cdot\sigma(n)+\sigma(m)\cdot 1, 1\cdot 1+\sigma(m)\sigma(n))]=[(1,\sigma(mn))]$, that is, $(\sigma(m)+\sigma(n), 1+\sigma(m)\sigma(n))\sim(1,\sigma(mn))$, or equivalently,

$$(\sigma(m)+\sigma(n))+\sigma(mn)=(1+\sigma(m)\sigma(n))+1.$$

But this is readily verified using the arithmetic rules in $\mathbb{N}$:

$$\begin{aligned}(\sigma(m)+\sigma(n))+\sigma(mn) &= ((m+1)+(n+1))+(mn+1)\\ &= mn+m\cdot 1+1\cdot n+1\cdot 1+1+1\\ &= m(n+1)+1(n+1)+1+1\\ &= (m+1)(n+1)+1+1\\ &= \sigma(m)\sigma(n)+1+1=(1+\sigma(m)\sigma(n))+1.\end{aligned}$$

□

In accordance with what we are used to, later on we will often skip writing $\cdot$ when denoting integer multiplication.

Exercise 3.5. For all $\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\in\mathbb{Z}$, show that

- (Associativity) $\boldsymbol{a}\cdot(\boldsymbol{b}\cdot\boldsymbol{c})=(\boldsymbol{a}\cdot\boldsymbol{b})\cdot\boldsymbol{c}$.
- (Commutativity) $\boldsymbol{a}\cdot\boldsymbol{b}=\boldsymbol{b}\cdot\boldsymbol{a}$.
- (Distributivity) $\boldsymbol{a}\cdot(\boldsymbol{b}+\boldsymbol{c})=\boldsymbol{a}\cdot\boldsymbol{b}+\boldsymbol{a}\cdot\boldsymbol{c}$.

Example 3.3. (Multiplicative identity).
We know that for any natural number $n \in \mathbb{N}$, $1 \times n = n = n \times 1$. We now show that if we view $1 \in \mathbb{N}$ as an integer, that is, $\mathbf{1} := [(1, \sigma(1))] = [(1, 2)]$, then it continues to serve as a multiplicative identity, but now in the bigger set $\mathbb{Z}$. So we want to check that for all $\boldsymbol{a} = [(m, n)] \in \mathbb{Z}$, $m, n \in \mathbb{N}$, we have

$$\boldsymbol{a} \cdot \mathbf{1} = \boldsymbol{a} = \mathbf{1} \cdot \boldsymbol{a}.$$

As $m \cdot 2 = m \cdot (1 + 1) = m \cdot 1 + m \cdot 1 = m + m$ and $n \cdot 2 = n + n$, we have

$$\begin{aligned} \boldsymbol{a} \cdot \mathbf{1} &= [(m, n)] \cdot [(1, 2)] \\ &= [(m \cdot 2 + n \cdot 1, m \cdot 1 + n \cdot 2)] \\ &= [(m + m + n, m + n + n)] \\ &= [(m, n)], \end{aligned}$$

where the last equality follows from $(m + m + n, m + n + n) \sim (m, n)$. ◇

Exercise 3.6. Let $\boldsymbol{a} \in \mathbb{Z}$. Prove that $-\boldsymbol{a} = (-\mathbf{1}) \cdot \boldsymbol{a}$.
In particular, $(-\mathbf{1}) \cdot (-\mathbf{1}) = -(-\mathbf{1}) = \mathbf{1}$.

Exercise 3.7. Let $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{Z}$, and suppose $\boldsymbol{a} = \mathbf{0}$ or $\boldsymbol{b} = \mathbf{0}$. Prove that $\boldsymbol{a} \cdot \boldsymbol{b} = \mathbf{0}$.

The converse of the result from Exercise 3.7 holds.

Theorem 3.8. *Let $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{Z}$ and $\boldsymbol{a} \cdot \boldsymbol{b} = \mathbf{0}$. Then $\boldsymbol{a} = \mathbf{0}$ or $\boldsymbol{b} = \mathbf{0}$.*

Proof. Let $\boldsymbol{a} = [(k, \ell)]$ and $\boldsymbol{b} = [(m, n)]$, where $k, \ell, m, n \in \mathbb{N}$. Then

$$[(1, 1)] = \mathbf{0} = \boldsymbol{a} \cdot \boldsymbol{b} = [(k, \ell)] \cdot [(m, n)] = [(kn + \ell m, km + \ell n)],$$

and so $1 + km + \ell n = 1 + kn + \ell m$, giving $km + \ell n = kn + \ell m$. By the trichotomy law, we have the following three mutually exclusive cases:

1° $k = \ell$. Then $\boldsymbol{a} = [(k, k)] = [(1, 1)] = \mathbf{0}$. (Here we used $(k, k) \sim (1, 1)$.)

2° $k > \ell$. Then $k = \ell + j$ for some $j \in \mathbb{N}$. Hence $km + \ell n = kn + \ell m$ gives $\ell m + jm + \ell n = \ell n + jn + \ell m$, i.e., $jm = jn$ by the additive cancellation rule. The multiplicative cancellation rule gives $m = n$. So we obtain that $\boldsymbol{b} = [(m, m)] = [(1, 1)] = \mathbf{0}$.

3° $k < \ell$. Then $\ell = k + j$ for some $j \in \mathbb{N}$. Hence $km + \ell n = kn + \ell m$ gives $km + kn + jn = kn + km + jm$, i.e., $jn = jm$ by the additive cancellation rule. The multiplicative cancellation rule gives $m = n$. So we obtain that $\boldsymbol{b} = [(m, m)] = [(1, 1)] = \mathbf{0}$. □

Exercise 3.8. In this exercise, we assume familiarity with one variable calculus. Let $[0, 1] = \{x \in \mathbb{R} : 0 \leqslant x \leqslant 1\}$. Consider the set $C[0, 1]$ of all continuous functions on $[0, 1]$. For $f, g \in C[0, 1]$, define $f \cdot g \in C[0, 1]$ by $(f \cdot g)(x) = f(x)g(x)$ for all $x \in [0, 1]$. Let $\mathbf{0} \in C[0, 1]$ be the constant function that is identically equal to 0 on $[0, 1]$. Give an example to show that $f \cdot g = \mathbf{0}$, but neither f nor g equals $\mathbf{0}$.

Exercise 3.9. (Multiplicative cancellation rule).
Let $\boldsymbol{a} \in \mathbb{Z} \backslash \{\mathbf{0}\}$, $\boldsymbol{b}, \boldsymbol{c} \in \mathbb{Z}$ be such that $\boldsymbol{a} \cdot \boldsymbol{b} = \boldsymbol{a} \cdot \boldsymbol{c}$. Show that $\boldsymbol{b} = \boldsymbol{c}$.

3.3 Order

We can extend the order relation from $\mathbb{N}$ to $\mathbb{Z}$ as follows.

Definition 3.4. (Order $<$ on $\mathbb{Z}$).
For integers $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{Z}$, we say that $\boldsymbol{a}$ *is less than* $\boldsymbol{b}$, written as $\boldsymbol{a} < \boldsymbol{b}$ if for some $n \in \mathbb{N}$, $\boldsymbol{b} = \boldsymbol{a} + [(1, \sigma(n))]$. The notation $\boldsymbol{b} > \boldsymbol{a}$ means $\boldsymbol{a} < \boldsymbol{b}$.

For all $n \in \mathbb{N}$, $[(1, \sigma(n))] > \mathbf{0}$: $[(1, \sigma(n))] = \mathbf{0} + [(1, \sigma(n))]$.

Exercise 3.10. (Order on $\mathbb{Z}$ extends the order on $\mathbb{N}$). Let $m, n \in \mathbb{N}$. Show that $m < n$ if and only if $[(1, \sigma(m))] < [(1, \sigma(n))]$. *Hint*: E.g. use Theorem 3.3, p.27.

Exercise 3.11. (Transitivity of $<$). Let $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c} \in \mathbb{Z}$ be such that $\boldsymbol{a} < \boldsymbol{b}$ and $\boldsymbol{b} < \boldsymbol{c}$. Show that $\boldsymbol{a} < \boldsymbol{c}$. *Hint*: Use Theorem 3.3, p.27.

Exercise 3.12. Let $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c} \in \mathbb{Z}$ be such that $\boldsymbol{a} < \boldsymbol{b}$. Show that $\boldsymbol{a} + \boldsymbol{c} < \boldsymbol{b} + \boldsymbol{c}$.

Exercise 3.13. Let $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c} \in \mathbb{Z}$ be such that $\boldsymbol{a} < \boldsymbol{b}$ and $\boldsymbol{c} > \mathbf{0}$. Show that $\boldsymbol{a} \cdot \boldsymbol{c} < \boldsymbol{b} \cdot \boldsymbol{c}$.

Exercise 3.14. Let $\boldsymbol{a} \in \mathbb{Z}$ and $\mathbf{1} = [(1, 2)]$. Show that there is no $\boldsymbol{b} \in \mathbb{Z}$ such that $\boldsymbol{a} < \boldsymbol{b} < \boldsymbol{a} + \mathbf{1}$.

Theorem 3.9. (Trichotomy law).
Let $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{Z}$. *Then exactly one of the following holds*:
1° $\boldsymbol{a} < \boldsymbol{b}$.
2° $\boldsymbol{a} = \boldsymbol{b}$.
3° $\boldsymbol{a} > \boldsymbol{b}$.

Proof. By Theorem 3.4, p.28, there exists a unique integer $\boldsymbol{x}$ such that $\boldsymbol{b} = \boldsymbol{a} + \boldsymbol{x}$. For this integer $\boldsymbol{x}$, it follows from Theorem 3.5, p.29, that exactly one of the following three mutually exclusive options is valid:
(*i*) $\boldsymbol{x} = [(1, \sigma(k))]$ for some $k \in \mathbb{N}$. Then $\boldsymbol{a} < \boldsymbol{b}$, that is, 1° holds.
(*ii*) $\boldsymbol{x} = \mathbf{0}$. Then $\boldsymbol{b} = \boldsymbol{a}$, that is, 2° holds.
(*iii*) $\boldsymbol{x} = [(\sigma(k), 1)]$ for some $k \in \mathbb{N}$. Then case 3° holds, i.e., $\boldsymbol{a} > \boldsymbol{b}$, since

$$\begin{aligned} \boldsymbol{b} + [(1, \sigma(k))] &= (\boldsymbol{a} + \boldsymbol{x}) + [(1, \sigma(k))] \\ &= \boldsymbol{a} + (\boldsymbol{x} + [(1, \sigma(k))]) \\ &= \boldsymbol{a} + [(\sigma(k) + 1, 1 + \sigma(k))] \\ &= \boldsymbol{a} + [(1, 1)] = \boldsymbol{a} + \mathbf{0} = \boldsymbol{a}. \end{aligned}$$

□

Exercise 3.15. Let $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{Z}$. Show that the following are equivalent:
(1) $\boldsymbol{a} \cdot \boldsymbol{b} > \mathbf{0}$.
(2) $(\boldsymbol{a} > \mathbf{0}$ and $\boldsymbol{b} > \mathbf{0})$ or $(\boldsymbol{a} < \mathbf{0}$ and $\boldsymbol{b} < \mathbf{0})$.

Exercise 3.16. Let $\boldsymbol{a} \in \mathbb{Z} \backslash \{\mathbf{0}\}$. Show that $\boldsymbol{a}^2 := \boldsymbol{a} \cdot \boldsymbol{a} > \mathbf{0}$.

If $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{Z}$, and either $\boldsymbol{a} = \boldsymbol{b}$ or $\boldsymbol{a} < \boldsymbol{b}$, then we write $\boldsymbol{a} \leqslant \boldsymbol{b}$. The symbol $\geqslant$ is defined similarly: $\boldsymbol{a} \geqslant \boldsymbol{b}$ if $\boldsymbol{a} = \boldsymbol{b}$ or $\boldsymbol{a} > \boldsymbol{b}$.

Exercise 3.17. (Failure of the well-ordering principle for $\mathbb{Z}$). If S is a nonempty subset of $\mathbb{Z}$, then an element $\boldsymbol{\ell} \in S$ is called a *least element of* S if for all $\boldsymbol{n} \in S$, $\boldsymbol{\ell} \leqslant \boldsymbol{n}$. Show that $S := \mathbb{Z}$ does not have a least element.

Exercise 3.18. (Validity of the well-ordering principle for subsets of $\mathbb{Z}$ that are *bounded below*). Let S be a nonempty subset of $\mathbb{Z}$, which has a *lower bound*, namely an element $\boldsymbol{\ell} \in \mathbb{Z}$ such that $\boldsymbol{\ell} \leqslant \boldsymbol{m}$ for all $\boldsymbol{m} \in S$. Show that S has a least element. *Hint*: Consider $\{\boldsymbol{m} - \boldsymbol{\ell} : \boldsymbol{m} \in S\}$.

The least element is unique, since if $\boldsymbol{\ell}_1, \boldsymbol{\ell}_2 \in S$ are least elements of S, then we have $\boldsymbol{\ell}_1 \leqslant \boldsymbol{\ell}_2$ (since $\boldsymbol{\ell}_1$ is a least element of S and $\boldsymbol{\ell}_2 \in S$), and also $\boldsymbol{\ell}_2 \leqslant \boldsymbol{\ell}_1$ (since $\boldsymbol{\ell}_2$ is a least element of S and $\boldsymbol{\ell}_1 \in S$), together implying that $\boldsymbol{\ell}_1 = \boldsymbol{\ell}_2$.

We claim that if T is a nonempty subset of $\mathbb{Z}$, which has an *upper bound* (namely an element $\boldsymbol{u} \in \mathbb{Z}$ such that $\boldsymbol{m} \leqslant \boldsymbol{u}$ for all $\boldsymbol{m} \in T$), then T has a greatest element (i.e., an element $\boldsymbol{t}_* \in T$ such that $\boldsymbol{m} \leqslant \boldsymbol{t}_*$ for all $\boldsymbol{m} \in T$). Consider the set $S := -T := \{-\boldsymbol{m} : \boldsymbol{m} \in T\}$. Then S has $-\boldsymbol{u}$ as a lower bound, and so possesses a least element, say $\boldsymbol{\ell} \in S$. Then $\boldsymbol{t}_* = -\boldsymbol{\ell} \in T$ is a greatest element in T. Again, the greatest element of T is easily seen to be unique.

Example 3.4. (Motivation for the rational number system).
There is no integer $\boldsymbol{x} \in \mathbb{Z}$ such that $\mathbf{2} \cdot \boldsymbol{x} = \mathbf{1}$, where $\mathbf{2} = [(1, \sigma(2))]$ and $\mathbf{1} = [(1, \sigma(1))]$. Indeed, as $\mathbf{1} > \mathbf{0}$ and $\mathbf{2} > \mathbf{0}$, it follows from Exercise 3.15, p.32, that $\boldsymbol{x} > \mathbf{0}$. So $\boldsymbol{x} = \mathbf{0} + [(1, \sigma(n))] = [(1, \sigma(n))]$ for some $n \in \mathbb{N}$. Then $n \neq 1$ (otherwise $\mathbf{2} \cdot \boldsymbol{x} = \mathbf{2} \cdot [(1, \sigma(1))] = \mathbf{2} \cdot \mathbf{1} = \mathbf{2} \neq \mathbf{1}$). So n has a predecessor, say $\ell \in \mathbb{N}$, that is, $n = \sigma(\ell) = \ell + 1$. Then

$$\begin{aligned} \boldsymbol{x} &= [(1, \sigma(n))] = [(1, \sigma(\ell + 1))] \\ &= [(1, \sigma(1))] + [(1, \sigma(\ell)] \\ &= [(1, 2)] + [(1, \sigma(\ell)] \\ &= \mathbf{1} + [(1, \sigma(\ell)], \end{aligned}$$

showing $\boldsymbol{x} > \mathbf{1}$. By Exercise 3.13, p.32, $\mathbf{2} \cdot \boldsymbol{x} > \mathbf{2} \cdot \mathbf{1} = \mathbf{2} > \mathbf{1}$. But then by the trichotomy law, $\mathbf{2} \cdot \boldsymbol{x} \neq \mathbf{1}$. ◇

The previous example shows that given $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{Z}$, with $\boldsymbol{a} \neq \mathbf{0}$, the equation $\boldsymbol{a} \cdot \boldsymbol{x} = \boldsymbol{b}$ in the unknown $\boldsymbol{x} \in \mathbb{Z}$ is not always solvable. The set $\mathbb{Q}$ of rational numbers remedies this 'flaw' with integers. In the next chapter, we will learn about the construction of $\mathbb{Q}$ and its arithmetic operations. From now on, for $n \in \mathbb{N}$, we will often denote the integer $[(1, \sigma(n))]$ by n, the integer $[(1, 1)]$ by 0, and the integer $[(\sigma(n), 1)]$ by $-n$. By Theorem 3.5, p.29, $\mathbb{Z} = \{\cdots, -3, -2, -1, 0, 1, 2, 3, \cdots\}$. The elements of $\{1, 2, 3, \cdots\}$ are called the *positive integers*, and those of $\{-1, -2, -3, \cdots\}$ the *negative integers*.

Exercise 3.19. Determine $3 - 5$ and $2 \cdot (-2)$ using the definitions.

Chapter 4

Rational numbers

Before we give the formal construction of the rational numbers let us make some intuitive motivational remarks which will justify this construction to some extent. For simplicity, we denote multiplication in $\mathbb{Z}$ by simply by placing the factors next to each other, i.e., we write mn instead of $m \cdot n$ for the integer product of $m, n \in \mathbb{Z}$.

We had seen in the previous chapter that the equation $2x = 1$ has no integer solution x. More generally, given an integer b and a nonzero[1] integer a, the equation $ax = b$ may fail to have an integer solution x. But we would like to extend the integers to a bigger number system, so that viewing a and b as elements of this new number system, we are able to solve for x. The x is meant to be the solution to $ax = b$, but what should it be? Well, it comes from the equation $ax = b$, so why not just associate x with the data which describes the equation, which is this pair (b, a) of integers, where the first one is an integer b, and the second one is a non-zero integer a. So we simply think of this x as the solution to the equation $ax = b$ which is described by the pair $(b, a) \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$. But we note that if x is a solution to $ax = b$, then if one multiplies this throughout by 2, we get $(2a)x = 2b$, i.e., the same x also satisfies the equation described by the data $(2b, 2a)$. So if we associate the rational number with the pair (b, a), then it should also be associated with the pair $(2b, 2a)$, because it is the same x which satisfies the equation $(2a)x = 2b$ too. Similarly, if one multiplies $ax = b$ by -1, then we get $(-a)x = -b$, so the same x should be associated with the pair $(-b, -a)$ too. We can keep playing this game, resulting in a whole list of ordered pairs, and somehow they should all be indistinguishable from each other when we are talking about the 'rational number x'. So this makes us suspect that perhaps there is an equivalence relation $\sim$ which can be given on the set $\mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$ such that this list associated with $ax = b$ is exactly an equivalence class $[(b, a)]$ containing (b, a) under that

[1] If $a = 0$, then this forces the left-hand side to be 0 and so b must also be 0, and we see that the solution x is no longer unique, as every $x \in \mathbb{Z}$ satisfies $0x = 0$.

equivalence relation, which can be thought of as the rational number x. What should this equivalence relation should be? That is, when would $(b', a'), (b, a)$ appear in the same list? From the above, we know that there is an integer k such that either $b' = kb$ and $a' = ka$, or $b = kb'$ and $a = ka'$. In either case, we do have $b'a = a'b$. So it is natural to define $(b', a') \sim (b, a)$ if $b'a = a'b$. We switch to using letters n, d instead of b, a, foreshadowing the labels 'numerator' and 'denominator'.

Definition 4.1. (Equivalence relation $\sim$ on $\mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$).
The relation $\sim$ on $\mathbb{Z}\times(\mathbb{Z}\backslash\{0\})$ is defined as follows:
If $(n, d), (n', d') \in \mathbb{Z}\times(\mathbb{Z}\backslash\{0\})$, then $(n, d) \sim (n', d')$ if $nd' = n'd$.

Proposition 4.1. *$\sim$ is an equivalence relation on $\mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$.*

Proof. We check that $\sim$ is reflexive, symmetric, and transitive.
Reflexivity: For $(n, d) \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$, clearly $(n, d) \sim (n, d)$ (as $nd = nd$).
Symmetry: Let $(n, d), (n', d') \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$ be such that $(n, d) \sim (n', d')$. Then $nd' = n'd$, i.e. $n'd = nd'$, showing $(n', d') \sim (n, d)$.
Transitivity: Suppose that $(n, d), (n', d'), (n'', d'') \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$ are such that $(n, d) \sim (n', d')$ and $(n', d') \sim (n'', d'')$. Then $nd' = n'd$ and $n'd'' = n''d'$.
1° Let $n' = 0$. From $nd' = n'd = 0d = 0$, we conclude that $n = 0$ (as $d' \neq 0$).
Similarly, $n''d' = n'd'' = 0d'' = 0$ implies $n'' = 0$ (as $d' \neq 0$).
But then $nd'' = 0d'' = 0 = 0d = n''d$, so that $(n, d) \sim (n'', d'')$.
2° Let $n' \neq 0$. We have $nd'(n'd'') = n'd(n''d')$, i.e., $nd''(n'd') = n''d(n'd')$. By the cancellation rule, $nd'' = n''d$, as $n' \neq 0$ and $d' \neq 0$. So $(n, d) \sim (n'', d'')$. □

Exercise 4.1. Let $(n, d) \in \mathbb{Z}\times(\mathbb{Z}\backslash\{0\})$ and $m \in \mathbb{Z}\backslash\{0\}$. Show that $(n, d) \sim (mn, md)$.

Definition 4.2. (The set $\mathbb{Q}$ of rational numbers). Each equivalence class of $\mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$ under the equivalence relation $\sim$ is called a *rational number*. The set of all rational numbers is denoted by $\mathbb{Q}$. Given $(n, d) \in \mathbb{Z}\times(\mathbb{Z}\backslash\{0\})$, the equivalence class $[(n, d)]$ of (n, d) is denoted by $\frac{n}{d}$, and the integer n is called the *numerator*, while d is called the *denominator*.

The choice of the letter $\mathbb{Q}$ is motivated by thinking of $\frac{n}{d}$ as a '**q**uotient'.

If $(n, d) \in \mathbb{Z}\times(\mathbb{Z}\backslash\{0\})$, then $(-n)d = ((-1)n)d = n((-1)d) = n(-d)$. So $(n, d) \sim (-n, -d)$ and $\boldsymbol{r} = \frac{n}{d} = \frac{-n}{-d}$. Hence, we can always write a given $\boldsymbol{r} \in \mathbb{Q}$, as $\boldsymbol{r} = \frac{n}{d}$, where $n \in \mathbb{Z}$ and $d \in \{1, 2, 3, \cdots\}$' = 'ℕ (viewed as integers). Let d_* be the least $d \in \mathbb{N}$ such that there exists an $n \in \mathbb{Z}$ with which $\boldsymbol{r} = [(n, d)]$. By the well-ordering principle in ℕ, d_* is well-defined. The corresponding $n \in \mathbb{Z}$ is unique: if $n, n' \in \mathbb{Z}$ are such that $[(n', d_*)] = \boldsymbol{r} = [(n, d_*)]$, then $n'd_* = nd_*$, and so $n = n'$. We denote the corresponding unique n by n_*.

4.1 Addition and multiplication

In elementary school, we learn that the sum of fractions $\frac{n}{d}, \frac{p}{q}$ is defined by $\frac{n}{d} + \frac{p}{q} = \frac{nq+pd}{dq}$. We formalise this below.

Definition 4.3. (Addition in $\mathbb{Q}$). Let the pairs $(n,d),(p,q) \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$. For $\boldsymbol{r} := \frac{n}{d} \in \mathbb{Q}$ and $\boldsymbol{s} := \frac{p}{q} \in \mathbb{Q}$, we define $\boldsymbol{r} + \boldsymbol{s} = \frac{n}{d} + \frac{p}{q} := \frac{nq+pd}{dq}$.

Note that if $d \neq 0$ and $q \neq 0$, then $dq \neq 0$.

Exercise 4.2. (Addition is well-defined).
Let the pairs $(n,d),(n',d'),(p,q),(p',q') \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$ be such that $(n,d) \sim (n',d')$ and $(p,q) \sim (p',q')$. Show that $(nq+pd, dq) \sim (n'q'+p'd', d'q')$.

Exercise 4.3. Show that addition in $\mathbb{Q}$ is associative and commutative.

Define the zero element $\mathbf{0} = \frac{0}{1} \in \mathbb{Q}$. Let $\boldsymbol{r} = \frac{n}{d} \in \mathbb{Q}$. Then

$$\boldsymbol{r} + \mathbf{0} = \boldsymbol{r} = \mathbf{0} + \boldsymbol{r},$$

because $\boldsymbol{r} + \mathbf{0} = \frac{n}{d} + \frac{0}{1} = \frac{n \cdot 1 + 0 \cdot d}{d \cdot 1} = \frac{n+0}{d} = \frac{n}{d} = \boldsymbol{r}$.

Exercise 4.4. For $\boldsymbol{r} = \frac{n}{d} \in \mathbb{Q}$, define $-\boldsymbol{r} = \frac{-n}{d} \in \mathbb{Q}$.
Show that this notion of $-\boldsymbol{r}$ is well-defined. Prove that $\boldsymbol{r} + (-\boldsymbol{r}) = \mathbf{0}$.

Definition 4.4. (Multiplication in $\mathbb{Q}$). Let $(n,d),(p,q) \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$. For $\boldsymbol{r} := \frac{n}{d}$ and $\boldsymbol{s} := \frac{p}{q}$ in $\mathbb{Q}$, we define $\boldsymbol{r} \cdot \boldsymbol{s} = \frac{n}{d} \cdot \frac{p}{q} := \frac{np}{dq}$.

Exercise 4.5. (Multiplication is well-defined).
Let the pairs $(n,d),(n',d'),(p,q),(p',q') \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$ be such that $(n,d) \sim (n',d')$ and $(p,q) \sim (p',q')$. Show that $(np, dq) \sim (n'p', d'q')$.

Exercise 4.6. Show that multiplication in $\mathbb{Q}$ is associative and commutative.

Exercise 4.7. (Distributive law in $\mathbb{Q}$).
Let $\boldsymbol{r}, \boldsymbol{r}', \boldsymbol{s} \in \mathbb{Q}$. Show that $(\boldsymbol{r} + \boldsymbol{r}') \cdot \boldsymbol{s} = \boldsymbol{r} \cdot \boldsymbol{s} + \boldsymbol{r}' \cdot \boldsymbol{s}$.

We show that $\mathbb{Z}$ can be thought of as a subset of $\mathbb{Q}$, and the addition and multiplication in $\mathbb{Q}$ are 'extensions of' the addition and multiplication in $\mathbb{Z}$.

Theorem 4.1. *The map* $\mathbb{Z} \ni n \mapsto \frac{n}{1} \in \mathbb{Q}$ *is injective.*

Moreover, for $m, n \in \mathbb{Z}$, $\frac{m}{1} + \frac{n}{1} = \frac{m+n}{1}$, *and* $\frac{m}{1} \cdot \frac{n}{1} = \frac{mn}{1}$.

Proof. Suppose that for $n, n' \in \mathbb{Z}$, we have $\frac{n}{1} = \frac{n'}{1}$. Then $(n,1) \sim (n',1)$, and so $n(1) = n'(1)$, i.e., $n = n'$. Hence the map $\mathbb{Z} \ni n \mapsto \frac{n}{1} \in \mathbb{Q}$ is injective.
Moreover, $\frac{m}{1} + \frac{n}{1} = \frac{m(1)+n(1)}{1(1)} = \frac{m+n}{1}$, and $\frac{m}{1} \cdot \frac{n}{1} = \frac{mn}{1(1)} = \frac{mn}{1}$. □

In particular, we note that $\mathbf{1} := \frac{1}{1} \neq \frac{0}{1} = \mathbf{0}$.

The rational number $\mathbf{1} = \frac{1}{1} \in \mathbb{Q}$ serves as the multiplicative identity in $\mathbb{Q}$: indeed, for all $\boldsymbol{r} = \frac{n}{d} \in \mathbb{Q}$, where $(n, d) \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$, we have

$$\boldsymbol{r} \cdot \mathbf{1} = \frac{n}{d} \cdot \frac{1}{1} = \frac{n1}{d1} = \frac{n}{d} = \boldsymbol{r} = \frac{n}{d} = \frac{1n}{1d} = \frac{1}{1} \cdot \frac{n}{d} = \mathbf{1} \cdot \boldsymbol{r}.$$

We also note that $(-\mathbf{1}) \cdot \boldsymbol{r} = \frac{-1}{1} \cdot \frac{n}{d} = \frac{(-1)n}{1d} = \frac{-n}{d} = -\boldsymbol{r}$.

Every nonzero rational number has a 'reciprocal/multiplicative inverse'.

Theorem 4.2. *Let $\boldsymbol{r} \in \mathbb{Q}\backslash\{\mathbf{0}\}$. Then there exists a unique rational number, denoted by $\boldsymbol{r}^{-1} \in \mathbb{Q}$, such that $\boldsymbol{r} \cdot \boldsymbol{r}^{-1} = \mathbf{1} = \boldsymbol{r}^{-1} \cdot \boldsymbol{r}$.*

Proof. *Existence*: Let $\boldsymbol{r} = \frac{n}{d}$ where $(n, d) \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$. Then $\frac{0}{d} = \frac{0}{1} = \mathbf{0}$, as $0(1) = 0 = d(0)$. Since $\boldsymbol{r} \neq \mathbf{0}$, we have $n \neq 0$ (otherwise $\boldsymbol{r} = \frac{n}{d} = \frac{0}{d} = \frac{0}{1} = \mathbf{0}$). Set $\boldsymbol{r}^{-1} = \frac{d}{n} \in \mathbb{Q}$. Then $\boldsymbol{r} \cdot \boldsymbol{r}^{-1} = \frac{n}{d} \cdot \frac{d}{n} = \frac{nd}{dn} \overset{(*)}{=} \frac{1}{1} = \mathbf{1}$, where $(*)$ holds since $(nd, dn) \sim (1, 1)$. By commutativity, also $\boldsymbol{r}^{-1} \cdot \boldsymbol{r} = \mathbf{1}$.

Uniqueness: Let $\boldsymbol{s} \in \mathbb{Q}$ be such that $\boldsymbol{r} \cdot \boldsymbol{s} = \mathbf{1} = \boldsymbol{s} \cdot \boldsymbol{r}$. Then

$$\boldsymbol{r}^{-1} = \mathbf{1} \cdot \boldsymbol{r}^{-1} = (\boldsymbol{s} \cdot \boldsymbol{r}) \cdot \boldsymbol{r}^{-1} = \boldsymbol{s} \cdot (\boldsymbol{r} \cdot \boldsymbol{r}^{-1}) = \boldsymbol{s} \cdot \mathbf{1} = \boldsymbol{s}. \qquad \square$$

Next we show that $\mathbb{Q}$ does the job that prompted its construction.

Theorem 4.3. *Let $\boldsymbol{a} \in \mathbb{Q}\backslash\{\mathbf{0}\}$ and $\boldsymbol{b} \in \mathbb{Q}$.*
Then there exists a unique solution $\boldsymbol{x} \in \mathbb{Q}$ to the equation $\boldsymbol{a} \cdot \boldsymbol{x} = \boldsymbol{b}$.

Proof. Set $\boldsymbol{x} = \boldsymbol{a}^{-1} \cdot \boldsymbol{b} \in \mathbb{Q}$. Then $\boldsymbol{a} \cdot \boldsymbol{x} = \boldsymbol{a} \cdot (\boldsymbol{a}^{-1} \cdot \boldsymbol{b}) = (\boldsymbol{a} \cdot \boldsymbol{a}^{-1}) \cdot \boldsymbol{b} = \mathbf{1} \cdot \boldsymbol{b} = \boldsymbol{b}$. If $\boldsymbol{x}'$ is a solution, then $\boldsymbol{x}' = \mathbf{1} \cdot \boldsymbol{x}' = (\boldsymbol{a}^{-1} \cdot \boldsymbol{a}) \cdot \boldsymbol{x}' = \boldsymbol{a}^{-1} \cdot (\boldsymbol{a} \cdot \boldsymbol{x}') = \boldsymbol{a}^{-1} \cdot \boldsymbol{b}$. $\square$

In particular, the equation $\mathbf{2} \cdot \boldsymbol{x} = \mathbf{1}$ has the unique rational solution given by $\boldsymbol{x} = \mathbf{2}^{-1} \cdot \mathbf{1} = \mathbf{2}^{-1} = \frac{1}{2}$.

4.2 Order

We now extend the order relation from $\mathbb{Z}$ to $\mathbb{Q}$.

Definition 4.5. (Order $<$ on $\mathbb{Q}$).
Let $(n, d), (p, q) \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$, where d, q are *positive* integers, and let $\boldsymbol{r} := \frac{n}{d}$ and $\boldsymbol{s} := \frac{p}{q}$. We say that $\boldsymbol{r}$ *is less than* $\boldsymbol{s}$, written as $\boldsymbol{r} < \boldsymbol{s}$ if $nq < pd$.
The notation $\boldsymbol{s} > \boldsymbol{r}$ means $\boldsymbol{r} < \boldsymbol{s}$.

We claim the notion is well-defined. Let $\frac{n'}{d'} = \frac{n}{d}$ and $\frac{p'}{q'} = \frac{p}{q}$, where $n', p' \in \mathbb{Z}$, and d', q' are positive integers. We want to show that $n'q' < p'd'$.

As $nq < pd$, and as d', q' are positive integers, using Exercise 3.13, p.32, we get $nqd'q' < pdd'q'$, i.e., $(nd')qq' < (pq')dd'$. As $n'd = nd'$ and $pq' = p'q$, we get $(n'd)qq' < (p'q)dd'$, i.e, $(n'q')dq < (p'd')dq$. Hence $n'q' < p'd'$. (Otherwise, by the trichotomy law $n'q' \geqslant p'd'$, and as d, q are positive integers, Exercise 3.13 implies $(n'q')dq \geqslant (p'd')dq$, a contradiction.)

It is clear that for $m, n \in \mathbb{Z}$, $m < n$ in $\mathbb{Z}$ if and only if $\frac{m}{1} < \frac{n}{1}$ in $\mathbb{Q}$.

Exercise 4.8. (Transitivity of $<$ in $\mathbb{Q}$).
Let $\boldsymbol{r}, \boldsymbol{s}, \boldsymbol{t} \in \mathbb{Q}$ be such that $\boldsymbol{r} < \boldsymbol{s}$ and $\boldsymbol{s} < \boldsymbol{t}$. Show that $\boldsymbol{r} < \boldsymbol{t}$.

Exercise 4.9. (Trichotomy law).
Let $\boldsymbol{r}, \boldsymbol{s} \in \mathbb{Q}$. Show that exactly one of the following holds:
1° $\boldsymbol{r} < \boldsymbol{s}$.
2° $\boldsymbol{r} = \boldsymbol{s}$.
3° $\boldsymbol{r} > \boldsymbol{s}$.

Exercise 4.10. Let $\boldsymbol{r}, \boldsymbol{s}, \boldsymbol{t} \in \mathbb{Q}$. Prove that if $\boldsymbol{r} < \boldsymbol{s}$, then $\boldsymbol{r} + \boldsymbol{t} < \boldsymbol{s} + \boldsymbol{t}$.
(It follows that if $\boldsymbol{r} < \boldsymbol{s}$, then $-\boldsymbol{s} < -\boldsymbol{r}$: indeed, first we add $-\boldsymbol{s}$ to both sides of $\boldsymbol{r} < \boldsymbol{s}$ to obtain $\boldsymbol{r} - \boldsymbol{s} < \boldsymbol{0}$, and then add $-\boldsymbol{r}$ to both sides to get $-\boldsymbol{s} < -\boldsymbol{r}$.)

Exercise 4.11. Let $\boldsymbol{t}, \boldsymbol{r}, \boldsymbol{s} \in \mathbb{Q}$ be such that $\boldsymbol{t} > \boldsymbol{0}$, and $\boldsymbol{r} < \boldsymbol{s}$.
Show that $\boldsymbol{r} \cdot \boldsymbol{t} < \boldsymbol{s} \cdot \boldsymbol{t}$.

Exercise 4.12. Show that if $\boldsymbol{t} \in \mathbb{Q}$ is such that $\boldsymbol{t} > \boldsymbol{0}$, then $\boldsymbol{t}^{-1} > \boldsymbol{0}$.

The previous two exercises imply that if $\boldsymbol{0} < \boldsymbol{r} < \boldsymbol{s}$, then $\boldsymbol{0} < \boldsymbol{s}^{-1} < \boldsymbol{r}^{-1}$.

Exercise 4.13. Let $\boldsymbol{r}, \boldsymbol{s} \in \mathbb{Q}$ and $\boldsymbol{r} < \boldsymbol{s}$.
Show that there exists a $\boldsymbol{t} \in \mathbb{Q}$ such that $\boldsymbol{r} < \boldsymbol{t} < \boldsymbol{s}$.

If $\boldsymbol{r}, \boldsymbol{s} \in \mathbb{Q}$, and either $\boldsymbol{r} = \boldsymbol{s}$ or $\boldsymbol{r} < \boldsymbol{s}$, then we write $\boldsymbol{r} \leqslant \boldsymbol{s}$.
The symbol $\geqslant$ is defined similarly: $\boldsymbol{r} \geqslant \boldsymbol{s}$ if $\boldsymbol{r} = \boldsymbol{s}$ or $\boldsymbol{r} > \boldsymbol{s}$.

Exercise 4.14. Let $\boldsymbol{r} \in \mathbb{Q}$. Show that $\boldsymbol{r}^2 := \boldsymbol{r} \cdot \boldsymbol{r} \geqslant \boldsymbol{0}$.

Exercise 4.15. Define $|\cdot| : \mathbb{Q} \to \mathbb{Q}$ as follows:

$$\text{For } \boldsymbol{r} \in \mathbb{Q}, \quad |\boldsymbol{r}| = \begin{cases} \boldsymbol{r} & \text{if } \boldsymbol{r} \geqslant \boldsymbol{0}, \\ -\boldsymbol{r} & \text{if } \boldsymbol{r} < \boldsymbol{0}. \end{cases}$$

For example, as $\boldsymbol{-1} = \frac{-1}{1} < \frac{0}{1} = \boldsymbol{0}$, we have $|-\boldsymbol{1}| = -(-\boldsymbol{1}) = \frac{-(-1)}{1} = \frac{1}{1} = \boldsymbol{1}$.

(1) If $\boldsymbol{r}, \boldsymbol{s} \in \mathbb{Q}$, then show that $|\boldsymbol{r} \cdot \boldsymbol{s}| = |\boldsymbol{r}| \cdot |\boldsymbol{s}|$.
(2) If $\boldsymbol{r} \in \mathbb{Q}$, then show that $\boldsymbol{r} \leqslant |\boldsymbol{r}|$ and $-\boldsymbol{r} \leqslant |\boldsymbol{r}|$.
(3) (*Triangle inequality*) If $\boldsymbol{r}, \boldsymbol{s} \in \mathbb{Q}$, then show that $|\boldsymbol{r} + \boldsymbol{s}| \leqslant |\boldsymbol{r}| + |\boldsymbol{s}|$.
(4) (*Reverse triangle inequality*) If $\boldsymbol{r}, \boldsymbol{s} \in \mathbb{Q}$, then show that $||\boldsymbol{r}| - |\boldsymbol{s}|| \leqslant |\boldsymbol{r} - \boldsymbol{s}|$.

We have seen in Exercise 1.8, p.10, that

- $\mathbb{Q}$ does not have the least upper bound property, and
- not all Cauchy sequences in $\mathbb{Q}$ are convergent with a limit in $\mathbb{Q}$.

But in order to do analysis, it is convenient to work with a number system which has these two properties listed above. This '(analytical) flaw' of $\mathbb{Q}$ is remedied by the set of real numbers.

From now onwards, we will denote rational numbers simply using ordinary font letters such as $r, s, t, \cdots$ (instead of boldface $\boldsymbol{r}, \boldsymbol{s}, \boldsymbol{t}, \cdots$).

Also, if $a, b \in \mathbb{Q}$ and $a \neq 0$, then we use the notation $\frac{b}{a}$ for $a^{-1}b = ba^{-1}$.

Exercise 4.16. (Binomial coefficients).

The *factorial* is defined by $0! = 1$ and for $n \in \mathbb{N}$, $n! = 1 \cdot 2 \cdots (n-1) \cdot n$.

Also, $\binom{n}{k} := \frac{n!}{k!(n-k)!}$. For all $n \in \mathbb{N}$, $\binom{n}{0} = 1 = \binom{n}{n}$.

(1) Show that for all $n \in \mathbb{N}$ and all $0 \leqslant k < n$, we have $\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}$.

(2) Show for all $n \in \mathbb{N}$ and $0 \leqslant k \leqslant n$, we have $\binom{n}{k} \in \mathbb{N}$.

Chapter 5

Real numbers

We have now reached the stage where we can learn about the construction of the most important number system from the point of view of mathematical analysis, namely the real number system $\mathbb{R}$. This is because as opposed to $\mathbb{Q}$, $\mathbb{R}$ has the property that all Cauchy sequences in $\mathbb{R}$ converge in $\mathbb{R}$, and every nonempty subset of $\mathbb{R}$ possesses a least upper bound in $\mathbb{R}$, both of which are crucial for proving existence results (e.g. existence of a maximiser, or a solution to a differential equation).

Before formally constructing the set of real numbers, we give the intuitive idea behind it. Roughly speaking, the set of real numbers are the numbers to which Cauchy sequences in $\mathbb{Q}$ 'want to converge to'. So associated with a Cauchy sequence $(a_n)_{n\in\mathbb{N}}$, we have a real number $\boldsymbol{x}$:

$$(a_n)_{n\in\mathbb{N}} \rightsquigarrow \boldsymbol{x} \qquad \boldsymbol{x} \text{ `` } = \lim_{n\to\infty} a_n \text{ ''}.$$

As these limits may not be rational, we just name/label these numbers by the whole Cauchy sequence in $\mathbb{Q}$ itself! But then two Cauchy sequences in $\mathbb{Q}$ might want to converge to the same thing. For example, think of $(a_n)_{n\in\mathbb{N}}$ and $(a_n + \frac{1}{n})_{n\in\mathbb{N}}$. If one is Cauchy, the other is too, and we would want both of these to correspond to the same real number, since formally we have

$$\lim_{n\to\infty}(a_n + \tfrac{1}{n}) = \lim_{n\to\infty} a_n + \lim_{n\to\infty} \tfrac{1}{n} = \lim_{n\to\infty} a_n + 0 = \lim_{n\to\infty} a_n.$$

So we ought not to distinguish between such two Cauchy sequences. And we can keep playing this game: one can perturb $(a_n)_{n\in\mathbb{N}}$ by any Cauchy sequence in $\mathbb{Q}$ whose limit is equal to 0 (e.g. $(\frac{3n+1}{n^2})_{n\in\mathbb{N}}$ is another allowed perturbation besides $(\frac{1}{n})_{n\in\mathbb{N}}$), and for the purpose of talking about this real number $\boldsymbol{x}$, all of these Cauchy sequences $(b_n)_{n\in\mathbb{N}}$ are then the 'same', because each of them 'has the same limit $\boldsymbol{x}$'. This makes us suspect that perhaps there is an equivalence relation $\sim$ on the set $\mathscr{C}$ of all Cauchy sequences in $\mathbb{Q}$, such that under $\sim$, all of these Cauchy sequences $(b_n)_{n\in\mathbb{N}}$ belong to the same equivalence class $[(a_n)_{n\in\mathbb{N}}]$, and this equivalence class *is* then the real number $\boldsymbol{x}$.

So we must build an equivalence relation $\sim$ on Cauchy sequences (so that

$$(a_n)_{n\in\mathbb{N}} \sim (b_n)_{n\in\mathbb{N}} \quad \text{if} \quad \lim_{n\to\infty}(a_n - b_n) = 0,$$

and consider the real numbers as equivalence classes of Cauchy sequences under this equivalence relation. We had met this equivalence relation $\sim$ in Exercise 1.8, p.10, where we checked that this is indeed an equivalence relation. We recall this below.

Definition 5.1.
A *Cauchy sequence in* $\mathbb{Q}$ is a sequence $(a_n)_{n\in\mathbb{N}}$ of rational numbers such that for every rational $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that whenever $m, n \in \mathbb{N}$ satisfy $m, n > N$, we have $|a_n - a_m| < \epsilon$. The set of all Cauchy sequences in $\mathbb{Q}$ is denoted by $\mathscr{C}$. A sequence $(a_n)_{n\in\mathbb{N}}$ in $\mathbb{Q}$ *converges in* $\mathbb{Q}$ *to* $r \in \mathbb{Q}$ if for every rational $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that whenever $n \in \mathbb{N}$ satisfies $n > N$, we have $|a_n - r| < \epsilon$. The relation $\sim$ on $\mathscr{C}$ is defined as follows: for $(a_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ in $\mathscr{C}$, $(a_n)_{n\in\mathbb{N}} \sim (b_n)_{n\in\mathbb{N}}$ if the sequence $(a_n - b_n)_{n\in\mathbb{N}}$ converges in $\mathbb{Q}$ to $0 \in \mathbb{Q}$.

For example, if $r \in \mathbb{Q}$, then the constant sequence $(r)_{n\in\mathbb{N}}$ converges in $\mathbb{Q}$ to r, and it is also a Cauchy sequence in $\mathbb{Q}$.

Exercise 5.1. Let $(a_n)_{n\in\mathbb{N}}$, $(b_n)_{n\in\mathbb{N}}$ be Cauchy sequences in $\mathbb{Q}$.
Show that $(a_n + b_n)_{n\in\mathbb{N}}$ is also a Cauchy sequence in $\mathbb{Q}$.

A sequence $(a_n)_{n\in\mathbb{N}}$ in $\mathbb{Q}$ is *bounded* if there exists a rational $M > 0$ such that for all $n \in \mathbb{N}$, we have $|a_n| \leqslant M$.

Proposition 5.1. *Every Cauchy sequence in* $\mathbb{Q}$ *is bounded.*

Proof. Let $(a_n)_{n\in\mathbb{N}}$ be a Cauchy sequence in $\mathbb{Q}$. Choose a rational $\epsilon > 0$, say $\epsilon = 1$. There exists an $N \in \mathbb{N}$ such that for all natural numbers $n, m > N$, we have $|a_n - a_m| < \epsilon = 1$. In particular, with $m = N+1 > N$, and $n > N$, $|a_n - a_{N+1}| < 1$. By the triangle inequality (see Exercise 4.15, p.39), for all $n > N$, we get $|a_n| = |a_n - a_{N+1} + a_{N+1}| \leqslant |a_n - a_{N+1}| + |a_{N+1}| < 1 + |a_{N+1}|$. On the other hand, for $n \leqslant N$, $|a_n| \leqslant |a_1| + \cdots + |a_N| + |a_{N+1}| + 1 =: M > 0$. Thus $|a_n| \leqslant M$ for all $n \in \mathbb{N}$, that is, the sequence $(a_n)_{n\in\mathbb{N}}$ is bounded. □

Exercise 5.2. Let $(a_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ be Cauchy sequences in $\mathbb{Q}$.
Show that $(a_n b_n)_{n\in\mathbb{N}}$ is also a Cauchy sequence in $\mathbb{Q}$.
(It follows that if $(a_n)_{n\in\mathbb{N}}$ is Cauchy in $\mathbb{Q}$, then so is $(-a_n)_{n\in\mathbb{N}}$.)

Exercise 5.3. Suppose that $(a_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ are sequences in $\mathbb{Q}$ such that $(a_n)_{n\in\mathbb{N}}$ (respectively $(b_n)_{n\in\mathbb{N}}$) converges in $\mathbb{Q}$ to $r_a \in \mathbb{Q}$ (respectively $r_b \in \mathbb{Q}$).
(1) Show uniqueness of the limit: If $(a_n)_{n\in\mathbb{N}}$ converges in $\mathbb{Q}$ to $r'_a \in \mathbb{Q}$, then $r_a = r'_a$.
(2) Show that $(-a_n)_{n\in\mathbb{N}}$ converges to $-r_a$.
(3) Show that $(a_n + b_n)_{n\in\mathbb{N}}$ converges to $r_a + r_b$.

Proposition 5.2. *$\sim$ is an equivalence relation on $\mathscr{C}$.*

Proof. We check that $\sim$ is reflexive, symmetric, and transitive.
Reflexivity: Let $(a_n)_{n\in\mathbb{N}} \in \mathscr{C}$. Then $(a_n)_{n\in\mathbb{N}} \sim (a_n)_{n\in\mathbb{N}}$ because $(a_n - a_n)_{n\in\mathbb{N}}$ is the constant sequence $0, 0, 0, \cdots$ which is clearly convergent in $\mathbb{Q}$ to 0.
Symmetry: Let $(a_n)_{n\in\mathbb{N}}, (b_n)_{n\in\mathbb{N}} \in \mathscr{C}$ be such that $(a_n)_{n\in\mathbb{N}} \sim (b_n)_{n\in\mathbb{N}}$. Then $(a_n - b_n)_{n\in\mathbb{N}}$ converges in $\mathbb{Q}$ to 0. But then $((-1)(a_n - b_n))_{n\in\mathbb{N}} = (b_n - a_n)_{n\in\mathbb{N}}$ converges in $\mathbb{Q}$ to $-0 = 0$. So $(b_n)_{n\in\mathbb{N}} \sim (a_n)_{n\in\mathbb{N}}$.
Transitivity: Let $(a_n)_{n\in\mathbb{N}}, (b_n)_{n\in\mathbb{N}}, (c_n)_{n\in\mathbb{N}} \in \mathscr{C}$ be such that $(a_n)_{n\in\mathbb{N}} \sim (b_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}} \sim (c_n)_{n\in\mathbb{N}}$. Then $(a_n - b_n)_{n\in\mathbb{N}}$ and $(b_n - c_n)_{n\in\mathbb{N}}$ both converge in $\mathbb{Q}$ to 0. So their termwise sum, $(a_n - b_n + b_n - c_n)_{n\in\mathbb{N}} = (a_n - c_n)_{n\in\mathbb{N}}$ converges in $\mathbb{Q}$ to $0 + 0 = 0$. Thus $(a_n)_{n\in\mathbb{N}} \sim (c_n)_{n\in\mathbb{N}}$. □

Definition 5.2. (The set of real numbers).
A *real number* is an equivalence class of $\mathscr{C}$ under the relation $\sim$.
If $(a_n)_{n\in\mathbb{N}} \in \mathscr{C}$, then $[(a_n)_{n\in\mathbb{N}}]$ denotes the real number which is the equivalence class under $\sim$ of $\mathscr{C}$ containing the sequence $(a_n)_{n\in\mathbb{N}}$.
The set of all real numbers is denoted by $\mathbb{R}$.

The set of real numbers is supposed to be an *extension* of the rational numbers $\mathbb{Q}$, that is, we want to see that $\mathbb{Q}$ '$\subset$' $\mathbb{R}$. Given a rational number $r \in \mathbb{Q}$, the constant sequence $r, r, r, \cdots$, that is, $(r)_{n\in\mathbb{N}}$, is a Cauchy sequence in $\mathbb{Q}$. Thus $[(r)_{n\in\mathbb{N}}]$ is a real number. We have the following.

Proposition 5.3. *The map $\mathbb{Q} \ni r \mapsto [(r)_{n\in\mathbb{N}}] \in \mathbb{R}$ is injective.*

Proof. Let $r, s \in \mathbb{Q}$ be such that $[(r)_{n\in\mathbb{N}}] = [(s)_{n\in\mathbb{N}}]$. Then $(r)_{n\in\mathbb{N}} \sim (s)_{n\in\mathbb{N}}$. So $\lim_{n\to\infty} (r - s) = 0$. The sequence $r - s, r - s, r - s, \cdots$ converges in $\mathbb{Q}$ to $r - s$. As the limit is unique (Exercise 5.3, p.42), $r - s = 0$, i.e., $r = s$. □

5.1 Addition and multiplication

Clearly, if addition in $\mathbb{R}$ is to respect the addition in $\mathbb{Q}$, we must have that for $r, s \in \mathbb{Q}$, $[(r)_{n\in\mathbb{N}}] + [(s)_{n\in\mathbb{N}}]$ should equal $[(r + s)_{n\in\mathbb{N}}]$. Similarly, $[(r)_{n\in\mathbb{N}}] \cdot [(s)_{n\in\mathbb{N}}]$ should equal $[(rs)_{n\in\mathbb{N}}]$. This motivates the following.

Definition 5.3. Let $[(a_n)_{n\in\mathbb{N}}], [(b_n)_{n\in\mathbb{N}}] \in \mathbb{R}$, where $(a_n)_{n\in\mathbb{N}}, (b_n)_{n\in\mathbb{N}} \in \mathscr{C}$.
The *sum* of the real numbers $[(a_n)_{n\in\mathbb{N}}]$ and $[(b_n)_{n\in\mathbb{N}}]$ is defined by

$$[(a_n)_{n\in\mathbb{N}}] + [(b_n)_{n\in\mathbb{N}}] = [(a_n + b_n)_{n\in\mathbb{N}}].$$

The *product* of the real numbers $[(a_n)_{n\in\mathbb{N}}]$ and $[(b_n)_{n\in\mathbb{N}}]$ is defined by

$$[(a_n)_{n\in\mathbb{N}}] \cdot [(b_n)_{n\in\mathbb{N}}] = [(a_n b_n)_{n\in\mathbb{N}}].$$

As usual, we have to check well-definedness. We leave this as an exercise for addition, but give an argument for multiplication as follows.

Let $[(a_n)_{n\in\mathbb{N}}] = [(a'_n)_{n\in\mathbb{N}}] \in \mathbb{R}$ and $[(b_n)_{n\in\mathbb{N}}] = [(b'_n)_{n\in\mathbb{N}}] \in \mathbb{R}$. The idea is to use the inequality

$$|a'_n b'_n - a_n b_n| = |a'_n b'_n - a'_n b_n + a'_n b_n - a_n b_n| \leqslant |a'_n||b'_n - b_n| + |a'_n - a_n||b_n|$$

and the boundedness of $(a'_n)_{n\in\mathbb{N}}, (b_n)_{n\in\mathbb{N}}$ to show $(a_n b_n)_{n\in\mathbb{N}} \sim (a'_n b'_n)_{n\in\mathbb{N}}$. As $(a'_n)_{n\in\mathbb{N}}$ is Cauchy, it is bounded. Let $A' > 0$ be a rational such that $|a'_n| < A'$ for all $n \in \mathbb{N}$. Similarly, $(b_n)_{n\in\mathbb{N}}$ is bounded, and let $B > 0$ be a rational such that $|b_n| < B$ for all $n \in \mathbb{N}$. Let $\epsilon > 0$ be rational. As $(a_n)_{n\in\mathbb{N}} \sim (a'_n)_{n\in\mathbb{N}}$, $(a_n - a'_n)_{n\in\mathbb{N}}$ converges in $\mathbb{Q}$ to 0. So for the rational $\frac{\epsilon}{2B} > 0$, there exists an $N_a \in \mathbb{N}$ such that for all natural numbers $n > N_a$, we have $|a'_n - a_n| < \frac{\epsilon}{2B}$. Similarly, as $(b_n)_{n\in\mathbb{N}} \sim (b'_n)_{n\in\mathbb{N}}$, for the rational $\frac{\epsilon}{2A'} > 0$, there exists an $N_b \in \mathbb{N}$ such that $|b'_n - b_n| < \frac{\epsilon}{2A'}$ for all natural numbers $n > N_b$. Set $N = N_a + N_b$. For all natural numbers $n > N$, we have

$$\begin{aligned} |a'_n b'_n - a_n b_n| &= |a'_n b'_n - a'_n b_n + a'_n b_n - a_n b_n| \\ &\leqslant |a'_n||b'_n - b_n| + |a'_n - a_n||b_n| \\ &\leqslant A'|b'_n - b_n| + |a'_n - a_n|B \\ &< A'\tfrac{\epsilon}{2A'} + \tfrac{\epsilon}{2B}B = \epsilon. \end{aligned}$$

Thus $(a_n b_n)_{n\in\mathbb{N}} \sim (a'_n b'_n)_{n\in\mathbb{N}}$.

Exercise 5.4. (Addition is well-defined).
Let $(a_n)_{n\in\mathbb{N}}, (a'_n)_{n\in\mathbb{N}}, (b_n)_{n\in\mathbb{N}}, (b'_n)_{n\in\mathbb{N}}$ be Cauchy sequences in $\mathbb{Q}$ such that $(a_n)_{n\in\mathbb{N}} \sim (a'_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}} \sim (b'_n)_{n\in\mathbb{N}}$. Show that $(a_n + b_n)_{n\in\mathbb{N}} \sim (a'_n + b'_n)_{n\in\mathbb{N}}$.

Example 5.1. (The real numbers $\mathbf{0}$ and $\mathbf{1}$). We define the real numbers $\mathbf{0} = [(0)_{n\in\mathbb{N}}]$ and $\mathbf{1} = [(1)_{n\in\mathbb{N}}]$. Then for every real number $\boldsymbol{x} \in \mathbb{R}$, we have $\mathbf{0} + \boldsymbol{x} = \boldsymbol{x} = \boldsymbol{x} + \mathbf{0}$, and $\mathbf{1} \cdot \boldsymbol{x} = \boldsymbol{x} = \boldsymbol{x} \cdot \mathbf{1}$. So $\mathbf{0}$ serves as the additive identity and $\mathbf{1}$ serves as the multiplicative identity. Clearly $\mathbf{1} \neq \mathbf{0}$ since $(1 - 0)_{n\in\mathbb{N}}$ converges in $\mathbb{Q}$ to 1 ($\neq 0$). ◇

The set $\mathbb{R}$, together with the operations $+, \cdot : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ forms a 'field', i.e., the following hold.

$+$	(F1)	(Associativity)	For all $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{R}$, $\boldsymbol{x} + (\boldsymbol{y} + \boldsymbol{z}) = (\boldsymbol{x} + \boldsymbol{y}) + \boldsymbol{z}$.
	(F2)	(Additive identity)	For all $\boldsymbol{x} \in \mathbb{R}$, $\boldsymbol{x} + \mathbf{0} = \boldsymbol{x} = \mathbf{0} + \boldsymbol{x}$.
	(F3)	(Inverses)	For all $\boldsymbol{x} \in \mathbb{R}$, there exists $-\boldsymbol{x} \in \mathbb{R}$ such that $\boldsymbol{x} + (-\boldsymbol{x}) = \mathbf{0} = -\boldsymbol{x} + \boldsymbol{x}$.
	(F4)	(Commutativity)	For all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}$, $\boldsymbol{x} + \boldsymbol{y} = \boldsymbol{y} + \boldsymbol{x}$.
$\cdot$	(F5)	(Associativity)	For all $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{R}$, $\boldsymbol{x} \cdot (\boldsymbol{y} \cdot \boldsymbol{z}) = (\boldsymbol{x} \cdot \boldsymbol{y}) \cdot \boldsymbol{z}$.
	(F6)	(Multiplicative identity)	$\mathbf{1} \neq \mathbf{0}$ and for all $x \in \mathbb{R}$, $\boldsymbol{x} \cdot \mathbf{1} = \boldsymbol{x} = \mathbf{1} \cdot \boldsymbol{x}$.
	(F7)	(Inverses)	For all $\boldsymbol{x} \in \mathbb{R} \setminus \{\mathbf{0}\}$, there exists $\boldsymbol{x}^{-1} \in \mathbb{R}$ such that $\boldsymbol{x} \cdot \boldsymbol{x}^{-1} = \mathbf{1} = \boldsymbol{x}^{-1} \cdot \boldsymbol{x}$.
	(F8)	(Commutativity)	For all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}$, $\boldsymbol{x} \cdot \boldsymbol{y} = \boldsymbol{y} \cdot \boldsymbol{x}$.
$+, \cdot$	(F9)	(Distributivity)	For all $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{R}$, $\boldsymbol{x} \cdot (\boldsymbol{y} + \boldsymbol{z}) = \boldsymbol{x} \cdot \boldsymbol{y} + \boldsymbol{x} \cdot \boldsymbol{z}$.

In fact, if we replace everywhere $\mathbb{R}$ by $\mathbb{Q}$ (and $\mathbf{1}, \mathbf{0}$ by the rational numbers $1, 0$, respectively), then the set $\mathbb{Q}$ of rational numbers with their addition and multiplication, also satisfy the same properties. We say that $(\mathbb{Q}, +, \cdot)$ is also a field. (However, $(\mathbb{Z}, +, \cdot)$ is not a field, because multiplicative inverses do not always exist: the equation $2x = 1$ has no solution $x \in \mathbb{Z}$.) Each of the above essentially follow by 'termwise verifications', and by using the corresponding properties from the field of rationals. For $\mathbb{R}$, we will show below fact (F7) on the existence of inverses for nonzero reals, while the verification of the remaining facts is left as an exercise.

Exercise 5.5. For $\mathbb{R}$, check each of (F1)-(F9) except for (F7).

Let us now show the existence of multiplicative inverses for nonzero reals. First we prove the following lemma.

Lemma 5.1. *Let $\boldsymbol{x} \in \mathbb{R}$ be such that $\boldsymbol{x} \neq \mathbf{0}$. If $(a_n)_{n\in\mathbb{N}} \in \boldsymbol{x}$, then there exists a rational $d > 0$ and an $N \in \mathbb{N}$ such that for all natural numbers $n > N$, we have $|a_n| > d$.*

Proof. As $[(a_n)_{n\in\mathbb{N}}] = \boldsymbol{x} \neq \mathbf{0} = [(0)_{n\in\mathbb{N}}]$, we have $\neg\big((a_n)_{n\in\mathbb{N}} \sim (0)_{n\in\mathbb{N}}\big)$, i.e.[1],
$\neg\big((a_n - 0)_{n\in\mathbb{N}}$ converges in $\mathbb{Q}$ to $0\big)$, i.e.,
$\neg\big(\forall$ rational $\epsilon > 0$, $\exists N \in \mathbb{N}$ such that $\forall n \in \mathbb{N}$ satisfying $n > N$, $|a_n - 0| < \epsilon\big)$, i.e.,
$\exists$ rational $\epsilon > 0$ such that $\forall N \in \mathbb{N}$, $\exists n \in \mathbb{N}$ such that $n > N$ and $|a_n - 0| \geqslant \epsilon$. $(\star)$

Since $(a_n)_{n\in\mathbb{N}} \in \mathscr{C}$, for the rational $\frac{\epsilon}{2} > 0$, there exists an $N_* \in \mathbb{N}$ such that for all natural numbers $n, m > N_*$, we have $|a_n - a_m| < \frac{\epsilon}{2}$. From $(\star)$, taking $N = N_*$, there exists a natural number $n_* > N_*$ such that $|a_{n_*} - 0| \geqslant \epsilon$. Hence, for all natural numbers $n > N_*$, we have by the reverse triangle inequality that $|a_n| = |a_n - a_{n_*} + a_{n_*}| \geqslant |a_{n_*}| - |a_n - a_{n_*}| \geqslant \epsilon - \frac{\epsilon}{2} = \frac{\epsilon}{2} =: d$. □

Proposition 5.4. *Let the real number $\boldsymbol{x} \neq \mathbf{0}$.*
Then there exists an $\boldsymbol{x}^{-1} \in \mathbb{R}$ such that $\boldsymbol{x} \cdot \boldsymbol{x}^{-1} = \mathbf{1} = \boldsymbol{x}^{-1} \cdot \boldsymbol{x}$.

Proof. Let $\boldsymbol{x} = [(a_n)_{n\in\mathbb{N}}]$. By Lemma 5.1 above, there exists a rational $d > 0$ and an $N \in \mathbb{N}$ such that $|a_n| > d$ for all natural numbers $n > N$. In particular, $a_n \neq 0$ for all $n > N$. Set[2]

$$b_n := \begin{cases} 0 & \text{if } 1 \leqslant n \leqslant N, \\ a_n^{-1} & \text{if } n > N. \end{cases}$$

[1] Below, the notation '$\forall$' stands for the phrase 'for all/every'. Also, recall that '$\exists$' means 'there exists', and '$\neg$' means 'it is not the case that'.

[2] Although we set $b_n = 0$ for $1 \leqslant n \leqslant N$ here, any arbitrary N rational numbers can be specified here.

We claim that $(b_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{Q}$. Firstly, for all natural numbers $n, m > N$, we have

$$|b_n - b_m| = |\frac{1}{a_n} - \frac{1}{a_m}| = \frac{|a_n - a_m|}{|a_n||a_m|} \leqslant \frac{|a_n - a_m|}{d^2}.$$

Secondly, as $(a_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{Q}$, given a rational $\epsilon > 0$, there exists an $M \in \mathbb{N}$ such that for all natural numbers $n, m > M$, we have $|a_n - a_m| < \epsilon d^2$. Hence for all natural numbers $n, m > N + M$, we have

$$|b_n - b_m| \leqslant \frac{|a_n - a_m|}{d^2} < \frac{\epsilon d^2}{d^2} = \epsilon.$$

Thus $(b_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{Q}$. We have[3]

$$a_n b_n := \begin{cases} 0 & \text{if } 1 \leqslant n \leqslant N, \\ 1 & \text{if } n > N. \end{cases}$$

So $(a_n b_n)_{n\in\mathbb{N}}$ converges in $\mathbb{Q}$ to 1. Thus $[(a_n b_n)_{n\in\mathbb{N}}] = \mathbf{1}$. Set $\boldsymbol{x}^{-1} = [(b_n)_{n\in\mathbb{N}}]$. Then we have $\boldsymbol{x} \cdot \boldsymbol{x}^{-1} = \mathbf{1} = \boldsymbol{x}^{-1} \cdot \boldsymbol{x}$. □

Henceforth we often skip writing the $\cdot$ to denote multiplication in $\mathbb{R}$.

Exercise 5.6. Show that there is no $\boldsymbol{x} \in \mathbb{R}$ such that $\boldsymbol{x}^2 + \mathbf{1} = \mathbf{0}$, where $\boldsymbol{x}^2 := \boldsymbol{x} \cdot \boldsymbol{x}$.

Exercise 5.7. (Binomial theorem). Show that $(\mathbf{1} + \boldsymbol{x})^n = \sum_{k=0}^{n} \binom{n}{k} \boldsymbol{x}^k$ for $\boldsymbol{x} \in \mathbb{R}$, $n \in \mathbb{N}$.

It follows that if $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}$ and $n \in \mathbb{N}$, then $(\boldsymbol{a} + \boldsymbol{b})^n = \sum_{k=0}^{n} \binom{n}{k} \boldsymbol{a}^{n-k} \boldsymbol{b}^k$:
If $\boldsymbol{a} = \mathbf{0}$, then this is clear. If $\boldsymbol{a} \neq \mathbf{0}$, then by the binomial theorem for $\boldsymbol{x} := \boldsymbol{b}\boldsymbol{a}^{-1}$, we get $(\boldsymbol{a} + \boldsymbol{b})^n = \boldsymbol{a}^n (\mathbf{1} + \boldsymbol{b}\boldsymbol{a}^{-1})^n = \boldsymbol{a}^n \sum_{k=0}^{n} \binom{n}{k} (\boldsymbol{b}\boldsymbol{a}^{-1})^k = \sum_{k=0}^{n} \binom{n}{k} \boldsymbol{a}^{n-k} \boldsymbol{b}^k$.
In the above, if $n \in \mathbb{N}$ and $\boldsymbol{a} \in \mathbb{R}$, then $\boldsymbol{a}^n = \boldsymbol{a} \cdots \boldsymbol{a}$ (n times). If $\boldsymbol{a} \neq \mathbf{0}$, then $\boldsymbol{a}^0 := 1$ and $\boldsymbol{a}^{-n} := (\boldsymbol{a}^{-1})^n$ for all $n \in \mathbb{N}$. Then for all $\boldsymbol{a} \neq \mathbf{0}$ and all $m, n \in \mathbb{Z}$, we have $\boldsymbol{a}^{m+n} = \boldsymbol{a}^m \boldsymbol{a}^n$ and $(\boldsymbol{a}^m)^n = \boldsymbol{a}^{mn}$ (shown by induction).

5.2 Order

To compare real numbers $\boldsymbol{x} = [(a_n)_{n\in\mathbb{N}}]$ and $\boldsymbol{y} = [(b_n)_{n\in\mathbb{N}}]$, we would like to use the order relation $<$ on $\mathbb{Q}$. If we try to define $\boldsymbol{x} < \boldsymbol{y}$ by saying that for *all* $n \in \mathbb{N}$, we have $a_n < b_n$, then this will not be a well-defined notion. Indeed, changing the first few terms of $(a_n)_{n\in\mathbb{N}}$ we could easily violate this, without changing $[(a_n)_{n\in\mathbb{N}}]$. Intuitively, $\boldsymbol{x}$ is the real number that $(a_n)_{n\in\mathbb{N}}$ converges to:

$$\text{`}\boldsymbol{x} = \lim_{n\to\infty} a_n\text{'}, \quad \text{`}\boldsymbol{y} = \lim_{n\to\infty} b_n\text{'}.$$

So $\boldsymbol{x} < \boldsymbol{y}$ means $\text{`} \lim_{n\to\infty} a_n < \lim_{n\to\infty} b_n\text{'}$ and formally manipulating this, we get

$$\text{`} \lim_{n\to\infty} (b_n - a_n) > 0\text{'}.$$

From our former intuition with limits, we know that this means that for all *large enough* $n \in \mathbb{N}$, $b_n - a_n$ stays away from 0 by some positive distance d, say. This motivates the following.

[3] Had we specified $b_1, \cdots, b_N$ arbitrarily, we would get a bunch of initial terms $a_n b_n$ for $1 \leqslant n \leqslant N$, but this won't affect the rest of the proof.

Definition 5.4. (Order $<$ on $\mathbb{R}$).
Let $\boldsymbol{x} = [(a_n)_{n\in\mathbb{N}}]$ and $\boldsymbol{y} = [(b_n)_{n\in\mathbb{N}}]$ be real numbers. Then we say that $\boldsymbol{x}$ *is less than* $\boldsymbol{y}$, written as $\boldsymbol{x} < \boldsymbol{y}$, if there exists a rational number $d > 0$ and an $N \in \mathbb{N}$ such that for all $n \in \mathbb{N}$ satisfying $n > N$, we have $b_n - a_n > d$.
If $\boldsymbol{x} < \boldsymbol{y}$, we write equivalently $\boldsymbol{y} > \boldsymbol{x}$.

Let us show that this is a well-defined notion.

Proposition 5.5. *Let* $[(a_n)_{n\in\mathbb{N}}] = [(a'_n)_{n\in\mathbb{N}}] \in \mathbb{R}$ *and* $[(b_n)_{n\in\mathbb{N}}] = [(b'_n)_{n\in\mathbb{N}}] \in \mathbb{R}$, *where* $(a_n)_{n\in\mathbb{N}}, (a'_n)_{n\in\mathbb{N}}, (b_n)_{n\in\mathbb{N}}, (b'_n)_{n\in\mathbb{N}} \in \mathscr{C}$. *Suppose that there exists a rational number* $d > 0$ *and an* $N \in \mathbb{N}$ *such that for all natural numbers* $n > N$, *we have* $b_n - a_n > d$. *Then there exists a rational number* $d' > 0$ *and an* $N' \in \mathbb{N}$ *such that for all natural numbers* $n > N'$, *we have* $b'_n - a'_n > d'$.

Proof. As $(a_n)_{n\in\mathbb{N}} \sim (a'_n)_{n\in\mathbb{N}}$, $(a_n - a'_n)_{n\in\mathbb{N}}$ converges in $\mathbb{Q}$ to 0. So for the rational $\frac{d}{4} > 0$, there exists an $N_a \in \mathbb{N}$ such that for all $n > N_a$, $|a_n - a'_n| < \frac{d}{4}$, i.e., $-\frac{d}{4} < a_n - a'_n < \frac{d}{4}$. In particular

$$a_n - a'_n > -\tfrac{d}{4} \text{ for all } n > N_a. \qquad (*)$$

Similarly, $(b_n)_{n\in\mathbb{N}} \sim (b'_n)_{n\in\mathbb{N}}$ yields the existence of an $N_b \in \mathbb{N}$ such that

$$b'_n - b_n > -\tfrac{d}{4} \text{ for all } n > N_b. \qquad (**)$$

Set $N' = N_a + N_b + N$. Then for all $n > N'$, using $(*)$ and $(**)$, we have

$$b'_n - a'_n = b_n - a_n + b'_n - b_n + a_n - a'_n > d - \tfrac{d}{4} - \tfrac{d}{4} = \tfrac{d}{2} > 0.$$

So with the rational $d' := \frac{d}{2} > 0$, for all $n > N'$, we have $b'_n - a'_n > d'$. □

Exercise 5.8. Show that if $r, s \in \mathbb{Q}$ and $r < s$, then $[(r)_{n\in\mathbb{N}}] < [(s)_{n\in\mathbb{N}}]$.
(In particular, for the real numbers $\mathbf{0}, \mathbf{1}$, we have $\mathbf{0} < \mathbf{1}$.)

Exercise 5.9. (Transitivity of $<$).
Let $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{R}$ be such that $\boldsymbol{x} < \boldsymbol{y}$ and $\boldsymbol{y} < \boldsymbol{z}$. Prove that $\boldsymbol{x} < \boldsymbol{z}$.

Exercise 5.10. Let $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{R}$ and $\boldsymbol{x} < \boldsymbol{y}$. Show that $\boldsymbol{x} + \boldsymbol{z} < \boldsymbol{y} + \boldsymbol{z}$.

Theorem 5.1. (Trichotomy Law).
Let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}$. *Then exactly one of the following hold*:
1° $\boldsymbol{x} < \boldsymbol{y}$.
2° $\boldsymbol{x} = \boldsymbol{y}$.
3° $\boldsymbol{x} > \boldsymbol{y}$.

Proof. Let $\boldsymbol{x} = [(a_n)_{n\in\mathbb{N}}]$ and $\boldsymbol{y} = [(b_n)_{n\in\mathbb{N}}]$. If $\boldsymbol{x} = \boldsymbol{y}$, then $(a_n - b_n)_{n\in\mathbb{N}}$ converges in $\mathbb{Q}$ to 0. Let us show that $\neg(\boldsymbol{x} > \boldsymbol{y})$. Indeed, otherwise there exists a rational $d > 0$ and an $N \in \mathbb{N}$ such that $a_n - b_n > d$ for all $n > N$. For the rational $\epsilon = \frac{d}{2} > 0$, there exists an $N' \in \mathbb{N}$ such that for all $n > N'$, $|a_n - b_n| < \frac{d}{2}$. So with $n = N + N'$, we arrive at the contradiction that $d < a_n - b_n \leqslant |a_n - b_n| < \frac{d}{2}$. So if $\boldsymbol{x} = \boldsymbol{y}$, then $\neg(\boldsymbol{x} > \boldsymbol{y})$. Interchanging the

roles of $\boldsymbol{x}, \boldsymbol{y}$, we also have that if $\boldsymbol{x} = \boldsymbol{y}$, then $\neg(\boldsymbol{x} < \boldsymbol{y})$. Let us also note that if $\boldsymbol{x} < \boldsymbol{y}$, then $\neg(\boldsymbol{y} < \boldsymbol{x})$: otherwise there exist rational $d, d' > 0$ and $N, N' \in \mathbb{N}$ such that for all $n > N$ we have $b_n - a_n > d$, and for all $n > N'$, we have $a_n - b_n > d'$, so that with $n = N + N'$, we get $d' < a_n - b_n < -d$, giving $d' < -d$, and so $0 > d + d' > d + 0 = d$, a contradiction.

Let $\boldsymbol{x} \neq \boldsymbol{y}$. Then it is not the case that $(a_n - b_n)_{n\in\mathbb{N}}$ converges in $\mathbb{Q}$ to 0. Thus there exists a rational $\epsilon_* > 0$ such that

$$\text{for all } N \in \mathbb{N}, \text{ there exists an } n > N \text{ such that } |a_n - b_n| \geqslant \epsilon_*. \qquad (\star)$$

As $(a_n)_{n\in\mathbb{N}}$ is Cauchy, there exists an $N_a \in \mathbb{N}$ such that for all $m, n > N_a$, we have $|a_n - a_m| < \frac{\epsilon_*}{4}$, i.e., $-\frac{\epsilon_*}{4} < a_n - a_m < \frac{\epsilon_*}{4}$. In particular, $a_n - a_m > -\frac{\epsilon_*}{4}$ for all $n, m > N_a$. Similarly, as $(b_n)_{n\in\mathbb{N}}$ is Cauchy, there exists an $N_b \in \mathbb{N}$ such that for all $m, n > N_b$, $|b_n - b_m| < \frac{\epsilon_*}{4}$, giving in particular $b_n - b_m > -\frac{\epsilon_*}{4}$. Now take $N = N_a + N_b$ in $(\star)$. Then there exists an $n_* > N$ such that $|a_{n_*} - b_{n_*}| \geqslant \epsilon_* > 0$. In particular, $a_{n_*} - b_{n_*} \neq 0$. So by the trichotomy law for $<$ in $\mathbb{Q}$, we have the following two mutually exclusive possible cases:

1° $a_{n_*} - b_{n_*} > 0$. So $a_{n_*} - b_{n_*} = |a_{n_*} - b_{n_*}| \geqslant \epsilon_*$. For all $m > n_* (> N = N_a + N_b)$,
$a_m - b_m = a_{n_*} - b_{n_*} + a_m - a_{n_*} + b_{n_*} - b_m > \epsilon_* - \frac{\epsilon_*}{4} - \frac{\epsilon_*}{4} = \frac{\epsilon_*}{2} =: d.$
So for all $m > n_*$, we have that $a_m - b_m > d$, showing $\boldsymbol{x} > \boldsymbol{y}$.

2° $a_{n_*} - b_{n_*} < 0$. So $b_{n_*} - a_{n_*} = |a_{n_*} - b_{n_*}| \geqslant \epsilon_*$. For all $m > n_* (> N = N_a + N_b)$,
$b_m - a_m = b_{n_*} - a_{n_*} + b_m - b_{n_*} + a_{n_*} - a_m > \epsilon_* - \frac{\epsilon_*}{4} - \frac{\epsilon_*}{4} = \frac{\epsilon_*}{2} =: d.$
So for all $m > n_*$, we have that $b_m - a_m > d$, showing $\boldsymbol{y} > \boldsymbol{x}$. □

Exercise 5.11. Define the *set* $\boldsymbol{P}$ *of positive reals* by $\boldsymbol{P} = \{\boldsymbol{x} \in \mathbb{R} : \boldsymbol{x} > \boldsymbol{0}\}$.
Suppose $\boldsymbol{x}, \boldsymbol{y} \in \boldsymbol{P}$. Show that $\boldsymbol{x} + \boldsymbol{y} \in \boldsymbol{P}$ and $\boldsymbol{x} \cdot \boldsymbol{y} \in \boldsymbol{P}$.
(Note that if $\boldsymbol{b}, \boldsymbol{c} \in \mathbb{R}$, then $\boldsymbol{c} - \boldsymbol{b} \in \boldsymbol{P}$ if and only if $\boldsymbol{b} < \boldsymbol{c}$, see Exercise 5.10, p.47. It follows from this and the above that if moreover $\boldsymbol{a} > \boldsymbol{0}$, then $\boldsymbol{a} \cdot \boldsymbol{b} < \boldsymbol{a} \cdot \boldsymbol{c}$: indeed, $\boldsymbol{a} \cdot \boldsymbol{c} - \boldsymbol{a} \cdot \boldsymbol{b} = \boldsymbol{a} \cdot (\boldsymbol{c} - \boldsymbol{b}) > \boldsymbol{0}$.)

Exercise 5.12. Let $\boldsymbol{x} \in \mathbb{R}$ be such that $\boldsymbol{x} > \boldsymbol{0}$.
Prove that there exists an $r \in \mathbb{Q}$ such that $\boldsymbol{0} < [(r)_{n\in\mathbb{N}}] < \boldsymbol{x}$.
We write this succinctly as $\boldsymbol{0} < r < \boldsymbol{x}$.

If $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}$, and either $\boldsymbol{x} = \boldsymbol{y}$ or $\boldsymbol{x} < \boldsymbol{y}$, then we write $\boldsymbol{x} \leqslant \boldsymbol{y}$.
The symbol $\geqslant$ is defined similarly: $\boldsymbol{x} \geqslant \boldsymbol{y}$ if $\boldsymbol{x} = \boldsymbol{y}$ or $\boldsymbol{x} > \boldsymbol{y}$.

Exercise 5.13. Let $(a_n)_{n\in\mathbb{N}}$ be a Cauchy sequence in $\mathbb{Q}$.
Suppose there exists an $N \in \mathbb{N}$ such that for all $n > N$, we have $a_n \geqslant 0$.
Show that $\boldsymbol{x} := [(a_n)_{n\in\mathbb{N}}] \geqslant \boldsymbol{0}$. In particular, for any real $\boldsymbol{y} \in \mathbb{R}$, we have $\boldsymbol{y}^2 \geqslant \boldsymbol{0}$.

Exercise 5.14. Define $|\cdot| : \mathbb{R} \to \mathbb{R}$ as follows: For $\boldsymbol{x} \in \mathbb{R}$, $|\boldsymbol{x}| = \begin{cases} \boldsymbol{x} & \text{if } \boldsymbol{x} \geqslant \mathbf{0}, \\ -\boldsymbol{x} & \text{if } \boldsymbol{x} < \mathbf{0}. \end{cases}$
E.g., $-\mathbf{1} = [(-1)_{n\in\mathbb{N}}] < [(0)_{n\in\mathbb{N}}] = \mathbf{0}$, and so $|-\mathbf{1}| = -(-\mathbf{1}) = [(-(-1))_{n\in\mathbb{N}}] = \mathbf{1}$.

(1) If $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}$, then show that $|\boldsymbol{x} \cdot \boldsymbol{y}| = |\boldsymbol{x}| \cdot |\boldsymbol{y}|$.
(2) If $\boldsymbol{x} \in \mathbb{R}$, then show that $\boldsymbol{x} \leqslant |\boldsymbol{x}|$ and $-\boldsymbol{x} \leqslant |\boldsymbol{x}|$.
(3) (Triangle inequality) If $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}$, then show that $|\boldsymbol{x} + \boldsymbol{y}| \leqslant |\boldsymbol{x}| + |\boldsymbol{y}|$.
(4) (Reverse triangle inequality) If $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}$, then show that $||\boldsymbol{x}| - |\boldsymbol{y}|| \leqslant |\boldsymbol{x} - \boldsymbol{y}|$.

Exercise 5.15. (When does equality hold in the triangle inequality?)
(1) Show the generalised triangle inequality:
if $n \in \mathbb{N}$ and $\boldsymbol{a}_1, \cdots, \boldsymbol{a}_n \in \mathbb{R}$, then $|\boldsymbol{a}_1 + \cdots + \boldsymbol{a}_n| \leqslant |\boldsymbol{a}_1| + \cdots + |\boldsymbol{a}_n|$.
(2) We say that $\boldsymbol{a}_1, \cdots, \boldsymbol{a}_n$ *have the same sign* if either of the following is true:
1° $\boldsymbol{a}_1 \geqslant 0, \cdots, \boldsymbol{a}_n \geqslant \mathbf{0}$.
2° $\boldsymbol{a}_1 \leqslant 0, \cdots, \boldsymbol{a}_n \leqslant \mathbf{0}$.
Thus the numbers have the same sign if on the number line either they all lie on the right of $\mathbf{0}$ including $\mathbf{0}$, or they all lie on the left of $\mathbf{0}$ including $\mathbf{0}$. Show that equality holds in the generalised triangle inequality if and only if the numbers have the same sign. *Hint:* Consider the $n = 2$ case first.

Exercise 5.16. (Arithmetic mean-geometric mean inequality).
Let $n \in \mathbb{N}$, and $\boldsymbol{a}_1, \cdots, \boldsymbol{a}_n$ be positive reals such that their product is $\mathbf{1}$.
Show that their sum is at least n, with equality if and only if each is equal to $\mathbf{1}$.
It follows that if $\boldsymbol{x}_1, \cdots, \boldsymbol{x}_n$ are positive, then

$$\text{(the arithmetic mean)} \quad \frac{\boldsymbol{x}_1 + \cdots + \boldsymbol{x}_n}{n} \leqslant \sqrt[n]{\boldsymbol{x}_1 \cdot \ldots \cdot \boldsymbol{x}_n} \text{ (the geometric mean)},$$

with equality if and only if $\boldsymbol{x}_1 = \cdots = \boldsymbol{x}_n$.
(Here for $\boldsymbol{a} > \mathbf{0}$, $\sqrt[n]{\boldsymbol{a}}$ denotes the unique positive number[4] satisfying $(\sqrt[n]{\boldsymbol{a}})^n = \boldsymbol{a}$.)

Exercise 5.17. (Cauchy-Schwarz inequality). Let $n \in \mathbb{N}$, $\boldsymbol{a}_1, \cdots, \boldsymbol{a}_n, \boldsymbol{b}_1, \cdots, \boldsymbol{b}_n \in \mathbb{R}$. Show that $(\boldsymbol{a}_1^2 + \cdots + \boldsymbol{a}_n^2) \cdot (\boldsymbol{b}_1^2 + \cdots + \boldsymbol{b}_n^2) \geqslant (\boldsymbol{a}_1\boldsymbol{b}_1 + \cdots + \boldsymbol{b}_n\boldsymbol{b}_n)^2$, and equality holds if and only if either $\boldsymbol{a}_1 = \cdots = \boldsymbol{a}_n = \mathbf{0}$ or there exists $\boldsymbol{k} \in \mathbb{R}$ such that $\boldsymbol{b}_i = \boldsymbol{k}\boldsymbol{a}_i$ for all $i \in \{1, \cdots, n\}$.

5.3 Least upper bound property

In Exercise 1.8, p.10 we had seen that $\mathbb{Q}$ does not possess the least upper bound property. This 'analytical flaw' of $\mathbb{Q}$ is remedied by the set of real numbers. We begin by proving a few preliminaries. A more general version of the following result will be met later in Exercise 5.19, p.54.

Lemma 5.2. ('Baby' Archimedean property).
If $\boldsymbol{x} \in \mathbb{R}$, then there exists a natural number $m \in \mathbb{N}$ such that $m > \boldsymbol{x}$.

Proof. If $\boldsymbol{x} \leqslant \mathbf{0}$, then we can take $m = 1$, since $\mathbf{0} < \mathbf{1}$ gives by transitivity that $\boldsymbol{x} < \mathbf{1}$. Now consider the remaining case, namely $\boldsymbol{x} = [(a_n)_{n\in\mathbb{N}}] > \mathbf{0}$.

[4] The existence the completeness of $\mathbb{R}$, see [Sasane (2015), Chapter 6]. Assuming existence, the uniqueness is an easy consequence of the trichotomy law in $\mathbb{R}$.

We have seen that every Cauchy sequence in $\mathbb{Q}$ is bounded. So there exists a rational $A > 0$ such that for all $n \in \mathbb{N}$, $(a_n \leqslant)\ |a_n| \leqslant A$. This implies $\boldsymbol{x} < [(A+1)_{n\in\mathbb{N}}]$ (since $A+1-a_n \geqslant A+1-A = 1 > 0$ for all $n \in \mathbb{N}$). We have $A+1 = \frac{p}{q}$ for some $p, q \in \mathbb{N}$. Set $m = p+1$. Then we get $A+1 = \frac{p}{q} < \frac{m}{1}$ (since $p < p+1 \leqslant (p+1)q = mq$). So $\boldsymbol{x} < [(A+1)_{n\in\mathbb{N}}] < [(\frac{m}{1})_{n\in\mathbb{N}}]$, i.e., $\boldsymbol{x} < m$. □

Lemma 5.3. (Density of $\mathbb{Q}$ in $\mathbb{R}$). *Let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}$ be such that $\boldsymbol{x} < \boldsymbol{y}$. Then there exists an $r \in \mathbb{Q}$ such that $\boldsymbol{y} < r < \boldsymbol{x}$.*

Proof. As $\boldsymbol{y} - \boldsymbol{x} > \boldsymbol{0}$, we have in particular that $\boldsymbol{y} - \boldsymbol{x} \neq \boldsymbol{0}$, and so $(\boldsymbol{y} - \boldsymbol{x})^{-1}$ exists in $\mathbb{R}$. By Lemma 5.2, there exists an $n \in \mathbb{N}$ such that $n > (\boldsymbol{y} - \boldsymbol{x})^{-1}$, and so $n(\boldsymbol{y} - \boldsymbol{x}) > (\boldsymbol{y} - \boldsymbol{x})^{-1}(\boldsymbol{y} - \boldsymbol{x}) = \boldsymbol{1}$, that is, $n\boldsymbol{x} + \boldsymbol{1} < n\boldsymbol{y}$. By Lemma 5.2, there exists an $m_1 \in \mathbb{N}$ such that $m_1 > n\boldsymbol{x}$, and there exists an $m_2 \in \mathbb{N}$ such that $m_2 > -n\boldsymbol{x}$. So $-m_2 < n\boldsymbol{x} < m_1$ for some integers m_1, m_2. Define $T = \{k \in \mathbb{Z} : k \leqslant n\boldsymbol{x}\} \subset \mathbb{Z}$. Then $T \neq \varnothing$ as $-m_2 \in T$ and has $m_1 \in \mathbb{Z}$ as an upper bound. By the well-ordering principle in $\mathbb{Z}$ (see Exercise 3.18, p.33), T has a greatest element, say $k_* \in T$. Hence $k_* \leqslant n\boldsymbol{x}$. Also $n\boldsymbol{x} < k_* + 1$ (since otherwise $n\boldsymbol{x} \geqslant k_* + 1$, implying $k_* + 1 \in T$, and as k_* is a greatest element for T, we get $k_* + 1 \leqslant k_*$, a contradiction). So $k_* \leqslant n\boldsymbol{x} < k_* + 1$. Thus[5] $\boldsymbol{x} < \frac{k_*+1}{n} \leqslant \frac{n\boldsymbol{x}+1}{n} < \frac{n\boldsymbol{y}}{n} = \boldsymbol{y}$. With $r := \frac{k_*+1}{n} \in \mathbb{Q}$, the proof is complete. □

If $T \subset S \subset \mathbb{R}$, then we say T *is dense*[6] *in* S if for all $\boldsymbol{x}, \boldsymbol{y} \in S$, with $\boldsymbol{x} < \boldsymbol{y}$, there exists an $\boldsymbol{t} \in T$ such that $\boldsymbol{x} < \boldsymbol{t} < \boldsymbol{y}$. Thus we have shown above that '$\mathbb{Q}$ is dense in $\mathbb{R}$'.

Exercise 5.18. Show that the set $\mathbb{R}\backslash\mathbb{Q}$ (of irrational numbers) is dense in $\mathbb{R}$. *Hint:* Use the fact that $\sqrt{2} \in \mathbb{R}\backslash\mathbb{Q}$, and 'shift' $\boldsymbol{a}, \boldsymbol{b}$ by $\sqrt{2}$.

Lemma 5.4. *Let $\boldsymbol{x} = [(a_n)_{n\in\mathbb{N}}] \in \mathbb{R}$. Given any rational $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all natural numbers $n > N$, we have $a_n + \epsilon \geqslant \boldsymbol{x} \geqslant a_n - \epsilon$.*

Proof. Let a rational $\epsilon > 0$ be given. As $(a_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{Q}$, there exists an $N \in \mathbb{N}$ such that for all $n, m > N$, we have $|a_n - a_m| < \frac{\epsilon}{2}$, that is, $-\frac{\epsilon}{2} < a_n - a_m < \frac{\epsilon}{2}$. Fix an $n > N$. For all $m > N$, we obtain $a_n + \epsilon = a_n - a_m + a_m + \epsilon > -\frac{\epsilon}{2} + a_m + \epsilon = a_m + \frac{\epsilon}{2}$, i.e., $(a_n + \epsilon) - a_m > \frac{\epsilon}{2} =: d > 0$. So $[(a_n + \epsilon, a_n + \epsilon, a_n + \epsilon, \cdots)] > [(a_m)_{m\in\mathbb{N}}] = \boldsymbol{x}$. For all $m > N$, we also have $a_n - \epsilon = a_n - a_m + a_m - \epsilon < \frac{\epsilon}{2} + a_m - \epsilon = a_m - \frac{\epsilon}{2}$, i.e., $a_m - (a_n - \epsilon) > \frac{\epsilon}{2} > 0$. So $\boldsymbol{x} = [(a_m)_{m\in\mathbb{N}}] > [(a_n - \epsilon, a_n - \epsilon, a_n - \epsilon, \cdots)]$. □

[5] For $n \in \mathbb{N}$, $n^{-1} \in \mathbb{R}$ is $[(\frac{1}{n})_{k\in\mathbb{N}}]$. Also, $[(\frac{1}{n})_{k\in\mathbb{N}}] > \boldsymbol{0}$ as $\frac{1}{n} - 0 > \frac{1}{2n} =: d > 0$ for all $k \in \mathbb{N}$.

[6] In everyday language, we may say for example that 'These woods have a dense growth of birch trees', and the picture we then have in mind is that in any small area of the woods, we find a birch tree. A similar thing is conveyed by the above: no matter what 'patch' (described by the two numbers $\boldsymbol{x}$ and $\boldsymbol{y}$) we take in S (thought of as the woods), we can find an element of T (analogous to birch trees) in that patch.

Lemma 5.5. *Let* $\boldsymbol{x} = [(a_n)_{n\in\mathbb{N}}]$, $\boldsymbol{\alpha}, \boldsymbol{\beta} \in \mathbb{R}$, $N \in \mathbb{N}$ *be such that for all natural numbers* $n > N$, *we have* $\boldsymbol{\alpha} \leqslant a_n \leqslant \boldsymbol{\beta}$. *Then* $\boldsymbol{\alpha} \leqslant \boldsymbol{x} \leqslant \boldsymbol{\beta}$.

Proof. By the density of $\mathbb{Q}$ in $\mathbb{R}$, for each $n \in \mathbb{N}$, there exist $\alpha_n, \beta_n \in \mathbb{Q}$ such that $\boldsymbol{\alpha} - \frac{1}{n} < \alpha_n < \boldsymbol{\alpha}$, and $\boldsymbol{\beta} < \beta_n < \boldsymbol{\beta} + \frac{1}{n}$.

We claim that $(\alpha_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{Q}$. Indeed, for any $n, m \in \mathbb{N}$, we have $\boldsymbol{\alpha} - \frac{1}{n} < \alpha_n < \boldsymbol{\alpha}$, and $-\boldsymbol{\alpha} < -\alpha_m < -\boldsymbol{\alpha} + \frac{1}{m}$, which together give $-\frac{1}{n} < \alpha_n - \alpha_m < \frac{1}{m}$. Given any rational $\epsilon > 0$, let $N' \in \mathbb{N}$ be such that $N' > \epsilon^{-1}$. Then for $n, m > N'$, we have $-\frac{1}{N'} < -\frac{1}{n} < \alpha_n - \alpha_m < \frac{1}{m} < \frac{1}{N'}$, so that $|\alpha_n - \alpha_m| < \frac{1}{N'} < \epsilon$. Thus $(\alpha_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{Q}$.

A similar proof shows that also $(\beta_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{Q}$. Indeed, for any $n, m \in \mathbb{N}$, $\boldsymbol{\beta} < \beta_n < \boldsymbol{\beta} + \frac{1}{n}$, and $-\boldsymbol{\beta} - \frac{1}{m} < -\beta_m < -\boldsymbol{\beta}$, which together give $-\frac{1}{m} < \beta_n - \beta_m < \frac{1}{n}$. Given any rational $\epsilon > 0$, let $N' \in \mathbb{N}$ be such that $N' > \epsilon^{-1}$. Then for $n, m > N'$, we have $-\frac{1}{N'} < -\frac{1}{m} < \beta_n - \beta_m < \frac{1}{n} < \frac{1}{N'}$, so that $|\beta_n - \beta_m| < \frac{1}{N'} < \epsilon$. Thus $(\beta_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{Q}$.

We now show that $\boldsymbol{\alpha} = [(\alpha_n)_{n\in\mathbb{N}}]$. To do this we eliminate the other possibilities that $\boldsymbol{\alpha} < [(\alpha_n)_{n\in\mathbb{N}}]$ or $\boldsymbol{\alpha} > [(\alpha_n)_{n\in\mathbb{N}}]$. Let $\boldsymbol{\alpha} = [(\widetilde{\alpha}_n)_{n\in\mathbb{N}}]$.

1° Let $\boldsymbol{\alpha} < [(\alpha_n)_{n\in\mathbb{N}}]$. Then there exists a rational $d > 0$ and an $M \in \mathbb{N}$ such that for all $n > M$, we have $\alpha_n - \widetilde{\alpha}_n > d$. By Lemma 5.4 with $\epsilon = \frac{d}{2}$, there exists an $M' \in \mathbb{N}$ such that for all $n > M'$, $\widetilde{\alpha}_n + \epsilon = \widetilde{\alpha}_n + \frac{d}{2} \geqslant \boldsymbol{\alpha}$. So for $n > M + M'$, $d < \alpha_n - \widetilde{\alpha}_n \leqslant \alpha_n - \boldsymbol{\alpha} + \frac{d}{2} \leqslant \boldsymbol{0} + \frac{d}{2} = \frac{d}{2}$, a contradiction.

2° Let $\boldsymbol{\alpha} > [(\alpha_n)_{n\in\mathbb{N}}]$. Then there exists a rational $d > 0$ and an $M \in \mathbb{N}$ such that for all $n > M$, we have $\widetilde{\alpha}_n - \alpha_n > d$. By Lemma 5.4 with $\epsilon = \frac{d}{4}$, there exists an $M' \in \mathbb{N}$ such that for all $n > M'$, $\widetilde{\alpha}_n - \epsilon = \widetilde{\alpha}_n - \frac{d}{4} \leqslant \boldsymbol{\alpha}$. Finally, there exists an $M'' \in \mathbb{N}$ such that $M'' > \frac{4}{d}$. Then for all $n > M + M' + M''$, we have $d < \widetilde{\alpha}_n - \alpha_n \leqslant \frac{d}{4} + \boldsymbol{\alpha} - \alpha_n < \frac{d}{4} + \frac{1}{n} < \frac{d}{4} + \frac{d}{4} = \frac{d}{2}$, a contradiction.

Similarly, $\boldsymbol{\beta} = [(\beta_n)_{n\in\mathbb{N}}]$, as shown below. Let $\boldsymbol{\beta} = [(\widetilde{\beta}_n)_{n\in\mathbb{N}}]$.

1° Let $\boldsymbol{\beta} > [(\beta_n)_{n\in\mathbb{N}}]$. Then there exists a rational $d > 0$ and an $K \in \mathbb{N}$ such that for all $n > K$, we have $\widetilde{\beta}_n - \beta_n > d$. By Lemma 5.4 with $\epsilon = \frac{d}{2}$, there exists an $K' \in \mathbb{N}$ such that for all $n > K'$, $\widetilde{\beta}_n - \epsilon \leqslant \boldsymbol{\beta}$. Thus for $n > K + K'$, we get $d < \widetilde{\beta}_n - \beta_n \leqslant \widetilde{\beta}_n - \boldsymbol{\beta} \leqslant \epsilon = \frac{d}{2}$, a contradiction.

2° Let $\boldsymbol{\beta} < [(\beta_n)_{n\in\mathbb{N}}]$. Then there exists a rational $d > 0$ and an $K \in \mathbb{N}$ such that for all $n > K$, we have $\beta_n - \widetilde{\beta}_n > d$. By Lemma 5.4 with $\epsilon = \frac{d}{4}$, there exists an $K' \in \mathbb{N}$ such that for all $n > K'$, $\boldsymbol{\beta} \leqslant \widetilde{\beta}_n + \epsilon = \widetilde{\beta}_n + \frac{d}{4}$. Finally, there exists an $K'' \in \mathbb{N}$ such that $K'' > \frac{4}{d}$. Then for all $n > K + K' + K''$, we have $d < \beta_n - \widetilde{\beta}_n \leqslant \boldsymbol{\beta} + \frac{1}{n} - \boldsymbol{\beta} + \frac{d}{4} < \frac{d}{4} + \frac{d}{4} = \frac{d}{2}$, a contradiction.

For all $n > N$, we have $a_n - \alpha_n \geqslant \boldsymbol{\alpha} - \alpha_n > 0$, and $\beta_n - a_n \geqslant \beta_n - \boldsymbol{\beta} > 0$. By Exercise 5.13, p.48, $[(a_n - \alpha_n)_{n\in\mathbb{N}}] \geqslant \boldsymbol{0}$ and $[(\beta_n - a_n)_{n\in\mathbb{N}}] \geqslant \boldsymbol{0}$, i.e., $\boldsymbol{x} - \boldsymbol{\alpha} \geqslant \boldsymbol{0}$ and $\boldsymbol{\beta} - \boldsymbol{x} \geqslant \boldsymbol{0}$. Rearranging, we obtain $\boldsymbol{\alpha} \leqslant \boldsymbol{x} \leqslant \boldsymbol{\beta}$. □

From now on, we use ordinary letters $x, y, z, \cdots$ (instead of bold letters $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z}, \cdots$ for real numbers). We are now ready to show that $\mathbb{R}$ has the least upper bound property. First we define the terms upper bound and least upper bound of a subset of $\mathbb{R}$.

Definition 5.5. Let $S \subset \mathbb{R}$. An element $u \in \mathbb{R}$ is an *upper bound of* S if for all $x \in S$, we have $x \leqslant u$. If S possesses an upper bound, then S is *bounded above*. An element $u_* \in \mathbb{R}$ is a *least upper bound of* S if u_* is an upper bound of S, and for any upper bound u of S, we have $u_* \leqslant u$.

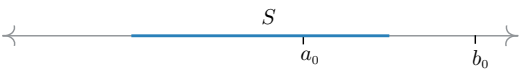
A set can have at most one least upper bound. Let $u_*, \widetilde{u}_*$ are two least upper bounds. As u_* is a least upper bound and $\widetilde{u}_*$ is an upper bound, we have $u_* \leqslant \widetilde{u}_*$. Also, since $\widetilde{u}_*$ is a least upper bound and u_* is an upper bound, we have $\widetilde{u}_* \leqslant u_*$. The trichotomy law then implies $u_* = \widetilde{u}_*$.

The least upper bound of a subset S of $\mathbb{R}$ is also called the *supremum of* S, and is denoted by $\sup S$. If $\sup S \in S$, then we call $\sup S$ *the maximum of* S, and denote it by $\max S$.

Theorem 5.2. (Least upper bound property of $\mathbb{R}$).
If $S \subset \mathbb{R}$, $S \neq \varnothing$, and S is bounded above, then S has a least upper bound.

(In the proof we have inserted pictures, which are of course not generic, but are only meant to serve as an aid to illustrate the steps in the argument.)

Proof. As $S \neq \varnothing$, there exists an element $a_0 \in S$, and since S is bounded above, there exists an upper bound $b_0 \in \mathbb{R}$, i.e., $a \leqslant b_0$ for all $a \in S$.



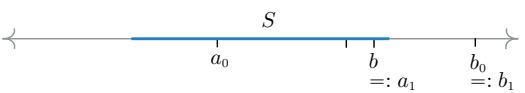
We define a_1, b_1 as follows:

1° If $\frac{a_0 + b_0}{2}$ is an upper bound of S, then define $a_1 := a_0$ and $b_1 := \frac{a_0 + b_0}{2}$.



So $a_0 \leqslant a_1$, $b_0 \geqslant b_1$, $a_1 \in S$, b_1 is an upper bound of S, $0 \leqslant b_1 - a_1 \leqslant \frac{b_0 - a_0}{2}$.

2° If $\frac{a_0 + b_0}{2}$ is not an upper bound of S, then there exists a $b \in S$ such that $\frac{a_0 + b_0}{2} < b$, and taking any such b, we define $a_1 := b$ and $b_1 = b_0$.



So $a_0 \leqslant \frac{a_0 + b_0}{2} < b = a_1$, $b_0 \geqslant b_1$, $a_1 \in S$, b_1 is an upper bound of S, and $0 \leqslant b_1 - a_1 \leqslant \frac{b_0 - a_0}{2}$.

Suppose for some $n \in \mathbb{N}$,

- $a_0, a_1, \cdots, a_{n-1} \in S$ and
- $b_0, b_1, \cdots, b_{n-1}$, which are upper bounds for S,

have been constructed such that

- $a_0 \leqslant a_1 \leqslant \cdots \leqslant a_{n-1}$
- $b_0 \geqslant b_1 \geqslant \cdots \geqslant b_{n-1}$, and
- $0 \leqslant b_k - a_k \leqslant \frac{b_0 - a_0}{2^k}$ for all $k \in \{1, \cdots, n-1\}$.

Now we construct a new $a_n \in S$ and a new upper bound b_n of S.

1° If $\frac{a_{n-1} + b_{n-1}}{2}$ is an upper bound of S, then $a_n := a_{n-1}$ and $b_n := \frac{a_{n-1} + b_{n-1}}{2}$. So $a_{n-1} \leqslant a_n$, $b_{n-1} \geqslant b_n$, $a_n \in S$, b_n is an upper bound of S, and we have $0 \leqslant b_n - a_n = \frac{b_{n-1} - a_{n-1}}{2} \leqslant \frac{b_0 - a_0}{2 \cdot 2^{n-1}} = \frac{b_0 - a_0}{2^n}$.

2° If $\frac{a_{n-1} + b_{n-1}}{2}$ is not an upper bound of S, then there exists a $b \in S$ such that $\frac{a_{n-1} + b_{n-1}}{2} < b$, and taking any such b, we set $a_n = b$ and $b_n = b_{n-1}$. So $a_{n-1} = \frac{a_{n-1} + a_{n-1}}{2} \leqslant \frac{a_{n-1} + b_{n-1}}{2} < b = a_n$, $b_{n-1} \geqslant b_n$, $a_n \in S$, b_n is an upper bound of S, $0 \leqslant b_n - a_n = b_{n-1} - b < b_{n-1} - \frac{a_{n-1} + b_{n-1}}{2} = \frac{b_{n-1} - a_{n-1}}{2} \leqslant \frac{b_0 - a_0}{2^n}$.

So we get sequences $a_0, a_1, \cdots$ in S, and $b_0, b_1, \cdots$ of upper bounds of S, such that

- $a_0 \leqslant a_1 \leqslant \cdots$,
- $b_0 \geqslant b_1 \geqslant \cdots$, and
- $0 \leqslant b_n - a_n \leqslant \frac{b_0 - a_0}{2^n}$ for all $n \in \mathbb{N}$.

If for some $n \geqslant 0$ we have $a_n = b_n$, then we claim that $u_* := b_n$ is the least upper bound of S. Indeed, firstly, $u_* = b_n$ is an upper bound of S by construction. Moreover, for any $u < u_* = b_n$, u cannot be an upper bound of S (because $u < u_* = b_n = a_n \in S$).

So we now have to consider the case that for all $n \geqslant 0$, we have $a_n < b_n$. By the density of $\mathbb{Q}$ in $\mathbb{R}$ for each $n \in \mathbb{N}$, there exists an $r_n \in \mathbb{Q}$ such that $a_n < r_n < b_n$. We claim that $(r_n)_{n \in \mathbb{N}}$ is a Cauchy sequence in $\mathbb{Q}$. To see this, let $\epsilon > 0$ be a given rational number. By the 'baby' Archimedean property (Lemma 5.2, p.49), there exists an $N \in \mathbb{N}$ such that $N > \frac{b_0 - a_0}{\epsilon}$, and so $\frac{b_0 - a_0}{2^N} \leqslant \frac{b_0 - a_0}{N} < \epsilon$ (using the inequality $n < 2^n$ for $n \in \mathbb{N}$: indeed, we have $1 < 2^1$, and if $n < 2^n$, then $n + 1 < 2^n + 1 \leqslant 2^n + n < 2^n + 2^n = 2 \cdot 2^n = 2^{n+1}$). If $n > m > N$, then $a_m < r_m < b_m$, $a_n < r_n < b_n$, $a_n \geqslant a_m$, which together give $r_m - r_n < b_m - r_n < b_m - a_n \leqslant b_m - a_m$. Moreover, since $b_n \leqslant b_m$, we also have that $r_m - r_n > a_m - r_n > a_m - b_n \geqslant a_m - b_m$. Hence

$$|r_m - r_n| < b_m - a_m \leqslant \frac{b_0 - a_0}{2^m} < \frac{b_0 - a_0}{2^N} \leqslant \frac{b_0 - a_0}{N} < \epsilon.$$

So $(r_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{Q}$. Define u_* to be the real number $[(r_n)_{n\in\mathbb{N}}]$ corresponding to $(r_n)_{n\in\mathbb{N}} \in \mathscr{C}$.

We claim u_* is the least upper bound of S. First, for any fixed $m \in \mathbb{N}$, and for all $n \geqslant m$, we have $a_m \leqslant a_n < r_n < b_n \leqslant b_m$. By Lemma 5.5, p.51, we get

$$a_m \leqslant u_* \leqslant b_m. \qquad (\star)$$

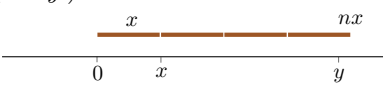
Suppose u_* is not an upper bound of S. Then there exists an $a \in S$ such that $a > u_*$. By the density of $\mathbb{Q}$ in $\mathbb{R}$, there exists an $r \in \mathbb{Q}$ such that

$$0 < r < a - u_*. \qquad (\star\star)$$

By the 'baby' Archimedean property, there exists an $m \in \mathbb{N}$ such that $m > \frac{b_0 - a_0}{r}$. So $0 \leqslant b_m - a_m \leqslant \frac{b_0 - a_0}{2^m} \leqslant \frac{b_0 - a_0}{m} < r$. Hence using $(\star)$ and $(\star\star)$, we get $b_m < a_m + r \leqslant u_* + r < a$, a contradiction to the fact that b_m is an upper bound of S.

Next, suppose $u < u_*$. Let $r \in \mathbb{Q}$ be such that $0 < r < u_* - u$. By the 'baby' Archimedean property, there exists an $m \in \mathbb{N}$ such that $m > \frac{b_0 - a_0}{r}$. So $0 \leqslant b_m - a_m \leqslant \frac{b_0 - a_0}{2^m} \leqslant \frac{b_0 - a_0}{m} < r$. Then $a_m > b_m - r \geqslant u_* - r > u$, showing u is not an upper bound of S. So u_* is the least upper bound of S. □

Exercise 5.19. (Archimedean property).
If $x, y \in \mathbb{R}$ and $x > 0$, then show that there exists an $n \in \mathbb{N}$ such that $y < nx$.
(If $y \leqslant 0$ to begin with, then this is just the trivial statement that $nx > 0 \geqslant y$, which works with every $n \in \mathbb{N}$. If $y > 0$, then the above tells us that no matter how small x is, if we keep 'tiling' the real line with multiples of the length x, then eventually we will surpass y.)



Exercise 5.20. (Greatest integer part $[\cdot]$).
Let $x \in \mathbb{R}$. Use the Archimedean property to show that there exists an integer $[x]$ with the property $[x] \leqslant x < [x] + 1$. Also show it is the unique integer k with the property that $k \leqslant x < k + 1$. If $m \in \mathbb{Z}$ satisfies $m \leqslant x$, then show that $m \leqslant [x]$.

Exercise 5.21. (Greatest lower bound property of $\mathbb{R}$).
The aim of this exercise is to show that $\mathbb{R}$ also has the 'greatest lower bound property'. Let $S \subset \mathbb{R}$. An element $\ell \in \mathbb{R}$ is a *lower bound of* S if for all $x \in S$, we have $\ell \leqslant x$. If S possesses a lower bound, then S is said to be *bounded below*. An element $\ell_* \in \mathbb{R}$ is a *greatest lower bound of* S if ℓ_* is a lower bound of S, and for any lower bound ℓ of S, we have $\ell \leqslant \ell_*$. Let S be a nonempty subset of real numbers which is bounded below. Show that S possesses a greatest lower bound. The greatest lower bound of a set S is unique, and is also called the *infimum of* S, denoted by $\inf S$. If $\inf S \in S$, then we call $\inf S$ the *minimum of* S, and denote it by $\min S$.

Exercise 5.22. Let $S = \{a_{nm} : m, n \in \mathbb{N}\}$ be a bounded subset of $\mathbb{R}$, that is, S has an upper bound and a lower bound. For $n, m \in \mathbb{N}$, define the sets

$$\text{Row}_n = \{a_{nm} : m \in \mathbb{N}\} = \{a_{n1}, a_{n2}, a_{n3}, \cdots\}, \text{ and}$$
$$\text{Col}_m = \{a_{nm} : n \in \mathbb{N}\} = \{a_{1m}, a_{2m}, a_{3m}, \cdots\}.$$

(1) Explain why $\sup \text{Row}_n$ and $\inf \text{Col}_m$ exist for all $n, m \in \mathbb{N}$.
(2) Let $n, m \in \mathbb{N}$. Prove that $\inf \text{Col}_m \leqslant \sup \text{Row}_n$. *Hint*: $a_{nm} \in \text{Row}_n \cap \text{Col}_m$.
(3) Explain why $\inf\{\sup \text{Row}_n : n \in \mathbb{N}\}$ and $\sup\{\inf \text{Col}_m : m \in \mathbb{N}\}$ exist.
(4) Prove that $\sup\{\inf \text{Col}_m : m \in \mathbb{N}\} \leqslant \inf\{\sup \text{Row}_n : n \in \mathbb{N}\}$. (In a full rectangular classroom, the tallest among the shortest students in the columns is at most as tall as the shortest among the tallest students in the rows.)
(5) Give an example of a set S for which we have a strict inequality in (4). *Hint*: Consider a 'diagonal' array.

5.4 Completeness of $\mathbb{R}$

We'd seen in Exercise 1.8, p.10 that not all Cauchy sequences in $\mathbb{Q}$ converge in $\mathbb{Q}$. In contrast, we will show that the set of 'Cauchy sequences in $\mathbb{R}$' coincides with the set of 'convergent sequences in $\mathbb{R}$'[7]. First, we carefully define these terms, and prove a few preliminary results on sequences.

Let S be a nonempty set. A *sequence in* S is a map $f : \mathbb{N} \to S$. Denoting the value of f at $n \in \mathbb{N}$ by $a_n \in S$, the entire sequence is referred to using the notation $(a_n)_{n\in\mathbb{N}}$. If $(n_k)_{k\in\mathbb{N}}$ is a strictly increasing sequence of natural numbers, i.e., $n_1 < n_2 < n_3 < \cdots$, then $(a_{n_k})_{k\in\mathbb{N}}$ is called a *subsequence of* $(a_n)_{n\in\mathbb{N}}$. We observe that for a strictly increasing sequence $(n_k)_{k\in\mathbb{N}}$ of natural numbers, we have $n_k \geqslant k$ for all $k \in \mathbb{N}$: indeed, $n_1 \geqslant 1$, and if $n_k \geqslant k$ for some $k \in \mathbb{N}$, then $n_{k+1} > n_k \geqslant k$ gives $n_{k+1} \geqslant k+1$ (Exercise 2.16, p.21).

Definition 5.6.
A sequence $(a_n)_{n\in\mathbb{N}}$ in $\mathbb{R}$ is a *Cauchy sequence in* $\mathbb{R}$ if for every real $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $(\mathbb{N} \ni) m, n > N$, we have $|a_m - a_n| < \epsilon$.
A sequence $(a_n)_{n\in\mathbb{N}}$ in $\mathbb{R}$ *converges in* $\mathbb{R}$ *to* $x \in \mathbb{R}$ if for every real $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $(\mathbb{N} \ni) n > N$, we have $|a_n - x| < \epsilon$.
We call x the *limit* of $(a_n)_{n\in\mathbb{N}}$, and write $\lim\limits_{n\to\infty} a_n = x$, or '$a_n \to x$ as $n \to \infty$'.
The notation $\lim\limits_{n\to\infty} a_n$ is justified by the following.

Exercise 5.23. (Uniqueness of limits).
Show that the limit of a real sequence which is convergent in $\mathbb{R}$ is unique.

Exercise 5.24. Suppose that $(a_n)_{n\in\mathbb{N}}$ is a real sequence such that for all $n \in \mathbb{N}$, we have $a_n \geqslant 0$. Prove that if $(a_n)_{n\in\mathbb{N}}$ is convergent in $\mathbb{R}$ with limit x, then $x \geqslant 0$. Show also that if $(\alpha_n)_{n\in\mathbb{N}}$ and $(\beta_n)_{n\in\mathbb{N}}$ are real sequences that are convergent in $\mathbb{R}$, and for all $n \in \mathbb{N}$, we have $a_n \leqslant b_n$, then $\lim\limits_{n\to\infty} a_n \leqslant \lim\limits_{n\to\infty} b_n$.

[7] This property is called the 'completeness of $\mathbb{R}$'. More generally, a metric space in which Cauchy sequences converge is called a *complete metric space*, see, e.g., [Rudin (1983)].

Theorem 5.3. *Every convergent sequence in* $\mathbb{R}$ *is a Cauchy sequence in* $\mathbb{R}$.

Proof. Let $(a_n)_{n\in\mathbb{N}}$ be a sequence of real numbers that converges to x. Let $(\mathbb{R} \ni)\ \epsilon > 0$. Then there exists an $N \in \mathbb{N}$ such that for all $n > N$, we have[8] $|a_n - x| < \frac{\epsilon}{2}$. Thus for $n, m > N$, we have

$$|a_n - a_m| = |a_n - x + x - a_m| \leqslant |a_n - x| + |a_m - x| < \tfrac{\epsilon}{2} + \tfrac{\epsilon}{2} = \epsilon.$$

So the sequence $(a_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{R}$. □

Definition 5.7. (Bounded sequence). A sequence $(a_n)_{n\in\mathbb{N}}$ in $\mathbb{R}$ is *bounded* if there exists a real $M > 0$ such that for all $n \in \mathbb{N}$, we have $|a_n| \leqslant M$.

Exercise 5.25. Show that a real sequence $(a_n)_{n\in\mathbb{N}}$ is bounded if and only if the set $S = \{a_n : n \in \mathbb{N}\}$ is a bounded subset of $\mathbb{R}$.

Lemma 5.6. *Every Cauchy sequence in* $\mathbb{R}$ *is bounded.*

Proof. Choose a positive ϵ, say $\epsilon = 1$. Then there exists an $N \in \mathbb{N}$ such that for all $n, m > N$, $|a_n - a_m| < \epsilon = 1$. In particular, with $m = N+1 > N$, and $n > N$, $|a_n - a_{N+1}| < 1$. Hence by the triangle inequality, for all $n > N$, $|a_n| = |a_n - a_{N+1} + a_{N+1}| \leqslant |a_n - a_{N+1}| + |a_{N+1}| < 1 + |a_{N+1}|$. On the other hand, for $n \leqslant N$, $|a_n| \leqslant |a_1| + \cdots + |a_N| + |a_{N+1}| + 1 =: M$. Then $M > 0$, and for all $n \in \mathbb{N}$, $|a_n| \leqslant M$. So the sequence $(a_n)_{n\in\mathbb{N}}$ is bounded. □

Exercise 5.26. (Algebra of limits).
Let $(a_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ be real sequences that are convergent in $\mathbb{R}$ with limits x and y, respectively.
(1) Show that $(a_n + b_n)_{n\in\mathbb{N}}$ is convergent in $\mathbb{R}$ with limit $x + y$.
(2) Show that $(a_n b_n)_{n\in\mathbb{N}}$ is convergent in $\mathbb{R}$ with limit xy.
For any $c \in \mathbb{R}$, since the constant sequence $(c)_{n\in\mathbb{N}}$ is convergent in $\mathbb{R}$ with limit c, it follows from (2) that the sequence $(cb_n)_{n\in\mathbb{N}}$ is convergent in $\mathbb{R}$ with the limit cy. In particular, $(-b_n)_{n\in\mathbb{N}}$ is convergent in $\mathbb{R}$ with the limit $-y$, and using (1), $(a_n - b_n)_{n\in\mathbb{N}}$ is convergent in $\mathbb{R}$ with limit $x - y$.

Definition 5.8. (Increasing, decreasing, and monotone sequences).
Let $(a_n)_{n\in\mathbb{N}}$ be a sequence of real numbers.
We say the sequence $(a_n)_{n\in\mathbb{N}}$ is *increasing* if for all $n \in \mathbb{N}$, we have $a_n \leqslant a_{n+1}$, i.e., if $a_1 \leqslant a_2 \leqslant a_3 \leqslant \cdots$.
We say the sequence $(a_n)_{n\in\mathbb{N}}$ is *decreasing* if for all $n \in \mathbb{N}$, we have $a_n \geqslant a_{n+1}$, i.e., if $a_1 \geqslant a_2 \geqslant a_3 \geqslant \cdots$.
We say $(a_n)_{n\in\mathbb{N}}$ is *monotone* if it is increasing or decreasing.

Proposition 5.6.
If a sequence in $\mathbb{R}$ *is monotone and bounded, then it is convergent in* $\mathbb{R}$.

[8] Here $\frac{\epsilon}{2}$ is the product of the real number ϵ with the rational number $\frac{1}{2}$ (viewed as a real number).

Proof. We just consider the case of *increasing* sequences which are bounded, leaving the case of decreasing sequences as an exercise. Denote the increasing and bounded sequence by $(a_n)_{n\in\mathbb{N}}$. Let $M>0$ be such that for all $n\in\mathbb{N}$, $|a_n|\leqslant M$. Let $S=\{a_n : n\in\mathbb{N}\}\subset\mathbb{R}$. Then $S\neq\varnothing$ (as $a_1\in S$), and M is an upper bound of S because for all $n\in\mathbb{N}$, $a_n\leqslant|a_n|\leqslant M$. By the least upper bound property of $\mathbb{R}$, S possesses a least upper bound, which we denote by x. We claim that $(a_n)_{n\in\mathbb{N}}$ converges in $\mathbb{R}$ to x. Let ϵ be a positive real. Since $x-\epsilon<x$, it follows that $x-\epsilon$ is not an upper bound for S, and so there exists an $N\in\mathbb{N}$ such that $x-\epsilon<a_N$, that is, $x-a_N<\epsilon$. For all $n>N$, we have $a_N\leqslant a_n$. As x is an upper bound for S, $a_n\leqslant x$, and so $|a_n-x|=x-a_n$. Thus for all $n>N$, $|a_n-x|=x-a_n\leqslant x-a_N<\epsilon$. □

Exercise 5.27. Fill in the blanks in the following proof of the following fact:

Every bounded decreasing sequence of real numbers converges in $\mathbb{R}$.

Let $(a_n)_{n\in\mathbb{N}}$ be a bounded decreasing sequence of real numbers. Let ℓ_* be the ____ lower bound of the nonempty bounded set $\{a_n : n\in\mathbb{N}\}$. The existence of ℓ_* is guaranteed by the ____ of the set of real numbers. We show that ℓ_* is the ____ of $(a_n)_{n\in\mathbb{N}}$. Taking $\epsilon>0$, we must show that there exists a positive integer N such that ____ for all $n>N$. Since $\ell_*+\epsilon>\ell_*$, $\ell_*+\epsilon$ is not ____ of $\{a_n : n\in\mathbb{N}\}$. Therefore there exists N with ____ $\leqslant a_N<$ ____. Since $(a_n)_{n\in\mathbb{N}}$ is ____, we have for all $n\geqslant N$ that $\ell_*-\epsilon<\ell_*\leqslant$ ____ $\leqslant a_N<\ell_*+\epsilon$, and so $|a_n-\ell_*|<\epsilon$. □

Lemma 5.7. *Any subsequence of a real sequence convergent in* $\mathbb{R}$ *is convergent in* $\mathbb{R}$ *with the same limit.*

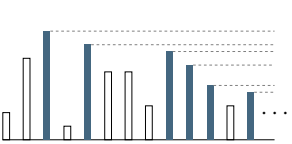
Proof. Let $(a_{n_k})_{k\in\mathbb{N}}$ be a subsequence of the sequence $(a_n)_{n\in\mathbb{N}}$ that converges in $\mathbb{R}$ with limit x. Given a real $\epsilon>0$, let $N\in\mathbb{N}$ be such that for all $n>N$, $|a_n-x|<\epsilon$. We have $n_N\geqslant N$, and so for all $k>N$, $n_k>n_N\geqslant N$. Thus for $k>N$, $|a_{n_k}-x|<\epsilon$. So $(a_{n_k})_{k\in\mathbb{N}}$ is convergent with limit x. □

Exercise 5.28. (Geometric series). Suppose that $a,r\in\mathbb{R}$, $a>0$, and $0<r<1$. Show that $(r^n)_{n\in\mathbb{N}}$ is decreasing and bounded, and is convergent in $\mathbb{R}$ to 0. Define $s_n=a+ar+\cdots+ar^{n-1}$ for all $n\in\mathbb{N}$. Prove that $(s_n)_{n\in\mathbb{N}}$ is increasing and bounded, and is convergent in $\mathbb{R}$ to $\frac{a}{1-r}$.

Proposition 5.7. *Every real sequence has a monotone subsequence.*

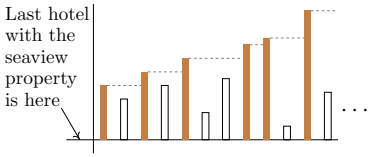
Before giving the formal proof, we give an illustration of the idea behind this proof. If $(a_n)_{n\in\mathbb{N}}$ is the given sequence, then imagine that there is an infinite chain of hotels along a line, where the nth hotel has height a_n, and at the horizon, there is a sea. A hotel has the *seaview property* if it is higher than all hotels following it (so that from the roof of the hotel, one can view the sea). There are only two possibilities, as shown in the following picture.

1° Infinitely many hotels have the seaview property



Then by taking successively the heights of the hotels the seaview property we get a *decreasing* subsequence.

2° Finitely many hotels have the seaview property



Then after the last hotel with the seaview property, one can start with any hotel and then always find one that is at least as high, which is taken as the next hotel, and then finding yet another that is at least as high as that one, and so on. The heights of these hotels form an *increasing* subsequence.

Proof. Let $(a_n)_{n\in\mathbb{N}}$ be the given sequence in $\mathbb{R}$. Define the subset S of $\mathbb{N}$ by $S=\{m\in\mathbb{N}:\text{for all } n>m,\ a_n<a_m\}$. We have the following two cases.

1° There exists an $M\in\mathbb{N}$ such that for all $m\in S$, we have $m\leqslant M$.
We define inductively an increasing subsequence $(a_{n_k})_{k\in\mathbb{N}}$ of $(a_n)_{n\in\mathbb{N}}$. If S is empty, then define $n_1=1$, and otherwise let $n_1=M+1$. Suppose for some $k\in\mathbb{N}$, $n_1<\cdots<n_k$ have been constructed such that $a_{n_1}\leqslant\cdots\leqslant a_{n_k}$. Let $S'=\{m\in\mathbb{N}:m>n_k \text{ and } a_m\geqslant a_{n_k}\}$ and define $n_{k+1}=\min S'$. The minimum exists by the well-ordering principle for $\mathbb{N}$ as S' is a nonempty subset of $\mathbb{N}$. (Otherwise if $S'=\varnothing$, then for all $m>n_k$, we have $a_m<a_{n_k}$, so that $n_k\in S$. But this is impossible if $S=\varnothing$, and also impossible if $S\neq\varnothing$, as $n_k\geqslant n_1=M+1>M\geqslant m$ for all $m\in S$.) By the definition of S', we get $a_{n_{k+1}}\geqslant a_{n_k}$. Then $(a_{n_k})_{k\in\mathbb{N}}$ is an increasing subsequence of $(a_n)_{n\in\mathbb{N}}$.

2° There does not exist[9] an $M\in\mathbb{N}$ such that for all $m\in S$, we have $m\leqslant M$. Then $S\neq\varnothing$ (otherwise $M=1$ would be an upper bound in $\mathbb{N}$ for S). Let n_1 be any element in S. Then $M=n_1$ is not an upper bound in $\mathbb{N}$ for S, and so there exists an $n_2\in S$ such that $n_2>n_1$. Suppose for some $k\in\mathbb{N}$, $n_1<\cdots<n_k$ have been constructed. Then $M=n_k$ is not an upper bound in $\mathbb{N}$ for S, and so there exists an $n_{k+1}\in S$ such that $n_{k+1}>n_k$. So there exists an increasing sequence $(n_k)_{k\in\mathbb{N}}$ in $\mathbb{N}$ such that $n_k\in S$ for each $k\in\mathbb{N}$. By the definition of S, the sequence $(a_{n_k})_{k\in\mathbb{N}}$ in $\mathbb{R}$ is then a (strictly) decreasing subsequence of $(a_n)_{n\in\mathbb{N}}$.

Thus every sequence $(a_n)_{n\in\mathbb{N}}$ has a monotone subsequence. □

[9] We abbreviate this by saying that S does not have an upper bound in $\mathbb{N}$.

Theorem 5.4. (Bolzano-Weierstrass).
Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.

Proof. Let $(a_n)_{n\in\mathbb{N}}$ be a bounded real sequence. Let M be a positive real such that for all $n \in \mathbb{N}$, we have $|a_n| \leqslant M$. By Proposition 5.7, p.57, $(a_n)_{n\in\mathbb{N}}$ has a monotone subsequence, say $(a_{n_k})_{k\in\mathbb{N}}$. But $(a_{n_k})_{k\in\mathbb{N}}$ is bounded too since for all $k \in \mathbb{N}$, we have $|a_{n_k}| \leqslant M$. As the real sequence $(a_{n_k})_{k\in\mathbb{N}}$ is monotone and bounded, it follows from Proposition 5.6, p.56, that it is convergent in $\mathbb{R}$. □

Theorem 5.5. *Every Cauchy sequence in $\mathbb{R}$ is convergent in $\mathbb{R}$.*

Proof. Let $(a_n)_{n\in\mathbb{N}}$ be Cauchy in $\mathbb{R}$. Lemma 5.6, p.56, implies that $(a_n)_{n\in\mathbb{N}}$ is bounded. As $(a_n)_{n\in\mathbb{N}}$ is bounded, it follows from the Bolzano-Weierstrass theorem that it possesses a convergent subsequence, say $(a_{n_k})_{k\in\mathbb{N}}$, which converges in $\mathbb{R}$ to, say, $x \in \mathbb{R}$. We claim that $(a_n)_{n\in\mathbb{N}}$ is also convergent with limit x. Let $\epsilon > 0$. Then there exists an $N \in \mathbb{N}$ such that

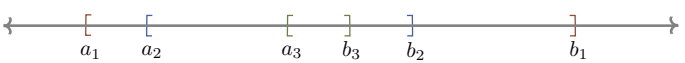
$$\text{for all } n, m > N,\ |a_n - a_m| < \tfrac{\epsilon}{2}. \qquad (\star)$$

As $(a_{n_k})_{k\in\mathbb{N}}$ converges to x, we can find[10] an $n_K > N$ such that $|a_{n_K} - x| < \frac{\epsilon}{2}$. Taking $m = n_K$ in $(\star)$, we get for all $n > N$ that

$$|a_n - x| = |a_n - a_{n_K} + a_{n_K} - x| \leqslant |a_n - a_{n_K}| + |a_{n_K} - x| < \tfrac{\epsilon}{2} + \tfrac{\epsilon}{2} = \epsilon.$$

Thus $(a_n)_{n\in\mathbb{N}}$ is convergent with limit x. □

Example 5.2. (Nested interval property). We will show the following: If for $m \in \mathbb{N}$, $I_m := \{x \in \mathbb{R} : a_m \leqslant x \leqslant b_m\}$ are such that $I_1 \supset I_2 \supset I_3 \supset \cdots$, then $\bigcap_{m\in\mathbb{N}} I_m \neq \varnothing$.



For $n \in \mathbb{N}$, $a_n \in I_n \subset I_1$, and so $a_1 \leqslant a_n \leqslant b_1$. So the sequence $(a_n)_{n\in\mathbb{N}}$ is bounded. By the Bolzano-Weierstrass theorem, $(a_n)_{n\in\mathbb{N}}$ has a convergent subsequence, say $(a_{n_k})_{k\in\mathbb{N}}$. Let a denote the limit of $(a_{n_k})_{k\in\mathbb{N}}$. We claim that for each $m \in \mathbb{N}$, we have $a_m \leqslant a$. Suppose, on the contrary, that there exists an $m \in \mathbb{N}$ such that $a_m > a$, i.e., $\epsilon := a_m - a > 0$. Since $(a_{n_k})_{k\in\mathbb{N}}$ is convergent with limit a, there exists a $K \in \mathbb{N}$ such that for all $k > K$, we have $|a_{n_k} - a| < \epsilon = a_m - a$. Then for $k > K$, we get $a_{n_k} - a \leqslant |a_{n_k} - a| < a_m - a$, and so $a_{n_k} < a_m$. For $k > m$, we have $n_k \geqslant k > m$, and so $I_{n_k} \subset I_m$. Thus for $k > m$, we obtain $a_{n_k} \geqslant a_m$. Hence, if $k > \max\{m, K\}$, then $a_m \leqslant a_{n_k} < a_m$, which is impossible. This proves the claim that for each $m \in \mathbb{N}$, $a_m \leqslant a$.

[10] If K' is such that for all $k > K'$, we have $|a_{n_k} - x| < \frac{\epsilon}{2}$, then we may take $K = K' + N$ (since then $n_K \geqslant K = N + K' \geqslant N + 1 > N$ and $n_K \geqslant K = K' + N \geqslant K' + 1 > K'$).

Next we claim that for all $m \in \mathbb{N}$, we have $a \leqslant b_m$. For $k \geqslant m$, we have $n_k \geqslant k \geqslant m$, and so $a_{n_k} \in I_{n_k} \subset I_m$. Thus for all $k \geqslant m$, we have $a_{n_k} \leqslant b_m$. Passing to the limit as $k \to \infty$ (see Exercise 5.24, p.55), we obtain $a \leqslant b_m$, proving the claim.

So for all $m \in \mathbb{N}$, $a_m \leqslant a \leqslant b_m$, i.e., $a \in I_m$. Thus $a \in \bigcap_{m\in\mathbb{N}} I_m \neq \varnothing$.

If the endpoints a_m, b_m of I_m are such that the 'length' $b_m - a_m$ of the interval I_m converges to 0 as m goes to infinity, then if $a' \in \bigcap_{m\in\mathbb{N}} I_m \neq \varnothing$, we have for all $m \in \mathbb{N}$ that $b_m - a_m \geqslant a' - a_m \geqslant 0$, and so $(a_m)_{m\in\mathbb{N}}$ converges in $\mathbb{R}$ to a'. (Moreover as $b_m = (b_m - a_m) + a_m$, also $(b_m)_{m\in\mathbb{N}}$ converges in $\mathbb{R}$ to a'.) But as the limit of the sequence $(a_m)_{m\in\mathbb{N}}$ is unique, it follows that the nonempty set $\bigcap_{m\in\mathbb{N}} I_m$ must be a singleton, equal to $\{a'\}$.[11] ◇

Exercise 5.29. Let M be a positive real number, and suppose that for each $n \in \mathbb{N}$, $(x_n, y_n) \in \mathbb{R}^2$ satisfies $|x_n| \leqslant M$ and $|y_n| \leqslant M$. Show that there exists a sequence $(m_\ell)_{\ell\in\mathbb{N}}$ of natural numbers which is strictly increasing (i.e., $m_1 < m_2 < m_3 < \cdots$), such that $(x_{m_\ell})_{\ell\in\mathbb{N}}$ and $(y_{m_\ell})_{\ell\in\mathbb{N}}$ converge in $\mathbb{R}$.

Exercise 5.30. (Kronecker's theorem).
Recall from p.50, the notion of a dense subset T of a set $S \subset \mathbb{R}$.
Let $\alpha \in \mathbb{R}\backslash\mathbb{Q}$, and $\langle 1, \alpha\rangle := \{m + n\alpha : m, n \in \mathbb{Z}\}$. Define $[0, 1] := \{x \in \mathbb{R} : 0 \leqslant x \leqslant 1\}$. Show that $\langle 1, \alpha\rangle \cap [0, 1]$ is dense in $[0, 1]$, and $\langle 1, \alpha\rangle$ is dense in $\mathbb{R}$.

Exercise 5.31. Using[12] $\pi \in \mathbb{R}\backslash\mathbb{Q}$, and Kronecker's theorem (Exercise 5.30 above), show that for all $x \in [-1, 1]$ and all $\epsilon > 0$, there is an $n \in \mathbb{N}$ such that $|\sin n - x| < \epsilon$.

Remark 5.1. (Alternative constructions of $\mathbb{R}$). While we have constructed the reals using Cauchy sequences in $\mathbb{Q}$, there are other equivalent ways of obtaining the reals. We outline one such construction in this remark, and yet another one is outlined in Exercise 5.32 below. The set $\mathbb{R}$ can be constructed from $\mathbb{Q}$ via 'Dedekind cuts', where we view real numbers as places where a line may be 'cut with scissors'. More precisely, a *cut* C is a subset of $\mathbb{Q}$ with the following three properties:
(C1) $C \neq \varnothing$ and $C \neq \mathbb{Q}$.
(C2) If $p \in C$, $q \in \mathbb{Q}$ and $q < p$, then $q \in C$.
(C3) If $p \in C$, then there exists an $r \in C$ such that $p < r$.
The set $\mathbb{R}$ is then defined to be the set of all cuts. For cuts C_1, C_2, we say $C_1 < C_2$ if $C_1 \subset C_2$. The arithmetic operations of addition and multiplication are defined as follows.

[11] Additionally, if $a_m, b_m \in \mathbb{Q}$ for each $m \in \mathbb{N}$, then this lends credibility to our intuitive picture that the real numbers, while extending the rationals, fill in any remaining holes between the rationals on the number line.
[12] For a proof of the irrationality of π, see, e.g., [Sasane (2015), Theorem 5.33].

For cuts C_1, C_2, we define $C_1 + C_2 = \{r + s : r \in C_1 \text{ and } s \in C_2\}$. Let the cut $C_* := \{q \in \mathbb{Q} : q < 0\}$. Defining multiplication in $\mathbb{R}$ is more cumbersome. One first defines multiplication of 'positive reals', which are elements of the set $\mathbb{R}_+ = \{C \in \mathbb{R} : C_* < C\}$. If $C_1, C_2 \in \mathbb{R}_+$, then

$$C_1 \cdot C_2 := \left\{ p \in \mathbb{Q} : \begin{array}{l} \text{there exist } q_1 \in C_1,\ q_2 \in C_2 \text{ such that} \\ 0 < q_1,\ 0 < q_2, \text{ and } p \leqslant q_1 q_2 \end{array} \right\}.$$

Finally, the definition of multiplication is completed as follows: $C \cdot C_* := C_*$, $C_* \cdot C := C_*$ for all $C \in \mathbb{R}$, and for cuts C_1, C_2, we have

$$C_1 \cdot C_2 := \begin{cases} (-C_1) \cdot (-C_2) & \text{if } C_1 < C_* \text{ and } C_2 < C_* \\ -((-C_1) \cdot C_2) & \text{if } C_1 < C_* \text{ and } C_* < C_2 \\ -(C_1 \cdot (-C_2)) & \text{if } C_* < C_1 \text{ and } C_2 < C_*. \end{cases}$$

Then[13] $\mathbb{R}$ is an ordered field[14], and has the least upper bound property. ∗

Exercise 5.32. Let S denote the set of all maps $\mu : \mathbb{Z} \to \mathbb{Z}$ for which the set $\{\mu(m+n) - \mu(m) - \mu(n) : m, n \in \mathbb{Z}\}$ is a finite set. Show that S is not empty. On S, define the relation $\sim$ as follows: For $\mu, \widetilde{\mu} \in S$, $\mu \sim \widetilde{\mu}$ if the set $\{\mu(n) - \widetilde{\mu}(n) : n \in \mathbb{Z}\}$ is a finite set. Show that $\sim$ is an equivalence relation. Show that if $\mu, \nu \in S$, then their pointwise sum $\mu + \nu$ belongs to S. We denote the equivalence class of $\mu \in S$ by $[\mu]$. Let $\mathbb{R} := \{[\mu] : \mu \in S\}$. Define addition $+$ on $\mathbb{R}$ as follows: for $\boldsymbol{a} = [\mu]$ and $\boldsymbol{b} = [\nu]$ in $\mathbb{R}$, we set $\boldsymbol{a} + \boldsymbol{b} = [\mu] + [\nu] = [\mu + \nu]$. Show that addition $+ : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ is well-defined. We define multiplication $\cdot : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ as follows: for $\boldsymbol{a} = [\mu]$ and $\boldsymbol{b} = [\nu]$ in $\mathbb{R}$, we set $\boldsymbol{a} \cdot \boldsymbol{b} = [\mu] \cdot [\nu] = [\mu \circ \nu]$. It can be shown that for $\mu, \nu \in S$, $\mu \circ \nu \in S$ too, and that the notion of multiplication in $\mathbb{R}$ is well-defined. Prove that there exists an element $\mathbf{1} \in \mathbb{R}$ such that for all $\boldsymbol{a} \in \mathbb{R}$, we have $\boldsymbol{a} \cdot \mathbf{1} = \boldsymbol{a} = \mathbf{1} \cdot \boldsymbol{a}$. Give an explicit $\mu : \mathbb{Z} \to \mathbb{Z}$ such that $\mathbf{1} = [\mu]$. We call $\mu \in S$ *positive* if the set $\{\mu(n) : n \in \mathbb{N} \text{ and } \mu(n) \leqslant 0\}$ is finite, and the set $\{\mu(n) : n \in \mathbb{Z}\}$ is infinite. An element $\boldsymbol{a} = [\mu] \in \mathbb{R}$ is *positive* if μ is positive. Then this is a well-defined notion. It can be proved[15] that the set $\mathbb{R}$ with addition, multiplication and the order relation (obtained from the notion of positive elements) is an ordered field, and has the least upper bound property.

[13] See, e.g., [Rudin (1983), Appendix, Chapter 1].
[14] See p.110.
[15] See [A'Campo (2021)].

Chapter 6

The ring of integers

In Chapter 4, we saw that the set of integers $\mathbb{Z}$ with its operations of integer addition and integer multiplication does not form a field (unlike the rationals and the reals). It fails narrowly from making the cut: nonzero integers do not necessarily have an integer multiplicative inverse. In general, such a structure is called a 'ring'. We will not study rings in general, but focus on the integers, and the ring structure of $\mathbb{Z}$.

6.1 The division algorithm

Theorem 6.1. (Division algorithm).
Given integers m, d with $d>0$, there exist unique integers q and r such that $m=qd+r$, and $0\leqslant r<d$.

Proof. Consider the set $S=\{m-nd : n\in\mathbb{Z} \text{ and } m-nd\geqslant 0\}$. Then $S\neq\varnothing$: $m-0d\in S$ if $m\geqslant 0$, and $m-md\in S$ if $m<0$ as $m-md=(-m)(d-1)\geqslant 0$.

1° If $0\in S$, then $m-qd=0=:r$ for some $q\in\mathbb{Z}$. So $m=qd+0$.

2° If $0\notin S$, then $S\subset\mathbb{N}$. By the well-ordering principle for $\mathbb{N}$, S possesses a least element, say r. Hence $r=m-qd$ for some $q\in\mathbb{Z}$, i.e., $m=qd+r$. Since $r\in S$, $r\geqslant 0$. It remains to show that $r<d$. Suppose $r\geqslant d$. Then $r':=r-d\geqslant 0$, and $r'=r-d=m-qd-d=m-(q+1)d\in S$. As r is the least element of S, $r\leqslant r'=r-d$, which yields $d\leqslant 0$, a contradiction.

Uniqueness: Let $q, q', r, r'\in\mathbb{Z}$ satisfy $0\leqslant r, r'<d$ and $n=qd+r=q'd+r'$. Let $r\neq r'$. Without loss of generality, let $r'>r$. Then $(q-q')d=r'-r>0$. As $d>0$, we have $q-q'>0$ (Exercise 3.15, p.32). So $r'-r=(q-q')d\geqslant 1d=d$. Thus $r'=r+d\geqslant 0+d=d$, a contradiction. So $r'=r$, and $(q-q')d=r'-r=0$. As $d\neq 0$, we get from Theorem 3.8, p.31 that $q-q'=0$, i.e., $q=q'$. □

Above, we assumed that d was positive. This was not essential. We have:

Corollary 6.1. *Given integers m, d with $d\neq 0$, there exist unique integers q and r such that $m=qd+r$, $0\leqslant r<|d|$.*

The number q is called the *quotient*, and r is called the *remainder*.

Proof. In light of Theorem 6.1, we only need to show this when $d < 0$. But then $-d > 0$, and so by Theorem 6.1, there exist $q', r \in \mathbb{Z}$ such that $m = q'(-d) + r$ and moreover $0 \leqslant r < -d = |d|$. Setting $q = -q' \in \mathbb{Z}$, we get $m = qd + r$, as wanted.

To show uniqueness, suppose $m = qd + r = \widetilde{q}d + \widetilde{r}$, for some $q, \widetilde{q} \in \mathbb{Z}$ and $0 \leqslant r, \widetilde{r} < |d| = -d$. Then we have $m = (-q)(-d) + r = (-\widetilde{q})(-d) + \widetilde{r}$. By the uniqueness part of Theorem 6.1, $-q = -\widetilde{q}$ (implying $\widetilde{q} = q$) and $r = \widetilde{r}$. □

Exercise 6.1.
(1) Show that all perfect squares (i.e., squares n^2 of integers n) have the form $4k$ or $4k + 1$ for some $k \in \mathbb{Z}$.
(2) Show that none of the numbers $11, 111, 1111, \cdots$ is a perfect square.

Exercise 6.2. Show that 1 cannot be written as a sum of reciprocals of an even number of odd integers.

Exercise 6.3. What are the last two digits of $(56789)^{41}$?

6.2 Divisibility, gcd, and the Euclid algorithm

Definition 6.1. (Divisor/multiple).
Let $m, n \in \mathbb{Z}$. Then we say
- *n is a divisor/factor of m*, or
- *n divides m*, or
- *m is a multiple of n*, or
- *m is divisible by n*

if there exists an integer $d \in \mathbb{Z}$ such that $m = nd$. We write
- $n \,|\, m$ if n is a divisor of m
- $n \nmid m$ if n is not a divisor of m.

Example 6.1. $3 \,|\, 12$ and $4 \,|\, 12$ as $12 = 3(4)$. Also $-3 \,|\, 12$ since $12 = -3(-4)$. But $0 \nmid 1$. For all $n \in \mathbb{Z}$, $n \,|\, 0$ since $0 = n0$, and $1 \,|\, n$ as $n = 1n$. ◇

Exercise 6.4. Let d, a, b are integers, $d \,|\, a$ and $d \,|\, b$.
Show that for all $x, y \in \mathbb{Z}$, we have $d \,|\, ax + by$.

Exercise 6.5. Let $a, b, c \in \mathbb{Z}$ be such that $a \,|\, b$ and $b \,|\, c$. Show that $a \,|\, c$.

Exercise 6.6. Let $a, b \in \mathbb{Z}$, $a \,|\, b$ and $b \neq 0$. Show that $|a| \leqslant |b|$.

Exercise 6.7. Let $a, b \in \mathbb{Z}$, $a \,|\, b$ and $b \,|\, a$. Show that $a = b$ or $a = -b$.

Exercise 6.8. Show that for all $n \in \mathbb{Z}$, $3 \,|\, n^3 - n$.

Exercise 6.9. Find all integers $n \in \mathbb{Z}$ such that $n^2 + 1$ is divisible by $n + 1$.
Hint: Consider $n^2 - 1$.

Exercise 6.10. Let $a \in \mathbb{Z}$ and $4 \,|\, a - 1$. Show that $4 \,|\, a^2 + 3$, but $8 \nmid a^2 + 3$.

Exercise 6.11. Show that if $n \in \mathbb{Z}$ is odd, then $16 \,|\, n^4 + 4n^2 + 11$.

Exercise 6.12. Let $a, b \in \mathbb{Z}$ and $n \in \mathbb{N}$. Prove that $a - b \,|\, a^n - b^n$.

Exercise 6.13. Let $a, b \in \mathbb{Z}$ and $k \in \mathbb{N}$. Prove that $a + b \,|\, a^{2k-1} + b^{2k-1}$.

Exercise 6.14. Show that 2025 divides $1^{2025} + 2^{2025} + 3^{2025} + \cdots + 2024^{2025}$.

Exercise 6.15. Suppose that p is a polynomial[1] with integer coefficients such that $p(5) = 2$. Show that $p(2025)$ cannot be a perfect square. *Hint:* For $x, y \in \mathbb{Z}$, $x - y$ divides $p(x) - p(y)$. Consider the remainder when divided by 4.

Exercise 6.16. (Infinite descent).
In this exercise we will show that there are no integer solutions to $x^3 + 3y^3 + 9z^3 = 0$ besides the trivial solution $x = y = z = 0$ as an illustration of the 'method of infinite descent' (where the idea is that if a statement holds for a bunch of integers, then the same statement is valid for smaller integers, leading to an infinite descent, and ultimately somehow a contradiction to the well-ordering principle). To use the well-ordering principle, we should have a set of integers bounded below, and so we define (with hindsight) the set

$$S = \{x^2 + y^2 + z^2 : (0,0,0) \neq (x,y,z) \in \mathbb{Z}^3 \text{ and } x^3 + 3y^3 + 9z^3 = 0\} \subset \mathbb{N}.$$

Suppose S is not empty. By the well-ordering principle, it has a smallest element, say $m \in \mathbb{N}$. Then $m = x_0^2 + y_0^2 + z_0^2$ for some integers x_0, y_0, z_0, not all zeros, such that $x_0^3 + 3y_0^3 + 9z_0^3 = 0$. Prove that then x_0, y_0, z_0 are each divisible by 3, allowing a new solution x_1, y_1, z_1 which is also nontrivial and for which $S \ni x_1^2 + y_1^2 + z_1^2 < m$, a contradiction. Conclude that there are no integer solutions to $x^3 + 3y^3 + 9z^3 = 0$ besides the trivial solution $x = y = z = 0$. *Hint:* To show the claimed divisibility by 3, start with x_0.

Definition 6.2. (Greatest common divisor).
Let $a, b \in \mathbb{Z}$. An integer $d > 0$ is the *greatest common divisor of* a, b if

- (common divisor) $d \,|\, a$, $d \,|\, b$, and
- whenever $d' \in \mathbb{Z}$ is such that $d' \,|\, a$ and $d' \,|\, b$, we have that $d' \,|\, d$.

We will denote the greatest common divisor of a, b by $\gcd(a, b)$.

Let us show uniqueness, justifying the use of the notation $\gcd(a, b)$. Suppose that d_1, d_2 both satisfy the conditions demanded. Then as d_2 is a common divisor of a and b, and since d_1 is a greatest common divisor, there holds $d_2 \mid d_1$. Also, since d_1 is a common divisor of a and b, and since d_2 is a greatest common divisor, there holds $d_1 \,|\, d_2$. Now Exercise 6.7, p.64, implies that $d_1 = d_2$ (since both are positive).

Note that if $a = b = 0$, then there is no $d > 0$ satisfying the conditions in the definition above: indeed, if there was such a $d > 0$, then with $d' := 2d$, we have $d' \,|\, a$ and $d' \,|\, b$, and so $2d = d' \,|\, d$, giving $2d \leqslant d$, a contradiction.

We now show that the $\gcd(a, b)$ exists whenever not both a, b are zero.

[1] See p.102 for the relevant definitions.

Theorem 6.2. *Let a and b be integers, not both 0. Then $\gcd(a,b)$ exists. Moreover, there exist $x, y \in \mathbb{Z}$ such that*[2] $\gcd(a,b) = ax_0 + by_0$.

Proof. Consider the set $S = \{ax + by : x, y \in \mathbb{Z}\}$. Since at least one of a, b is nonzero, and as S contains them, there exist nonzero integers in S. Also, if $ax + by \in S$, then $-(ax + by) = a(-x) + b(-y) \in S$, and so S contains some positive integers. Thus the set $S_+ = \{n \in S : n > 0\} \subset \mathbb{N}$ is nonempty, and by the well-ordering principle, S_+ has a least element d. As $d \in S_+$, we have $d = ax_0 + by_0$ for some $x_0, y_0 \in \mathbb{Z}$, and $d > 0$. We claim that $d = \gcd(a,b)$. (Thus the Bezout equation is then satisfied too: $\gcd(a,b) = d = ax_0 + by_0$.)

For any element $n = ax + by \in S$, by the division algorithm, there exist $q \in \mathbb{Z}$ and $r \in \mathbb{Z}$ such that $n = qd + r$ and $0 \leqslant r < d$. Thus $ax + by = n = qd + r$, that is, $0 \leqslant r = ax + by - q(ax_0 + by_0) = a(x - qx_0) + b(y - qy_0) \in S$. If $r \neq 0$, then $0 < r$, so that $r \in S_+$, and as $r < d$, we have a contradiction, since d was the least element in S_+. So $r = 0$. Hence $d \mid n$. So d divides each element of S. But as $a = a1 + b0$ and $b = a0 + b1$, we have that $a, b \in S$. Hence d divides a and b. Let d' divide a and b. So d' divides $ax + by$ for all $x, y \in \mathbb{Z}$ (Exercise 6.4, p.64). In particular, d' divides $ax_0 + by_0 = d$. □

Let $x_0, y_0 \in \mathbb{Z}$ satisfy $\gcd(a,b) = ax_0 + by_0$. Then for any $n \in \mathbb{Z}$, we have that $x := x_0 + bn$ and $y := y_0 - an$ also satisfy the Bezout equation:
$ax + by = a(x_0 + bn) + b(y_0 - an) = ax_0 + abn + by_0 - ban = ax_0 + by_0 = \gcd(a,b)$.
So the coefficients x_0, y_0 of a, b solving the Bezout equation are not unique.

Exercise 6.17. Let $a, b \in \mathbb{Z}$, not both zero, and let $d := \gcd(a,b) = ax_0 + by_0$ for some $x_0, y_0 \in \mathbb{Z}$. Let $q_a, q_b \in \mathbb{Z}$ be such that $a = q_a d$, $b = q_b d$. Let $x, y \in \mathbb{Z}$. Show that the following are equivalent:
(1) The integers x, y satisfy $ax + by = d$.
(2) There exists an $n \in \mathbb{Z}$ such that $x = x_0 + q_b n$, $y = y_0 - q_a n$.

Exercise 6.18. The *Fibonacci sequence* $(F_n)_{n \in \mathbb{N}}$ is defined recursively as follows: $F_1 = F_2 = 1$, and for $n \geqslant 2$, $F_{n+1} = F_n + F_{n-1}$. It has the terms $1, 1, 2, 3, 5, 8, 13, \cdots$. Show that $\gcd(F_n, F_{n+1}) = 1$ for all $n \in \mathbb{N}$.

Exercise 6.19. Let a, b be integers, not both zero. From the definition of the greatest common divisor, clearly $\gcd(a,b) = \gcd(b,a)$ (as the order of a, b does not matter). Show also that $\gcd(a,b) = \gcd(-a,b) = \gcd(a,-b) = \gcd(-a,-b)$.

Exercise 6.20. Let $n, N \in \mathbb{N}$ be such that $N > n > 1$. Prove that $\frac{\gcd(n,N)}{N}\binom{N}{n} \in \mathbb{Z}$.

To compute the greatest common divisor of integers a, b, not both zero, there is no loss of generality (by Exercise 6.19), in assuming at least one, say a, is positive. Moreover, if $a > 0$, then $\gcd(a,0) = a$ and $\gcd(a,a) = a$.

[2] This equation is called the *Bezout equation*.

We now learn an algorithm for finding the greatest common divisor supposing $b > a > 0$. The key result is the following, allowing us to use the division algorithm successively to keep reducing the numbers.

Lemma 6.1. *Let $a, b \in \mathbb{Z}$, with $a \neq 0$. For all $x \in \mathbb{Z}$, $\gcd(a, b) = \gcd(a, b + ax)$.*

Proof. As $a \neq 0$, we have that $d := \gcd(a, b)$ and $d' := \gcd(a, b + ax)$ exist. Since $d = \gcd(a, b)$, we have $d \mid a$ and $d \mid b$. So $d \mid a$ and $d \mid b + ax$. Since $d' = \gcd(a, b + ax)$, we have $d \mid d'$. As $d' = \gcd(a, b + ax)$, we have $d' \mid a$ and $d' \mid b + ax$. So $d' \mid (b + ax) + a(-x)$, that is, $d' \mid b$. Since $d = \gcd(a, b)$, and as d' divides a and b, we have $d' \mid d$. As $d, d' > 0$, and they divide each other, it follows from Exercise 6.7, p.64, that $d = d'$. □

We now describe Euclid's algorithm for determining the greatest common divisor. Let a, b be integers, where $b > a > 0$. We divide b by a obtaining integers q_1, r_1, such that $b = q_1 a + r_1$, and $0 \leqslant r_1 < a$.

1° If $r_1 = 0$, then $\gcd(a, b) = \gcd(a, b - q_1 a) = \gcd(a, r_1) = \gcd(a, 0) = a$.

2° If $r_1 > 0$, then we repeat the process replacing a, b with $a' := r_1 > 0$ and $b' := a > r_1 = a' > 0$, respectively.

We note that if $a + b =: N \in \mathbb{N}$, then in the case 2° eventuality, we have

$$a' + b' = r_1 + a \leqslant a - 1 + b - 1 = a + b - 2 = N - 2.$$

So, by the well-ordering principle, we cannot keep obtaining case 2° forever. Hence we must end up in case 1° at some point, and then the previous remainder (i.e., the current a) is the wanted gcd. To see this more explicitly, let us suppose that we keep getting case 2° $n - 1$ times for an $n \in \mathbb{N}$ and then in the nth step, we get remainder 0, that is:

$$\begin{aligned}
b &= q_1 a + r_1, && 0 < r_1 < a,\\
a &= q_2 r_1 + r_2, && 0 < r_2 < r_1,\\
r_1 &= q_3 r_2 + r_3, && 0 < r_3 < r_2,\\
&\vdots && \vdots\\
r_{n-3} &= q_{n-1} r_{n-2} + r_{n-1}, && 0 < r_{n-1} < r_{n-2},\\
r_{n-2} &= q_n r_{n-1}.
\end{aligned}$$

We have by a repeated application of Lemma 6.1, p.67, that

$$\begin{aligned}
\gcd(a, b) &= \gcd(a, b - q_1 a) = \gcd(a, r_1) = \gcd(r_1, a)\\
&= \gcd(r_1, a - q_2 r_1) = \gcd(r_1, r_2) = \gcd(r_2, r_1)\\
&= \gcd(r_2, r_1 - q_3 r_2) = \gcd(r_2, r_3) = \gcd(r_3, r_2)\\
&= \quad \cdots\\
&= \gcd(r_{n-2}, r_{n-3} - q_{n-1} r_{n-2}) = \gcd(r_{n-2}, r_{n-1}) = \gcd(r_{n-1}, r_{n-2})\\
&= \gcd(r_{n-1}, r_{n-2} - q_n r_{n-1}) = \gcd(r_{n-1}, 0) = r_{n-1}.
\end{aligned}$$

What about expressing $\gcd(a,b)$ as a 'linear combination' of a,b, which we know was guaranteed to be possible (the Bezout equation)? By writing each remainder in a later step using the ones from the previous step, and working backwards, this can be done. Rather than explain this abstractly, we consider an example.

Example 6.2. Let us find $\gcd(1976, 2375)$. We have

$$\begin{aligned} 2375 &= 1 \cdot 1976 + 399, \\ 1976 &= 4 \cdot 399 + 380, \\ 399 &= 1 \cdot 380 + 19, \\ 380 &= 20 \cdot 19. \end{aligned}$$

Thus $\gcd(1976, 2375) = 19$. To find a solution to the Bezout equation, we start with the penultimate remainder (i.e., $19 = \gcd(1976, 2375)$), and work backwards, each time writing the remainder in terms of the other data, until we reach a 'linear combination' with integer coefficients of the given numbers $a = 1976$ and $b = 2375$:

$$\begin{aligned} 19 &= 399 - 1 \cdot 380 \\ &= 399 - 1 \cdot (1976 - 4 \cdot 399) \\ &= -1976 + 5 \cdot 399 \\ &= -1976 + 5 \cdot (2375 - 1 \cdot 1976) \\ &= 5 \cdot 2375 - 6 \cdot 1976. \end{aligned}$$

Thus $1976 \cdot (-6) + 2375 \cdot 5 = \gcd(1976, 2375)$. By Exercise 6.17, p.66, we then know *all* the solutions to the Bezout equation. ◇

Exercise 6.21. Show that for every $n \in \mathbb{N}$, we have $\gcd(21n+4, 14n+3) = 1$. Conclude that $\frac{21n+4}{14n+3}$ is irreducible (that is, the numerator and denominator have no common factor besides 1 or -1).

Definition 6.3. (Relatively prime/coprime integers).
Integers a, b are called *relatively prime* or *coprime* if $\gcd(a,b) = 1$.

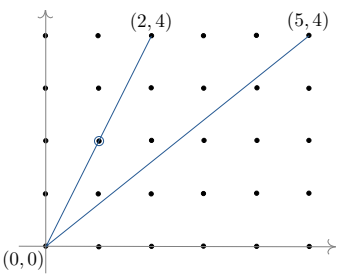
Proposition 6.1. *Let $a, b \in \mathbb{Z}$. Then the following are equivalent*:
(1) *a, b are relatively prime.*
(2) *There exist integers x_0, y_0 such that $ax_0 + by_0 = 1$.*

Proof. (1)⇒(2) follows immdediately from Theorem 6.2, p.66.
(2)⇒(1): Let there exist $x_0, y_0 \in \mathbb{Z}$ such that $ax_0 + by_0 = 1$. Then not both a and b are zero (otherwise $1 = ax_0 + by_0 = a0 + b0 = 0$, a contradiction). So $\gcd(a,b)$ exists. As $\gcd(a,b) \mid a$ and $\gcd(a,b) \mid b$, also $\gcd(a,b) \mid ax_0 + by_0 = 1$. By Exercise 6.6, p.64, $(0 <)\ \gcd(a,b) \leqslant 1$. So $\gcd(a,b) = 1$, that is, a, b are relatively prime. □

Proposition 6.2. *If* $a, b, n \in \mathbb{Z}$ *satisfy* $\gcd(a, n) = 1$ *and* $n \mid ab$, *then* $n \mid b$.

Proof. As $\gcd(a, n) = 1$, by Proposition 6.1, p.68, there exist $x, y \in \mathbb{Z}$ such that $ax + ny = 1$. Multiplying by b, $abx + nyb = b$. As $n \mid ab$, $ab = nm$ for some $m \in \mathbb{Z}$. Thus $b = abx + nyb = nmx + nyb = n(mx + yb)$, and so $n \mid b$. □

Example 6.3. (Geometric interpretation of coprimeness).
We will show that $a, b \in \mathbb{N}$ are coprime if and only if the point (a, b) is 'visible' from $(0, 0)$ in $\mathbb{Z} \times \mathbb{Z}$ ('lattice' of points in the plane, the plane being identified with $\mathbb{R} \times \mathbb{R}$ in the usual way). The 'visibility' means that there is no lattice point on the line segment joining $(0, 0)$ to (a, b), i.e., there is no 'obstructing' lattice point. For example $(5, 4)$ is visible from the origin, since there is no lattice point on the line segment joining $(0, 0)$ and $(5, 4)$, and we also do have that $\gcd(5, 4) = 1$. On the other hand, $\gcd(2, 4) = 2$, so that $2, 4$ are not coprime, and indeed, the view of $(2, 4)$ from the origin is blocked by the lattice point $(1, 2)$. So the statement seems to be true in the case of these two examples. Let us now prove the statement.



Suppose first that a, b are coprime and that (a, b) is not visible from $(0, 0)$. Thus some point, say (x, y), blocks the view of (a, b) from the origin, where $0 < x < a$ and $0 < y < b$. The point (x, y) lies on the line joining $(0, 0)$ to (a, b), and so the slope of the line joining (x, y) to $(0, 0)$ coincides with the slope of the line joining $(0, 0)$ to (a, b), that is, $\frac{b-0}{a-0} = \frac{y-0}{x-0}$, giving $ay = bx$. As a, b are coprime, there exist integers $x_0, y_0 \in \mathbb{Z}$ such that $1 = ax_0 + by_0$. Thus $y = y1 = y(ax_0 + by_0) = yax_0 + yby_0 = bxx_0 + yby_0 = b(xx_0 + yy_0)$. As $y > 0$, it follows from here that $y = |y| \geqslant |b| = b$, a contradiction.

Conversely, suppose that (a, b) is visible from $(0, 0)$. Since $a, b \in \mathbb{N}$, it follows that $\gcd(a, b)$ exists. Let $d = \gcd(a, b) > 1$. As $d \mid a$ and $d \mid b$, there exist $x, y \in \mathbb{N}$ such that $xd = a$ and $yd = b$. But as $d > 1$, we also have that $a = xd > x1 = x$ and similarly, $b > y$. We have $\frac{b-0}{a-0} = \frac{yd}{dx} = \frac{y}{x} = \frac{y-0}{x-0}$, and so (x, y) lies on the line joining $(0, 0)$ to (a, b), blocking the view, a contradiction. This contradiction shows that our original assumption that $\gcd(a, b) > 1$ is not true. Hence $\gcd(a, b) = 1$, that is, a, b are co-prime. ◇

Exercise 6.22. Suppose $m \in \mathbb{Z}$, $n, c_0 \in \mathbb{Z}\backslash\{0\}$, $d \in \mathbb{N}$, $\gcd(m,n)=1$ and $m \mid c_0 n^d$. Show that $m \mid c_0$.

Exercise 6.23. Let $a, b, n \in \mathbb{Z}$, $a \mid n$, $b \mid n$ and $\gcd(a,b)=1$. Show that $ab \mid n$. *Hint:* $n = ad$ for a $d \in \mathbb{Z}$, $b \mid n = ad$. Use Proposition 6.2, p.69.

Exercise 6.24. Show that if $a, b \in \mathbb{Z}$ are relatively prime, then every $n \in \mathbb{Z}$ can be written as a 'linear combination of a, b with integer coefficients', i.e., $n = ax + by$ for some $x, y \in \mathbb{Z}$.

Exercise 6.25. Let $a, b, n \in \mathbb{Z}$ and $\gcd(a,n) = 1 = \gcd(b,n)$. Show that $\gcd(ab, n) = 1$.

Exercise 6.26. Let $a, b \in \mathbb{Z}$, not both zero, $d = \gcd(a,b)$, and $q_a, q_b \in \mathbb{Z}$ be such that $a = q_a d$, $b = q_b d$. Show that $\gcd(q_a, q_b) = 1$.

Exercise 6.27. Let a, b be relatively prime. Show that a^2, b^2 are relatively prime.

Exercise 6.28. Show that $\gcd(n!+1, (n+1)!+1) = 1$ for all $n \in \mathbb{N}$.

Exercise 6.29.
(1) Show that if $a, b, a', b' \in \mathbb{Z}$ and $a + b\sqrt{2} = a' + b'\sqrt{2}$, then $a = a'$ and $b = b'$.
(2) Define a_n, b_n via the relation $a_n + b_n\sqrt{2} = (1+\sqrt{2})^n$ for all $n \in \mathbb{N}$.
Show that a_n, b_n are well-defined, they belong to $\mathbb{N}$, and that $\gcd(a_n, b_n) = 1$.

Exercise 6.30. (Catalan numbers).
(1) Show that for all $n \in \mathbb{N}$, $\gcd(2n+1, n+1) = 1$. *Hint:* $2(n+1) - (2n+1) = 1$.
(2) The numbers $C_n := \frac{1}{n+1}\binom{2n}{n}$, $n \in \mathbb{N}$, are called the *Catalan numbers*.
Prove that for all $n \in \mathbb{N}$, we have[3] $C_n \in \mathbb{N}$. *Hint:* First show $\frac{2n+1}{n+1}\binom{2n}{n} \in \mathbb{Z}$.

Exercise 6.31. Given an angle of 11°, show that one can divide it into 11 equal parts using a straight edge and a compass. *Hint:* It is enough to construct 1°.

Exercise 6.32. Let $n, N \in \mathbb{N}$ and $n < N$. Prove that $\gcd(2^{2^n}+1, 2^{2^N}+1) = 1$.

Exercise 6.33. (Bezout equation in $C[a,b]$). For $a, b \in \mathbb{R}$ with $a < b$, let $C[a,b]$ denote the set of all real-valued continuous functions on $[a,b] := \{x \in \mathbb{R} : a \leqslant x \leqslant b\}$. For $f \in C[a,b]$, the *zero set* Z_f *of* f is $Z_f = \{x \in [a,b] : f(x) = 0\}$. For $f, g \in C[a,b]$, define $f+g, fg \in C[a,b]$ by $(f+g)(x) = f(x) + g(x)$ and $(fg)(x) = f(x)g(x)$ for all $x \in [a,b]$. Also, define $\mathbf{1} \in C[a,b]$ by $\mathbf{1}(x) = 1$ for all $x \in [a,b]$. Let $f_1, f_2 \in C[a,b]$. Show that the following are equivalent:
(1) There exist $g_1, g_2 \in C[a,b]$ such that $f_1 g_1 + f_2 g_2 = \mathbf{1}$.
(2) $Z_{f_1} \cap Z_{f_2} = \varnothing$. (That is, f_1, f_2 have no common zero.)
Hint: For (2)$\Rightarrow$(1), $f_1^2 + f_2^2 > 0$ everywhere, and so $g_i := \frac{f_i}{f_1^2 + f_2^2} \in C[a,b]$, $i = 1, 2$.

[3] The Catalan number C_n is the number of different ways $n+1$ terms $a_1, \cdots, a_{n+1}$ can be bracketed if their order is not changed. (This for instance gives the number of equivalent expressions in Exercise 2.9, p.19.) For example, if $n = 3$, then the number of ways of bracketing a, b, c, d is 5: $(a(b(cd)))$, $(ab)(cd)$, $((ab)c)d$, $a((bc)d)$, $(a(bc))d$, and indeed, $C_3 = \frac{1}{4}\binom{6}{2} = 5$. This follows from the recursion $C_{n+1} = C_n + C_1 C_{n-1} + \cdots + C_{n-1} C_1 + C_n$ with $C_1 = 1$, see, e.g. [Comtet (1974)].

6.3 Fundamental theorem of arithmetic

Definition 6.4. (Prime numbers).
An integer p is called *a prime* if $p>1$ and the only positive divisors of p are 1 and p. Natural numbers that are not prime are called *composite numbers*.

Thus 2 is a prime number[4]. This is the only even prime number. The first few terms of the subsequent list of primes are $3, 5, 7, 11, 13, 17, 19, 23, 29, \cdots$.

Proposition 6.3.
If p is a prime number, and $n \in \mathbb{Z}$ is such that $p \nmid n$, then $\gcd(p, n) = 1$.

Proof. Let $d := \gcd(p, n)$. Then $d \mid p$ and $d \mid n$. But p is prime, and so $d = 1$ or $d = p$. But $p \nmid n$, so that d cannot be p. Thus $d = 1$. □

In particular, distinct primes are coprime. Also, for any $n \in \mathbb{Z}$, and any prime p, we have the dichotomy that either $p \mid n$ or $\gcd(p, n) = 1$.

Exercise 6.34. If p and $8p-1$ are prime, then show that $8p+1$ is composite. *Hint*: Consider the remainder left by p when divided by 3.

Exercise 6.35.
(1) Let $a, b \in \mathbb{Z}$. Let p be a prime such that $p \mid ab$. Show that $p \mid a$ or $p \mid b$.
Hint: Proposition 6.2, p.69.
(2) Let $a_1, \cdots, a_n \in \mathbb{Z}$ for some $n \geqslant 2$. Let p be a prime such that $p \mid a_1 \cdots a_n$. Prove that p divides at least one of the factors $a_1, \cdots, a_n$. *Hint*: Induction.

Exercise 6.36. Consider the polynomial[5] p_{Euler} given by $p_{\text{Euler}}(x) = x^2 + x + 41$ for all $x \in \mathbb{R}$. Then $p_{\text{Euler}}(1) = 43$ is a prime, $p_{\text{Euler}}(2) = 47$ is a prime, $p_{\text{Euler}}(3) = 53$ is a prime. In fact, $p_{\text{Euler}}(n)$ is a prime for all $1 \leqslant n \leqslant 39$. But $p_{\text{Euler}}(40) = 41 \times 41$ is not a prime. Show that there is no nonconstant polynomial with integer coefficients such that for all $n \in \mathbb{N}$, $p(n)$ is a prime number.

Theorem 6.3. (Fundamental theorem of arithmetic).
Every integer $n \neq 0$ can be expressed as a product $n = cp_1 \cdots p_k$,
where $c \in \{1, -1\}$, each p_i is a prime number, and $k \geqslant 0$.
(If $k = 0$, the 'empty product' $p_1 \cdots p_k$ is defined to be 1.)
This expression is unique except for the ordering of the primes.

Proof. *Existence*: It is enough to consider $n \in \mathbb{N}$. We use induction on n. If $n = 1$, then set $c = 1$ and $k = 0$. The claim is true for $n = 2$, as $n = 2 = 1(2)$, and 2 is a prime number. Suppose for some $n \in \mathbb{N}$ each natural number less than or equal to n possesses the claimed factorisation. If $n + 1$ is prime, then we are done. Otherwise, it has a divisor $d \in \mathbb{N}$ which is not 1 or $n + 1$.

[4] If $2 = mn$ for positive integers m and n, then they cannot both be equal to 1. Let $n > 1$. Then $1 \leqslant m = m1 < mn = 2$, giving $1 \leqslant m < 2$. So $m = 1$ and $n = 2$.
[5] See p.102 for the relevant definitions.

So $n+1=md$ for some $m\in\mathbb{N}$. Then m is not 1 or $n+1$ either. But then $1<m=m1<md=n+1$. Also, $1<d=d1<dm=n+1$. Hence $1<m,d\leqslant n$. By the induction hypothesis[6], $(0<)\ m=p_1\cdots p_k$ and $(0<)\ d=q_1\cdots q_\ell$, where $p_1,\cdots,p_k,q_1,\cdots,q_\ell$ are primes. So $n+1=md=p_1\cdots p_kq_1\cdots q_\ell$ is the desired factorisation of $n+1$. By induction, the proof is complete.

Uniqueness: Let S be the set of all integers that fail to have a unique factorisation. Suppose $S\neq\varnothing$. We claim that S contains natural numbers. If $0>m\in S$ has two different factorisations $cp_1\cdots p_k=m=c'q_1\cdots q_\ell$, then (as the primes are all positive) $c=c'=-1$, and so $n:=-m\in\mathbb{N}$ has two different factorisations $p_1\cdots p_k=q_1\cdots q_\ell$. Thus the set S_+ of all natural numbers that fail to have a unique factorisation is not empty as well. By the well-ordering principle, S_+ has a least element, say n_*. Suppose that $cp_1\cdots p_k=n_*=c'q_1\cdots q_\ell$ (where $c,c'\in\{-1,1\}$ and $p_1,\cdots,p_k,\ q_1,\cdots,q_\ell$ are primes) are distinct factorisations of n_*. As $n_*>0$, we have $c=c'=1$. If $k=0$, so that $p_1\cdots p_k$ is (the empty product) 1, then $q_1\cdots q_\ell$ cannot be the empty product (otherwise the factorisations were identical), and so $q_1\cdots q_\ell>1$, and cannot equal (the empty product $p_1\cdots p_k=)\ 1$. Hence k cannot be zero. Similarly $\ell\neq0$. Applying Exercise 6.35, p.71, with $p=p_1$, we conclude that p_1 divides some q_i. But as q_i is prime, $p_1=q_i$. Cancelling p_1 and q_i, we obtain

$$\mathbb{N}\ni n':=p_2\cdots p_k=\frac{n_*}{p_1}=\frac{n_*}{q_i}=q_1\cdots q_{i-1}q_{i+1}\cdots q_\ell<n_*,$$

showing that the natural number n' $(<n_*)$ has the distinct factorisations $p_2\cdots p_k=n'=q_1\cdots q_{i-1}q_{i+1}\cdots q_\ell$, contradicting the minimality of n_*. So $S=\varnothing$. □

Thus every $n\in\mathbb{Z}\backslash\{0\}$ can be expressed as $n=cp_1^{\alpha_1}\cdots p_k^{\alpha_k}$, where $c\in\{-1,1\}$, $k\in\mathbb{N}\cup\{0\}$, $p_1<\cdots<p_k$ are primes, and the integers $\alpha_1,\cdots,\alpha_k\geqslant0$. By using the exponent zero to raise a prime, we can expand any two numbers using the same list of primes, for example, $20=2^23^05^1$ and $18=2^13^25^0$. We now give a different way of computing the gcd of two nonzero integers.

Lemma 6.2. *Let $d,a\in\mathbb{N}$, $d=p_1^{\alpha_1}\cdots p_k^{\alpha_k}$, $a=p_1^{\beta_1}\cdots p_k^{\beta_k}$, where $p_1<\cdots<p_k$ are primes, and $\alpha_1,\cdots,\alpha_k$ and $\beta_1,\cdots,\beta_k$ are nonnegative integers. If $d\,|\,a$, then $\alpha_i\leqslant\beta_i$ for all $1\leqslant i\leqslant k$.*

Proof. As $p_1^{\alpha_1}\cdots p_k^{\alpha_k}=d\,|\,a=p_1^{\beta_1}\cdots p_k^{\beta_k}$, in particular, $p_1^{\alpha_1}\,|\,p_1^{\beta_1}\cdots p_k^{\beta_k}$. Suppose $\alpha_1>\beta_1$. Then $p_1^{\alpha_1-\beta_1}\,|\,p_2^{\beta_2}\cdots p_k^{\beta_k}$. In particular, $p_1\,|\,p_2^{\beta_2}\cdots p_k^{\beta_k}$, a contradiction to Exercise 6.35, p.71. Hence $\alpha_1\leqslant\beta_1$. Similarly, $\alpha_i\leqslant\beta_i$ for all $2\leqslant i\leqslant k$ as well. □

[6] As $m>0$, and all primes are >0, if $m=cp_1\cdots p_k$, where $c\in\{-1,1\}$, then $c=1$.

Theorem 6.4. *Let* $a, b \in \mathbb{Z}\backslash\{0\}$, $a = cp_1^{\alpha_1} \cdots p_k^{\alpha_k}$, *and* $b = c' p_1^{\beta_1} \cdots p_k^{\beta_k}$, *where* $c, c' \in \{-1, 1\}$, $k \geqslant 0$, $p_1 < \cdots < p_k$ *are primes, and* $\alpha_1, \cdots, \alpha_k, \beta_1, \cdots, \beta_k$ *are nonnegative integers. Then* $\gcd(a, b) = p_1^{\min\{\alpha_1, \beta_1\}} \cdots p_k^{\min\{\alpha_k, \beta_k\}}$.

Proof. Let $d = \gcd(a, b)$. If p is a prime that divides d, then p also divides a and b. Hence p can only be one of $p_1, \cdots, p_k$ by Exercise 6.35, p.71. So $d = p_1^{\gamma_1} \cdots p_k^{\gamma_k}$ for some nonnegative integers $\gamma_1, \cdots, \gamma_k$. By Lemma 6.2 above, $\gamma_i \leqslant \alpha_i$, and $\gamma_i \leqslant \beta_i$ for $1 \leqslant i \leqslant k$. Hence $\gamma_i \leqslant \min\{\alpha_i, \beta_i\}$ for $1 \leqslant i \leqslant k$.

Next, let $d' = p_1^{\min\{\alpha_1, \beta_1\}} \cdots p_k^{\min\{\alpha_k, \beta_k\}}$. Then $d' \mid a$ (as $\min\{\alpha_i, \beta_i\} \leqslant \alpha_i$ for all i) and $d' \mid b$ (as $\min\{\alpha_i, \beta_i\} \leqslant \beta_i$ for all i). As $d = \gcd(a, b)$, $d' \mid d$. By Lemma 6.2 above, $\min\{\alpha_i, \beta_i\} \leqslant \gamma_i$ for $1 \leqslant i \leqslant k$.
Thus $\gamma_i = \min\{\alpha_i, \beta_i\}$ for $1 \leqslant i \leqslant k$. So $\gcd(a, b) = p_1^{\min\{\alpha_1, \beta_1\}} \cdots p_k^{\min\{\alpha_k, \beta_k\}}$. □

The above is mostly of theoretical value, and the method is good to use when we have small numbers, so that we can check if the number has a certain prime factor easily. For example, since $20 = 2^2 3^0 5^1$ and $18 = 2^1 3^2 5^0$, it is immediate that $\gcd(20, 18) = 2^1 3^0 5^0 = 2$.

Example 6.4. Let us recalculate $\gcd(1976, 2375)$ using the above. We have

$$\begin{aligned} 2375 &= 5(475) = 5^2(95) = 5^3 19^1, \\ 1976 &= 2(988) = 2^2(494) = 2^3(247) = 2^3 13^1 19^1, \end{aligned}$$

and so $\gcd(1976, 2375) = 19^1 = 19$. ◇

Theorem 6.5. (Euclid). *There are infinitely many primes.*

Proof. Suppose that $p_1, \cdots, p_n$ are the only prime numbers. Consider the integer $N = p_1 \cdots p_n + 1$. Then N is not divisible by any of $p_1, \cdots, p_n$. But N cannot be a prime itself, since it is strictly bigger than each of $p_1, \cdots, p_n$. Thus N is not a prime number. By the fundamental theorem of arithmetic, N has a factorisation into primes, and in particular N (> 1) must be divisible by some prime p'. This p' cannot be any of $p_1, \cdots, p_n$, as none of them divide N. This contradicts our assumption that $p_1, \cdots, p_n$ were the only primes. □

Exercise 6.37. (The prime counting function). Arrange the primes in increasing order as $p_1 < p_2 < \cdots$. For any real $x > 0$, define $\pi(x)$ to be the number of elements in the set $\{n \in \mathbb{N} : p_n \leqslant x\}$. The function $\pi(\cdot)$ is called the *prime counting function.* Euclid's proof of Theorem 6.5 above shows that the $(n+1)^{\text{st}}$ prime must be at most $p_1 \cdots p_n + 1$. Using this inequality and induction on n, show that $p_n < 2^{2^n}$ for all $n \in \mathbb{N}$. For any real $x \geqslant 2$, we can find an $n \in \mathbb{N}$ such that $2^n \leqslant \log_2 x < 2^{n+1}$. Show that $\log_e \log_e x \leqslant \pi(x) \leqslant x$ for all $x \geqslant 2$.
(We remark that the *prime number theorem* states that $\lim\limits_{x \to \infty} \frac{\log_e x}{x} \pi(x) = 1$. See, e.g., [Chandrasekharan (1968), Chapter XI].)

Exercise 6.38. Use Exercise 6.32, p.70, to show there are infinitely many primes. *Hint:* Otherwise for some distinct n, N, $2^{2^n}+1$, $2^{2^N}+1$ would share a prime factor.

Exercise 6.39. (Twin primes: Maybe. Prime triples: No!).
(A *twin prime* is a pair $(n, n+2)$, where n and $n+2$ are primes. For example, $(3,5),(5,7),(11,13)$ are twin primes. A famous open problem is that of establishing the truth or falsity of the *twin prime conjecture* stating that there are infinitely many twin primes.) Motivated by this, we define a *prime triple* as a triple $(n, n+2, n+4)$, where n, $n+2$ and $n+4$ are all primes. For example $(3,5,7)$ is a prime triple. Show that there are no others. *Hint:* Divide n by 3.

Exercise 6.40. Can a right angled triangle with integer sides have the nonhypotenuse sides equal to twin primes?

Exercise 6.41. (Arbitrary large gaps between primes). Let $n \in \mathbb{N}$. Show that the set $\{(n+1)!+2, \cdots, (n+1)!+n+1\}$ of n consecutive numbers has no primes.

Exercise 6.42. (Fermat primes).
If 2^n+1 is prime for an $n \in \mathbb{N}$, then show that n is a power of 2, that is, $n = 2^m$ for some nonnegative integer m. *Hint:* Use Exercise 6.13, p.65.
(A prime number of the form $2^{2^m}+1$ is called a *Fermat prime*. Not all numbers $f_m := 2^{2^m}+1$, $m \in \mathbb{Z}$, are primes: $f_0 = 3$, $f_1 = 5$, $f_2 = 17$, $f_3 = 257$, $f_4 = 65537$ are primes, but Euler showed $f_5 = 2^{32}+1 = 4924967297 = 641 \times 6700417$. It is conjectured that f_0, f_1, f_2, f_3, f_4 are the only Fermat primes.)

Exercise 6.43. To show that $n \in \mathbb{N}$ is a prime number, prove that it is sufficient to check that n is not divisible by all prime numbers p satisfying $p \leqslant \sqrt{n}$.

Exercise 6.44. Show that if the integers a, b are relatively prime, then ab, $a+b$ are relatively prime. *Hint:* Let the prime p be a common divisor of ab and $a+b$.

Exercise 6.45. (Least common multiple).
Let $a, b \in \mathbb{N}$. A *least common multiple of* a, b is a positive integer m such that
- $a \,|\, m$, $b \,|\, m$, and
- whenever $m' \in \mathbb{N}$ is such that $a \,|\, m'$ and $b \,|\, m'$, we have $m \,|\, m'$.

(1) Show that the least common multiple always exists and is unique (denoted by $\mathrm{lcm}(a,b)$). *Hint:* Consider $S = \{m \in \mathbb{N} : a \,|\, m \text{ and } b \,|\, m\}$.
(2) Show that $\gcd(a,b) \cdot \mathrm{lcm}(a,b) = ab$.
Hint: First show that a, b divide $m_* := \frac{ab}{\gcd(a,b)} \in \mathbb{Z}$ is an integer. Next if a, b divide an $m \in \mathbb{N}$, then use the Bezout equation (for a, b) to show that $m_* \,|\, m$.
(3) If $a = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$, $b = p_1^{\beta_1} \cdots p_k^{\beta_k}$, for prime numbers $p_1 < \cdots < p_k$, and integers $\alpha_1, \cdots, \alpha_k, \beta_1, \cdots, \beta_k \geqslant 0$, then show that $\mathrm{lcm}(a,b) = p_1^{\max\{\alpha_1,\beta_1\}} \cdots p_k^{\max\{\alpha_k,\beta_k\}}$.

Exercise 6.46. ($\mathbb{Z}$-linear independence of $1, \sqrt{2}, \sqrt{3}$).
(1) Show that if $a, b \in \mathbb{Z}$ and $2a^2 = 3b^2$, then $a = b = 0$.
Hint: Write $a = 2^\alpha 3^\beta A$ and $b = 2^{\alpha'} 3^{\beta'} B$ for nonnegative integers $\alpha, \beta, \alpha', \beta'$ and A, B integers with prime factorisations not containing the primes $2, 3$.
(2) Let $a, b, c \in \mathbb{Z}$ be such that $a\sqrt{2} + b\sqrt{3} + c = 0$. Prove that $a = b = c = 0$.
Hint: Square $a\sqrt{2} + c = -b\sqrt{3}$.

Chapter 7

Modular arithmetic

Now we'll learn about Gauss's definition of congruence of integers, modular arithmetic and the ring $\mathbb{Z}/n\mathbb{Z}$, which are useful notions in number theory.

Definition 7.1. (mod n).
Let $n \in \mathbb{N}$ be fixed. Integers a, b are *congruent modulo* n, if $n \mid a - b$.
We then write $a \equiv b \bmod n$, read as 'a and b are congruent modulo n' or 'a is congruent b mod n'.

Exercise 7.1. Let $n \in \mathbb{N}$. Suppose that $a, b, c, d \in \mathbb{Z}$ are such that $a \equiv b \bmod n$ and $c \equiv d \bmod n$. Prove that $(a + c) \equiv (b + d) \bmod n$ and $ac \equiv bd \bmod n$.

Exercise 7.2. Let $n \in \mathbb{N}$. Suppose that $a, b, c, d \in \mathbb{Z}$ are such that $ab \equiv ac \bmod n$ and $\gcd(a, n) = 1$. Show that $b \equiv c \bmod n$. Give an example to show that the hypothesis $\gcd(a, n) = 1$ is not superfluous.

Proposition 7.1. *Let $n \in \mathbb{N}$. Congruency modulo n is an equivalence relation on $\mathbb{Z}$, and for this equivalence relation, the number of distinct equivalence classes is n.*

Proof. (1) Reflexivity: For all $a \in \mathbb{Z}$, $a \equiv a \bmod n$ because $n \mid 0 = a - a$.
Symmetry: Let $a, b \in \mathbb{Z}$ be such that $a \equiv b \bmod n$. Then $n \mid a - b$, i.e., $a - b = nk$ for some $k \in \mathbb{Z}$. So $b - a = n(-k)$. Thus $n \mid b - a$. So $b \equiv a \bmod n$.
Transitivity: Let $a, b, c \in \mathbb{Z}$ be such that $a \equiv b \bmod n$ and $b \equiv c \bmod n$. Then $n \mid a - b$ and $n \mid b - c$. So $n \mid (a - b) + (b - c) = a - c$. Thus $a \equiv c \bmod n$.
(2) We claim $[0], [1], \cdots, [n-1]$ are the only equivalence classes, and these are distinct. If $a \in \mathbb{Z}$, then by the division algorithm, there exist integers q and r such that $a = qn + r$, where $0 \leqslant r < n$. Since $n \mid qn = a - r$, we have $a \equiv r \bmod n$, i.e., $[a] = [r] \in \{[0], [1], \cdots, [n-1]\}$.

Now we will show the equivalence classes $[0], \cdots, [n-1]$ are distinct. Suppose, on the contrary, $0 \leqslant r < r' \leqslant n-1$ and $[r] = [r']$. Then $r' \equiv r \bmod n$, and so $n \mid r' - r$. So there exists a $q \in \mathbb{Z}$ such that $0 < r' - r = qn$. Thus $q > 0$, i.e., $q \geqslant 1$. So $r' - r = qn \geqslant n$. Hence $n \leqslant r' - r \leqslant (n-1) - 0 = n - 1$, which gives $1 \leqslant 0$, a contradiction. □

Exercise 7.3. Use modular arithmetic to show that $7 \mid 2222^{5555} + 5555^{2222}$.
Hint: The following can be useful: $2222 = 317 \cdot 7 + 3$, $5555 = 793 \cdot 7 + 4$, $3 \equiv -4 \bmod 7$, $4^3 = 64 = 9 \cdot 7 + 1$, $2222 = 740 \cdot 3 + 2$, $5555 = 1851 \cdot 3 + 2$.

Exercise 7.4. Let $n \in \mathbb{N}$ and $a \in \mathbb{Z}$. Show that $[a] = \{a + qn : q \in \mathbb{Z}\}$.

Exercise 7.5. Show that if $n \in \mathbb{N}$, then n is congruent modulo 9 to the sum of the decimal digits of n.

Exercise 7.6. The natural number 2^{29} has a decimal representation consisting of nine digits, and each of $0, 1, 2, \cdots, 9$ appear, except for one of these. Without computing 2^{29}, find which of these is missing.

Let $\mathbb{Z}/n\mathbb{Z}$ denote the set of equivalence classes under the relation of congruency modulo n on $\mathbb{Z}$, that is,

$$\mathbb{Z}/n\mathbb{Z} = \{[0], [1], \cdots, [n-1]\}.$$

The set $\mathbb{Z}/n\mathbb{Z}$ is called the *set of integers modulo n*. Each of its members is actually a set:

$$\begin{aligned}[a] &= \{b \in \mathbb{Z} : b \equiv a \bmod n\} \\ &= \{a + qn : q \in \mathbb{Z}\} \\ &= \{\cdots, a-3n, a-2n, a-n, a, a+n, a+2n, a+3n, \cdots\}.\end{aligned}$$

For example, if $n = 1$, then $\mathbb{Z}/1\mathbb{Z} = \{[0]\}$ and

$$[0] = \{\cdots, -3, -2, -1, 0, 1, 2, 3, \cdots\} = \mathbb{Z}.$$

If $n = 2$, then $\mathbb{Z}/2\mathbb{Z} = \{[0], [1]\}$, where

$$\begin{aligned}[0] &= \{2q : q \in \mathbb{Z}\} \\ &= \{\text{even integers}\} \\ &= \{\cdots, -6, -4, -2, 0, 2, 4, 6, \cdots\}, \\ [1] &= \{2q+1 : q \in \mathbb{Z}\} \\ &= \{\text{odd integers}\} \\ &= \{\cdots, -5, -3, -1, 1, 3, 5, \cdots\}.\end{aligned}$$

We now define addition and multiplication in $\mathbb{Z}/n\mathbb{Z}$ making it a 'ring'.

Definition 7.2. (Addition and multiplication modulo n).
Let $n \in \mathbb{N}$. We define *addition modulo n*, $+ : (\mathbb{Z}/n\mathbb{Z}) \times (\mathbb{Z}/n\mathbb{Z}) \to (\mathbb{Z}/n\mathbb{Z})$, and *multiplication modulo n*, $\cdot : (\mathbb{Z}/n\mathbb{Z}) \times (\mathbb{Z}/n\mathbb{Z}) \to (\mathbb{Z}/n\mathbb{Z})$ by

$$\begin{aligned}[a] + [b] &= [a + b] \text{ and} \\ [a] \cdot [b] &= [ab]\end{aligned}$$

for all $[a], [b] \in \mathbb{Z}/n\mathbb{Z}$.

We need to check the well-definition, that is, if $[a] = [a']$ and $[b] = [b']$, then $[a + b] = [a' + b']$ and $[ab] = [a'b']$. We have $a' \equiv a \bmod n$ and $b' \equiv b \bmod n$,

and so by Exercise 7.1, p.75, $(a'+b') \equiv (a+b) \bmod n$ and $a'b' \equiv ab \bmod n$. Thus $[a+b]=[a'+b']$ and $[ab]=[a'b']$.

Using the arithmetic properties of addition and multiplication in $\mathbb{Z}$, it is easy to see that the following properties hold for all $[a],[b],[c] \in \mathbb{Z}/n\mathbb{Z}$:

- $[a]+([b]+[c])=([a]+[b])+[c]$.
- $[a]+[0]=[a]=[0]+[a]$.
- $[a]+[-a]=[0]=[-a]+[a]$.
- $[a]+[b]=[b]+[a]$.
- $[a]\cdot([b]\cdot[c])=([a]\cdot[b])\cdot[c]$.
- $[a]\cdot[1]=[a]=[1]\cdot[a]$.
- $[a]\cdot[b]=[b]\cdot[a]$.
- $[a]\cdot([b]+[c])=[a]\cdot[b]+[a]\cdot[c]$.

As an example, let us show the last one, namely the distributive property:

$$\begin{aligned}[a]\cdot([b]+[c]) &= [a]\cdot[b+c]=[a(b+c)]\\ &= [ab+ac]=[ab]+[ac]\\ &= [a]\cdot[b]+[a]\cdot[c].\end{aligned}$$

In the above list, while we have stated that additive inverses always exist, we haven't said anything about the existence of multiplicative inverses. Not every element in $\mathbb{Z}/n\mathbb{Z}$ may have a multiplicative inverse. We will show that $[a] \in \mathbb{Z}/n\mathbb{Z}$ possesses a multiplicative inverse if and only if a is relatively prime to n. In particular, if $n=p$, a prime, then *every* nonzero element of $\mathbb{Z}/p\mathbb{Z}$ has a multiplicative inverse in $\mathbb{Z}/p\mathbb{Z}$; see Corollary 7.1, p.78.

Proposition 7.2. *Let $n \in \mathbb{N}$ and $a \in \mathbb{Z}$. Then the following are equivalent:*
(1) *$a \in \mathbb{Z}$ is relatively prime to n.*
(2) *There exists an $x \in \mathbb{Z}$ such that $[a]\cdot[x]=[1]=[x]\cdot[a]$ in $\mathbb{Z}/n\mathbb{Z}$.*

Proof. (1)⇒(2): Let a be relatively prime to n, that is, $\gcd(a,n)=1$. By Proposition 6.1, p.68, there exist integers x, y such that $ax+ny=1$. As $n \equiv 0 \bmod n$, we have $[0]=[n]$ in $\mathbb{Z}/n\mathbb{Z}$. Hence

$$\begin{aligned}[1] &= [ax+ny]=[ax]+[ny]=[a]\cdot[x]+[n]\cdot[y]\\ &= [a]\cdot[x]+[0]\cdot[y]=[a]\cdot[x]+[0y]=[a]\cdot[x]+[0]\\ &= [a]\cdot[x].\end{aligned}$$

As multiplication in $\mathbb{Z}/n\mathbb{Z}$ is commutative, also $[x]\cdot[a]=[1]$.

(2)⇒(1): Let there exist an integer x such that $[a]\cdot[x]=[1]=[x]\cdot[a]$ in $\mathbb{Z}/n\mathbb{Z}$. Then $[ax]=[1]$. So $ax \equiv 1 \bmod n$, that is, $n \mid ax-1$. Thus there exists some $y \in \mathbb{Z}$ such that $ax-1=ny$, i.e., $ax+n(-y)=1$. By Proposition 6.1, p.68, $\gcd(a,n)=1$. Hence a is relatively prime to n. □

If a multiplicative inverse of $[a] \in \mathbb{Z}/n\mathbb{Z}$ exists, then it must be unique: if $[x], [y] \in \mathbb{Z}/n\mathbb{Z}$ are such that $[a]\cdot[x] = [x]\cdot[a] = [1]$ and $[a]\cdot[y] = [y]\cdot[a] = [1]$, then $[x] = [1]\cdot[x] = ([y]\cdot[a])\cdot[x] = [y]\cdot([a]\cdot[x]) = [y]\cdot[1] = [y]$. We denote the unique multiplicative inverse of $[a]$ in $\mathbb{Z}/n\mathbb{Z}$, if it exists, by $[a]^{-1}$.

Corollary 7.1. *If p is a prime, then $\mathbb{Z}/p\mathbb{Z}$ is a field*[1].

Proof. We only have to show that every nonzero element in $\mathbb{Z}/p\mathbb{Z}$ has a multiplicative inverse. Let $a \in \mathbb{Z}$ be such that $[a] \neq [0]$ in $\mathbb{Z}/p\mathbb{Z}$. Then $p \nmid a - 0 = a$. By Proposition 6.3, p.71, $\gcd(a, p) = 1$. From Proposition 7.2, p.77, there exists an $x \in \mathbb{Z}$ such $[a]\cdot[x] = [1] = [x]\cdot[a]$ in $\mathbb{Z}/p\mathbb{Z}$. □

Lemma 7.1. *Let p be a prime, and $a \in \mathbb{Z}$.*
Then $[a]^2 = [1]$ in $\mathbb{Z}/p\mathbb{Z}$ if and only if $[a] = [1]$ or $[a] = [-1]$ in $\mathbb{Z}/p\mathbb{Z}$.

Proof. (If part:) We have $[1]^2 = [1]\cdot[1] = [1]$, and $[-1]^2 = [-1(-1)] = [1]$. (Only if part:) Suppose $[a]^2 = [1]$, i.e., $[a^2] = [1]$. So $a^2 \equiv 1 \bmod p$, i.e., $p \mid a^2 - 1 = (a-1)(a+1)$. By Exercise 6.35, p.71, $p \mid a-1$ or $p \mid a+1$. Thus $a \equiv 1 \bmod p$ or $a \equiv -1 \bmod p$, i.e., $[a] = [1]$ or $[a] = [-1]$. □

Lemma 7.2. *Let $p > 3$ be a prime. Then $[2]\cdot[3]\cdot\ldots\cdot[p-2] = [1]$ in $\mathbb{Z}/p\mathbb{Z}$.*

Proof. As p is odd, there are an even number of terms in the product on the left-hand side. For $[a] \in \{[2], \cdots, [p-2]\}$, we have $[a] \neq [1]$ and $[a] \neq [-1]$. So, by Lemma 7.1, $[a]^2 \neq [1]$. So each element $[a] \in \{[2], \cdots, [p-2]\}$ pairs up together with another member (namely its inverse $[a]^{-1}$) in the same list $[2], \cdots, [p-2]$, such that their product gives $[1]$. □

Theorem 7.1. *Let p be a prime. Then $[1]\cdot[2]\cdot\ldots\cdot[p-1] = [-1]$ in $\mathbb{Z}/p\mathbb{Z}$.*

Proof. If $p = 2$, then $[1] = [-1]$ in $\mathbb{Z}/2\mathbb{Z}$: $1 \equiv -1 \bmod 2$ (as $2 \mid 1 - (-1) = 2$).
If $p = 3$, then $[1]\cdot[2] = [2] = [-1]$ in $\mathbb{Z}/3\mathbb{Z}$: $2 \equiv -1 \bmod 3$ (as $3 \mid 2 - (-1) = 3$).
If $p > 3$, then by Lemma 7.2 above, $[2]\cdot[3]\cdot\ldots\cdot[p-2] = [1]$. Consequently, we get $[1]\cdot[2]\cdot\ldots\cdot[p-2]\cdot[p-1] = [1]\cdot[1]\cdot[p-1] = [p-1] = [-1]$. □

[1] Recall that a *field* is a set $\mathbb{F}$ together with two maps, addition, $+ : \mathbb{F} \times \mathbb{F} \to \mathbb{F}$, and multiplication $\cdot : \mathbb{F} \times \mathbb{F} \to \mathbb{F}$, such that:
(F1) addition is associative, i.e., for all $a, b, c \in \mathbb{F}$, $a + (b + c) = (a + b) + c$
(F2) there exists an additive identity $0 \in \mathbb{F}$, i.e., for all $a \in \mathbb{F}$, $a + 0 = a = 0 + a$
(F3) every element has an additive inverse, i.e.,
for all $a \in \mathbb{F}$, there exists an element $-a \in \mathbb{F}$ such that $a + (-a) = 0 = -a + a$
(F4) addition is commutative, i.e., for all $a, b \in \mathbb{F}$, $a + b = b + a$
(F5) multiplication is associative, for all $a, b, c \in \mathbb{F}$, $a \cdot (b \cdot c) = (a \cdot b) \cdot c$
(F6) there exists a multiplicative identity $1 \in \mathbb{F}\backslash\{0\}$, i.e., for all $a \in \mathbb{F}$, $a \cdot 1 = a = 1 \cdot a$
(F7) every element $a \in \mathbb{F}\backslash\{0\}$ has a multiplicative inverse, i.e.,
there exists an element $a^{-1} \in \mathbb{F}$ such that $a \cdot a^{-1} = 1 = a^{-1} \cdot a$
(F8) multiplication is commutative, i.e., for all $a, b \in \mathbb{F}$, $a \cdot b = b \cdot a$ and
(F9) multiplication distributes over addition, i.e., for all $a, b, c \in \mathbb{F}$, $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$.

Corollary 7.2. (Wilson's Theorem). *Let p be a prime. Then $p\,|\,(p-1)!+1$.*

Proof. From Theorem 7.1, p.78, we have

$$[(p-1)!] = [1\cdot 2\cdot 3\cdot \ldots \cdot (p-2)\cdot (p-1)] = [1]\cdot[2]\cdot \ldots \cdot [p-2]\cdot[p-1] = [-1].$$

So $(p-1)! \equiv -1 \bmod p$, that is, $p\,|\,(p-1)!+1$. □

Exercise 7.7. We had seen in Exercise 4.16, p.40, that if $n \geqslant k \geqslant 0$, then the binomial coefficient $\binom{n}{k} = \frac{n(n-1)\cdots(n-k+1)}{k!} \in \mathbb{Z}$. The aim of this exercise is to show that the product of any k consecutive integers is divisible by $k!$. In other words, for *any* $n \in \mathbb{Z}$ and all $k \in \mathbb{N}$, the product $n(n-1)\cdots(n-k+1)$ is divisible by $k!$. Proceed as follows. Choose a large enough $m \in \mathbb{N}$ such that $n + m(k!) \geqslant k$. Set $N = n + m(k!) \in \mathbb{N}$. By Exercise 4.16, p.40, $N(N-1)\cdots(N-k+1) \equiv 0 \bmod (k!)$. Conclude that $n(n-1)\cdots(n-k+1) \equiv 0 \bmod (k!)$. *Hint:* $(N-\ell) \equiv (n-\ell) \bmod (k!)$.

Exercise 7.8. (ISBN numbers). An International Standard Book Number (ISBN) is a number used for the purpose of uniquely identifying books. For instance, the ISBN number of the book *I Want to be a Mathematician* by Paul Halmos is 0-387-96078-3. The initial digit 0 indicates that the book is published in an English-speaking region. The next block 387 identifies the publisher. The third block 96078 is assigned by the publisher and identifies this book. The last digit, 3, is called the *check digit*. Sometimes the check digit happens to be 'X', in which case, it has the numerical value 10. A valid ISBN number $d_1 d_2 \cdots d_9 d_{10}$ has a check digit which satisfies $d_{10} \equiv (d_1 + 2d_2 + 3d_3 + \cdots + 9d_9) \bmod 11$. The check digit is used for 'error detection', to find out if one of the two most common errors, namely

- typing a digit wrong, or
- transposing adjacent digits,

has occurred. For example, 0-387-96078-3 is a valid ISBN number, since we have that $1\cdot 0 + 2\cdot 3 + 3\cdot 8 + 4\cdot 7 + 5\cdot 9 + 6\cdot 6 + 7\cdot 0 + 8\cdot 7 + 9\cdot 8 = 267 = 264 + 3 = 11\cdot 24 + 3$.

(1) Show that if any two adjacent digits among $d_1 \cdots d_9$ are swapped, then the check digit flags that an error has been committed.
(2) Prove that if any one digit $d_1 \cdots d_9$ is incorrect, then the check digit flags that an error has been committed.

Exercise 7.9. What are the last two digits of 3^{1234}? *Hint:* Show $3^{20} \equiv 1 \bmod 100$.

Exercise 7.10.

(1) Show that for any $m \in \mathbb{Z}$, $m^3 \equiv n \bmod 9$ for some $n \in \{-1, 0, 1\}$.
(2) Show that there is no solution in integers to $x^3 + y^3 + z^3 = 1976$.

Exercise 7.11. Solve the equation $[2]\cdot x = [3]$ in $\mathbb{Z}/5\mathbb{Z}$.

Exercise 7.12. For which $n \in \mathbb{N}$ does the linear system of equations

$$\begin{aligned} x + y &= [2], \\ [2]\cdot x - [3]\cdot y &= [3] \end{aligned}$$

have a solution in $\mathbb{Z}/n\mathbb{Z}$?

Exercise 7.13. (Chinese remainder theorem).
(1) Show that there is no integer x such that

$$x \equiv 4 \bmod 6,$$
$$x \equiv 5 \bmod 10.$$

Hint: Consider the 'parity of x' (that is, the cases x is even/odd).
(2) Let m, n be relatively prime, and $a, b \in \mathbb{Z}$.
Show that there exists an $x \in \mathbb{Z}$ such that

$$x \equiv a \bmod m,$$
$$x \equiv b \bmod n.$$

Moreover show that for any two solutions x, x', we have $x' \equiv x \bmod (mn)$.
Hint: Let $x_0, y_0 \in \mathbb{Z}$ satisfy $mx_0 + ny_0 = 1$.
Multiply by $b - a$ to get $mk + n\ell = b - a$, and set $x = a + mk = b - n\ell$.

Chapter 8

Cardinality

Recall that a set S is *finite* if it is empty or there exists an $n \in \mathbb{N}$ and a bijection $f : S \to \{1, \cdots, n\}$. We can compare the sizes of finite sets by 'counting' the number of elements, namely via the cardinal number. The cardinal number of the empty set is defined to be 0. For example, the set $\{A, B, C, \cdots, Z\}$ of alphabet letters in the English language has cardinal number 26, while the cardinal number of $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ is 10. Finite sets with the same cardinal number can be put in a one-to-one correspondence, i.e., we can define a bijection between the two sets.

Recall also that sets that are not finite are called infinite sets. For example, in Exercise 2.23, p.24, we had seen that the set $\mathbb{N}$ is infinite. One can then ask the natural question:

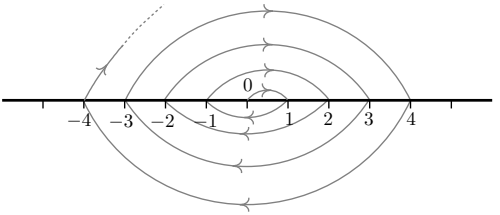
Can any two infinite sets always be put in a one-to-one correspondence?

For example, we know that the set $\mathbb{N}$ is infinite, and now suppose we have another infinite set S. Then can we always establish a bijection between the elements of $\mathbb{N}$ and those of S? In other words can we 'list' the elements of S, as the first element of S, the second element of S, and so on? The answer, perhaps surprisingly, is: No! For example, such a bijection fails to exist if we take $S = \mathbb{R}$, and this is the content of Theorem 8.3, p.86. This motivates the following definition.

Definition 8.1. (Countable and uncountable sets).
Let S be an infinite set.
The set S is *countable* if there is a bijective map from $\mathbb{N}$ onto S.
The set S is *uncountable* if S is not countable.

Example 8.1. ($\mathbb{N}$ is countable).
The identity map $n \mapsto n : \mathbb{N} \to \mathbb{N}$ is bijective, and so $\mathbb{N}$ is countable. $\diamond$

Example 8.2. ($\mathbb{Z}$ is countable).
The set $\mathbb{Z}$ of integers is countable. This is easily seen by means of the following picture.



Clearly the resulting map from $\mathbb{N}$ to $\mathbb{Z}$ is injective (since each integer is crossed by the spiral path *only once* ever – having crossed an integer, the subsequent distance of the path to the origin *increases*) and surjective (since every integer will be crossed by the spiral path *some*time). This is made rigorous as follows. The picture describes the map $f : \mathbb{N} \to \mathbb{Z}$, where

$$f(n) = \begin{cases} \frac{n}{2} & \text{if } n \text{ is even,} \\ \frac{1-n}{2} & \text{if } n \text{ is odd.} \end{cases}$$

Then[1] f is injective and surjective. A different argument using the fundamental theorem of arithmetic is given in Exercise 8.2, p.83. ◇

Exercise 8.1. Prove that the set $S = \{m \in \mathbb{N} : \exists\, n \in \mathbb{N} \text{ such that } m = n^2\}$ $(\subset \mathbb{N})$ of squares is countable.

Next, we will show the set $\mathbb{Q}$ of rational numbers is countable. To this end, we show the following two auxiliary results, interesting in their own right.

Proposition 8.1. *Every infinite subset of a countable set is countable.*

Proof. Let us first show that any infinite subset S of $\mathbb{N}$ is countable. The idea is simple. Imagine the elements of S displayed in circles (with gray lettering) amongst the natural numbers. See the picture below. Scanning the natural numbers starting from 1 moving rightwards, the moment we hit a circled element, we label it (in boldface) as **1**, and we continue scanning till we hit the next circled element, which we label as **2**, and so on. The map $f : \mathbb{N} \to S$ is obtained by sending the boldface points to the corresponding circled points. We make this rigorous below.



[1] Injectivity: Suppose $m, n \in \mathbb{N}$ are distinct, and $f(m) = f(n)$. Then:
1° m, n even: $\frac{n}{2} = \frac{m}{2}$ implies $m = n$, a contradiction.
2° m is even, n is odd: $\frac{m}{2} = \frac{1-n}{2}$. So $m + n = 1$. But $m + n \geqslant 1 + n > 1$, a contradiction.
3° m is odd, n is odd: $\frac{1-m}{2} = \frac{n}{2}$. So $m + n = 1$. But $m + n \geqslant 1 + n > 1$, a contradiction.
4° m, n odd: $\frac{1-m}{2} = \frac{1-n}{2}$ implies $m = n$, a contradiction.
Surjectivity: $f(0) = 0$, and so $0 \in \operatorname{ran} f$. If $n \in \mathbb{N}$, then $f(2n) = \frac{2n}{2} = n$, and so $\mathbb{N} \subset \operatorname{ran} f$.
If $n \in \mathbb{N}$, then $f(2n+1) = \frac{1-(2n+1)}{2} = -n$. So $-\mathbb{N} \subset \operatorname{ran} f$. By Theorem 3.5,p.29, $\mathbb{Z} \subset \operatorname{ran} f$.

Let $a_1 = \min S$. Suppose that for some $k \in \mathbb{N}$, $a_1 < \cdots < a_k$ have been constructed. Define $a_{k+1} = \min(S\backslash\{a_1, \cdots, a_k\})$. As S is not finite, the set $S\backslash\{a_1, \cdots, a_k\}$ is a nonempty subset of $\mathbb{N}$, and by the well-ordering principle, possesses a unique least element, showing that a_{k+1} is well-defined. Since $a_2 \in S\backslash\{a_1\} \subset S$, we get $a_1 = \min S \leqslant a_2$, and $a_2 \neq a_1$. For all $k > 1$, as $S\backslash\{a_1, \cdots, a_k\} \subset S\backslash\{a_1, \cdots, a_{k-1}\}$, $a_{k+1} \geqslant a_k$. As $a_{k+1} \in S\backslash\{a_1, \cdots, a_k\}$, $a_{k+1} \neq a_k$. So $a_{k+1} > a_k$ $(> a_{k-1} > \cdots > a_1)$. Define $f : \mathbb{N} \to S$ by $f(n) = a_n$, $n \in \mathbb{N}$. Then f is injective (since if $n < m$, then $f(n) < f(m)$).

We claim f is surjective. Let $m \in S$. As $\mathbb{N}$ is infinite, and f is injective, it follows from Exercise 2.27, p.24, that $f(\mathbb{N})$ cannot be a subset of $\{1, \cdots, m\}$. So there exists an $n \in \mathbb{N}$ such that $f(n) > m$. Let $n_* = \min\{n \in \mathbb{N} : f(n) \geqslant m\}$. Thus $f(n_*) \geqslant m$ and $f(1), \cdots, f(n_* - 1) < m$. Now we will show $f(n_*) \leqslant m$, which together with $f(n_*) \geqslant m$ yields $f(n_*) = m$, establishing surjectivity. Since $f(1), \cdots, f(n_* - 1) < m$, we have $m \notin \{f(1), \cdots, f(n_* - 1)\}$. Thus $f(n_*) = a_{n_*} = \min(S\backslash\{a_1, \cdots, a_{n_*-1}\}) = \min(S\backslash\{f(1), \cdots, f(n_* - 1)\}) \leqslant m$, where the last equality is justified as follows:

- $m \in S\backslash\{f(1), \cdots, f(n_* - 1)\}$ (as $m \in S$ and $m \notin \{f(1), \cdots, f(n_* - 1)\}$),
- the minimum $\min(S\backslash\{f(1), \cdots, f(n_* - 1)\})$ is less than or equal to each member of $S\backslash\{f(1), \cdots, f(n_* - 1)\}$, and so in particular the member m.

Since $f(n_*) \geqslant m$ and $f(n_*) \leqslant m$, we have $f(n_*) = m$. Hence f is surjective. Thus f is bijective. Consequently, any infinite subset S of $\mathbb{N}$ is countable.

Now let S be countable and T be an infinite subset of S. Let $\varphi : S \to \mathbb{N}$ be a bijection. Then the restriction $\varphi|_T : T \to \mathbb{N}$ of φ to T is an injective map from T to $\mathbb{N}$. By changing its codomain to $\mathrm{ran}(\varphi|_T)$, we get a bijection $\widetilde{\varphi} : T \to \mathrm{ran}(\varphi|_T)$. But as T is not finite, it now follows that $\mathrm{ran}(\varphi|_T)$ is not finite either. Hence $\mathrm{ran}(\varphi|_T)$ is an infinite subset of $\mathbb{N}$, and so it is countable. Consequently T is countable too. □

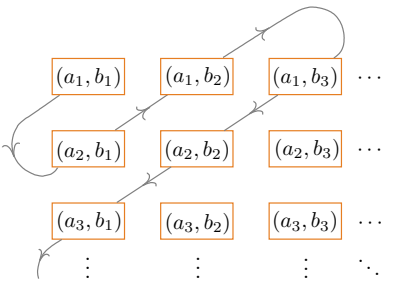
Exercise 8.2. Show that $f : \mathbb{Z} \to \mathbb{N}$, given by $f(n) = \left\{ \begin{matrix} 2^n & \text{if } n \geqslant 0 \\ 3^{-n} & \text{if } n < 0 \end{matrix} \right\}$, is injective. Conclude that $\mathbb{Z}$ is countable.

Exercise 8.3. An integer n is *even* if there exists a $m \in \mathbb{Z}$ such that $n = 2m$. Show that the set of all even integers is countable.

Exercise 8.4. The aim of this exercise is to show that a finite union of countable sets is countable. It suffices to prove this for just two countable sets, say A and B. The proof is essentially the same as in Exercise 8.2 above, where we think of elements of A as the nonnegative integers, and those of B as the negative integers. Let $\alpha : A \to \mathbb{N}$ and $\beta : B \to \mathbb{N}$ be bijections. Show that the map $f : A \cup B \to \mathbb{N}$, given by $f(x) = \left\{ \begin{matrix} 2^{\alpha(x)} & \text{if } x \in A \\ 3^{\beta(x)} & \text{if } x \in B\backslash A \end{matrix} \right\}$, is injective, and that $f(A \cup B)$ is infinite. Conclude that $A \cup B$ is countable.

Proposition 8.2. *If A, B are countable, then $A \times B$ is also countable.*

Proof. Since A and B are countable, we can list their elements via bijections $\mathbb{N} \ni n \mapsto a_n \in A$ and $\mathbb{N} \ni n \mapsto b_n \in B$. Arrange the elements of $A \times B$ in an array, and list them following the zig-zag path shown:



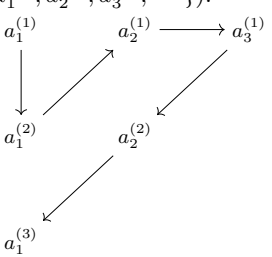
The resulting map from $\mathbb{N}$ to $A \times B$ is clearly surjective, since every element (a_n, b_m) is hit by the zig-zag path *some*time. The map is also injective, since the zig-zag path never hits a point after having crossed it because it moves on to a parallel antidiagonal below. (We will give a different proof in Exercise 8.5 below.) □

Exercise 8.5. Let A and B be countable sets, and let $m : A \to \mathbb{N}$, $n : B \to \mathbb{N}$ be bijections. Define $f : A \times B \to \mathbb{N}$ by $f(a, b) = 2^{m(a)}3^{n(b)}$ for all $(a, b) \in A \times B$. Show that f is injective. Conclude that $A \times B$ is countable. *Hint:* To show that f is injective, use the fundamental theorem of arithmetic.

Exercise 8.6. The aim of this exercise is to show that a countable union of countable sets is countable: If A_n, $n \in \mathbb{N}$, is a collection of sets such that each A_n is countable, then their union, namely

$$A := \bigcup_{n \in \mathbb{N}} A_n,$$

is countable. Analogous to the proof of Proposition 8.2, p.84, it is visually clear from the diagram below that there exists an enumeration of elements by following a zig-zag path (here $A_n = \{a_1^{(n)}, a_2^{(n)}, a_3^{(n)}, \cdots\}$):



We give a different proof which is more explicit. For $n \in \mathbb{N}$, let $f_n : A_n \to \mathbb{N}$ be a bijection. Define $f : A \to \mathbb{N}$ as follows: If $x \in A$, then it belongs to some A_n, and let $n(x)$ be the least/first $n \in \mathbb{N}$ such that $x \in A_n$, and define $f(x) = 2^{n(x)}3^{f_{n(x)}(x)}$. Prove that f is injective. Conclude that A is countable.

Exercise 8.7. Let A_n, $n \in \mathbb{N}$, be finite sets (some may be empty), such that $A := \bigcup_{n\in\mathbb{N}} A_n$ is an infinite set. Prove that A is countable. Proceed as follows:
For an $A_n \neq \varnothing$, let $m_n \in \mathbb{N}$ be its cardinal number. Write $A_n = \{a_1^{(n)}, \cdots, a_{m_n}^{(n)}\}$. For $x \in A$, let $n(x)$ be the least $n \in \mathbb{N}$ such that $x \in A_n$. As $x \in A$ belongs to $A_{n(x)}$, there exists a unique $k(x) \in \{1, \cdots, m_{n(x)}\}$ such that $x = a_{k(x)}^{(n(x))}$. Define $f : A \to \mathbb{N}$ by $f(x) = 2^{n(x)}3^{k(x)}$ for all $x \in A$. Prove that f is injective, and so A is countable.

Exercise 8.8. If A_n, $n \in \mathbb{N}$, is such that for each $n \in \mathbb{N}$, A_n is finite or countable, and $A := \bigcup_{n\in\mathbb{N}} A_n$ is infinite, then show that A is countable.

The set of rationals is an infinite set since it contains the infinite set of integers as a subset.

Theorem 8.1. *$\mathbb{Q}$ is countable.*

Proof. Each $q \in \mathbb{Q}$ can be written as $q = \frac{n}{d}$, where $n, d \in \mathbb{Z}$, $d > 0$. Among such representations of q, take the smallest possible positive denominator $d =: d_*(q) \in \mathbb{N}$. There is a unique corresponding numerator $n \in \mathbb{Z}$ such that $q = \frac{n}{d_*(q)}$, and we denote this unique n by $n_*(q)$. Hence $q = \frac{n_*(q)}{d_*(q)}$.

Define $f : \mathbb{Q} \to \mathbb{Z} \times \mathbb{N}$ by $f(q) = (n_*(q), d_*(q))$ for all $q \in \mathbb{Q}$. Then f is well-defined and injective. As $\mathbb{Z}$ and $\mathbb{N}$ are countable, it follows from Proposition 8.2, p.84, that $\mathbb{Z} \times \mathbb{N}$ is countable. Using Exercise 2.28, p.24, we get $f(\mathbb{Q})$ is infinite. Being an infinite subset of the countable set $\mathbb{Z}\times\mathbb{N}$, $f(\mathbb{Q})$ is countable. By restricting the codomain of f to $f(\mathbb{Q})$, we get a bijection $\widetilde{f} : \mathbb{Q} \to f(\mathbb{Q})$. Thus $\mathbb{Q}$ is countable. □

Exercise 8.9. Show that in the plane $\mathbb{R}^2$, the set of all circles whose centre belongs to $\mathbb{Q}^2$ (that is, the centre is at a point $\boldsymbol{x} = (a, b)$, whose x- and y-coordinates a, b are both rational numbers), and whose radii are rational, is a countable set.

Theorem 8.2. *The set $\{f : f$ is a function from $\mathbb{N}$ to $\{0, 1\}\}$, consisting of all $\{0, 1\}$-valued sequences, is uncountable.*

Proof. Let there exist an enumeration $f_1, f_2, f_3, \cdots$ of these sequences. The idea is to arrive at a contradiction by showing that this list misses out on a sequence f, by constructing an f which differs from each of these sequences. A way to construct such an f is to 'flip/toggle' the value of the terms occurring along the diagonal:

$$\begin{aligned} f_1 &\equiv \mathbf{f_1(1)},\ f_1(2),\ f_1(3), \cdots \\ f_2 &\equiv f_2(1),\ \mathbf{f_2(2)},\ f_2(3), \cdots \\ f_3 &\equiv f_3(1),\ f_3(2),\ \mathbf{f_3(3)}, \cdots \\ &\qquad\qquad \vdots \qquad \ddots \end{aligned}$$

We make this rigorous now.

We define an $f:\mathbb{N}\to\{0,1\}$ which does not appear in this list. For $n\in\mathbb{N}$, set

$$f(n) = \begin{cases} 0 & \text{if } f_n(n)=1, \\ 1 & \text{if } f_n(n)=0. \end{cases}$$

Then

$$\begin{aligned} f &\neq f_1 \text{ since } f(1)\neq f_1(1), \\ f &\neq f_2 \text{ since } f(2)\neq f_2(2), \\ f &\neq f_3 \text{ since } f(3)\neq f_3(3), \\ &\vdots \end{aligned}$$

showing that f differs from each of $f_1, f_2, f_3, \cdots$, a contradiction. □

Exercise 8.10. Show that the set of all $\{0,1\}$-valued sequences with infinitely many ones is uncountable. *Hint:* Let A (respectively B) be the set of all $\{0,1\}$-valued sequences with infinitely many ones (respectively zeroes). What is $A\cup B$?

Exercise 8.11. (How many subsequences?)
Show that $(n)_{n\in\mathbb{N}}$ has uncountably many subsequences. *Hint:* Given a subsequence, construct a $\{0,1\}$-valued sequence but putting a 1 where the subsequence term appears and 0 whenever it does not. Use Exercise 8.10 above.

Theorem 8.3. $\mathbb{R}$ *is uncountable.*

Proof. Let $S=\{f : f \text{ is a function from } \mathbb{N} \text{ to } \{0,1\}\}$. We'll construct an injective map $\varphi : S \to \mathbb{R}$. Then by restricting the codomain of φ to $\operatorname{ran}\varphi$ gives a bijection $\widetilde{\varphi} : S \to \varphi(S)$. As S is uncountable (Theorem 8.2, p.85), $\varphi(S)$ is uncountable. Hence $\mathbb{R}$ is uncountable (otherwise, its infinite subset $\varphi(S)$ would be countable).

So it remains to construct the promised injection $\varphi : S \to \mathbb{R}$. We send an $f \in S$ to the limit of the convergent sequence $(\frac{f(1)}{3} + \cdots + \frac{f(n)}{3^n})_{n\in\mathbb{N}}$. To see that this sequence is convergent, we note that it is increasing and bounded. It is increasing because the difference of the $(n+1)^{\text{st}}$ term and the nth term is $\frac{f(n+1)}{3^{n+1}} \geqslant 0$ for all $n \in \mathbb{N}$. Moreover, the sequence is bounded, as for all $n \in \mathbb{N}$, we have

$$0 \leqslant \frac{f(1)}{3} + \cdots + \frac{f(n)}{3^n} \leqslant \frac{1}{3} + \cdots + \frac{1}{3^n} = \frac{1}{3}\frac{(1-\frac{1}{3^n})}{1-\frac{1}{3}} \leqslant \frac{1}{3}\frac{1}{\frac{2}{3}} = \frac{1}{2}.$$

Now we will show that f is injective. Let $f, g \in S$ be distinct sequences. Then there is a least $n_* \in \mathbb{N}$ such that $f(n_*)\neq g(n_*)$. Suppose without loss generality that $f(n_*) > g(n_*)$ (otherwise just relabel f, g). Then $f(n_*)=1$ and $g(n_*)=0$. If $n_*=1$, then set $\sigma=0$, and if $n_*>1$, set

$$\sigma = \frac{f(1)}{3} + \cdots + \frac{f(n_*-1)}{3^{n_*-1}} = \frac{g(1)}{3} + \cdots + \frac{g(n_*-1)}{3^{n_*-1}}.$$

For $n>n_*$, we have $\frac{f(1)}{3} + \cdots + \frac{f(n_*)}{3^{n_*}} + \frac{f(n_*+1)}{3^{n_*+1}} + \cdots + \frac{f(n)}{3^n} \geqslant \sigma + \frac{1}{3^{n_*}} + 0.$

Passing to the limit as $n \to \infty$ (Exercise 5.24, p.55), we obtain

$$\varphi(f) \geqslant \sigma + \tfrac{1}{3^{n_*}}. \qquad (\star)$$

For $n > n_*$, we have

$$\begin{aligned} &\tfrac{g(1)}{3} + \cdots + \tfrac{g(n_*)}{3^{n_*}} + \tfrac{g(n_*+1)}{3^{n_*+1}} + \cdots + \tfrac{g(n)}{3^n} \\ &\leqslant \sigma + \tfrac{0}{3^{n_*}} + \tfrac{1}{3^{n_*+1}} + \cdots + \tfrac{1}{3^n} \leqslant \sigma + 0 + \tfrac{1}{3^{n_*+1}} \tfrac{(1 - \frac{1}{3^{n-n_*}})}{1 - \frac{1}{3}} \\ &\leqslant \sigma + \tfrac{1}{3^{n_*+1}} \tfrac{1}{\frac{2}{3}} = \sigma + \tfrac{1}{2 \cdot 3^{n_*}}. \end{aligned}$$

Passing to the limit as $n \to \infty$, we obtain

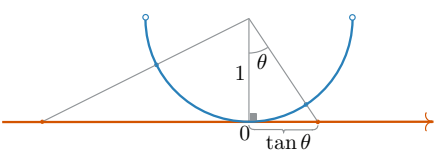
$$\varphi(g) \leqslant \sigma + \tfrac{1}{2 \cdot 3^{n_*}}. \qquad (\star\star)$$

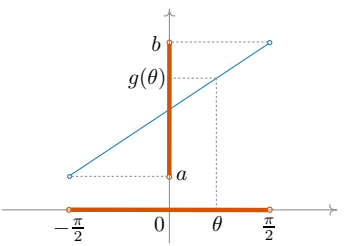
The inequalities $(\star)$ and $(\star\star)$ imply that $\varphi(f) \geqslant \sigma + \frac{1}{3^{n_*}} > \sigma + \frac{1}{2 \cdot 3^{n_*}} \geqslant \varphi(g)$. So $\varphi(f) \neq \varphi(g)$. Hence φ is injective, completing the proof. □

For reals $a < b$, we introduce the sets $(a, b), [a, b), (a, b], [a, b]$ ('intervals'):

$$\begin{aligned} (a, b) &= \{x \in \mathbb{R} : a < x < b\} \quad \text{(open interval)} \\ [a, b) &= \{x \in \mathbb{R} : a \leqslant x < b\} \\ (a, b] &= \{x \in \mathbb{R} : a < x \leqslant b\} \\ [a, b] &= \{x \in \mathbb{R} : a \leqslant x \leqslant b\} \quad \text{(closed interval)}. \end{aligned}$$

Example 8.3. (Uncountability of intervals). We will show that $(-\frac{\pi}{2}, \frac{\pi}{2})$ is uncountable using the uncountability of $\mathbb{R}$. A bijection between points of the semicircle of radius 1 and the real line is visualised below on the left:



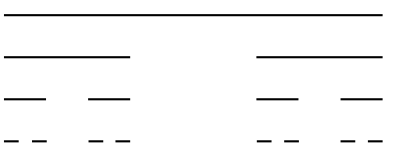


We claim that $f : (-\frac{\pi}{2}, \frac{\pi}{2}) \to \mathbb{R}$ given by $f(\theta) = \tan\theta$ for all $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$, is a bijection. As tan is continuous on $(-\frac{\pi}{2}, \frac{\pi}{2})$, and $\tan\theta \to \pm\infty$ as $\theta \to \pm\frac{\pi}{2}$, by the intermediate value theorem[2], f is surjective. It is also injective, since $f'(\theta) = \frac{1}{(\cos\theta)^2} > 0$, showing that f is strictly increasing on $(-\frac{\pi}{2}, \frac{\pi}{2})$. Thus f is a bijection. Since $\mathbb{R}$ is uncountable, so is $(-\frac{\pi}{2}, \frac{\pi}{2})$.

For any $a, b \in \mathbb{R}$ with $a < b$, we claim that (a, b) is uncountable: indeed, there is a bijection $g : (-\frac{\pi}{2}, \frac{\pi}{2}) \to (a, b)$, e.g. using the picture on the right above, given explicitly by $g(\theta) = (\theta + \frac{\pi}{2}) \frac{(b-a)}{\pi} + a$ for all $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$. ◇

[2] See, e.g., [Sasane (2015), Theorem 3.4, p.99].

Example 8.4. (Cantor set). The Cantor set is constructed as follows. First, denote the interval $[0,1]$ by F_1. Next, delete from F_1 the interval $(\frac{1}{3},\frac{2}{3})$ which is its middle third, and denote the remaining set by F_2. Clearly, $F_2 = [0,\frac{1}{3}] \cup [\frac{2}{3},1]$. Next, delete from F_2 the intervals $(\frac{1}{9},\frac{2}{9})$ and $(\frac{7}{9},\frac{8}{9})$, which are the middle thirds of its two pieces, and denote the remaining set by F_3. Then $F_3 = [0,\frac{1}{9}] \cup [\frac{2}{9},\frac{1}{3}] \cup [\frac{2}{3},\frac{7}{9}] \cup [\frac{8}{9},1]$. If we continue this process, at each stage deleting the open middle third of each closed interval remaining from the previous stage, we obtain a sequence of sets F_n, each of which contains all of its successors. The picture below illustrates this.



The *Cantor set* is[3] $C = \bigcap_{n=1}^{\infty} F_n$.

The set $C \subset [0,1]$, and consists of those points in the closed interval $[0,1]$ which 'ultimately remain' after the removal of the open intervals $(\frac{1}{3},\frac{2}{3}), (\frac{1}{9},\frac{2}{9}), (\frac{7}{9},\frac{8}{9}), \cdots$. What points do remain? The set C contains the end-points of the closed intervals which make up each set F_n: $0, 1, \frac{1}{3}, \frac{2}{3}, \frac{1}{9}, \frac{2}{9}, \frac{7}{9}, \frac{8}{9}, \cdots$. Does C contain any other points? In fact, C contains many more points than the above list of end points. Indeed, the above list of endpoints is countable, but we now show that C is uncountable, by giving a bijection φ from C and $S = \{f : f \text{ is a function from } \mathbb{N} \text{ to } \{0,1\}\}$. Any point x in C is sent to a sequence $\varphi(x)$ taking values in $\{0,1\}$ as follows. Let $x \in C$. Then for any n, $x \in F_n$, and when the middle thirds of each subinterval in F_n is removed, x is present either in the left part or the right part of the subinterval, and the nth term in the sequence of letters is 0 (left) or 1 (right) accordingly. For example,

$$\begin{aligned} 0 &\equiv 0,0,0,0,0,0,\cdots \\ 1 &\equiv 1,1,1,1,1,1,\cdots \\ \tfrac{1}{3} &\equiv 0,1,1,1,1,1,\cdots \\ \tfrac{2}{9} &\equiv 0,1,0,0,0,0,\cdots \\ \tfrac{20}{27} &\equiv 1,0,1,0,0,0,\cdots . \end{aligned}$$

We claim that $\varphi : C \to S$ is injective. Let $x, x' \in C$ be distinct elements such

[3]The Cantor set is a very intricate mathematical object, and is often a source of interesting examples/counterexamples in analysis: for example, as the sum of the lengths of the intervals removed is $\frac{1}{3} + 2\frac{1}{3^2} + 4\frac{1}{3^3} + \ldots = 1$, (factor out $\frac{1}{3}$ and sum the resulting geometric series), the '(Lebesgue length) measure' of F is $1 - 1 = 0$. So C is an example of an uncountable set with 'Lebesgue measure' 0.

that $\varphi(x) = \varphi(x')$. Then $|x - x'| > 0$. Let $n \in \mathbb{N}$ be such that $3^n > |x - x'|^{-1}$, and so $\frac{1}{3^n} < |x - x'|$. But as $\varphi(x) = \varphi(x')$, in particular, x, x' belong to the same subinterval of F_{n+1}, and each of these subintervals has a length of $\frac{1}{3^n}$, showing that $|x - x'| < \frac{1}{3^n}$, a contradiction. Thus $\varphi : C \to S$ is injective.

To show the surjectivity of φ, we will use the nested interval property (Example 5.2, p.59). Let $(a_n)_{n\in\mathbb{N}}$ in S. We will inductively construct as follows a sequence of nested intervals $[\alpha_n, \beta_n]$ of length $\frac{1}{3^{n-1}}$, $n \in \mathbb{N}$, such that α_n, β_n are the endpoints of one of the subintervals of F_n. Set $\alpha_1 = 0$ and $\beta_1 = 1$. Then $\beta_1 - \alpha_1 = \frac{1}{3^{1-1}}$. Suppose that for some $n \in \mathbb{N}$, $[\alpha_n, \beta_n] \subset \cdots \subset [\alpha_1, \beta_1]$ have been constructed such that α_k, β_k are the endpoints of one of the subintervals of F_k, and $\beta_k - \alpha_k = \frac{1}{3^{k-1}}$ for all $1 \leqslant k \leqslant n$. If $a_n = 0$, then $\alpha_{n+1} := \alpha_n$ and $\beta_{n+1} := \alpha_n + \frac{1}{3^n}$, while if $a_n = 1$, then $\alpha_{n+1} := \beta_n - \frac{1}{3^n}$ and $\beta_{n+1} := \beta_n$. Note that $\alpha_{n+1}, \beta_{n+1}$ are then the endpoints of a subinterval of F_{n+1}, and $\beta_{n+1} - \alpha_{n+1} = \frac{1}{3^n} = \frac{1}{3^{(n+1)-1}}$. This completes the induction step. Since $\frac{1}{3^n}$ goes to 0 as n goes to infinity, it follows that $\bigcap_{n\in\mathbb{N}} [\alpha_n, \beta_n]$ is a singleton, say $x \in \mathbb{R}$. Since $[\alpha_n, \beta_n] \subset F_n$ for each $n \in \mathbb{N}$, we conclude that $x \in C$. We now check that $\varphi(x) = (a_n)_{n\in\mathbb{N}}$. Let $(\widetilde{a}_n)_{n\in\mathbb{N}}$ be the sequence $\varphi(x)$. We want to show that $(\widetilde{a}_n)_{n\in\mathbb{N}} = (a_n)_{n\in\mathbb{N}}$. Let $n \in \mathbb{N}$. We have $x \in [\alpha_n, \beta_n] \subset F_n$. If $a_n = 0$, then $\alpha_n = \alpha_{n+1} \leqslant x \leqslant \beta_{n+1} = \alpha_{n+1} + \frac{1}{3^n}$, and so $\widetilde{a}_n = 0 = a_n$. If $a_n = 1$, then $\beta_n - \frac{1}{3^n} = \alpha_{n+1} \leqslant x \leqslant \beta_{n+1} = \beta_n$, and so $\widetilde{a}_n = 1 = a_n$.

As S is uncountable, it now follows that so is C. ◇

Exercise 8.12. Show that the set $\mathbb{R}\backslash\mathbb{Q}$ of all irrational numbers is uncountable.

Exercise 8.13. Show that for all $d \in \mathbb{N}$, there is no bijection $\varphi : \mathbb{R} \to \mathbb{Q}^d$.
(This can be used to show that $\mathbb{R}$ is not a finite-dimensional vector space over $\mathbb{Q}$.)

Exercise 8.14. The result from Example 8.3, p.87, can be used to show that $[0, 1]$ is uncountable. Here we give a different proof. Show that $[0, 1]$ is uncountable using Exercise 8.6, p.84, and the fact that $\mathbb{R} = \bigcup_{m\in\mathbb{Z}} [m, m+1)$.

Exercise 8.15. Let $C = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x^2 + y^2 = 1\}$, and $C_{\mathbb{Q}} = C \cap (\mathbb{Q} \times \mathbb{Q})$.
(1) Show that C is infinite and uncountable.
(2) Show that $C_{\mathbb{Q}}$ is an infinite set. *Hint:* For $n \in \mathbb{N}$, show that $(\frac{1-n^2}{1+n^2}, \frac{2n}{1+n^2}) \in C_{\mathbb{Q}}$.
(3) Show that $C_{\mathbb{Q}}$ is countable.

Exercise 8.16. Define a relation $\sim$ on $\mathbb{R}$ by setting $a \sim b$ if $a - b \in \mathbb{Q}$. Then $\sim$ is an equivalence relation. The equivalence class of $a \in \mathbb{R}$ is denoted by $[a]$.
If $a \in \mathbb{R}$, then prove that $[a]$ is infinite and countable.
Show that the set $S = \{[a] : a \in \mathbb{R}\}$ is infinite and uncountable.

Exercise 8.17. Show that there exists a bijection between $(0, 1)$ and $[0, 1]$.
Hint: First consider the rationals in these sets.

The power set $\mathcal{P}(S)$ of a set S is the collection of all the subsets of S. E.g., if $S=\{a,b\}$, then $\mathcal{P}(S)=\{\varnothing,\{a\},\{b\},\{a,b\}\}$. If $S=\varnothing$, then $\mathcal{P}(\varnothing)=\{\varnothing\}$.

Theorem 8.4. (Cantor).
There does not exist a surjection from a set S onto its power set $\mathcal{P}(S)$.

Proof. Let S be a set such that there exists a surjective map $f : S \to \mathcal{P}(S)$. Define the set[4] $X=\{x \in S : x \notin f(x)\}$. Note that $X \in \mathcal{P}(S)$ (as X consists of elements taken only from S). We claim that for all $x \in S$, $f(x) \neq X$, which means that $X \in \mathcal{P}(S)$ fails to be in the range of the map f, showing f is not surjective, a contradiction. We have two possible cases:

1° $x \in X$. By the definition of X, $x \notin f(x)$. So the element x belongs to X but not to $f(x)$, so that the two sets $f(x)$ and X cannot be the same.

2° $x \notin X$. Then $\neg(x \notin f(x))$, i.e., $x \in f(x)$. So x does not belong to X but belongs to $f(x)$, showing that $f(x)$ and X cannot be the same.

So for all $x \in S$, we have $f(x) \neq X$, completing the proof. □

Corollary 8.1. *There is no bijection between a set and its power set.*

Exercise 8.18. Let S be a finite, nonempty set with cardinal number $n \in \mathbb{N}$.
(1) Show that the number of subsets of S with k ($\leqslant n$) elements is $\binom{n}{k}$.
(2) Show that $\mathcal{P}(S)$ has 2^n elements.
(3) Deduce that $n<2^n$ for all $n \in \mathbb{N}$.

Exercise 8.19.
(1) Show that the set of all two element subsets of $\mathbb{N}$ is countable.
(2) Show that the set of all nonempty finite subsets of $\mathbb{N}$ is countable.
(3) Show that the set of all subsets of $\mathbb{N}$ is uncountable.

Remark 8.1. (Continuum hypothesis). For sets A and B, we say that A has *smaller cardinality* than B if there is an injection from A to B, but there is no bijection from A to B. Thus $\mathbb{N}$ has smaller cardinality than $\mathbb{R}$. It can be shown that there is a bijection between $\mathcal{P}(\mathbb{N})$ and $\mathbb{R}$. A famous conjecture of set theory[5], advanced by Cantor, was:

> There does not exist a set S such that $\mathbb{N}$ has cardinality smaller than S, and S has cardinality smaller than $\mathbb{R}$.

This is the *continuum hypothesis* (as $\mathbb{R}$ was called the 'continuum'). Results by Gödel and Cohen showed that the continuum hypothesis is independent of the ZFC axioms of set theory, i.e., it can neither be proved nor disproved within the ZFC axiomatic set theory system; see e.g. [Feferman (1999)]. ∗

[4] This makes sense as a set. Indeed, for each $x \in S$, $f(x) \in \mathcal{P}(S)$ is a subset of S, and so we can ask whether or not the element x of S belongs to this subset. X is simply the collection of those elements of S for which $x \notin f(x)$.

[5] First problem in Hilbert's list of 23 problems!

Chapter 9

Real numbers revisited

9.1 Irrational numbers

While $\mathbb{Q} \subset \mathbb{R}$ and is a smaller field inside the bigger field $\mathbb{R}$, its complement $\mathbb{R}\backslash\mathbb{Q}$ of irrational numbers does not form a field[1], because the sum/product of two irrational numbers is need not be irrational: $\sqrt{2} + (-\sqrt{2}) = 0 \in \mathbb{Q}$ and $\sqrt{2} \cdot \sqrt{2} = 2 \in \mathbb{Q}$. Thus the restriction to irrationals of the real number addition/multiplication are not maps[2] from $(\mathbb{R}\backslash\mathbb{Q}) \times (\mathbb{R}\backslash\mathbb{Q})$ *to* $(\mathbb{R}\backslash\mathbb{Q})$. In Exercise 8.12, p.89, we showed that the set of irrational numbers is uncountable, and in that sense there are 'many more irrational numbers than rational numbers'. Later in this chapter, will learn an elementary result, called the rational zeroes theorem, which can be useful for proving irrationality of some real numbers. But we begin by showing the irrationality of the Euler's number e.

Theorem 9.1. *The sequence* $(1 + \frac{1}{1!} + \cdots + \frac{1}{n!})_{n\in\mathbb{N}}$ *converges in* $\mathbb{R}$, *and its limit* (*denoted by* e, *called* Euler's number) *is irrational.*

Proof. Let $a_n := 1 + \frac{1}{1!} + \cdots + \frac{1}{n!}$ for all $n \in \mathbb{N}$.
As $a_{n+1} - a_n = \frac{1}{(n+1)!} > 0$ for all $n \in \mathbb{N}$, the sequence $(a_n)_{n\in\mathbb{N}}$ is increasing.
For all $n \in \mathbb{N}$ such that $n \geqslant 2$, $n! = 1 \cdot 2 \cdot 3 \cdot \ldots \cdot n > 1 \cdot 2 \cdot 2 \cdot \ldots \cdot 2 = 2^{n-1}$. So

$$\begin{aligned} 0 \leqslant a_n &= 1 + \tfrac{1}{1!} + \tfrac{1}{2!} + \cdots + \tfrac{1}{n!} \\ &\leqslant 1 + 1 + \tfrac{1}{2} + \cdots + \tfrac{1}{2^{n-1}} = 1 + \frac{1 - \frac{1}{2^n}}{1 - \frac{1}{2}} = 1 + 2(1 - \tfrac{1}{2^n}) \\ &< 1 + 2 = 3. \end{aligned}$$

Hence $(a_n)_{n\in\mathbb{N}}$ is also bounded. By Proposition 5.6, p.56, $(a_n)_{n\in\mathbb{N}}$ converges in $\mathbb{R}$, and we denote its limit by e. Next, we will show that $e \notin \mathbb{Q}$.

Fix an $m \in \mathbb{N}$. Then $(a_{m+k})_{k\in\mathbb{N}}$ is a subsequence of $(a_n)_{n\in\mathbb{N}}$, and so is convergent with the same limit e. So $(a_{m+k} - a_m)_{k\in\mathbb{N}}$ is convergent with limit $e - a_m$.

[1] See page 78 for the definition of a field.
[2] Recall that the field additon and multiplication are both maps from $\mathbb{F} \times \mathbb{F}$ *to* $\mathbb{F}$.

But

$$\begin{aligned} a_{m+k} - a_m &= \tfrac{1}{(m+1)!} + \tfrac{1}{(m+2)!} + \cdots + \tfrac{1}{(m+k)!} \\ &= \tfrac{1}{(m+1)!}\big(1 + \tfrac{1}{m+2} + \cdots + \tfrac{1}{(m+2)\cdots(m+k)}\big) \\ &\leqslant \tfrac{1}{(m+1)!}\big(1 + \tfrac{1}{2} + \cdots + \tfrac{1}{2^{k-1}}\big) \\ &= \tfrac{1}{(m+1)!}\,\tfrac{(1-\frac{1}{2^k})}{(1-\frac{1}{2})} = \tfrac{1}{(m+1)!}(2 - \tfrac{1}{2^{k-1}}) \\ &< \tfrac{2}{(m+1)!}. \end{aligned}$$

The first equality also shows that $\frac{1}{(m+1)!} \leqslant a_{m+k} - a_m$. So the above gives

$$\tfrac{1}{(m+1)!} \leqslant a_{m+k} - a_m < \tfrac{2}{(m+1)!}.$$

Passing to the limit as $k \to \infty$, we obtain

$$\tfrac{1}{(m+1)!} \leqslant e - (1 + \tfrac{1}{1!} + \tfrac{1}{2!} + \tfrac{1}{3!} + \cdots + \tfrac{1}{m!}) \leqslant \tfrac{2}{(m+1)!} \qquad (*)$$

The choice of $m \in \mathbb{N}$ was arbitrary, and so $(*)$ holds for all $m \in \mathbb{N}$. Suppose $e \in \mathbb{Q}$, and write $(0<)\ e = \frac{p}{q}$, where $p, q \in \mathbb{N}$. Take $m = q + 2$, and multiply $(*)$ by $m!$ throughout. Since $m!$ contains q as a factor (as $m > q$), we obtain that the integer

$$N := m!e - (m! + \tfrac{m!}{1!} + \tfrac{m!}{2!} + \tfrac{m!}{3!} + \cdots + \tfrac{m!}{m!})$$

satisfies $0 < \frac{1}{m+1} \leqslant N \leqslant \frac{2}{m+1} < \frac{2}{2+1} < 1$, a contradiction. So $e \notin \mathbb{Q}$. □

Exercise 9.1. Show that[3] $\log_2 3$ is irrational.

Exercise 9.2. Show that there exist irrational $a, b \in \mathbb{R}$ such that a^b is rational.

Hint: Consider $\sqrt{2}^{\sqrt{2}}$: The two possibilities are $\sqrt{2}^{\sqrt{2}} \in \mathbb{Q}$ or $\sqrt{2}^{\sqrt{2}} \notin \mathbb{Q}$. Alternatively, use Exercise 9.1 above.

Exercise 9.3. Show that $\tan 1^\circ$ is irrational.

Hint: Use[4] $\tan(\alpha+\beta) = \dfrac{\tan\alpha + \tan\beta}{1 - (\tan\alpha)(\tan\beta)}$ for $\alpha, \beta \in [0, \frac{\pi}{4})$.

Exercise 9.4. Suppose $\ell, m, n \in \mathbb{N}$ are odd numbers and $d \in \mathbb{N}$ is at least 2. Let p be the polynomial given by $p(x) = \ell x^d + mx + n$ for all $x \in \mathbb{R}$. Prove that the real zeroes[5] of p, if they exist, are all irrational.

[3] For the definition of logarithm and the exponential functions (starting from the least upper bound property of $\mathbb{R}$), see e.g. [Sasane (2015), Chapter 5]. For real $a, b > 0$, we define $a^b = e^{b \log_e a}$. It is shown in [Sasane (2015), Chapter 5] that $b^{\log_b a} = a$, and for real $c > 0$, there holds $\log_b(a^c) = c \log_b a$ and $a^{bc} = (a^b)^c$.

[4] For the construction of trigonometric functions and the proof of their elementary properties, see e.g. [Sasane (2015), Chapter 5].

[5] See p.102 for the relevant terminology.

9.2 Decimal representation

By a *decimal representation*

$$N.d_1d_2d_3\cdots$$

where $N \in \mathbb{N} \cup \{0\}$, and $d_n \in \{0,1,2,3,4,5,6,7,8,9\}$ for all $n \in \mathbb{N}$, we mean the real number given by

$$x := N + \lim_{n\to\infty}(\tfrac{d_1}{10} + \tfrac{d_2}{10^2} + \tfrac{d_3}{10^3} + \cdots + \tfrac{d_n}{10^n}).$$

A negative sign in front, i.e., $-N.d_1d_2d_3\cdots$, denotes $-x$, the unique additive inverse in $\mathbb{R}$ of $x = N.d_1d_2d_3\cdots$.

The following result shows that every decimal representation does correspond to a real number.

Theorem 9.2. *Let $(d_n)_{n\in\mathbb{N}}$ be such that $d_n \in \{0,1,2,3,4,5,6,7,8,9\}$ for all $n \in \mathbb{N}$. Then $(\tfrac{d_1}{10} + \tfrac{d_2}{10^2} + \tfrac{d_3}{10^3} + \cdots + \tfrac{d_n}{10^n})_{n\in\mathbb{N}}$ is convergent in $\mathbb{R}$.*

Proof. Let $a_n := \tfrac{d_1}{10} + \cdots + \tfrac{d_n}{10^n}$ for all $n \in \mathbb{N}$.
The sequence $(a_n)_{n\in\mathbb{N}}$ is bounded, since for all $n \in \mathbb{N}$ we have that

$$0 < \tfrac{d_1}{10} + \cdots + \tfrac{d_n}{10^n} \leqslant \tfrac{9}{10} + \cdots + \tfrac{9}{10^n} = (\tfrac{9}{10})\frac{1-(\frac{1}{10})^n}{1-\frac{1}{10}} = 1 - (\tfrac{1}{10})^n < 1.$$

Also, $(a_n)_{n\in\mathbb{N}}$ is increasing because $a_{n+1} - a_n = \tfrac{d_{n+1}}{10^{n+1}} \geqslant 0$ for all $n \in \mathbb{N}$.
Monotone and bounded sequences converge in $\mathbb{R}$ (Proposition 5.6, p.56), and so $(a_n)_{n\in\mathbb{N}}$ converges in $\mathbb{R}$. □

Example 9.1. $(0.999\cdots = 1.000\cdots)$. We have

$$\begin{aligned} 0.999\cdots &= \lim_{n\to\infty}(\tfrac{9}{10} + \cdots + \tfrac{9}{10^n}) \\ &= \tfrac{9}{10}\lim_{n\to\infty}(1 + \tfrac{1}{10} + \cdots + \tfrac{1}{10^{n-1}}) = \tfrac{9}{10}\frac{1}{(1-\frac{1}{10})} \\ &= 1 = 1 + \lim_{n\to\infty}(\tfrac{0}{10} + \cdots + \tfrac{0}{10^n}) = 1.000\cdots. \end{aligned}$$

◇

By Theorem 9.2, every decimal representation corresponds to a real number. Vice versa, we now show that every real number admits a decimal representation. An integer $d \in \{0,1,2,3,4,5,6,7,8,9\}$ is called a *digit*.

Theorem 9.3. *Let $x \in \mathbb{R}$. Then there exist $N \in \mathbb{N} \cup \{0\}$ and a sequence $(d_n)_{n\in\mathbb{N}}$ with $d_n \in \{0,1,2,3,4,5,6,7,8,9\}$ for all $n \in \mathbb{N}$ such that*

$$x = \begin{cases} N.d_1d_2d_3\cdots & \text{if } x \geqslant 0, \\ -N.d_1d_2d_3\cdots & \text{if } x < 0. \end{cases}$$

Proof. If $x = 0$, then set $N = 0$ and $(d_n)_{n\in\mathbb{N}} = (0)_{n\in\mathbb{N}}$, i.e., $0 = 0.000\cdots$. It is enough to consider the case $x > 0$, since if $x < 0$, then $-x$ admits a decimal representation $-x = N.d_1d_2d_3\cdots$, giving $x = -N.d_1d_2d_3\cdots$.

Let the real number $x > 0$. Define $N = [x]$, the greatest integer less than or equal to x (see Exercise 5.20, p.54). Then $N \leqslant x < N+1$. Also, as $0 < x$, we have $0 < N+1$ and so $0 \leqslant N$, i.e., $N \in \mathbb{N} \cup \{0\}$.

We will construct the sequence $(d_n)_{n\in\mathbb{N}}$ of digits of x inductively. Define $d_1 = [10(x-N)]$. As $0 \leqslant 10(x-N)$, $0 \leqslant d_1$. Also, $d_1 \leqslant 10(x-N) < 10(1) = 10$. Thus $0 \leqslant d_1 < 10$, i.e., d_1 is a digit. From $d_1 \leqslant 10(x-N) < d_1 + 1$, we obtain $N + \frac{d_1}{10} \leqslant x < N + \frac{d_1}{10} + \frac{1}{10}$. Suppose digits $d_1, \cdots, d_n$ have been constructed for some $n \in \mathbb{N}$, and that

$$N + \frac{d_1}{10} + \cdots + \frac{d_n}{10^n} \leqslant x < N + \frac{d_1}{10} + \cdots + \frac{d_n}{10^n} + \frac{1}{10^n}.$$

Set

$$d_{n+1} = [10^{n+1}\big(x - (N + \frac{d_1}{10} + \cdots + \frac{d_n}{10^n})\big)].$$

Since $0 \leqslant x - (N + \frac{d_1}{10} + \cdots + \frac{d_n}{10^n})$, we have $0 \leqslant d_{n+1}$. Also,

$$d_{n+1} \leqslant 10^{n+1}\big(x - (N + \frac{d_1}{10} + \cdots + \frac{d_n}{10^n})\big) < 10^{n+1}\frac{1}{10^n} = 10.$$

So d_{n+1} is a digit. As $d_{n+1} \leqslant 10^{n+1}\big(x - (N + \frac{d_1}{10} + \cdots + \frac{d_n}{10^n})\big) < d_{n+1} + 1$,

$$N + \frac{d_1}{10} + \cdots + \frac{d_{n+1}}{10^{n+1}} \leqslant x < N + \frac{d_1}{10} + \cdots + \frac{d_{n+1}}{10^{n+1}},$$

completing the induction step. So we have obtained $N \in \mathbb{N} \cup \{0\}$ and a sequence $(d_n)_{n\in\mathbb{N}}$ of digits such that

$$\text{for all } n \in \mathbb{N}, \quad N + \frac{d_1}{10} + \cdots + \frac{d_n}{10^n} \leqslant x < N + \frac{d_1}{10} + \cdots + \frac{d_n}{10^n} + \frac{1}{10^n}. \qquad (\star)$$

Define $\widetilde{x} = N.d_1d_2d_3\cdots \in \mathbb{R}$. Noting that $(\frac{1}{10^n})_{n\in\mathbb{N}}$ converges to 0, and passing to the limit as $n \to \infty$ in $(\star)$, we get $\widetilde{x} \leqslant x \leqslant \widetilde{x} + 0$, i.e., $x = \widetilde{x}$. □

A real number x is said to have a *terminating decimal representation* if we have $x = N.d_1d_2d_3\cdots$, and there exists an $N \in \mathbb{N}$ such that $d_n = 0$ for all $n > N$. Then $x = N + \frac{d_1}{10} + \cdots + \frac{d_N}{10^N} \in \mathbb{Q}$. If nonzero digits in the decimal representation of x persist (i.e., for all $N \in \mathbb{N}$ there exists an $n \in \mathbb{N}$ such that $n > N$ and $d_n \neq 0$), then we call the decimal representation *nonterminating*.

Theorem 9.4. *Let $n, m \in \mathbb{N}$, and let $d_{n+1}, \cdots, d_{n+m}$ be digits, not all zero. Let the real x have a decimal representation which is nonterminating and repeating, i.e., of the form $x = N.d_1\cdots d_n\, d_{n+1}\cdots d_{n+m}\, d_{n+1}\cdots d_{n+m}\cdots$, where the block of m digits $d_{n+1}\cdots d_{n+m}$ keeps repeating. Then $x \in \mathbb{Q}$.*

Proof. Let $s_k = N + \frac{d_1}{10} + \cdots + \frac{d_k}{10^k}$ for $k \in \mathbb{N}$. We know $(s_k)_{n\in\mathbb{N}}$ converges. Hence its subsequence $s_n, s_{n+m}, s_{n+2m}, \cdots$ converges with the same limit. So we have

$$\begin{aligned} x &= N + \tfrac{d_1}{10} + \cdots + \tfrac{d_n}{10^n} \\ &\quad + \lim_{k\to\infty}\left(\left(\tfrac{d_{n+1}}{10^{n+1}} + \cdots + \tfrac{d_{n+m}}{10^{n+m}}\right) + \cdots + \tfrac{1}{10^{mk}}\left(\tfrac{d_{n+1}}{10^{n+1}} + \cdots + \tfrac{d_{n+m}}{10^{n+m}}\right)\right) \\ &= N + \tfrac{d_1}{10} + \cdots + \tfrac{d_n}{10^n} + \left(\tfrac{d_{n+1}}{10^{n+1}} + \cdots + \tfrac{d_{n+m}}{10^{n+m}}\right)\lim_{k\to\infty}\left(1 + \tfrac{1}{10^m} + \cdots + \tfrac{1}{10^{mk}}\right) \\ &= N + \tfrac{d_1}{10} + \cdots + \tfrac{d_n}{10^n} + \left(\tfrac{d_{n+1}}{10^{n+1}} + \cdots + \tfrac{d_{n+m}}{10^{n+m}}\right)\frac{1}{1 - \frac{1}{10^m}}. \end{aligned}$$

Thus x is a rational number. □

Example 9.2. As $e \notin \mathbb{Q}$, its decimal representation cannot be terminating, nor can it be nonterminating and repeating, and so it must be nonterminating and nonrepeating. ◇

We now show that every rational number has a terminating or a nonterminating but repeating decimal representation.

Theorem 9.5. *Every rational number has a decimal representation that is either terminating or is nonterminating and repeating.*

Proof. It suffices to consider positive rationals. Let $x = \frac{p}{q}$, where $p, q \in \mathbb{N}$. Then q admits a factorisation $q = 2^i 5^j q'$, where i, j are nonnegative integers, and the primes $2, 5$ do not divide $q' \in \mathbb{N}$. Choose a natural number $n > i, j$. Then we have that $\frac{10^n}{2^i 5^j}$ is a natural number, and so

$$10^n x = 10^n \tfrac{p}{q} = \tfrac{p'}{q'},$$

where $p' = 2^{n-i}5^{n-j}p \in \mathbb{N}$, and the integers $q', 10$ are coprime. But if we look at the remainders we get when we divide the numbers in the infinite sequence $10, 10^2, 10^3, \cdots$ by q', then for some integers $K > k \geqslant 1$, the integers $10^K, 10^k$ leave the same remainder when divided by q' (e.g. using the result from Exercise 2.27, p.24). So $10^K - 10^k = 10^k(10^{K-k} - 1)$ is divisible by q'. But the integers $q', 10$ are coprime, and hence also $q', 10^k$ are coprime (e.g. by a repeated application of Exercise 6.25, p.70, with $n = q'$ and $a = 10$, and b taken as successive powers of 10). Thus q' must divide $10^{K-k} - 1$. So with $m := K - k$, we have

$$10^n(10^m - 1)x = (10^{K-k} - 1)\tfrac{p'}{q'} \in \mathbb{N}.$$

Now let $x = N.d_1d_2d_3\cdots$. Then[6]

$$\begin{aligned} & 10^{n+m}x - 10^n x \\ = \; & Nd_1\cdots d_{n+m} - Nd_1\cdots d_n \\ & +(\tfrac{d_{n+m+1}}{10} + \tfrac{d_{n+m+2}}{10^2} + \tfrac{d_{n+m+3}}{10^3} + \cdots - \tfrac{d_{n+1}}{10} - \tfrac{d_{n+2}}{10^2} - \tfrac{d_{n+3}}{10^3} - \cdots), \end{aligned}$$

belongs to $\mathbb{N}$. This implies that the part in the brackets above, henceforth denoted by Δ, is also an integer:

$$\Delta = \tfrac{d_{n+m+1} - d_{n+1}}{10} + \tfrac{d_{n+m+2} - d_{n+2}}{10^2} + \tfrac{d_{n+m+3} - d_{n+3}}{10^3} + \cdots \in \mathbb{Z}.$$

We claim that $d_{n+m+\ell} = d_{n+\ell}$ for all $\ell \in \mathbb{N}$, giving the desired conclusion.

1° It cannot be the case that $(d_{n+m+\ell} - d_{n+\ell})_{\ell\in\mathbb{N}}$ is eventually the constant sequence $9, 9, 9, \cdots$. Otherwise there is an ℓ_* such that $d_{n+m+\ell} - d_{n+\ell} = 9$ for all $\ell > \ell_*$. This implies $d_{n+m+\ell} = 9$ and $d_{n+\ell} = 0$ for all $\ell > \ell_*$. So $d_{(n+m)+(\ell_*+1)} = 9$, but $d_{n+(m+\ell_*+1)} = 0$, a contradiction.

2° Similar to 1°, it cannot be the case that the sequence $(d_{n+m+\ell} - d_\ell)_{\ell\in\mathbb{N}}$ is eventually the constant sequence $-9, -9, -9, \cdots$ either.

3° Suppose that $\ell_* \in \mathbb{N}$ is the smallest number such that $d_{n+m+\ell_*} \neq d_{n+\ell_*}$.

Consider first $d_{n+m+\ell_*} > d_{n+\ell_*}$. By 2°, there exists a number $L \in \mathbb{N}$ such that $d_{n+m+\ell_*+L} - d_{n+\ell_*+L} \neq -9$, and so

$$\Delta \geqslant \tfrac{1}{10^{\ell_*}} - \tfrac{9}{10^{\ell_*+1}} - \cdots - \tfrac{9}{10^{\ell_*+L-1}} - \tfrac{8}{10^{\ell_*+L}} - \tfrac{9}{10^{\ell_*+L+1}} \cdots = \tfrac{1}{10^{\ell_*+L}} > 0.$$

We also have (the rightmost strict inequality below follows from 1°):

$$1 \geqslant \tfrac{1}{10^{\ell_*-1}} = \tfrac{9}{10^{\ell_*}} + \tfrac{9}{10^{\ell_*+1}} + \tfrac{9}{10^{\ell_*+2}} + \cdots > \Delta.$$

Hence $1 > \Delta > 0$, a contradiction to the fact that Δ is an integer.

Consider the only remaining possibility, namely, $d_{n+m+\ell_*} < d_{n+\ell_*}$. Then by 1°, there exists an $L \in \mathbb{N}$ such that $d_{n+m+\ell_*+L} - d_{n+\ell_*+L} \neq 9$, and so

$$\Delta \leqslant \tfrac{-1}{10^{\ell_*}} + \tfrac{9}{10^{\ell_*+1}} + \cdots + \tfrac{9}{10^{\ell_*+L-1}} + \tfrac{8}{10^{\ell_*+L}} + \tfrac{9}{10^{\ell_*+L+1}} + \cdots = -\tfrac{1}{10^{\ell_*+L}}.$$

So $\Delta < 0$. We also have (the rightmost strict inequality follows from 2°):

$$-1 \leqslant -\tfrac{1}{10^{\ell_*-1}} = -\tfrac{9}{10^{\ell_*}} - \tfrac{9}{10^{\ell_*+1}} - \tfrac{9}{10^{\ell_*+2}} + \cdots < \Delta.$$

Thus $-1 < \Delta < 0$, again a contradiction to Δ being an integer.

Consequently, $d_{n+m+\ell} = d_{n+\ell}$ for all $\ell \in \mathbb{N}$, which means that the block of digits $d_{n+1}\cdots d_{n+m}$ keeps repeating in the decimal representation of x. □

Exercise 9.5. Is $0.12345678910111213\cdots$ a rational number?

[6] Here by $Nd_1\cdots d_n$ we mean the number which in decimal notation has the digits of N followed by the digits $d_1, \cdots, d_n$. The number $Nd_1\cdots d_{n+m}$ has a similar connotation.

9.3 Continued fraction representation

It is a natural question to ask for the closest/best rational approximation $\frac{p}{q}$, $p, q \in \mathbb{N}$, to a positive real x, subject to the condition that $q \leqslant N$ for some given bound $N \in \mathbb{N}$. We will now learn about 'continued fractions', and see that they give a method for finding the best such rational approximations. A *finite continued fraction* is an expression of the form

$$a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cfrac{1}{\ddots \cfrac{}{a_{n-1} + \cfrac{1}{a_n + r_n}}}}}$$

where $a_1 \geqslant 0$ and $a_2, \cdots, a_n \in \mathbb{N}$. For notational ease, we denote the above finite continued fraction simply by $[a_1, a_2, \cdots, a_n]$.

Example 9.3. We have

$$\frac{3}{28} = \frac{1}{\frac{28}{3}} = \frac{1}{9+\frac{1}{3}} = [0, 9, 3], \text{ and}$$

$$\frac{33}{23} = 1 + \frac{10}{23} = 1 + \frac{1}{2+\frac{3}{10}} = 1 + \frac{1}{2 + \frac{1}{3+\frac{1}{3}}} = [1, 2, 3, 3].$$

Also,

$$[0, 1, 2, 3] = 0 + \frac{1}{1 + \frac{1}{2+\frac{1}{3}}} = \frac{7}{10}, \text{ and}$$

$$[2, 3, 2, 3] = 2 + \frac{1}{3 + \frac{1}{2+\frac{1}{3}}} = \frac{55}{24}.$$

◇

Consider the rational number $\frac{p}{q}$, $p, q \in \mathbb{N}$, such that $\gcd(p, q) = 1$. Recall the Euclidean algorithm from p.67 for computing the greatest common divisor. Suppose that the Euclidean algorithm in the case of p, q gives the following sequence of remainders:

$$\begin{aligned} p &= a_1 q + r_1 \\ q &= a_2 r_1 + r_2 \\ r_1 &= a_3 r_2 + r_3 \\ &\vdots \\ r_{n-3} &= a_{n-1} r_{n-2} + 1 \\ r_{n-2} &= a_n. \end{aligned}$$

We get the remainder 1 in the second last step because $\gcd(p, q) = 1$.

Then

$$\frac{p}{q} = \frac{a_1 q + r_1}{q} = a_1 + \frac{r_1}{q} = a_1 + \frac{r_1}{a_2 r_1 + r_2} = a_1 + \frac{1}{a_2 + \frac{r_2}{r_1}} = a_1 + \frac{1}{a_2 + \frac{r_2}{a_3 r_2 + r_3}}$$

$$= a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{r_3}{r_2}}} = \cdots = a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cfrac{1}{\ddots \cfrac{}{a_{n-1} + \cfrac{r_{n-1}}{a_n r_{n-1}}}}}}$$

Thus $\frac{p}{q} = a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cfrac{1}{\ddots \cfrac{}{a_{n-1} + \cfrac{1}{a_n}}}}} = [a_1, a_2, a_3, \cdots, a_n].$

So every rational number can be represented as a finite continuous fraction.

Exercise 9.6. Using the method described above, find the continued fraction representation of $\frac{225}{157}$.

If $x \in \mathbb{R}\backslash\mathbb{Q}$ and $x > 0$, then we define a sequence $(r_n)_{n\in\mathbb{N}}$ of nonzero remainders as follows. Let $a_1 := [x] \geqslant 0$, and $r_1 := x - [x]$ and $x = a_1 + r_1$. Then $0 < r_1 < 1$. We define successively the natural numbers $a_2, a_3, \cdots, a_n$ and the 'remainders' $r_2, r_3, \cdots, r_n \in [0, 1)$ by

$$\frac{1}{r_1} = a_2 + r_2\ , \quad \cdots \quad , \ \frac{1}{r_{n-1}} = a_n + r_n.$$

Then x can be written in the form

$$x = a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cfrac{1}{\ddots \cfrac{}{a_{n-1} + \cfrac{1}{a_n + r_n}}}}} = \frac{P_n + r_n S_n}{Q_n + r_n T_n}, \qquad (\star)$$

for suitable integers P_n, Q_n, S_n, T_n. In particular,

$P_1 = a_1 \geqslant 0,$	$S_1 = 1,$	$Q_1 = 1,$	$T_1 = 0,$
$P_2 = a_1 a_2 + 1 \in \mathbb{N},$	$S_2 = a_1 \geqslant 0,$	$Q_2 = a_2 \in \mathbb{N},$	$T_2 = 1,$
$P_3 = a_1 a_2 a_3 + a_1 + a_3 \in \mathbb{N},$	$S_3 = 1 + a_1 a_2 \in \mathbb{N},$	$Q_3 = 1 + a_2 a_3 \in \mathbb{N},$	$T_3 = a_2 \in \mathbb{N}.$

Each r_n satisfies $0 < r_n < 1$ (since if $r_n = 0$ for some $n \in \mathbb{N}$, then x would be rational, a contradiction).

Define the 2×2 matrix by $M_n = \left[\begin{smallmatrix} P_n & S_n \\ Q_n & T_n \end{smallmatrix}\right]$. Then M_n has integer entries, and $M_1 = \left[\begin{smallmatrix} P_1 & S_1 \\ Q_1 & T_1 \end{smallmatrix}\right] = \left[\begin{smallmatrix} a_1 & 1 \\ 1 & 0 \end{smallmatrix}\right]$. For convenience, in the right-hand side of $(\star)$,

let us define the 'denominator' $D_n = Q_n + r_n T_n$. We have

$$x = \frac{P_{n-1} + r_{n-1}S_{n-1}}{Q_{n-1} + r_{n-1}T_{n-1}} = \frac{P_{n-1}(a_n + r_n) + S_{n-1}}{Q_{n-1}(a_n + r_n) + T_{n-1}} = \frac{P_n + r_n S_n}{Q_n + r_n T_n},$$

and so

$$\begin{aligned} P_n &= a_n P_{n-1} + S_{n-1} \\ Q_n &= a_n Q_{n-1} + T_{n-1} \\ S_n &= P_{n-1} \\ T_n &= Q_{n-1} \end{aligned}$$

Using the last two equations, we can rewrite the first two as

$$\begin{aligned} P_n &= a_n P_{n-1} + P_{n-2} \\ Q_n &= a_n Q_{n-1} + Q_{n-2}. \end{aligned}$$

Since $P_2, P_3, Q_2, Q_3 \in \mathbb{N}$, it follows from these recurrence relations that $P_n, Q_n \in \mathbb{N}$ for all $n \geqslant 2$. Also,

$$x = \frac{P_n + r_n S_n}{Q_n + r_n T_n} = \frac{P_n + r_n P_{n-1}}{Q_n + r_n Q_{n-1}}.$$

Moreover,

$$\begin{aligned} D_n &= Q_n + r_n T_n = Q_n + r_n Q_{n-1} = a_n Q_{n-1} + Q_{n-2} + r_n Q_{n-1} \\ &= (a_n + r_n)Q_{n-1} + Q_{n-2} = \frac{1}{r_{n-1}} Q_{n-1} + Q_{n-2} \\ &= \frac{Q_{n-1} + r_{n-1}Q_{n-2}}{r_{n-1}} = \frac{Q_{n-1} + r_{n-1}T_{n-1}}{r_{n-1}} \\ &= \frac{D_{n-1}}{r_{n-1}}. \end{aligned} \tag{$*$}$$

As $D_1 = 1$, and since each $r_k \neq 0$ $(k \in \mathbb{N})$, we get

$$D_n = \frac{D_n}{D_{n-1}} \frac{D_{n-1}}{D_{n-2}} \cdots \frac{D_2}{D_1} D_1 = \frac{1}{r_{n-1}} \frac{1}{r_{n-2}} \cdots \frac{1}{r_1} 1 = \frac{1}{r_{n-1} r_{n-2} \cdots r_1}.$$

We have

$$\begin{aligned} M_n &= \begin{bmatrix} P_n & S_n \\ Q_n & T_n \end{bmatrix} = \begin{bmatrix} P_n & P_{n-1} \\ Q_n & Q_{n-1} \end{bmatrix} = \begin{bmatrix} a_n P_{n-1} + P_{n-2} & P_{n-1} \\ a_n Q_{n-1} + Q_{n-2} & Q_{n-1} \end{bmatrix} \\ &= \begin{bmatrix} P_{n-1} & P_{n-2} \\ Q_{n-1} & Q_{n-2} \end{bmatrix} \begin{bmatrix} a_n & 1 \\ 1 & 0 \end{bmatrix} = M_{n-1} \begin{bmatrix} a_n & 1 \\ 1 & 0 \end{bmatrix}. \end{aligned}$$

Hence $M_n = \begin{bmatrix} P_n & P_{n-1} \\ Q_n & Q_{n-1} \end{bmatrix} = \begin{bmatrix} a_1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a_2 & 1 \\ 1 & 0 \end{bmatrix} \cdots \begin{bmatrix} a_{n-1} & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a_n & 1 \\ 1 & 0 \end{bmatrix}$ for all $n \in \mathbb{N}$.

The determinant of each matrix factor $\begin{bmatrix} a_k & 1 \\ 1 & 0 \end{bmatrix}$ $(1 \leqslant k \leqslant n)$ is -1. Thus

$$P_n Q_{n-1} - P_{n-1} Q_n = \det M_n = (-1)^n. \tag{$**$}$$

Dividing throughout by $Q_n Q_{n-1}$, we also obtain that

$$\frac{P_n}{Q_n} - \frac{P_{n-1}}{Q_{n-1}} = \frac{(-1)^n}{Q_{n-1} Q_n}. \tag{$***$}$$

Finally, using $(*)$ and $(**)$, we get

$$\begin{aligned} x - \frac{P_n}{Q_n} &= \frac{P_n + r_n P_{n-1}}{Q_n + r_n Q_{n-1}} - \frac{P_n}{Q_n} \\ &= \frac{r_n(P_{n-1}Q_n - P_n Q_{n-1})}{Q_n(Q_n + r_n Q_{n-1})} \\ &= \frac{r_n(-1)^{n-1}}{Q_n D_n} \\ &= \frac{(-1)^{n-1}}{Q_n D_{n+1}}. \end{aligned} \tag{$\star\star$}$$

Theorem 9.6. *Let $x \in \mathbb{R}\backslash\mathbb{Q}$, $x>0$, and P_n, Q_n $(n \in \mathbb{N})$ as above. Then*:
(1) *For all $n \in \mathbb{N}$, $\frac{P_{2n-1}}{Q_{2n-1}} < x < \frac{P_{2n}}{Q_{2n}}$.*
(2) $\lim_{n\to\infty}(\frac{P_{n+1}}{Q_{n+1}} - \frac{P_n}{Q_n}) = 0$.
(3) $\lim_{n\to\infty} \frac{P_n}{Q_n} = x$.

Proof. (1) follows from $(\star\star)$ since $Q_k \in \mathbb{N}$ and $D_k > 0$ for all $k \in \mathbb{N}$.

For $n \geqslant 3$, we have $Q_n = a_n Q_{n-1} + Q_{n-2} \geqslant 1Q_{n-1} + 1 > Q_{n-1}$, and so the sequence $(Q_n)_{n\in\mathbb{N}}$ of natural numbers is strictly increasing for $n \geqslant 3$. Thus (2) now follows from $(***)$.

We have $D_n = Q_n + r_n Q_{n-1} \geqslant Q_n + 0 = Q_n$. Thus from $(\star\star)$, we get

$$|x - \frac{P_n}{Q_n}| = \frac{1}{Q_n D_{n+1}} \leqslant \frac{1}{Q_n Q_{n+1}} \to 0 \text{ as } n \to \infty.$$

This shows (3). □

Exercise 9.7. Let $x \in \mathbb{R}\backslash\mathbb{Q}$ be positive, and let the sequence $(r_n)_{n\in\mathbb{N}}$ of remainders described above be periodic, that is, there exist an $N \in \mathbb{N}$ such that $r_{N+1} = r_1$. Prove that x is an algebraic number[7] with a minimal polynomial of degree 2.

We now show the claimed 'best rational approximation' property.

Theorem 9.7. *Let $x \in \mathbb{R}\backslash\mathbb{Q}$, $x>0$, and P_n, Q_n $(n \in \mathbb{N})$ as above. Then*:
(1) *For all $n \in \mathbb{N}$, $|x - \frac{P_n}{Q_n}| < \frac{1}{Q_n Q_{n+1}}$.*
(2) *If $A, B \in \mathbb{Z}$, and $0 < B < Q_n$, then $|x - \frac{A}{B}| \geqslant |x - \frac{P_n}{Q_n}|$.*

Proof. As $r_{n+1} > 0$ and $Q_n \in \mathbb{N}$, we have $D_{n+1} = Q_{n+1} + r_{n+1}Q_n > Q_{n+1}$. From $(\star\star)$ on p.100, we get $|x - \frac{P_n}{Q_n}| = \frac{1}{Q_n D_{n+1}} < \frac{1}{Q_n Q_{n+1}}$. This shows (1).

[7] See p.103 for the definition of an algebraic number and its minimal polynomial.

Suppose that $A, B \in \mathbb{Z}$ are such that $0 < B < Q_n$, and $|x - \frac{A}{B}| < |x - \frac{P_n}{Q_n}|$. Then

$$|(-1)^n(x - \tfrac{A}{B})| = |x - \tfrac{A}{B}| < |x - \tfrac{P_n}{Q_n}| = \frac{1}{Q_n D_{n+1}},$$

and so

$$-\frac{1}{Q_n D_{n+1}} < (-1)^n(x - \tfrac{A}{B}) < \frac{1}{Q_n D_{n+1}}. \qquad (\star)$$

Adding $\frac{1}{Q_n D_{n+1}} = (-1)^{n-1}(x - \frac{P_n}{Q_n}) = (-1)^n(\frac{P_n}{Q_n} - x)$ throughout $(\star)$ yields

$$0 < (-1)^n(x - \tfrac{A}{B}) + (-1)^n(\tfrac{P_n}{Q_n} - x) = (-1)^n(\tfrac{P_n}{Q_n} - \tfrac{A}{B}) < \frac{2}{Q_n D_{n+1}} < \frac{2}{Q_n Q_{n+1}}.$$

Hence

$$0 < (-1)^n(BP_n - AQ_n) < \frac{2B}{Q_{n+1}}. \qquad (\star\star\star)$$

As $B < Q_n$ and $Q_n < Q_{n+1}$, the right-hand side bound is $\frac{2B}{Q_{n+1}} < \frac{2Q_n}{Q_{n+1}} < 2$. So the integer $(-1)^n(BP_n - AQ_n)$ lies strictly between 0 and 2, and hence must be equal to 1, i.e., $(-1)^n(BP_n - AQ_n) = 1$. We note that as $Q_1 = 1$, the condition that the integer B lies between 0 and Q_n implies that $n \geqslant 2$. From $(**)$ on p.99, it follows that with $A = P_{n-1}$ and $B = Q_{n-1}$, we have $0 < B < Q_n$ and $(-1)^n(BP_n - AQ_n) = (-1)^n(-1)^n = 1$. We claim no other B satisfies $0 < B < Q_n$ and $(-1)^n(BP_n - AQ_n) = 1$. By Proposition 6.1, p.68, $\gcd(P_n, Q_n) = 1$. By Exercise 6.17, p.66, for any solution to the Bezout equation $(-1)^n(BP_n - AQ_n) = 1$ we have $(-1)^n B = (-1)^n Q_{n-1} + NQ_n$ for some $N \in \mathbb{Z}$, and so $B = Q_{n-1} + \widetilde{N}Q_n$, where $\widetilde{N} = (-1)^n N \in \mathbb{Z}$. To show $B = Q_{n-1}$ we want $\widetilde{N} = 0$. Consider the following possible cases:

1° If $\widetilde{N} > 0$, then $B < Q_n$ gives $Q_{n-1} + \widetilde{N}Q_n < Q_n$.
So $1 \leqslant Q_{n-1} < (1 - \widetilde{N})Q_n \leqslant 0$, a contradiction.

2° If $\widetilde{N} < 0$, then $0 < B$ gives $0 < Q_{n-1} + \widetilde{N}Q_n$.
So $Q_{n-1} > (-\widetilde{N})Q_n \geqslant 1Q_n = Q_n$, a contradiction.

Thus $B = Q_{n-1}$. But then from $(\star\star\star)$, we get

$$1 = (-1)^n(BP_n - AQ_n) < \frac{2B}{Q_{n+1}} = \frac{2Q_{n-1}}{Q_{n+1}},$$

i.e., $Q_{n+1} < 2Q_{n-1}$. But (recalling that $n \geqslant 2$ so that $a_n \in \mathbb{N}$)

$$Q_{n+1} = a_n Q_n + Q_{n-1} \geqslant 1Q_n + Q_{n-1} \geqslant Q_{n-1} + Q_{n-1} = 2Q_{n-1},$$

a contradiction. □

9.4 Rational zeroes theorem and algebraic numbers

In Theorem 1.1, p.3, we had given a 'geometric' proof of the irrationality of $\sqrt{2}$. We now learn about a useful result, called the rational zeroes theorem, which gives a tool for showing irrationality, in particular for surds[8]. The proof of the rational zeroes theorem relies on the following fact shown in Proposition 6.2, p.69: if the integers m, n have no common factor, and m divides nk, where $k \in \mathbb{Z}$, then m must divide k. We begin by introducing some terminology related to 'polynomials'.

A *polynomial*[9] (*function*) is a map $p : \mathbb{R} \to \mathbb{R}$ that is a 'linear combination' of the power functions, that is, there exists an integer $d \geqslant 0$, and real numbers $c_0, \cdots, c_d$ such that $p(x) = c_0 + c_1 x + \cdots + c_d x^d$ for all $x \in \mathbb{R}$.
Some terminology:

- The numbers $c_0, \cdots, c_d$ are called the *coefficients of the polynomial* p. If $c_0, \cdots, c_d \in \mathbb{Z}$, then we say the *polynomial* p *has integer coefficients*. The set of all polynomials with integer coefficients is denoted by $\mathbb{Z}[x]$.
- If $c_d \neq 0$, then d is called the *degree of the polynomial* p, and c_d is called the *leading coefficient of* p. If $c_d = 1$, then p is called a *monic polynomial*.
- A real number ζ is called a (*real*) *zero of the polynomial* p if $p(\zeta) = 0$.

Theorem 9.8. (Rational zeroes theorem).
Let $d \in \mathbb{N}$. *Let* $c_0, c_1, \cdots, c_d \in \mathbb{Z}$ *be such that* $c_d \neq 0$ *and*[10] $c_0 \neq 0$.
Let m, n *be integers such that* $n \neq 0$, *and* $\gcd(m, n) = 1$.
Let $\frac{m}{n} \in \mathbb{Q}$ *be a zero of the polynomial* $p = c_0 + c_1 x + \cdots + c_d x^d \in \mathbb{Z}[x]$.
Then n *divides* c_d *and* m *divides* c_0.

Proof. We have $c_0 + c_1 \frac{m}{n} + \cdots + c_d \frac{m^d}{n^d} = 0$. Multiplying throughout by n^d,

$$c_d m^d = -(c_0 n^d + c_1 m n^{d-1} + \cdots + c_{d-1} m^{d-1} n). \qquad (*)$$

But n divides the right-hand side, and n divides $c_d m^d$. Since $\gcd(m, n) = 1$, Exercise 6.22, p.70, implies n divides c_d. Also, by rearranging $(*)$, we obtain $c_0 n^d = -(c_1 m n^{d-1} + \cdots + c_{d-1} m^{d-1} n + c_d m^d)$. As m divides the right-hand side, m must divide $c_0 n^d$. But m and n have no common factor. So m must divide c_0. □

[8] 'Surds' refer to irrational numbers which arise as the nth root of a natural number. The mathematician al-Khwarizmi (around 820 AD) called irrational numbers 'inaudible', which was later translated to the Latin *surdus* for 'mute'.

[9] If c is a constant and k is a nonnegative integer, then the function $\mathbb{R} \ni x \mapsto cx^k \in \mathbb{R}$ is called a *monomial*. The name 'polynomial' makes sense since it is a pointwise sum of *many* monomials.

[10] The proof does not use that $c_0 \neq 0$, but the theorem does not give useful information on deducing the nonzero rational zero in case $c_0 = 0$. One can always factor out a suitable power of x in the case when $c_0 = 0$, and then use the rational zeroes theorem.

Example 9.4. ($\sqrt{2} \notin \mathbb{Q}$). We now show $\sqrt{2}$ is irrational using the rational zeroes theorem. Suppose $\sqrt{2}$ is rational. Let $m, n \in \mathbb{Z}$ be such that $n > 0$, m, n have no common factor, and $\sqrt{2} = \frac{m}{n}$. Then $\frac{m}{n}$ is a rational zero of the polynomial $x^2 - 2 \in \mathbb{Z}[x]$. By the rational zeroes theorem, m divides -2 and n (> 0) divides 1. So $m \in \{2, -2, 1, -1\}$ and $n = 1$. Hence $\frac{m}{n} \in \{2, -2, 1, -1\}$. But $\sqrt{2} = \frac{m}{n}$ is not equal to any of the values $2, -2, 1, -1$ (as $(\pm 2)^2 = 4 \neq 2$ and $(\pm 1)^2 = 1 \neq 2$). This contradiction shows that $\sqrt{2} \notin \mathbb{Q}$. ◇

Exercise 9.8. Show that $\sqrt[3]{6}$ is irrational using the rational zeroes theorem.

Exercise 9.9. Show that $\sqrt{2} + \sqrt{3}$ is irrational using the rational zeroes theorem.

Exercise 9.10. Show that $(2 + \sqrt{5})^{\frac{1}{3}} - (-2 + \sqrt{5})^{\frac{1}{3}}$ is rational. What is its value?
Hint: Calling the number α, and cubing, show that $\alpha^3 + 3\alpha - 4 = 0$.
Factorise the polynomial $x^3 + 3x - 4$ assuming it has a rational root.

Algebraic and transcendental numbers.
Complex zeroes of nonzero polynomials in $\mathbb{Z}[x]$ are called *algebraic numbers*. Nonalgebraic numbers are called *transcendental*. Clearly all rational numbers are algebraic numbers, but some irrational numbers are algebraic too, for example $\sqrt{2}$ is a zero of $x^2 - 2 \in \mathbb{Z}[x]$. The number $i \in \mathbb{C}$ is algebraic since it is a zero of $x^2 + 1 \in \mathbb{Z}[x]$.

Given an algebraic number α, there are infinitely many polynomials in $\mathbb{Z}[x]$ that have α as a zero, since if $p(\alpha) = 0$, then also[11] $(pq)(\alpha) = 0$ for any polynomial $q \in \mathbb{Z}[x]$. Let $p = c_0 + c_1 x + \cdots + c_{d-1} x^{d-1} + c_d x^d \in \mathbb{Z}[x]$ be a polynomial of degree $d \in \mathbb{N}$, having the algebraic number $\alpha \in \mathbb{C}$ as a zero. Then dividing throughout by the leading coefficient c_d, we get a monic polynomial

$$\widetilde{p} = \frac{c_0}{c_d} + \frac{c_1}{c_d} x + \cdots + \frac{c_{d-1}}{c_d} x^{d-1} + x^d \in \mathbb{Q}[x]$$

with *rational* coefficients that has α as a zero. Vice versa, the complex zeroes of a monic polynomial in $\mathbb{Q}[x]$ are algebraic. A *minimal polynomial of an algebraic number* α is a monic polynomial $p \in \mathbb{Q}[x]$ of least degree that has α as a zero. The minimal polynomial of an algebraic number α is unique, since otherwise by subtracting the two minimal polynomials, the leading term x^d cancels, leaving a nonconstant polynomial of degree lesser than d, with rational coefficients, that has α as a zero, and by diving by the new leading coefficient, we can make it monic, a contradiction.

[11] Here $pq : \mathbb{R} \to \mathbb{R}$ denotes the polynomial function which is the pointwise product of the polynomials p and q: if $p(x) = a_0 + \cdots + a_m x^m$ and $q(x) = b_0 + \cdots + b_n x^n$ for all $x \in \mathbb{R}$, then $(pq)(x) = p(x)q(x) = c_0 + \cdots + c_{m+n} x^{m+n}$, where for $0 \leqslant k \leqslant m+n$, $c_k := \sum_{\ell=0}^{k} a_\ell b_{k-\ell}$.

Remark 9.1. It can be shown[12] that the set of algebraic numbers forms a field. It can be proved that e is transcendental[13]. In 1900, Hilbert listed 23 open problems, which proved to be quite influential[14] in Mathematics. The 7th one is:

Is a^b transcendental, for algebraic $a \neq 0, 1$ and irrational algebraic b?

This was settled in 1934 by Gelfond, and independently by Schneider, by showing the following, but its proof[15] is beyond the scope of the book.

Theorem 9.9. (Gelfond-Schneider theorem).
If a, b are algebraic numbers with $a \notin \{0, 1\}$, and $b \notin \mathbb{Q}$, then a^b is transcendental. ∗

Exercise 9.11. Show that $\mathbb{Z}[x]$ is a countable set.
We'll see in the next chapter that a nonzero polynomial with complex coefficients of degree d has at most d complex zeroes. We also know that an infinite set which is the countable union of finite sets is countable. Thus the set of all complex zeroes of all polynomials in $\mathbb{Z}[x]$ is countable, i.e., the set of algebraic numbers is countable. In particular, the infinite set of all algebraic numbers that are real is also countable. As the set of real numbers is uncountable (Theorem 8.3, p.86), we conclude that the set of all transcendental real numbers is uncountable.

Although there are uncountably many transcendental numbers, proving an irrational number is transcendental can be very difficult, for example, at the moment, it is not known whether the numbers $e+\pi$ and $e\pi$ are transcendental or not. However, one can concoct many transcendental numbers based on the following result, and we give such an example below in Exercise 9.12.

Theorem 9.10. (Liouville's approximation theorem).
Suppose α is an algebraic number with a minimal polynomial of degree d. Then there exists a constant $A = A(\alpha) > 0$ such that for all $a \in \mathbb{Z}$ and $b \in \mathbb{N}$, satisfying $0 < |\alpha - \frac{a}{b}| < 1$, we have $|\alpha - \frac{a}{b}| \geqslant \frac{A}{b^d}$.

Proof. Let $a \in \mathbb{Z}$ and $b \in \mathbb{N}$ be such that $0 < |\alpha - \frac{a}{b}| < 1$. Set $\beta = \frac{a}{b}$. Let $p = c_0 + c_1 x + \cdots + x^d \in \mathbb{Q}[x]$ be the minimal polynomial of α. Consider first the case when $d = 1$. Then $p = c_0 + x$. Since $p(\alpha) = 0$, $\alpha = -c_0 = \frac{a'}{b'}$ for some $a' \in \mathbb{Z}$ and $b' \in \mathbb{N}$. As $\alpha \neq \beta$, we have $ab' - a'b \neq 0$, and so $|ab' - a'b| \in \mathbb{N}$. Thus $|\alpha - \frac{a}{b}| = |\frac{a'}{b'} - \frac{a}{b}| = \frac{|ab' - a'b|}{bb'} \geqslant \frac{1}{bb'} = \frac{A}{b}$, where $A := \frac{1}{b'}$.

[12] See, e.g., [Niven (2006), Theorem 7.2] for a proof.
[13] See, e.g., [Niven (2006), Theorem 2.12] for a proof of the transcendence of e.
Accepting the fact that e is transcendental allows us to show $\log_e 2 \notin \mathbb{Q}$ (which was mentioned in the Preface, p.vi): if $(0 <) \log_e 2 = \frac{p}{q}$ for $p, q \in \mathbb{N}$, then $2^q = e^p$, so that e^p is an algebraic number, and so e is an algebraic number, a contradiction.
[14] E.g., the 8th one is the Riemann hypothesis, a famous currently unsolved problem.
[15] See, e.g., [Niven (2006), Chapter X] for a proof.

Next consider the case $d>1$. We claim that $p(\beta)\neq 0$. We have

$$\begin{aligned} p(\alpha)-p(\beta) &= (\alpha^d-\beta^d)+c_{d-1}(\alpha^{d-1}-\beta^{d-1})+\cdots+c_1(\alpha-\beta)\\ &= (\alpha-\beta)\big((\alpha^{d-1}+\alpha^{d-2}\beta+\cdots+\alpha\beta^{d-2}+\beta^{d-1})\\ &\qquad +c_{d-1}(\alpha^{d-2}+\alpha^{d-3}\beta+\cdots+\alpha\beta^{d-3}+\beta^{d-2})\\ &\qquad +\cdots+c_1(1)\big)\\ &= (\alpha-\beta)q(\alpha), \qquad (\star) \end{aligned}$$

for a suitable nonzero monic polynomial $q\in\mathbb{Q}[x]$ of degree $d-1$. If $p(\beta)=0$, then $\alpha\neq\beta$ implies $q(\alpha)=0$, a contradiction to the fact that the minimal polynomial of α has degree d.

Write each of the rational numbers $c_0,\cdots,c_{d-1}$ as the ratio of integers with nonzero denominators. Let $D\in\mathbb{Z}\backslash\{0\}$ be the product of the denominators of $c_0,\cdots,c_{d-1}$. Then $\widetilde{p}:=Dp\in\mathbb{Z}[x]$ has α as a zero. Also, since $p(\beta)\neq 0$, we have $\widetilde{p}(\beta)\neq 0$. We have

$$|\widetilde{p}(\beta)| = \underbrace{|Dc_0b^d+Dc_1ab^{d-1}+\cdots+c_{d-1}a^{d-1}b+Da^d|}_{\text{a natural number}}\frac{1}{b^d}\geqslant\frac{1}{b^d}.$$

From $(\star)$, $\widetilde{p}(\beta)=\widetilde{p}(\beta)-\widetilde{p}(\alpha)=(\beta-\alpha)Dq(\alpha)$. As $|\beta|\leqslant|\alpha|+1=:M$, we get $|q(\alpha)|\leqslant dM^{d-1}+|c_{d-1}|(d-1)M^{d-2}+\cdots+|c_1|=:M'$. Hence we obtain that $|\alpha-\frac{a}{b}|=|\widetilde{p}(\beta)|\frac{1}{|D||q(\alpha)|}\geqslant\frac{1}{b^d}\frac{1}{|D|M'}=\frac{A}{b^d}$, where $A:=\frac{1}{|D|M'}$. □

Exercise 9.12. (Liouville's number).

Consider the increasing sequence $(s_n)_{n\in\mathbb{N}}$, where $s_n=\sum_{m=1}^{n}\frac{1}{2^{m!}}$ for all $n\in\mathbb{N}$.

Since $0\leqslant s_n\leqslant\sum_{m=1}^{n}\frac{1}{2^m}=1-\frac{1}{2^m}<1$ for all $n\in\mathbb{N}$, $(s_n)_{n\in\mathbb{N}}$ is bounded.

Hence $(s_n)_{n\in\mathbb{N}}$ is convergent, with the limit $L:=\sup\{s_n:n\in\mathbb{N}\}\in\mathbb{R}$.

Show that L is transcendental.

9.5 Extended real number system $\overline{\mathbb{R}}$

In analysis (for example while discussing limiting processes), it is convenient to extend the real number system by adjoining two points, denoted by $+\infty$ (*plus infinity*) and $-\infty$ (*minus infinity*) with certain properties. We define the *extended real number system* to be the set $\overline{\mathbb{R}}=\mathbb{R}\cup\{-\infty,+\infty\}$.

Note that $-\infty\notin\mathbb{R}$, $+\infty\notin\mathbb{R}$, and $-\infty\neq+\infty$. Moreover, we extend the earlier real number addition and multiplication, as follows.

For all $x\in\mathbb{R}$,

$$\begin{aligned} x+(+\infty) &= +\infty=(+\infty)+x\\ x+(-\infty) &= -\infty=(-\infty)+x\\ (+\infty)+(+\infty) &= +\infty\\ (-\infty)+(-\infty) &= -\infty. \end{aligned}$$

We remark that $+\infty+(-\infty)$ and $(-\infty)+(+\infty)$ are left undefined.

If $x \in \mathbb{R}$ and $x > 0$, then

$$x(+\infty) = +\infty = (+\infty)x$$
$$x(-\infty) = -\infty = (-\infty)x.$$

If $x \in \mathbb{R}$ and $x < 0$, then

$$x(+\infty) = -\infty = (+\infty)x$$
$$x(-\infty) = +\infty = (-\infty)x.$$

We also define

$$(+\infty)(+\infty) = (+\infty)$$
$$(-\infty)(-\infty) = +\infty$$
$$(+\infty)(-\infty) = -\infty = (-\infty)(+\infty).$$

It is also convenient to define $\frac{x}{+\infty} = 0 = \frac{x}{-\infty}$ for all $x \in \mathbb{R}$.

We extend the order relation by defining $-\infty < +\infty$ and $\neg(+\infty < -\infty)$. Also, for every $x \in \mathbb{R}$, we define

$$-\infty < x, \quad \text{and} \quad \neg(x < -\infty),$$
$$x < +\infty, \quad \text{and} \quad \neg(+\infty < x).$$

The elements of $\mathbb{R}$ are said to be *finite*, as opposed to the symbols $+\infty$ and $-\infty$. For $a, b \in \overline{\mathbb{R}}$, we write $a \leqslant b$ if $a < b$ or $a = b$.

Exercise 9.13. Let $S \subset \overline{\mathbb{R}}$. An element $u \in \overline{\mathbb{R}}$ is an *upper bound of* S if for all $x \in S$, $x \leqslant u$. An upper bound $u_* \in \overline{\mathbb{R}}$ of S is called a *least upper bound of* S if for any upper bound u of S, $u_* \leqslant u$.

(1) Show that $-\infty$ is a least upper bound of the empty set.

(2) If S is a nonempty subset of $\mathbb{R}$ which does not have an upper bound in $\mathbb{R}$, then show that $+\infty$ is a least upper bound of S.

(3) Show that every subset of $\mathbb{R}$ has a least upper bound in $\overline{\mathbb{R}}$.

This illustrates a use of the extended real number system: one can talk about the least upper bound of subsets S of the real line without any restrictions on S.

We end this chapter with an exercise on the p-adic numbers. This is a different completion of $\mathbb{Q}$, by using an alternative way of defining the distance between two rational numbers. The p-adic numbers are useful in mathematics, and there are speculations that it might also be relevant to the real world (see, e.g., [Varadarajan (2002)]).

Exercise 9.14. (p-adic numbers).
Let p be any prime number. If $a \in \mathbb{Q}\backslash\{0\}$, then we can write it as

$$a = p^k \frac{n}{d},$$

where $k, n, d \in \mathbb{Z}$ and $\gcd(n, p) = 1 = \gcd(d, p)$. We define

$$\operatorname{ord}_p a = \begin{cases} k & \text{if } a \in \mathbb{Q}\backslash\{0\}, \\ +\infty & \text{if } a = 0, \end{cases} \quad \text{and} \quad |a|_p = p^{-\operatorname{ord}_p a} = \begin{cases} p^{-k} & \text{if } a \in \mathbb{Q}\backslash\{0\}, \\ 0 & \text{if } a = 0. \end{cases}$$

Show that if $a, b \in \mathbb{Q}$, then $|ab|_p = |a|_p |b|_p$, and $|a+b|_p \leqslant \max\{|a|_p, |b|_p\} \leqslant |a|_p + |b|_p$.

Also show that $|a|_p = 0$ if and only if $a = 0$. Thus $|\cdot|_p$ is analogous[16] to the usual absolute value, and so also can be used to define a distance function d_p between rationals: $d_p(a,b) := |a-b|_p$ for $a, b \in \mathbb{Q}$. Now two rationals will be close to each other if a big power of p 'divides' $a - b$.

Just as the field $\mathbb{R}$ completes $\mathbb{Q}$ with the usual absolute value, one can complete $\mathbb{Q}$ with respect to $|\cdot|_p$ to get a field $\mathbb{Q}_p$, as outlined below. A sequence $(a_n)_{n\in\mathbb{N}}$ in $\mathbb{Q}$ is *Cauchy with respect to* $|\cdot|_p$ if for every rational $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $n, m \in \mathbb{N}$ satisfying $n, m > N$, we have $|a_n - a_m|_p < \epsilon$. Let $\mathscr{C}_p$ be the set of all sequences that are Cauchy with respect to $|\cdot|_p$. Show that if $(a_n)_{n\in\mathbb{N}} \in \mathscr{C}_p$, then it is bounded, that is, there exists a positive rational M such that for all $n \in \mathbb{N}$, $|a_n|_p \leqslant M$. It can be shown that if $(a_n)_{n\in\mathbb{N}}, (b_n)_{n\in\mathbb{N}} \in \mathscr{C}_p$, then $(a_n + b_n)_{n\in\mathbb{N}} \in \mathscr{C}_p$. Show that also $(a_n b_n)_{n\in\mathbb{N}} \in \mathscr{C}_p$. Define the relation $\sim$ on $\mathscr{C}_p$ as follows: If $(a_n)_{n\in\mathbb{N}}, (b_n)_{n\in\mathbb{N}} \in \mathscr{C}_p$, then $(a_n)_{n\in\mathbb{N}} \sim (b_n)_{n\in\mathbb{N}}$ if for every rational $\epsilon > 0$ there exists an $N \in \mathbb{N}$ such that for all $n \in \mathbb{N}$ satisfying $n > N$, we have $|a_n - b_n|_p < \epsilon$. It can be shown that $\sim$ is an equivalence relation. Check the transitivity of $\sim$. The equivalence classes of $\mathscr{C}_p$ under $\sim$ are called *p-adic numbers*, and the set of p-adic numbers is denoted by $\mathbb{Q}_p$.

Define $+, \cdot : \mathbb{Q}_p \times \mathbb{Q}_p \to \mathbb{Q}_p$ by $\boldsymbol{a} + \boldsymbol{b} = [(a_n + b_n)_{n\in\mathbb{N}}]$ and $\boldsymbol{a} \cdot \boldsymbol{b} = [(a_n b_n)_{n\in\mathbb{N}}]$ for $\boldsymbol{a} = [(a_n)_{n\in\mathbb{N}}], \boldsymbol{b} = [(b_n)_{n\in\mathbb{N}}] \in \mathbb{Q}_p$. Then these are well-defined notions. Check this for multiplication. It can be shown that all the field axioms are satisfied by $\mathbb{Q}_p$ with these two operations. In particular, check that $[(1)_{n\in\mathbb{N}}]$ is a multiplicative identity, that every nonzero element in $\mathbb{Q}_p$ possesses a multiplicative inverse, and the distributive law.

Check that the map $a \mapsto [(a)_{n\in\mathbb{N}}] : \mathbb{Q} \to \mathbb{Q}_p$ is injective. This map also respects addition and multiplication. In this sense $\mathbb{Q}$ is contained in $\mathbb{Q}_p$, and so, besides $\mathbb{R}$, each $\mathbb{Q}_p$, p a prime, gives a different completion of $\mathbb{Q}$.

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[16] Call a map $\|\cdot\| : \mathbb{Q} \to \mathbb{Q}$ is an *absolute value on* $\mathbb{Q}$ if:
(A1) for all $a \in \mathbb{Q}$, $\|a\| \geqslant 0$.
(A2) for all $a \in \mathbb{Q}$, $\|a\| = 0$ if and only if $a = 0$.
(A3) for all $a, b \in \mathbb{Q}$, $\|ab\| = \|a\|\|b\|$.
(A4) for all $a, b \in \mathbb{Q}$, $\|a + b\| \leqslant \|a\| + \|b\|$.
An absolute value on $\mathbb{Q}$ is *non-Archimedean* if $\|a + b\| \leqslant \max\{\|a\|, \|b\|\}$.
Two absolute values on $\mathbb{Q}$ are *equivalent* if they induce the same topology on $\mathbb{Q}$.
Ostrowski's theorem states that every absolute value on $\mathbb{Q}$ is equivalent to either the usual absolute value (p.39) or one of the p-adic absolute values. See [Ostrowski (1916)].

Chapter 10

Complex numbers

A *complex number* is an ordered pair of real numbers. The set $\mathbb{R} \times \mathbb{R}$ of all complex numbers is denoted by $\mathbb{C}$. So $\mathbb{C} = \{\boldsymbol{z} = (x, y) : x \in \mathbb{R} \text{ and } y \in \mathbb{R}\}$. For a complex number $\boldsymbol{z} = (x, y) \in \mathbb{C}$, where $x, y \in \mathbb{R}$, the real number x is called the *real part of* $\boldsymbol{z}$, and y is called the *imaginary part of* $\boldsymbol{z}$.

10.1 Addition and multiplication

We define the operations of *addition* and *multiplication* on $\mathbb{C}$ by:

$$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2),$$
$$(x_1, y_1) \cdot (x_2, y_2) = (x_1 x_2 - y_1 y_2, x_1 y_2 + x_2 y_1),$$

for complex numbers (x_1, y_1), (x_2, y_2). With these operations, $\mathbb{C}$ is a field. In particular, $+ : \mathbb{C} \times \mathbb{C} \to \mathbb{C}$ satisfies the following properties:

(F1) (Associativity) For all $\boldsymbol{z}_1, \boldsymbol{z}_2, \boldsymbol{z}_3 \in \mathbb{C}$, $(\boldsymbol{z}_1 + \boldsymbol{z}_2) + \boldsymbol{z}_3 = \boldsymbol{z}_1 + (\boldsymbol{z}_2 + \boldsymbol{z}_3)$.

(F2) (Additive identity) Let $\mathbf{0} = (0, 0) \in \mathbb{C}$. For all $\boldsymbol{z} \in \mathbb{C}$, $\boldsymbol{z} + \mathbf{0} = \boldsymbol{z} = \mathbf{0} + \boldsymbol{z}$.

(F3) (Additive inverse) For all $\boldsymbol{z} = (x, y) \in \mathbb{C}$, with $-\boldsymbol{z} := (-x, -y) \in \mathbb{C}$, we have $\boldsymbol{z} + (-\boldsymbol{z}) = \mathbf{0} = -\boldsymbol{z} + \boldsymbol{z}$.

(F4) (Commutativity) For all $\boldsymbol{z}_1, \boldsymbol{z_2} \in \mathbb{C}$, $\boldsymbol{z}_1 + \boldsymbol{z}_2 = \boldsymbol{z}_2 + \boldsymbol{z}_1$.

Also, $\cdot : \mathbb{C} \times \mathbb{C} \to \mathbb{C}$ satisfies the following properties:

(F5) (Associativity) For all $\boldsymbol{z}_1, \boldsymbol{z}_2, \boldsymbol{z}_3 \in \mathbb{C}$, $(\boldsymbol{z}_1 \cdot \boldsymbol{z}_2) \cdot \boldsymbol{z}_3 = \boldsymbol{z}_1 \cdot (\boldsymbol{z}_2 \cdot \boldsymbol{z}_3)$.

(F6) (Multiplicative identity) Let $\mathbf{1} = (1, 0) \in \mathbb{C}$. For all $\boldsymbol{z} \in \mathbb{C}$, $\boldsymbol{z} \cdot \mathbf{1} = \boldsymbol{z} = \mathbf{1} \cdot \boldsymbol{z}$.

(F7) (Multiplicative inverse) Let $\boldsymbol{z} = (x, y) \in \mathbb{C} \backslash \{\mathbf{0}\}$. Then $x^2 + y^2 \neq 0$. With $\boldsymbol{z}^{-1} := (\frac{x}{x^2 + y^2}, -\frac{y}{x^2 + y^2}) \in \mathbb{C}$, we have $\boldsymbol{z} \cdot (\boldsymbol{z}^{-1}) = \mathbf{1} = \boldsymbol{z}^{-1} \cdot \boldsymbol{z}$.

(F8) (Commutativity) For all $\boldsymbol{z}_1, \boldsymbol{z_2} \in \mathbb{C}$, $\boldsymbol{z}_1 \cdot \boldsymbol{z}_2 = \boldsymbol{z}_2 \cdot \boldsymbol{z}_1$.

Finally the two operations obey the distributive law:

(F9) (Distributivity) For all $\boldsymbol{z}_1, \boldsymbol{z}_2, \boldsymbol{w} \in \mathbb{C}$, $\boldsymbol{w} \cdot (\boldsymbol{z}_1 + \boldsymbol{z}_2) = \boldsymbol{w} \cdot \boldsymbol{z}_1 + \boldsymbol{w} \cdot \boldsymbol{z}_2$.

Exercise 10.1. Check (F7).

For $\boldsymbol{z} \in \mathbb{C} \backslash \{0\}$, we often denote its inverse $\boldsymbol{z}^{-1}$ by $\frac{\mathbf{1}}{\boldsymbol{z}}$, and if $\boldsymbol{w} \in \mathbb{C}$, then $\frac{\boldsymbol{w}}{\boldsymbol{z}} := \boldsymbol{w} \cdot \frac{\mathbf{1}}{\boldsymbol{z}} = \frac{\mathbf{1}}{\boldsymbol{z}} \cdot \boldsymbol{w}$.

$\mathbb{R}$ is 'contained' in $\mathbb{C}$. We can embed $\mathbb{R}$ inside $\mathbb{C}$, and view $\mathbb{R}$ as a 'subfield' of $\mathbb{C}$, that is, the map $x \mapsto \boldsymbol{x} := (x, 0)$ sending the real number x to the complex number $\boldsymbol{x}$ is an injective field homomorphism. This means that the operations of addition and multiplication are preserved by this map, and distinct real numbers are sent to distinct complex numbers.

$\mathbb{R}$		$\mathbb{C}$
x	$\mapsto$	$(x, 0)$
$x_1 + x_2$	$\mapsto$	$(x_1 + x_2, 0) = (x_1, 0) + (x_2, 0)$
$x_1 \cdot x_2$	$\mapsto$	$(x_1 \cdot x_2, 0) = (x_1, 0) \cdot (x_2, 0)$
1	$\mapsto$	$(1, 0)$
0	$\mapsto$	$(0, 0)$

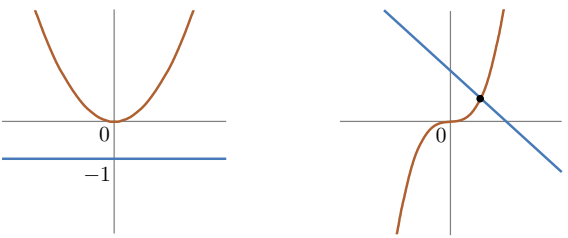
So we can view real numbers as complex numbers, via this identification. For example, the real number $\sqrt{2}$ can be viewed as the complex number $(\sqrt{2}, 0)$. If this makes one uneasy, one should note that we have been doing such identifications in the previous chapters too, where for instance, we identified integers with rational numbers, for example, $\mathbb{Z} \ni 3 = \frac{3}{1} \in \mathbb{Q}$.

Let $\boldsymbol{i} = (0, 1) \in \mathbb{C}$. If $\boldsymbol{z} = (x, y) \in \mathbb{C}$, then with $\boldsymbol{x} = (x, 0)$ and $\boldsymbol{y} = (0, y)$, we have $\boldsymbol{z} = (x, y) = (x, 0) + (0, y) = (x, 0) + (y, 0) \cdot (0, 1) = \boldsymbol{x} + \boldsymbol{y} \cdot \boldsymbol{i} = \boldsymbol{x} + \boldsymbol{i} \cdot \boldsymbol{y}$. An advantage of working with $\mathbb{C}$ is that while in $\mathbb{R}$ there is no solution $x \in \mathbb{R}$ to the equation $x^2 + 1 = 0$, there is a complex number satisfying this equation by viewing 0 and 1 as complex numbers. Indeed, if $\boldsymbol{x} := \boldsymbol{i} = (0, 1) \in \mathbb{C}$, then $\boldsymbol{x}^2 + \mathbf{1} = (0, 1) \cdot (0, 1) + (1, 0) = (-1, 0) + (1, 0) = (0, 0) = \mathbf{0}$, where $\mathbf{1} = (1, 0)$, $\mathbf{0} = (0, 0)$ and $\boldsymbol{x}^2 = \boldsymbol{x} \cdot \boldsymbol{x}$. Similarly, with $\boldsymbol{x} = -\boldsymbol{i} = (0, -1)$, $\boldsymbol{x}^2 + \mathbf{1} = \mathbf{0}$ again.

Exercise 10.2. (Ordered fields; $\mathbb{C}$ is not ordered). A field $\mathbb{F}$ is *ordered* if there is a subset $P \subset \mathbb{F}$, called the *set of positive elements of* $\mathbb{F}$, satisfying the following:
(P1) For all $x, y \in P$, $x + y \in P$.
(P2) For all $x, y \in P$, $x \cdot y \in P$.
(P3) For each $x \in \mathbb{F}$, exactly one of the following is true:
1° $x = 0$. 2° $x \in P$. 3° $-x \in P$.
For example, the field $\mathbb{R}$ is ordered: $P = \{x \in \mathbb{R} : 0 < x\}$ satisfies (P1), (P2), (P3).
(1) Given an ordered field $\mathbb{F}$, define the 'order relation' $<$ on $\mathbb{F}$ as follows: for $x, y \in \mathbb{F}$, $x < y$ if $y + (-x) \in P$. Show that the following hold:
(O1) If $x, y \in \mathbb{F}$ satisfy $x < y$, then for all $z \in \mathbb{F}$, $x + z < y + z$.
(O2) If $x, y, z \in \mathbb{F}$ satisfy $x < y$ and $0 < z$, then $x \cdot z < y \cdot z$.
(O3) If $x, y, z \in \mathbb{F}$ satisfy $x < y$ and $y < z$, then $x < z$.
(O4) For $x, y \in \mathbb{F}$, exactly one of the following hold:
1° $x = y$. 2° $y < x$. 3° $x < y$.
(2) Vice versa, show that if $<$ is a relation on $\mathbb{F}$ satisfying (O1), (O2), (O3), (O4), then $P := \{x \in \mathbb{F} : 0 < x\}$ satisfies (P1), (P2), (P3).
(3) Show that $\mathbb{C}$ is not an ordered field. *Hint:* Consider $\boldsymbol{x} := \boldsymbol{i}$ and $\boldsymbol{x} \cdot \boldsymbol{x}$.

Historical development of complex numbers.

Contrary to popular belief, historically, it was not the need for solving *quadratic* equations, but it was *cubic* equations, that led mathematicians to take complex numbers seriously. Around the 16th century, solving equations like $x^2 + bx + c = 0$ was viewed *geometrically* as the problem of finding the intersection point of the parabola $y = x^2$ with the line $y = -bx - c$. Based on this geometric interpretation, it was easy to dismiss the lack of solvability in reals of a quadratic such as $x^2 + 1 = 0$, since that just reflected the geometric fact that parabola $y = x^2$ did not meet the line $y = -1$. See the left image in the following picture.



Cardano[1] gave a formula for solving the cubic equation $x^3 = 3px + 2q$:

$$x = \sqrt[3]{q + \sqrt{q^2 - p^3}} + \sqrt[3]{q - \sqrt{q^2 - p^3}}.$$

For example, one can check that for the equation $x^3 = 6x + 6$, with $p = 2$ and $q = 3$, Cardano's formula yields a solution $x = \sqrt[3]{4} + \sqrt[3]{2}$.

Using the intermediate value theorem[2], it can be shown that the cubic $y = x^3$ *always* intersects the line $y = 3px + 2q$. See the picture on the right above. For $x^3 = 15x + 4$, when $p = 5$ and $q = 2$, we get $q^2 - p^3 = -121 < 0$, and so Cardano's formula fails with real arithmetic. But we do have a real root, namely $x = 4$, since $4^3 = 64 = 60 + 4 = 15(4) + 4$. Three decades after the appearance of Cardano's work, Bombelli suggested that with the use of *complex* arithmetic, Cardano's formula gives the desired real root. So we ask if

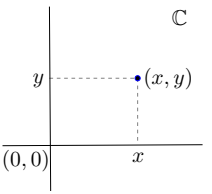
$$\boldsymbol{x = \sqrt[3]{2 + 11i} + \sqrt[3]{2 - 11i} \stackrel{?}{=} 4}.$$

As $\boldsymbol{(2 + i)^3 = 2 + 11i}$ and $\boldsymbol{(2 - i)^3 = 2 - 11i}$, the above does work, with appropriate choices of cube roots. Thus Bombelli's work established that even for *real* problems, complex arithmetic can be relevant. From then on, complex numbers entered mainstream mathematics.

[1] A 16th century Italian mathematician.
[2] See, e.g. [Sasane (2015), Theorem 3.4].

10.2 Geometric representation of complex numbers

Since $\mathbb{C} = \mathbb{R} \times \mathbb{R}$, we can identify complex numbers with points in the plane.

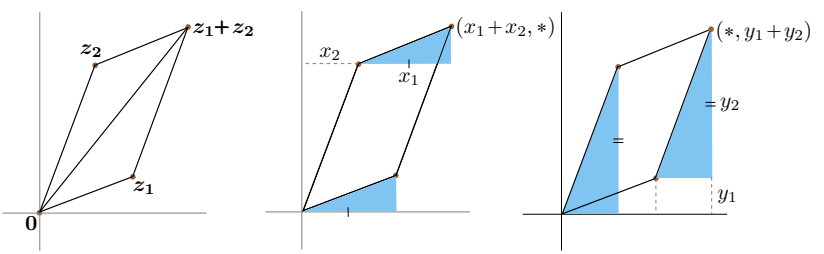


Exercise 10.3. Depict $\mathbf{0}$, $\mathbf{1}$, $-\frac{\mathbf{3}}{\mathbf{2}}$, $\boldsymbol{i}$, $-\sqrt{\mathbf{2}}\boldsymbol{i}$, $\mathbf{1} - \boldsymbol{i}$ in the complex plane.

Hence we can identify $\mathbb{C}$ as a *set* with the set of points in the plane $\mathbb{R} \times \mathbb{R}$. Do the field operations in $\mathbb{C}$ have some geometric meaning in the plane? We begin with addition in $\mathbb{C}$, and show that it corresponds to 'vector addition' in the plane.

Geometric meaning of complex addition.

With the identification of complex numbers with points in the plane, complex addition corresponds to addition of vectors in $\mathbb{R}^2$. By addition of vectors, we mean the usual way of combining two vectors, that is, by completing the parallelogram formed by the line segments joining $(0, 0)$ to each of the two complex numbers as sides, and then taking the endpoint of the diagonal from $(0, 0)$ as the sum of the two given complex numbers.

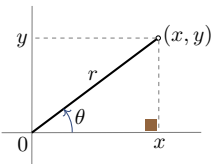


The middle picture shows that addition of $\boldsymbol{z_1}$ and $\boldsymbol{z_2}$ as vectors in the plane yields the correct x-coordinate of their sum as complex numbers by looking at the two congruent shaded triangles. Similarly, from the rightmost picture, we see that the y-coordinate is also correct.

Geometric meaning of complex multiplication.

We will now see the special geometric meaning of complex multiplication. In order to do this, it will be convenient to use polar coordinates. Let the point $(x, y) \in \mathbb{R}^2 \backslash \{(0, 0)\}$ have polar coordinates $r > 0$ and $\theta \in (-\pi, \pi]$. This means that the Euclidean distance of (x, y) to $(0, 0)$ is r (> 0), and the ray

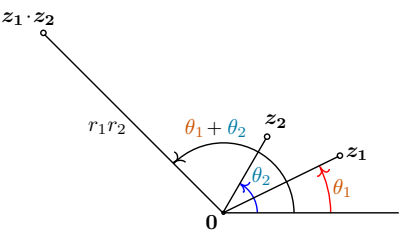
joining $(0,0)$ to (x,y) makes an angle[3] $\theta \in (-\pi, \pi]$ with the positive real axis. For $(x,y) = (0,0)$, $r := 0$ and $\theta := 0$. We have $(x,y) = (r\cos\theta, r\sin\theta)$.



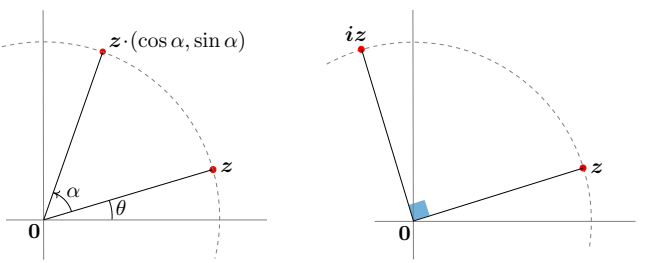
If $\boldsymbol{z_1} = (r_1\cos\theta_1, r_1\sin\theta_1)$ and $\boldsymbol{z_2} = (r_2\cos\theta_2, r_2\sin\theta_2)$, then

$$\begin{aligned}\boldsymbol{z_1}\cdot\boldsymbol{z_2} &= (r_1\cos\theta_1, r_1\sin\theta_1)\cdot(r_2\cos\theta_2, r_2\sin\theta_2)\\ &= (r_1r_2(\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2), r_1r_2(\cos\theta_1\sin\theta_2 + \cos\theta_2\sin\theta_1))\\ &= (r_1r_2\cos(\theta_1+\theta_2), r_1r_2\sin(\theta_1+\theta_2)),\end{aligned}$$

using the trigonometric identities for angle addition. So the angles that $\boldsymbol{z_1}$ and $\boldsymbol{z_2}$ make with the positive real axis are *added* in order to get the angle[4] $\boldsymbol{z_1}\cdot\boldsymbol{z_2}$ makes with the positive real axis, and their distances to the origin are *multiplied* to get the distance of $\boldsymbol{z_1}\cdot\boldsymbol{z_2}$ to the origin.



As a special case, suppose $\alpha \in (-\pi, \pi]$, and consider multiplication by the complex number $(\cos\alpha, \sin\alpha)$, which is at a distance of 1 from the origin. If $\boldsymbol{z} \in \mathbb{C}$, then $\boldsymbol{z}\cdot(\cos\alpha, \sin\alpha)$ is obtained by rotating the line joining $\mathbf{0}$ to $\boldsymbol{z}$ anticlockwise through an angle of α. In particular, multiplying $\boldsymbol{z}$ by $\boldsymbol{i} = (0,1) = (\cos\frac{\pi}{2}, \sin\frac{\pi}{2})$ produces a counterclockwise rotation of 90°.



[3] For a precise definition of the angle made with the positive real axis, see, e.g., [Sasane (2015), Chapter 6].

[4] One may need to subtract 2π from $\theta_1 + \theta_2$ in order to view $\theta_1 + \theta_2$ as the angle made with the positive real axis. This does not change $\boldsymbol{z_1}\cdot\boldsymbol{z_2}$ since cos and sin are 2π-periodic.

De Moivre's formula and n^{th} roots.

de Moivre's formula states $(\cos\theta, \sin\theta)^n = (\cos(n\theta), \sin(n\theta))$ for all $n \in \mathbb{N}$, and $\theta \in \mathbb{R}$, and is easily shown by using induction on n.

Exercise 10.4. Recover $\cos(3\theta) = 4(\cos\theta)^3 - 3\cos\theta$ using de Moivre's formula.

Exercise 10.5. Without expanding, express $(\mathbf{1}+\boldsymbol{i})^{10}$ in the form $\boldsymbol{x}+\boldsymbol{iy}$, $x, y \in \mathbb{R}$.

Exercise 10.6. Considering $(\mathbf{2}+\boldsymbol{i})\cdot(\mathbf{3}+\boldsymbol{i})$, show that $\frac{\pi}{4} = \tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3}$.

Exercise 10.7.
(1) Show that $\sin\frac{\pi}{5}$ is a real zero of $16x^4 - 20x^2 + 5$.
Conclude that $2\sin\frac{\pi}{5}$ is a real zero of the polynomial $p := x^4 - 5x^2 + 5$.
(2) Prove that $2\sin\frac{\pi}{5}$ is irrational using the rational zeroes theorem.
(3) Show that if a regular pentagon is inscribed in a circle with radius 1, then its side has an irrational length.

Exercise 10.8. *Gaussian integers* are complex numbers of the form (m, n), where $m, n \in \mathbb{Z}$. Thus they are integral lattice points in the complex plane. Show that it is impossible to draw an equilateral triangle such that all vertices are Gaussian integers. *Hint:* Rotation of one of the sides should give the other. Use the fact that $\sqrt{3} \notin \mathbb{Q}$ (seen easily e.g. using the rational zeroes theorem).

We can find the nth roots of a complex number using de Moivre's formula.

Proposition 10.1. *Let* $\boldsymbol{z} = (r\cos\theta, r\sin\theta)$, $r > 0$, $\theta \in (-\pi, \pi]$, $n \in \mathbb{N}$.
Then $\boldsymbol{w} \in \mathbb{C}$ *satisfies* $\boldsymbol{w}^n = \boldsymbol{z}$ *if and only if*

$$\boldsymbol{w} \in \{(\sqrt[n]{r}\cos(\tfrac{\theta}{n} + k\tfrac{2\pi}{n}),\ \sqrt[n]{r}\sin(\tfrac{\theta}{n} + k\tfrac{2\pi}{n})) : k = 0, 1, \cdots, n-1\}.$$

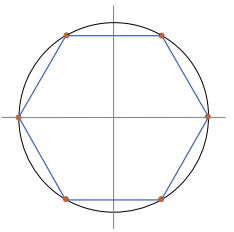
Proof.
(If:) If $\boldsymbol{w} \in \{(\sqrt[n]{r}\cos(\frac{\theta}{n} + k\frac{2\pi}{n}), \sqrt[n]{r}\sin(\frac{\theta}{n} + k\frac{2\pi}{n})) : k = 0, \cdots, n-1\}$, then

$$\begin{aligned}\boldsymbol{w}^n &= ((\sqrt[n]{r})^n\cos(n(\tfrac{\theta}{n} + k\tfrac{2\pi}{n})), (\sqrt[n]{r})^n\sin(n(\tfrac{\theta}{n} + k\tfrac{2\pi}{n})))\\ &= (r\cos(\theta + 2\pi k), r\sin(\theta + 2\pi k)) = \boldsymbol{z}\end{aligned}$$

(Only if:) Let $\boldsymbol{w} = (\rho\cos\alpha, \rho\sin\alpha)$, where $\rho \geqslant 0$ and $\alpha \in \mathbb{R}$. As $\boldsymbol{w}^n = \boldsymbol{z}$, we have $(\rho^n\cos(n\alpha), \rho^n\sin(n\alpha)) = (r\cos\theta, r\sin\theta)$. Equating the distance to $\mathbf{0}$, $\rho^n = r$, i.e., $\rho = \sqrt[n]{r}$ (as $\rho, r \geqslant 0$). So $\cos(n\alpha) = \cos\theta$ and $\sin(n\alpha) = \sin\theta$. For some $k \in \mathbb{Z}$, we have $n\alpha - 2\pi k \in (-\pi, \pi]$, and so[5] $n\alpha - 2\pi k = \theta$, i.e., $\alpha \in \{\frac{\theta}{n} + k\frac{2\pi}{n} : k \in \mathbb{Z}\}$. For $\alpha \in \{\frac{\theta}{n}, \frac{\theta}{n} + \frac{2\pi}{n}, \frac{\theta}{n} + 2\frac{2\pi}{n}, \cdots, \frac{\theta}{n} + (n-1)\frac{2\pi}{n}\}$, we get distinct $\boldsymbol{w}$. (Indeed, for any integer k, the division algorithm gives $q, r \in \mathbb{Z}$ such that $k = nq + r$ and $0 \leqslant r < n$, so that $\frac{2\pi}{n}k = \frac{2\pi}{n}r + 2\pi q$.) □

[5] By looking at the graphs of cos and sin in the interval $(-\pi, \pi]$, it is easy to see that if for angles $\beta_1, \beta_2 \in (-\pi, \pi]$ we have $\sin\beta_1 = \sin\beta_2$ and $\cos\beta_1 = \cos\beta_2$, then $\beta_1 = \beta_2$.

In particular, if $\boldsymbol{z}=1$, we get the nth roots of unity, which are located at the vertices of an n-sided regular polygon inscribed in a circle. For example, the following picture depicts the six 6th roots of unity.



Exercise 10.9. Find $\boldsymbol{w}\in\mathbb{C}$ such that $\boldsymbol{w}^4=-\mathbf{1}$. Depict them in the complex plane.

Exercise 10.10. If $a,b,c\in\mathbb{R}$ satisfy $a^2+b^2+c^2=ab+bc+ca$, then $a=b=c$: doubling and rearranging gives $(a-b)^2+(b-c)^2+(c-a)^2=0$, and as each summand is nonnegative, each must then be 0. Show that if $\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\in\mathbb{C}$ are such that $\boldsymbol{a}^2+\boldsymbol{b}^2+\boldsymbol{c}^2=\boldsymbol{a}\cdot\boldsymbol{b}+\boldsymbol{b}\cdot\boldsymbol{c}+\boldsymbol{c}\cdot\boldsymbol{a}$, then they must lie on the vertices of an equilateral triangle in the complex plane. Explain the real result in light of this. *Hint:* Compute $((\boldsymbol{b}-\boldsymbol{a})\cdot\omega+(\boldsymbol{b}-\boldsymbol{c}))\cdot((\boldsymbol{b}-\boldsymbol{a})\cdot\boldsymbol{\omega}^2+(\boldsymbol{b}-\boldsymbol{c}))$, where $\boldsymbol{\omega}$ is a nonreal cube root of unity.

From now onwards, for notational ease, we use ordinary letters instead of boldface letters for complex numbers, and denote the complex multiplication $a\cdot b$ of $a,b\in\mathbb{C}$ simply by ab.

A (*complex*) *polynomial* is a map $p:\mathbb{C}\to\mathbb{C}$ such that there exists an integer $d\geqslant 0$, and complex numbers $c_0,\cdots,c_d$ such that for all $z\in\mathbb{C}$, we have $p(z)=c_0+c_1z+\cdots+c_dz^d$. The numbers $c_0,\cdots,c_d$ are called the *coefficients of* p. If $c_d\neq 0$, then d is called the *degree of* p, and c_d is called the *leading coefficient of* p. We define the degree of the zero polynomial to be 0. A complex number ζ is a *zero of* p if $p(\zeta)=0$. The set of all complex polynomials is denoted by $\mathbb{C}[z]$.

Exercise 10.11. Let $p\in\mathbb{C}[z]$ have degree $d\in\mathbb{N}$, and $z_0\in\mathbb{C}$. Show that there exists a $q\in\mathbb{C}[z]$ of degree $d-1$ such that for all $z\in\mathbb{C}$, $p(z)=q(z)(z-z_0)+p(z_0)$. In particular, if $p(z_0)=0$, then $p(z)=q(z)(z-z_0)$. Conclude that if $n\in\mathbb{N}$, and $z_1,\cdots,z_n$ are the n distinct n^{th} roots of unity, then $z^n-1=(z-z_1)\cdots(z-z_n)$.

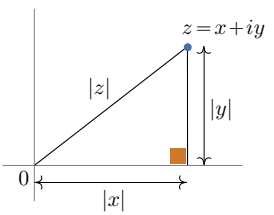
Exercise 10.12. Show that if $d\in\mathbb{N}\cup\{0\}$, and $c_0,\cdots,c_d\in\mathbb{C}$ are such that for all $z\in\mathbb{C}$, we have $c_0+c_1z+\cdots+c_dz^d=0$, then $c_0=\cdots=c_d=0$.

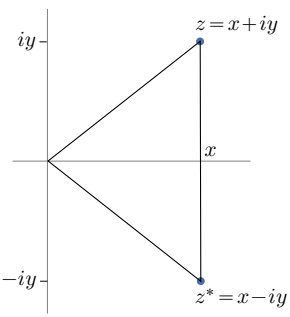
Exercise 10.13. (Division algorithm for polynomials). Let h be a polynomial of degree $d\in\mathbb{N}$. Show that for any polynomial p, there exist unique polynomials q (the quotient) and r (the remainder) such that for all $z\in\mathbb{C}$, $p(z)=q(z)h(z)+r(z)$, where the degree of r is strictly less than d.

Absolute value and complex conjugate.

The (*complex*) *absolute value* or *modulus* $|z|$ of a complex number $z = x + iy$, where $x, y \in \mathbb{R}$, is defined by $|z| = \sqrt{x^2 + y^2}$. By Pythagoras's theorem, $|z|$ is the distance of the complex number z to 0 in the complex plane. See the picture on the left below. By expressing $z_1, z_2 \in \mathbb{C}$ in terms of polar coordinates, or by a direct calculation, it can be verified that $|z_1 z_2| = |z_1||z_2|$. If $z \neq 0$, then $1 = |1| = |z\frac{1}{z}| = |z||\frac{1}{z}|$ and so $|\frac{1}{z}| = \frac{1}{|z|}$, and for $w \in \mathbb{C}$, $|\frac{w}{z}| = \frac{|w|}{|z|}$.

Exercise 10.14. Verify the identity $|z_1 z_2| = |z_1||z_2|$ by expressing $z_1, z_2 \in \mathbb{C}$ using Cartesian coordinates.





The *complex conjugate* z^* of $z = x + iy$ where $x, y \in \mathbb{R}$, is $z^* := x - iy$. In the complex plane, z^* is the reflection in the real axis of z. With this geometric interpretation, it can be seen that for all $z_1, z_2 \in \mathbb{C}$, $(z_1 + z_2)^* = z_1^* + z_2^*$ and $(z_1 z_2)^* = z_1^* z_2^*$. Also, it can be checked that the following hold:

$$(z^*)^* = z, \quad zz^* = |z|^2, \quad \operatorname{Re} z = \frac{z + z^*}{2}, \quad \operatorname{Im} z = \frac{z - z^*}{2i}.$$

Exercise 10.15. Verify that the four equalities above.

Exercise 10.16. Prove that for all $z \in \mathbb{C}$, $|z| = |z^*|$, $|\operatorname{Re} z| \leqslant |z|$, and $|\operatorname{Im} z| \leqslant |z|$. Give geometric interpretations of each.

Exercise 10.17. Let $z, w \in \mathbb{C}$ be distinct complex numbers. Show that the 'convex combination of z, w', namely, $L := \{(1-t)z + tw : t \in [0,1]\}$ is the line segment joining z and w. *Hint:* $(1-t)z + tw = z + t(w - z)$.

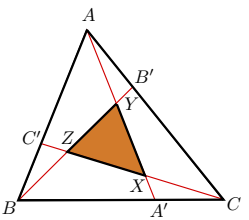
Exercise 10.18. Let p be a polynomial of degree $d \in \mathbb{N}$ having real coefficients, i.e., there exist $c_0, c_1, \cdots, c_d \in \mathbb{R}$, such that for all $z \in \mathbb{C}$, $p(z) = c_0 + c_1 z + \cdots + c_d z^d$. Show that if $w \in \mathbb{C}$ is such that $p(w) = 0$, then $p(w^*) = 0$.

Exercise 10.19. (Area of triangle).
Let $a, b \in \mathbb{C}$ such that $a, b, 0$ do not lie on a straight line in the complex plane. Show that the area of the triangle with vertices at $0, a, b$ is equal to $|\frac{\operatorname{Im}(ab^*)}{2}|$.

Exercise 10.20. On a visit to Cornell University, Richard Feynman gave a colloquium on the history of the Feynman-Kac formula. After the talk, Kai Lai Chung entertained Feynman to dinner, at which much of the discussion centred on the following result.

> **Theorem.** For a triangle in the plane, if each vertex is joined to the point one-third along the opposite side (see p.117), then the area of the inner triangle formed by these lines is exactly one-seventh of the area of the initial triangle.

Feynman couldn't believe that the ratio of the areas of the triangles was $\frac{1}{7}$ as this had nothing to do with the number 3. He spent most of the evening trying to disprove it, but finally proved it in the case when the triangle was equilateral.



Use the result of the previous exercise to prove it in general.

Exercise 10.21. Let $z, w \in \mathbb{C}\backslash\{0\}$. Show that $|z+w| = |\frac{|w|}{|z|}z + \frac{|z|}{|w|}w|$. What is the geometric interpretation?

Exercise 10.22. Show that $S^1 = \{z \in \mathbb{C} : |z| = 1\}$ is an Abelian group[6] with respect to complex multiplication. For any $n \in \mathbb{N}$, show that S^1 has a subgroup[7] with n distinct elements. Does S^1 have an infinite subgroup which is countable?

Exercise 10.23. Let $n \in \mathbb{N}$ be such that $n \geqslant 2$, and $z_1, \cdots, z_n$ be the n distinct n^{th} roots of unity. Prove that $z_1 + \cdots + z_n = 0$.

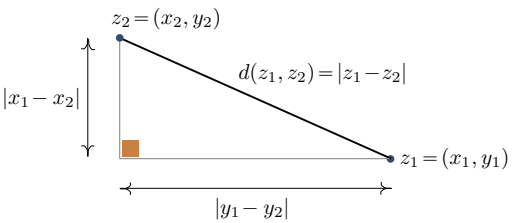
10.3 Topology of $\mathbb{C}$

The concepts in ordinary calculus in the setting of $\mathbb{R}$, like convergence of sequences, or continuity and differentiability of functions, all rely on the notion of closeness of points in $\mathbb{R}$. For example, when we talk about the convergence of a real sequence $(a_n)_{n\in\mathbb{N}}$ to its limit $L \in \mathbb{R}$, we mean that

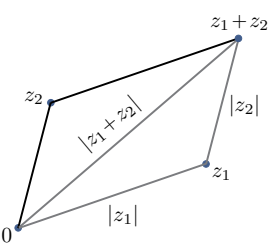
[6] A *group* is a set G together with a map $\cdot : G \times G \to G$ (called the *group operation*) such that the following properties hold:
(G1) (Associativity) For all $g_1, g_2, g_3 \in G$, $g_1 \cdot (g_2 \cdot g_3) = (g_1 \cdot g_2) \cdot g_3$.
(G2) (Identity) There exists an element $e \in G$ such that for all $g \in G$, $g \cdot e = g = e \cdot g$.
(G3) (Inverses) For all $g \in G$, there exists an element $g^{-1} \in G$ such that $g \cdot g^{-1} = e = g^{-1} \cdot g$.
If in addition, for all $g_1, g_2 \in G$, we have $g_1 \cdot g_2 = g_2 \cdot g_1$, then the group is called *Abelian*.

[7] A *subgroup* H of a group G is a subset of G which is itself a group with the restriction to H of the group operation on G. It can be checked that if H is a nonempty subset of G such that for all $h_1, h_2 \in H$ we have $h_1 \cdot h_2^{-1} \in H$, then H is a subgroup of G.

given any positive ϵ, there is a large enough index N such that beyond that index, the corresponding terms a_n all have a *distance* to L which is at most ϵ. This 'distance of a_n to L' is taken as $|a_n - L|$, and this is the length of the line segment joining the numbers a_n and L on the real number line. In order to do calculus with *complex* numbers, we need a notion of distance $d(z_1, z_2)$ between complex numbers z_1 and z_2. As $\mathbb{C}$ is $\mathbb{R}^2$, we use the usual Euclidean distance in the plane, i.e., the length of the straight line segment joining the two complex numbers. Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ be two complex numbers. Then by Pythagoras's theorem the length of the line segment joining the points $(x_1, y_1), (x_2, y_2)$ in $\mathbb{R}^2$ is given by $d(z_1, z_2) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = |z_1 - z_2|$.



Using the geometric meaning of addition of complex numbers, and fact (from Euclidean geometry) that the sum of the lengths of any two sides of a triangle is at least as big as the length of the third side, we obtain the *triangle inequality* for the absolute value: $|z_1 + z_2| \leqslant |z_1| + |z_2|$ for $z_1, z_2 \in \mathbb{C}$. The triangle inequality can alternatively be verified analytically as follows: $|z_1 + z_2|^2 = (z_1 + z_2)(z_1^* + z_2^*) = |z_1|^2 + |z_2|^2 + 2\mathrm{Re}(z_1 z_2^*) \leqslant |z_1|^2 + |z_2|^2 + 2|z_1||z_2|$.



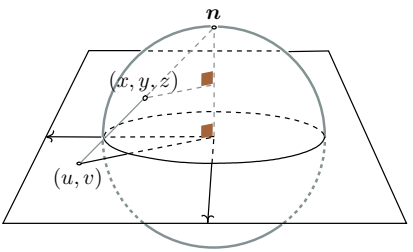
Exercise 10.24. Show that for all $z_1, z_2 \in \mathbb{C}$, $|z_1 - z_2| \geqslant ||z_1| - |z_2||$.

Exercise 10.25. Let $a \in \mathbb{C}$ and $r > 0$. Show that $\{z \in \mathbb{C} : |z|^2 - 2\mathrm{Re}(a^* z) + |a|^2 = r^2\}$ is a circle centred at a with radius r.

Exercise 10.26. Show that for all $z_1, z_2 \in \mathbb{C}$, $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$. What is the geometric interpretation of this equality?

Exercise 10.27. (Cosine formula). Let $z, w \in \mathbb{C}$ be distinct and nonzero. Show that if A, O, B correspond to $z, 0, w$, then $\cos \angle AOB = \frac{AO^2+BO^2-AB^2}{2(AO)(BO)}$.

Exercise 10.28. (Riemann sphere). Let $S^2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$ be the unit sphere in $\mathbb{R}^3$, and denote the 'north pole' by $\boldsymbol{n} = (0, 0, 1)$. For each $\boldsymbol{p} \in S^2 \backslash \{\boldsymbol{n}\}$, define the 'stereographic' projection of $\boldsymbol{p} = (x, y, z)$ to be the point $\varphi(\boldsymbol{p}) = (u, v)$ where the ray starting from $\boldsymbol{n}$, and passing through $\boldsymbol{p}$, meets the (x, y)-plane (thought of as the complex plane). See the picture below. From the two similar triangles shown there, we obtain $(1 - z) : 1 = x : u$ and $(1 - z) : 1 = y : v$. Thus we get a map φ, given by $S^2 \backslash \{\boldsymbol{n}\} \ni (x, y, z) \mapsto \varphi(x, y, z) = \frac{1}{1-z}(x, y) \in \mathbb{R}^2$. From the geometric construction of φ, clearly $\varphi : S^2 \backslash \{\boldsymbol{n}\} \to \mathbb{C}$ is a bijection.



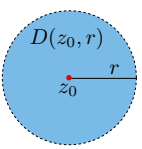
Show that $(u, v) \mapsto (\frac{2u}{u^2+v^2+1}, \frac{2v}{u^2+v^2+1}, \frac{u^2+v^2-1}{u^2+v^2+1}) : \mathbb{R}^2 \to S^2 \backslash \{\boldsymbol{n}\}$ is the inverse of φ. Via the bijection φ, we can equip $\mathbb{C}$ with another 'distance function' d_c, called the *chordal metric*, defined by $d_c(w, w') = d_2(\varphi^{-1}w, \varphi^{-1}w')$ for all $w, w' \in \mathbb{C}$. Show that $d_c(w, w') = \frac{2|w - w'|}{\sqrt{(1 + |w|^2)(1 + |w'|^2)}} \leqslant 2$.

Hint: Use the cosine formula in two triangles: $\boldsymbol{n}, \varphi^{-1}w, \varphi^{-1}w'$, and $\boldsymbol{n}, w, w'$.

If $w, w' \in \mathbb{C} \backslash \{0\}$, then show that $d_c(w, w') = d_c(\frac{1}{w}, \frac{1}{w'})$.

Open discs, open sets, closed sets and compact sets.

To talk about sets of points 'near' a given point, we introduce the following. If $z_0 \in \mathbb{C}$ and $r > 0$, then the *open disc* $D(z_0, r)$ *with centre* z_0 *and radius* r is the set $D(z_0, r) = \{z \in \mathbb{C} : |z - z_0| < r\}$.



A subset U of $\mathbb{C}$ is called *open* if for every $z \in U$, there exists an $r_z > 0$ such that $D(z, r_z) \subset U$. In other words, no matter what point we choose in U, there is always some 'room' around that point comprising points only from U. For example, it can be checked that an open disc $D(z_0, r)$ is an open set. So using the adjective *open* in the name for $D(z_0, r)$ makes sense.

It is also convenient to give a special name to sets whose complement is open: a set $F \subset \mathbb{C}$ is *closed* if $\mathbb{C} \backslash F$ is open. A subset S of $\mathbb{C}$ is *bounded* if there exists an $M > 0$ such that for all $z \in S$, $|z| \leqslant M$. A subset $K \subset \mathbb{C}$ is called *compact* if it is both closed and bounded.

Convergence and continuity.

A sequence $(z_n)_{n\in\mathbb{N}}$ of complex numbers is *convergent with limit* $L \in \mathbb{C}$ if for every $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that for every $n \in \mathbb{N}$ satisfying $n > N$, we have $|z_n - L| < \epsilon$. Using the triangle inequality, it can be seen that for a convergent sequence the limit is unique, and we write $\lim\limits_{n\to\infty} z_n = L$.

Example 10.1. If $z \in \mathbb{C}$ and $|z| < 1$, then $(z^n)_{n\in\mathbb{N}}$ converges in $\mathbb{C}$ to $0 \in \mathbb{C}$: $|z^n - 0| = ||z|^n - 0|$, and as $|z| < 1$, the real sequence $(|z|^n)_{n\in\mathbb{N}}$ converges in $\mathbb{R}$ to $0 \in \mathbb{R}$. $\diamond$

Let S be a subset of $\mathbb{C}$, $z_0 \in S$ and $f : S \to Y$, where $Y = \mathbb{C}$ or $Y = \mathbb{R}$. Then f is *continuous at* z_0 if for every $\epsilon > 0$, there exists a $\delta > 0$ such that whenever $z \in S$ satisfies $|z - z_0| < \delta$, we have $|f(z) - f(z_0)| < \epsilon$. The function f is *continuous* (on S) if for every $z \in S$, f is continuous at z.

Example 10.2. Complex conjugation, $z \mapsto z^* : \mathbb{C} \to \mathbb{C}$, is continuous: $|z^* - z_0^*| = |(z - z_0)^*| = |z - z_0|$ for all $z, z_0 \in \mathbb{C}$. Thus complex conjugation is continuous at each $z_0 \in \mathbb{C}$. As $z_0 \in \mathbb{C}$ was arbitrary, $\cdot^*$ is continuous. $\diamond$

Exercise 10.29. Let $d \in \mathbb{N}$, $c_0, \cdots, c_d \in \mathbb{C}$, $c_d \neq 0$ and $p \in \mathbb{C}[z]$ be given by $p(z) = c_0 + c_1 z + \cdots + c_d z^d$ for all $z \in \mathbb{C}$. Show that there exist $m, M, R > 0$ such that for all $z \in \mathbb{C}$ such that $|z| > R$, we have $M|z|^d \geqslant |p(z)| \geqslant m|z|^d$.

Exercise 10.30. Show that a sequence $(z_n)_{n\in\mathbb{N}}$ in $\mathbb{C}$ is convergent to $L \in \mathbb{C}$ if and only if $(\operatorname{Re} z_n)_{n\in\mathbb{N}}$ and $(\operatorname{Im} z_n)_{n\in\mathbb{N}}$ are convergent in $\mathbb{R}$ to $\operatorname{Re} L$ and $\operatorname{Im} L$, respectively.

Exercise 10.31. Prove that $\mathbb{C}$ is complete, that is, every Cauchy sequence in $\mathbb{C}$ converges in $\mathbb{C}$. (A sequence $(z_n)_{n\in\mathbb{N}}$ is *Cauchy* if for every real $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $m, n \in \mathbb{N}$ satisfying $m, n > N$, we have $|z_n - z_m| < \epsilon$.)

Exercise 10.32. Let $S \subset \mathbb{C}$, $z_0 \in S$, and f be a real-valued or complex-valued function on S. Show that f is continuous at z_0 if and only if for every sequence $(z_n)_{n\in\mathbb{N}}$ in S convergent to z_0, we have that $(f(z_n))_{n\in\mathbb{N}}$ is convergent to $f(z_0)$.

Exercise 10.33. (Algebra of continuous functions).
Suppose that $S \subset \mathbb{C}$, $Y = \mathbb{R}$ or $\mathbb{C}$, $z_0 \in S$, and $f, g : S \to Y$ are continuous at z_0. Define $|f|, f + g, f \cdot g : S \to Y$ by $|f|(z) = |f(z)|$, $(f + g)(z) = f(z) + g(z)$, and $(f \cdot g)(z) = f(z)g(z)$ for all $z \in S$. Show that $|f|, f + g, f \cdot g$ are continuous at z_0.
(It follows that if $p \in \mathbb{C}[z]$, then p and $|p|$ are continuous on $\mathbb{C}$ since the constant function $z \mapsto c$ and the identity function $z \mapsto z$ are continuous on $\mathbb{C}$.)

Exercise 10.34. A sequence $(z_n)_{n\in\mathbb{N}}$ in $\mathbb{C}$ is said to *converge to* ∞ if for all $R>0$, there exists an $N\in\mathbb{N}$ such that for all $n\in\mathbb{N}$ satisfying $n>N$, we have $|z_n|>R$.

(1) Show that $(ni^n)_{n\in\mathbb{N}}$ converges to ∞.

(2) Recall Exercise 10.28, p.119.
Show $(z_n)_{n\in\mathbb{N}}$ converges to ∞ if and only if $\lim\limits_{n\to\infty} d_2(\varphi^{-1}z_n, \boldsymbol{n})=0$.

It is sometimes convenient to adjoin a point called *infinity*, denoted by ∞, to the complex plane to form the *extended complex plane* $\widehat{\mathbb{C}}=\mathbb{C}\cup\{\infty\}$, and extend $\varphi: S^2\backslash\{\boldsymbol{n}\}\to\mathbb{C}$ to a bijection $\varphi: S^2\to\widehat{\mathbb{C}}$ by setting $\varphi(\boldsymbol{n})=\infty$. If we use this bijection φ and the Euclidean metric on S^2 to obtain a distance function on $\widehat{\mathbb{C}}$ (which is the chordal distance d_c we had seen in Exercise 10.28, p.119, together with the extra definition $d_c(z,\infty)=d_2(\varphi^{-1}z,\boldsymbol{n})=2/\sqrt{1+|z|^2}$ for all $z\in\mathbb{C}$), then it can be shown that the metric space[8] $(\widehat{\mathbb{C}}, d_c)$ is complete and compact. This space is called the *one-point compactification* of $\mathbb{C}$.

Exercise 10.35. Prove that $z\mapsto \mathrm{Re}\, z$ and $z\mapsto|z|$ from $\mathbb{C}$ to $\mathbb{R}$ are continuous.

10.4 Fundamental theorem of algebra

Theorem 10.1. (Fundamental theorem of algebra).
Every polynomial $p\in\mathbb{C}[z]$ of degree at least 1 has a zero in $\mathbb{C}$.

We will first show the following.

Lemma 10.1. *If $p\in\mathbb{C}[z]$ is a zero-free polynomial of degree at least 1, then $|p|$ has no minimiser in $\mathbb{C}$.*

Proof. Let p have degree $d\in\mathbb{N}$, and $z_0\in\mathbb{C}$ be a minimiser of $|p|$. Define

$$q(z)=\frac{p(z+z_0)}{p(z_0)} \quad \text{for all } z\in\mathbb{C}.$$

As $(z+z_0)^m$ is a polynomial of degree m for each $m\in\{1,\cdots,d\}$, it follows that $q\in\mathbb{C}[z]$ has degree d. Moreover, we have $q(0)=1$, and $|q(z)|\geqslant 1=|q(0)|$ for all $z\in\mathbb{C}$. We will show that there exists a $w\in\mathbb{C}$ such that $|q(w)|<1$, giving the desired contradiction. Let $c_1,\cdots,c_d\in\mathbb{C}$ be such that $c_d\neq 0$ and

[8] A *metric space* is a set X together with a function $d: X\times X\to\mathbb{R}$ such that:
(D1) (Positive definiteness) For all $x,y\in X$, $d(x,y)\geqslant 0$. For all $x\in X$, $d(x,x)=0$. If $x,y\in X$ are such that $d(x,y)=0$, then $x=y$.
(D2) (Symmetry) For all $x,y\in X$, $d(x,y)=d(y,x)$.
(D3) (Triangle inequality) For all $x,y,z\in X$, $d(x,y)+d(y,z)\geqslant d(x,z)$.
A sequence $(a_n)_{n\in\mathbb{N}}$ of points in a metric space (X,d) is *convergent with limit* $L\in X$ if for every $\epsilon>0$, there exists an $N\in\mathbb{N}$ such that whenever $n\in\mathbb{N}$ satisfies $n>N$, $d(a_n,L)<\epsilon$.
A sequence $(a_n)_{n\in\mathbb{N}}$ of points in a metric space (X,d) is *Cauchy* if for every $\epsilon>0$, there exists an $N\in\mathbb{N}$ such that whenever $m,n\in\mathbb{N}$ satisfy $m,n>N$, we have $d(a_m,a_n)<\epsilon$.
A metric space in which every Cauchy sequence converges is called *complete*. A subset K of X is said to be *compact* if every sequence in K has a convergent subsequence with limit in K, that is, if $(x_n)_{n\in\mathbb{N}}$ is a sequence such that $x_n\in K$ for each $n\in\mathbb{N}$, then there exists a subsequence $(x_{n_k})_{k\in\mathbb{N}}$ which converges to some $L\in K$.

$q(z) = 1 + c_1 z + \cdots + c_d z^d$ for all $z \in \mathbb{C}$. Let $m \in \{1, \cdots, d\}$ be the least index such that $c_m \neq 0$.

If $m = d$, then $q(z) = 1 + c_d z^d$. Let $v \in \mathbb{C}$ be a dth root of $\frac{-c_d^*}{|c_d|}$, and

$$w := \frac{v}{\sqrt[d]{2|c_d|}} \in \mathbb{C}.$$

Then $|q(w)| = |1 + c_d w^d| = |1 + c_d \frac{1}{2|c_d|}(\frac{-c_d^*}{|c_d|})| = |1 - \frac{1}{2}| = \frac{1}{2} < 1$, a contradiction.

It remains to consider the case $m < d$. We have $q(z) = 1 + c_m z^m + r(z)$, where $r(z) := z^{m+1}(c_{m+1} + c_{m+2} z + \cdots + c_d z^{d-m-1})$. Let

$$r_1 := \frac{|c_m|}{|c_{m+1}| + \cdots + |c_d|} \quad \text{and} \quad r_2 := \frac{1}{\sqrt[m]{|c_m|}}.$$

We have

$$\begin{aligned} |r(z)| &\leqslant |z|^{m+1}(|c_{m+1}| + |c_{m+2}||z| + \cdots + |c_d||z^{d-m-1}|) \\ &< |z|^{m+1}(|c_{m+1}| + |c_{m+2}| + \cdots + |c_d|) && \text{if } 0 < |z| < 1 \\ &< |z|^m |c_m| = |c_m z^m| && \text{if moreover } |z| < r_1 \\ &< 1 && \text{if moreover } |z| < r_2. \end{aligned}$$

Hence, if $0 < |z| \leqslant \frac{1}{2}\min\{1, r_1, r_2\} =: R$, then $|r(z)| < |c_m z^m| < 1$. Let $v \in \mathbb{C}$ be an mth root of $\frac{-c_m^*}{|c_m|}$, and $\widetilde{w} = Rv$. Then $|v| = 1$, $0 < |\widetilde{w}| = R$, and

$$c_m \widetilde{w}^m = c_m R^m (\frac{-c_m^*}{|c_m|}) = -|c_m| R^m.$$

Thus

$$\begin{aligned} |q(\widetilde{w})| &\leqslant |1 + c_m \widetilde{w}^m| + |r(\widetilde{w})| \\ &< |1 - |c_m| R^m| + |c_m \widetilde{w}^m| \\ &= 1 - |c_m| R^m + |c_m| R^m = 1, \end{aligned}$$

a contradiction. So $|p|$ does not have a minimiser in $\mathbb{C}$. □

Lemma 10.2. *Let $p \in \mathbb{C}[z]$, $R > 0$, and $K = \{z \in \mathbb{C} : |z| \leqslant R\}$. Then $|p|$ restricted to K has a minimiser in K.*

Proof. If p is a constant, then the claim is obvious since every point $z \in K$ serves as a minimiser of $|p|$. Suppose p has a degree $d \in \mathbb{N}$, and coefficients $c_0, \cdots, c_d \in \mathbb{C}$, so that for all $z \in \mathbb{C}$, we have $p(z) = c_0 + c_1 z + \cdots + c_d z^d$. The subset $S = \{|p(z)| : z \in K\}$ has 0 as a lower bound, and $S \neq \varnothing$ since $0 \in K$ and $|p(0)| \in S$. By the greatest lower bound property of $\mathbb{R}$ (see Exercise 5.21, p.54), S has a greatest lower bound, say ℓ_*. If $n \in \mathbb{N}$, then $\ell_* + \frac{1}{n} > \ell_*$, and so $\ell_* + \frac{1}{n}$ cannot be a lower bound of S, i.e., there exists an element $z_n \in K$ such that $|p(z_n)| < \ell_* + \frac{1}{n}$. In this manner we get a sequence $(z_n)_{n \in \mathbb{N}}$ in K. Let $z_n = (x_n, y_n)$, where $x_n, y_n \in \mathbb{R}$. For all $n \in \mathbb{N}$, we have $|x_n| \leqslant |z_n| \leqslant R$, and so by the Bolzano-Weierstrass theorem, p.59, the bounded sequence

$(x_n)_{n\in\mathbb{N}}$ has a convergent subsequence, say, $(x_{n_k})_{k\in\mathbb{N}}$, with a limit, say $a\in\mathbb{R}$. For all $n\in\mathbb{N}$, we have $|y_n|\leqslant|z_n|\leqslant R$, and in particular, for all $k\in\mathbb{N}$, we also have $|y_{n_k}|\leqslant R$. By the Bolzano-Weierstrass theorem, the bounded sequence $(y_{n_k})_{k\in\mathbb{N}}$ has a convergent subsequence, say, $(y_{n_{k_\ell}})_{\ell\in\mathbb{N}}$, with limit, say $b\in\mathbb{R}$. As $(x_{n_{k_\ell}})_{\ell\in\mathbb{N}}$ is a subsequence of the convergent sequence $(x_{n_k})_{k\in\mathbb{N}}$ with limit a, it is convergent too, with the same limit, i.e., a. Hence $(z_{n_{k_\ell}})_{\ell\in\mathbb{N}}$ has convergent real and imaginary parts, and hence is convergent in $\mathbb{C}$, with limit $w:=(a,b)$ (see Exercise 10.30, p.120). We also have for all $\ell\in\mathbb{N}$, that $x_{n_{k_\ell}}^2+y_{n_{k_\ell}}^2\leqslant R^2$, giving $a^2+b^2\leqslant R^2$ (by Exercises 5.26, p.56 and 5.24, p.55). So $w=(a,b)\in K$. As $|p|$ is continuous, $(|p(z_{n_{k_\ell}})|)_{\ell\in\mathbb{N}}$ is convergent as well with limit $|p(w)|$. But from the inequalities $\ell_*\leqslant|p(z_n)|<\ell_*+\frac{1}{n}$ $(n\in\mathbb{N})$, it follows that $|p(w)|=\ell_*$. So for all $z\in K$, $|p(w)|=\ell_*\leqslant|p(z)|$, that is, $w\in K$ is a minimiser of $|p|$. □

We are now ready to prove the fundamental theorem of algebra.

Proof. (Of Theorem 10.1, p.121.) Let $d\in\mathbb{N}$ and $c_0,\cdots,c_d\in\mathbb{C}$ be such that $c_d\neq0$ and $p(z)=c_0+c_1z+\cdots+c_dz^d$ for all $z\in\mathbb{C}$. Let p be zero-free. Then $\widetilde{p}:=\frac{1}{c_d}p$ is zero-free too. For $z\in\mathbb{C}\backslash\{0\}$, $\widetilde{p}(z)=z^d(1+r(z))$, where

$$r(z):=\frac{\widetilde{c}_0}{z^d}+\frac{\widetilde{c}_1}{z^{d-1}}+\cdots+\frac{\widetilde{c}_{d-1}}{z}\quad\text{and}\quad\widetilde{c}_0:=\frac{c_0}{c_d},\ \widetilde{c}_1:=\frac{c_1}{c_d},\ \cdots,\widetilde{c}_{d-1}:=\frac{c_{d-1}}{c_d}.$$

Let $R=2d\max\{|\widetilde{c}_0|,\cdots,|\widetilde{c}_{d-1}|,1\}$. Then $R\geqslant2d1>1$. For $0\leqslant k\leqslant d-1$ and $|z|\geqslant R$, we have

$$|\frac{\widetilde{c}_k}{z^{d-k}}|=\frac{|\widetilde{c}_k|}{|z|^{d-k}}\leqslant\frac{|\widetilde{c}_k|}{R^{d-k}}\leqslant\frac{|\widetilde{c}_k|}{R}\leqslant\frac{1}{2d},$$

which yields

$$|r(z)|\leqslant|\frac{\widetilde{c}_0}{z^d}|+|\frac{\widetilde{c}_1}{z^{d-1}}|+\cdots+|\frac{\widetilde{c}_{d-1}}{z}|\leqslant\frac{1}{2d}d=\frac{1}{2}.$$

Hence for $|z|\geqslant R$,

$$|1+r(z)|\geqslant|1-|r(z)||\geqslant1-|r(z)|\geqslant1-\frac{1}{2}=\frac{1}{2},$$

and so $|\widetilde{p}(z)|=|z^d(1+r(z))|\geqslant\frac{R^d}{2}$. Since $R\geqslant2d|\widetilde{c}_0|=2d|\widetilde{p}(0)|$, we have

$$\text{for all } z \text{ with } |z|\geqslant R,\quad |\widetilde{p}(0)|\leqslant\frac{R}{2d}\leqslant\frac{R^d}{2(1)}\leqslant|\widetilde{p}(z)|. \qquad (\star)$$

By Lemma 10.2, p.122, $|\widetilde{p}|$ restricted to the set $K=\{w\in\mathbb{C}:|w|\leqslant R\}$ has a minimiser $w_*\in K$. Thus if $|z|\leqslant R$, we have $|\widetilde{p}(z)|\geqslant|\widetilde{p}(w_*)|$. The inequality $(\star)$ implies that if $|z|\geqslant R$, then we have $|\widetilde{p}(z)|\geqslant|\widetilde{p}(0)|\geqslant|\widetilde{p}(w_*)|$, where the last inequality follows from the fact that $0\in K$, and w_* is a minimiser of $|\widetilde{p}|$ on K. So for all $z\in\mathbb{C}$, we have $|\widetilde{p}(z)|\geqslant|\widetilde{p}(w_*)|$, i.e., w_* is a minimiser of $|\widetilde{p}|$ on all of $\mathbb{C}$. This contradicts Lemma 10.1, p.121, as $\widetilde{p}$ has degree (equal to that of p, namely) $d\in\mathbb{N}$. □

Exercise 10.36. (Möbius transformations).
Let $a, b, c, d \in \mathbb{C}$ be such that $ad - bc \neq 0$. Define $f : D \to \mathbb{C}$ by

$$f(z) = \frac{az+b}{cz+d} \text{ for all } z \in D := \{z \in \mathbb{C} : cz+d \neq 0\}.$$

The constants c and d cannot both be zero since $ad - bc \neq 0$. If $c = 0$, then f is defined on $\mathbb{C}$, while if $c \neq 0$, then f is defined in $\mathbb{C}\backslash\{-\frac{d}{c}\}$. It will be convenient to extend f to the extended complex plane $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$. If $c = 0$, then $f(\infty) := \infty$, while if $c \neq 0$, then $f(-\frac{d}{c}) := \infty$ and $f(\infty) := \frac{a}{c}$. Such a map $f : \widehat{\mathbb{C}} \to \widehat{\mathbb{C}}$ is called a *Möbius transformation*.

(1) Show that f is bijective, with inverse given by the Möbius transformation g, where $g(w) = \frac{-dw+b}{cw-a}$. Note that $(-d)(-a) - bc = ad - bc \neq 0$.

(2) Prove that the composition of two Möbius transformations is again a Möbius transformation. For simplicity, consider only the case of transformations

$$f(z) = \frac{az+b}{cz+d} \text{ and } g(z) = \frac{a'z+b'}{c'z+d'}, \text{ where } c, c' \text{ are both nonzero.}$$

(The set of all Möbius transformations forms a group with composition of maps taken as the group operation.)

(3) Describe the fate of circles and lines under the following Möbius maps:

(Translation) $f(z) = z + b$ (Dilation) $f(z) = az$ $(a > 0)$
(Rotation) $f(z) = e^{i\theta}z$ $(\theta \in \mathbb{R})$ (Inversion) $f(z) = \frac{1}{z}$.

For the image under inversion of a line not passing through 0, it can be assumed that the line is $L = \{z_0 + tiz_0 : t \in \mathbb{R}\}$, where $z_0 \neq 0$. For such a line, show that the image lies on a circle with centre $\frac{1}{2z_0}$ and radius $\frac{1}{2|z_0|}$.

(If $c \neq 0$, then $\frac{az+b}{cz+d} = \frac{a}{c} + \frac{bc-ad}{c(cz+d)}$. This allows us to conclude that: every Möbius transformation is the composition of the special ones above. Imagining straight lines as circles of infinite radius, define a *sircle* to be a line or a circle. Möbius transformations map sircles to sircles.)

Complex numbers can also be introduced as expressions $a+b\boldsymbol{i}$ where $a, b \in \mathbb{R}$, with addition given by $a + b\boldsymbol{i} + (c + d\boldsymbol{i}) = a + c + (b + d)\boldsymbol{i}$, and multiplication being based on distributivity of multiplication over addition, together with the rule that $\boldsymbol{i}^2 = -1$: thus

$$(a + b\boldsymbol{i})(c + d\boldsymbol{i}) := ac + bd\boldsymbol{i}^2 + ad\boldsymbol{i} + bc\boldsymbol{i} = ac - bd + (ad + bc)\boldsymbol{i}.$$

We have seen that complex number multiplication can be used to describe rotations in the plane. One might wonder if a strategy similar to this can produce an object that can be used to describe rotations in $\mathbb{R}^3$. In linear algebra one learns that a rotation about an axis $\boldsymbol{q} \in \mathbb{R}^3$ can be represented by a linear transformation corresponding to matrix multiplication by a 3×3 real orthogonal matrix A with determinant 1. However, given the angle of rotation and the axis, forming the matrix A is not immediate. Moreover, given two rotations about axes $\boldsymbol{p}, \boldsymbol{q} \in \mathbb{R}^3$ through angles α, β, composing the two also produces a rotation, but computing the new axis of rotation and the

angle is again not straightforward. We will see in the following exercise that analogous to the field $\mathbb{C}$ of complex numbers, there is a 'division algebra'[9] $\mathbb{H}$ of quaternions, that can be used to easily describe rotations in $\mathbb{R}^3$.

Exercise 10.37. (Quaternions). In $\mathbb{R}^3$, let $\boldsymbol{i} = (1,0,0)$, $\boldsymbol{j} = (0,1,0)$, $\boldsymbol{k} = (0,0,1)$. (These are thus the standard basis unit-length[10] vectors in the real vector space $\mathbb{R}^3$ equipped with the usual Euclidean dot product.) A *quaternion* q is an expression of the form $q = q_0 + \boldsymbol{q} = q_0 + q_1\boldsymbol{i} + q_2\boldsymbol{j} + q_3\boldsymbol{k}$, where $q_0 \in \mathbb{R}$ and $\boldsymbol{q} = (q_1, q_2, q_3) \in \mathbb{R}^3$. The set of quaternions is denoted[11] by $\mathbb{H}$. We define addition of quaternions componentwise, i.e., $p + q = p_0 + q_0 + (p_1 + q_1)\boldsymbol{i} + (p_2 + q_2)\boldsymbol{j} + (p_3 + q_3)\boldsymbol{k}$ where besides the quaternion q above, we have $p = p_0 + \boldsymbol{p}$, $\boldsymbol{p} = (p_1, p_2, p_3) \in \mathbb{R}^3$. The product of p, q is based on the distributivity of multiplication over addition, using the rules $\boldsymbol{i}^2 = \boldsymbol{j}^2 = \boldsymbol{k}^2 = \boldsymbol{ijk} = -1$, $\boldsymbol{ij} = \boldsymbol{k} = -\boldsymbol{ji}$, $\boldsymbol{jk} = \boldsymbol{i} = -\boldsymbol{kj}$, $\boldsymbol{ki} = \boldsymbol{j} = -\boldsymbol{ik}$. Thus multiplication is non-commutative. A concise expression for the product of quaternions p, q is given by $pq = p_0q_0 - \boldsymbol{p}\cdot\boldsymbol{q} + p_0\boldsymbol{q} + q_0\boldsymbol{p} + \boldsymbol{p}\times\boldsymbol{q}$, where for the vectors $\boldsymbol{p}, \boldsymbol{q} \in \mathbb{R}^3$, $\boldsymbol{p}\cdot\boldsymbol{q}$ is the dot product and $\boldsymbol{p}\times\boldsymbol{q}$ is the cross product. If $c \in \mathbb{R}$ and $q \in \mathbb{H}$, then $cq = qc$.

(1) A real polynomial $p \in \mathbb{R}[x]$ of degree $d \in \mathbb{N}$ has *at most* d real zeroes. A consequence of the fundamental theorem of algebra and the division algorithm for polynomials (Exercise 10.13, p.115) is that a complex polynomial $p \in \mathbb{C}[z]$ of degree $d \in \mathbb{N}$ has *exactly* d complex zeroes. We will now show that a quaternionic polynomial $p \in \mathbb{H}[q]$ of degree d can have infinitely many zeroes in $\mathbb{H}$! This is a manifestation of the noncommutativity of multiplication in $\mathbb{H}$. Define the polynomial $p \in \mathbb{H}[q]$ of degree 2 by $p(q) = q^2 + 1$ for all $q \in \mathbb{H}$. If $\boldsymbol{v} = (a,b,c) \in \mathbb{R}^3$ with $\|\boldsymbol{v}\|_2 := \sqrt{a^2+b^2+c^2} = 1$, then show that $p(\boldsymbol{v}) = 0$.

(2) The *complex conjugate* of q is $q^* = q_0 - \boldsymbol{q}$. Check that $(q^*)^* = q$, $q + q^* = 2q_0$, and $q^*q = q_0^2 + q_1^2 + q_2^2 + q_3^2 = qq^*$. Also check that $(pq)^* = q^*p^*$. Define the *norm* $|q|$ of q by $|q| = \sqrt{q^*q}$. Check that $|pq|^2 = |p|^2|q|^2$. If $|q| \neq 0$, then the *inverse* q^{-1} of q is $q^{-1} = \dfrac{q^*}{|q|^2}$. Show that $q^{-1}q = qq^{-1} = 1$.

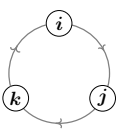
(3) Quaternions lie in $\mathbb{R}^4$, but can be used to efficiently describe rotations in $\mathbb{R}^3$. Suppose $q_0 \in \mathbb{R}$, $\boldsymbol{q} \in \mathbb{R}^3\backslash\{\boldsymbol{0}\}$ are such that $q := q_0 + \boldsymbol{q}$ has norm $|q| = 1$, that is, $q_0^2 + \|\boldsymbol{q}\|_2^2 = 1$. There exists a unique $\theta \in (-\pi, \pi]$ such that $q_0 = \cos\theta$ and $\|\boldsymbol{q}\|_2 = \sin\theta$. As $\|\boldsymbol{q}\|_2 > 0$, in fact $\theta \in (0, \pi)$. Define $L_q : \mathbb{R}^3 \to \mathbb{R}^3$ by $L_q(\boldsymbol{v}) = q\boldsymbol{v}q^*$ for all $\boldsymbol{v} \in \mathbb{R}^3$. Check that L_q is a well-defined linear transformation. Show that L_q is an isometry, i.e., $\|L_q(\boldsymbol{v})\|_2 = \|\boldsymbol{v}\|_2$ for all $\boldsymbol{v} \in \mathbb{R}^3$. Verify that $\boldsymbol{q}$ is an eigenvector of L_q with eigenvalue 1. Show that L_q is a rotation through an angle 2θ about $\boldsymbol{q}$ as the axis of rotation. Composition of non co-axial rotations is done using multiplication in $\mathbb{H}$: show that if $p = p_0 + \boldsymbol{p}$, $q = q_0 + \boldsymbol{q} \in \mathbb{H}$ with $|p| = 1 = |q|$, and $\boldsymbol{p}, \boldsymbol{q} \in \mathbb{R}^3\backslash\{\boldsymbol{0}\}$ not collinear, we have $L_p \circ L_q = L_{pq}$.

[9] Roughly, an *algebra* over a field is a vector space with a multiplication rule for vectors with the certain nice properties (like distributivity of multiplication over vector addition and compatibility of multiplication with multiplication by field elements), and a *division algebra* is an algebra over a field in which division by nonzero elements is possible.

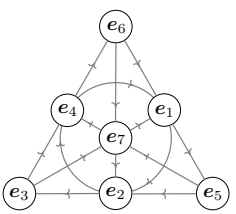
[10] The *length* of $\boldsymbol{v} = (a,b,c) \in \mathbb{R}^3$ is $\|\boldsymbol{v}\|_2 := \sqrt{a^2+b^2+c^2}$.

[11] After the mathematical physicist W.R. Hamilton (1805–1865), who introduced them.

Exercise 10.38. (Octonions). A *real division algebra* is a vector space V over $\mathbb{R}$ with a bilinear product[12], such that division by nonzero elements is possible[13]. A *normed division algebra* is a real division algebra with a norm[14] $|\cdot|$ satisfying $|u \cdot v| = |u||v|$ for all $u, v \in V$. It turns out[15] that there are only four finite-dimensional normed division algebras: the real numbers, the complex numbers, the quaternions, and the octonions. In this exercise we describe the arithmetic of octonions. The octonions are the elements of an 8-dimensional real division algebra $\mathbb{O}$ with the basis $1, \boldsymbol{e}_n$, $1 \leqslant n \leqslant 7$. We now describe the multiplication in $\mathbb{O}$ (and we denote the multiplication of $p, q \in \mathbb{O}$ simply by pq). Recall that the quaternions are the elements of a 4-dimensional algebra $\mathbb{H}$ with the basis $1, \boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k}$, and the multiplication rules we met earlier can be summarised by saying that $\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k}$ are square roots of -1, and the remaining rules can be read off from the picture below as follows. Multiplying two elements clockwise results in the next one, and going anticlockwise produces an extra minus sign. Thus $\boldsymbol{jk} = \boldsymbol{i}$, but $\boldsymbol{kj} = -\boldsymbol{i}$.



Similarly, for the octonions, the multiplication is obtained by specifying the rules for multiplication of the basis elements: $\boldsymbol{e}_k^2 = -1$ $(1 \leqslant n \leqslant 7)$, and the remaining rules can be read off from the picture below as follows. The 'lines' are the sides of the triangle, its altitudes, and the circle passing through the midpoints of the sides. Any two points lie on a unique such line. Each such line has exactly three points, which can be ordered in a cyclic manner using the arrows shown: e.g., we have the order $\boldsymbol{e}_3 \to \boldsymbol{e}_7 \to \boldsymbol{e}_1 \to \boldsymbol{e}_3$. If $(\boldsymbol{e}_i, \boldsymbol{e}_j, \boldsymbol{e}_k)$ are in such a cyclic order, then we set $\boldsymbol{e}_i\boldsymbol{e}_j = \boldsymbol{e}_k$ and $\boldsymbol{e}_j\boldsymbol{e}_i = -\boldsymbol{e}_k$. In our example above, we get $\boldsymbol{e}_1\boldsymbol{e}_3 = \boldsymbol{e}_7$ and $\boldsymbol{e}_3\boldsymbol{e}_1 = -\boldsymbol{e}_7$.



Write the multiplication table for the basis elements $\boldsymbol{e}_1, \cdots, \boldsymbol{e}_7$ of the octonions. Check that $\boldsymbol{e}_1(\boldsymbol{e}_2\boldsymbol{e}_3) \neq (\boldsymbol{e}_1\boldsymbol{e}_2)\boldsymbol{e}_3$. Thus multiplication in $\mathbb{O}$ is not associative.

[12] That is, for all vectors $u, v, w \in V$ and all $\alpha, \beta \in \mathbb{R}$, we have $u \cdot (v + w) = u \cdot v + u \cdot w$, $(u + v) \cdot w = u \cdot w + v \cdot w$, and $(\alpha u) \cdot (\beta v) = (\alpha\beta)(u \cdot v)$.

[13] That is, for any $b \in V$ and any nonzero $a \in V$ in there are unique elements $x, y \in V$ such that $a \cdot x = b$ and $x \cdot a = b$.

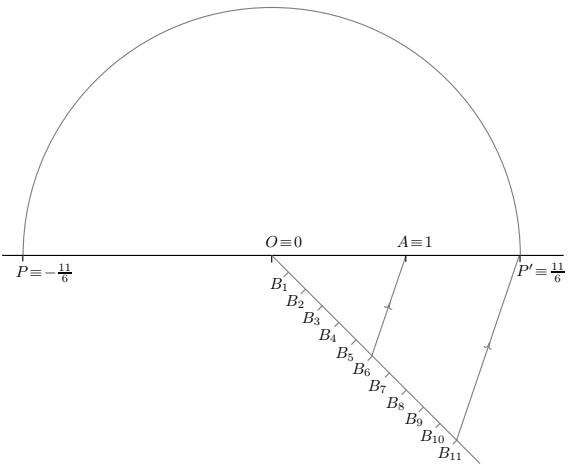
[14] A *norm* is a map $|\cdot| : V \to \mathbb{R}$ satisfying the following properties:
(N1) For all $v \in V$, $|v| \geqslant 0$. If $v \in V$ is such that $|v| = 0$, then $v = 0$.
(N2) For all $\alpha \in \mathbb{R}$ and all $v \in V$, $|\alpha v| = |\alpha||v|$.
(N3) For all $u, v \in V$, $|u + v| \leqslant |u| + |v|$.

[15] Hurwitz' theorem [Hurwitz (1922)].

Solutions

Chapter 1

Solution to Exercise 1.1

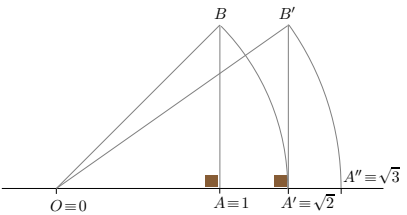


Steps for the construction of $-11/6$ on the real number line: Draw a ray OA and take $\ell(OA) = 1$. Draw a ray OB_1 not parallel to OA. Cut equal lengths $\ell(OB_1) = \ell(B_1B_2) = \cdots = \ell(B_{10}B_{11})$ along the ray OB_1. Join A to B_6. Construct $B_{11}P'$ parallel to B_6A, meeting OA (extended) at P'. Then[16] $P' \equiv \frac{11}{6}$. With O as center and radius $\ell(OP')$, draw a circle which meets the line passing through O and A at the point P which is different from P'. Then $P \equiv -\frac{11}{6}$.

See the following picture for the construction of $\sqrt{3}$ on the real number line. The steps are: Draw a ray OA and take $\ell(OA) = 1$. Construct AB perpendicular to OA such that $\ell(AB) = 1$. Then $\ell(OB) = \sqrt{1^2 + 1^2} = \sqrt{2}$. With centre O and radius $\ell(OB)$ draw a circular arc meeting OA extended at A'. Then $A' \equiv \sqrt{2}$. Construct $A'B'$ perpendicular to OA' such that $\ell(A'B') = 1$. Then $\ell(OB') = \sqrt{(\sqrt{2})^2 + 1^2} = \sqrt{2+1} = \sqrt{3}$. With center O

[16] Here '$\equiv$' stands for 'corresponding to'.

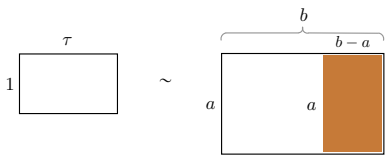
and radius $\ell(OB')$ draw a circular arc meeting OA' extended at A''. Then $A'' \equiv \sqrt{3}$.



(In fact, we can construct $\sqrt{n}$ for all $n \in \mathbb{N}$: Given the length $\sqrt{n}$, the hypotenuse of the right-angled triangle with sides $\sqrt{n}$ and 1 is equal to $\sqrt{n+1}$. So knowing the unit length, we can construct $\sqrt{2}, \sqrt{3}, \sqrt{4} = 2, \sqrt{5}, \cdots$.)

Solution to Exercise 1.2

Let $\tau \in \mathbb{Q}$. Then some scaling of the golden rectangle with side lengths 1 and τ will produce a golden rectangle with integer sides.



Arrange all such similar golden rectangles in increasing order of their longest side lengths. Consider the first rectangle R in this list. Then R is a golden rectangle with integer side lengths, say a, b, in which the longest side b $(= \tau a > a)$ is the smallest possible. But if we cut out a square with side length equal to a, then we are left with a smaller golden rectangle, with integer sides, and with longest side a $(< b)$, contradicting our choice of R. Thus τ is not rational. Hence $\sqrt{5} = 2\tau - 1$ is not rational either.

Solution to Exercise 1.3

(1) R is reflexive: As $0 < 1$, adding $x \in \mathbb{R}$ to both sides, we obtain $x < x + 1$. So xRx for all $x \in \mathbb{R}$.
R is not symmetric: E.g., $0R2$ (as $0 < 2+1$), but $\neg(2R0)$ (as $2 \nless 0+1$).
R is not transitive: E.g., $1.1R0.9$ (as $1.1 < 0.9+1$) and $0.9R0$ (as $0.9 < 0+1$), but $\neg(1.1R0)$ (as $1.1 \nless 0+1$).

(2) R is reflexive, since for all $m \in \mathbb{Z}$, $m = 1m$ and so $m \mid m$.
R is not symmetric: E.g., $1 \mid 2$, but $2 \nmid 1$.
R is transitive: If mRn and nRk, then $m \mid n$ and $n \mid k$. So there exist integers d, d' such that $n = md$, $k = nd'$. Thus $k = nd' = (md)d' = m(dd')$. So $m \mid k$, i.e., mRk.

(3) R is reflexive: For all $x \in \mathbb{R}\backslash\{0\}$, $xx^{-1} = 1 \in \mathbb{Q}$. So xRx for all $x \in \mathbb{R}$.
R is symmetric: If xRy, then $xy^{-1} = q \in \mathbb{Q}$. Also, as x, y are nonzero, so is q. But then $yx^{-1} = (xy^{-1})^{-1} = q^{-1} \in \mathbb{Q}$. So yRx.
R is transitive: Let xRy and yRz. Then $xy^{-1} = q \in \mathbb{Q}$ and $yz^{-1} = q' \in \mathbb{Q}$. Thus $xz^{-1} = (xy^{-1})(yz^{-1}) = qq' \in \mathbb{Q}$. So xRz.
As R is reflexive, symmetric and transitive, it is an equivalence relation.

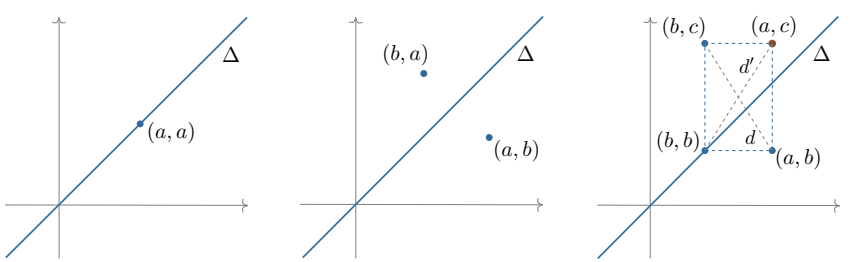
(4) R is reflexive: For all $n \in \mathbb{N}$, $n = 1n = 2^0 n$. So nRn for all $n \in \mathbb{N}$.
R is symmetric: If nRn', then $n = 2^m n'$ for some $m \in \mathbb{Z}$.
But then $-m \in \mathbb{Z}$, and $n' = 2^{-m} n$. So $n'Rn$.
R is transitive: Suppose nRn' and $n'Rn''$. Then $n = 2^m n'$ and $n' = 2^{m'} n''$ for some $m, m' \in \mathbb{Z}$. As $m + m' \in \mathbb{Z}$ and $n = 2^m n' = 2^m 2^{m'} n'' = 2^{m+m'} n''$, we conclude that nRn''.
As R is reflexive, symmetric and transitive, it is an equivalence relation.

Solution to Exercise 1.4

Reflexivity says that for all $a \in \mathbb{R}$, aRa, i.e., $(a, a) \in R$. Thus the 'diagonal line' $\Delta := \{(a, a) \in \mathbb{R}^2 : a = a\} \subset R$. Consequently, in the plane, reflexivity means that the region described by R contains Δ. See the left picture.

Symmetry says that whenever aRb, we have bRa. So $(a, b) \in R$ implies that the point $(b, a) \in R$. Consequently, in the plane, symmetry of R means that the region given by R is symmetric when reflected in the diagonal line Δ. See the middle picture below.

Transitivity means that whenever aRb and bRc, we have that aRc. Thus $(a, b), (b, c) \in R$ implies $(a, c) \in R$. Geometrically, this means that the region given by R has the property that whenever we have two distinct points in R such that the rectangle for which they form a diagonal d is such that one of the endpoints of the other diagonal d' lies on Δ, then the other endpoint of d' belongs to R. See the rightmost picture below.



Solution to Exercise 1.5

(1) (Geometrically we see that R contains the diagonal line and is symmetric with respect to reflection in the diagonal. So we anticipate that R is an equivalence relation. We verify this now.) R is an equivalence relation:
(ER1) For all $x \in \mathbb{R}$, $(x, x) \in R$, and so xRx. Thus R is reflexive.
(ER2) Let $x, y \in \mathbb{R}$ be such that xRy. Then $(x, y) \in R = \{(s, s) : s \in \mathbb{R}\}$. So $x = y$ and $(y, x) = (x, x) \in R$. Hence yRx. Thus R is symmetric.
(ER3) Let $x, y, z \in \mathbb{R}$ be such that xRy and yRx. Then $(x, y), (y, z) \in R$. So $x = y$ and $y = z$. Then $x = z$, and so xRz. Thus R is transitive.

(2) R is not an equivalence relation, since reflexivity fails:
E.g. $0 \in \mathbb{R}$, but of course $(0, 0) \notin \varnothing$, and so $\neg(0R0)$.

(3) (Geometrically the set described by R in the plane is just the x-axis which clearly does not contain the diagonal. So R cannot be an equivalence relation. We show this now.) R is not an equivalence relation since reflexivity fails: For example, $1 \in \mathbb{R}$, but $\neg(1R1)$ because $(1, 1) \notin R$.

(4) (Geometrically the set described by R in the (x, y)-plane is the hyperbolic curve described by $xy = -1$, lying in the second and the fourth quadrants. Although R is symmetric about the diagonal line, R does not contain the diagonal, and so R cannot be an equivalence relation.) R is not an equivalence relation since reflexivity fails: e.g. $0 \in \mathbb{R}$, but $\neg(0R0)$ since $(0, 0) \notin R$ $(0 \cdot 0 + 1 = 1 \neq 0)$.

(5) Factorising, $x^2y - xy^2 - x + y = xy(x - y) - (x - y) = (xy - 1)(x - y)$, and so

$$\begin{aligned} R &= \{(x, y) \in \mathbb{R}^2 : x^2y - xy^2 - x + y = 0\} \\ &= \{(x, y) \in \mathbb{R}^2 : (xy - 1)(x - y) = 0\} \\ &= \{(x, y) \in \mathbb{R}^2 : xy - 1 = 0 \text{ or } x - y = 0\} \\ &= \{(x, y) \in \mathbb{R}^2 : xy = 1\} \cup \{(x, y) \in \mathbb{R}^2 : x = y\}. \end{aligned}$$

(So geometrically, the set described by R in the (x, y)-plane is the union of the hyperbolic curve described by $xy = 1$ in the first and the third quadrants, and the diagonal line. Clearly R contains the diagonal line and is symmetric with respect to reflection in the diagonal line. Hence we anticipate that R is an equivalence relation. We check this below.) R is an equivalence relation:
(ER1) If $x \in \mathbb{R}$, then $(x, x) \in R$: $x^2x - xx^2 - x + x = 0$. So R is reflexive.
(ER2) Let $x, y \in \mathbb{R}$ be such that xRy. Then $x^2y - xy^2 - x + y = 0$. Multiplying by -1 gives $-x^2y + xy^2 + x - y = 0$, and rearranging, $xy^2 - x^2y - y + x = 0$. So $(y, x) \in R$, that is, yRx. Thus R is symmetric.

(ER3) Let $x, y, z \in \mathbb{R}$ be such that xRy and yRx, i.e., $(x, y) \in R$ and $(y, z) \in R$. Then [$xy = 1$ or $x = y$] and [$yz = 1$ or $y = z$]. We have:
- $xy = 1$ and $yz = 1$. Then $x = y^{-1} = z$. So [$xz = 1$ or $x = z$], i.e., $(x, z) \in R$.
- $xy = 1$ and $y = z$. Then $xz = xy = 1$. So [$xz = 1$ or $x = z$], i.e., $(x, z) \in R$.
- $x = y$ and $yz = 1$. Then $xz = yz = 1$. So [$xz = 1$ or $x = z$], i.e., $(x, z) \in R$.
- $x = y$ and $y = z$. Then $x = y = z$. So [$xz = 1$ or $x = z$], i.e., $(x, z) \in R$.

So in each case, $(x, z) \in R$, that is, xRz. So R is transitive.

Solution to Exercise 1.6

(1) We have:

(ER1) For all $x \in X$, $f(x) = f(x)$, and so $x \sim x$.

(ER2) If $x \sim x'$, then $f(x) = f(x')$, and so $x' \sim x$.

(ER3) If $x \sim x'$ and $x' \sim x''$, then $f(x) = f(x') = f(x'')$, and so $x \sim x''$.

(2) Let $x \in X$. If $x' \in [x]$, then $x' \sim x$. So $f(x') = f(x)$, i.e., $x' \in f^{-1}(\{f(x)\})$. So $[x] \subset f^{-1}(\{f(x)\})$.

If $x'' \in f^{-1}(\{f(x)\})$, then $f(x'') \in \{f(x)\}$, i.e., $f(x'') = f(x)$. So $x'' \sim x$. Thus $x'' \in [x]$. Hence $f^{-1}(\{f(x)\}) \subset [x]$.

As $[x] \subset f^{-1}(\{f(x)\})$ and $f^{-1}(\{f(x)\}) \subset [x]$, we have $[x] = f^{-1}(\{f(x)\})$.

(3) If $x, x' \in S$, then $x \sim x'$, and so $f(x) = f(x')$. It follows that there is no ambiguity in the definition of $f(S)$, since we get the same element in the image of f, no matter which element is used from S to be acted upon by f.

(4) Injectivity: Let $\overline{f}(S) = \overline{f}(S')$ for some equivalence classes $S, S' \in \overline{X}$. If $x \in S$ and $x' \in S'$, then $S = [x]$ and $S' = [x']$. We have $\overline{f}([x]) = \overline{f}([x'])$. By the definition of $\overline{f}$, $f(x) = f(x')$. So $x \sim x'$ and $[x] = [x']$, i.e., $S = S'$.

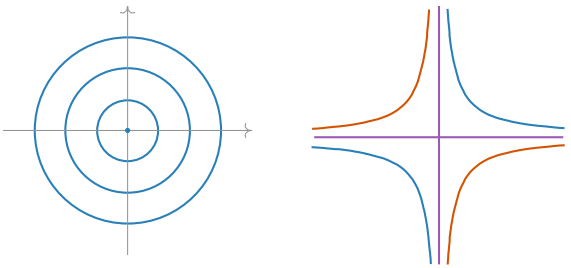
Surjectivity: Let $y \in \operatorname{ran} f$. Then $y = f(x)$ for some $x \in X$. So $[x] \in \overline{X}$, and $\overline{f}([x]) = f(x) = y$.

As $\overline{f}$ is injective and surjective, it is bijective.

(5)(a) For $z, w \in \mathbb{C}$, $z \sim w$ if $|z| = |w|$. So if $z \in \mathbb{C}$, and $|z| = r \geqslant 0$, then

$$\begin{aligned}[z] &= \{w = x + iy \in \mathbb{C} : x, y \in \mathbb{R} \text{ and } |w| = |z|\} \\ &= \{x + iy : x, y \in \mathbb{R} \text{ and } \sqrt{x^2 + y^2} = r\} \\ &= \{x + iy : x, y \in \mathbb{R} \text{ and } x^2 + y^2 = r^2\}.\end{aligned}$$

If $r = 0$, then clearly $[z] = \{0+0i\}$. If $r > 0$, then $[z]$ is a circle of radius r with centre $0+0i$. Thus the equivalence classes are given by concentric circles C_r about the origin; see the left picture below. (Clearly these partition the complex plane.)



(b) Let $(a, b) \in \mathbb{R}^2$ and let $z := ab$. Then

$$[(a, b)] = \{(x, y) \in \mathbb{R}^2 : f(x, y) = f(a, b)\} = \{(x, y) \in \mathbb{R}^2 : xy = z\}.$$

If $z > 0$, this is a hyperbola lying in the first and third quadrants.
If $z < 0$, this is a hyperbola lying in the second and fourth quadrants.
If $z = 0$, this is the union of the two axes:
$\{(x, 0) : x \in \mathbb{R}\} \cup \{(0, y) : y \in \mathbb{R}\}$.
These hyperbolas are shown in the rightmost picture above. (Clearly they partition the plane.)

Solution to Exercise 1.7

For all $a \in \mathbb{R}$, $a - a = 0 \in \mathbb{Z}$ and so $a \sim a$. So $\sim$ is reflexive.

If $a \sim b$ where $a, b \in \mathbb{R}$, then $m = a - b \in \mathbb{Z}$. We have $b - a = -(a - b) = -m \in \mathbb{Z}$, and so $b \sim a$. Hence $\sim$ is symmetric.

Suppose $a, b, c \in \mathbb{R}$ are such that $a \sim b$ and $b \sim c$. Then $m = a - b \in \mathbb{Z}$ and $n = b - c \in \mathbb{Z}$. We have $a - c = a - b + b - c = m + n \in \mathbb{Z}$, and so $a \sim c$.
Thus $\sim$ is transitive.

As $\sim$ is reflexive, symmetric and transitive, it is an equivalence relation.

Let $b \in [a]$. Then $a \sim b$, and so $m := a - b \in \mathbb{Z}$. Thus with $n_* := -m \in \mathbb{Z}$, we have $b = a + (-m) = a + n_* \in \{a + n : n \in \mathbb{Z}\}$. Hence $[a] \subset \{a + n : n \in \mathbb{Z}\}$.
Vice versa, if $b \in \{a + n : n \in \mathbb{Z}\}$, then $b = a + n_*$ for some $n_* \in \mathbb{Z}$, and so $a - b = a - (a + n_*) = -n_* \in \mathbb{Z}$, showing that $a \sim b$ and $b \in [a]$.
Thus $\{a + n : n \in \mathbb{Z}\} \subset [a]$.

Solution to Exercise 1.8

(1) Given a rational $\epsilon > 0$, let $N \in \mathbb{N}$ be such that $N > \frac{2}{\epsilon}$. If $n, m \in \mathbb{N}$ are such that $n, m > N$, then $|\frac{1}{n} - \frac{1}{m}| \leqslant |\frac{1}{n}| + |-\frac{1}{m}| = \frac{1}{n} + \frac{1}{m} < \frac{1}{N} + \frac{1}{N} = \frac{2}{N} < \epsilon$. So $(\frac{1}{n})_{n \in \mathbb{N}}$ is a Cauchy sequence in $\mathbb{Q}$.

(2) Let $(a_n)_{n\in\mathbb{N}}$ be a sequence in $\mathbb{Q}$ that converges in $\mathbb{Q}$ to $L \in \mathbb{Q}$. Let $\epsilon > 0$ be a rational number. (We want to find an $N \in \mathbb{N}$ which guarantees that whenever $m, n \in \mathbb{N}$ satisfy for $n, m > N$, we have $|a_n - a_m| < \epsilon$. But we *do* know that the terms a_n, a_m can both be made close to L if n, m are large enough. As $|a_n - a_m| = |a_n - L + L - a_m|$, we can use the triangle inequality to complete the argument, as follows.)
There exists an $N \in \mathbb{N}$ such that whenever $n \in \mathbb{N}$ is such that $n > N$, we have $|a_n - L| < \frac{\epsilon}{2}$. Then for $n, m \in \mathbb{N}$ such that $n, m > N$, we have $|a_n - a_m| = |a_n - L + L - a_m| \leqslant |a_n - L| + |a_m - L| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$.
So the sequence $(a_n)_{n\in\mathbb{N}}$ is a Cauchy sequence.

(3) We have $a_n := \frac{(1-\frac{1}{m})(1+\frac{1}{m}+\cdots+\frac{1}{m^{n-1}})}{1-\frac{1}{m}} = \frac{1-\frac{1}{m^n}}{1-\frac{1}{m}}$ for all $n \in \mathbb{N}$. So we get $|a_n - \frac{1}{1-\frac{1}{m}}| = \frac{1}{m^n(1-\frac{1}{m})}$. Given a rational $\epsilon > 0$, let $N \in \mathbb{N}$ be such that $m^N > \frac{1}{\epsilon(1-\frac{1}{m})}$. Then for $n > N$, $|a_n - \frac{1}{1-\frac{1}{m}}| = \frac{1}{m^n(1-\frac{1}{m})} < \frac{1}{m^N(1-\frac{1}{m})} < \epsilon$.

(4) Suppose that $r < c$. Taking $\epsilon := \frac{c-r}{2} > 0$, there exists an $N' \in \mathbb{N}$ such that whenever $n \in \mathbb{N}$ satisfies $n > N'$, we have $|a_n - r| < \frac{c-r}{2}$. Hence we have $a_{N+N'} - r \leqslant |a_{N+N'} - r| < \frac{c-r}{2}$, and so $a_{N+N'} < \frac{c+r}{2} < \frac{c+c}{2} = c$, a contradiction.

Suppose that $r > d$. Taking $\epsilon := \frac{r-d}{2} > 0$, there exists an $N'' \in \mathbb{N}$ such that whenever $n \in \mathbb{N}$ satisfies $n > N''$, we have $|a_n - r| < \frac{r-d}{2}$. Hence we have $r - a_{N+N''} \leqslant |a_{N+N''} - r| < \frac{r-d}{2}$, and so $a_{N+N''} > \frac{r+d}{2} > \frac{d+d}{2} = d$, a contradiction.

(5) If $k \geqslant 3$, then $k! = 1 \cdot 2 \cdot 3 \cdot \ldots \cdot k \geqslant 1 \cdot 2 \cdot 2 \cdot \ldots \cdot 2 = 2^{k-1}$ and it holds for $k \in \{1, 2\}$ too. Denote the nth term of the given sequence by a_n. For $n > m \geqslant 1$, $|a_n - a_m| = \sum_{k=m+1}^{n} \frac{1}{k!} \leqslant \sum_{k=m+1}^{n} \frac{1}{2^{k-1}} = \frac{1}{2^m} \frac{1-\frac{1}{2^{n-m}}}{1-\frac{1}{2}} < \frac{1}{2^m} \frac{1}{\frac{1}{2}} < \frac{1}{2^{m-1}}$.
Given a rational $\epsilon > 0$, let $N \in \mathbb{N}$ be such that $2^N > \frac{1}{\epsilon}$. If $n > m > N$, then $|a_n - a_m| < \frac{1}{2^{m-1}} \leqslant \frac{1}{2^N} < \epsilon$.

(6) Suppose it converges in $\mathbb{Q}$ to $L \in \mathbb{Q}$. Then $L = \frac{p}{q}$ for some $p \in \mathbb{Z}$ and $q \in \mathbb{N}$. Also, we may choose $q \geqslant 2$ (by multiplying, if necessary, both p and q by 2). Denote the nth term of the sequence by a_n, and consider the sequence $(b_n)_{n\in\mathbb{N}}$ given by

$$b_n = a_{n+q} - \sum_{k=1}^{q} \frac{1}{k!}.$$

It follows that $(b_n)_{n\in\mathbb{N}}$ converges in $\mathbb{Q}$ to $L - \sum_{k=1}^{q} \frac{1}{k!}$.

Then

$$\begin{aligned} q!b_n &= \tfrac{1}{q+1} + \tfrac{1}{(q+1)(q+2)} + \cdots + \tfrac{1}{(q+1)\cdots(q+n)} \\ &\leqslant \tfrac{1}{3} + \tfrac{1}{3^2} + \cdots + \tfrac{1}{3^n} = \tfrac{1}{3} \cdot \tfrac{1-\frac{1}{3^n}}{1-\frac{1}{3}} < \tfrac{1}{3} \cdot \tfrac{1}{1-\frac{1}{3}} = \tfrac{1}{2}. \end{aligned}$$

Hence we obtain that for all $n \in \mathbb{N}$, we have $0 < \frac{1}{q+1} \leqslant q!b_n \leqslant \frac{1}{2}$.

So $0 < \frac{1}{q+1} \leqslant q!(L - \sum_{k=1}^{q} \frac{1}{k!}) =$ (an integer) $\leqslant \frac{1}{2} < 1$, a contradiction.

(7) Suppose, on the contrary, that $u_* \in \mathbb{Q}$ is a least upper bound of S. We claim that the sequence $(\frac{1}{1!} + \cdots + \frac{1}{n!})_{n\in\mathbb{N}}$ then converges in $\mathbb{Q}$ to u_* (and which will contradict the result from the previous part). Let ϵ be any positive rational number. Then $u := u_* - \epsilon < u_*$, and so u cannot be an upper bound of S. So there must exist an $a_N = \frac{1}{1!} + \cdots + \frac{1}{N!} \in S$ which prevents u from being an upper bound of S, that is, $u < a_N$. Then for all $n > N$, we have

$$\begin{aligned} &a_n > a_N > u = u_* - \epsilon, \quad \text{and} \\ &a_n \leqslant u_* < u_* + \epsilon. \end{aligned}$$

From the above, we get $-\epsilon < a_n - u_* < \epsilon$, that is, $|a_n - u_*| < \epsilon$ for all $n > N$. So $(\frac{1}{1!} + \cdots + \frac{1}{n!})_{n\in\mathbb{N}}$ converges in $\mathbb{Q}$ to u_*, a contradiction to the previously proved result from part (6).

(8) $\sim$ is an equivalence relation:

Reflexivity: If $(a_n)_{n\in\mathbb{N}} \in \mathscr{C}$, then $(a_n)_{n\in\mathbb{N}} \sim (a_n)_{n\in\mathbb{N}}$ since $(a_n - a_n)_{n\in\mathbb{N}}$ is the constant sequence $0, 0, 0, \cdots$ which is convergent in $\mathbb{Q}$ to $0 \in \mathbb{Q}$.

Symmetry: If $(a_n)_{n\in\mathbb{N}}, (b_n)_{n\in\mathbb{N}} \in \mathscr{C}$ and $(a_n)_{n\in\mathbb{N}} \sim (b_n)_{n\in\mathbb{N}}$, then we have $(a_n - b_n)_{n\in\mathbb{N}}$ converges to 0. Then $(b_n - a_n)_{n\in\mathbb{N}}$ converges to 0 too.

Transitivity: If $(a_n)_{n\in\mathbb{N}}, (b_n)_{n\in\mathbb{N}}, (c_n)_{n\in\mathbb{N}} \in \mathscr{C}$ and $(a_n)_{n\in\mathbb{N}} \sim (b_n)_{n\in\mathbb{N}}$, $(b_n)_{n\in\mathbb{N}} \sim (c_n)_{n\in\mathbb{N}}$, then $(a_n - b_n)_{n\in\mathbb{N}}$ converges to 0, and $(b_n - c_n)_{n\in\mathbb{N}}$ converges to 0. Then it follows that $(a_n - b_n + b_n - c_n)_{n\in\mathbb{N}} = (a_n - c_n)_{n\in\mathbb{N}}$ converges to 0 too. Thus $(a_n)_{n\in\mathbb{N}} \sim (c_n)_{n\in\mathbb{N}}$.

(9) If $q, q' \in \mathbb{Q}$, and $[(q)_{n\in\mathbb{N}}] = [(q')_{n\in\mathbb{N}}]$, then $(q)_{n\in\mathbb{N}} \in [(q)_{n\in\mathbb{N}}] = [(q')_{n\in\mathbb{N}}]$, so that $(q)_{n\in\mathbb{N}} \sim (q')_{n\in\mathbb{N}}$. Thus $(q - q')_{n\in\mathbb{N}}$ converges to 0. If $q \neq q'$, then with $\epsilon := \frac{|q-q'|}{2} > 0$, there exists an $N \in \mathbb{N}$ such that for all $n > N$, we have $0 < |q - q' - 0| < \epsilon = \frac{|q-q'|}{2}$, giving $2 < 1$, a contradiction. Thus $q = q'$. So the map $\mathbb{Q} \ni q \mapsto [(q)_{n\in\mathbb{N}}]$ is injective.

Chapter 2

Solution to Exercise 2.1

From (N1), in particular, $\sigma(1) \neq 1$. So $1 \in S$. If $n \in S$, then $\sigma(n) \neq n$. By the injectivity of σ, we then have that $\sigma(\sigma(n)) \neq \sigma(n)$. Thus $\sigma(n) \in S$. Consequently, $S = \mathbb{N}$ by (N3). So for all $n \in \mathbb{N} = S$, we have $\sigma(n) \neq n$.

Solution to Exercise 2.2

We have

$$\begin{aligned} 2 + 3 &= 2 + \sigma(2) = \sigma(2 + 2) = \sigma(2 + \sigma(1)) = \sigma(\sigma(2 + 1)) \\ &= \sigma(\sigma(\sigma(2))) = \sigma(\sigma(3)) = \sigma(4) = 5. \end{aligned}$$

Also,

$$\begin{aligned} 1 + 4 &= 1 + \sigma(3) = \sigma(1 + 3) = \sigma(1 + \sigma(2)) \\ &= \sigma(\sigma(1 + 2)) = \sigma(\sigma(1 + \sigma(1))) \\ &= \sigma(\sigma(\sigma(1 + 1))) = \sigma(\sigma(\sigma(\sigma(1)))) \\ &= \sigma(\sigma(\sigma(2))) = \sigma(\sigma(3)) = \sigma(4) = 5. \end{aligned}$$

Solution to Exercise 2.3

We have

$$\begin{aligned} 2 \times 3 &= 2 \times \sigma(2) = (2 \times 2) + 2 = (2 \times \sigma(1)) + 2 = ((2 \times 1) + 2) + 2 \\ &= (2 + 2) + 2 = (2 + \sigma(1)) + 2 = \sigma(2 + 1) + 2 = \sigma(\sigma(2)) + 2 \\ &= \sigma(3) + 2 = 4 + 2 = 4 + \sigma(1) = \sigma(4 + 1) = \sigma(\sigma(4)) = \sigma(5) = 6. \end{aligned}$$

Solution to Exercise 2.4

(1) $1 \in S$ because $1 + 1 = 1 + 1$. If $n \in S$, then $n + 1 = 1 + n$, and so

$$\begin{aligned} \sigma(n) + 1 &= (n + 1) + 1 && \text{(definition of addition)} \\ &= (1 + n) + 1 && \text{(since } n \in S\text{)} \\ &= 1 + (n + 1) && \text{(addition is associative)} \\ &= 1 + \sigma(n) && \text{(definition of addition).} \end{aligned}$$

Thus $\sigma(n) \in S$ too. By (N3), $S = \mathbb{N}$.

(2) $1 \in S'$ because $1 + n = n + 1$ for all $n \in \mathbb{N}$ by part (1) above.

If $m \in S'$, then for all $n \in \mathbb{N}$, $n + m = m + n$, and so for $n \in \mathbb{N}$, we have

$$\begin{aligned} \sigma(m) + n &= (m + 1) + n && \text{(definition of addition)} \\ &= m + (1 + n) && \text{(addition is associative)} \\ &= m + (n + 1) && \text{(by part (1))} \\ &= (n + 1) + m && \text{(since } m \in S'\text{)} \\ &= n + (1 + m) && \text{(addition is associative)} \\ &= n + (m + 1) && \text{(by part (1))} \\ &= n + \sigma(m) && \text{(definition of addition).} \end{aligned}$$

Thus $\sigma(m) \in S'$ too. By (N3), $S' = \mathbb{N}$.

Solution to Exercise 2.5

$1 \in S$ since otherwise $1 = 1 + m = m + 1 = \sigma(m)$ for some $m \in \mathbb{N}$, contradicting (N1). If $n \in S$, then $n + m \neq n$ for all $m \in \mathbb{N}$, and so $\sigma(n+m) \neq \sigma(n)$ by the injectivity of σ, giving $\sigma(n) + m = m + \sigma(n) = \sigma(m + n) = \sigma(n + m) \neq \sigma(n)$ for all $m \in \mathbb{N}$, and so $\sigma(n) \in S$ too. Thus $S = \mathbb{N}$ by (N3).

Solution to Exercise 2.6

Let $S = \{n \in \mathbb{N} : 1 \times n = n\}$. By definition $1 \times 1 = 1$, and so $1 \in S$.
Suppose $n \in S$, that is, $1 \times n = n$. Then we have

$$\begin{aligned} 1 \times \sigma(n) &= (1 \times n) + 1 && \text{(definition of multiplication)} \\ &= n + 1 && (n \in S) \\ &= \sigma(n) && \text{(definition of addition).} \end{aligned}$$

Thus $\sigma(n) \in S$ too. By (N3), $S = \mathbb{N}$. Hence $1 \times n = n$ for all $n \in \mathbb{N}$.

Solution to Exercise 2.7

Let $S = \{\ell \in \mathbb{N} : m \times (n + \ell) = (m \times n) + (m \times \ell) \text{ for all } m, n \in \mathbb{N}\}$. For all $m, n \in \mathbb{N}$, $m \times (n + 1) = (m \times n) + m = (m \times n) + (m \times 1)$, and so $1 \in S$.
If $\ell \in S$, then for all $m, n \in \mathbb{N}$,

$$\begin{aligned} m \times (n + \sigma(\ell)) &= m \times (n + (\ell + 1)) && \text{(addition definition)} \\ &= m \times ((n + \ell) + 1) && \text{(addition is associative)} \\ &= m \times \sigma(n + \ell) && \text{(addition definition)} \\ &= (m \times (n + \ell)) + m && \text{(multiplication definition)} \\ &= ((m \times n) + (m \times \ell)) + m && (\ell \in S) \\ &= (m \times n) + ((m \times \ell) + m) && \text{(addition is associative)} \\ &= (m \times n) + (m \times \sigma(\ell)) && \text{(multiplication definition)} \end{aligned}$$

and so $\sigma(\ell) \in S$ too. By (N3), $S = \mathbb{N}$.

Solution to Exercise 2.8

Let $S = \{\ell \in \mathbb{N} : (m \times n) \times \ell = m \times (n \times \ell) \text{ for all } m, n \in \mathbb{N}\}$.
For all $m, n \in \mathbb{N}$, $(m \times n) \times 1 = m \times n = m \times (n \times 1)$, and so $1 \in S$.
If $\ell \in S$, then for all $m, n \in \mathbb{N}$,

$$\begin{aligned} (m \times n) \times \sigma(\ell) &= ((m \times n) \times \ell) + (m \times n) && \text{(definition of multiplication)} \\ &= (m \times (n \times \ell)) + (m \times n) && (\ell \in S) \\ &= m \times ((n \times \ell) + n) && \text{(distributive law)} \\ &= m \times (n \times \sigma(\ell)) && \text{(definition of multiplication)} \end{aligned}$$

and so $\sigma(\ell) \in S$ too. By (N3), $S = \mathbb{N}$.

Solution to Exercise 2.9

In this solution, we use the notion of order in $\mathbb{N}$, which is covered in Section 2.4, p.20. We prove the claim using induction on n. There is nothing to prove if $n = 1, 2$. If $n = 3$, then there are only two possible bracketings

possible, namely, $a_1 \cdot (a_2 \cdot a_3)$ and $(a_1 \cdot a_2) \cdot a_3$, and these are equal since the associativity property is known to hold for $\cdot$. Now suppose that we have shown the claim for some $n \in \mathbb{N}$ such that $n \geqslant 3$. Let $a_1, \cdots, a_{n+1} \in S$. In any expression A with a given bracketing, consider the last time we combine elements using $\cdot$, and call the two 'factors' as A_1 and A_2, so that $A = A_1 \cdot A_2$, where A_1 involves, say, the elements $a_1, \cdots, a_k$, and A_2 involves the remaining elements $a_{k+1}, \cdots, a_{n+1}$, for some $k \in \{1, \cdots, n\}$. Note that A_2 has $n - k + 1 \leqslant n - 1 + 1 = n$ elements. As A_1 and A_2 each have $\leqslant n$ factors, by the induction hypothesis, we can rearrange the brackets within each of these factors to have them 'nested left to right' as follows:
$A_1 = (\ldots((a_1 \cdot a_2) \cdot a_3) \cdot \ldots \cdot a_k)$ and $A_2 = (\ldots((a_{k+1} \cdot a_{k+2}) \cdot a_{k+3}) \cdot \ldots \cdot a_{n+1})$.
We further split A_2 by separating its last factor, and write $A_2 = a \cdot a_{n+1}$, where $a := (\ldots((a_{k+1} \cdot a_{k+2}) \cdot a_{k+3}) \cdot \ldots \cdot a_n)$. Using the associativity property, we get $A = A_1 \cdot (a \cdot a_{n+1}) = (A_1 \cdot a) \cdot a_{n+1}$. As $A_1 \cdot a$ has $k + n - k = n$ elements, it follows from the induction hypothesis that $A_1 \cdot a = (\ldots((a_1 \cdot a_2) \cdot a_3) \cdot \ldots \cdot a_n)$.
Thus $A = (A_1 \cdot a) \cdot a_{n+1} = (\ldots((a_1 \cdot a_2) \cdot a_3) \cdot \ldots \cdot a_n) \cdot a_{n+1}$.

Solution to Exercise 2.10

The postfix notation for $(a+b) \times (c+d+e/f-g/h)$ is $ab+cdef/gh/-++\times$.
The usual notation for $ab \times cd/+efgh-/\times\times$ is $((a \times b)+c/d) \times (e \times f/(g-h))$.

Solution to Exercise 2.11

As $m < n$ and $n < k$, there exist $\ell, \ell' \in \mathbb{N}$ such that $n = m + \ell$ and $k = n + \ell'$.
Then $k = n + \ell' = (m + \ell) + \ell' = m + (\ell + \ell')$. Since $\ell + \ell' \in \mathbb{N}$, we get $m < k$.

Solution to Exercise 2.12

Since $m < n$, there exists an $\ell \in \mathbb{N}$ such that $n = m + \ell$.
Then $n + k = (m + \ell) + k = m + (\ell + k) = m + (k + \ell) = (m + k) + \ell$, using the associativity and commutativity of $+$. So $m + k < n + k$.
Moreover, using the distributive law, $n \times k = (m + \ell) \times k = (m \times k) + (\ell \times k)$, showing that $m \times k < n \times k$.

Solution to Exercise 2.13

Suppose that there exists such an element $m \in \mathbb{N}$ satisfying $n < m < n + 1$.
Then as $m > n$, there exists a $k \in \mathbb{N}$ such that $m = n + k$.

1° $k = 1$. Then $m = n + 1$. But we also have $m < n + 1$ (by our assumption above). This contradicts the trichotomy law.

2° $k \neq 1$. Then k has a predecessor: $k = \sigma(\ell)$ for some $\ell \in \mathbb{N}$. Then we have $m = n + k = n + \sigma(\ell) = n + (\ell + 1) = n + (1 + \ell) = (n + 1) + \ell$. So $m > n + 1$. But $m < n + 1$ (assumption above), contradicting the trichotomy law.

Solution to Exercise 2.14

Let $m \neq n$. By the trichotomy law, we must have that $m > n$ or $m < n$. We just show $m > n$ is impossible. The proof of the impossibility of $m < n$ is analogous (via the swap $m \leftrightarrow n$).

(1) If $m > n$, then by Exercise 2.12, $m + k > n + k$. By the trichtomy law, this is a contradiction to $m + k = n + k$.

(2) If $m > n$, then by Exercise 2.12, $m \times k > n \times k$. By the trichtomy law, this is a contradiction to $m \times k = n \times k$.

Solution to Exercise 2.15

If $n \neq 1$, then n has a predecessor: $n = \sigma(m)$ for some element $m \in \mathbb{N}$, and so $n = \sigma(m) = m + 1 = 1 + m$, showing $n > 1$. Thus either $n = 1$ or (if $n \neq 1$ then) $n > 1$. Hence $1 \leqslant n$.

Solution to Exercise 2.16

As $m < n$, we have $n = m + k$ for some $k \in \mathbb{N}$.

1° If $k = 1$, then $n = m + 1$. So $m + 1 \leqslant n$.

2° If $k \neq 1$, then k has a predecessor: $\sigma(\ell) = k$ for some $\ell \in \mathbb{N}$, and so we have $n = m + k = m + \sigma(\ell) = m + (\ell + 1) = m + (1 + \ell) = (m + 1) + \ell$, showing $m + 1 < n$. So $m + 1 \leqslant n$.

Solution to Exercise 2.17

(1)⇒(2): Suppose $m < n$. Then there exists an $\ell \in \mathbb{N}$ such that $m + \ell = n$. Let $k \in \mathbb{N}$. Adding k on both sides of $m + \ell = n$, we get $(m + \ell) + k = n + k$, and so $(m + k) + \ell = m + (k + \ell) = m + (\ell + k) = (m + \ell) + k = n + k$. Hence $m + k < n + k$.

(2)⇒(3): Obvious.

(3)⇒(1): Let $k \in \mathbb{N}$ be such that $m + k < n + k$. We want to show that $m < n$. Suppose $\neg(m < n)$. Then by the trichotomy law, $m \geqslant n$.

1° $m = n$. Adding k, we get $m + k = n + k$, contradicting $m + k < n + k$.

2° $m > n$. Then adding k to both sides gives $m + k > n + k$ (thanks to the implication (1)⇒(2) proved above). But this contradicts $m + k < n + k$.

So $\neg(m < n)$ is impossible, and hence we have $m < n$.

Solution to Exercise 2.18

(If part:) Suppose $m = n = 1$. Then $m \times n = m \times 1 = m = 1$.

(Only if part:) Let $m \times n = 1$. Suppose that $n \neq 1$. Then there exists a $k \in \mathbb{N}$ such that $\sigma(k) = n$. So $1 = m \times n = m \times \sigma(k) = m \times k + m$. Hence $m < 1$. But we know that as $m \in \mathbb{N}$, by Exercise 2.15, p.21, $m \geqslant 1$. So we have arrived at a contradiction to the trichotomy law. Hence $n \neq 1$ is not possible, that is, $n = 1$. Hence $1 = m \times n = m \times 1 = m$. So $m = 1$ too.

Solution to Exercise 2.19

If $m=1$, then $n\times m=n\times 1=n$. So $n\times m\geqslant n$.
If $m\neq 1$, then m has a predecessor, that is, there is some $\ell\in\mathbb{N}$ such that $\sigma(\ell)=m$, and so $n\times m=n\times\sigma(\ell)=(n\times\ell)+n$, showing that $n\times m>n$ (and so also $n\times m\geqslant n$).

Solution to Exercise 2.20

Let $S:=\{n\in\mathbb{N}: P(n) \text{ is true for all } k\in\mathbb{N} \text{ such that } k\leqslant n\}$.

Then $1\in S$: if $k\in\mathbb{N}$ and $k\leqslant 1$, then $k=1$ by Exercise 2.15, p.21, and the trichotomy law, and $P(1)$ is true.

Suppose that $n\in S$. Then $P(k)$ is true for all $k\leqslant n$, and so also $P(n+1)$ is true. Now let $k\in\mathbb{N}$ be such that $k\leqslant\sigma(n)=n+1$.

1° If $k\leqslant n$, then since $n\in S$, we know that $P(k)$ is true.

2° If $k>n$, then $k\geqslant n+1$ by Exercise 2.16, p.21. Since also $k\leqslant n+1$, $k=n+1$ by the trichotomy law. Thus $P(k)$ is true (as $P(n+1)$ is true).

So $\sigma(n)\in S$. By (N3), $S=\mathbb{N}$. So $P(n)$ is true for all $n\in\mathbb{N}$.

Solution to Exercise 2.21

Suppose that $u\in\mathbb{N}$ is a greatest element of $\mathbb{N}$. Then as $\sigma(u)\in\mathbb{N}$, we have $\sigma(u)\leqslant u$, i.e., $\sigma(u)=u$ or $\sigma(u)<u$. But neither of these are possible by the trichotomy law, as we know that $\sigma(u)>u$ (since $\sigma(u)=u+1$).

Solution to Exercise 2.22

Let $\ell,\ell'\in S$ be least elements. As ℓ is a least element of S, and since $\ell'\in S$, we have $\ell\leqslant\ell'$, i.e.,

$$\ell<\ell' \text{ or } \ell=\ell'. \qquad (\star)$$

Suppose that $\ell<\ell'$. As ℓ' is a least element of S, and since $\ell\in S$, $\ell'\leqslant\ell$, that is, $\ell'<\ell$ or $\ell'=\ell$. But by the trichotomy law, and the assumption that $\ell<\ell'$, none of the cases $\ell'<\ell$ or $\ell'=\ell$ are possible. So our assumption that $\ell<\ell'$ is not possible. By $(\star)$, $\ell=\ell'$.

Solution to Exercise 2.23

Let $\mathbb{N}$ be a finite set. Since $1\in\mathbb{N}$, $\mathbb{N}$ is not empty. Thus there exists an $n\in\mathbb{N}$ and a bijection $f:\mathbb{N}\to\{1,\cdots,n\}$. The map $i:\{1,\cdots,\sigma(n)\}\to\mathbb{N}$ given by $i(k)=k$ for all $k\in\{1,\cdots,\sigma(n)\}$ is injective. Since f and i are injective, their composition $f\circ i:\{1,\cdots,\sigma(n)\}\to\{1,\cdots,n\}$ is injective. By Lemma 2.2, p.23, $\sigma(n)\leqslant n$, a contradiction to the trichotomy law, since we know that $\sigma(n)=n+1>n$.

Solution to Exercise 2.24

(1)⇒(2): Let $f:S\to\{1,\cdots,n+1\}$ be bijective. Suppose $f(a)=n+1$. Let $a'\in S$ be such that $f(a')=1$. Then $f(a')=1\neq n+1=f(a)$, and so $a'\neq a$.

Thus $a' \in S\backslash\{a\}$ and $S\backslash\{a\} \neq \varnothing$. Define $g : S\backslash\{a\} \to \{1, \cdots, n\}$ by $g(x) = f(x)$ for $x \in S\backslash\{a\}$. Then g is well-defined, as the range of values assumed by g does not include $n+1$: by the injectivity of f, we have, for all $x \in S$ with $x \neq a$, $f(x) \neq f(a) = n+1$. Also, g is injective, since if $x, x' \in S\backslash\{a\}$ and $g(x) = g(x')$, then $f(x) = f(x')$, and by the injectivity of f, $x = x'$. Finally, g is surjective, since for any $k \in \{1, \cdots, n\}$, there exists an $x \in S$ such that $f(x) = k$, and this $x \neq a$ (otherwise $n+1 = f(a) = f(x) = k \in \{1, \cdots, n\}$, a contradiction).

Now let $f(a) \in \{1, \cdots, n\}$, and $a' \in S$ be such that $f(a') = n+1$. Then $a' \neq a$ (otherwise $f(a) = f(a') = n+1 \notin \{1, \cdots, n\}$, a contradiction). In particular, $S\backslash\{a\} \neq \varnothing$ since it contains a'. By the injectivity of f, a' is the *only* element in S whose image is $n+1$. Define $g : S\backslash\{a\} \to \{1, \cdots, n\}$ by $g(x) = f(x)$ if $x \in S\backslash\{a, a'\}$, and $g(a') = f(a)$. Then g is well-defined as the range of values taken by g is contained in $\{1, \cdots, n\}$. Moreover, g is injective, since if $x, x' \in S\backslash\{a, a'\}$ are such that $g(x) = g(x')$, then $f(x) = f(x')$ and so $x = x'$ by the injectivity of f, and also, there is no $x'' \in S\backslash\{a, a'\}$ such that $g(x'') = g(a')$ (since otherwise $f(x'') = f(a)$, which implies $x'' = a$, a contradiction). Finally g is surjective: we have $g(a') = f(a)$, and for $k \in \{1, \cdots, n\}\backslash\{f(a)\}$, if $a'' \in S$ is such that $f(a'') = k$, then $a'' \neq a$ (otherwise $k = f(a'') = f(a)$, a contradiction), and $g(a'') = f(a'') = k$.

(2)⇒(1): Let $S\backslash\{a\} \neq \varnothing$ and $g : S\backslash\{a\} \to \{1, \cdots, n\}$ be bijective. Define $f : S \to \{1, \cdots, n+1\}$ by $f(x) = g(x)$ if $x \in S\backslash\{a\}$, and $f(a) = n+1$. If $x, x' \in S\backslash\{a\}$ are such that $f(x) = f(x')$, then $g(x) = g(x')$ and so $x = x'$. If $x \in S\backslash\{a\}$, then $f(x) = g(x) \in \{1, \cdots, n\}$, and since $n+1 \notin \{1, \cdots, n\}$, $f(x) \neq n+1 = f(a)$. Hence f is injective. Moreover, f is surjective: if $k \in \{1, \cdots, n\}$, and $a' \in S\backslash\{a\}$ is such that $g(a') = k$, then $f(a') = g(a') = k$, and $f(a) = n+1$.

Solution to Exercise 2.25

If $A = \varnothing$, then the only subset of A is the empty set, which is finite.

Let S be the set of all $n \in \mathbb{N}$ such that each subset B of a finite *nonempty* set A of cardinal number n is finite, and if B is nonempty, then the cardinal number of B is at most n. We want to show that $S = \mathbb{N}$.

We claim that $1 \in S$. Indeed, if A has cardinal number 1, then there is a bijection $f : A \to \{1\}$, and so A is a singleton set $\{a\}$, whose only subsets are $\{a\} = A$ and the empty set, and in either case, B is finite. Moreover, if $B = A = \{a\}$, then the cardinal number of B is 1 which is equal to the cardinal number of A.

Next, suppose that some natural number n belongs to S. Let A be a finite nonempty set with cardinal number $n+1$, and B be a subset of A. If $B=\varnothing$, then B is finite. Now suppose B is nonempty. If $B=A$, then B is finite, and has cardinality equal to that of A, and there is nothing to prove. Thus we assume that $B \subsetneq A$, that is, there exists an $a \in A\backslash B$. Since there exists a bijection $f : A \to \{1, \cdots, n+1\}$, by Exercise 2.24, p.24, $A\backslash\{a\} \neq \varnothing$ and there exists an bijective map $g : A\backslash\{a\} \to \{1, \cdots, n\}$. So $A\backslash\{a\}$ has cardinal number n. But $B \subset A\backslash\{a\}$, and so by the induction hypothesis, B is a finite set, and as B was nonempty, we have that the cardinal number m of B satisfies $m \leqslant n$. But $n < n+1$, and so we get $m < n+1$. By (N3), $S=\mathbb{N}$, as wanted.

Solution to Exercise 2.26

If A or B is the empty set, then $A \cup B = A$, which is finite, and we are done. Suppose that neither A nor B is the empty set. Let the cardinal numbers of A, B be n_A, n_B, respectively, and let $f_A : A \to \{1, \cdots, n_A\}$ and $f_B : B \to \{1, \cdots, n_B\}$ be bijections. Define $f : A \cup B \to \{1, \cdots, n_A + n_B\}$ by

$$f(x) = \begin{cases} f_A(x) & \text{if } x \in A, \\ n_A + f_B(x) & \text{if } x \in B. \end{cases}$$

Then f is well-defined, as $A \cap B = \varnothing$, and $f_B(x) \leqslant n_B$. Also, f is injective:

1° If $a, a' \in A$, and $f(a) = f(a')$, then $f_A(a) = f_A(a')$ implies $a = a'$ since f_A is bijective and in particular injective.

2° If $b, b' \in B$, and $f(b) = f(b')$, then $n_A + f_B(b) = n_A + f_A(b')$ implies that $f_B(b) = f_A(b')$ by the cancellation law in $\mathbb{N}$, giving further $b = b'$, since f_B is bijective and in particular injective.

3° If $a \in A$ and $b \in B$, then $f(a) = f_A(a) \leqslant n_A < n_A + f_B(b) = f(b)$, and so $f(a) \neq f(b)$ by the trichotomy law.

Moreover, we claim that f is surjective. If $k \in \{1, \cdots, n_A\}$, then there exists an $a \in A$ such that $f_A(a) = k$, and so $f(a) = f_A(a) = k$. If $n_A < k \leqslant n_A + n_B$, then the first inequality implies $k = n_A + \ell$ for some $\ell \in \mathbb{N}$, and the second inequality implies $\ell \leqslant n_B$ (otherwise, if $\ell > n_B$, then we get the contradiction that $k = n_A + \ell > n_A + n_B$). Hence $\ell \in \{1, \cdots, n_B\}$. By the surjectivity of f_B, there exists a $b \in B$ such that $f_B(b) = \ell$. Hence $f(b) = n_A + f_B(b) = n_A + \ell = k$. So f is bijective, and so $A \cup B$ is a finite set with cardinal number $n_A + n_B$.

In the general case (when A, B may not be disjoint), we have that $A \cup B = A \cup (B\backslash A)$, and we note that $A \cap (B\backslash A) = \varnothing$. So to show that $A \cup B$ is finite, in light of the previous part, it is enough to show that $B\backslash A$ is finite (since A is already assumed finite). But $B\backslash A$ is a subset of the finite set B, and hence is finite by Exercise 2.25, p.24.

Solution to Exercise 2.27

(1)⇒(2): Let $f : \{1, \cdots, n\} \to A$ be surjective. Given $a \in A$, the inverse image $f^{-1}(\{a\})$ is a nonempty subset of $\{1, \cdots, n\} \subset \mathbb{N}$ because f is surjective, and so by the well-ordering principle, possesses a (unique) least element, which we denote by $g(a)$. Thus we obtain a map $A \ni a \mapsto g(a) \in \{1, \cdots, n\}$. If $a, a' \in A$ are distinct elements, then $f^{-1}(\{a\})$ and $f^{-1}(\{a'\})$ are disjoint (otherwise, if x is a common element, then $a = f(x) = a'$, a contradiction), and so their least elements are different. Hence g is injective.

(2)⇒(3): Let $g : A \to \{1, \cdots, n\}$ be injective. Then $\widetilde{g} : A \to \operatorname{ran} g$ obtained by changing the codomain of g is bijective. But $\operatorname{ran} g$ is a nonempty subset of the finite set $\{1, \cdots, n\}$ with cardinal number n. By Exercise 2.26, p.24, $\operatorname{ran} g$ is a finite set with cardinal number $m \leqslant n$. Thus there is a bijection $h : \operatorname{ran} g \to \{1, \cdots, m\}$. The composition $h \circ \widetilde{g} : A \to \{1, \cdots, m\}$ of the bijective maps $h, \widetilde{g}$ is bijective. This shows that A is finite and has cardinal number (equal to m) which is at most n.

(3)⇒(1): Let A be finite and have cardinal number $m \leqslant n$. Then there exists a bijection $h : A \to \{1, \cdots, m\}$. Its inverse, $h^{-1} : \{1, \cdots, m\} \to A$ is also a bijection. If $m = n$, then $f := h^{-1}$ serves as the wanted surjection $f : \{1, \cdots, n\} \to A$. Suppose that $m < n$. Define $f : \{1, \cdots, n\} \to A$ by $f(k) = h^{-1}(k)$ for $1 \leqslant k \leqslant m$, and $f(k) = h^{-1}(m)$ for $m < k \leqslant n$. Then f is surjective since h^{-1} already was surjective, and f extends h^{-1}.

Solution to Exercise 2.28

Let B be finite. Since $B \neq \varnothing$, there exists bijective map $h : B \to \{1, \cdots, m\}$, where $m \in \mathbb{N}$ is the cardinal number of B. As $h^{-1} : \{1, \cdots, m\} \to B$ is a bijection, and in particular a surjection, and $f : B \to A$ is a surjection, their composition $f \circ h^{-1} : \{1, \cdots, m\} \to A$ is a surjection, implying by Exercise 2.27, p.24, that A is finite, a contradiction. So B is infinite.

If $\operatorname{ran} g$ $(\neq \varnothing)$ is finite, then there exists a bijection $k : \operatorname{ran} g \to \{1, \cdots, \ell\}$ for some $\ell \in \mathbb{N}$. Restricting the codomain of g to $\operatorname{ran} g$, we get a bijection $\widetilde{g} : A \to \operatorname{ran} g$. The composition $k \circ \widetilde{g} : A \to \{1, \cdots, \ell\}$ of the bijections k and $\widetilde{g}$ is a bijection, implying A is finite, a contradiction. Thus $\operatorname{ran} g$ is infinite. As the infinite set $\operatorname{ran} g$ is a subset of C, by Exercise 2.25, p.24, also C is infinite.

Chapter 3

Solution to Exercise 3.1

If $[(1,\sigma(n))]=[(1,\sigma(m))]$ for some $n,m\in\mathbb{N}$, then $(1,\sigma(n))\in[(1,\sigma(m))]$. So $(1,\sigma(n))\sim(1,\sigma(m))$, that is, $1+\sigma(m)=1+\sigma(n)$. By the addition cancellation rule (Exercise 2.14, p.21), $\sigma(n)=\sigma(m)$. By the injectivity of the successor function σ, we obtain $n=m$. So the given map is injective.

Solution to Exercise 3.2

$(m+k,n+k)\sim(m,n)$ as $(m+k)+n=(n+k)+m$ using the associativity and commutativity of addition in $\mathbb{N}$. Thus $[(m+k,n+k)]=[(m,n)]$.

Solution to Exercise 3.3

Let $\boldsymbol{a}=[(m,n)],\boldsymbol{b}=[(k,\ell)],\boldsymbol{c}=[(p,q)]\in\mathbb{Z}$. Then using the associativity of addition in $\mathbb{N}$, we get the associativity of addition in $\mathbb{Z}$:

$$\begin{aligned}\boldsymbol{a}+(\boldsymbol{b}+\boldsymbol{c}) &= [(m,n)]+([(k,\ell)]+[(p,q)]) = [(m,n)]+[(k+p,\ell+q)]\\ &= [(m+(k+p),n+(\ell+q))] = [((m+k)+p,(n+\ell)+q)]\\ &= [(m+k,n+\ell)]+[(p,q)] = ([(m,n)]+[(k,\ell)])+[(p,q)]\\ &= (\boldsymbol{a}+\boldsymbol{b})+\boldsymbol{c}.\end{aligned}$$

Let $\boldsymbol{a}=[(m,n)],\boldsymbol{b}=[(k,\ell)]\in\mathbb{Z}$. Then using the commutativity of addition in $\mathbb{N}$, we get the commutativity of addition in $\mathbb{Z}$:

$$\begin{aligned}\boldsymbol{a}+\boldsymbol{b} &= [(m,n)]+[(k,\ell)] = [(m+k,n+\ell)] = [(k+m,\ell+n)]\\ &= [(k,\ell)]+[(m,n)] = \boldsymbol{b}+\boldsymbol{a}.\end{aligned}$$

Solution to Exercise 3.4

Let $\boldsymbol{a}+\boldsymbol{c}=\boldsymbol{b}+\boldsymbol{c}$. Thus

$$\begin{aligned}\boldsymbol{a} &= \boldsymbol{a}+\boldsymbol{0} = \boldsymbol{a}+(\boldsymbol{c}+(-\boldsymbol{c})) = (\boldsymbol{a}+\boldsymbol{c})+(-\boldsymbol{c})\\ &= (\boldsymbol{b}+\boldsymbol{c})+(-\boldsymbol{c}) = \boldsymbol{b}+(\boldsymbol{c}+(-\boldsymbol{c})) = \boldsymbol{b}+\boldsymbol{0} = \boldsymbol{b}.\end{aligned}$$

Solution to Exercise 3.5

Let $\boldsymbol{a}=[(k,\ell)]$, $\boldsymbol{b}=[(m,n)]$, $\boldsymbol{c}=[(p,q)]$, where $k,\ell,m,n,p,q\in\mathbb{N}$.
Associativity: We have

$$\begin{aligned}\boldsymbol{a}\cdot(\boldsymbol{b}\cdot\boldsymbol{c}) &= [(k,\ell)]\cdot([(m,n)]\cdot[(p,q)]) = [(k,\ell)]\cdot[(mq+np,\ mp+nq)]\\ &= [(k(mp+nq)+\ell(mq+np),\ k(mq+np)+\ell(mp+nq))]\\ &= [(kmp+knq+\ell mq+\ell np,\ kmq+knp+\ell mp+\ell nq)], \qquad (\star)\\ (\boldsymbol{a}\cdot\boldsymbol{b})\cdot\boldsymbol{c} &= ([(k,\ell)]\cdot[(m,n)])\cdot[(p,q)] = [(kn+\ell m,\ km+\ell n)]\cdot[(p,q)]\\ &= [((kn+\ell m)q+(km+\ell n)p,\ (kn+\ell m)p+(km+\ell n)q)]\\ &= [(knq+\ell mq+kmp+\ell np,\ knp+\ell mp+kmq+\ell nq)]. \qquad (\star\star)\end{aligned}$$

Comparing the right-hand sides of $(\star)$ and $(\star\star)$, we conclude $\boldsymbol{a}(\boldsymbol{bc})=(\boldsymbol{ab})\boldsymbol{c}$.

Commutativity: We have

$$\begin{aligned}\boldsymbol{a}\cdot\boldsymbol{b} &= [(k,\ell)]\cdot[(m,n)] = [(kn+\ell m, km+\ell n)]\\ &= [(m\ell+nk, mk+n\ell)] = [(m,n)]\cdot[(k,\ell)] = \boldsymbol{b}\cdot\boldsymbol{a}.\end{aligned}$$

Distributivity: We have

$$\begin{aligned}\boldsymbol{a}\cdot(\boldsymbol{b}+\boldsymbol{c}) &= [(k,\ell)]\cdot([(m,n)]+[(p,q)]) = [(k,\ell)]\cdot[(m+p,\ n+q)]\\ &= [(k(n+q)+\ell(m+p),\ k(m+p)+\ell(n+q))]\\ &= [(kn+kq+\ell m+\ell p,\ km+kp+\ell n+\ell q)], \qquad (*)\\ \boldsymbol{a}\cdot\boldsymbol{b}+\boldsymbol{a}\cdot\boldsymbol{c} &= [(k,\ell)]\cdot[(m,n)]+[(k,\ell)]\cdot[(p,q)]\\ &= [(kn+\ell m,\ km+\ell n)]+[(kq+\ell p,\ kp+\ell q)]\\ &= [(kn+\ell m+kq+\ell p,\ km+\ell n+kp+\ell q)]. \qquad (**)\end{aligned}$$

Comparing the right-hand sides of $(*)$ and $(**)$, we get $\boldsymbol{a}\cdot(\boldsymbol{b}+\boldsymbol{c})=\boldsymbol{a}\cdot\boldsymbol{b}+\boldsymbol{a}\cdot\boldsymbol{c}$.

Solution to Exercise 3.6

If $\boldsymbol{a}=[(m,n)]$, where $m,n\in\mathbb{N}$. Then $-\boldsymbol{a}=[(n,m)]$. Also, we have that $\mathbf{1}=[(1,2)]$, and so $-\mathbf{1}=[(2,1)]$. Thus

$$\begin{aligned}(-\mathbf{1})\cdot\boldsymbol{a} &= [(2,1)]\cdot[(m,n)] = [(2n+m, 2m+n)]\\ &= [(n+(m+n), m+(m+n))] = [(n,m)] = -\boldsymbol{a}.\end{aligned}$$

Solution to Exercise 3.7

Suppose $\boldsymbol{a}=\mathbf{0}=[(1,1)]$ and $\boldsymbol{b}=[(m,n)]$, where $m,n\in\mathbb{N}$. We have

$$\begin{aligned}\boldsymbol{a}\cdot\boldsymbol{b} &= [(1,1)]\cdot[(m,n)] = [(1n+1m, 1m+1n)]\\ &= [(n+m, m+n)] = [(1,1)] = \mathbf{0}.\end{aligned}$$

The case when $\boldsymbol{b}=\mathbf{0}$ follows from the above by using the commutativity of multiplication.

Solution to Exercise 3.8

Consider for example the 'ramp' functions

$$f(x)=\left\{\begin{array}{ll}0 & \text{if } 0\leqslant x\leqslant\frac{1}{2}\\ x-\frac{1}{2} & \text{if } \frac{1}{2}\leqslant x\leqslant 1\end{array}\right\} \text{ and } g(x)=\left\{\begin{array}{ll}\frac{1}{2}-x & \text{if } 0\leqslant x\leqslant\frac{1}{2}\\ 0 & \text{if } \frac{1}{2}\leqslant x\leqslant 1\end{array}\right\}$$

Then f,g are continuous on $[0,1]$, and for $x\in[0,1]$,

$$(f\cdot g)(x)=\left\{\begin{array}{ll}0\cdot(\frac{1}{2}-x) & \text{if } 0\leqslant x\leqslant\frac{1}{2}\\ (x-\frac{1}{2})\cdot 0 & \text{if } \frac{1}{2}\leqslant x\leqslant 1\end{array}\right\}=0=\mathbf{0}(x).$$

There are of course other examples. The precise shape is not important, as long as the (nonempty) 'support' sets (places where the functions are nonzero) are contained in the 'zero' sets (places where the functions are zero) of the other function.

Solution to Exercise 3.9

Adding $\boldsymbol{a}\cdot(-\boldsymbol{c})$ to both sides of $\boldsymbol{a}\cdot\boldsymbol{b}=\boldsymbol{a}\cdot\boldsymbol{c}$, we obtain

$$\boldsymbol{a}\cdot\boldsymbol{b}+\boldsymbol{a}\cdot(-\boldsymbol{c}) = \boldsymbol{a}\cdot\boldsymbol{c}+\boldsymbol{a}\cdot(-\boldsymbol{c}).$$

It follows from distributivity that $\boldsymbol{a}\cdot(\boldsymbol{b}+(-\boldsymbol{c}))=\boldsymbol{a}\cdot(\boldsymbol{c}+(-\boldsymbol{c}))=\boldsymbol{a}\cdot\boldsymbol{0}=\boldsymbol{0}$. As $\boldsymbol{a}\neq\boldsymbol{0}$, it follows from Theorem 3.8, p.31, that $\boldsymbol{b}+(-\boldsymbol{c})=\boldsymbol{0}$. Adding $\boldsymbol{c}$ on both sides, and using the associativity property of $+$, we get

$$\boldsymbol{b}=\boldsymbol{b}+\boldsymbol{0}=\boldsymbol{b}+(-\boldsymbol{c}+\boldsymbol{c})=(\boldsymbol{b}+(-\boldsymbol{c}))+\boldsymbol{c}=\boldsymbol{0}+\boldsymbol{c}=\boldsymbol{c}.$$

Solution to Exercise 3.10

(Only if part:) Let $m<n$. Then $n=m+k$ for some $k\in\mathbb{N}$. By Theorem 3.3, p.27, we have that $[(1,\sigma(n))]=[(1,\sigma(m+k))]=[(1,\sigma(m))]+[(1,\sigma(k))]$. So $[(1,\sigma(m))]<[(1,\sigma(n))]$.

(If part:) Suppose that $[(1,\sigma(m))]<[(1,\sigma(n))]$. Then there exists a $k\in\mathbb{N}$ such that $[(1,\sigma(n))]=[(1,\sigma(m))]+[(1,\sigma(k))]$. Again by Theorem 3.3, the right-hand side is $[(1,\sigma(m+k))]$, and by the injectivity of the map $\mathbb{N}\ni q\mapsto[(1,\sigma(q))]\in\mathbb{Z}$, $m+k=n$, that is, $m<n$.

Solution to Exercise 3.11

Since $\boldsymbol{a}<\boldsymbol{b}$, there exists an $m\in\mathbb{N}$ such that $\boldsymbol{b}=\boldsymbol{a}+[(1,\sigma(m))]$.
As $\boldsymbol{b}<\boldsymbol{c}$, there exists an $n\in\mathbb{N}$ such that $\boldsymbol{c}=\boldsymbol{b}+[(1,\sigma(n))]$. Hence

$$\begin{aligned}\boldsymbol{c}&=\boldsymbol{b}+[(1,\sigma(n))]=(\boldsymbol{a}+[(1,\sigma(m))])+[(1,\sigma(n))]\\&=\boldsymbol{a}+([(1,\sigma(m))]+[(1,\sigma(n))])=\boldsymbol{a}+[(1,\sigma(m+n))],\end{aligned}$$

showing $\boldsymbol{a}<\boldsymbol{c}$. Here the last equality follows from Theorem 3.3, p.27.

Solution to Exercise 3.12

As $\boldsymbol{a}<\boldsymbol{b}$, we have $\boldsymbol{b}=\boldsymbol{a}+[(1,\sigma(m))]$ for some $m\in\mathbb{N}$.
So $\boldsymbol{b}+\boldsymbol{c}=(\boldsymbol{a}+[(1,\sigma(m))])+\boldsymbol{c}=(\boldsymbol{a}+\boldsymbol{c})+[(1,\sigma(m))]$. Thus $\boldsymbol{a}+\boldsymbol{c}<\boldsymbol{b}+\boldsymbol{c}$.

Solution to Exercise 3.13

As $\boldsymbol{a}<\boldsymbol{b}$, we have $\boldsymbol{b}=\boldsymbol{a}+[(1,\sigma(m))]$ for some $m\in\mathbb{N}$. Since $\boldsymbol{c}>\boldsymbol{0}$, there exists an $n\in\mathbb{N}$ such that $\boldsymbol{c}=\boldsymbol{0}+[(1,\sigma(n))]=[(1,\sigma(n))]$. Hence

$$\begin{aligned}\boldsymbol{b}\cdot\boldsymbol{c}&=(\boldsymbol{a}+[(1,\sigma(m))])\cdot\boldsymbol{c}\\&=\boldsymbol{a}\cdot\boldsymbol{c}+[(1,\sigma(m))]\cdot\boldsymbol{c}&&\text{(distributive law)}\\&=\boldsymbol{a}\cdot\boldsymbol{c}+[(1,\sigma(m))]\cdot[(1,\sigma(n))]\\&=\boldsymbol{a}\cdot\boldsymbol{c}+[(1,\sigma(mn))]&&\text{(Theorem 3.7, p.30).}\end{aligned}$$

Consequently, $\boldsymbol{a}\cdot\boldsymbol{c}<\boldsymbol{b}\cdot\boldsymbol{c}$.

Solution to Exercise 3.14

Let there exist such an integer $\boldsymbol{b}\in\mathbb{Z}$. Then there exist $m,n\in\mathbb{N}$ such that $\boldsymbol{b}=\boldsymbol{a}+[(1,\sigma(m))]$ and $\boldsymbol{a}+\boldsymbol{1}=\boldsymbol{b}+[(1,\sigma(n))]$. So

$$\begin{aligned}\boldsymbol{a}+[(1,2)]&=\boldsymbol{a}+\boldsymbol{1}=\boldsymbol{b}+[(1,\sigma(n))]\\&=(\boldsymbol{a}+[(1,\sigma(m))])+[(1,\sigma(n))]\\&=\boldsymbol{a}+([(1,\sigma(m))]+[(1,\sigma(n))])\\&=\boldsymbol{a}+[(1,\sigma(m+n))].\end{aligned}$$

By adding $-\boldsymbol{a}$ to both sides, we get $[(1,\sigma(1)] = [(1,2)] = [(1,\sigma(m+n))]$. By Exercise 3.1, p.26, $1 = m+n$. Now if $n=1$, we get $1 = m+1 = \sigma(m)$, which is impossible by (N1). So $n \neq 1$. But then n has a predecessor, say $\ell \in \mathbb{N}$. Then $1 = m+n = m+\sigma(\ell) = \sigma(m+\ell)$, which is again impossible by (N1). So our assumption about the existence of $\boldsymbol{b}$ is false, i.e., there is no $\boldsymbol{b} \in \mathbb{Z}$ such that $\boldsymbol{a} < \boldsymbol{b} < \boldsymbol{a}+\mathbf{1}$.

Solution to Exercise 3.15

(1)⇒(2): Let $\boldsymbol{a}\cdot\boldsymbol{b} > 0$. If $\boldsymbol{a}=\mathbf{0}$ or $\boldsymbol{b}=\mathbf{0}$, then $\boldsymbol{a}\cdot\boldsymbol{b}=\mathbf{0}$ by Exercise 3.7, p.31, and contradicting the trichotomy law. So $\neg(\boldsymbol{a}=\mathbf{0})$ and $\neg(\boldsymbol{b}=\mathbf{0})$. By the trichotomy law, we have the following four cases:

1° $\boldsymbol{a}>\mathbf{0}$ and $\boldsymbol{b}>\mathbf{0}$.

2° $\boldsymbol{a}>\mathbf{0}$ and $\boldsymbol{b}<\mathbf{0}$.

3° $\boldsymbol{a}<\mathbf{0}$ and $\boldsymbol{b}>\mathbf{0}$.

4° $\boldsymbol{a}<\mathbf{0}$ and $\boldsymbol{b}<\mathbf{0}$.

We will show that the cases 2°, 3° are impossible, leaving only cases 1° or 4° as the only valid possibilities. In other words, (2) holds.

Proof that the cases 2°, 3° are impossible:

2° $\boldsymbol{a}>\mathbf{0}$ and $\boldsymbol{b}<\mathbf{0}$. Then we get $\mathbf{0} = \mathbf{0}\cdot\boldsymbol{a} > \boldsymbol{b}\cdot\boldsymbol{a}$ by Exercise 3.13, p.32, contradicting the trichotomy law. So this case is impossible.

3° $\boldsymbol{a}<\mathbf{0}$ and $\boldsymbol{b}>\mathbf{0}$. Then $\boldsymbol{a}\cdot\boldsymbol{b} < \mathbf{0}\cdot\boldsymbol{b} = \mathbf{0}$ by Exercise 3.13, contradicting the trichotomy law. So this case is impossible.

(2)⇒(1): Let $(\boldsymbol{a}>\mathbf{0}$ and $\boldsymbol{b}>\mathbf{0})$ or $(\boldsymbol{a}<\mathbf{0}$ and $\boldsymbol{b}<\mathbf{0})$.

If $\boldsymbol{a}>\mathbf{0}$ and $\boldsymbol{b}>\mathbf{0}$, then $\boldsymbol{a}\cdot\boldsymbol{b} > \mathbf{0}\cdot\boldsymbol{b} = \mathbf{0}$ by Exercise 3.13.

If $\boldsymbol{a}<\mathbf{0}$ and $\boldsymbol{b}<\mathbf{0}$, then $\mathbf{0} = \boldsymbol{a} + [(1,\sigma(m))]$ and $\mathbf{0} = \boldsymbol{b} + [(1,\sigma(n))]$ for some $m,n \in \mathbb{N}$. Adding $[(\sigma(m),1)]$ on both sides of the first equation, we get $\boldsymbol{a} = [(\sigma(m),1)]$. Similarly $\boldsymbol{b} = [(\sigma(n),1)]$. Hence

$$\begin{aligned}
\boldsymbol{a}\cdot\boldsymbol{b} &= [(\sigma(m),1)]\cdot[(\sigma(n),1)] \\
&= [(\sigma(m)\cdot 1 + 1\cdot\sigma(n),\ \sigma(m)\sigma(n) + 1\cdot 1)] \\
&= [(\sigma(m)+\sigma(n),\ (m+1)(n+1)+1)] \\
&= [(m+1+n+1,\ mn+m+n+1+1)] \\
&= [((m+n+1)+1,\ (m+n+1)+mn+1)] \\
&= [(1,\ mn+1)] = [(1,\sigma(mn))] \\
&= \mathbf{0} + [(1,\sigma(mn))],
\end{aligned}$$

and so $\boldsymbol{a}\cdot\boldsymbol{b} > \mathbf{0}$. (Alternatively, e.g., one could use $(-\mathbf{1})\cdot(-\mathbf{1}) = \mathbf{1}$.)

Solution to Exercise 3.16

By the trichotomy law, $\boldsymbol{a}>\mathbf{0}$ or $\boldsymbol{a}<\mathbf{0}$. The claim that $\boldsymbol{a}^2 > \mathbf{0}$ now follows by using the result from Exercise 3.15, p.32.

Solution to Exercise 3.17

Let $\boldsymbol{\ell} \in \mathbb{Z}$ be a least element of $S = \mathbb{Z}$. We have $\boldsymbol{n} := \boldsymbol{\ell} + (-\mathbf{1}) \in \mathbb{Z} = S$. Then $\boldsymbol{n} + [(1, \sigma(1))] = \boldsymbol{n} + \mathbf{1} = (\boldsymbol{\ell} + (-\mathbf{1})) + \mathbf{1} = \boldsymbol{\ell} + (-\mathbf{1} + \mathbf{1}) = \boldsymbol{\ell} + \mathbf{0} = \boldsymbol{\ell}$, and so by the definition of $<$, we get $\boldsymbol{n} < \boldsymbol{\ell}$. By the trichotomy law, $\neg(\boldsymbol{\ell} \leqslant \boldsymbol{n})$. On the other hand, we know that $\boldsymbol{\ell} \leqslant \boldsymbol{n}$ because $\boldsymbol{n} \in S$ and $\boldsymbol{\ell}$ is a lower bound for S. So we have arrived at a contradiction.

Solution to Exercise 3.18

If $\mathbf{0} \in \{\boldsymbol{m} - \boldsymbol{\ell} : \boldsymbol{m} \in S\}$, then $\mathbf{0} = \boldsymbol{m} - \boldsymbol{\ell}$ for some $\boldsymbol{m} \in S$, i.e., (adding $\boldsymbol{\ell}$ on both sides) $\boldsymbol{\ell} = \boldsymbol{m} \in S$. Also $\boldsymbol{\ell}$, being a lower bound, satisfies $\boldsymbol{\ell} \leqslant \boldsymbol{m}$ for all $\boldsymbol{m} \in S$. So $\boldsymbol{\ell}$ is a least element of S, and we are done.

Suppose $\mathbf{0} \notin \{\boldsymbol{m} - \boldsymbol{\ell} : \boldsymbol{m} \in S\}$. Thus, if $\boldsymbol{m} \in S$, then $\boldsymbol{m} \neq \boldsymbol{\ell}$. If $\boldsymbol{m} \in S$, then as $\boldsymbol{m} \geqslant \boldsymbol{\ell}$, we get $\boldsymbol{m} > \boldsymbol{\ell}$, and adding $-\boldsymbol{\ell}$, we get $\boldsymbol{m} - \boldsymbol{\ell} > \mathbf{0}$. Define the set $S' = \{n \in \mathbb{N} : \exists\, \boldsymbol{m} \in S$ such that $\boldsymbol{m} - \boldsymbol{\ell} = \mathbf{0} + [(1, \sigma(n))]\} \neq \varnothing$. Then $\{\boldsymbol{m} - \boldsymbol{\ell} : \boldsymbol{m} \in S\} = \{[(1, \sigma(n))] : n \in S'\}$. The well-ordering principle for $\mathbb{N}$ implies that S' has a least element, say $n_* \in S' \subset \mathbb{N}$. Let $\boldsymbol{m}_* \in S$ be such that $\boldsymbol{m}_* - \boldsymbol{\ell} = [(1, \sigma(n_*))]$. We claim $\boldsymbol{m}_* \in S$ is a least element of S. Let $\boldsymbol{m} \in S$. Then $\boldsymbol{m} - \boldsymbol{\ell} = [(1, \sigma(n))]$ for some $n \in S'$. As n_* is the least element of S', it follows that $n_* \leqslant n$.

1° If $n_* = n$, then $\boldsymbol{m} - \boldsymbol{\ell} = [(1, \sigma(n))] = [(1, \sigma(n_*))] = \boldsymbol{m}_* - \boldsymbol{\ell}$. By the additive cancellation law for $\mathbb{Z}$, we get $\boldsymbol{m} = \boldsymbol{m}_*$.

2° If $n_* < n$, then there exists a $k \in \mathbb{N}$ such that $n = n_* + k$, and so

$$\begin{aligned} \boldsymbol{m} - \boldsymbol{\ell} &= [(1, \sigma(n))] = [(1, \sigma(n_* + k))] = [(1, \sigma(n_*))] + [(1, \sigma(k))] \\ &= \boldsymbol{m}_* - \boldsymbol{\ell} + [(1, \sigma(k))]. \end{aligned}$$

Again, by the additive cancellation law, we obtain from the above that $\boldsymbol{m} = \boldsymbol{m}_* + [(1, \sigma(k))]$, showing that $\boldsymbol{m}_* < \boldsymbol{m}$.

So $\boldsymbol{m}_* \leqslant \boldsymbol{m}$. Thus $\boldsymbol{m}_* \in S$ is a least element of S.

Solution to Exercise 3.19

Since $3 = [(1, \sigma(3))] = [(1, 4)]$, $5 = [(1, \sigma(5))] = [(1, 6)]$ and $-5 = [(6, 1)]$,

$$\begin{aligned} 3 - 5 &= 3 + (-5) = [(1, 4)] + [(6, 1)] = [(6 + 1, 4 + 1)] = [(4 + 2 + 1, 4 + 1)] \\ &= [(2 + 1, 1)] = [(\sigma(2), 1)] = -[(1, \sigma(2))] = -2, \end{aligned}$$

where the last equality follows by noting that $2 = [(1, \sigma(2))]$. Also,

$$\begin{aligned} 2 \cdot (-2) &= [(1, 3)] \cdot [(3, 1)] = [(1 \cdot 1 + 3 \cdot 3, 1 \cdot 3 + 3 \cdot 1)] \\ &= [(1 + 3 \cdot 2 + 3, 3 + 3)] = [(1 + 3 \cdot 2, 3)] = [(1 + 3 \cdot 1 + 3, 3)] \\ &= [(1 + 3 + 2 + 1, 2 + 1)] = [(1 + 3 + 1, 1)] \\ &= -[(1, 3 + 1 + 1)] = -[(1, 4 + 1)] = -[(1, \sigma(4))] = -4. \end{aligned}$$

Chapter 4

Solution to Exercise 4.1

As $m \neq 0$ and $d \neq 0$, we have by Theorem 3.8, p.31 that $md \neq 0$.
Also, $n(md) = (mn)d$, and so $(n, d) \sim (mn, md)$.

Solution to Exercise 4.2

As $(n, d) \sim (n', d')$ and $(p, q) \sim (p', q')$, we have $nd' = n'd$ and $pq' = p'q$. Thus

$$\begin{aligned}(nq+pd)(d'q') &= nd'qq' + pq'dd' \\ &= n'dqq' + p'qdd' \\ &= (n'q')(dq) + (p'd')(dq) \\ &= (n'q' + p'd')(dq).\end{aligned}$$

Hence $(nq + pd, dq) \sim (n'q' + p'd', d'q')$.

Solution to Exercise 4.3

Let $\boldsymbol{a} = \frac{n}{d}$, $\boldsymbol{b} = \frac{p}{q}$, $\boldsymbol{c} = \frac{r}{s} \in \mathbb{Q}$, where $(n, d), (p, q), (r, s) \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$. Then

$$\begin{aligned}\boldsymbol{a} + (\boldsymbol{b} + \boldsymbol{c}) &= \tfrac{n}{d} + (\tfrac{p}{q} + \tfrac{r}{s}) = \tfrac{n}{d} + \tfrac{ps + rq}{qs} = \tfrac{n(qs) + (ps + rq)d}{d(qs)} \\ &= \tfrac{(nq)s + (pd)s + r(dq)}{(dq)s} = \tfrac{(nq + pd)s + r(dq)}{(dq)s} \\ &= \tfrac{nq + pd}{dq} + \tfrac{r}{s} = (\tfrac{n}{d} + \tfrac{p}{q}) + \tfrac{r}{s} = (\boldsymbol{a} + \boldsymbol{b}) + \boldsymbol{c},\end{aligned}$$

showing that addition in $\mathbb{Q}$ is associative. Moreover, we have

$$\boldsymbol{a} + \boldsymbol{b} = \tfrac{n}{d} + \tfrac{p}{q} = \tfrac{nq + pd}{dq} = \tfrac{pd + nq}{qd} = \tfrac{p}{q} + \tfrac{n}{d} = \boldsymbol{b} + \boldsymbol{a},$$

showing that addition in $\mathbb{Q}$ is commutative.

Solution to Exercise 4.4

Since $\frac{n}{d} = \frac{n'}{d'}$, we have $(n, d) \sim (n', d')$, i.e., $nd' = n'd$. So $(-n, d) \sim (-n', d')$, as $(-n)d' = ((-1)n)d' = (-1)(nd') = (-1)(n'd) = ((-1)n')d = (-n')d$. Thus $(-n, d) \sim (-n', d')$, showing that $\frac{-n}{d} = \frac{-n'}{d'}$.

Also, $\boldsymbol{r} + (-\boldsymbol{r}) = \frac{n}{d} + \frac{-n}{d} = \frac{nd + (-n)d}{dd} = \frac{(n + (-n))d}{dd} = \frac{0d}{dd} = \frac{0}{dd} \overset{(*)}{=} \frac{0}{1} = \mathbf{0}$,

where $(*)$ holds as $0(1) = 0 = 0(dd)$, showing that $(0, dd) \sim (0, 1)$.

Solution to Exercise 4.5

Since $(n, d) \sim (n', d')$ and $(p, q) \sim (p', q')$, we have $nd' = n'd$ and $pq' = p'q$. Thus we obtain $(np)(d'q') = (nd')(pq') = (n'd)(p'q) = (n'p')(dq)$, and so $(np, dq) \sim (n'p', d'q')$.

Solution to Exercise 4.6

Let $\boldsymbol{a}=\frac{n}{d}$, $\boldsymbol{b}=\frac{p}{q}$, $\boldsymbol{c}=\frac{r}{s}\in\mathbb{Q}$, where $(n,d),(p,q),(r,s)\in\mathbb{Z}\times(\mathbb{Z}\backslash\{0\})$. Then

$$\begin{aligned}\boldsymbol{a}\cdot(\boldsymbol{b}\cdot\boldsymbol{c}) &= \frac{n}{d}\cdot(\frac{p}{q}\cdot\frac{r}{s})=\frac{n}{d}\cdot\frac{pr}{qs}=\frac{n(pr)}{d(qs)}\\ &= \frac{(np)r}{(dq)s}=\frac{np}{dq}\cdot\frac{r}{s}=(\frac{n}{d}\cdot\frac{p}{q})\cdot\frac{r}{s}=(\boldsymbol{a}\cdot\boldsymbol{b})\cdot\boldsymbol{c},\end{aligned}$$

showing that mutiplication in $\mathbb{Q}$ is associative. Moreover, we have

$$\boldsymbol{a}\cdot\boldsymbol{b}=\frac{n}{d}\cdot\frac{p}{q}=\frac{np}{dq}=\frac{pn}{qd}=\frac{p}{q}\cdot\frac{n}{d}=\boldsymbol{b}\cdot\boldsymbol{a},$$

showing that multiplication in $\mathbb{Q}$ is commutative.

Solution to Exercise 4.7

Let $\boldsymbol{r}=\frac{n}{d}$, $\boldsymbol{r'}=\frac{n'}{d'}$ and $\boldsymbol{s}=\frac{p}{q}$, where $(n,d),(n',d'),(p,q)\in\mathbb{Z}\times(\mathbb{Z}\backslash\{0\})$. Then

$$(\boldsymbol{r}+\boldsymbol{r'})\cdot\boldsymbol{s}=(\frac{n}{d}+\frac{n'}{d'})\cdot\frac{p}{q}=\frac{nd'+n'd}{dd'}\cdot\frac{p}{q}=\frac{(nd'+n'd)p}{(dd')q}. \qquad (\star)$$

Moreover,

$$\boldsymbol{r}\cdot\boldsymbol{s}+\boldsymbol{r'}\cdot\boldsymbol{s}=\frac{n}{d}\cdot\frac{p}{q}+\frac{n'}{d'}\cdot\frac{p}{q}=\frac{np}{dq}+\frac{n'p}{d'q}=\frac{npd'q+n'pdq}{dqd'q}. \qquad (\star\star)$$

To show that the right-hand sides of $(\star)$ and $(\star\star)$ are the same, note that

$$\begin{aligned}((nd'+n'd)p)(dqd'q) &= nd'pdqd'q+n'dpdqd'q\\ &= (npd'q)(dd'q)+(n'pdq)(dd'q)\\ &= (npd'q+n'pdq)(dd'q).\end{aligned}$$

So $((nd'+n'd)p,(dd')q)\sim(npd'q+n'pdq,dqd'q)$ and $\frac{(nd'+n'd)p}{(dd')q}=\frac{npd'q+n'pdq}{dqd'q}$.

In light of $(\star)$ and $(\star\star)$, we conclude that $(\boldsymbol{r}+\boldsymbol{r'})\cdot\boldsymbol{s}=\boldsymbol{r}\cdot\boldsymbol{s}+\boldsymbol{r'}\cdot\boldsymbol{s}$.

Solution to Exercise 4.8

Let $\boldsymbol{r}=\frac{n}{d}$, $\boldsymbol{s}=\frac{p}{q}$, $\boldsymbol{t}=\frac{x}{y}$, where $n,p,x\in\mathbb{Z}$ and d,q,y are positive integers. Then $nq<dp$ and $py<qx$. We want to show $ny<dx$.

Suppose that $ny\geqslant dx$. As q is a positive integer, $nyq\geqslant dxq$. Since y is a positive integer, from $nq<dp$ we get $nqy<dpy$, and as d is a positive integer, from $py<qx$ we get $dpy<dqx$. By transitivity of the order relation in $\mathbb{Z}$, we have $nqy<dqx$, which contradicts $nyq\geqslant dxq$. Hence our assumption that $ny\geqslant dx$ is false. Consequently, $ny<dx$, as wanted.

Solution to Exercise 4.9

Let $\boldsymbol{r}=\frac{n}{d}$ and $\boldsymbol{s}=\frac{p}{q}$, where $n,p\in\mathbb{Z}$, and d,q are positive integers.

By the trichotomy law for the order relation in $\mathbb{Z}$, we have that exactly one of the following mutually exclusive cases holds:

1° $nq<dp$. This is equivalent to $\boldsymbol{r}<\boldsymbol{s}$ by the definition of $<$ in $\mathbb{Q}$.

2° $nq=dp$. This is equivalent to $\boldsymbol{r}=\boldsymbol{s}$ by the definition of $(n,d)\sim(p,q)$.

3° $nq>dp$. This is equivalent to $\boldsymbol{r}>\boldsymbol{s}$ by the definition of $<$ in $\mathbb{Q}$.

Solution to Exercise 4.10

Let $\boldsymbol{r}=\frac{n}{d}$, $\boldsymbol{s}=\frac{p}{q}$, $\boldsymbol{t}=\frac{x}{y}$, where $n,p,x\in\mathbb{Z}$, and d,q,y are positive integers. As $d>0$, $q>0$, and $y>0$, we have $dy>0y=0$ and $qy>0y=0$. Moreover, $\boldsymbol{r}<\boldsymbol{s}$ yields $nq<pd$. We have $\boldsymbol{r}+\boldsymbol{t}=\frac{ny+xd}{dy}$ and $\boldsymbol{s}+\boldsymbol{t}=\frac{py+xq}{qy}$. As $y>0$, also $y^2=yy>y0=0$. Multiplying $nq<pd$ by y^2 yields $nqy^2<pdy^2$. Adding $dxqy$ on both sides of $nqy^2<pdy^2$, we obtain $nqy^2+dxqy<pdy^2+dxqy$, that is, $(ny+xd)qy<(py+xq)dy$. Since $dy>0$ and $qy>0$, it follows from $(ny+xd)qy<(py+qx)dy$ that $\frac{ny+xd}{dy}<\frac{py+xq}{qy}$, that is, $\boldsymbol{r}+\boldsymbol{t}<\boldsymbol{s}+\boldsymbol{t}$.

Solution to Exercise 4.11

Let $\boldsymbol{r}=\frac{n}{d}$, $\boldsymbol{s}=\frac{p}{q}$, $\boldsymbol{t}=\frac{x}{y}$, where $n,p,x\in\mathbb{Z}$, and d,q,y are positive integers. As $d>0$, $q>0$, and $y>0$, we have $dy>0y=0$ and $qy>0y=0$. As $\boldsymbol{r}<\boldsymbol{s}$, we have $nq<pd$. Also, as $\frac{x}{y}=\boldsymbol{t}>\boldsymbol{0}=\frac{0}{1}$, it follows that $x=x(1)>y(0)=0$. We have $\boldsymbol{r}\cdot\boldsymbol{t}=\frac{nx}{dy}$ and $\boldsymbol{s}\cdot\boldsymbol{t}=\frac{px}{qy}$. Since $x>0$ and $y>0$, we have $xy>0y=0$. Multiplying both sides of $nq<pd$ by $xy>0$, we obtain $nqxy<pdxy$, i.e., $(nx)qy<(px)dy$. Since $dy>0$ and $qy>0$, it follows from $(nx)qy<(px)dy$ that $\frac{nx}{dy}<\frac{px}{qy}$, that is, $\boldsymbol{r}\cdot\boldsymbol{t}<\boldsymbol{s}\cdot\boldsymbol{t}$.

Solution to Exercise 4.12

It is impossible that $\boldsymbol{t}^{-1}=\boldsymbol{0}$ (otherwise $\boldsymbol{1}=\boldsymbol{t}\cdot\boldsymbol{t}^{-1}=\boldsymbol{t}\cdot\boldsymbol{0}=\boldsymbol{0}$, a contradiction). Also the case $\boldsymbol{t}^{-1}<\boldsymbol{0}$ is impossible, since otherwise by multiplying by $\boldsymbol{t}>0$, we get $\boldsymbol{1}=\boldsymbol{t}\cdot\boldsymbol{t}^{-1}<\boldsymbol{t}\cdot\boldsymbol{0}=\boldsymbol{0}$, a contradiction. By the trichotomy law, $\boldsymbol{t}^{-1}>\boldsymbol{0}$.

Solution to Exercise 4.13

(We know the average of the two rational numbers lies between the two, but we write a careful argument using the rules learnt so far.)
As $\boldsymbol{r}<\boldsymbol{s}$, it follows from Exercise 4.10, p.39, that $\boldsymbol{r}+\boldsymbol{r}<\boldsymbol{r}+\boldsymbol{s}$, and $\boldsymbol{r}+\boldsymbol{s}<\boldsymbol{s}+\boldsymbol{s}$. But $\boldsymbol{r}+\boldsymbol{r}=\boldsymbol{1}\cdot\boldsymbol{r}+\boldsymbol{1}\cdot\boldsymbol{r}=(\boldsymbol{1}+\boldsymbol{1})\cdot\boldsymbol{r}$. Now $\boldsymbol{1}+\boldsymbol{1}=\frac{2}{1}\neq\frac{0}{1}=\boldsymbol{0}$, and $(\boldsymbol{1}+\boldsymbol{1})^{-1}=\frac{1}{2}>\frac{0}{1}=\boldsymbol{0}$ (as $1(1)=1>0=0(2)$). By Exercise 4.11, p.39, and $(\boldsymbol{1}+\boldsymbol{1})\cdot\boldsymbol{r}=\boldsymbol{r}+\boldsymbol{r}<\boldsymbol{r}+\boldsymbol{s}$, we get

$$\boldsymbol{r}<(\boldsymbol{1}+\boldsymbol{1})^{-1}\cdot(\boldsymbol{r}+\boldsymbol{s}). \qquad (\star)$$

Similarly, $\boldsymbol{r}+\boldsymbol{s}<\boldsymbol{s}+\boldsymbol{s}=(\boldsymbol{1}+\boldsymbol{1})\cdot\boldsymbol{s}$ yields

$$(\boldsymbol{1}+\boldsymbol{1})^{-1}\cdot(\boldsymbol{r}+\boldsymbol{s})<\boldsymbol{s}. \qquad (\star\star)$$

$(\star)$ and $(\star\star)$ show that $\boldsymbol{t}:=(\boldsymbol{1}+\boldsymbol{1})^{-1}\cdot(\boldsymbol{r}+\boldsymbol{s})\in\mathbb{Q}$ is a rational number such that $\boldsymbol{r}<\boldsymbol{t}<\boldsymbol{s}$. (Successive bisection in this manner shows the existence of infinitely many rational numbers between $\boldsymbol{r}$ and $\boldsymbol{s}$.)

Solution to Exercise 4.14

Let $\boldsymbol{r} = \frac{n}{d}$, where $(n, d) \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$.

1° If $n = 0$, then $\boldsymbol{r}^2 = \frac{0}{d} \cdot \frac{0}{d} = \frac{0}{d^2} = \frac{0}{1} = \mathbf{0}$ (as $0(1) = 0 = 0d^2$).

2° If $n \neq 0$, then $\boldsymbol{r}^2 = \frac{n^2}{d^2} > \frac{0}{1} = \mathbf{0}$ (as $d^2 > 0$ and $n^2 1 = n^2 > 0 = 0d^2$).

Solution to Exercise 4.15

(1) We consider the various possible cases separately.

1° $\boldsymbol{r} = \mathbf{0}$ or $\boldsymbol{s} = \mathbf{0}$. Then $\boldsymbol{r} \cdot \boldsymbol{s} = \mathbf{0}$, and so $|\boldsymbol{r} \cdot \boldsymbol{s}| = \mathbf{0}$.
Also, $|\boldsymbol{r}| = \mathbf{0}$ or $|\boldsymbol{s}| = \mathbf{0}$, and so $|\boldsymbol{r}| \cdot |\boldsymbol{s}| = \mathbf{0}$ as well.

2° $\boldsymbol{r} > \mathbf{0}$ and $\boldsymbol{s} > \mathbf{0}$. Then $\boldsymbol{r} \cdot \boldsymbol{s} > \mathbf{0} \cdot \boldsymbol{s} = \mathbf{0}$, and so $|\boldsymbol{r} \cdot \boldsymbol{s}| = \boldsymbol{r} \cdot \boldsymbol{s} = |\boldsymbol{r}| \cdot |\boldsymbol{s}|$.

3° $\boldsymbol{r} > \mathbf{0}$ and $\boldsymbol{s} < \mathbf{0}$. Then $\boldsymbol{r} \cdot \boldsymbol{s} < \boldsymbol{r} \cdot \mathbf{0} = \mathbf{0}$.
So $|\boldsymbol{r} \cdot \boldsymbol{s}| = -(\boldsymbol{r} \cdot \boldsymbol{s}) = (-\mathbf{1}) \cdot \boldsymbol{r} \cdot \boldsymbol{s} = \boldsymbol{r} \cdot ((-\mathbf{1}) \cdot \boldsymbol{s}) = \boldsymbol{r} \cdot (-\boldsymbol{s}) = |\boldsymbol{r}| \cdot |\boldsymbol{s}|$.

4° $\boldsymbol{r} < \mathbf{0}$ and $\boldsymbol{s} > \mathbf{0}$. Follows from 3° by swapping $\boldsymbol{r}, \boldsymbol{s}$ and the commutativity of multiplication in $\mathbb{Q}$: $|\boldsymbol{r} \cdot \boldsymbol{s}| = |\boldsymbol{s} \cdot \boldsymbol{r}| \overset{3^\circ}{=} |\boldsymbol{s}| \cdot |\boldsymbol{r}| = |\boldsymbol{r}| \cdot |\boldsymbol{s}|$.

5° $\boldsymbol{r} < \mathbf{0}$ and $\boldsymbol{s} < \mathbf{0}$. Then $-\boldsymbol{r} > \mathbf{0}$ and so $(-\boldsymbol{r}) \cdot \boldsymbol{s} < (-\boldsymbol{r}) \cdot \mathbf{0} = \mathbf{0}$. Hence $(-\mathbf{1}) \cdot (\boldsymbol{r} \cdot \boldsymbol{s}) < \mathbf{0}$, i.e., $-(\boldsymbol{r} \cdot \boldsymbol{s}) < \mathbf{0}$ and so $\boldsymbol{r} \cdot \boldsymbol{s} > \mathbf{0}$. Thus $|\boldsymbol{r} \cdot \boldsymbol{s}| = \boldsymbol{r} \cdot \boldsymbol{s}$.
Also, $|\boldsymbol{r}| \cdot |\boldsymbol{s}| = (-\boldsymbol{r}) \cdot (-\boldsymbol{s}) = (-\mathbf{1}) \cdot \boldsymbol{r} \cdot (-\mathbf{1}) \cdot \boldsymbol{s} = -(-\mathbf{1}) \cdot \boldsymbol{r} \cdot \boldsymbol{s} = \mathbf{1} \cdot \boldsymbol{r} \cdot \boldsymbol{s} = \boldsymbol{r} \cdot \boldsymbol{s}$.

(2) 1° If $\boldsymbol{r} \geqslant \mathbf{0}$, then $|\boldsymbol{r}| = \boldsymbol{r}$.

2° If $\boldsymbol{r} < \mathbf{0}$, then adding $-\boldsymbol{r}$ on both sides gives $-\boldsymbol{r} > \mathbf{0}$.
So $|\boldsymbol{r}| = -\boldsymbol{r} > \mathbf{0} > \boldsymbol{r}$.

So for all $\boldsymbol{r} \in \mathbb{Q}$, we have $\boldsymbol{r} \leqslant |\boldsymbol{r}|$.

Thus also $-\boldsymbol{r} \leqslant |-\boldsymbol{r}| = |(-\mathbf{1}) \cdot \boldsymbol{r}| = |-\mathbf{1}| \cdot |\boldsymbol{r}| = \mathbf{1} \cdot |\boldsymbol{r}| = |\boldsymbol{r}|$.

(3) 1° Let $\boldsymbol{r} + \boldsymbol{s} \geqslant \mathbf{0}$. We have $\boldsymbol{r} \leqslant |\boldsymbol{r}|$ and so $\boldsymbol{r} + |\boldsymbol{s}| \leqslant |\boldsymbol{r}| + |\boldsymbol{s}|$.
Similarly, $\boldsymbol{s} \leqslant |\boldsymbol{s}|$ yields $\boldsymbol{r} + \boldsymbol{s} \leqslant \boldsymbol{r} + |\boldsymbol{s}|$. Putting these together, we obtain $|\boldsymbol{r} + \boldsymbol{s}| = \boldsymbol{r} + \boldsymbol{s} \leqslant \boldsymbol{r} + |\boldsymbol{s}| \leqslant |\boldsymbol{r}| + |\boldsymbol{s}|$.

2° Let $\boldsymbol{r} + \boldsymbol{s} < \mathbf{0}$. We have $-\boldsymbol{r} \leqslant |\boldsymbol{r}|$ and so $-\boldsymbol{r} + (-\boldsymbol{s}) \leqslant |\boldsymbol{r}| + (-\boldsymbol{s})$.
Similarly, as $-\boldsymbol{s} \leqslant |\boldsymbol{s}|$, we get $|\boldsymbol{r}| + (-\boldsymbol{s}) \leqslant |\boldsymbol{r}| + |\boldsymbol{s}|$.
We have $|\boldsymbol{r} + \boldsymbol{s}| = -(\boldsymbol{r} + \boldsymbol{s}) = (-\mathbf{1}) \cdot (\boldsymbol{r} + \boldsymbol{s})$. Using distributivity, we get $|\boldsymbol{r} + \boldsymbol{s}| = (-\mathbf{1}) \cdot \boldsymbol{r} + (-\mathbf{1}) \cdot \boldsymbol{s} = -\boldsymbol{r} + (-\boldsymbol{s}) \leqslant |\boldsymbol{r}| + (-\boldsymbol{s}) \leqslant |\boldsymbol{r}| + |\boldsymbol{s}|$.

(4) We have $|\boldsymbol{r}| = |\boldsymbol{s} + \boldsymbol{r} - \boldsymbol{s}| \leqslant |\boldsymbol{s}| + |\boldsymbol{r} - \boldsymbol{s}|$. Adding $-|\boldsymbol{s}|$ on both sides yields

$$|\boldsymbol{r}| - |\boldsymbol{s}| \leqslant |\boldsymbol{r} - \boldsymbol{s}|. \qquad (*)$$

Swapping $\boldsymbol{r}$ and $\boldsymbol{s}$, $|\boldsymbol{s}| - |\boldsymbol{r}| \leqslant |\boldsymbol{s} - \boldsymbol{r}| = |(-\mathbf{1})(\boldsymbol{r} - \boldsymbol{s})| = |-\mathbf{1}||\boldsymbol{r} - \boldsymbol{s}| = |\boldsymbol{r} - \boldsymbol{s}|$.
So $(-\mathbf{1}) \cdot (|\boldsymbol{r}| - |\boldsymbol{s}|) = |\boldsymbol{s}| - |\boldsymbol{r}| \leqslant |\boldsymbol{r} - \boldsymbol{s}|$, i.e.,

$$-(|\boldsymbol{r}| - |\boldsymbol{s}|) \leqslant |\boldsymbol{r} - \boldsymbol{s}|. \qquad (**)$$

From $(*)$ and $(**)$, we get $||\boldsymbol{r}| - |\boldsymbol{s}|| \leqslant |\boldsymbol{r} - \boldsymbol{s}|$.

Solution to Exercise 4.16

(1) We have

$$\begin{aligned}\binom{n}{k}+\binom{n}{k+1} &= \frac{n!}{k!(n-k)!}+\frac{n!}{(k+1)!(n-(k+1))!}\\ &= \frac{n!}{k!(n-(k+1))!}\left(\frac{1}{n-k}+\frac{1}{k+1}\right)\\ &= \frac{n!}{k!(n-(k+1))!}\frac{(n+1)}{(n-k)(k+1)} = \frac{(n+1)!}{(k+1)!(n-k)!}\\ &= \frac{(n+1)!}{(k+1)!((n+1)-(k+1))!} = \binom{n+1}{k+1}.\end{aligned}$$

(2) We use induction on n. We have $\binom{1}{1}=1=\binom{1}{0}$.

Suppose that the claim is true for some $n\in\mathbb{N}$.

We have $\binom{n+1}{0}=\frac{(n+1)!}{0!(n+1)!}=1\in\mathbb{N}$ and $\binom{n+1}{n+1}=\frac{(n+1)!}{(n+1)!0!}=1\in\mathbb{N}$.

Also, $\binom{n+1}{n}=\frac{(n+1)!}{n!1!}=n+1\in\mathbb{N}$.

For all $1\leqslant k<n$, we have by part (1) and the induction hypothesis that $\binom{n+1}{k}=\binom{n}{k-1}+\binom{n}{k}\in\mathbb{N}$.

(Alternatively, $\binom{n}{k}$ is a natural number since it is the number of ways of choosing k objects from n distinct objects.)

Chapter 5

Solution to Exercise 5.1

Let $\epsilon > 0$ be rational. There exists an $N_a \in \mathbb{N}$ such that for all natural numbers $n, m > N_a$, we have $|a_n - a_m| < \frac{\epsilon}{2}$. Also, there exists an $N_b \in \mathbb{N}$ such that for all natural numbers $n, m > N_b$, we have $|b_n - b_m| < \frac{\epsilon}{2}$. Set $N := N_a + N_b$. Then for all natural numbers $m, n > N$, we have

$$|a_n + b_n - (a_m + b_m)| = |a_n - a_m + b_n - b_m| \leqslant |a_n - a_m| + |b_n - b_m| \leqslant \tfrac{\epsilon}{2} + \tfrac{\epsilon}{2} = \epsilon.$$

So $(a_n + b_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{Q}$.

Solution to Exercise 5.2

The sequences $(a_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ are bounded (as they are Cauchy). Let $A, B > 0$ be rational, such that $|a_n| < A$ and $|b_n| < B$ for all $n \in \mathbb{N}$. Let $\epsilon > 0$ be rational. There exists an $N_a \in \mathbb{N}$ such that for all natural numbers $n, m > N_a$, we have $|a_n - a_m| < \frac{\epsilon}{2B}$. Also, there exists an $N_b \in \mathbb{N}$ such that for all natural numbers $n, m > N_b$, we have $|b_n - b_m| < \frac{\epsilon}{2A}$. Set $N := N_a + N_b$. Then for all natural numbers $m, n > N$, we have

$$\begin{aligned} |a_n b_n - a_m b_m| &= |a_n b_n - a_m b_n + a_m b_n - a_m b_m| \\ &\leqslant |a_n b_n - a_m b_n| + |a_m b_n - a_m b_m| \\ &= |a_n - a_m||b_n| + |a_m||b_n - b_m| \\ &\leqslant \tfrac{\epsilon}{2B} \cdot B + A \cdot \tfrac{\epsilon}{2A} = \epsilon. \end{aligned}$$

So $(a_n b_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{Q}$.

Solution to Exercise 5.3

(1) Let $r_a \neq r'_a$. Let $\epsilon = \frac{|r_a - r'_a|}{2} \in \mathbb{Q}$. As $r_a \neq r_{a'}$, it follows from the trichotomy law that $r_a - r'_a > 0$ or $r_a - r'_a < 0$. In either case $|r_a - r'_a| > 0$. For the rational $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all natural numbers $n > N$, we have $|a_n - r_a| < \epsilon$. Also, there exists an $N' \in \mathbb{N}$ such that for all natural numbers $n > N'$, we have $|a_n - r'_a| < \epsilon$. Let $n = N + N'$. So $n > N$ and $n > N'$, and (by the triangle inequality in $\mathbb{Q}$)

$$\begin{aligned} |r_a - r'_a| &= |r_a - a_n + a_n - r'_a| \\ &\leqslant |r_a - a_n| + |a_n - r'_a| \\ &< \epsilon + \epsilon = |r_a - r'_a|, \end{aligned}$$

a contradiction to the trichotomy law. Thus $r_a = r'_a$.

(2) Given any rational $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all natural numbers $n > N$, we have $|a_n - r_a| < \epsilon$. Then for all natural numbers $n > N$, we have $|-a_n - (-r_a)| = |-(a_n - r_a)| = |-1||a_n - r_a| = |a_n - r_a| < \epsilon$. So $(-a_n)_{n\in\mathbb{N}}$ converges in $\mathbb{Q}$ to $-r_a$.

(3) Let the rational $\epsilon > 0$. Then $\frac{\epsilon}{2} > 0$. There exists an $N_a \in \mathbb{N}$ such that for all natural numbers $n > N_a$, we have $|a_n - r_a| < \frac{\epsilon}{2}$. Also, there exists an $N_b \in \mathbb{N}$ such that for all natural numbers $n > N_b$, we have $|b_n - r_b| < \frac{\epsilon}{2}$. Let $N = N_a + N_b$. Then for all natural numbers $n > N$, we have

$$\begin{aligned} |(a_n + b_n) - (r_a + r_b)| &= |a_n - r_a + (b_n - r_b)| \\ &\leqslant |a_n - r_a| + |b_n - r_b| < \tfrac{\epsilon}{2} + \tfrac{\epsilon}{2} = \epsilon. \end{aligned}$$

So $(a_n + b_n)_{n\in\mathbb{N}}$ converges in $\mathbb{Q}$ to $r_a + r_b$.

Solution to Exercise 5.4

As $(a_n)_{n\in\mathbb{N}} \sim (a'_n)_{n\in\mathbb{N}}$, $(a_n - a'_n)_{n\in\mathbb{N}}$ converges in $\mathbb{Q}$ to 0. So for a rational $\epsilon > 0$, there exists an $N_a \in \mathbb{N}$ such that $|a'_n - a_n| < \frac{\epsilon}{2}$ for all natural numbers $n > N_a$. Similarly, as $(b_n)_{n\in\mathbb{N}} \sim (b'_n)_{n\in\mathbb{N}}$, it follows that there exists an $N_b \in \mathbb{N}$ such that $|b'_n - b_n| < \frac{\epsilon}{2}$ for all natural numbers $n > N_b$. Set $N = N_a + N_b$. For all natural numbers $n > N$,

$$|a'_n + b'_n - (a_n + b_n)| = |a'_n - a_n + b'_n - b_n| \leqslant |a'_n - a_n| + |b'_n - b_n| < \tfrac{\epsilon}{2} + \tfrac{\epsilon}{2} = \epsilon.$$

Thus $(a_n + b_n)_{n\in\mathbb{N}} \sim (a'_n + b'_n)_{n\in\mathbb{N}}$.

Solution to Exercise 5.5

Let $\boldsymbol{x} = [(a_n)_{n\in\mathbb{N}}], \boldsymbol{y} = [(b_n)_{n\in\mathbb{N}}], \boldsymbol{z} = [(c_n)_{n\in\mathbb{N}}] \in \mathbb{R}$.
(Here $(a_n)_{n\in\mathbb{N}}, (b_n)_{n\in\mathbb{N}}, (c_n)_{n\in\mathbb{N}} \in \mathscr{C}$.)

(F1) We have

$$\begin{aligned} \boldsymbol{x} + (\boldsymbol{y} + \boldsymbol{z}) &= [(a_n)_{n\in\mathbb{N}}] + \big([(b_n)_{n\in\mathbb{N}}] + [(c_n)_{n\in\mathbb{N}}]\big) \\ &= [(a_n)_{n\in\mathbb{N}}] + [(b_n + c_n)_{n\in\mathbb{N}}] \\ &= [(a_n + (b_n + c_n))_{n\in\mathbb{N}}] = [((a_n + b_n) + c_n)_{n\in\mathbb{N}}] \\ &= [(a_n + b_n)_{n\in\mathbb{N}}] + [(c_n)_{n\in\mathbb{N}}] \\ &= \big([(a_n)_{n\in\mathbb{N}}] + [(b_n)_{n\in\mathbb{N}}]\big) + [(c_n)_{n\in\mathbb{N}}] \\ &= (\boldsymbol{x} + \boldsymbol{y}) + \boldsymbol{z}. \end{aligned}$$

(F2) We have

$$\boldsymbol{x} + \boldsymbol{0} = [(a_n)_{n\in\mathbb{N}}] + [(0)_{n\in\mathbb{N}}] = [(a_n + 0)_{n\in\mathbb{N}}] = [(a_n)_{n\in\mathbb{N}}] = \boldsymbol{x}, \text{ and}$$
$$\boldsymbol{0} + \boldsymbol{x} = [(0)_{n\in\mathbb{N}}] + [(a_n)_{n\in\mathbb{N}}] = [(0 + a_n)_{n\in\mathbb{N}}] = [(a_n)_{n\in\mathbb{N}}] = \boldsymbol{x}.$$

(F3) Define $-\boldsymbol{x} = [(-a_n)_{n\in\mathbb{N}}] \in \mathbb{R}$. This is well-defined, since if $(a'_n)_{n\in\mathbb{N}} \in \mathscr{C}$ and $(a_n)_{n\in\mathbb{N}} \sim (a'_n)_{n\in\mathbb{N}}$, then $(-a_n - (-a'_n))_{n\in\mathbb{N}} = (a'_n - a_n)_{n\in\mathbb{N}}$ converges in $\mathbb{Q}$ to 0, showing that $(-a_n)_{n\in\mathbb{N}} \sim (-a'_n)_{n\in\mathbb{N}}$. We have

$$\boldsymbol{x} + (-\boldsymbol{x}) = [(a_n)_{n\in\mathbb{N}}] + [(-a_n)_{n\in\mathbb{N}}] = [(0)_{n\in\mathbb{N}}] = \boldsymbol{0}, \text{ and}$$
$$-\boldsymbol{x} + \boldsymbol{x} = [(-a_n)_{n\in\mathbb{N}}] + [(a_n)_{n\in\mathbb{N}}] = [(0)_{n\in\mathbb{N}}] = \boldsymbol{0}.$$

(F4) We have

$$\begin{aligned} \boldsymbol{x} + \boldsymbol{y} &= [(a_n)_{n\in\mathbb{N}}] + [(b_n)_{n\in\mathbb{N}}] = [(a_n + b_n)_{n\in\mathbb{N}}] \\ &= [(b_n + a_n)_{n\in\mathbb{N}}] = [(b_n)_{n\in\mathbb{N}}] + [(a_n)_{n\in\mathbb{N}}] = \boldsymbol{y} + \boldsymbol{x}. \end{aligned}$$

(F5) We have

$$\begin{aligned}\boldsymbol{x}\cdot(\boldsymbol{y}\cdot\boldsymbol{z}) &= [(a_n)_{n\in\mathbb{N}}]\cdot\big([(b_n)_{n\in\mathbb{N}}]\cdot[(c_n)_{n\in\mathbb{N}}]\big)\\ &= [(a_n)_{n\in\mathbb{N}}]\cdot[(b_nc_n)_{n\in\mathbb{N}}] = [(a_n(b_nc_n))_{n\in\mathbb{N}}]\\ &= [((a_nb_n)c_n)_{n\in\mathbb{N}}] = [(a_nb_n)_{n\in\mathbb{N}}]\cdot[(c_n)_{n\in\mathbb{N}}]\\ &= \big([(a_n)_{n\in\mathbb{N}}]\cdot[(b_n)_{n\in\mathbb{N}}]\big)\cdot[(c_n)_{n\in\mathbb{N}}] = (\boldsymbol{x}\cdot\boldsymbol{y})\cdot\boldsymbol{z}.\end{aligned}$$

(F6) We have seen that $\mathbf{1}\neq\mathbf{0}$. Moreover,

$$\begin{aligned}\boldsymbol{x}\cdot\mathbf{1} &= [(a_n)_{n\in\mathbb{N}}]\cdot[(1)_{n\in\mathbb{N}}] = [(a_n1)_{n\in\mathbb{N}}] = [(a_n)_{n\in\mathbb{N}}] = \boldsymbol{x},\ \text{and}\\ \mathbf{1}\cdot\boldsymbol{x} &= [(1)_{n\in\mathbb{N}}]\cdot[(a_n)_{n\in\mathbb{N}}] = [(1a_n)_{n\in\mathbb{N}}] = [(a_n)_{n\in\mathbb{N}}] = \boldsymbol{x}.\end{aligned}$$

(F8) We have

$$\begin{aligned}\boldsymbol{x}\cdot\boldsymbol{y} &= [(a_n)_{n\in\mathbb{N}}]\cdot[(b_n)_{n\in\mathbb{N}}] = [(a_nb_n)_{n\in\mathbb{N}}]\\ &= [(b_na_n)_{n\in\mathbb{N}}] = [(b_n)_{n\in\mathbb{N}}]\cdot[(a_n)_{n\in\mathbb{N}}] = \boldsymbol{y}\cdot\boldsymbol{x}.\end{aligned}$$

(F9) We have

$$\begin{aligned}\boldsymbol{x}\cdot(\boldsymbol{y}+\boldsymbol{z}) &= [(a_n)_{n\in\mathbb{N}}]\cdot\big([(b_n)_{n\in\mathbb{N}}]+[(c_n)_{n\in\mathbb{N}}]\big)\\ &= [(a_n)_{n\in\mathbb{N}}]\cdot[(b_n+c_n)_{n\in\mathbb{N}}]\\ &= [(a_n(b_n+c_n))_{n\in\mathbb{N}}] = [(a_nb_n+a_nc_n)_{n\in\mathbb{N}}]\\ &= [(a_nb_n)_{n\in\mathbb{N}}]+[(a_nc_n)_{n\in\mathbb{N}}]\\ &= [(a_n)_{n\in\mathbb{N}}]\cdot[(b_n)_{n\in\mathbb{N}}]+[(a_n)_{n\in\mathbb{N}}]\cdot[(c_n)_{n\in\mathbb{N}}]\\ &= \boldsymbol{x}\cdot\boldsymbol{y}+\boldsymbol{x}\cdot\boldsymbol{z}.\end{aligned}$$

Solution to Exercise 5.6

Let $\boldsymbol{x}=[(a_n)_{n\in\mathbb{N}}]$, $(a_n)_{n\in\mathbb{N}}\in\mathscr{C}$, be such that $\boldsymbol{x}^2+\mathbf{1}=\mathbf{0}$. Then

$$\begin{aligned}[(0)_{n\in\mathbb{N}}] = \mathbf{0} = \boldsymbol{x}^2+\mathbf{1} &= [(a_n)_{n\in\mathbb{N}}]\cdot[(a_n)_{n\in\mathbb{N}}]+[(1)_{n\in\mathbb{N}}]\\ &= [(a_n^2)_{n\in\mathbb{N}}]+[(1)_{n\in\mathbb{N}}] = [(a_n^2+1)_{n\in\mathbb{N}}].\end{aligned}$$

So $(a_n^2+1)_{n\in\mathbb{N}}\sim(0)_{n\in\mathbb{N}}$, i.e., $(a_n^2+1-0)_{n\in\mathbb{N}}$ converges in $\mathbb{Q}$ to 0. Taking $\epsilon=\frac{1}{2}>0$, there exists an $N\in\mathbb{N}$ such that for all natural numbers $n>N$, we have $|(a_n^2+1-0)-0|<\epsilon=\frac{1}{2}$. Thus for all natural numbers $n>N$, we have $a_n^2+1=|(a_n^2+1-0)-0|<\epsilon=\frac{1}{2}$. In particular, for $n=N+1$, $1=0+1\leqslant a_{N+1}^2+1<\frac{1}{2}$, which is absurd since $\frac{1}{2}<\frac{1}{1}$.

Solution to Exercise 5.7

We use induction on n. For $n=1$, of course $(\mathbf{1}+\boldsymbol{x})^1=\mathbf{1}+\boldsymbol{x}=\binom{1}{0}\mathbf{1}+\binom{1}{1}\boldsymbol{x}$. If the result holds for some $n\in\mathbb{N}$, then

$$\begin{aligned}(\mathbf{1}+\boldsymbol{x})^{n+1} &= (\mathbf{1}+\boldsymbol{x})^n(\mathbf{1}+\boldsymbol{x}) = \Big(\sum_{k=0}^{n}\binom{n}{k}\boldsymbol{x}^k\Big)(\mathbf{1}+\boldsymbol{x}) = \sum_{k=0}^{n}\binom{n}{k}\boldsymbol{x}^k+\sum_{k=0}^{n}\binom{n}{k}\boldsymbol{x}^{k+1}\\ &= \binom{n}{0}\mathbf{1}+\binom{n}{1}\boldsymbol{x}+\binom{n}{2}\boldsymbol{x}^2+\cdots+\binom{n}{n-1}\boldsymbol{x}^{n-1}+\binom{n}{n}\boldsymbol{x}^n\\ &\quad+\binom{n}{0}\boldsymbol{x}+\binom{n}{1}\boldsymbol{x}^2+\cdots+\binom{n}{n-2}\boldsymbol{x}^{n-1}+\binom{n}{n-1}\boldsymbol{x}^n+\binom{n}{n}\boldsymbol{x}^n\\ &= \binom{n+1}{0}\mathbf{1}+\binom{n+1}{1}\boldsymbol{x}+\binom{n+1}{2}\boldsymbol{x}^2+\cdots+\binom{n+1}{n-1}\boldsymbol{x}^{n-1}+\binom{n+1}{n}\boldsymbol{x}^n+\binom{n+1}{n+1}\boldsymbol{x}^{n+1}.\end{aligned}$$

Here we used $\binom{n}{0}=1=\binom{n+1}{0}$, $\binom{n}{n}=1=\binom{n+1}{n+1}$, and Exercise 4.16, p.40.

Solution to Exercise 5.8

Let $a_n := r$ and $b_n := s$ for all $n \in \mathbb{N}$. Take the rational $d := \frac{s-r}{2} > 0$.
Then for all $n \in \mathbb{N}$, we have $b_n - a_n = s - r > \frac{s-r}{2} = d$.
So $[(r)_{n\in\mathbb{N}}] < [(s)_{n\in\mathbb{N}}]$.

Solution to Exercise 5.9

Let $\boldsymbol{x} = [(a_n)_{n\in\mathbb{N}}]$, $\boldsymbol{y} = [(b_n)_{n\in\mathbb{N}}]$, $\boldsymbol{z} = [(c_n)_{n\in\mathbb{N}}]$.
As $\boldsymbol{x} < \boldsymbol{y}$, there exists a rational $d > 0$ and an $N \in \mathbb{N}$ such that

$$b_n - a_n > d \quad \text{for all } n > N. \tag{$\star$}$$

As $\boldsymbol{y} < \boldsymbol{z}$, there exists a rational $d' > 0$ and an $N' \in \mathbb{N}$ such that

$$c_n - b_n > d' \quad \text{for all } n > N'. \tag{$\star\star$}$$

Set $N'' = N + N'$, and let the rational number $d'' = d + d' > 0$.
Then ($\star$) and ($\star\star$) give for all $n > N''$ that

$$c_n - a_n = c_n - b_n + b_n - a_n > d' + d = d''.$$

Thus $\boldsymbol{x} < \boldsymbol{z}$.

Solution to Exercise 5.10

Suppose that $\boldsymbol{x} = [(a_n)_{n\in\mathbb{N}}]$, $\boldsymbol{y} = [(b_n)_{n\in\mathbb{N}}]$, and $\boldsymbol{z} = [(c_n)_{n\in\mathbb{N}}]$, where the sequences $(a_n)_{n\in\mathbb{N}}, (b_n)_{n\in\mathbb{N}}, (c_n)_{n\in\mathbb{N}} \in \mathscr{C}$. Then $\boldsymbol{x} + \boldsymbol{z} = [(a_n + c_n)_{n\in\mathbb{N}}]$ and $\boldsymbol{y} + \boldsymbol{c} = [(b_n + c_n)_{n\in\mathbb{N}}]$. Since $\boldsymbol{x} < \boldsymbol{y}$, there exist an $N \in \mathbb{N}$ and a rational $d > 0$ such that for all natural numbers $n > N$, we have $b_n - a_n > d$, and so $(b_n + c_n) - (a_n + c_n) > d$ too, showing that $\boldsymbol{x} + \boldsymbol{z} < \boldsymbol{y} + \boldsymbol{z}$.

Solution to Exercise 5.11

Let $\boldsymbol{x} = [(a_n)_{n\in\mathbb{N}}]$ and $\boldsymbol{y} = [(b_n)_{n\in\mathbb{N}}]$. As $\boldsymbol{x} > \boldsymbol{0}$, there exists a rational $d > 0$ and an $N \in \mathbb{N}$ such that for all $n > N$, we have $a_n - 0 = a_n > d$. Similarly, as $\boldsymbol{y} > \boldsymbol{0}$, there exists a rational $d' > 0$ and an $N' \in \mathbb{N}$ such that for all $n > N'$, we have $b_n - 0 = b_n > d'$. So with $N'' = N + N'$, we have for all $n > N''$ that $a_n + b_n - 0 = a_n + b_n > d + b_n > d + d' =: d'' > 0$. Thus $\boldsymbol{x} + \boldsymbol{y} > \boldsymbol{0}$. Also, for $n > N''$, we have $a_n b_n - 0 = a_n b_n > d b_n > d d' =: \widetilde{d} > 0$. So $\boldsymbol{x} \cdot \boldsymbol{y} > \boldsymbol{0}$.

Solution to Exercise 5.12

Let $\boldsymbol{x} = [(a_n)_{n\in\mathbb{N}}]$. As $\boldsymbol{x} > \boldsymbol{0}$, there exists a rational $d > 0$, and an $N \in \mathbb{N}$ such that for all $n > N$, $a_n - 0 > d$. Define $r = \frac{d}{2} \in \mathbb{Q}$. Then $r > 0$, and for all $n > N$, $a_n - r = a_n - \frac{d}{2} > d - \frac{d}{2} = \frac{d}{2}$. Thus $\boldsymbol{x} = [(a_n)_{n\in\mathbb{N}}] > [(r)_{n\in\mathbb{N}}] > [(0)_{n\in\mathbb{N}}] = \boldsymbol{0}$.

Solution to Exercise 5.13

Suppose $\neg(\boldsymbol{x} \geqslant \boldsymbol{0})$. Then by the trichotomy law, $\boldsymbol{x} < \boldsymbol{0}$. So there exists an $N \in \mathbb{N}$ and a rational $d > 0$, such that for all $n > N$, we have $0 - a_n > d$. Thus for $n = N + 1$, we get $d < -a_n = -a_n + 0 \leqslant -a_n + a_n = 0$, a contradiction to $d > 0$. So $\boldsymbol{x} \geqslant 0$.

Solution to Exercise 5.14

(1) We consider the various possible cases separately.

1° $\boldsymbol{x}=\boldsymbol{0}$ or $\boldsymbol{y}=\boldsymbol{0}$. Then $\boldsymbol{x}\cdot\boldsymbol{y}=\boldsymbol{0}$, and so $|\boldsymbol{x}\cdot\boldsymbol{y}|=\boldsymbol{0}$.
Also, $|\boldsymbol{x}|=\boldsymbol{0}$ or $|\boldsymbol{y}|=\boldsymbol{0}$, and so $|\boldsymbol{x}|\cdot|\boldsymbol{y}|=\boldsymbol{0}$ as well.

2° $\boldsymbol{x}>\boldsymbol{0}$ and $\boldsymbol{y}>\boldsymbol{0}$. Then $\boldsymbol{x}\cdot\boldsymbol{y}>\boldsymbol{0}$, and so $|\boldsymbol{x}\cdot\boldsymbol{y}|=\boldsymbol{x}\cdot\boldsymbol{y}=|\boldsymbol{x}|\cdot|\boldsymbol{y}|$.

3° $\boldsymbol{x}>\boldsymbol{0}$ and $\boldsymbol{y}<\boldsymbol{0}$. Then $\boldsymbol{x}\cdot\boldsymbol{y}<\boldsymbol{x}\cdot\boldsymbol{0}=\boldsymbol{0}$ using Exercise 5.11, p.48.
So $|\boldsymbol{x}\cdot\boldsymbol{y}|=-(\boldsymbol{x}\cdot\boldsymbol{y})=(-\boldsymbol{1})\cdot\boldsymbol{x}\cdot\boldsymbol{y}=\boldsymbol{x}\cdot((-\boldsymbol{1})\cdot\boldsymbol{y})=\boldsymbol{x}\cdot(-\boldsymbol{y})=|\boldsymbol{x}|\cdot|\boldsymbol{y}|$.

4° $\boldsymbol{x}<\boldsymbol{0}$ and $\boldsymbol{y}>\boldsymbol{0}$. Follows from 3°: $|\boldsymbol{x}\cdot\boldsymbol{y}|=|\boldsymbol{y}\cdot\boldsymbol{x}|=|\boldsymbol{y}|\cdot|\boldsymbol{x}|=|\boldsymbol{x}|\cdot|\boldsymbol{y}|$.

5° $\boldsymbol{x}<\boldsymbol{0}$ and $\boldsymbol{y}<\boldsymbol{0}$. As $\boldsymbol{x}<\boldsymbol{0}$, adding $-\boldsymbol{x}$ on both sides gives $-\boldsymbol{x}>\boldsymbol{0}$.
So $-\boldsymbol{x}\cdot\boldsymbol{y}<-\boldsymbol{x}\cdot\boldsymbol{0}=\boldsymbol{0}$, giving $-(\boldsymbol{x}\cdot\boldsymbol{y})=(-\boldsymbol{1})\cdot(\boldsymbol{x}\cdot\boldsymbol{y})<\boldsymbol{0}$, i.e., $\boldsymbol{x}\cdot\boldsymbol{y}>\boldsymbol{0}$.
Hence $|\boldsymbol{x}\cdot\boldsymbol{y}|=\boldsymbol{x}\cdot\boldsymbol{y}$.
Also, $|\boldsymbol{x}|\cdot|\boldsymbol{y}|=(-\boldsymbol{x})\cdot(-\boldsymbol{y})=(-\boldsymbol{1})\cdot\boldsymbol{x}\cdot(-\boldsymbol{1})\cdot\boldsymbol{y}=\boldsymbol{1}\cdot\boldsymbol{x}\cdot\boldsymbol{y}=\boldsymbol{x}\cdot\boldsymbol{y}$.

(2) 1° If $\boldsymbol{x}\geqslant\boldsymbol{0}$, then $|\boldsymbol{x}|=\boldsymbol{x}$.

2° If $\boldsymbol{x}<\boldsymbol{0}$, then adding $-\boldsymbol{x}$ on both sides gives $-\boldsymbol{x}>\boldsymbol{0}$.
So $|-\boldsymbol{x}|=-\boldsymbol{x}>\boldsymbol{0}>\boldsymbol{x}$.

Hence for all $\boldsymbol{x}\in\mathbb{R}$, we have $\boldsymbol{x}\leqslant|\boldsymbol{x}|$.
Thus also $-\boldsymbol{x}\leqslant|-\boldsymbol{x}|=|(-\boldsymbol{1})\cdot\boldsymbol{x}|=|-\boldsymbol{1}|\cdot|\boldsymbol{x}|=\boldsymbol{1}\cdot|\boldsymbol{x}|=|\boldsymbol{x}|$.

(3) 1° Let $\boldsymbol{x}+\boldsymbol{y}\geqslant\boldsymbol{0}$. We have $\boldsymbol{x}\leqslant|\boldsymbol{x}|$ and so $\boldsymbol{x}+|\boldsymbol{y}|\leqslant|\boldsymbol{x}|+|\boldsymbol{y}|$.
Similarly, $\boldsymbol{y}\leqslant|\boldsymbol{y}|$ yields $\boldsymbol{x}+\boldsymbol{y}\leqslant\boldsymbol{x}+|\boldsymbol{y}|$. Putting these together, we obtain $|\boldsymbol{x}+\boldsymbol{y}|=\boldsymbol{x}+\boldsymbol{y}\leqslant\boldsymbol{x}+|\boldsymbol{y}|\leqslant|\boldsymbol{x}|+|\boldsymbol{y}|$.

2° Let $\boldsymbol{x}+\boldsymbol{y}<\boldsymbol{0}$. We have $-\boldsymbol{x}\leqslant|\boldsymbol{x}|$ and so $-\boldsymbol{x}+(-\boldsymbol{y})\leqslant|\boldsymbol{x}|+(-\boldsymbol{y})$.
Similarly, as $-\boldsymbol{y}\leqslant|\boldsymbol{y}|$, we get $|\boldsymbol{x}|+(-\boldsymbol{y})\leqslant|\boldsymbol{x}|+|\boldsymbol{y}|$.
We have $|\boldsymbol{x}+\boldsymbol{y}|=-(\boldsymbol{x}+\boldsymbol{y})=(-\boldsymbol{1})\cdot(\boldsymbol{x}+\boldsymbol{y})$. Using distributivity, we get $|\boldsymbol{x}+\boldsymbol{y}|=(-\boldsymbol{1})\cdot\boldsymbol{x}+(-\boldsymbol{1})\cdot\boldsymbol{y}=-\boldsymbol{x}+(-\boldsymbol{y})\leqslant|\boldsymbol{x}|+(-\boldsymbol{y})\leqslant|\boldsymbol{x}|+|\boldsymbol{y}|$.

(4) We have $|\boldsymbol{x}|=|\boldsymbol{y}+\boldsymbol{x}-\boldsymbol{y}|\leqslant|\boldsymbol{y}|+|\boldsymbol{x}-\boldsymbol{y}|$. Adding $-|\boldsymbol{y}|$ yields

$$|\boldsymbol{x}|-|\boldsymbol{y}|\leqslant|\boldsymbol{x}-\boldsymbol{y}|. \qquad (*)$$

Swapping $\boldsymbol{x}$ and $\boldsymbol{y}$, $|\boldsymbol{y}|-|\boldsymbol{x}|\leqslant|\boldsymbol{y}-\boldsymbol{x}|=|-(\boldsymbol{x}-\boldsymbol{y})|=|-\boldsymbol{1}|\cdot|\boldsymbol{x}-\boldsymbol{y}|=|\boldsymbol{x}-\boldsymbol{y}|$. So

$$-(|\boldsymbol{x}|-|\boldsymbol{y}|)\leqslant|\boldsymbol{x}-\boldsymbol{y}|. \qquad (**)$$

From $(*)$ and $(**)$, we get $||\boldsymbol{x}|-|\boldsymbol{y}||\leqslant|\boldsymbol{x}-\boldsymbol{y}|$.

Solution to Exercise 5.15

(1) We use induction on n. If $n=1$, then equality holds ($|\boldsymbol{a}_1|=|\boldsymbol{a}_1|$).
Suppose the claim is true for some $n\in\mathbb{N}$. Let $\boldsymbol{a}_1,\cdots,\boldsymbol{a}_n,\boldsymbol{a}_{n+1}\in\mathbb{R}$.
By the triangle inequality and the induction hypothesis, we obtain that

$$\begin{aligned}|\boldsymbol{a}_1+\cdots+\boldsymbol{a}_n+\boldsymbol{a}_{n+1}|&\leqslant|\boldsymbol{a}_1+\cdots+\boldsymbol{a}_n|+|\boldsymbol{a}_{n+1}|\\&\leqslant|\boldsymbol{a}_1|+\cdots+|\boldsymbol{a}_n|+|\boldsymbol{a}_{n+1}|,\end{aligned}$$

and so the result follows by induction.

(2) Let $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}$ and $|\boldsymbol{a}| + |\boldsymbol{b}| = |\boldsymbol{a} + \boldsymbol{b}|$. Suppose $\boldsymbol{a}, \boldsymbol{b}$ do not have the same sign. Then one must be positive and the other must be negative. Without loss of generality, we assume $\boldsymbol{a} < \mathbf{0} < \boldsymbol{b}$. We have two cases:

1° If $\boldsymbol{a} + \boldsymbol{b} \geqslant \mathbf{0}$, then $-\boldsymbol{a} + \boldsymbol{b} = |\boldsymbol{a}| + |\boldsymbol{b}| = |\boldsymbol{a} + \boldsymbol{b}| = \boldsymbol{a} + \boldsymbol{b}$.
So $\boldsymbol{a} = \mathbf{0}$, a contradiction.

2° If $\boldsymbol{a} + \boldsymbol{b} < 0$, then $-\boldsymbol{a} + \boldsymbol{b} = |\boldsymbol{a}| + |\boldsymbol{b}| = |\boldsymbol{a} + \boldsymbol{b}| = -(\boldsymbol{a} + \boldsymbol{b}) = -\boldsymbol{a} + (-\boldsymbol{b})$.
So $\boldsymbol{b} = \mathbf{0}$, a contradiction.

So $\boldsymbol{a}, \boldsymbol{b}$ must have the same sign. Conversely, if $\boldsymbol{a}, \boldsymbol{b}$ have the same sign, then clearly we have equality.

For the general equality case, we use induction on n again. Suppose that the result holds for some $n \in \mathbb{N}$ and that

$$|\boldsymbol{a}_1| + \cdots + |\boldsymbol{a}_n| + |\boldsymbol{a}_{n+1}| = |\boldsymbol{a}_1 + \cdots + \boldsymbol{a}_n + \boldsymbol{a}_{n+1}|.$$

By the triangle inequality, we have

$$\begin{aligned} |\boldsymbol{a}_1| + \cdots + |\boldsymbol{a}_n| + |\boldsymbol{a}_{n+1}| &= |\boldsymbol{a}_1 + \cdots + \boldsymbol{a}_n + \boldsymbol{a}_{n+1}| \\ &\leqslant |\boldsymbol{a}_1 + \cdots + \boldsymbol{a}_n| + |\boldsymbol{a}_{n+1}| \\ &\leqslant |\boldsymbol{a}_1| + \cdots + |\boldsymbol{a}_n| + |\boldsymbol{a}_{n+1}|, \end{aligned}$$

and so all the inequalities above must be equalities. In particular,

$$|\boldsymbol{a}_1 + \cdots + \boldsymbol{a}_n| + |\boldsymbol{a}_{n+1}| = |\boldsymbol{a}_1| + \cdots + |\boldsymbol{a}_n| + |\boldsymbol{a}_{n+1}|$$

gives $|\boldsymbol{a}_1 + \cdots + \boldsymbol{a}_n| = |\boldsymbol{a}_1| + \cdots + |\boldsymbol{a}_n|$. By the induction hypothesis, $\boldsymbol{a}_1, \cdots, \boldsymbol{a}_n$ must have the same sign. Also

$$|\boldsymbol{a}_1 + \cdots + \boldsymbol{a}_n + \boldsymbol{a}_{n+1}| = |\boldsymbol{a}_1 + \cdots + \boldsymbol{a}_n| + |\boldsymbol{a}_{n+1}|$$

shows that $\boldsymbol{a}_{n+1}$ must have the same sign as $\boldsymbol{a}_1 + \cdots + \boldsymbol{a}_n$. But since $\boldsymbol{a}_1, \cdots, \boldsymbol{a}_n$ all have the same sign, $\boldsymbol{a}_1 + \cdots + \boldsymbol{a}_n$ has the same sign as $\boldsymbol{a}_1, \cdots, \boldsymbol{a}_n$. Consequently, $\boldsymbol{a}_1, \cdots, \boldsymbol{a}_n, \boldsymbol{a}_{n+1}$ have the same sign.

Solution to Exercise 5.16

The claim is trivial if $n = 1$, as $\boldsymbol{a}_1 = \mathbf{1}$. Suppose that the claim holds for some $n \geqslant 1$. Let $\boldsymbol{a}_1, \cdots, \boldsymbol{a}_{n+1}$ be positive real numbers whose product is $\mathbf{1}$. Without loss of generality, we may assume that[17] $\boldsymbol{a}_1 = \max\{\boldsymbol{a}_1, \cdots, \boldsymbol{a}_n\}$ and $\boldsymbol{a}_2 = \min\{\boldsymbol{a}_1, \cdots, \boldsymbol{a}_2\}$. Then $\boldsymbol{a}_1^n \geqslant \boldsymbol{a}_1 \cdot \ldots \cdot \boldsymbol{a}_n = \mathbf{1}$, and so it follows that $\boldsymbol{a}_1 \geqslant \mathbf{1}$. Similarly, $\boldsymbol{a}_2 \leqslant \mathbf{1}$. Thus $\boldsymbol{a}_1 - \mathbf{1} \geqslant \mathbf{0}$ and $\mathbf{1} - \boldsymbol{a}_2 \geqslant \mathbf{0}$, implying $(\boldsymbol{a}_1 - \mathbf{1}) \cdot (\mathbf{1} - \boldsymbol{a}_2) \geqslant \mathbf{0}$, i.e.,

$$\boldsymbol{a}_1 + \boldsymbol{a}_2 - \boldsymbol{a}_1\boldsymbol{a}_2 \geqslant \mathbf{1}. \qquad (\alpha)$$

The induction hypothesis applied to $\widetilde{\boldsymbol{a}}_1 := \boldsymbol{a}_1\boldsymbol{a}_2$, $\widetilde{\boldsymbol{a}}_2 = \boldsymbol{a}_3$, $\cdots$, $\widetilde{\boldsymbol{a}}_n = \boldsymbol{a}_{n+1}$, whose product is $\mathbf{1}$, implies that $\widetilde{\boldsymbol{a}}_1 + \cdots + \widetilde{\boldsymbol{a}}_n \geqslant \mathbf{1}$, i.e.,

$$\boldsymbol{a}_1\boldsymbol{a}_2 + \boldsymbol{a}_3 + \cdots + \boldsymbol{a}_{n+1} \geqslant n. \qquad (\beta)$$

[17] For the definitions of max, min, see p.52 and p.54.

From (α) and (β), we get

$$\boldsymbol{a}_1 + \boldsymbol{a}_2 + \boldsymbol{a}_3 + \cdots + \boldsymbol{a}_{n+1} \geqslant n+1. \qquad (\gamma)$$

If $\boldsymbol{a}_1 = \cdots = \boldsymbol{a}_{n+1} = \mathbf{1}$, then equality holds in (γ). On the other hand, if equality holds in (γ), then equality must hold in (α) and in (β). Equality in (β) holds if and only if $\boldsymbol{a}_1\boldsymbol{a}_2 = \boldsymbol{a}_3 = \cdots = \boldsymbol{a}_{n+1} = \mathbf{1}$. Equality in (α) holds if and only if $\boldsymbol{a}_1 = \mathbf{1}$ or $\boldsymbol{a}_2 = \mathbf{1}$. If $\boldsymbol{a}_1 = \mathbf{1}$, then we get $\mathbf{1}\boldsymbol{a}_2 = \boldsymbol{a}_3 = \cdots = \boldsymbol{a}_{n+1} = \mathbf{1}$, and similarly, if $\boldsymbol{a}_2 = \mathbf{1}$, then we get $\boldsymbol{a}_1\mathbf{1} = \cdots = \boldsymbol{a}_{n+1} = \mathbf{1}$, showing that in either case we have $\boldsymbol{a}_1 = \boldsymbol{a}_2 = \cdots = \boldsymbol{a}_{n+1} = \mathbf{1}$.

Solution to Exercise 5.17

For all $1 \leqslant i, j \leqslant n$, we have $(\boldsymbol{a}_i\boldsymbol{b}_j - \boldsymbol{a}_j\boldsymbol{b}_i)^2 \geqslant \mathbf{0}$. Adding these, we get

$$\begin{aligned}
\mathbf{0} &\leqslant \sum_{i=1}^{n}\sum_{j=1}^{n}(\boldsymbol{a}_i\boldsymbol{b}_j - \boldsymbol{a}_j\boldsymbol{b}_i)^2 \\
&= \sum_{i=1}^{n}\sum_{j=1}^{n}(\boldsymbol{a}_i^2\boldsymbol{b}_j^2 - 2\boldsymbol{a}_i\boldsymbol{b}_j\boldsymbol{a}_j\boldsymbol{b}_i + \boldsymbol{a}_j^2\boldsymbol{b}_i^2) \\
&= \sum_{i=1}^{n}\boldsymbol{a}_i^2\sum_{j=1}^{n}\boldsymbol{b}_j^2 - 2\sum_{i=1}^{n}\boldsymbol{a}_i\boldsymbol{b}_i\sum_{j=1}^{n}\boldsymbol{a}_j\boldsymbol{b}_j + \sum_{i=1}^{n}\boldsymbol{b}_i^2\sum_{j=1}^{n}\boldsymbol{a}_j^2 \\
&= 2\Big(\sum_{i=1}^{n}\boldsymbol{a}_i^2\sum_{j=1}^{n}\boldsymbol{b}_j^2 - \big(\sum_{i=1}^{n}\boldsymbol{a}_i\boldsymbol{b}_i\big)^2\Big),
\end{aligned}$$

proving the inequality.

If we have equality, then from the above, we see that for $1 \leqslant i, j \leqslant n$, we have $\boldsymbol{a}_i\boldsymbol{b}_j - \boldsymbol{a}_j\boldsymbol{b}_i = \mathbf{0}$. If it is not the case that $\boldsymbol{a}_1 = \cdots = \boldsymbol{a}_n = \mathbf{0}$, then there exists an $i_* \in \{1, \cdots, n\}$ such that $a_{i_*} \neq 0$. Then for each $j \in \{1, \cdots, n\}$, $\boldsymbol{a}_{i_*}\boldsymbol{b}_j - \boldsymbol{a}_j\boldsymbol{b}_{i_*} = 0$ can be rewritten as $\boldsymbol{b}_j = \boldsymbol{k}\boldsymbol{a}_j$, where

$$\boldsymbol{k} := \frac{\boldsymbol{b}_{i_*}}{\boldsymbol{a}_{i_*}} \in \mathbb{R}.$$

If $\boldsymbol{a}_1 = \cdots = \boldsymbol{a}_n = \mathbf{0}$, then both sides equal $\mathbf{0}$, and so equality holds. Also, if there exists $\boldsymbol{k} \in \mathbb{R}$ such that $\boldsymbol{b}_i = \boldsymbol{k}\boldsymbol{a}_i$ for all $i \in \{1, \cdots, n\}$, then the left-hand side is $\boldsymbol{k}^2 \cdot (\boldsymbol{a}_1^2 + \cdots + \boldsymbol{a}_n^2)^2$, while the right-hand side is $(\boldsymbol{k} \cdot (\boldsymbol{a}_1^2 + \cdots + \boldsymbol{a}_n^2))^2$, and so again equality holds.

Solution to Exercise 5.18

Let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}$ and $\boldsymbol{x} < \boldsymbol{y}$. Then $\boldsymbol{x} + (-\sqrt{2}) < \boldsymbol{y} + (-\sqrt{2})$. By the density of $\mathbb{Q}$ in $\mathbb{R}$, there exists an $r \in \mathbb{Q}$ such that $\boldsymbol{x} + (-\sqrt{2}) < r < \boldsymbol{y} + (-\sqrt{2})$. So $\boldsymbol{x} < r + \sqrt{2} < \boldsymbol{y}$. We have $r + \sqrt{2} \in \mathbb{R}\backslash\mathbb{Q}$. (Indeed, if $r + \sqrt{2} \in \mathbb{Q}$, then since $-r \in \mathbb{Q}$, one gets $\sqrt{2} = (r + \sqrt{2}) + (-r) \in \mathbb{Q}$, a contradiction.)

Solution to Exercise 5.19

Suppose it is not the case that there exists an $n \in \mathbb{N}$ such that $nx > y$. So for *every* $n \in \mathbb{N}$, we have $nx \leqslant y$. Let $S = \{nx : n \in \mathbb{N}\} \subset \mathbb{R}$. Then $S \neq \varnothing$ (e.g. $x = 1x \in S$). Also, y is an upper bound of S. Thus by the least upper bound property of $\mathbb{R}$, S has a least upper bound u_*. As $x > 0$, we

have $u_* - x < u_*$. Hence $u_* - x$ cannot be an upper bound of S, and so there exists an element $mx \in S$, for some $m \in \mathbb{N}$, such that $mx > u_* - x$. Rearranging, we obtain $u_* < mx + x = (m+1)x \in S$, contradicting the fact that u_* is an upper bound of S. Thus our original claim is false, i.e., there *does exist* an $n \in \mathbb{N}$ such that $nx > y$.

Solution to Exercise 5.20

By the Archimedean property of $\mathbb{R}$, there exists an $n_1 \in \mathbb{N}$ such that $n_1 1 > x$. Also, there exists an $n_2 \in \mathbb{N}$ such that $n_2 1 > -x$. So we have $-n_2 < x < n_1$. Define $S_x = \{m \in \mathbb{Z} : m \leqslant x\}$. Then $m := -n_2 \in S_x$ (as $-n_2 \leqslant x$), and so S_x is not empty. Also, S_x has n_1 as an upper bound (if $m \in S_x$, then $m \leqslant x < n_1$). By Exercise 3.18, p.33, S_x has a unique greatest element, which we denote by $[x]$. Since $[x] \in S_x$, we have $[x] \leqslant x$. Moreover, if $[x] + 1 \leqslant x$, then $[x] + 1 \in S_x$, and so $[x] + 1 \leqslant [x]$, a contradiction. Thus $x < [x] + 1$.

Uniqueness: Let k_1, k_2 be distinct integers satisfying $k_1 \leqslant x < k_1 + 1$ and $k_2 \leqslant x < k_2 + 1$. Assume, without loss of generality that $k_1 < k_2$. Then $0 < k_2 - k_1 \leqslant x - k_1 < 1$, i.e., the integer $\ell := k_2 - k_1$ satisfies $0 < \ell < 1$, a contradiction to Exercise 3.14, p.32.

Let $m \in \mathbb{Z}$ satisfy $m \leqslant x$. Then $m \in S_x$ and so $m \leqslant [x]$.

Solution to Exercise 5.21

Define $-S = \{-x : x \in S\} \subset \mathbb{R}$. As S is not empty, there exists an $x \in S$. Then $-x \in -S$, showing that $-S \neq \varnothing$. Let ℓ be a lower bound of S. Then for all $x \in S$, we have $\ell \leqslant x$. So for all $x \in S$, we have $-x \leqslant -\ell$, i.e., for all $y \in -S$, we have $y \leqslant -\ell$. Thus $-S$ is bounded above (as $-\ell$ is an upper bound of $-S$). By the least upper bound property of $\mathbb{R}$, $-S$ has a least upper bound, which we denote by u_*. We claim that $-u_*$ is a greatest lower bound of S. Since u_* is an upper bound of $-S$, for all $y \in -S$, we have $y \leqslant u_*$, i.e., for all $x \in S$, we have $-x \leqslant u_*$. Hence for all $x \in S$, we have $-u_* \leqslant x$. So $-u_*$ is a lower bound of S. Suppose ℓ is a lower bound of S such that $-u_* < \ell$. Then for all $x \in S$, we have $-u_* < \ell \leqslant x$, i.e., for all $x \in S$, we have $-x \leqslant -\ell < u_*$. Hence for all $y \in -S$, we have $y \leqslant -\ell < u_*$. So $-\ell$ is an upper bound of $-S$, and $-\ell < u_*$, which contradicts the fact that u_* is the least upper bound of $-S$. Hence $\ell \leqslant -u_*$.

Solution to Exercise 5.22

(1) Let u be an upper bound of S, and ℓ be a lower bound of S. Since $\mathrm{Row}_n \subset S$, u is an upper bound of Row_n. As $a_{n1} \in \mathrm{Row}_n$, $\mathrm{Row}_n \neq \varnothing$. By the least upper bound property of $\mathbb{R}$, $\sup \mathrm{Row}_n$ exists. As $\mathrm{Col}_m \subset S$, ℓ is a lower bound of Col_m. Since $a_{1m} \in \mathrm{Col}_m$, $\mathrm{Col}_m \neq \varnothing$. By the greatest lower bound property of $\mathbb{R}$, $\inf \mathrm{Col}_m$ exists.

(2) As $a_{nm} \in \text{Row}_n$, $a_{nm} \leqslant \sup \text{Row}_n$. Since $a_{nm} \in \text{Col}_m$, $\inf \text{Col}_m \leqslant a_{nm}$. Hence $\inf \text{Col}_m \leqslant a_{nm} \leqslant \sup \text{Row}_n$. In particular $\inf \text{Col}_m \leqslant \sup \text{Row}_n$.

(3) From part (2), the nonempty set $\{\sup \text{Row}_n : n \in \mathbb{N}\}$ is bounded below by $\inf \text{Col}_1$. By the greatest lower bound property of $\mathbb{R}$, it follows that $\inf\{\sup \text{Row}_n : n \in \mathbb{N}\}$ exists. Also, the nonempty set $\{\inf \text{Col}_m : m \in \mathbb{N}\}$ is bounded above by $\sup \text{Row}_1$. The least upper bound property of $\mathbb{R}$ implies that $\sup\{\inf \text{Col}_m : m \in \mathbb{N}\}$ exists.

(4) Fix $n \in \mathbb{N}$. From (2), $\{\inf \text{Col}_m : m \in \mathbb{N}\}$ is bounded above by $\sup \text{Row}_n$. Thus $\sup\{\inf \text{Col}_m : m \in \mathbb{N}\} \leqslant \sup \text{Row}_n$. As $n \in \mathbb{N}$ was arbitrary, we have that for all $n \in \mathbb{N}$, $L := \sup\{\inf \text{Col}_m : m \in \mathbb{N}\} \leqslant \sup \text{Row}_n$. So L is a lower bound for $\{\sup \text{Row}_n : n \in \mathbb{N}\}$. Thus $L \leqslant \inf\{\sup \text{Row}_n : n \in \mathbb{N}\}$, i.e., $\sup\{\inf \text{Col}_m : m \in \mathbb{N}\} \leqslant \inf\{\sup \text{Row}_n : n \in \mathbb{N}\}$.

(5) Let $a_{nn} = 1$ for all $n \in \mathbb{N}$, and $a_{nm} = 0$ if $n \neq m$:

$$\begin{array}{|cccc} 1 & 0 & 0 & \cdots \\ 0 & 1 & 0 & \cdots \\ 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{array}$$

For $n, m \in \mathbb{N}$, $\text{Row}_n = \{0, 1\} = \text{Col}_m$. Thus $\sup \text{Row}_n = 1$ and $\inf \text{Col}_m = 0$. So $\sup\{\inf \text{Col}_m : m \in \mathbb{N}\} = \sup\{0\} = 0 < 1 = \inf\{1\} = \inf\{\sup \text{Row}_n : n \in \mathbb{N}\}$.

Solution to Exercise 5.23

Let $(a_n)_{n\in\mathbb{N}}$ be a real sequence that is convergent in $\mathbb{R}$ to limits x_1 and x_2, with $x_1 \neq x_2$. Then $\epsilon = \frac{|x_1 - x_2|}{3} > 0$. As x_1 is a limit of $(a_n)_{n\in\mathbb{N}}$, there exists an $N_1 \in \mathbb{N}$ such that for all $n \in \mathbb{N}$ satisfying $n > N_1$, we have $|a_n - x_1| < \epsilon$. As x_2 is a limit of $(a_n)_{n\in\mathbb{N}}$, there exists an $N_2 \in \mathbb{N}$ such that for all $n \in \mathbb{N}$ satisfying $n > N_2$, we have $|a_n - x_2| < \epsilon$. For $n > N_1 + N_2$, we have $n > N_1$ and $n > N_2$, and so $|x_1 - x_2| = |x_1 - a_n + a_n - x_2| \leqslant |x_1 - a_n| + |a_n - x_2| < \epsilon + \epsilon = \frac{2}{3}|x_1 - x_2|$. So we arrive at the contradiction that $1 < \frac{2}{3}$. Hence our original assumption was incorrect, and so a convergent sequence must have a unique limit.

Solution to Exercise 5.24

Suppose $x < 0$. Then $\epsilon := -\frac{x}{2} > 0$, and so there exists an $N \in \mathbb{N}$ such that for all $n > N$, we have $|a_n - x| < \epsilon = -\frac{x}{2}$. Hence with $n = N + 1$ $(> N)$, we have $a_{N+1} - x \leqslant |a_{N+1} - x| < -\frac{x}{2}$, i.e., $a_{N+1} < \frac{x}{2} < 0$, a contradiction.

Denote the limits of $(\alpha_n)_{n\in\mathbb{N}}$ and $(\beta_n)_{n\in\mathbb{N}}$ by a, b, respectively. In light of the above, it is enough to show that $(\alpha_n - \beta_n)_{n\in\mathbb{N}}$ converges in $\mathbb{R}$ with limit $a - b$. Let $\epsilon > 0$. As $(\alpha_n)_{n\in\mathbb{N}}$ is convergent in $\mathbb{R}$ to a, there exists an $N_1 \in \mathbb{N}$ such that for all $n > N_1$, $|\alpha_n - a| < \frac{\epsilon}{2}$. Similarly, as $(\beta_n)_{n\in\mathbb{N}}$ is convergent in $\mathbb{R}$ to b, there exists an $N_2 \in \mathbb{N}$ such that for all $n > N_2$, $|\beta_n - b| = |b - \beta_n| < \frac{\epsilon}{2}$. Then for all $n > N := N_1 + N_2$, we have in particular $n > N_1$ and $n > N_2$, and so $|(\alpha_n - \beta_n) - (a - b)| = |\alpha_n - a + b - \beta_n| \leqslant |\alpha_n - a| + |b - \beta_n| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$.

Solution to Exercise 5.25

If the sequence is bounded, then for all $n \in \mathbb{N}$, we have $|a_n| \leqslant M$, i.e., $-M \leqslant a_n \leqslant M$, showing that M is an upper bound of S, and $-M$ is a lower bound of S. Thus S is bounded.

Vice versa, let S be bounded. Let u be an upper bound of S and ℓ be a lower bound of S. For all $x \in S$, we have $\ell \leqslant x \leqslant u$. Thus $x \leqslant u \leqslant \max\{u, -\ell\}$ and $-x \leqslant -\ell \leqslant \max\{u, -\ell\}$. So $0 \leqslant |x| \leqslant \max\{u, -\ell\} < \max\{u, -\ell\} + 1 =: M$. (Note that $M > |x| \geqslant 0$. So $M > 0$. We added 1 to $\max\{u, -\ell\}$ to make M positive, as demanded in the definition of a bounded sequence.) Thus $(a_n)_{n\in\mathbb{N}}$ is bounded.

Solution to Exercise 5.26

(1) Let $\epsilon > 0$ be given.
Then there exists an $N_1 \in \mathbb{N}$ such that for all $n > N_1$, we have $|a_n - x| < \frac{\epsilon}{2}$.
Also, there exists an $N_2 \in \mathbb{N}$ such that for all $n > N_2$, we have $|b_n - y| < \frac{\epsilon}{2}$.
For all $n > N := N_1 + N_2$, we have

$$|a_n + b_n - (x + y)| \leqslant |a_n - x| + |b_n - y| < \tfrac{\epsilon}{2} + \tfrac{\epsilon}{2} = \epsilon.$$

Hence $(a_n + b_n)_{n\in\mathbb{N}}$ is convergent in $\mathbb{R}$ with limit $x + y$.

(2) Let $\epsilon > 0$. Since $(b_n)_{n\in\mathbb{N}}$ is convergent in $\mathbb{R}$, it is Cauchy in $\mathbb{R}$, and hence bounded. Let $M > 0$ be such that for all $n \in \mathbb{N}$, we have $|b_n| \leqslant M$.
Let $N_1 \in \mathbb{N}$ be such that for $n > N_1$, we have $|a_n - x| < \frac{\epsilon}{2M}$.
Let $N_2 \in \mathbb{N}$ be such that for all $n > N_2$, we have $|b_n - y| < \frac{\epsilon}{2(|x|+1)}$.
Thus for $n > N := N_1 + N_2$, we have

$$\begin{aligned} |a_n b_n - xy| &= |a_n b_n - x b_n + x b_n - xy| \\ &\leqslant |a_n b_n - x b_n| + |x b_n - xy| \\ &= |a_n - x||b_n| + |x||b_n - y| \\ &< \tfrac{\epsilon}{2M} M + |x| \tfrac{\epsilon}{2(|x|+1)} < \tfrac{\epsilon}{2} + \tfrac{\epsilon}{2} = \epsilon. \end{aligned}$$

So $(a_n b_n)_{n\in\mathbb{N}}$ is convergent in $\mathbb{R}$ with limit xy.

Solution to Exercise 5.27

Let $(a_n)_{n\in\mathbb{N}}$ be a bounded decreasing sequence of real numbers. Let ℓ_* be the greatest lower bound of the nonempty bounded set $\{a_n : n \in \mathbb{N}\}$. The existence of ℓ_* is guaranteed by the greatest lower bound property of the set of real numbers. We show that ℓ_* is the limit of $(a_n)_{n\in\mathbb{N}}$. Taking $\epsilon > 0$, we must show that there exists a positive integer N such that $|a_n - \ell_*| < \epsilon$ for all $n > N$. Since $\ell_* + \epsilon > \ell_*$, $\ell_* + \epsilon$ is not a lower bound of $\{a_n : n \in \mathbb{N}\}$. Therefore there exists N with $\ell_* \leqslant a_N < \ell_* + \epsilon$. Since $(a_n)_{n\in\mathbb{N}}$ is decreasing, we have for all $n \geqslant N$ that $\ell_* - \epsilon < \ell_* \leqslant a_n \leqslant a_N < \ell_* + \epsilon$, and so $|a_n - \ell_*| < \epsilon$.

Solution to Exercise 5.28

Define $a_n = r^n$ for all $n \in \mathbb{N}$. For all $n \in \mathbb{N}$, $a_n - a_{n+1} = (1-r)r^n > 0$, showing that $(a_n)_{n\in\mathbb{N}}$ is decreasing. For all $n \in \mathbb{N}$, $|a_n| = a_n = r^n \leqslant 1$, showing that $(a_n)_{n\in\mathbb{N}}$ is bounded. By Proposition 5.6, p.56, $(a_n)_{n\in\mathbb{N}}$ is convergent in $\mathbb{R}$ with limit, say, $x \in \mathbb{R}$. Then $(r^{n+1})_{n\in\mathbb{N}} = (rr^n)_{n\in\mathbb{N}}$ converges in $\mathbb{R}$ with limit rx: indeed, given $\epsilon > 0$, if $N \in \mathbb{N}$ is such that for all $n > N$ we have $|r^n - x| < \frac{\epsilon}{r}$, then $|r^{n+1} - rx| = r|r^n - x| < r\frac{\epsilon}{r} = \epsilon$. But $(r^{n+1})_{n\in\mathbb{N}} = (a_{n+1})_{n\in\mathbb{N}}$ being a subsequence of the convergent sequence $(a_n)_{n\in\mathbb{N}}$ with limit x, is also convergent in $\mathbb{R}$ with the same limit x. By the uniqueness of limits, $rx = x$, i.e., $(1-r)x = 0$, and so $x = 0$.

We have $|s_n| = a + ar + \cdots + ar^{n-1} = \frac{a(1+r+\cdots+r^{n-1})(1-r)}{1-r} = \frac{a(1-r^n)}{1-r} \leqslant \frac{a}{1-r}$ for all $n \in \mathbb{N}$, and so $(s_n)_{n\in\mathbb{N}}$ is bounded. For all $n \in \mathbb{N}$, $s_{n+1} - s_n = ar^n > 0$, and so $(s_n)_{n\in\mathbb{N}}$ is increasing. By Proposition 5.6, p.56, $(s_n)_{n\in\mathbb{N}}$ is convergent in $\mathbb{R}$, and we claim that its limit is $\frac{a}{1-r}$. Given $\epsilon > 0$, let $N \in \mathbb{N}$ be such that for all $n > N$, we have $r^n = |r^n - 0| < \frac{(1-r)\epsilon}{a}$. Such an N exists thanks to the fact that $(r^n)_{n\in\mathbb{N}}$ converges in $\mathbb{R}$ to 0 (shown above). Thus for $n > N$, we have $|s_n - \frac{a}{1-r}| = |\frac{a(1-r^n)}{1-r} - \frac{a}{1-r}| = \frac{ar^n}{1-r} < \epsilon$.

Solution to Exercise 5.29

By the Bolzano-Weierstrass theorem, the bounded sequence $(x_n)_{n\in\mathbb{N}}$ has a convergent subsequence, say $(x_{n_k})_{k\in\mathbb{N}}$. As $|y_{n_k}| \leqslant M$ for all $k \in \mathbb{N}$, $(y_{n_k})_{k\in\mathbb{N}}$ is bounded. By the Bolzano-Weierstrass theorem, the bounded sequence $(y_{n_k})_{k\in\mathbb{N}}$ has a convergent subsequence, say $(y_{n_{k_\ell}})_{\ell\in\mathbb{N}}$. But then $(x_{n_{k_\ell}})_{\ell\in\mathbb{N}}$, being a subsequence of the convergent subsequence $(x_{n_k})_{k\in\mathbb{N}}$, is convergent too. Set $m_\ell := n_{k_\ell}$ for all $\ell \in \mathbb{N}$. Then for all $\ell \in \mathbb{N}$, we have $k_{\ell+1} > k_\ell$, and so $n_{k_{\ell+1}} > n_{k_\ell}$, i.e., $m_{\ell+1} > m_\ell$. Furthermore, $(x_{m_\ell})_{\ell\in\mathbb{N}} = (x_{n_{k_\ell}})_{\ell\in\mathbb{N}}$ and $(y_{m_\ell})_{\ell\in\mathbb{N}} = (y_{n_{k_\ell}})_{\ell\in\mathbb{N}}$ converge in $\mathbb{R}$.

Solution to Exercise 5.30

If $[x]$ denotes the greatest integer part of $x \in \mathbb{R}$, then define its fractional part $\{x\} := x - [x]$. Then $0 \leqslant \{x\} < 1$. For distinct $k, \ell \in \mathbb{N}$, $\{k\alpha\} \neq \{\ell\alpha\}$ (otherwise $k\alpha - [k\alpha] = \ell\alpha - [\ell\alpha]$ gives $\alpha = \frac{[k\alpha] - [\ell\alpha]}{k-\ell} \in \mathbb{Q}$, a contradiction). The sequence $(\{n\alpha\})_{n\in\mathbb{N}}$ lies in $[0,1]$, and so by the Bolzano-Weierstrass theorem, it has a convergent subsequence, say $(\{n_k\alpha\})_{k\in\mathbb{N}}$, and in particular, $(\{n_k\alpha\})_{k\in\mathbb{N}}$ is Cauchy. So given any $\epsilon > 0$, there exist distinct $k, \ell \in \mathbb{N}$ such that $|\{k\alpha\} - \{\ell\alpha\}| < \epsilon$. Without loss of generality, suppose $\{k\alpha\} > \{\ell\alpha\}$. Then we have $0 < \{k\alpha\} - \{\ell\alpha\} < 1$. Now let a, b be real numbers satisfying $0 \leqslant a < b \leqslant 1$. Let $\epsilon := \frac{b-a}{2} > 0$, and let k, ℓ be as above for this ϵ, so that

$0<\delta:=\{k\alpha\}-\{\ell\alpha\}<\epsilon$. Choose the smallest $N\in\mathbb{N}$ so that $N\delta>\frac{a+b}{2}$. Then $(N-1)\delta\leqslant\frac{a+b}{2}<N\delta$. So $a<\frac{a+b}{2}<N\delta<\frac{a+b}{2}+\delta<\frac{a+b}{2}+\epsilon=\frac{a+b}{2}+\frac{b-a}{2}=b$, and we note that $N\delta=N(\{k\alpha\}-\{\ell\alpha\})=N([\ell\alpha]-[k\alpha])+N(k-\ell)\alpha=m+\alpha n$, where $m:=N([\ell\alpha]-[k\alpha])\in\mathbb{Z}$ and $n:=N(k-\ell)\in\mathbb{Z}$.

Density of $\langle 1,\alpha\rangle$ in $\mathbb{R}$: Suppose that $a,b\in\mathbb{R}$ and $a<b$.

1° If $b-[a]>1$, then with $m=1+[a]\in\mathbb{Z}$ and $n=0$, we have $m+n\alpha=1+[a]$, and $a<1+[a]<b$, and we are done.

2° If $b-[a]\leqslant 1$, then $0\leqslant a-[a]<b-[a]\leqslant 1$, and so it follows from the above that there exist $m,n\in\mathbb{Z}$ such that $a-[a]<m+n\alpha<b-[a]$. But then with $\widetilde{m}=m+[a]\in\mathbb{Z}$ we have $a<\widetilde{m}+n\alpha<b$.

This completes the proof.

Solution to Exercise 5.31

First consider the case when $x\in[-1,1]$ is nonzero. Let $\theta\in\mathbb{R}$ be such that $\sin\theta=x$. For $k\in\mathbb{N}$, $\theta<\theta+\frac{1}{k}$. Since $2\pi\in\mathbb{R}\backslash\mathbb{Q}$, Kronecker's theorem implies that $\theta<m_k+n_k(2\pi)<\theta+\frac{1}{k}$ for some $m_k,n_k\in\mathbb{Z}$, and so $|m_k+2\pi n_k-\theta|<\frac{1}{k}$. Hence $(m_k+2\pi n_k)_{k\in\mathbb{N}}$ converges to θ. As sin is continuous, we get[18] that $(\sin(m_k+2\pi n_k))_{k\in\mathbb{N}}$ converges to $\sin\theta$, i.e., $(\sin m_k)_{k\in\mathbb{N}}$ converges to x. If for infinitely many $k\in\mathbb{N}$ we have $m_k=0$, then we would get the contradiction that $x=0$. So there can be at most finitely many $k\in\mathbb{N}$ for which $m_k=0$. We 'start' the sequence after that last term with $m_k=0$ has passed, and so below we suppose that for each $k\in\mathbb{N}$, $m_k\neq 0$. Now we consider separately the two possible cases below.

Suppose that $m_k>0$ for infinitely many $k\in\mathbb{N}$. Then by extracting the subsequence of $(\sin m_k)_{k\in\mathbb{N}}$ corresponding to those terms, we get a subsequence $(\sin m_{k_\ell})_{\ell\in\mathbb{N}}$, such that each $m_{k_\ell}\in\mathbb{N}$ for all $\ell\in\mathbb{N}$. Moreover, since $(\sin m_{k_\ell})_{\ell\in\mathbb{N}}$ converges to x, we are done.

Suppose $m_k>0$ for at most finitely many $k\in\mathbb{N}$. We start the sequence after that last term with $m_k>0$ has passed, and so we may assume $m_k<0$ for all $k\in\mathbb{N}$. Again using Kronecker's theorem, there exists a sequence $(\widetilde{m}_k+2\pi\widetilde{n}_k)_{k\in\mathbb{N}}$ that converges to π. Suppose there are infinitely many terms such that $\widetilde{m}_k\geqslant 0$. Then extracting that subsequence, we may assume that $\widetilde{m}_k\geqslant 0$ for all $k\in\mathbb{N}$. Define $m'_k=\widetilde{m}_k-m_k\geqslant 0-m_k=-m_k\in\mathbb{N}$ for all $k\in\mathbb{N}$. Then

$$\begin{aligned}\sin m'_k &= \sin(\widetilde{m}_k-m_k)=\sin(\widetilde{m}_k-m_k+2\pi(\widetilde{n}_k-n_k))\\ &= \sin(\widetilde{m}_k+2\pi\widetilde{n}_k-(m_k+2\pi\widetilde{n}_k)))\\ &\to \sin(\pi-\theta)=\sin\theta=x\end{aligned}$$

[18] See, e.g., Exercise 10.32, p.120.

as $k \to \infty$, and so $(\sin m'_k)_{k\in\mathbb{N}}$ converges to x, and we are done. On the other hand, if there are only finitely many $k \in \mathbb{N}$ such that $\widetilde{m}_k \geqslant 0$, then there is no loss of generality in assuming $\widetilde{m}_k < 0$ for all $k \in \mathbb{N}$. Set $m'_k = -\widetilde{m}_k - m_k$ for all $k \in \mathbb{N}$. Then $m'_k > 0$ and so $m'_k \in \mathbb{N}$ for all $k \in \mathbb{N}$. Then we have

$$\begin{aligned}\sin m'_k &= \sin(-\widetilde{m}_k - m_k) = \sin(-(\widetilde{m}_k + 2\pi\widetilde{n}_k) - (m_k + 2\pi\widetilde{n}_k)) \\ &\to \sin(-\pi - \theta) = \sin\theta = x\end{aligned}$$

as $k \to \infty$. So again $(\sin m'_k)_{k\in\mathbb{N}}$ converges to x, and we are done.

It remains to consider the case $x = 0$. For $k \in \mathbb{N}$, $\epsilon_k := \frac{1}{2k} > 0$, and $x_k := \frac{1}{k}$ is such that $0 < x_k \leqslant 1$. By the preceding case, there is an $n_k \in \mathbb{N}$ such that $|\sin n_k - x_k| < \epsilon_k$, i.e, $\frac{1}{2k} < \sin n_k < \frac{3}{2k}$. Then $(\sin n_k)_{k\in\mathbb{N}}$ converges to 0: indeed, $|\sin n_k - 0| = |\sin n_k| = \sin n_k < \frac{3}{2k} \to 0$ as $k \to \infty$.

Solution to Exercise 5.32

S is not empty: For example, the identity map ι defined by $\iota(n) = n$ for all $n \in \mathbb{Z}$ belongs to S, since $\iota(n+m) - \iota(n) - \iota(m) = n + m - n - m = 0$ for all $n, m \in \mathbb{Z}$. So $\{\iota(n+m) - \iota(n) - \iota(m) : n, m \in \mathbb{Z}\} = \{0\}$, which is a finite set. There are other simple examples, e.g., the function that is identically zero.

$\sim$ is an equivalence relation:

Reflexive: For $\mu \in S$, $\mu \sim \mu$, because for all $n \in \mathbb{Z}$, $\mu(n) - \mu(n) = 0$, i.e., $\{\mu(n) - \mu(n) : n \in \mathbb{Z}\} = \{0\}$, a finite set.

Symmetric: Let $\mu, \nu \in S$ and $\mu \sim \nu$, i.e., $F := \{\mu(n) - \nu(n) : n \in \mathbb{Z}\}$ is finite. Then $\{\nu(n) - \mu(n) : n \in \mathbb{Z}\} = -F := \{-f : f \in F\}$ is finite too. Thus $\nu \sim \mu$.

Transitive: Let $\lambda, \mu, \nu \in S$, $\lambda \sim \mu$ and $\mu \sim \nu$, i.e., $F_1 := \{\lambda(n) - \mu(n) : n \in \mathbb{Z}\}$ and $F_2 := \{\mu(n) - \nu(n) : n \in \mathbb{Z}\}$ are finite. For all $n \in \mathbb{Z}$, we have $\lambda(n) - \nu(n) = \lambda(n) - \mu(n) + \mu(n) - \nu(n) \in F_1 + F_2 := \{a + b : a \in F_1, b \in F_2\}$. Since F_1, F_2 are finite (and nonempty), say with $N_1, N_2 \in \mathbb{N}$ number of elements respectively, $F_1 + F_2$ can have at most $N_1 N_2$ number of elements. So $\{\lambda(n) - \mu(n) : n \in \mathbb{Z}\} \subset F_1 + F_2$, is finite too. Hence $\lambda \sim \nu$.

Next we show that $\mu, \nu \in S \Rightarrow \mu + \nu \in S$:

Since $\mu, \nu \in S$, we have that $F_1 := \{\mu(m+n) - \mu(m) - \mu(n) : m, n \in \mathbb{Z}\}$ and $F_2 := \{\nu(m+n) - \nu(m) - \nu(n) : m, n \in \mathbb{Z}\}$ are finite. But for $m, n \in \mathbb{Z}$,

$$\begin{aligned}&(\mu+\nu)(m+n) - (\mu+\nu)(m) - (\mu+\nu)(n) \\ &= \mu(m+n) + \nu(m+n) - \mu(m) - \nu(m) - \mu(n) - \nu(n) \\ &= \mu(m+n) - \mu(m) - \nu(m) + \nu(m+n) - \nu(m) - \nu(n) \in F_1 + F_2.\end{aligned}$$

As F_1, F_2 are finite (and nonempty), say with $N_1, N_2 \in \mathbb{N}$ number of elements respectively, $F_1 + F_2$ is finite with at most $N_1 N_2$ number of elements. Hence $\{(\mu+\nu)(m+n) - (\mu+\nu)(m) - (\mu+\nu)(n) : m, n \in \mathbb{Z}\} \subset F_1 + F_2$, is a finite set. Thus $\mu + \nu \in S$.

Addition in $\mathbb{R}$ is well-defined: Let $\mu, \nu \in S$. Then $\mu + \nu \in S$. Let $\widetilde{\mu}, \widetilde{\nu} \in S$ be such that $\widetilde{\mu} \sim \mu$ and $\widetilde{\nu} \sim \nu$, i.e., the set $F_1 := \{\widetilde{\mu}(n) - \mu(n) : n \in \mathbb{Z}\}$, and the set $F_2 := \{\widetilde{\nu}(n) - \nu(n) : n \in \mathbb{Z}\}$ are finite. For $n \in \mathbb{Z}$, we have

$$\begin{aligned}(\widetilde{\mu} + \widetilde{\nu})(n) - (\mu + \nu)(n) &= \widetilde{\mu}(n) + \widetilde{\nu}(n) - \mu(n) - \nu(n) \\ &= \widetilde{\mu}(n) - \mu(n) + \widetilde{\nu}(n) - \nu(n) \in F_1 + F_2.\end{aligned}$$

Since F_1, F_2 are finite (and nonempty), say with $N_1, N_2 \in \mathbb{N}$ number of elements respectively, $F_1 + F_2$ can have at most $N_1 N_2$ number of elements. Hence $\{(\widetilde{\mu} + \widetilde{\nu})(n) - (\mu + \nu)(n) : n \in \mathbb{Z}\}$, being a subset of the finite set $F_1 + F_2$, is a finite set. Thus $(\widetilde{\mu} + \widetilde{\nu}) \sim (\mu + \nu)$, that is, $[\widetilde{\mu} + \widetilde{\nu}] = [\mu + \nu]$.

Multiplicative identity: Above, we had seen that the identity function ι, $\mathbb{Z} \ni n \mapsto \iota(n) = n$, belongs to S. Set $\mathbf{1} = [\iota] \in \mathbb{R}$. Then for all $\boldsymbol{a} = [\nu] \in \mathbb{R}$,

$$\begin{aligned}\boldsymbol{a} \cdot \mathbf{1} &= [\nu] \cdot [\iota] = [\nu \circ \iota] = [\nu] = \boldsymbol{a}, \text{ and} \\ \mathbf{1} \cdot \boldsymbol{a} &= [\iota] \cdot [\nu] = [\iota \circ \nu] = [\nu] = \boldsymbol{a}.\end{aligned}$$

Chapter 6

Solution to Exercise 6.1

(1) Let $n \in \mathbb{Z}$. Taking $d = 2$ in the division algorithm, there exists a $q \in \mathbb{Z}$ and $r \in \{0, 1\}$ such that $n = 2q + r$. Thus $n^2 = (2q + r)^2 = 4q^2 + 4qr + r^2$.
1° If $r = 0$, then $n^2 = 4q^2 = 4k$, where $k = q^2 \in \mathbb{Z}$.
2° If $r = 1$, then $n^2 = 4q^2 + 4q + 1 = 4(q^2 + q) + 1 = 4k + 1$, where $k = q^2 + q \in \mathbb{Z}$.

(2) We have $t = 11 = 4(2) + 3$, and so t leaves the remainder 3 when divided by 4. A number t with $m \geqslant 3$ ones can be written as $t = 100n + 11$, where n is the integer with $m - 2$ ones. So $t = 100n + 11 = 4(25n + 2) + 3$. Hence t leaves remainder 3 when divided by 4. By part (1), a perfect square must leave a remainder 0 or 1 when divided by 4. Thus t cannot be a perfect square.

Solution to Exercise 6.2

Suppose on the contrary there exist $k \in \mathbb{N}$ and odd integers $m_1, \cdots, m_{2k}$ such that

$$1 = \frac{1}{m_1} + \cdots + \frac{1}{m_{2k}}. \qquad (\star)$$

Let $P := m_1 \cdots m_{2k} \in \mathbb{Z}$. As the product of odd integers is odd[19], P is odd. Multiplying $(\star)$ by P,

$$P = \frac{P}{m_1} + \cdots + \frac{P}{m_{2k}}. \qquad (\star\star)$$

For $1 \leqslant \ell \leqslant 2k$, $n_\ell := \dfrac{P}{m_\ell} = m_1 \cdots m_{\ell-1} m_{\ell+1} \cdots m_{2k}$ is an odd integer.
Then $(\star\star)$ becomes $P = (n_1 + n_2) + (n_3 + n_4) + \cdots + (n_{2k-1} + n_{2k})$. But each bracketed summand is even (being the sum of two odd integers), and as the sum of even numbers is even, it follows that P is even, a contradiction.

Solution to Exercise 6.3

Using the binomial theorem, we get for suitable $N, N', N'' \in \mathbb{N}$ that

$$\begin{aligned} 56789^{41} &= (56790 - 1)^{41} = 100N + 41(56790)^1(-1)^{40} + (-1)^{41} \\ &= 100N + (40 + 1)(56790) - 1 \\ &= 100N' + 56790 - 1 \\ &= 100N'' + 90 - 1 = 100N'' + 89, \end{aligned}$$

and so $(56789)^{41}$ ends in 89.

Solution to Exercise 6.4

Let $a', b' \in \mathbb{Z}$ be such that $a = da'$ and $b = db'$.
Then $ax + by = (da')x + (db')y = d(a'x + b'y)$. As $a'x + b'y \in \mathbb{Z}$, $d \,|\, ax + by$.

[19] Since for all $k, \ell \in \mathbb{Z}$, we have $(2k + 1)(2\ell + 1) = 2(2k\ell + k + \ell) + 1$.

Solution to Exercise 6.5
Let $d, d' \in \mathbb{Z}$ be such that $b = ad$ and $c = bd'$. Then $c = bd' = (ad)d' = a(dd')$. As $dd' \in \mathbb{Z}$, we conclude that $a \mid c$.

Solution to Exercise 6.6
Let $d \in \mathbb{Z}$ be such that $b = ad$. Then $d \neq 0$ (otherwise $b = ad = a0 = 0$, a contradiction). Hence either $d > 0$ or $d < 0$, that is, either $d \in \mathbb{N}$ or $-d \in \mathbb{N}$. In either case, $|d| \in \mathbb{N}$. Thus $|d| \geqslant 1$. So $|b| = |ad| = |a||d| \geqslant |a|1 = |a|$.

Solution to Exercise 6.7
Let $d, d' \in \mathbb{Z}$ be such that $a = bd$ and $b = ad'$.
1° If $a = 0$, then $b = ad' = 0d' = 0$. So $a = 0 = b$.
2° If $b = 0$, then $a = bd = 0d = 0$. So $a = 0 = b$.
3° If $a \neq 0$ and $b \neq 0$, then by Exercise 6.6, $|a| \leqslant |b|$ and $|b| \leqslant |a|$, so that $|a| = |b|$. Hence $a = b$ or $a = -b$.

Solution to Exercise 6.8
For all $n \in \mathbb{Z}$, $n^3 - n = n(n^2 - 1) = (n-1)n(n+1)$. By the division algorithm, there exist integers q, r such that $n = 3q + r$ and $0 \leqslant r < 3$ (i.e., $r \in \{0, 1, 2\}$).
1° If $r = 0$, then $r = 3q$, and so $3 \mid n$. Thus also $3 \mid (n-1)n(n+1) = n^3 - n$.
2° If $r = 1$, then $n - 1 = 3q$. So $3 \mid n - 1$, and also $3 \mid (n-1)n(n+1) = n^3 - n$.
3° If $r = 2$, then $n + 1 = 3(q+1)$. So $3 \mid n+1$, and also $3 \mid (n-1)n(n+1) = n^3 - n$.
(More generally, *Fermat's little theorem*[20] states that if p is any prime, then for all $n \in \mathbb{Z}$, $p \mid n^p - n$. The assumption that p is prime is not superfluous: e.g. with $p = 9$, and $n = 2$, $n^p - n = 2^9 - 2 = 2^6 2^3 - 2 = (63+1)8 - 2 = 9(*) + 6$.)

Solution to Exercise 6.9
Suppose $(n+1) \mid n^2 + 1$. We have $(n+1) \mid n^2 - 1$ since $n^2 - 1 = (n+1)(n-1)$. Hence $(n+1) \mid n^2 + 1 - (n^2 - 1) = 2$. But the divisors of 2 are $-2, -1, 1, 2$. So these are the possibilities for $n + 1$, giving $n = -3, -2, 0, 1$. Moreover, in each of these instances, $n + 1$ does divide $n^2 + 1$, as is easily checked: $-2 \mid 10$, $-1 \mid 5$, $1 \mid 1$, $2 \mid 2$.

Solution to Exercise 6.10
As $4 \mid a - 1$, there exists an integer d such that $a - 1 = 4d$, i.e., $a = 4d + 1$. Hence $a^2 + 3 = (4d + 1)^2 + 3 = 16d^2 + 8d + 1 + 3 = 4(4d^2 + 2d + 1)$, and so $4 \mid a^2 + 3$.

[20] This can be shown as follows. If $p \mid n$, then $p \mid n^p$, and so $p \mid n^p - n$ too. Now suppose p does not divide n. So $[n] \neq [0]$ in $\mathbb{Z}/p\mathbb{Z}$. Now $G := (\mathbb{Z}/p\mathbb{Z}) \setminus \{[0]\}$ is an Abelian group with respect to multiplication modulo p, and its order is $p - 1$. As the order of $[n] \in G$ divides the order of the group, we have $[n]^{p-1} = [1]$, i.e., $[n^{p-1} - 1] = [0]$, that is, $p \mid n^{p-1} - 1$. But then also $p \mid (n^{p-1} - 1)n = n^p - n$, completing the proof.

Let us suppose that $8 \mid a^2 + 3$. Then there exists an integer d' such that $8d' = a^2 + 3 = 4(4d^2 + 2d + 1)$, i.e., $2d' = 4d^2 + 2d + 1$. So $2(d' - 2d^2 - d) = 1$. Thus $2 \mid 1$. By Exercise 6.6, $|2| \leqslant |1|$, i.e., $2 \leqslant 1$, a contradiction. So $8 \nmid a^2 + 3$.

Solution to Exercise 6.11

Let $n = 2k + 1$ for some $k \in \mathbb{Z}$. Then we have that $n^2 = 4k^2 + 4k + 1$, and $n^4 = 16k^4 + 16k^2 + 1 + 32k^3 + 8k^2 + 8k$. Thus

$$\begin{aligned} n^4 + 4n^2 + 11 &= 16k^4 + 16k^2 + 1 + 32k^3 + 8k^2 + 8k + 16k^2 + 16k + 4 + 11 \\ &= 16(k^4 + k^2 + 2k^3 + k^2 + k + 1) + 8k(k + 1). \end{aligned}$$

We claim that $k(k + 1)$ is divisible by 2: indeed, $k = 2\ell + r$ for some $\ell, r \in \mathbb{Z}$ with $r \in \{0, 1\}$. If $r = 1$, then $2 \mid (k + 1) = 2\ell + 1 + 1 = 2(\ell + 1)$. If $r = 0$, then $2 \mid k = 2\ell$. Thus $16 \mid 8k(k + 1)$, and so from the above, $16 \mid n^4 + 4n^2 + 11$.

Solution to Exercise 6.12

We use induction on n. If $n = 1$, then $a - b = 1(a - b)$ and so $a - b \mid a - b$. Suppose the claim holds for some $n \in \mathbb{N}$. Let $a, b \in \mathbb{Z}$. Then $a^n - b^n = q(a - b)$ for some $q \in \mathbb{Z}$. Then

$$\begin{aligned} a^{n+1} - b^{n+1} &= a^{n+1} - a^n b + a^n b - b^{n+1} = a^n(a - b) + b(a^n - b^n) \\ &= a^n(a - b) + bq(a - b) = (a^k + bq)(a - b). \end{aligned}$$

As $a^k + bq \in \mathbb{Z}$, we get $a - b \mid a^{n+1} - b^{n+1}$. So for all $a, b \in \mathbb{Z}$, $a - b \mid a^{n+1} - b^{n+1}$, completing the induction step. Thus the result holds for all $n \in \mathbb{N}$.

Solution to Exercise 6.13

If $k = 1$, then $a^{2k-1} + b^{2k-1} = a + b$, and so the claim is true in this case. Suppose for some $k \in \mathbb{N}$, $(a + b) \mid a^{2k-1} + b^{2k-1}$. Then $a^{2k-1} + b^{2k-1} = (a + b)d$ for some $d \in \mathbb{Z}$. Thus

$$\begin{aligned} a^{2k+1} + b^{2k+1} &= a^{2k+1} + b^{2k-1}b^2 \\ &= a^{2k+1} + b^2((a + b)d - a^{2k-1}) \\ &= a^{2k-1}a^2 + (a + b)b^2 d - b^2 a^{2k-1} \\ &= a^{2k-1}(a^2 - b^2) + (a + b)b^2 d \\ &= a^{2k-1}(a - b)(a + b) + (a + b)b^2 d \\ &= (a + b)(a^{2k-1}(a - b) + b^2 d). \end{aligned}$$

So $(a + b) \mid a^{2k+1} + b^{2k+1}$. By induction, $(a + b) \mid a^{2k-1} + b^{2k-1}$ for all $k \in \mathbb{N}$.

Solution to Exercise 6.14

We regroup the summands in given sum as follows:

$(1^{2025} + 2024^{2025}) + \cdots + (k^{2025} + (2025 - k)^{2025}) + \cdots + (1012^{2025} + 1013^{2021})$.

By Exercise 6.13, each summand $(k^{2025} + (2025 - k)^{2025})$, $1 \leqslant k \leqslant 1012$, is divisible by $k + (2025 - k) = 2025$. So 2025 divides the given sum.

Solution to Exercise 6.15

Let $p(x) = c_0 + c_1 x + \cdots + c_d x^d$, where $c_0, \cdots, c_d \in \mathbb{Z}$. For $x, y \in \mathbb{Z}$, we have $p(x) - p(y) = c_1(x-y) + \cdots + c_d(x^d - y^d)$. As $x - y$ divides $x^k - y^k$ for all $1 \leqslant k \leqslant d$, we get that $x - y$ divides $p(x) - p(y)$. So with $x = 2025$ and $y = 5$, we obtain $2020 \,|\, p(2025) - p(5) = p(2025) - 2$. Hence $p(2025) = 2020k + 2$ for some $k \in \mathbb{Z}$. In particular, $p(2025)$ is even, and so if it is a perfect square, it must be divisible by 4 (see Exercise 6.1, p.64). But $4 \mid 2020k$, and so $4 \,|\, p(2025) - 2020k = 2$, a contradiction.

Solution to Exercise 6.16

Let S be nonempty. By the well-ordering principle, it has a smallest element, say $m \in \mathbb{N}$. So $m = x_0^2 + y_0^2 + z_0^2$ for some integers x_0, y_0, z_0, not all zeroes, such that $x_0^3 + 3y_0^3 + 9z_0^3 = 0$. As $x_0^3 = 3(-y_0^3 - 3z_0^3)$, we get $3 \,|\, x_0^3$. Thus $3 \,|\, x_0$. (Otherwise, by the division algorithm, $x_0 = 3q + r$ for some integers q, r with $0 < r < 3$, so that $x_0^3 = (3q+r)^3 = 27q^3 + 27q^2 r + 9qr^2 + r^3$, and $3 \nmid r^3$. But $r = 1$ or 2, and so $r^3 = 1$ or 8, and we know $3 \nmid 1$ and $3 \nmid 8$.) So $x_0 = 3x_1$ for some $x_1 \in \mathbb{Z}$. Then $x_0^3 + 3y_0^3 + 9z_0^3 = 0$ gives $9x_1^3 + y_0^3 + 3z_0^2 = 0$. By the same argument as above, $3 \,|\, y_0^3$ and so $3 \,|\, y_0$. Hence $y_0 = 3y_1$ for some $y_1 \in \mathbb{Z}$. So $9x_1^3 + y_0^3 + 3z_0^2 = 0$ gives $3x_1^3 + 9y_1^3 + z_0^3 = 0$. Thus $3 \,|\, z_0^3$ and so $3 \,|\, z_0$. So $z_0 = 3z_1$ for some $z_1 \in \mathbb{Z}$. Hence $x_1^3 + 3y_1^3 + 9z_1^3 = 0$. We note that $(x_1, y_1, z_1) \in \mathbb{Z}^3$, and $(x_1, y_1, z_1) \neq (0,0,0)$ (otherwise $(x_0, y_0, z_0) = (3x_1, 3y_1, 3z_1) = (0,0,0)$, a contradiction). So $x_1^2 + y_1^2 + z_1^2 \in S$. Thus $x_1^2 + y_1^2 + z_1^2 \geqslant m$. Since $1 < 9$, we get $m < 9m \leqslant 9(x_1^2 + y_1^2 + z_1^2) = x_0^2 + y_0^2 + z_0^2 = m$, a contradiction. This contradiction shows that our assumption that S is nonempty was false, and so $x = y = z = 0$ is the only solution in integers to $x^3 + 3y^3 + 9z^3 = 0$.

Solution to Exercise 6.17

(1)⇒(2): We have $d1 = d = ax_0 + by_0 = (q_a d)x_0 + (q_b d)y_0 = d(q_a x_0 + q_b y_0)$, and since $d > 0$, in particular nonzero, we obtain $1 = q_a x_0 + q_b y_0$. Similarly, $d1 = d = ax + by = (q_a x + q_b y)d$ gives $1 = q_a x + q_b y$. Hence

$$\begin{aligned} x &= 1x = (q_a x_0 + q_b y_0)x = q_a x x_0 + q_b x y_0 = (1 - q_b y)x_0 + q_b x y_0 \\ &= x_0 - q_b x_0 y + q_b x y_0 = x_0 + q_b(x y_0 - x_0 y) = x_0 + q_b n, \end{aligned}$$

where $n := xy_0 - x_0 y \in \mathbb{Z}$. Moreover,

$$\begin{aligned} y_0 - q_a n &= y_0 - q_a(x y_0 - x_0 y) = y_0 - q_a x y_0 + q_a x_0 y \\ &= y_0(1 - q_a x) + q_a x_0 y = y_0(q_b y) + q_a x_0 y \\ &= y(y_0 q_b + q_a x_0) = y1 = y. \end{aligned}$$

(2)⇒(1): We have

$$\begin{aligned} ax + by &= a(x_0 + q_b n) + b(y_0 - q_a n) = a x_0 + a q_b n + b y_0 - b q_a n \\ &= a x_0 + b y_0 + d q_a q_b n - d q_b q_a n = d + 0 = d. \end{aligned}$$

Solution to Exercise 6.18

Since $F_1 = F_2 = 1$, $\gcd(F_1, F_2) = \gcd(1,1) = 1$. Suppose $\gcd(F_n, F_{n+1}) = 1$ for some $n \in \mathbb{N}$. We show that $d := \gcd(F_{n+1}, F_{n+2}) = 1$. As d is the gcd of F_{n+1}, F_{n+2}, it divides each of them that is, $d \mid F_{n+2}$ and $d \mid F_{n+1}$. But thanks to the recursion $F_{n+2} = F_{n+1} + F_n$, we have $F_n = F_{n+2} - F_{n+1}$, and so we have $d \mid F_n$ too. Now $d \mid F_n$ and $d \mid F_{n+1}$, so that $d \mid \gcd(F_n, F_{n+1})$, that is, $d \mid 1$. As $d > 0$, we have $d = 1$, and by induction the proof is complete.

Solution to Exercise 6.19

We first show that $\gcd(a,b) = \gcd(a,-b)$ and use this to show the rest. (Clearly if $(a,b) \neq (0,0)$, then also $(a,-b), (-a,-b), (-a,b)$ are not equal to $(0,0)$. So $\gcd(a,-b), \gcd(-a,-b), \gcd(-a,b)$ all exist.) Let $d := \gcd(a,b)$ and $d' := \gcd(a,-b)$. As $d = \gcd(a,b)$, we have $d \mid a$ and $d \mid b$. Hence also $d \mid a$ and $d \mid -b$. Since $d' = \gcd(a,-b)$, we conclude that $d \mid d'$. As $d' = \gcd(a,-b)$, we have $d' \mid a$ and $d' \mid -b$. Hence also $d' \mid a$ and $d' \mid b$. Since $d = \gcd(a,b)$, we conclude that $d' \mid d$. From the facts that $d \mid d'$ and $d' \mid d$, in light of $d, d' > 0$, it follows from Exercise 6.7, p.64, that $d = d'$. Using the observations that $\gcd(\cdot,\cdot)$ can swap arguments, and can put a $-$ sign in front of the second argument, it follows that

$$\begin{aligned}\gcd(a,b) &= \gcd(a,-b) = \gcd(-b,a) = \gcd(-b,-a) = \gcd(-a,-b)\\ &= \gcd(-a,-(-b)) = \gcd(-a,b).\end{aligned}$$

Solution to Exercise 6.20

There exist $x, y \in \mathbb{Z}$ such $nx + Ny = \gcd(n,N)$. Hence

$$\begin{aligned}\frac{\gcd(n,N)}{N}\binom{N}{n} &= \frac{nx+Ny}{N}\binom{N}{n} = \frac{nx}{N}\frac{N!}{n!(N-n)!} + y\binom{N}{n}\\ &= x\frac{(N-1)!}{(n-1)!(N-n)!} + y\binom{N}{n}\\ &= x\binom{N-1}{n-1} + y\binom{N}{n}.\end{aligned}$$

As binomial coefficients are integers, and $x, y \in \mathbb{Z}$, we have $\frac{\gcd(n,N)}{N}\binom{N}{n} \in \mathbb{Z}$.

Solution to Exercise 6.21

To show $\gcd(21n+4, 14n+3) = 1$, we use the Euclidean algorithm:

$$\begin{aligned}21n+4 &= 1(14n+3) + 7n+1, \quad && (0 < 7n+1 < 14n+3),\\ 14n+3 &= 2(7n+1) + 1, && (0 < 1 < 7n+1),\\ 7n+1 &= (7n+1)1.\end{aligned}$$

So $\gcd(21n+4, 14n+3) = 1$. It follows that in $\frac{21n+4}{14n+3}$, any common factor for the numerator $21n+4$ and the denominator $14n+3$ will have to divide 1, and hence is forced to be -1 or 1.

Solution to Exercise 6.22

We use induction on d. If $d=1$, then $m \mid c_0 n$. By Proposition 6.2, p.69, $m \mid c_0$. Let the claim be true for some $d \in \mathbb{N}$. Suppose $m \mid c_0 n^{d+1} = (c_0 n^d)n$. As $\gcd(m,n)=1$, by Proposition 6.2, $m \mid c_0 n^d$. By the induction hypothesis, $m \mid c_0$. So the claim follows by induction.

Solution to Exercise 6.23

As $a \mid n$, there exists an integer d such that $n = ad$. Since $b \mid n$, we have $b \mid ad$. But $\gcd(a,b)=1$, and so by Proposition 6.2, p.69, $b \mid d$. Hence there is an integer k such that $d = bk$. Thus $n = ad = a(bk) = (ab)k$. So $ab \mid n$.

Solution to Exercise 6.24

As a, b are relatively prime, by Proposition 6.1, p.68, there exist $x_0, y_0 \in \mathbb{Z}$ such that $1 = ax_0 + by_0$. Multiplying by n, we get

$$n = n1 = n(ax_0 + by_0) = a(nx_0) + b(ny_0) = ax + by,$$

where $x := nx_0 \in \mathbb{Z}$ and $y := ny_0 \in \mathbb{Z}$.

Solution to Exercise 6.25

As $\gcd(a,n) = 1$, by Proposition 6.1, p.68, there exist $x, y \in \mathbb{Z}$ such that $ax + ny = 1$. Similarly, there exist $u, v \in \mathbb{Z}$ such that $bu + nv = 1$. So we get

$$\begin{aligned} 1 = 1(1) &= (ax + ny)(bu + nv) = abxu + axnv + nybu + nynv \\ &= (ab)(xu) + n(axv + ybu + ynv). \end{aligned}$$

By Proposition 6.1, we conclude that $\gcd(ab,n) = 1$.

Solution to Exercise 6.26

By Theorem 6.2, p.66, we have $d = ax_0 + by_0$ for some $x_0, y_0 \in \mathbb{Z}$. Since $a = dq_a$ and $b = dq_b$, we get $d1 = d = ax_0 + by_0 = dq_a x_0 + dq_b y_0 = d(q_a x_0 + q_b y_0)$. As $d > 0$, we obtain $1 = q_a x_0 + q_b y_0$. Hence by Proposition 6.1, p.68, q_a, q_b are relatively prime, i.e., $\gcd(q_a, q_b) = 1$.

Solution to Exercise 6.27

As a, b are coprime, by Proposition 6.1, p.68, $ax + by = 1$ for some $x, y \in \mathbb{Z}$. Hence $1 = 1^2 = (ax + by)^2 = a^2x^2 + b^2y^2 + 2abxy = a^2x^2 + b(by^2 + 2axy)$. So $\gcd(a^2, b) = 1$. Thus if $\gcd(a,b) = 1$, then $\gcd(a^2, b) = 1$ (i.e., if we have a coprime pair, then squaring one of the members of the coprime pair maintains coprimeness). Applying this, we get from $\gcd(a,b) = 1$, that $\gcd(a^2, b) = 1$, i.e., $\gcd(b, a^2) = 1$, implying $\gcd(b^2, a^2) = 1$, i.e., $\gcd(a^2, b^2) = 1$. (Alternatively, a slick solution can be obtained by cubing $ax + by = 1$:

$$\begin{aligned} 1 = 1^3 &= (ax + by)^3 = a^3x^3 + 3a^2x^2by + 3axb^2y^2 + b^3y^3 \\ &= a^2(ax^3 + 3x^2by) + b^2(3axy^2 + by^3). \end{aligned}$$

By Proposition 6.1, p.68, a^2, b^2 are coprime.)

Solution to Exercise 6.28

Let $d := \gcd(n!+1, (n+1)!+1)$. Then $d \mid n!+1$. So $n!+1 = qd$ for some $q \in \mathbb{Z}$. Then

$$\begin{aligned}(n+1)!+1 &= (n!)(n+1)+1 = (qd-1)(n+1)+1 = nqd+qd-n \\ &= d(nq+q)-n.\end{aligned}$$

As $d \mid (n+1)!+1$, the above implies $d \mid n$ too. Since $n \mid n!$, we obtain $d \mid n!$. But $d \mid n!+1$. So $d \mid (n!+1)-(n!) = 1$. As $d > 0$, we get $d = 1$.

Solution to Exercise 6.29

(1) The equation $a + b\sqrt{2} = a' + b'\sqrt{2}$ gives $a - a' = (b'-b)\sqrt{2}$.
If $b' - b \neq 0$, then $\sqrt{2} = \frac{a-a'}{b'-b} \in \mathbb{Q}$, a contradiction. So $b' - b = 0$, i.e., $b = b'$. Then $a - a' = (b'-b)\sqrt{2} = 0\sqrt{2} = 0$, and so $a = a'$.

(2) As $a_1 + b_1\sqrt{2} = (1+\sqrt{2})^1$, we get $a_1 = 1 \in \mathbb{N}$, $b_1 = 1 \in \mathbb{N}$, and $\gcd(a_1, b_1) = 1$.
For an $n \in \mathbb{N}$, let $a_n, b_n \in \mathbb{N}$ and $\gcd(a_n, b_n) = 1$. Then

$$\begin{aligned}a_{n+1} + b_{n+1}\sqrt{2} &= (1+\sqrt{2})^{n+1} = (1+\sqrt{2})(a_n + b_n\sqrt{2}) \\ &= a_n + 2b_n + (a_n + b_n)\sqrt{2}.\end{aligned}$$

By part (1), we get $a_{n+1} = a_n + 2b_n$ and $b_{n+1} = a_n + b_n$. As $a_n, b_n \in \mathbb{N}$, we now get $a_{n+1}, b_{n+1} \in \mathbb{N}$. Moreover, if $d = \gcd(a_{n+1}, b_{n+1})$, then $d \mid (a_{n+1} - b_{n+1}) = a_n + 2b_n - (a_n + b_n) = b_n$. Thus we conclude that $d \mid b_{n+1} - b_n = (a_n + b_n) - b_n = a_n$. As $d \mid a_n$ and $d \mid b_n$, it follows that d is a common divisor of a_n and b_n, and so $d \mid \gcd(a_n, b_n) = 1$, i.e., $d = 1$. So $\gcd(a_{n+1}, b_{n+1}) = 1$. The claim holds for all $n \in \mathbb{N}$ by induction.

Solution to Exercise 6.30

(1) As $(n+1)2 - (2n+1) = 1$, by Proposition 6.1, p.68, $\gcd(2n+1, n+1) = 1$.

(2) The binomial coefficients are natural numbers (Exercise 4.16, p.40). We have $\frac{2n+1}{n+1}\binom{2n}{n} = \frac{2n+1}{n+1}\frac{(2n)!}{n!n!} = \frac{(2n+1)!}{(n+1)!n!} = \binom{2n+1}{n} =: b \in \mathbb{N}$. Set $\binom{2n}{n} =: a \in \mathbb{N}$. Then $\frac{2n+1}{n+1}a = b$, that is, $(2n+1)a = (n+1)b$. So $(2n+1) \mid (n+1)b$.
As $\gcd(2n+1, n+1) = 1$, by Proposition 6.2, p.69, $(2n+1) \mid b$. Hence $b = (2n+1)b'$ for some $b' \in \mathbb{N}$. So $(2n+1)a = (n+1)b = (n+1)(2n+1)b'$, giving $a = (n+1)b' \in \mathbb{N}$, i.e., $C_n = \frac{1}{n+1}\binom{2n}{n} = \frac{1}{n+1}a = b' \in \mathbb{N}$.

Solution to Exercise 6.31

The 45° angle is constructible using a straightedge and compass, by starting with 180° and bisecting it twice. Also $45 \cdot 1 - 11 \cdot 4 = 1$. As angles can be reproduced using a straightedge and compass, we can construct the 44° angle by doubling 11°, and then doubling it again. Then we can construct the 45° angle with a ray in common with the 44° angle, giving the $(45-44=)$ 1° angle. Now we can reproduce 1° eleven times within the 11° angle, hence dividing it into 11 equal parts.

Solution to Exercise 6.32

Let $d = \gcd(2^{2^n}+1,\ 2^{2^N}+1)$. As $d \mid 2^{2^n}+1$, there exists an integer m such that $2^{2^n}+1 = md$, and so $2^{2^n} = md-1$. Thus

$$2^{2^N}+1 = 2^{2^n 2^{N-n}}+1 = (2^{2^n})^{2^{N-n}}+1 = (md-1)^{2^{N-n}}+1$$

By Exercise 6.13, p.65, $(md-1)^{\text{odd}}+1$ is divisible by $md-1+1 = md$, and in particular by d. So $(md-1)^{2^{N-n}-1}+1 = dK$ for some $K \in \mathbb{Z}$. Thus

$$\begin{aligned} 2^{2^N}+1 &= (md-1)^{2^{N-n}}+1 = (md-1)^{2^{N-n}-1}(md-1)+1 \\ &= (dK-1)(md-1)+1 = d(Kmd-K-m)+2. \end{aligned}$$

As $d \mid 2^{2^N}+1$, it follows from the above that $d \mid 2$. Since $d>0$, $d=1$ or $d=2$. But $2^{2^n}+1$ is odd, and hence not divisible by 2. As d must divide $2^{2^n}+1$, d cannot be 2. So $d=1$. This completes the proof.

Solution to Exercise 6.33

(1)⇒(2): Let $g_1, g_2 \in C[a,b]$ satisfy $f_1g_1+f_2g_2 = \mathbf{1}$ and $Z_{f_1} \cap Z_{f_2} \neq \varnothing$. Then there exists an $x_0 \in [a,b]$ such that $f_1(x_0)=0$ and $f_2(x_0)=0$. Evaluating $f_1g_1+f_2g_2=\mathbf{1}$ at x_0 yields $1 = \mathbf{1}(x_0) = f_1(x_0)g_1(x_0) + f_2(x_0)g_2(x_0) = 0$, a contradiction. Hence $Z_{f_1} \cap Z_{f_2} = \varnothing$, that is, (2) holds.

(2)⇒(1): Suppose $Z_{f_1} \cap Z_{f_2} = \varnothing$. We claim that $f := f_1^2+f_2^2 \in C[a,b]$ is everywhere positive on $[a,b]$. Indeed, it is nonnegative (the sum of squares $(f_1(x))^2+(f_2(x))^2 \geqslant 0$ for all $x \in [a,b]$), and if $x_0 \in [a,b]$ is a point where it vanishes, then $(f_1(x_0))^2+(f_2(x_0))^2 = 0$ implies $f_1(x_0) = f_2(x_0) = 0$, so that $x_0 \in Z_{f_1} \cap Z_{f_2} = \varnothing$, a contradiction. Thus $\frac{1}{f} \in C[a,b]$, and hence we have that $g_i := \frac{1}{f} f_i \in C[a,b]$, $i=1,2$. We have that (1) holds, because

$$f_1g_1+f_2g_2 = f_1\frac{f_1}{f_1^2+f_2^2} + f_2\frac{f_2}{f_1^2+f_2^2} = \mathbf{1}.$$

Solution to Exercise 6.34

If $p, 8p-1$ are prime, then $p \neq 2$ (otherwise $8p-1=15$, which is not prime). If $p=3$, then $8p+1 = 25 = 5(5)$ is indeed composite. If $p>3$, then $3 \nmid p$. So $p = 3k+r$, for some $k, r \in \mathbb{Z}$ with $0<r<3$.

1° If $r=1$, then $8p+1 = 8(3k+1)+1 = 3(8k+3)$. As $8p+1 > 8+1 > 3$ and since $3 \mid 8p+1$, it follows that $8p+1$ is composite.

2° If $r=2$, then $8p-1 = 8(3k+2)-1 = 3(8k+5)$. As $8p-1 > 8-1 > 3$ and since $3 \mid 8p-1$, $8p-1$ cannot be prime. Thus this case is impossible.

Solution to Exercise 6.35

(1) We have the following cases:

1° If $p \mid a$, then there is nothing to prove.

2° If $p \nmid a$, then by Proposition 6.3, p.71, $\gcd(p,a)=1$.
As $p \mid ab$, Proposition 6.2, p.69, implies $p \mid b$.

(2) If $n=2$, then the claim follows from part (1).

Suppose for some $n\in\mathbb{N}$ with $n\geqslant 2$, whenever we have that the prime p divides a product of n integers, p must divide at least one of these integers. Now suppose that $a_1,\cdots,a_n,a_{n+1}\in\mathbb{Z}$, and $p\,|\,a_1\cdots a_na_{n+1}$. Then $p\,|\,Aa_{n+1}$, where A is the integer $a_1\cdots a_n$. By part (1), only the following two cases are possible:

1° If $p\,|\,a_{n+1}$, then there is nothing to prove.

2° If $p\,|\,A$, then by the induction hypothesis, p divides at least one of $a_1,\cdots,a_n$.

So p divides some integer in the list $a_1,\cdots,a_n,a_{n+1}$. By induction, the claim holds for $n\geqslant 2$.

Solution to Exercise 6.36

We use the capital letter P for the polynomial to avoid confusion with the letter p used for a prime below. Let P be a nonconstant polynomial with integer coefficients such that for all $n\in\mathbb{N}$, $P(n)$ is a prime number. Let p be the prime number $P(1)$. Let $d\in\mathbb{N}$, $c_0,c_1,\cdots,c_d\in\mathbb{Z}$, $c_d\neq 0$, and $P(x)=c_0+c_1x+\cdots+c_dx^d$ for all $x\in\mathbb{R}$.

We claim that $c_d>0$. For all $n\in\mathbb{N}$, we have

$$0<P(n)=c_0+c_1n+\cdots+c_dn^d=n^d\left(\tfrac{c_0}{n^d}+\cdots+\tfrac{c_{d-1}}{n}+c_d\right).$$

As $\lim\limits_{n\to\infty}\left(\tfrac{c_0}{n^d}+\cdots+\tfrac{c_{d-1}}{n}\right)=0$, there exists an $N\in\mathbb{N}$ such that for all $n>N$,

$$-\tfrac{1}{2}<\tfrac{c_0}{n^d}+\cdots+\tfrac{c_{d-1}}{n}<\tfrac{1}{2}.$$

Then for all $n>N$, we get

$$0<\frac{P(n)}{n^d}=\tfrac{c_0}{n^d}+\cdots+\tfrac{c_{d-1}}{n}+c_d<\tfrac{1}{2}+c_d,$$

and so $c_d>-\frac{1}{2}$. Thus $c_d\geqslant 0$ (otherwise $c_d\in\mathbb{Z}$ satisfies $-1<-\frac{1}{2}<c_d<0$, a contradiction). Since $c_d\neq 0$, $c_d>0$. So $c_d\geqslant 1$.

Let $N'\in\mathbb{N}$ be such that $N'>\dfrac{p}{c_d-\frac{1}{2}}$. Then for $n>\max\{N,N'\}$, we have

$$P(n)=n^d\left(\tfrac{c_0}{n^d}+\cdots+\tfrac{c_{d-1}}{n}+c_d\right)>n^d\left(-\tfrac{1}{2}+c_d\right)=n^{d-1}n\left(c_d-\tfrac{1}{2}\right)>1p,$$

where the last inequality holds since $n>N'$. So:

$$\text{For all } n>\widetilde{N}:=\max\{N,N'\},\text{ we have } P(n)>p. \tag{$\star$}$$

But for $n>\widetilde{N}$, we have $1+pn>pn>n>\widetilde{N}$, and so from ($\star$), we get

$$P(1+pn)>p. \tag{$\star\star$}$$

Now for $n > \widetilde{N}$, we obtain

$$\begin{aligned} P(1+pn) &= c_0 + c_1(1+pn) + c_2(1+pn)^2 + \cdots + c_d(1+pn)^d \\ &= c_0 + c_1(1+pQ_1) + c_2(1+pQ_2) + \cdots + c_d(1+pQ_d), \end{aligned}$$

for suitable integers $Q_1, \cdots, Q_d$. (Reason: By the binomial theorem, we have that $(1+pn)^m = 1 + \binom{m}{1}pn + \binom{m}{2}p^2n^2 + \cdots + \binom{m}{m}p^m n^m = 1 + pQ_m$, where $Q_m := \binom{m}{1}n + \binom{m}{2}pn^2 + \cdots + \binom{m}{m}p^{m-1}n^m \in \mathbb{Z}$). Thus for $n > \widetilde{N}$, with $Q := c_1Q_1 + \cdots + c_dQ_d \in \mathbb{Z}$, we have

$$\begin{aligned} P(1+pn) &= c_0 + c_1 + \cdots + c_d + p(c_1Q_1 + \cdots + c_dQ_d) \\ &= P(1) + pQ = p + pQ = p(1+Q). \end{aligned}$$

So $p \,|\, P(1+pn)$. As $p, P(1+pn)$ are primes, $p = P(1+pn)$, contradicting $(\star\star)$.

Solution to Exercise 6.37

For $n = 1$, $p_1 = 2 < 4 = 2^{2^1}$. If for some $n \in \mathbb{N}$, we have $p_k < 2^{2^k}$ for all $k \leqslant n$, then $p_{n+1} \leqslant p_1 \cdots p_n + 1$ implies

$$p_{n+1} < p_1 \cdots p_n < 2^{2^1} \cdots 2^{2^n} = 2^{2^1 + \cdots + 2^n} = 2^{2(2^n - 1)} = 2^{2^{n+1} - 2} < 2^{2^{n+1}},$$

completing the induction step.

The fact that $p_n < 2^{2^n}$ implies that the number of primes up to 2^{2^n} is at least n, and so $\pi(2^{2^n}) \geqslant n$ for all $n \in \mathbb{N}$.

For $x \geqslant 2$, let $n \in \mathbb{N}$ be such that $2^n \leqslant \log_2 x < 2^{n+1}$. Then $x \geqslant 2^{2^n}$. As $\pi(\cdot)$ is increasing, $\pi(x) \geqslant \pi(2^{2^n}) \geqslant n > (\log_2 \log_2 x) - 1$. Thus we obtain $\pi(x) \geqslant \log_2 \log_2 x$ for all $x \geqslant 2$. As $1 < 2 < e$, we have $0 < \log_e 2 < 1$. Thus for all $y > 0$, $\log_2 y = \frac{\log_e y}{\log_e 2} > \log_e y$. Hence $\pi(x) \geqslant \log_2 \log_2 x \geqslant \log_e \log_e x$ for all $x \geqslant 2$. Also, $\pi(x) \leqslant x$ for all $x > 0$.

Solution to Exercise 6.38

Let $p_1 < \cdots < p_k$ be the only primes. The set $A = \{f_n := 2^{2^n} + 1 : n \in \mathbb{N}\}$ is not finite. (Otherwise, suppose the nonempty set A has cardinal number m. Then by Exercise 2.27, p.24, there exists a surjection $f : \{1, \cdots, m\} \to A$. If $f_{n_1}, \cdots, f_{n_m}$ are $f(1), \cdots, f(m)$, then with $k := n_1 + \cdots + n_m$, we have $f_k = 2^{2^k} + 1 > 2^{2^i} + 1$ for each $i \in \{n_1, \cdots, n_m\}$, showing that $f_k \in A$ is distinct from $f(1), \cdots, f(m)$, a contradiction.) Each $a \in A$ satisfies $a > 1$, and has a factorisation into primes by the fundamental theorem of arithmetic. In the factorisation of each such element of A, choose the index of the smallest prime appearing. This gives a well-defined map $g : A \to \{1, \cdots, k\}$. By Exercise 2.27, p.24, g cannot be injective (otherwise A would be finite). So there exist $n, m \in \mathbb{N}$, $n \neq m$, such that the prime $p_{g(n)} = p_{g(m)} > 1$ divides both f_n and f_m, contradicting the result from Exercise 6.32, p.70, where we established that $\gcd(f_n, f_m) = 1$ whenever $n, m \in \mathbb{N}$ are distinct.

Solution to Exercise 6.39

Let $n \geqslant 4$. Then n, being prime, is not divisible by 3.
By the division algorithm, $n = 3k + r$, where $r \in \{1, 2\}$ and $k \in \mathbb{N}$.
1° $r = 1$. Then $n + 2 = 3k + 1 + 2 = 3(k + 1)$. So $3 \mid n + 2$, and $n + 2$ is not prime.
2° $r = 2$. Then $n + 4 = 3k + 2 + 4 = 3(k + 2)$. So $3 \mid n + 4$, and $n + 4$ is not prime.
Hence for all $n \geqslant 4$, $(n, n + 2, n + 4)$ is not a prime triple.

Solution to Exercise 6.40

Let the hypotenuse be k, and the other two sides be the primes $p, p + 2$. By Pythagoras's theorem, $k^2 = p^2 + (p + 2)^2 = 2(p^2 + 2p + 2)$, showing that k^2, and hence also k, must be even. Let $k = 2k'$. Then $p^2 = 2(k'^2 - p - 1)$, and so p must be even. As p is prime, $p = 2$. But then $p + 2 = 2 + 2 = 4$, which is not prime, a contradiction. So there cannot exist a right-angled triangle with integer sides having the nonhypotenuse sides equal to twin primes.

Solution to Exercise 6.41

For $2 \leqslant k \leqslant n + 1$, $k \mid (n + 1)! + k = 1 \cdot 2 \cdots (k - 1) \cdot k \cdot (k + 1) \cdots (n + 1) + k$. Also, clearly $k \neq 1$ (as $k \geqslant 2$) and $k \neq (n + 1)! + k$ (since $k < (n + 1)! + k$). So $(n + 1)! + k$ is not a prime for all $2 \leqslant k \leqslant n + 1$. So the list of n consecutive numbers given by $(n + 1)! + 2, \cdots, (n + 1)! + n + 1$ has no primes.

Solution to Exercise 6.42

We have $2^1 + 1 = 3$ is prime, and $n = 1 = 2^0$. Suppose $n > 1$ is not a power of 2. By the fundamental theorem of arithmetic, there exists an odd prime, say p, that divides n: $n = pk$, where $k \in \mathbb{N}$. Thus $2^n + 1 = 2^{pk} + 1 = (2^k)^p + 1^p$. By Exercise 6.13, p.65, $2^k + 1 \mid (2^k)^p + 1^p = 2^n + 1$. But

$$1 < 2^k + 1 < (2^k)^p + 1 = 2^n + 1.$$

Hence $2^n + 1$ is not a prime number. Consequently, if $2^n + 1$ is a prime number, then n must be a power of 2.

Solution to Exercise 6.43

We want to show: $(\forall$ prime p such that $p \leqslant \sqrt{n}$, $\neg(p \mid n)) \Rightarrow (n$ is prime$)$.
This is equivalent to:
$(n$ composite$) \Rightarrow (\exists$ a prime p such that $p \leqslant \sqrt{n}$ and $p \mid n)$.
Let n be composite, and $n = ab$, where $1 < a, b < n$. If $a > \sqrt{n}$ and $b > \sqrt{n}$, then $n = ab > \sqrt{n}\sqrt{n} = n$, a contradiction. So at least one of the factors, say a, is $\leqslant \sqrt{n}$. Then $1 < a \leqslant \sqrt{n}$, and any prime factor p_* of a will be $\leqslant \sqrt{n}$. As $p_* \mid a$, and $a \mid n$, we have $p_* \mid n$ too. So we have shown that if n is composite, then there exists a prime, namely p_*, such that $p_* \leqslant \sqrt{n}$ and $p_* \mid n$.

Solution to Exercise 6.44

We know that as a, b are relatively prime, they are not both zero. If $ab = 0$, then one of them must be zero, in which case $a + b$ is nonzero. So not both ab and $a + b$ are zero. Thus $d := \gcd(ab, a + b)$ exists. If $d > 1$, then it has a prime factor, say p. As p divides d, and hence also ab, p divides either a or b (Exercise 6.35, p.71), say it divides a. As $d \mid a + b$, and as $p \mid d$, we have $p \mid a + b$. But since $p \mid a$, it now follows that $p \mid (a + b) - a = b$. So $p > 1$ is a common divisor of a and b, a contradiction to $\gcd(a, b) = 1$. Consequently, $d = 1$, that is, $\gcd(ab, a + b) = 1$, and so $ab, a + b$ are relatively prime.

Solution to Exercise 6.45

(1) Let $S = \{m \in \mathbb{N} : a \mid m \text{ and } b \mid m\}$. Then $S \neq \varnothing$ as e.g. $ab \in S$. By the well-ordering principle, S has a least element, say $m_* \in \mathbb{N}$. Then $a \mid m_*$ and $b \mid m_*$. Suppose $m \in \mathbb{N}$ is such that $a \mid m$ and $b \mid m$. Then $m \in S$. By the division algorithm $m = qm_* + r$ for some $q \in \mathbb{Z}$ and $0 \leqslant r < m_*$. We claim $r = 0$. Otherwise, $(0 <)\ r = m - qm_*$ is divisible by a and b (as m, m_* are). Thus $r \in S$, contradicting the definition of m_* as the smallest element of S. Hence $r = 0$. This means $m_* \mid m$. So m_* is a least common multiple of a, b.

Uniqueness: Let m, m' be least common multiples of a, b. Then $a \mid m$ and $b \mid m$. As m' is a least common multiple, it follows that $m' \mid m$. Also, $a \mid m'$ and $b \mid m'$. As m is a least common multiple, we have $m \mid m'$. Exercise 6.7, p.64 now implies $m = m'$ or $m = -m'$. But m, m' are both positive. Hence $m = m'$.

(2) Let $d := \gcd(a, b)$. As $d \mid a$ and $d \mid b$, we have $a = q_a d$ and $b = q_b d$ for some $q_a, q_b \in \mathbb{Z}$. Set $m_* := q_a q_b d \in \mathbb{Z}$. Then $aq_b = m_*$ and so $a \mid m_*$. Moreover, $bq_a = m_*$ and so $b \mid m_*$. Let $a \mid m$ and $b \mid m$, and let $q, q' \in \mathbb{Z}$ be such that $m = qa = q'b$. By Theorem 6.2, p.66, there exist $x, y \in \mathbb{Z}$ such that $ax + by = d$. Multiplying by m, we get

$$dm = max + mby = q'bax + qaby = ab(q'x + qy) = q_a d q_b d(q'x + q'y)$$

and so $m = q_a q_b d(q'x + qy) = m_*(q'x + qy)$. Thus $m_* \mid m$.

Consequently, $\operatorname{lcm}(a, b) = m_* = q_a q_b d = \dfrac{q_a d q_b d}{d} = \dfrac{ab}{\gcd(a, b)}$.

(3) We have $\min\{\alpha_i, \beta_i\} + \max\{\alpha_i, \beta_i\} = \alpha_i + \beta_i$ for all i, and so

$$\operatorname{lcm}(a, b) = \frac{ab}{\gcd(a, b)} = \frac{(p_1^{\alpha_1} \cdots p_k^{\alpha_k})(p_1^{\beta_1} \cdots p_k^{\beta_k})}{p_1^{\min\{\alpha_1, \beta_1\}} \cdots p_k^{\min\{\alpha_k, \beta_k\}}} = p_1^{\max\{\alpha_1, \beta_1\}} \cdots p_k^{\max\{\alpha_k, \beta_k\}}$$

as wanted.

Solution to Exercise 6.46

(1) Let $a \neq 0$. Then $2a^2 = 3b^2$ shows $b \neq 0$. By the fundamental theorem of arithmetic, $a = 2^{\alpha}3^{\beta}A$ and $b = 2^{\alpha'}3^{\beta'}B$ for some integers $\alpha, \beta, \alpha', \beta' \geqslant 0$ and integers A, B whose prime factorisations don't contain the primes $2, 3$ (i.e., $A = 5^{\gamma}7^{\delta}\cdots p_r^k$, $B = 5^{\gamma'}7^{\delta'}\cdots p_r^{k'}$ where $\gamma, \gamma', \delta, \delta', \ldots, k, k' \geqslant 0$ are integers and $5 < 7 < \cdots < p_r$ are primes).
Substituting this in $2a^2 = 3b^2$ gives $2 \cdot 2^{2\alpha}3^{2\beta}A^2 = 3 \cdot 2^{2\alpha'}3^{2\beta'}B^2$, i.e., $2^{2\alpha+1}3^{2\beta}A^2 = 2^{2\alpha'}3^{2\beta'+1}B^2$. By the uniqueness part of the fundamental theorem of arithmetic, we get $2\alpha + 1 = 2\alpha'$. But this is impossible, since otherwise $2 \mid 1$, which is false (e.g. by Exercise 6.6, p.64). So our original assumption that $a \neq 0$ must be false. Hence $a = 0$. Then $3b^2 = 2a^2 = 0$, implying that $b = 0$ as well.

(2) Suppose $a, b, c \in \mathbb{Z}$ and $a\sqrt{2} + b\sqrt{3} + c = 0$, that is, $a\sqrt{2} + c = -b\sqrt{3}$. Squaring, $2a^2 + c^2 + 2ac\sqrt{2} = 3b^2$. Since $\sqrt{2} \notin \mathbb{Q}$, we must have $ac = 0$. Thus $a = 0$ or $c = 0$.
1° If $a = 0$, then $b\sqrt{3} = -c$. As $\sqrt{3} \notin \mathbb{Q}$, $b = 0$. So $c = -b\sqrt{3} = -0\sqrt{3} = 0$.
2° If $c = 0$, then $2a^2 + c^2 + 2ac\sqrt{2} = 3b^2$ gives $2a^2 = 3b^2$. By (1), $a = b = 0$.

Chapter 7

Solution to Exercise 7.1

Since $a \equiv b \bmod n$, we have that $n \mid a-b$, and so $a-b=qn$ for some $q \in \mathbb{Z}$. Similarly $c \equiv d \bmod n$ implies that $c-d=q'n$ for some $q' \in \mathbb{Z}$. Adding, we obtain $(a-b)+(c-d)=qn+q'n$, that is, $(a+c)-(b+d)=(q+q')n$. Thus we have $n \mid (a+c)-(b+d)$, and so $(a+c) \equiv (b+d) \bmod n$.

Moreover, $ac=(nq+b)(nq'+d)=n^2qq'+nqd+nbq'+bd$. So we have $ac-bd=n(nqq'+qd+bq')$. Hence $n \mid ac-bd$, that is, $ac \equiv bd \bmod n$.

Solution to Exercise 7.2

As $ab \equiv ac \bmod n$, we have $n \mid (ab-ac)=a(b-c)$. But $\gcd(a,n)=1$. Hence by Proposition 6.2, p.69, $n \mid b-c$. So $b \equiv c \bmod n$.

To show the condition $\gcd(a,n)=1$ is essential, consider for example $n=2$, $a=2$, $b=3$, $c=2$: Then

- $6 \equiv 4 \bmod 2$, i.e., $ab \equiv ac \bmod n$,
- $\gcd(a,n)=\gcd(2,2)=2 \neq 1$,
- $\neg(b \equiv c \bmod n)$ because $n=2 \nmid 1=3-2=b-c$.

Solution to Exercise 7.3

$2222 = 317 \cdot 7 + 3$ gives $2222 \equiv 3 \bmod 7$. But $3 \equiv -4 \bmod 7$, and so we obtain $2222 \equiv -4 \bmod 7$. Hence $2222^{5555} \equiv -4^{5555} \bmod 7$. Moreover, since $5555=793 \cdot 7+4$, we get $5555 \equiv 4 \bmod 7$. Thus $5555^{2222} \equiv 4^{2222} \bmod 7$. So $(2222^{5555}+5555^{2222}) \equiv (-4^{5555}+4^{2222}) \bmod 7$. As $4^3 \equiv 1 \bmod 7$, and since $4^{5555}=(4^3)^{1851}4^2$, we get $4^{5555} \equiv 4^2 \bmod 7$. Also, $4^{2222}=(4^3)^{740}4^2$, and so $4^{2222} \equiv 4^2 \bmod 7$. Thus $(-4^{5555}+4^{2222}) \equiv 0 \bmod 7$. So $7 \mid 2222^{5555}+5555^{2222}$.

Solution to Exercise 7.4

- For $q \in \mathbb{Z}$, $(a+qn) \equiv a \bmod n$, as $n \mid ((a+qn)-a)=qn$.
  So $\{a+qn : q \in \mathbb{Z}\} \subset [a]$.
- For $b \in [a]$, $b \equiv a \bmod n$. So $n \mid b-a$, i.e., $b-a=q_*n$ for some $q_* \in \mathbb{Z}$.
  Thus $b=a+q_*n \in \{a+qn : q \in \mathbb{Z}\}$. So $[a] \subset \{a+qn : q \in \mathbb{Z}\}$.

Consequently, $[a]=\{a+qn : q \in \mathbb{Z}\}$.

Solution to Exercise 7.5

Suppose that $n \in \mathbb{N}$ has the decimal representation $d_K \cdots d_1 d_0$ where the integers $d_0, \cdots, d_K$ satisfy $0 \leqslant d_0, \cdots, d_K \leqslant 9$. So $n=10^K d_K+\cdots+10d_1+d_0$. We want to show that $n \equiv (d_K+\cdots+d_1+d_0) \bmod 9$. For all $1 \leqslant k \leqslant K$, the binomial theorem gives $10^k=(1+9)^k=1+\binom{k}{1}9+\cdots+\binom{k}{k}9^k=1+9m$, where $m=\binom{k}{1}+\binom{k}{2}9+\cdots+\binom{k}{k}9^{k-1}$. So $10^k \equiv 1 \bmod 9$. (This can also be seen using Exercise 6.12, 65, since $9=10-1 \mid 10^k-1^k$.) Also $d_k \equiv d_k$

mod 9. Hence (by 'multiplying these') we get $10^k d_k \equiv 1 d_k \bmod 9$ for each $1 \leqslant k \leqslant K$. 'Adding these', together with $d_0 \equiv d_0 \bmod 9$, we arrive at $(10^K d_K + \cdots + 10 d_1 + d_0) \equiv (d_K + \cdots + d_1 + d_0) \bmod 9$, that is, we have $n \equiv (d_K + \cdots + d_1 + d_0) \bmod 9$.

(The above gives a 'divisibility-by-9 test':
$9 \,|\, n \Leftrightarrow 9 \,|\, \sigma$, where σ is the sum of the decimal digits of n.
($\Rightarrow$) Let $9 \,|\, n$, i.e., $n \equiv 0 \bmod 9$. As $\sigma \equiv n \bmod 9$, we get $\sigma \equiv 0 \bmod 9$. So $9 \,|\, \sigma$.
($\Leftarrow$) Let $9 \,|\, \sigma$, i.e., $\sigma \equiv 0 \bmod 9$. As $n \equiv \sigma \bmod 9$, we get $n \equiv 0 \bmod 9$. So $9 \,|\, n$.

Solution to Exercise 7.6

We have $2^3 = 8$ and $8 \equiv -1 \bmod 9$. Thus we obtain $(2^3)^9 \equiv (-1)^9 \bmod 9$, i.e., $2^{27} \equiv -1 \bmod 9$. Also, $2^2 \equiv 4 \bmod 9$. Hence $2^{29} \equiv -4 \bmod 9$. Suppose that $k \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ is the missing digit. Then the sum σ of the decimal digits of 2^{29} is equal to $(0 + 1 + 2 + \cdots + 9) - k = \frac{9(10)}{2} - k = 9(5) - k$. We have $\sigma \equiv -k \bmod 9$ and $2^{29} \equiv \sigma \bmod 9$. Thus $2^{29} \equiv -k \bmod 9$. But we had seen above that $2^{29} \equiv -4 \bmod 9$. Consequently, $-k \equiv -4 \bmod 9$. Together with $-1 \equiv -1 \bmod 9$, we obtain $k \equiv 4 \bmod 9$, i.e., $9 \,|\, k - 4$. Hence $k = 4$. (We remark that in fact $2^{29} = 536870912$.)

Solution to Exercise 7.7

Choose a large enough $m \in \mathbb{N}$ so that $n + m(k!) \geqslant k$. Set $N = n + m(k!) \in \mathbb{N}$. Then $N \geqslant k$. By Exercise 4.16, p.40, $N(N-1)\cdots(N-k+1) \equiv 0 \bmod (k!)$. From $N = n + m(k!)$, we have $N \equiv n \bmod (k!)$. So, for $0 \leqslant \ell \leqslant k-1$, we also have $(N - \ell) \equiv (n - \ell) \bmod (k!)$. Using the result from Exercise 7.1, p.75, it now follows ('by multiplying these') that

$$N(N-1)\cdots(N-k+1) \equiv n(n-1)\cdots(n-k+1) \bmod (k!). \qquad (\star)$$

By transitivity, we get from $N(N-1)\cdots(N-k+1) \equiv 0 \bmod (k!)$ and $(\star)$, that $n(n-1)\cdots(n-k+1) \equiv 0 \bmod (k!)$, i.e., $k! \,|\, n(n-1)\cdots(n-k+1)$.

Solution to Exercise 7.8

Let $d_1 \cdots d_{10}$ be a valid ISBN number. Then

$$d_{10} \equiv (d_1 + 2d_2 + \cdots + 9d_9) \bmod 11. \qquad (*)$$

(1) Let the d_i be swapped with d_{i+1} $(\neq d_i)$ for some $i \in \{1, \cdots, 8\}$. Suppose that

$$d_{10} \equiv (d_1 + 2d_2 + \cdots + id_{i+1} + (i+1)d_i + \cdots + 9d_9) \bmod 11. \qquad (**)$$

Subtracting $(*)$ and $(**)$, we get

$$\big((id_i + (i+1)d_{i+1}) - ((i+1)d_i + id_{i+1})\big) \equiv 0 \bmod 11,$$

that is, $11 \,|\, (d_{i+1} - d_i)$. But this is impossible, since $1 \leqslant |d_{i+1} - d_i| \leqslant 9$. Thus $(**)$ cannot hold, showing that the check digit flags that there is an error committed.

(2) Let $(*)$ hold again. We assume that for some $i \in \{1, \cdots, 9\}$, the correct digit d_i has been replaced by an incorrect digit d_i' $(\neq d_i)$. Suppose that

$$d_{10} \equiv (d_1 + 2d_2 + \cdots + id_i' + \cdots + 9d_9) \bmod 11. \qquad (***)$$

Subtracting $(*)$ and $(***)$, we get $i(d_i - d_i') \equiv 0 \bmod 11$, i.e., $11 \mid i(d_i - d_i')$. But $1 \leqslant i \leqslant 9$ and $1 \leqslant |d_i - d_i'| \leqslant 9$, so that $11 \nmid i$ and $11 \nmid (d_i - d_i')$. This contradicts Exercise 6.35, p.71 since 11 is a prime number. Thus $(***)$ cannot hold, i.e., the check digit flags that there is an error committed.

Solution to Exercise 7.9

We have $3^4 \equiv 81 \bmod 100$. So $3^5 \equiv 43 \bmod 100$, as $81 \cdot 3 = 243 = 2 \cdot 100 + 43$. Since $43^2 = (40+3)^2 = 40^2 + 2 \cdot 40 \cdot 3 + 9 = 40^2 + 249$, we get $3^{10} \equiv 49 \bmod 100$. As $(40+9)^2 = 40^2 + 2 \cdot 40 \cdot 9 + 81 = 40^2 + 720 + 81 = 40^2 + 801$, $3^{20} \equiv 1 \bmod 100$. Now $1234 = 20 \cdot 61 + 14$, and so

$$\begin{aligned} 3^{1234} &\equiv (1)^{61} 3^{14} \bmod 100 \\ &\equiv (3^{10} 3^4) \bmod 100 \\ &\equiv (49 \cdot 81) \bmod 100 \\ &\equiv 3969 \bmod 100 \\ &\equiv 69 \bmod 100. \end{aligned}$$

So 3^{1234} ends in 69. (There are many alternative ways of arriving of this.)

Solution to Exercise 7.10

(1) By the division algorithm, $m = 3k + r$, for some $k, r \in \mathbb{Z}$, with $0 \leqslant r < 3$. Thus $r \in \{0, 1, 2\}$. We have $m^3 = (3k + r)^3 = 27k^3 + 27k^2 r + 9kr^2 + r^3$. Hence $m^3 \equiv r^3 \bmod 9$. If $r = 0, 1, 2$, then respectively $r^3 = 0, 1, 8$, and so $r^3 \equiv 0 \bmod 9$, $r^3 \equiv 1 \bmod 9$, $r^3 \equiv -1 \bmod 9$ (as $8 = 9 - 1$). Thus, we have $m^3 \equiv n \bmod 9$ for some $n \in \{-1, 0, 1\}$.

(2) Let $(x, y, z) \in \mathbb{Z}^3$ satisfy $x^3 + y^3 + z^3 = 1976$. From (1), $x^3 \equiv a \bmod 9$, $y^3 \equiv b \bmod 9$, $z^3 \equiv c \bmod 9$ for some $a, b, c \in \{-1, 0, 1\}$. So their sum $a + b + c \in \{-3, -2, -1, 0, 1, 2, 3\}$. Hence $x^3 + y^3 + z^3 \equiv n \bmod 9$, where $n \in \{-3, -2, -1, 0, 1, 2, 3\}$. Moreover, $1976 = 9 \cdot 219 + 5$, which implies $1976 \equiv 5 \bmod 9$. So $5 \equiv n \bmod 9$ and $9 \mid (5 - n)$. But the possible values of $5 - n$ are $8, 7, 6, 5, 4, 3, 2$, and clearly 9 does not divide these numbers, a contradiction. So $x^3 + y^3 + z^3 = 1976$ doesn't have have integer solutions.

Solution to Exercise 7.11

We want to find $[2]^{-1}$ in $\mathbb{Z}/5\mathbb{Z}$. Since $\gcd(2, 5) = 1$, there exist $x_0, y_0 \in \mathbb{Z}$ such that $2x_0 + 5y_0 = 1$. To find such x_0, y_0, we use the Euclidean algorithm (or inspection for these small numbers):

$$\begin{aligned} 5 &= 2 \cdot 2 + 1 \\ 2 &= 2 \cdot 1 + 0. \end{aligned}$$

So $1 = 5\cdot 1 + 2\cdot(-2)$. So $[2]\cdot[-2] = [1 - 5\cdot 1] = [1]$. If x is a solution, then $[2]\cdot x = [3]$ gives (upon premultiplying by $[2]^{-1}$) that

$$\begin{aligned} x &= [1]\cdot x = ([2]^{-1}\cdot[2])\cdot x \\ &= [2]^{-1}\cdot([2]\cdot x) = [2]^{-1}\cdot[3] \\ &= [-2]\cdot[3] = [-6] \in \mathbb{Z}/5\mathbb{Z}. \end{aligned}$$

Vice versa, we check that $[2]\cdot[-6] = [-12] = [3\cdot 5 - 12] = [3]$ in $\mathbb{Z}/5\mathbb{Z}$.
(We remark that $x = [-6] = [2\cdot 5 - 6] = [4] \in \mathbb{Z}/5\mathbb{Z} = \{[0], [1], [2], [3], [4]\}$.)

Solution to Exercise 7.12

We first seek a necessary condition. So let us suppose that $x, y \in \mathbb{Z}/nZ$ satisfy the given equations. Then multiplying the first equation by $[2]$, and subtracting the second from it, we obtain:

$$\begin{aligned} [2]\cdot x + [2]\cdot y &= [4] \\ \underline{[2]\cdot x - [3]\cdot y &= [3]} \\ [5]\cdot y &= [1]. \end{aligned}$$

So $[5]$ is invertible in $\mathbb{Z}/n\mathbb{Z}$. This implies $\gcd(5, n) = 1$, and so this is a necessary condition for solvability of the given system of linear equations in $\mathbb{Z}/n\mathbb{Z}$. We now show below that $\gcd(5, n) = 1$ is also sufficient.

Suppose $\gcd(5, n) = 1$. Then $[5] \in \mathbb{Z}/n\mathbb{Z}$ is invertible in $\mathbb{Z}/n\mathbb{Z}$. Define $y = [5]^{-1} \in \mathbb{Z}/n\mathbb{Z}$, and $x = [2] - [5]^{-1} \in \mathbb{Z}/n\mathbb{Z}$. Then we have

$$\begin{aligned} x + y &= [2] - [5]^{-1} + [5]^{-1} = [2] + [0] = [2], \\ [2]\cdot x - [3]\cdot y &= [2]\cdot([2] - [5]^{-1}) - [3]\cdot[5]^{-1} = [4] - [2]\cdot[5]^{-1} - [3]\cdot[5]^{-1} \\ &= [4] - [5]\cdot[5]^{-1} = [4] - [1] = [3]. \end{aligned}$$

So the linear system has a solution in $\mathbb{Z}/n\mathbb{Z}$.

We remark that $\gcd(5, n) = 1 \Leftrightarrow 5 \nmid n$:

($\Rightarrow$) Let $\gcd(5, n) = 1$. Then there exist $a, b \in \mathbb{Z}$ such that $5a + nb = 1$. If $5 \mid n$, then also $5 \mid 5a + nb = 1$, i.e., $5 \mid 1$, a contradiction. So $5 \nmid n$.

($\Leftarrow$) Let $5 \nmid n$. Suppose $\gcd(5, n) =: d > 1$. Then since 5 is prime and $1 < d \mid 5$, we get $d = 5$. So $5 = d = \gcd(5, n) \mid n$, i.e., $5 \mid n$, a contradiction.

Solution to Exercise 7.13

(1) Let x be a solution. As $x \equiv 4 \bmod 6$, there exists a $k \in \mathbb{Z}$ such that $x - 4 = 6k$, i.e., $x = 4 + 6k = 2(2 + 3k)$. So x is even. Since $x \equiv 5 \bmod 10$, there exists an $\ell \in \mathbb{Z}$ such that $x - 5 = 10\ell$, i.e., $x = 5 + 10\ell = 2(2 + 5\ell) + 1$. Thus x is odd, a contradiction.

(2) As m, n are relatively prime, we have $mx_0 + ny_0 = 1$ for some $x_0, y_0 \in \mathbb{Z}$. Multiplying by $b - a$, and setting $k = (b - a)x_0$, $\ell = y_0(b - a)$, we get $mk + n\ell = b - a$. Hence $x := a + mk = b - n\ell$. Thus $x \equiv a \bmod m$, and $x \equiv b \bmod n$.

Uniqueness mod (mn): Let x, x' be solutions. Then $x \equiv a \bmod m$ implies $x - a = md$ for some $d \in \mathbb{Z}$. Similarly, $x' \equiv a \bmod m$ implies $x' - a = md'$ for some $d' \in \mathbb{Z}$. Subtracting these yields $x' - x = m(d' - d)$. An analogous argument shows that $x \equiv b \bmod n$ and $x' \equiv b \bmod n$ together imply that $x' - x = n(q' - q)$ for some $q, q' \in \mathbb{Z}$. So $x' - x = m(d' - d) = n(q' - q)$. In particular, $m \mid n(q' - q)$. But $\gcd(m, n) = 1$. By Proposition 6.2, p.69, $m \mid (q' - q)$. So $q' - q = mr$ for some $r \in \mathbb{Z}$. Hence $x' - x = n(q' - q) = nmr$, and so $(mn) \mid x' - x$. Thus $x' \equiv x \bmod (mn)$.

Chapter 8

Solution to Exercise 8.1

The map $f : \mathbb{N} \to S$ given by $f(n) = n^2$ is surjective. It is injective, since if $n, m \in \mathbb{N}$ are such that $f(n) = f(m)$, then $n^2 = m^2$, i.e., $(n-m)(n+m) = 0$, and as $n + m > 1$, and in particular $n + m \neq 0$, it follows that $n - m = 0$, i.e., $n = m$. Consequently, f is bijective, and S is countable.

Solution to Exercise 8.2

Suppose $f(n) = f(m)$ for some $n, m \in \mathbb{Z}$. We will use the uniqueness part of the fundamental theorem of arithmetic in each case below:

1° $n \geqslant 0$ and $m \geqslant 0$. Then $2^n = f(n) = f(m) = 2^m$. So $n = m$.

2° $n < 0$ and $m < 0$. Then $3^{-n} = 3^{-m}$, and so $-n = -m$, giving $n = m$.

3° $n \geqslant 0$ and $m < 0$. Then $2^n = 3^{-m}$. So $n = 0$ and $-m = 0$, a contradiction.

4° $n < 0$ and $m \geqslant 0$. Then $3^{-n} = 2^m$. So $m = 0$ and $-n = 0$, a contradiction.

Thus f is injective. So by changing the codomain of f to $\operatorname{ran} f$, we get a bijection $\widetilde{f} : \mathbb{Z} \to \operatorname{ran} f$. As the infinite set $\mathbb{N}$ is contained in $\mathbb{Z}$, $\mathbb{Z}$ is infinite. By Exercise 2.28, p.24, $\operatorname{ran} f$ is infinite. By Proposition 8.1, p.82, $\operatorname{ran} f$ is countable. Since $\operatorname{ran} f$ is in one-to-one correspondence with $\mathbb{Z}$ (under the bijective map $\widetilde{f}$), $\mathbb{Z}$ is countable too.

Solution to Exercise 8.3

Let $E = \{2n : n \in \mathbb{Z}\}$. Define $f : E \to \mathbb{Z}$ by $f(m) = \frac{m}{2}$ for all $m \in E$. Then f is bijective. As $\mathbb{Z}$ is countable, so is E.

Solution to Exercise 8.4

Suppose $f(x) = f(x')$ for some $x, x' \in A \cup B$. We have the following cases:

1° $x, x' \in A$. Then $2^{\alpha(x)} = 2^{\alpha(x')}$, and so $\alpha(x) = \alpha(x')$. Hence $x = x'$.

2° $x, x' \in B \backslash A$. Then $3^{\beta(x)} = 3^{\beta(x')}$, and so $\beta(x) = \beta(x')$. Thus $x = x'$.

3° $x \in A$, $x' \in B \backslash A$. Then $f(x) = 2^{\alpha(x)}$ is even, while $f(x') = 3^{\beta(x')}$ is odd, and so $f(x) \neq f(x')$, a contradiction.

4° $x \in B \backslash A$, $x' \in A$. Then $f(x) = 3^{\beta(x)}$ is odd, while $f(x') = 2^{\alpha(x')}$ is even, and so $f(x) \neq f(x')$, a contradiction.

So f is injective. As $A \cup B$ contains the infinite set A, it follows that $A \cup B$ is infinite too. By Exercise 2.28, p.24, $\operatorname{ran} f = f(A \cup B)$ is infinite. Since $f(A \cup B)$ is an infinite subset of $\mathbb{N}$, it is countable, by Proposition 8.1, p.82. The map $\widetilde{f} : A \cup B \to f(A \cup B)$, obtained from f by restricting its codomain to $\operatorname{ran} f$, is a bijection. Hence $A \cup B$ is countable.

Solution to Exercise 8.5

Suppose that $f(a,b) = f(a',b')$ for some $(a,b),(a',b') \in A \times B$. Then we get $2^{n(a)}3^{m(b)} = 2^{n(a')}3^{m(b')}$. By the uniqueness part of the fundamental theorem of arithmetic, we obtain $n(a) = n(a')$ and $m(b) = m(b')$. As n, m are injective, we conclude that $a = a'$ and $b = b'$. Hence $(a,b) = (a',b')$. So f is injective. By restricting the codomain of f to its range $f(A \cup B)$, we get a bijection $\widetilde{f} : A \times B \to f(A \times B)$. We claim $f(A \times B)$ is infinite. For a fixed $b_* \in B$, $f(A \times B)$ contains $f(A \times \{b_*\}) = \{2^{m(a)}3^{n(b_*)} : a \in A\}$, which is an infinite set (since it is the injective image of A under the map $A \ni a \mapsto 2^{m(a)}3^{n(b_*)} \in \mathbb{N}$). As $f(A \times B)$ is an infinite subset of $\mathbb{N}$, $f(A \times B)$ is countable. Consequently, $A \times B$ is countable too.

Solution to Exercise 8.6

Let $x, x' \in A$ be such that $f(x) = f(x')$, i.e., $2^{n(x)}3^{f_{n(x)}(x)} = 2^{n(x')}3^{f_{n(x')}(x')}$. By the uniqueness part of the fundamental theorem of arithmetic, we get

$$n(x) = n(x') \qquad (*)$$
$$f_{n(x)}(x) = f_{n(x')}(x'). \qquad (**)$$

So $f_{n(x)}(x) \overset{(**)}{=} f_{n(x')}(x') \overset{(*)}{=} f_{n(x)}(x')$. Since $f_{n(x)}$ injective, we get $x = x'$. Consequently, f is injective. For all $x \in A_1$, we have $f(x) = 2^1 3^{f_1(x)} \in f(A)$, so that $f(A)$ in infinite (as A_1 is infinite and f_1 is injective). Since $f(A)$ is an infinite subset of $\mathbb{N}$, $f(A)$ is countable. The map f with its codomain restricted to $f(A)$ gives a bijection $\widetilde{f} : A \to f(A)$. Thus A is countable.

Solution to Exercise 8.7

Let $x, x' \in A$ be such that $f(x) = f(x')$, that is, $2^{n(x)}3^{k(x)} = 2^{n(x')}3^{k(x')}$. The uniqueness part of the fundamental theorem of arithmetic gives $n(x) = n(x')$ and $k(x) = k(x')$. So $x = a_{k(x)}^{(n(x))} = a_{k(x')}^{(n(x'))} = x'$. So f is injective.

As A is infinite and f is injective, $f(A)$ is infinite. Being an infinite subset of $\mathbb{N}$, $f(A)$ is countable. The map f with its codomain restricted to $f(A)$ gives a bijection $\widetilde{f} : A \to f(A)$. Thus A is countable.

Solution to Exercise 8.8

For a finite $A_n \neq \varnothing$ with cardinal number $m_n \in \mathbb{N}$, let $f_n : \{1, \cdots, m_n\} \to A_n$ be a bijection. For an infinite A_n, as it is countable, there exists a bijection $f_n : \mathbb{N} \to A_n$. For $x \in A$, let $n(x)$ be the least $n \in \mathbb{N}$ such that $x \in A_n$. If $x \in A$, then it belongs to $A_{n(x)}$. We have the following two possibilities:

1° $A_{n(x)}$ is the finite set $\{f_{n(x)}(1), \cdots, f_{n(x)}(m_{n(x)})\}$.
Then there exists a unique $k(x) \in \{1, \cdots, m_{n(x)}\}$ such that $x = f_{n(x)}(k(x))$.

2° $A_{n(x)}$ is the infinite countable set $\{f_{n(x)}(k) : k \in \mathbb{N}\}$.
Then there exists a unique $k(x) \in \mathbb{N}$ such that $x = f_{n(x)}(k(x))$.

Define $f : A \to \mathbb{N}$ by $f(x) = 2^{n(x)}3^{k(x)}$ for all $x \in A$. Let $x, x' \in A$ be such that $f(x) = f(x')$, i.e., $2^{n(x)}3^{k(x)} = 2^{n(x')}3^{k(x')}$. By the uniqueness part of the fundamental theorem of arithmetic, we get $n(x) = n(x')$ and $k(x) = k(x')$. Hence $x = f_{n(x)}(k(x)) = f_{n(x')}(k(x')) = x'$. So f is injective.

As A is infinite and f is injective, $f(A)$ is infinite. Being an infinite subset of $\mathbb{N}$, $f(A)$ is countable. The map f with its codomain restricted to $f(A)$ gives a bijection $\widetilde{f} : A \to f(A)$. Thus A is countable.

Solution to Exercise 8.9

As $\mathbb{Q}$ is countable, by Proposition 8.2, p.84, $\mathbb{Q}^2$ is countable. Let $C(\boldsymbol{x}, r)$ be the circle with centre at $\boldsymbol{x}$ and radius $r > 0$. Then $\mathbb{Q}_+ = \{q \in \mathbb{Q} : q > 0\}$ contains the infinite set of natural numbers, and hence is an infinite set. As $\mathbb{Q}$ is countable, its infinite subset $\mathbb{Q}_+$ is countable (Proposition 8.1, p.82). So for each fixed $\boldsymbol{x} \in \mathbb{Q}^2$, the set $\{C(\boldsymbol{x}, r) : r \in \mathbb{Q}_+\}$ is countable. The set of all circles with rational radii and centres in $\mathbb{Q}^2$ is the countable union $\bigcup_{\boldsymbol{x} \in \mathbb{Q}^2} \{C(\boldsymbol{x}, r) : r \in \mathbb{Q}_+\}$ of countable sets, and so is countable.

Solution to Exercise 8.10

Let A be the set of all $\{0, 1\}$-valued sequences that have infinitely many ones, and B be the set of all $\{0, 1\}$-valued sequences that have infinitely many zeros. We can create a bijection between A and B by just swapping 0 with 1. Now suppose that A is countable. Then B is countable too. By Exercise 8.4, p.83, $A \cup B$ is countable. But $A \cup B$ is the set of all $\{0, 1\}$-valued sequences, which is uncountable (Theorem 8.2, p.85). So we have arrived at a contradiction. Thus A (and also B) is uncountable.

Solution to Exercise 8.11

We create a bijection between the set S of all subsequences of $(n)_{n \in \mathbb{N}}$ and the set A of all $\{0, 1\}$-valued sequences having infinitely many ones as follows. Define $f : S \to A$ by setting $f((n_k)_{k \in \mathbb{N}}) = (b_n)_{n \in \mathbb{N}}$, where

$$b_n = \begin{cases} 1 & \text{if } n = n_k \text{ for some } k \in \mathbb{N}, \\ 0 & \text{if for all } k \in \mathbb{N},\ n \neq n_k. \end{cases}$$

(For example, $(2n)_{n \in \mathbb{N}}$ is sent by f to the sequence $0, 1, 0, 1, \cdots$.) Then f is injective and surjective. As A is uncountable (Exercise 8.10, p.86), so is S.

Solution to Exercise 8.12

Suppose $\mathbb{R} \backslash \mathbb{Q}$ is countable. As $\mathbb{Q}$ is countable, by Exercise 8.4, p.83, we get $\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \backslash \mathbb{Q})$ is countable, a contradiction. Thus $\mathbb{R} \backslash \mathbb{Q}$ is not countable.

Solution to Exercise 8.13

As $\mathbb{Q}$ is countable, a repeated application of Proposition 8.2, p.84 (Cartesian product of countable sets is countable) shows that $\mathbb{Q}^d = \mathbb{Q}\times\cdots\times\mathbb{Q}$ (d times) is also countable. Suppose there exists a bijection $f:\mathbb{R}\to\mathbb{Q}^d$. Then $\mathbb{R}$ would be countable too, a contradiction to the fact that $\mathbb{R}$ is uncountable.

Solution to Exercise 8.14

Suppose $[0,1)$ is countable. For any $m\in\mathbb{Z}$, we claim $\varphi_m:[0,1)\to[m,m+1)$ given by $\varphi_m(x)=x+m$ for all $x\in[0,1)$, is a bijection. Indeed, it has the inverse $\varphi_m^{-1}:[m,m+1)\to[0,1)$ given by $\varphi_m^{-1}(y)=y-m$ for all $y\in[m,m+1)$: for $x\in[0,1)$ and $y\in[m,m+1)$, we have

$$(\varphi_m\circ\varphi_m^{-1})(y) = \varphi_m(y-m) = (y-m)+m = y,$$
$$(\varphi_m^{-1}\circ\varphi_m)(x) = \varphi_m^{-1}(x+m) = (x+m)-m = x.$$

Our assumption that $[0,1)$ is countable then yields that for each $m\in\mathbb{Z}$, $[m,m+1)$ is countable. But then, as

$$\mathbb{R} = \bigcup_{m\in\mathbb{Z}}[m,m+1),$$

$\mathbb{R}$ is a countable union of countable sets. So $\mathbb{R}$ is countable, a contradiction. Hence $[0,1)$ is uncountable. Thus $[0,1]$ ($\supset[0,1)$) is uncountable.

Solution to Exercise 8.15

(1) For all $x\in[-1,1]$, $(x,\sqrt{1-x^2})\in C$, and $[-1,1]\ni x\mapsto(x,\sqrt{1-x^2})\in C$ is injective. As $[-1,1]$ is uncountable, C is infinite and uncountable.

(2) For all $n\in\mathbb{N}$,

$$(\tfrac{1-n^2}{1+n^2})^2+(\tfrac{2n}{1+n^2})^2 = \tfrac{1-2n^2+n^4+4n^2}{(1+n^2)^2} = \tfrac{1+2n^2+n^4}{(1+n^2)^2} = \tfrac{(1+n^2)^2}{(1+n^2)^2} = 1.$$

The map $\mathbb{N}\ni n\mapsto(\frac{1-n^2}{1+n^2},\frac{2n}{1+n^2})\in C_{\mathbb{Q}}$ is injective, and so $C_{\mathbb{Q}}$ is infinite.

(3) $\mathbb{Q}\times\mathbb{Q}$ is countable. As $C_{\mathbb{Q}}\subset\mathbb{Q}\times\mathbb{Q}$ is an infinite subset of the countable set $\mathbb{Q}\times\mathbb{Q}$, $C_{\mathbb{Q}}$ is countable.

Solution to Exercise 8.16

For $a\in\mathbb{R}$, $[a]$ is infinite and countable:
We claim $[a]=a+\mathbb{Q}:=\{a+r:r\in\mathbb{Q}\}$. Then $a+\mathbb{Q}$ is infinite, as it contains the distinct elements $a+n$ for all $n\in\mathbb{N}\subset\mathbb{Q}$. For all $r\in\mathbb{Q}$, $(a+r)-a=r\in\mathbb{Q}$, so that $a+r\sim a$, i.e., $a+r\in[a]$. Thus $a+\mathbb{Q}\subset[a]$. Vice versa, if $b\in[a]$, then $b\sim a$, and so $b-a=:r\in\mathbb{Q}$, i.e., $b=a+r\in a+\mathbb{Q}$, showing $[a]\subset a+\mathbb{Q}$. Consequently, $[a]=a+\mathbb{Q}$. We claim $a+\mathbb{Q}$ is countable. If $f:\mathbb{N}\to\mathbb{Q}$ is a bijection, then define $g:\mathbb{N}\to a+\mathbb{Q}$ by $g(n)=a+f(n)\in a+\mathbb{Q}$ for all $n\in\mathbb{N}$. Then g is injective: if $g(n)=g(n')$ for $n,n'\in\mathbb{N}$, then $a+f(n)=a+f(n')$,

giving $f(n)=f(n')$ and so $n=n'$. Also, if $b\in a+\mathbb{Q}$, then there exists an $r\in\mathbb{Q}$ such that $b=a+r$, and for this $r\in\mathbb{Q}$, there exists an $n\in\mathbb{N}$ such that $f(n)=r$, so that $g(n)=a+f(n)=a+r$, showing that g is surjective. Thus g is a bijection, and $a+\mathbb{Q}$ $(=[a])$ is countable.

The set $S=\{[a]:a\in\mathbb{R}\}$ is infinite and uncountable:
We first show that $S=\{[a]:a\in\mathbb{R}\}$ is infinite. For primes p,p' with $p'>p$, $[\sqrt{p'}]\neq[\sqrt{p}]$. (Otherwise, $0\neq\sqrt{p'}-\sqrt{p}=r\in\mathbb{Q}$, giving $p'=r^2+p+2r\sqrt{p}$, which is impossible since[21] $\sqrt{p}\notin\mathbb{Q}$.) The equivalence classes partition $\mathbb{R}$, and each equivalence class is countable. If $S=\{[a]:a\in\mathbb{R}\}$ were countable, then (since the countable union of countable sets is countable), $\mathbb{R}=\bigcup_{[a]\in S}[a]$ would be countable, a contradiction. Hence S is uncountable.

Solution to Exercise 8.17

Let $A:=(0,1)\cap\mathbb{Q}$ and $B:=[0,1]\cap\mathbb{Q}$. As A is an infinite set contained in the countable set $\mathbb{Q}$, A is countable. Similarly B is countable too. So there exists a bijection $\alpha:A\to B$. Define $f:(0,1)\to[0,1]$ by

$$f(x)=\begin{cases}\alpha(x) & \text{if } x\in A,\\ x & \text{if } x\notin A.\end{cases}$$

Note that

$$\big(x\in(0,1)\text{ and }x\notin\mathbb{Q}\big)\Leftrightarrow\big(x\in[0,1]\text{ and }x\notin\mathbb{Q}\big).\qquad(*)$$

We claim f is surjective. Let $y\in[0,1]$. Then there are two possible cases:
1° $y\in\mathbb{Q}$. Then $y\in B$. As α is surjective, there exists an $x\in A$ such that $\alpha(x)=y$, i.e., $f(x)=\alpha(x)=y$.
2° $y\notin\mathbb{Q}$. Then by $(*)$, $y\in(0,1)$ and $y\notin\mathbb{Q}$, i.e., $y\in(0,1)\backslash A$. So $f(y)=y$.
We claim f is injective. Suppose $x,x'\in(0,1)$ are such that $f(x)=f(x')$.
1° If $x,x'\in A$, then $\alpha(x)=f(x)=f(x')=\alpha(x')$. As α is injective, $x=x'$.
2° If $x,x'\notin A$, then $x=f(x)=f(x')=x'$.
3° If $x\in A$ and $x'\notin A$, then $\mathbb{Q}\supset B\ni\alpha(x)=x'\notin\mathbb{Q}$, a contradiction.
4° If $x\notin A$ and $x'\in A$, then $\mathbb{Q}\not\ni x=\alpha(x')\in B\subset\mathbb{Q}$, a contradiction.
As f is injective and surjective, f is a bijection from $(0,1)$ to $[0,1]$.

Solution to Exercise 8.18

(1) We claim that the number of ways of choosing k distinct objects from a collection of n $(\geqslant k)$ distinct objects is $\frac{n(n-1)\cdots(n-k+1)}{k!}=\frac{n!}{k!(n-k)!}=\binom{n}{k}$. The 1st object can be chosen in n ways, the 2nd in $n-1$ ways and so on till the kth object can be chosen in $n-(k-1)$ ways. So a sequential

[21] By $\sqrt{p}$ is a zero of $x^2-p\in\mathbb{Z}[x]$, and by the rational zeros theorem, p.102, if $\sqrt{p}=\frac{m}{n}$ for $m,n\in\mathbb{N}$, then $m\,|\,p$ and $n\,|\,1$, so that $\sqrt{p}$ is either p or 1, which is clearly false.

choice of k objects can be done in $n(n-1)\cdots(n-k+1)$ ways. But if we choose them all at once, then the sequential order of these k objects does not matter, i.e., we need to divide by $k!$, since $k!$ is the number of ways of arranging k objects sequentially. So $\binom{n}{k}$ is the number of subsets of a set of n elements having k elements.

(2) The sum $\binom{n}{0}+\cdots+\binom{n}{n}$ gives the number of all subsets of a set with n elements, i.e., it is the number of elements in $\mathcal{P}(S)$, and using the binomial theorem, this equals $\sum_{k=0}^{n}\binom{n}{k}=\sum_{k=0}^{n}\binom{n}{k}1^k1^{n-k}=(1+1)^n=2^n$.

(3) By Cantor's theorem, p.90, $\mathcal{P}(S)$ cannot have n elements (otherwise there exists a bijection, and hence also a surjection from S to $\mathcal{P}(S)$). Thus $2^n<n$ or $2^n>n$. If $(\varnothing\neq)\ S=\{a_1,\cdots,a_n\}$, then $\mathcal{P}(S)$ has more than n elements, since it contains $\varnothing,\{a_1\},\cdots,\{a_n\}$. So $2^n>n$.

Solution to Exercise 8.19

(1) Let $\mathcal{S}_2$ denote the collection all two element subsets of $\mathbb{N}$.
For any two element subset $S\in\mathcal{S}_2$ of $\mathbb{N}$, its elements are distinct, and we can arrange its members in increasing order. This shows that there is an injective map $f:\mathcal{S}_2\to\mathbb{N}\times\mathbb{N}$ that sends $S\in\mathcal{S}_2$ to the ordered pair obtained by ordering the members of S in increasing order. For example, if $S=\{399,76\}$, then $f(S)=(76,399)$. By restricting the codomain of f to $f(\mathcal{S}_2)$, we get a bijection $\widetilde{f}:\mathcal{S}_2\to f(\mathcal{S}_2)$. Clearly, the set $f(\mathcal{S}_2)$ is infinite (for instance, $f(\{1,n\})=(1,n)\in f(\mathcal{S}_2)$ for all $n\in\{2,3,\cdots\}$). Moreover, $f(\mathcal{S}_2)$ is a subset of the countable set $\mathbb{N}\times\mathbb{N}$. So $f(\mathcal{S}_2)$, and hence also $\mathcal{S}_2$, is countable.

(2) Let $d\in\mathbb{N}$. As in part (1), it can be seen that the set $\mathcal{S}_d$ of all subsets of $\mathbb{N}$ with d elements is countable. (For any such subset $S\in\mathcal{S}_d$ of $\mathbb{N}$, we arrange its members in increasing order, and hence associate it uniquely with an ordered d-tuple in $\mathbb{N}^d$, giving an injection from $\mathcal{S}_d$ to $\mathbb{N}^d$. Since $\mathbb{N}^d$ is countable, it follows that $\mathcal{S}_d$ is countable.) The set $\mathcal{S}_1$ of all singleton subsets of $\mathbb{N}$, is in one-to-one correspondence with $\mathbb{N}$. The set of all finite nonempty subsets of $\mathbb{N}$, i.e., $\mathcal{S}=\bigcup_{d\in\mathbb{N}}\mathcal{S}_d$, is a countable union of the countable sets $\mathcal{S}_d$. So $\mathcal{S}$ is countable.

(3) Let us suppose that the infinite set $\mathcal{P}(\mathbb{N})$ is countable. Then there exists a bijection $f:\mathbb{N}\to\mathcal{P}(\mathbb{N})$. In particular, f is a surjection. But this contradicts Cantor's theorem, p.90. Hence $\mathcal{P}(\mathbb{N})$ is uncountable.

Chapter 9

Solution to Exercise 9.1

Suppose $\log_2 3 = \frac{n}{d}$, where $n, d \in \mathbb{Z}$ and $d \neq 0$. As $\log_2 3 > 0$, we can assume that $n, d \in \mathbb{N}$. Then $n = d \log_2 3 = \log_2(3^d)$, giving $2^n = 3^d$. Since $n \neq 0$, 2^n is even. But 3^d is odd. Thus we have arrived at a contradiction. Hence $\log_2 3$ is irrational.

Solution to Exercise 9.2

We give two proofs. The first one is 'non-constructive', and goes as follows. We have two possible cases:

1° If $\sqrt{2}^{\sqrt{2}} \in \mathbb{Q}$, then we are done, since we take $a = b = \sqrt{2} \in \mathbb{R}\backslash\mathbb{Q}$.

2° If $\sqrt{2}^{\sqrt{2}} \notin \mathbb{Q}$, then taking $a = \sqrt{2}^{\sqrt{2}}$ and $b = \sqrt{2}$, we have

$$a^b = (\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = \sqrt{2}^2 = 2 \in \mathbb{Q}.$$

We remark that by Gelfond-Schneider's result (Theorem 9.9, p.104), $\sqrt{2}^{\sqrt{2}}$ is not algebraic, much less rational, and so case 1° does not hold. But we did not use this fact above.

Next we give an elementary constructive proof using Exercise 9.1, p.92. Take $a = \sqrt{2} \in \mathbb{R}\backslash\mathbb{Q}$, and $b = 2\log_2 3 \in \mathbb{R}\backslash\mathbb{Q}$. Then

$$a^b = \sqrt{2}^{2\log_2 3} = (\sqrt{2}^2)^{\log_2 3} = 2^{\log_2 3} = 3 \in \mathbb{Q}.$$

By the Gelfond-Schneider Theorem, the case 2° of the first proof is a constructive proof too, but we'd have to appeal to this non-elementary result.

Solution to Exercise 9.3

Suppose $\tan 1^\circ$ is rational. Write $\tan 1^\circ = \frac{n}{d}$, where n, d are integers and $d > 0$. Then as $0^\circ < 1^\circ < 45^\circ$, and since tan is strictly increasing[22] on $[0, \frac{\pi}{4})$, we know that $0 < \frac{n}{d} < 1$. Thus

$$\tan 2^\circ = \tan(1^\circ + 1^\circ) = \frac{\tan 1^\circ + \tan 1^\circ}{1 - (\tan 1^\circ)(\tan 1^\circ)} = \frac{2\frac{n}{d}}{1 - (\frac{n}{d})^2} \in \mathbb{Q}.$$

Suppose $k \in \mathbb{N}$ is such that $k, k+1 < 45$, and $\tan k^\circ \in \mathbb{Q}$. Since tan is strictly increasing, $0 < \tan k^\circ < 1$, and in particular $0 < (\tan k^\circ)(\tan 1^\circ) < 1$. Thus

$$\tan(k+1)^\circ = \frac{\tan k^\circ + \tan 1^\circ}{1 - (\tan k^\circ)(\tan 1^\circ)} \in \mathbb{Q}.$$

So we see inductively that if $\tan 1^\circ \in \mathbb{Q}$, then $\tan 2^\circ, \tan 3^\circ, \cdots, \tan 30^\circ \in \mathbb{Q}$. But $\tan 30^\circ = \frac{1}{\sqrt{3}} \notin \mathbb{Q}$, a contradiction. Hence $\tan 1^\circ \notin \mathbb{Q}$.

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[22] Because $\frac{d \tan}{dx}(x) = \frac{1}{1+x^2} > 0$.

Solution to Exercise 9.4

Let $x_0 \in \mathbb{Q}$ and $p(x_0) = 0$. Write $x_0 = \frac{a}{b}$, where $a, b \in \mathbb{Z}$, $b \neq 0$, and $\gcd(a, b) = 1$. Since $p(x_0) = 0$, we obtain

$$\ell \frac{a^d}{b^d} + m\frac{a}{b} + n = 0.$$

Multiplying throughout by b^d gives $\ell a^d + mab^{d-1} + nb^d = 0$.
We consider the following cases:

1° a is even and b is even. This contradicts $\gcd(a, b) = 1$.
So this case is not possible.

2° a is even and b is odd. Then $nb^d = -\ell a^d - mab^{d-1}$ is impossible, since the left-hand side is odd, while the right-hand side is even.
So this case is not possible.

3° a is odd and b is even. Then $\ell a^d = -nb^d - mab^{d-1}$ is impossible, since the left-hand side is odd, while the right-hand side is even.
So this case is not possible.

4° a is odd and b is odd. Then $\ell a^d + mab^{d-1}$ is even (being the sum of two odd integers). Thus $(\ell a^d + mab^{d-1}) + nb^d$ is odd (being the sum of an even integer with an odd integer), and hence it cannot equal 0.
So this case is not possible.

Thus we have arrived at a contradiction. So there is no rational zero of p.

Solution to Exercise 9.5

The real number x with decimal representation $0.12345678910111213\cdots$ is not rational, because its string of digits after the decimal point will contain the digits of 10^k for all $k \in \mathbb{N}$, having a '1' followed by a block of '0' of increasing size k, showing that the decimal representation of x is nonterminating and nonrepeating. As every rational number has either a terminating or nonterminating but repeating decimal representation, it follows that $x = 0.12345678910111213\cdots$ is irrational.

Solution to Exercise 9.6

We have

$$\begin{aligned} 225 &= 1 \cdot 157 + 68 \\ 157 &= 2 \cdot 68 + 21 \\ 68 &= 3 \cdot 21 + 5 \\ 21 &= 4 \cdot 5 + 1 \\ 5 &= 5 \cdot 1. \end{aligned}$$

Thus $\frac{225}{157} = [1, 2, 3, 4, 5]$.

Solution to Exercise 9.7

We have $r_1 = x - [x] = x - a_1$. Also $x = \dfrac{P_{N+1} + r_{N+1}P_N}{Q_{N+1} + r_{N+1}Q_N}$, giving

$$r_{N+1}(xQ_N - P_N) = P_{N+1} - xQ_{N+1} \neq 0$$

(where the last non-equality holds because $x \neq \dfrac{P_{N+1}}{Q_{N+1}} \in \mathbb{Q}$). Thus

$$r_{N+1} = \frac{P_{N+1} - xQ_{N+1}}{xQ_N - P_N}.$$

The periodicity condition that $r_{N+1} = r_1$ yields

$$\frac{P_{N+1} - xQ_{N+1}}{xQ_N - P_N} = x - a_1,$$

i.e., $Q_N x^2 + (Q_{N+1} - a_1 Q_N)x + (a_1 P_N - P_{N+1}) = 0$. As $Q_N \in \mathbb{N}$, the degree of the polynomial $p := Q_N x^2 + (Q_{N+1} - a_1 Q_N)x + (a_1 P_N - P_{N+1}) \in \mathbb{Z}[x]$ is 2. Also, since x is irrational, its minimal polynomial cannot have degree 1. Hence x is algebraic with a minimal polynomial having degree 2.

Solution to Exercise 9.8

Suppose $\sqrt[3]{6} \in \mathbb{Q}$. Let $m, n \in \mathbb{Z}$ have no common factor, $n > 0$, and $\sqrt[3]{6} = \frac{m}{n}$. Then $\frac{m}{n}$ is a rational zero of $x^3 - 6 \in \mathbb{Z}[x]$. By the rational zeros theorem, m divides -6 and n (> 0) divides 1. So $m \in \{6, -6, 3, -3, 2, -2, 1, -1\}$ and $n = 1$, giving $\frac{m}{n} \in \{6, -6, 3, -3, 2, -2, 1, -1\}$. However, none of these values satisfy $x^3 - 6 = 0$, a contradiction. So $\sqrt[3]{6} \notin \mathbb{Q}$.

Solution to Exercise 9.9

Let $\alpha = \sqrt{2} + \sqrt{3}$. Then $\alpha - \sqrt{2} = \sqrt{3}$. Squaring, $\alpha^2 - 2\sqrt{2}\alpha + 2 = 3$, i.e., $\alpha^2 - 1 = 2\sqrt{2}\alpha$. Squaring again, $\alpha^4 - 2\alpha^2 + 1 = 8\alpha^2$, i.e., $\alpha^4 - 10\alpha^2 + 1 = 0$. Thus α is a zero of $x^4 - 10x^2 + 1 \in \mathbb{Z}[x]$. Suppose that $\alpha \in \mathbb{Q}$. Let $m, n \in \mathbb{Z}$ have no common factor, $n \neq 0$, and $\alpha = \frac{m}{n}$. By the rational zeros theorem, m divides 1 and n divides 1, forcing them both to belong to the set $\{-1, 1\}$. Thus $\alpha = \frac{m}{n} \in \{-1, 1\}$. But $\alpha = \sqrt{2} + \sqrt{3} > 1$ (as $\sqrt{2} > \sqrt{1} = 1$). So we have arrived at a contradiction. Hence $\sqrt{2} + \sqrt{3} \notin \mathbb{Q}$.

Solution to Exercise 9.10

Let $\alpha = (2 + \sqrt{5})^{\frac{1}{3}} - (-2 + \sqrt{5})^{\frac{1}{3}}$. Cubing, we get

$$\begin{aligned}\alpha^3 &= 2 + \sqrt{5} - 3(2+\sqrt{5})^{\frac{2}{3}}(-2+\sqrt{5})^{\frac{1}{3}} + 3(2+\sqrt{5})^{\frac{1}{3}}(-2+\sqrt{5})^{\frac{2}{3}} - (-2+\sqrt{5}) \\ &= 4 - 3((2+\sqrt{5})(-2+\sqrt{5}))^{\frac{1}{3}}((2+\sqrt{5})^{\frac{1}{3}} - (-2+\sqrt{5})^{\frac{1}{3}}) \\ &= 4 - 3(1)(\alpha).\end{aligned}$$

Thus α is a zero of $p = x^3 + 3x - 4 \in \mathbb{Z}[x]$.

Let $m, n \in \mathbb{Z}$ have no common factor, $n > 0$, and suppose $\frac{m}{n}$ is a zero of p. The rational zeros theorem then implies that n divides 1, and m divides

4, so that $\frac{m}{n} \in \{1, -1, 2, -2, 4, -4\}$. We now try these values as arguments of p, and get $p(1) = 1 + 3 - 4 = 0$. Motivated by this, we divide p by $x - 1$:

$$\begin{array}{r|l} & x^2 + x + 4 \\ \hline x - 1 & x^3 + 3x - 4 \\ & \underline{-x^3 + x^2} \\ & \quad x^2 + 3x \\ & \quad \underline{-x^2 + x} \\ & \quad\quad 4x - 4 \\ & \quad\quad \underline{-4x + 4} \\ & \quad\quad\quad 0 \end{array}$$

Hence $p = x^3 + 3x - 4 = (x - 1)(x^2 + x + 4)$. Now the quadratic $x^2 + x + 4$ is never zero for any real number: (for example) because

$$x^2 + x + 4 = x^2 + 2x \cdot \tfrac{1}{2} + \tfrac{1}{4} - \tfrac{1}{4} + 4 = (x + \tfrac{1}{2})^2 + \tfrac{15}{4} > 0 \text{ for all } x \in \mathbb{R}.$$

So the only real zero of p is $x = 1$. As we know that $\alpha \in \mathbb{R}$ is a zero of p, we conclude that $\alpha = 1 \in \mathbb{Q}$.

Solution to Exercise 9.11

For $d \in \mathbb{N}$, let $\mathcal{P}_d = \{p \in \mathbb{Z}[x] : \deg(p) = d\}$ and $\mathcal{C}_d = \underbrace{\mathbb{Z} \times \cdots \times \mathbb{Z}}_{d \text{ times}} \times (\mathbb{Z} \setminus \{0\})$.

Define $f : \mathcal{P}_d \to \mathcal{C}_d$ by $f(p) = (c_0, c_1, \cdots, c_d)$ for $p = c_0 + c_1 x + \cdots + c_d x^d \in \mathcal{P}_d$. Then f is a bijection. As $\mathcal{C}_d$ is countable (which can be seen by a repeated application of Proposition 8.2, p.84), it follows that $\mathcal{P}_d$ is countable. But

$$\mathbb{Z}[x] = \mathbb{Z} \cup \bigcup_{d \in \mathbb{N}} \mathcal{P}_d.$$

By Exercise 8.6, p.84, $\bigcup_{d \in \mathbb{N}} \mathcal{P}_d$ is countable. So $\mathbb{Z}[x]$ is countable.

Solution to Exercise 9.12

Fix an $n \in \mathbb{N}$. Then for all $N > n$, we have

$$\begin{aligned} s_N &= s_n + \tfrac{1}{2^{(n+1)!}} + \cdots + \tfrac{1}{2^{N!}} \\ &< s_n + \sum_{m=(n+1)!}^{N!} \tfrac{1}{2^m} = s_n + \tfrac{1}{2^{(n+1)!}} 2(1 - (\tfrac{1}{2})^{N! - (n+1)! + 1}) < s_n + \tfrac{2}{2^{(n+1)!}}. \end{aligned}$$

Passing to the limit as $N \to \infty$, we get $L \leqslant s_n + \frac{2}{2^{(n+1)!}}$ for all $n \in \mathbb{N}$.

As $(n+1)! - 1 \geqslant (n+1)! - n! = n!n$ and $L \geqslant s_n$, for all $n \in \mathbb{N}$, we get

$$0 \leqslant L - s_n \leqslant \tfrac{2}{2^{(n+1)!}} \leqslant \tfrac{1}{2^{n!n}} = \tfrac{1}{(2^{n!})^n}. \qquad (*)$$

Define the natural numbers $a_n = \sum_{m=1}^{n} 2^{n! - m!}$ and $b_n = 2^{n!}$. Then $s_n = \frac{a_n}{b_n}$.

If L was an algebraic number with a minimal polynomial of degree d, then by Liouville's approximation theorem, there exists a constant $A > 0$ such that for all large enough n (so that the positive distance of s_n to L is

less than 1), we have

$$L - s_n = |L - s_n| \geqslant \frac{A}{b_n^d} = \frac{A}{(2^{n!})^d}. \qquad (**)$$

The inequalities $(*)$ and $(**)$ imply for all large enough n that $A < \frac{1}{(2^{n!})^{n-d}}$. Passing to the limit as $n \to \infty$, we get $A \leqslant 0$, a contradiction.

Solution to Exercise 9.13

(1) As there are no elements x in the empty set which can violate $x \leqslant -\infty$, it is vacuously true that:

$$\text{For all } x \in \varnothing,\ x \leqslant -\infty.$$

Hence $-\infty$ is an upper bound of $\varnothing$. We claim that $u_* := -\infty$ is a least upper bound of $\varnothing$. Let $u \in \overline{\mathbb{R}}$ be an upper bound of $\varnothing$. If $u = -\infty$, then $u_* = u$, and in particular, $u_* \leqslant u$ holds. So suppose $u \neq -\infty$. Then $u \in \mathbb{R}$ or $u = +\infty$. In either case $u_* = -\infty < u$, and so $u \leqslant u_*$ holds again.

(2) For all $x \in S$, we have that $x < +\infty$, and so $+\infty$ is an upper bound of S. Let $u \in \overline{\mathbb{R}}$ be an upper bound of S. Then $u \notin \mathbb{R}$ because S does not have an upper bound in $\mathbb{R}$. Also, as S is nonempty, there exists an $x \in S$, and so $u \neq -\infty$ (indeed, if $u = -\infty$, then we have $x \leqslant -\infty$, and as $x \neq -\infty$, we get $x < -\infty$, a contradiction). Hence $u = +\infty$, and in particular, $+\infty \leqslant u$ holds.

(3)(a) If $S = \varnothing$, then $-\infty$ is a least upper bound of S by part (1) above.

(b) If $S \neq \varnothing$, and S has an upper bound in $\mathbb{R}$, then S has a least upper bound u_* in $\mathbb{R}$ by Theorem 5.2, p.52. Then for all $x \in S$, $x \leqslant u_*$. Let $u \in \overline{\mathbb{R}}$ be an upper bound of S. Consider the following cases:

1° $u = -\infty$. As $S \neq \varnothing$, there exists an $x \in S \subset \mathbb{R}$. So $x \leqslant -\infty$, a contradiction.

2° If $u \in \mathbb{R}$, then $u_* \leqslant u$.

3° If $u = +\infty$, then $u_* < +\infty$.

Thus u_* is a least upper bound[23] of S.

(c) If $S \neq \varnothing$, but does not have an upper bound in $\mathbb{R}$, then by part (2) above, $+\infty$ is a least upper bound of S.

Solution to Exercise 9.14

Suppose a or b is 0. Then $ab = 0$, and so $|ab|_p = 0$. On the other hand, at least one of $|a|_p, |b|_p$ is 0, and so $|a|_p|b|_p = 0$ too. Also, if $a = 0$, then we have $|a+b|_p = |0+b|_p = |b|_p = \max\{0, |b|_p\} = \max\{|a|_p, |b|_p\}$, and similarly if $b = 0$, and $|a+b|_p \leqslant \max\{|a|_p, |b|_p\}$.

[23] That is, also with our new definition of least upper bound of S, which admits extended values for upper bounds.

Now let a, b be both nonzero. Then we can write

$$a = p^{k_a}\frac{n_a}{d_a} \text{ and } b = p^{k_b}\frac{n_b}{d_b},$$

where $k_a := \mathrm{ord}_p a, k_b := \mathrm{ord}_p b, n_a, d_a, n_b, d_b \in \mathbb{Z}$, $\gcd(n_a, p) = 1 = \gcd(d_a, p)$, and $\gcd(n_b, p) = 1 = \gcd(d_b, p)$. Then $1 = \gcd(n_a n_b, p) = \gcd(d_a d_b, p)$ by Exercise 6.25, p.70. As

$$ab = p^{k_a}\frac{n_a}{d_a}p^{k_b}\frac{n_b}{d_b} = p^{k_a+k_b}\frac{n_a n_b}{d_a d_b},$$

it follows that $\mathrm{ord}_p(ab) = k_a + k_b = \mathrm{ord}_p a + \mathrm{ord}_p b$, and so

$$|ab|_p = \frac{1}{p^{\mathrm{ord}_p(ab)}} = \frac{1}{p^{\mathrm{ord}_p a + \mathrm{ord}_p b}} = \frac{1}{p^{\mathrm{ord}_p a}}\frac{1}{p^{\mathrm{ord}_p b}} = |a|_p|b|_p.$$

If $a + b = 0$, then $|a + b|_p = |0|_p = 0 \leqslant \max\{|a|_p, |b|_p\}$. So let us suppose that $a + b \neq 0$. We have

$$a + b = p^{k_a}\frac{n_a}{d_a} + p^{k_b}\frac{n_b}{d_b} = p^{\min\{k_a,k_b\}}\frac{p^{k_a-\min\{k_a,k_b\}}n_a d_b + p^{k_b-\min\{k_a,k_b\}}n_b d_a}{d_a d_b}.$$

Write $p^{k_a-\min\{k_a,k_b\}}n_a d_b + p^{k_b-\min\{k_a,k_b\}}n_b d_a = p^k q$, where q is coprime to p and the integer $k \geqslant 0$. Then $\mathrm{ord}_p(a + b) = \min\{k_a, k_b\} + k \geqslant \min\{k_a, k_b\}$. If $k_a \geqslant k_b$, then $\min\{k_a, k_b\} = k_b$, and so

$$|a + b|_p = \frac{1}{p^{\mathrm{ord}_p(a+b)}} \leqslant \frac{1}{p^{\min\{k_a,k_b\}}} = \frac{1}{p^{k_b}} \leqslant \max\{\frac{1}{p^{k_a}}, \frac{1}{p^{k_b}}\} = \max\{|a|_p, |b|_p\}.$$

Similarly, if $k_a \leqslant k_b$, then $|a + b|_p \leqslant \max\{|a|_p, |b|_p\}$.

Finally, as $|a|_p, |b|_p \geqslant 0$, we have $\max\{|a|_p, |b|_p\} \leqslant |a|_p + |b|_p$.

If $a = 0$, then $|a|_p = 0$ by definition. On the other hand, if $a \in \mathbb{Q}\backslash\{0\}$, then $\mathrm{ord}_p a \in \mathbb{Z}$, giving $|a|_p = \frac{1}{p^{\mathrm{ord}_p a}} > 0$, and so $|a|_p \neq 0$. Hence $|a|_p = 0$ if and only if $a = 0$.

Elements of $\mathscr{C}_p$ are bounded: Let $(a_n)_{n\in\mathbb{N}} \in \mathscr{C}_p$. For $\epsilon = 1 > 0$, there exists an $N \in \mathbb{N}$ such that whenever $n, m > N$, we have $|a_n - a_m|_p < \epsilon = 1$. In particular, for all $n > N$,

$|a_n|_p = |a_n - a_{N+1} + a_{N+1}|_p \leqslant \max\{|a_n - a_{N+1}|_p, |a_{N+1}|_p\} \leqslant \max\{1, |a_{N+1}|_p\}$.

Also, for $n \in \{1, \cdots, N\}$, we have $|a_n|_p \leqslant \max\{|a_1|_p, \cdots, |a_N|_p\}$. Hence for all $n \in \mathbb{N}$, $|a_n|_p \leqslant \max\{|a_1|_p, \cdots, |a_N|_p, |a_{N+1}|_p, 1\} =: M$, and M is clearly a positive rational (being at least 1, which is positive).

$(a_n)_{n\in\mathbb{N}}, (b_n)_{n\in\mathbb{N}} \in \mathscr{C}_p \Rightarrow (a_n b_n)_{n\in\mathbb{N}} \in \mathscr{C}_p$:

There exist rationals $M_a, M_b > 0$ such that for all $n \in \mathbb{N}$, $|a_n|_p \leqslant M_a$ and $|b_n|_p \leqslant M_b$. Let ϵ be any positive rational number.

There exists an $N_a \in \mathbb{N}$ such that for all $n, m > N_a$, we have $|a_n - a_m|_p < \frac{\epsilon}{M_b}$.

There exists an $N_b \in \mathbb{N}$ such that for all $n, m > N_b$, we have $|b_n - b_m|_p < \frac{\epsilon}{M_a}$.

So for all $n, m > N := \max\{N_a, N_b\}$, we have

$$\begin{aligned} |a_n b_n - a_m b_m|_p &= |a_n b_n - a_n b_m + a_n b_m - a_m b_m|_p \\ &\leqslant \max\{|a_n b_n - a_n b_m|_p, |a_n b_m - a_m b_m|_p\} \\ &= \max\{|a_n|_p |b_n - b_m|_p, |b_m|_p |a_n - a_m|_p\} \\ &< \max\{M_a \tfrac{\epsilon}{M_a}, M_b \tfrac{\epsilon}{M_b}\} = \epsilon. \end{aligned}$$

So $(a_n b_n)_{n\in\mathbb{N}} \in \mathscr{C}_p$.

Transitivity of $\sim$:

Let $(a_n)_{n\in\mathbb{N}}, (b_n)_{n\in\mathbb{N}}, (c_n)_{n\in\mathbb{N}} \in \mathscr{C}_p$, $(a_n)_{n\in\mathbb{N}} \sim (b_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}} \sim (c_n)_{n\in\mathbb{N}}$. Let ϵ be any given positive rational number.

There exists an $N_1 \in \mathbb{N}$ such that whenever $n > N_1$, we have $|a_n - b_n|_p < \epsilon$.

There exists an $N_2 \in \mathbb{N}$ such that whenever $n > N_2$, we have $|b_n - c_n|_p < \epsilon$.

So for all $n > N := \max\{N_1, N_2\}$, we have

$|a_n - c_n|_p = |a_n - b_n + b_n - c_n|_p \leqslant \max\{|a_n - b_n|_p, |b_n - c_n|_p\} < \max\{\epsilon, \epsilon\} = \epsilon$.

Hence $(a_n)_{n\in\mathbb{N}} \sim (c_n)_{n\in\mathbb{N}}$.

Multiplication is well-defined: Let $(a_n)_{n\in\mathbb{N}}, (\widetilde{a}_n)_{n\in\mathbb{N}}, (b_n)_{n\in\mathbb{N}}, (\widetilde{b}_n)_{n\in\mathbb{N}} \in \mathscr{C}_p$ be such that $(a_n)_{n\in\mathbb{N}} \sim (\widetilde{a}_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}} \sim (\widetilde{b}_n)_{n\in\mathbb{N}}$.

There exist rationals $M_{\tilde{a}}, M_b > 0$ such that for all $n \in \mathbb{N}$, $|\widetilde{a}_n|_p \leqslant M_{\tilde{a}}$ and $|b_n|_p \leqslant M_b$. Let ϵ be a given positive rational.

There exists an $N_a \in \mathbb{N}$ be such that if $n > N_a$, the $|a_n - \widetilde{a}_n|_p < \frac{\epsilon}{M_b}$.

There exists an $N_b \in \mathbb{N}$ be such that if $n > N_b$, then $|b_n - \widetilde{b}_n|_p < \frac{\epsilon}{M_{\tilde{a}}}$.

So $n > N := \max\{N_a, N_b\}$, we have

$$\begin{aligned} |a_n b_n - \widetilde{a}_n \widetilde{b}_n|_p &= |a_n b_n - \widetilde{a}_n b_n + \widetilde{a}_n b_n - \widetilde{a}_n \widetilde{b}_n|_p \\ &\leqslant \max\{|a_n b_n - \widetilde{a}_n b_n|_p, |\widetilde{a}_n b_n - \widetilde{a}_n \widetilde{b}_n|_p\} \\ &= \max\{|b_n|_p |a_n - \widetilde{a}_n|_p, |\widetilde{a}_n|_p |b_n - \widetilde{b}_n|_p\} \\ &< \max\{M_b \tfrac{\epsilon}{M_b}, M_{\tilde{a}} \tfrac{\epsilon}{M_{\tilde{a}}}\} = \epsilon. \end{aligned}$$

So the given description of multiplication in $\mathbb{Q}_p$ is well-defined.

$\mathbf{1} := [(1)_{n\in\mathbb{N}}]$ is the multiplicative identity in $\mathbb{Q}_p$: The constant sequence $(1)_{n\in\mathbb{N}} \in \mathscr{C}_p$, and $[(1)_{n\in\mathbb{N}}] \in \mathbb{Q}_p$. For any $\boldsymbol{a} = [(a_n)_{n\in\mathbb{N}}] \in \mathbb{Q}_p$, we have

$$\boldsymbol{a} \cdot \mathbf{1} = [(a_n)_{n\in\mathbb{N}}] \cdot [(1)_{n\in\mathbb{N}}] = [(a_n 1)_{n\in\mathbb{N}}] = [(a_n)_{n\in\mathbb{N}}] = \boldsymbol{a}, \text{ and}$$
$$\mathbf{1} \cdot \boldsymbol{a} = [(1)_{n\in\mathbb{N}}] \cdot [(a_n)_{n\in\mathbb{N}}] = [(1 a_n)_{n\in\mathbb{N}}] = [(a_n)_{n\in\mathbb{N}}] = \boldsymbol{a}.$$

Nonzero elements of $\mathbb{Q}_p$ possess a multiplicative inverse in $\mathbb{Q}_p$:

The constant sequence $(0)_{n\in\mathbb{N}} \in \mathscr{C}_p$. The zero element (i.e., additive identity) in $\mathbb{Q}_p$ is $\mathbf{0} := [(0)_{n\in\mathbb{N}}]$. Let $\boldsymbol{a} = [(a_n)_{n\in\mathbb{N}}] \in \mathbb{Q}_p \backslash \{\mathbf{0}\}$. Then it is not the case that $(a_n)_{n\in\mathbb{N}} \sim (0)_{n\in\mathbb{N}}$. Thus, it is not the case that for every rational

$\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that whenever $n \in \mathbb{N}$ satisfies $n > N$ we have $|a_n - 0|_p < \epsilon$. In other words:

$$\left.\begin{array}{l}\text{There exists a rational } \epsilon_* > 0 \text{ such that for all } N \in \mathbb{N},\\ \text{there exists an } n \in \mathbb{N} \text{ such that } n > N \text{ and } |a_n|_p \geqslant \epsilon.\end{array}\right\} \ (\star)$$

As $(a_n)_{n\in\mathbb{N}} \in \mathscr{C}_p$, for the rational $\frac{\epsilon_*}{2}$, there exists an $N_* \in \mathbb{N}$ such that for all $n, m > N_*$, we have $|a_n - a_m|_p < \frac{\epsilon_*}{2}$. Taking $N = N_*$ in $(\star)$, we get an $n_* > N_*$ such that $|a_{n_*}|_p \geqslant \epsilon_*$. Hence for $n > N_*$, we have

$$\epsilon_* \leqslant |a_{n_*}|_p = |a_{n_*} - a_n + a_n|_p \leqslant \max\{|a_{n_*} - a_n|_p, |a_n|_p\} \leqslant \max\{\tfrac{\epsilon_*}{2}, |a_n|_p\}.$$

If $|a_n|_p < \frac{\epsilon_*}{2}$, then we get the contradiction that $\epsilon_* \leqslant \max\{\frac{\epsilon_*}{2}, |a_n|_p\} = \frac{\epsilon_*}{2}$. Hence $|a_n|_p \geqslant \frac{\epsilon_*}{2} > 0$ for all $n > N_*$. In particular, $a_n \in \mathbb{Q}\backslash\{0\}$ for all $n > N_*$.

Set $b_n := \left\{\begin{array}{ll} 0 & \text{if } 1 \leqslant n \leqslant N_* \\ a_n^{-1} & \text{if } n > N_* \end{array}\right\}$. Then $b_n \in \mathbb{Q}$ for all $n \in \mathbb{Q}$.

We claim that $(b_n)_{n\in\mathbb{N}} \in \mathscr{C}_p$. Let ϵ be a given positive rational.

Firstly, for $n, m > N_*$, $|b_n - b_m|_p = |\frac{1}{a_n} - \frac{1}{a_m}|_p = \frac{|a_n - a_m|_p}{|a_n|_p |a_m|_p} \leqslant 4\frac{|a_n - a_m|_p}{\epsilon_*^2}$.

Secondly, as $(a_n)_{n\in\mathbb{N}} \in \mathscr{C}_p$, there exists an $\widetilde{N}$ such that for all $n, m > \widetilde{N}$, we have $|a_n - a_m|_p < \frac{1}{4}\epsilon_*^2\epsilon$. So for all $n, m > N := \max\{N_*, \widetilde{N}\}$, we have

$$|b_n - b_m|_p \leqslant 4\frac{|a_n - a_m|_p}{\epsilon_*^2} < 4\frac{1}{4}\frac{\epsilon_*^2\epsilon}{\epsilon_*^2} = \epsilon.$$

Thus $(b_n)_{n\in\mathbb{N}} \in \mathscr{C}_p$.

As $a_n b_n = \left\{\begin{array}{ll} 0 & \text{if } 1 \leqslant n \leqslant N_* \\ 1 & \text{if } n > N_* \end{array}\right\}$, $(a_n b_n)_{n\in\mathbb{N}} \sim (1)_{n\in\mathbb{N}}$, i.e., $[(a_n b_n)_{n\in\mathbb{N}}] = [(1)_{n\in\mathbb{N}}]$.

Since $a_n b_n = b_n a_n$ for all $n \in \mathbb{N}$, also $[(b_n a_n)_{n\in\mathbb{N}}] = [(1)_{n\in\mathbb{N}}]$.

Hence with $\boldsymbol{b} := [(b_n)_{n\in\mathbb{N}}] \in \mathbb{Q}_p$, we have

$$\begin{aligned}\boldsymbol{a}\cdot\boldsymbol{b} &= [(a_n)_{n\in\mathbb{N}}]\cdot[(b_n)_{n\in\mathbb{N}}] = [(a_n b_n)_{n\in\mathbb{N}}] = [(1)_{n\in\mathbb{N}}] = \mathbf{1}, \text{ and}\\ \boldsymbol{b}\cdot\boldsymbol{a} &= [(b_n)_{n\in\mathbb{N}}]\cdot[(a_n)_{n\in\mathbb{N}}] = [(b_n a_n)_{n\in\mathbb{N}}] = [(1)_{n\in\mathbb{N}}] = \mathbf{1}.\end{aligned}$$

Thus $\boldsymbol{a}$ is invertible in $\mathbb{Q}_p$ with the inverse $\boldsymbol{a}^{-1} = \boldsymbol{b} = [(b_n)_{n\in\mathbb{N}}] \in \mathbb{Q}_p$.

Distributive law: Let $\boldsymbol{a} = [(a_n)_{n\in\mathbb{N}}], \boldsymbol{b} = [(b_n)_{n\in\mathbb{N}}], \boldsymbol{c} = [(c_n)_{n\in\mathbb{N}}] \in \mathbb{Q}_p$. Then

$$\begin{aligned}\boldsymbol{a}\cdot(\boldsymbol{b}+\boldsymbol{c}) &= [(a_n)_{n\in\mathbb{N}}]\cdot([(b_n)_{n\in\mathbb{N}}] + [(c_n)_{n\in\mathbb{N}}])\\ &= [(a_n)_{n\in\mathbb{N}}]\cdot[(b_n+c_n)_{n\in\mathbb{N}}] = [(a_n(b_n+c_n))_{n\in\mathbb{N}}]\\ &= [(a_n b_n + a_n c_n)_{n\in\mathbb{N}}] = [(a_n b_n)_{n\in\mathbb{N}}] + [(a_n c_n)_{n\in\mathbb{N}}]\\ &= [(a_n)_{n\in\mathbb{N}}]\cdot[(b_n)_{n\in\mathbb{N}}] + [(a_n)_{n\in\mathbb{N}}]\cdot[(c_n)_{n\in\mathbb{N}}] = \boldsymbol{a}\cdot\boldsymbol{b} + \boldsymbol{a}\cdot\boldsymbol{c}.\end{aligned}$$

The map $a \mapsto [(a)_{n\in\mathbb{N}}] : \mathbb{Q} \to \mathbb{Q}_p$ is injective: If not, there exist distinct $a, b \in \mathbb{Q}$ be such that $[(a)_{n\in\mathbb{N}}] = [(b)_{n\in\mathbb{N}}]$, i.e., $(a)_{n\in\mathbb{N}} \sim (b)_{n\in\mathbb{N}}$, and so for the positive rational $\epsilon := \frac{1}{2}|a - b|_p$, there exists an $N \in \mathbb{N}$ such that for all $n > N$, we have $|a - b|_p < \epsilon := \frac{1}{2}|a - b|_p$, which is clearly false.

Chapter 10

Solution to Exercise 10.1

Let $\boldsymbol{z} = (x, y) \neq \mathbf{0} = (0, 0)$. Then at least one of x, y is nonzero. So $x^2 + y^2 \neq 0$. Thus $(\frac{x}{x^2+y^2}, \frac{-y}{x^2+y^2}) \in \mathbb{R}^2 = \mathbb{C}$. Moreover,

$$\begin{aligned}(x, y) \cdot (\tfrac{x}{x^2+y^2}, \tfrac{-y}{x^2+y^2}) &= (x\tfrac{x}{x^2+y^2} - y(\tfrac{-y}{x^2+y^2}),\ x(\tfrac{-y}{x^2+y^2}) + y\tfrac{x}{x^2+y^2}) \\ &= (\tfrac{x^2+y^2}{x^2+y^2},\ \tfrac{-xy+xy}{x^2+y^2}) = (1, 0).\end{aligned}$$

Hence for $\boldsymbol{z} \cdot (\boldsymbol{z}^{-1}) = \mathbf{1}$. Similarly, $\boldsymbol{z}^{-1} \cdot \boldsymbol{z} = \mathbf{1}$.

Solution to Exercise 10.2

We first prove some generalities that are valid in a field, which will be used below. We claim for all $x \in \mathbb{F}$, $0 \cdot x = 0$. We have $0 \cdot x = (0+0) \cdot x = 0 \cdot x + 0 \cdot x$, and adding the additive inverse of $0 \cdot x$ on both sides yields $0 = 0 \cdot x$. Next, we claim the additive inverse of an element is unique. If x_1, x_2 are additive inverses of x, then $x_1 = x_1 + 0 = x_1 + (x + x_2) = (x_1 + x) + x_2 = 0 + x_2 = x_2$. For any $x \in \mathbb{F}$, $-x = (-1) \cdot x$ as $x + (-1) \cdot x = 1 \cdot x + (-1) \cdot x = (1 + (-1)) \cdot x = 0 \cdot x = 0$.

(1) If $x < y$, then $y + (-x) \in P$. We have

$$\begin{aligned}(y + z) + (-(x + z)) &= (y + z) + (-1) \cdot (x + z) \\ &= (y + z) + ((-1) \cdot x + (-1) \cdot z) \\ &= y + z + ((-x) + (-z)) \\ &= y + (-x) + (z + (-z)) = y + (-x) \in P,\end{aligned}$$

and so $x + z < y + z$, i.e., (O1) holds.

Next, let $x < y$ and $0 < z$. Then $y + (-x) \in P$. Also, $z + (-0) \in P$, i.e., $z \in P$. Hence $z \cdot (y + (-x)) \in P$, i.e., $z \cdot y + z \cdot (-x) = z \cdot y + (-(z \cdot x)) \in P$, and so $z \cdot x < z \cdot y$, i.e., (O2) holds.

Let $x < y$ and $y < z$. Then $y + (-x) \in P$ and $z + (-y) \in P$. By (P1), we get $y + (-x) + z + (-y) \in P$, i.e., $z + (-x) \in P$. So $x < z$. Thus (O3) holds.

Let $x, y \in \mathbb{R}$. Consider $z = y + (-x) \in \mathbb{F}$. By (P3), exactly one of the following hold for z: 1° $z = 0$, 2° $z \in P$, 3° $-z \in P$.

1° If $z = 0$, then $y + (-x) = 0$, and so adding x to both sides gives $y = x$. Vice versa, if $y = x$, then adding $-x$ to both sides gives $y + (-x) = 0$ and so $z = 0$. Thus $z = 0 \Leftrightarrow x = y$.

2° If $z \in P$, then $y + (-x) \in P$, and so $x < y$. Vice versa, if $x < y$, then $y + (-x) \in P$, i.e., $z \in P$. Thus $z \in P \Leftrightarrow x < y$.

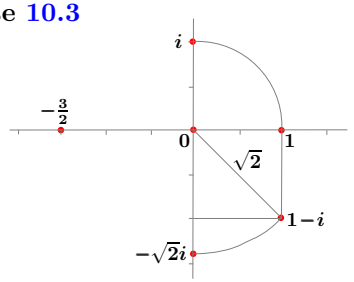
3° If $-z \in P$, then $-(y + (-x)) \in P$, i.e., $x + (-y) \in P$, and so $y < x$. Vice versa, if $y < x$, then $x + (-y) = -(y + (-x)) \in P$, i.e., $-z \in P$. Thus $-z \in P \Leftrightarrow y < x$.

So (O4) holds.

(2) Let $x, y \in P$, i.e., $0<x$ and $0<y$. By (O1), $y=0+y<x+y$. Together with $0<y$, (O3) now yields $0<x+y$, i.e., $x+y \in P$. So (P1) holds.
Again, let $x, y \in P$, i.e., $0<x$ and $0<y$. By (O2), $0=0\cdot y<x\cdot y$. So $x\cdot y \in P$. Hence (P2) holds.
Let $x \in \mathbb{F}$. By (O4), for the pair $x, 0 \in \mathbb{F}$, exactly one of the following is true: $1^\circ\ x=0$, $2^\circ\ 0<x$, $3^\circ\ x<0$.
2° If $0<x$, then $x \in P$. Vice versa, if $x \in P$, then $0<x$. So $0<x \Leftrightarrow x \in P$.
3° If $x<0$, then by (O1), $-x+x<0+(-x)$, i.e., $0<-x$, and so $-x \in P$. Vice versa, if $-x \in P$, then $0<-x$, and by (O1), $0+x<-x+x$, i.e., $x<0$. Hence $x<0 \Leftrightarrow -x \in P$.
So (P3) holds.

(3) Let $P \subset \mathbb{C}$ be a set of positive elements of $\mathbb{C}$. Since $\boldsymbol{i} \neq \mathbf{0}$, by (P3), either $\boldsymbol{i} \in P$ or ($\boldsymbol{i} \notin P$ and $-\boldsymbol{i} \in P$). In either case, by (P2), we have firstly $\mathbf{-1}=\boldsymbol{i}\cdot\boldsymbol{i}=(-\boldsymbol{i})\cdot(-\boldsymbol{i}) \in P$, and next also $\mathbf{1}=(\mathbf{-1})\cdot(\mathbf{-1}) \in P$. But $\mathbf{1}\neq\mathbf{0}$, and so we have a contradiction to (P3) for $\boldsymbol{x}=\mathbf{1}$.

Solution to Exercise 10.3



Solution to Exercise 10.4

For $\theta \in \mathbb{R}$, by de Moivre's formula, $(\cos\theta, \sin\theta)^3 = (\cos(3\theta), \sin(3\theta))$. But

$$\begin{aligned}(\cos\theta, \sin\theta)^3 &= (\cos\theta, \sin\theta)\cdot((\cos\theta)^2-(\sin\theta)^2, 2\cos\theta\sin\theta)\\ &= (\cos\theta((\cos\theta)^2-(\sin\theta)^2) - \sin\theta(2\cos\theta\sin\theta), *).\end{aligned}$$

Equating the real parts, we obtain

$$\begin{aligned}\cos(3\theta) &= \mathrm{Re}(\cos(3\theta), \sin(3\theta)) = \mathrm{Re}((\cos\theta, \sin\theta)^3)\\ &= \cos\theta((\cos\theta)^2-(\sin\theta)^2) - 2\cos\theta(\sin\theta)^2\\ &= \cos\theta((\cos\theta)^2-1+(\cos\theta)^2) - 2\cos\theta(1-(\cos\theta)^2)\\ &= (\cos\theta)^3-\cos\theta+(\cos\theta)^3-2\cos\theta+2(\cos\theta)^3\\ &= 4(\cos\theta)^3-3\cos\theta.\end{aligned}$$

Solution to Exercise 10.5

As $\mathbf{1}+\boldsymbol{i}=(\sqrt{2}\frac{1}{\sqrt{2}}, \sqrt{2}\frac{1}{\sqrt{2}}) = (\sqrt{2}\cos\frac{\pi}{4}, \sqrt{2}\sin\frac{\pi}{4})$, we have

$$\begin{aligned}(\mathbf{1}+\boldsymbol{i})^{10} &= (\sqrt{2}\cos\tfrac{\pi}{4}, \sqrt{2}\sin\tfrac{\pi}{4})^{10} = (\sqrt{2}^{10}\cos(10\tfrac{\pi}{4}), \sqrt{2}^{10}\sin(10\tfrac{\pi}{4}))\\ &= (2^5\cos(2\pi+\tfrac{\pi}{2}), 2^5\sin(2\pi+\tfrac{\pi}{2})) = (32\cos\tfrac{\pi}{2}, 32\sin\tfrac{\pi}{2}) = (0, 32) = \mathbf{32}\boldsymbol{i}.\end{aligned}$$

Solution to Exercise 10.6

The angle made by $\mathbf{2}+\boldsymbol{i}$ with the positive real axis is $\tan^{-1}\frac{1}{2}$, and the angle made by $\mathbf{3}+\boldsymbol{i}$ with the positive real axis is $\tan^{-1}\frac{1}{3}$. Thus the angle made by $(\mathbf{2}+\boldsymbol{i})\cdot(\mathbf{3}+\boldsymbol{i})$ with the positive real axis is $\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{3}$. As $(\mathbf{2}+\boldsymbol{i})\cdot(\mathbf{3}+\boldsymbol{i})=\mathbf{5}+\mathbf{5}\boldsymbol{i}$, the angle made by $(\mathbf{2}+\boldsymbol{i})\cdot(\mathbf{3}+\boldsymbol{i})$ with the positive real axis is $\tan^{-1}\frac{5}{5}=\tan^{-1}1=\frac{\pi}{4}$. Consequently, $\frac{\pi}{4}=\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{3}$.

Solution to Exercise 10.7

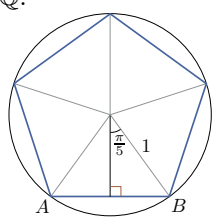
(1) Let $\alpha:=\sin\frac{\pi}{5}$. Then $\cos\frac{\pi}{5}=\sqrt{1-\alpha^2}$. By de Moivre's formula,

$$\begin{aligned}\mathbf{-1} &= (\cos 5\tfrac{\pi}{5}, i\sin 5\tfrac{\pi}{5}) = (\cos\tfrac{\pi}{5},\sin\tfrac{\pi}{5})^5 = (\sqrt{1-\alpha^2},\alpha)^5\\ &= (\sqrt{1-\alpha^2},\alpha)\cdot(1-2\alpha^2,2\alpha\sqrt{1-\alpha^2})^2\\ &= (\sqrt{1-\alpha^2},\alpha)\cdot(1-8\alpha^2+8\alpha^4,4\alpha(1-2\alpha^2)\sqrt{1-\alpha^2})\\ &= (*,5\alpha-20\alpha^3+16\alpha^5).\end{aligned}$$

Equating imaginary parts, we obtain $16\alpha^5-20\alpha^3+5\alpha=0$. Since $\alpha\neq 0$, we get $16\alpha^4-20\alpha^2+5=0$, i.e., $(2\alpha)^4-5(2\alpha)^2+5=0$. Consequently, $p(2\alpha)=0$, where $p:=x^4-5x^2+5\in\mathbb{Z}[x]$.

(2) If $m,n\in\mathbb{Z}$ are such that $n\neq 0$, and $2\alpha=\frac{m}{n}$, then by the rational zeros theorem, m divides 5 and n divides 1. Hence $2\alpha=\frac{m}{n}\in\{-1,1,-5,5\}$. But $p(\pm 1)=1-5+5=1\neq 0$ and $p(\pm 5)=5^4-5\cdot 5^2+5=5^3\cdot 4+5>0$. So 2α cannot be rational.

(3) We have $AB=2\sin\frac{\pi}{5}\notin\mathbb{Q}$.



Solution to Exercise 10.8

If the vertices A,B,C of an equilateral triangle are at the complex numbers $\boldsymbol{z_A},\boldsymbol{z_B},\boldsymbol{z_C}$, and they are labelled in the anticlockwise fashion, then since $AC=AB$ and $\angle CAB=\frac{\pi}{3}$, we have

$$\boldsymbol{z_C}-\boldsymbol{z_A}=(\cos\tfrac{\pi}{3},\sin\tfrac{\pi}{3})\cdot(\boldsymbol{z_B}-\boldsymbol{z_A}). \qquad (*)$$

Suppose $p,q,m,n\in\mathbb{Z}$ are such that $\boldsymbol{z_C}-\boldsymbol{z_A}=(p,q)$, and $\boldsymbol{z_B}-\boldsymbol{z_A}=(m,n)$. Then $(*)$ becomes $(p,q)=(\frac{1}{2},\frac{\sqrt{3}}{2})\cdot(m,n)$, i.e., $p=\frac{m}{2}-\frac{\sqrt{3}}{2}n$, and $q=\frac{m\sqrt{3}}{2}+\frac{n}{2}$. Multiplying the expression for p by $-n$, the expression for q by m, and adding, we get $qm-pn=\frac{\sqrt{3}}{2}(m^2+n^2)$. But $m^2+n^2\neq 0$ (as $\boldsymbol{z_B}\neq\boldsymbol{z_A}$), and so $\sqrt{3}=\frac{2(qm-pn)}{m^2+n^2}\in\mathbb{Q}$, a contradiction.

Solution to Exercise 10.9

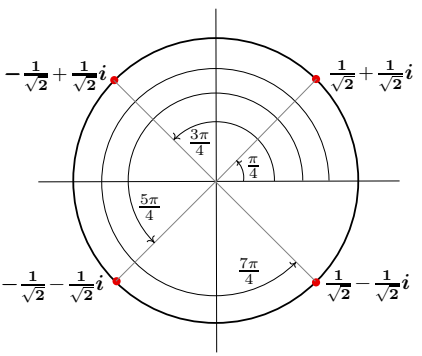
Let $\boldsymbol{w} = (\rho\cos\alpha, \rho\sin\alpha)$, where $\rho \geqslant 0$ and $\alpha \in \mathbb{R}$. Then

$$(\rho^4\cos(4\alpha), \rho^4\sin(4\alpha)) = \boldsymbol{w}^4 = -\mathbf{1} = (1\cos\pi, 1\sin\pi).$$

So $\rho^4 = 1$. As $\rho \geqslant 0$, $\rho = 1$. Thus $\boldsymbol{w} = (\rho\cos\alpha, \rho\sin\alpha) = (1\cos\alpha, 1\sin\alpha)$ belongs to the set

$$\begin{aligned}&\{(\cos\tfrac{\pi}{4}, \sin\tfrac{\pi}{4}), (\cos\tfrac{3\pi}{4}, \sin\tfrac{3\pi}{4}), (\cos\tfrac{5\pi}{4}, \sin\tfrac{5\pi}{4}), (\cos\tfrac{7\pi}{4}, \sin\tfrac{7\pi}{4})\}\\ = &\{\tfrac{\mathbf{1}}{\sqrt{\mathbf{2}}} + \tfrac{\mathbf{1}}{\sqrt{\mathbf{2}}}\boldsymbol{i}, -\tfrac{\mathbf{1}}{\sqrt{\mathbf{2}}} + \tfrac{\mathbf{1}}{\sqrt{\mathbf{2}}}\boldsymbol{i}, -\tfrac{\mathbf{1}}{\sqrt{\mathbf{2}}} - \tfrac{\mathbf{1}}{\sqrt{\mathbf{2}}}\boldsymbol{i}, \tfrac{\mathbf{1}}{\sqrt{\mathbf{2}}} - \tfrac{\mathbf{1}}{\sqrt{\mathbf{2}}}\boldsymbol{i}\}.\end{aligned}$$

The location in the complex plane of the four 4th roots of $-\mathbf{1}$ is shown below.

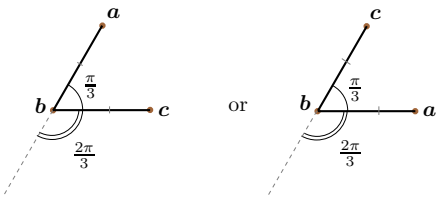


Solution to Exercise 10.10

Let $\boldsymbol{\omega} \in \mathbb{C}\backslash\mathbb{R}$ be such that $\boldsymbol{\omega}^3 = \mathbf{1}$. Then $\mathbf{0} = \boldsymbol{\omega}^3 - \mathbf{1} = (\boldsymbol{\omega} - \mathbf{1})\cdot(\boldsymbol{\omega}^2 + \boldsymbol{\omega} + \mathbf{1})$, and since $\boldsymbol{\omega} \neq \mathbf{1}$, we have $\boldsymbol{\omega}^2 + \boldsymbol{\omega} + \mathbf{1} = \mathbf{0}$. Hence

$$\begin{aligned}&((\boldsymbol{b}-\boldsymbol{a})\cdot\boldsymbol{\omega} + (\boldsymbol{b}-\boldsymbol{c}))\cdot((\boldsymbol{b}-\boldsymbol{a})\cdot\boldsymbol{\omega}^2 + (\boldsymbol{b}-\boldsymbol{c}))\\ = &\,(\boldsymbol{b}-\boldsymbol{a})^2\cdot\boldsymbol{\omega}^3 + (\boldsymbol{b}-\boldsymbol{a})\cdot(\boldsymbol{b}-\boldsymbol{c})\cdot(-\mathbf{1}) + (\boldsymbol{b}-\boldsymbol{c})^2\\ = &\,(\boldsymbol{b}-\boldsymbol{a})^2\cdot\mathbf{1} + (\boldsymbol{b}-\boldsymbol{a})\cdot(\boldsymbol{b}-\boldsymbol{c})\cdot(-\mathbf{1}) + (\boldsymbol{b}-\boldsymbol{c})^2\\ = &\,(\boldsymbol{b}-\boldsymbol{a})\cdot(\boldsymbol{b}-\boldsymbol{a}-\boldsymbol{b}+\boldsymbol{c}) + (\boldsymbol{b}-\boldsymbol{c})^2\\ = &\,(\boldsymbol{b}-\boldsymbol{a})\cdot(\boldsymbol{c}-\boldsymbol{a}) + (\boldsymbol{b}-\boldsymbol{c})^2\\ = &\,\boldsymbol{b}\cdot\boldsymbol{c} - \boldsymbol{c}\cdot\boldsymbol{a} - \boldsymbol{a}\cdot\boldsymbol{b} + \boldsymbol{a}^2 + \boldsymbol{b}^2 - \mathbf{2}\cdot\boldsymbol{b}\cdot\boldsymbol{c} + \boldsymbol{c}^2\\ = &\,\boldsymbol{a}^2 + \boldsymbol{b}^2 + \boldsymbol{c}^2 - \boldsymbol{a}\cdot\boldsymbol{b} - \boldsymbol{b}\cdot\boldsymbol{c} - \boldsymbol{c}\cdot\boldsymbol{a} = \mathbf{0}.\end{aligned}$$

Hence $(\boldsymbol{b}-\boldsymbol{a})\cdot\boldsymbol{\omega} = \boldsymbol{c}-\boldsymbol{b}$ or $(\boldsymbol{b}-\boldsymbol{a})\cdot\boldsymbol{\omega}^2 = \boldsymbol{c}-\boldsymbol{b}$. But the latter case is the same as $(\boldsymbol{b}-\boldsymbol{a})\cdot\boldsymbol{\omega}^3 = (\boldsymbol{c}-\boldsymbol{b})\cdot\boldsymbol{\omega}$, i.e., $(\boldsymbol{c}-\boldsymbol{b})\cdot\boldsymbol{\omega} = \boldsymbol{b}-\boldsymbol{a}$. In either of these cases, it follows that the distance of $\boldsymbol{b}-\boldsymbol{a}$ to the origin is the same as the distance of $\boldsymbol{c}-\boldsymbol{b}$ to the origin, and moreover, the angle between the line segments joining $\boldsymbol{a}$ to $\boldsymbol{b}$ and $\boldsymbol{c}$ to $\boldsymbol{b}$ is $\frac{\pi}{3}$. So the triangle formed by $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$ is equilateral. See the following picture.



If $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$ are all real, then the equilateral triangle must degenerate to a point $r \in \mathbb{R}$, and so $\boldsymbol{a} = \boldsymbol{b} = \boldsymbol{c}$ $(= \boldsymbol{r})$. Thus we recover the real case result.

Solution to Exercise 10.11

For $k \in \mathbb{N}$, $(z - z_0)(z^{k-1} + z^{k-2}z_0 + \cdots + zz_0^{k-2} + z_0^{k-1})$ when expanded 'telescopes' to give $z^k - z_0^k$. We have:

$$\begin{aligned} p(z) &= c_0 + c_1 z + c_2 z^2 + \cdots + c_d z^d \\ p(z_0) &= c_0 + c_1 z_0 + c_2 z_0^2 + \cdots + c_d z_0^d \\ \hline p(z) - p(z_0) &= c_1(z - z_0) + c_2(z^2 - z_0^2) + \cdots + c_d(z^d - z_0^d). \end{aligned}$$

With $q(z) := c_1 + c_2(z + z_0) + \cdots + c_d(z^{d-1} + z^{k-2}z_0 + \cdots + zz_0^{k-2} + z_0^{k-1}) \in \mathbb{C}[z]$, we have $p(z) - p(z_0) = (z - z_0)q(z)$, i.e., $p(z) = (z - z_0)q(z) + p(z_0)$. As $c_d \neq 0$, q has degree $d - 1$. If $p(z_0) = 0$, then $p(z) = (z - z_0)q(z) + 0 = (z - z_0)q(z)$.

If $p(z) := z^n - 1$, then since $p(z_1) = 0$, we have $p(z) = (z - z_1)q_1(z)$ for some polynomial q_1 of degree $n-1$. But $0 = p(z_2) = (z_2 - z_1)q_1(z_2)$, and since $z_2 \neq z_1$, we get $q_1(z_2) = 0$. So $q_1(z) = (z - z_2)q_2(z)$ for some polynomial q_2 of degree $n-2$. Hence $p(z) = (z - z_1)(z - z_2)q_2(z)$. Proceeding in this manner, we eventually obtain $p(z) = (z - z_1)(z - z_2)\cdots(z - z_{n-1})q_{n-1}(z)$, for some $q_{n-1} \in \mathbb{C}[z]$ of degree 1. Moreover, $q_{n-1}(z_n) = 0$, so that $q_{n-1}(z) = C(z - z_n)$ for some constant C. But

$$\begin{aligned} -1 = 0^n - 1 = p(0) &= C(0 - z_1)\cdots(0 - z_n) = C(-1)^n z_1 \cdots z_n \\ &= C(-1)^n \prod_{k=0}^{n-1} (\cos(\tfrac{2\pi}{n} + k\tfrac{2\pi}{n}) + i\sin(\tfrac{2\pi}{n} + k\tfrac{2\pi}{n})) \\ &= C(-1)^n(\cos(2\pi + (n-1)\pi) + \sin(2\pi + (n-1)\pi)) \\ &= C(-1)^n(-1)^{n-1} = -C, \end{aligned}$$

and so $C = 1$.

Solution to Exercise 10.12

In particular, for all $x \in \mathbb{R}$, $c_0 + c_1 x + \cdots + c_d x^d = 0$. Setting $x = 0$, we obtain $c_0 = 0$. Suppose for some $n \in \{0, 1, \cdots, d\}$ we have shown $c_0 = \cdots = c_n = 0$. Hence for all $x \in \mathbb{R}$, $c_{n+1}x^{n+1} + \cdots + c_d x^d = 0$. Differentiating $n+1$ times with respect to x, and then setting $x = 0$, we obtain $c_{n+1}(n+1)! + 0 = 0$, and so $c_{n+1} = 0$. By induction, it follows that $c_0 = \cdots = c_d = 0$.

Solution to Exercise 10.13

First we show uniqueness. Let $p = qh + r = \widetilde{q}h + \widetilde{r}$ for polynomials $q, \widetilde{q}, r, \widetilde{r}$, where $r, \widetilde{r}$ each have a degree strictly less than d. Then $(q - \widetilde{q})h = \widetilde{r} - r$. If $q \neq \widetilde{q}$, then the left-hand side has degree at least d, while the right-hand side has degree strictly less than d, a contradiction. So $q = \widetilde{q}$. This then yields $\widetilde{r} - r = 0h = 0$, and so $\widetilde{r} = r$ as well.

We use induction on the degree of p to show existence. If p is constant, then with $q := 0$ and $r := p(0)$, $p(z) = p(0) = 0h(z) + p(0) = q(z)h(z) + r(z)$. Suppose for some $n \in \mathbb{N} \cup \{0\}$, the claim is true for all polynomials p of degree $\leqslant n$. Let p be a polynomial of degree $n + 1$ ($\geqslant 0 + 1 = 1$). By the division algorithm for integers, $n + 1 = dk + \ell$, for integers k, ℓ, where $0 \leqslant \ell < d$. We note that if $k < 0$, then $\ell = n + 1 + d(-k) \geqslant 1 + d(1) > d$, a contradiction. So $k \geqslant 0$. Let $p(z) = c_{n+1}z^{n+1} + \cdots + c_0$, and $h(z) = a_d z^d + \cdots + a_0$. Set $h^0 := 1$ and $a_d^0 := 1$. The coefficient of the leading term of $\widetilde{p}$ given by

$$\widetilde{p} := \frac{c_{n+1}}{a_d^k} h^k z^\ell$$

is c_{n+1}, and so $p - \widetilde{p}$ has degree $\leqslant n$. By the induction hypothesis, there exist polynomials $\widetilde{q}, \widetilde{r}$ such that $p - \widetilde{p} = h\widetilde{q} + \widetilde{r}$, and such that the degree of $\widetilde{r}$ is strictly less than d.

- If $k = 0$, then $q := \widetilde{q}$, and absorb $\widetilde{p} = c_{n+1}z^\ell$ in $\widetilde{r}$, i.e., $r := \widetilde{r} + \widetilde{p}$.
- If $k > 0$, then $r := \widetilde{r}$, and $q := \widetilde{q} + \dfrac{c_{n+1}}{a_d^k} h^{k-1} z^\ell$.

In either case, $p = qh + r$, for suitable polynomials $q, r \in \mathbb{C}[z]$, with the polynomial r having degree $< d$. This completes the induction step.

Solution to Exercise 10.14

Let $z_1 = x_1 + iy_1$, and $z_2 = x_2 + iy_2$, where $x_1, x_2, y_1, y_2 \in \mathbb{R}$.
Then $z_1 z_2 = x_1 x_2 - y_1 y_2 + i(x_1 y_2 + y_1 x_2)$, and

$$\begin{aligned} |z_1 z_2|^2 &= (x_1 x_2 - y_1 y_2)^2 + (x_1 y_2 + y_1 x_2)^2 \\ &= x_1^2 x_2^2 - 2x_1 x_2 y_1 y_2 + y_1^2 y_2^2 + x_1^2 y_2^2 + 2x_1 y_2 y_1 x_2 + y_1^2 x_2^2 \\ &= x_1^2(x_2^2 + y_2^2) + y_1^2(y_2^2 + x_2^2) \\ &= (x_1^2 + y_1^2)(x_2^2 + y_2^2) = |z_1|^2 |z_2|^2. \end{aligned}$$

Since $|z_1|, |z_2|, |z_1 z_2|$ are all nonnegative, it follows that $|z_1 z_2| = |z_1||z_2|$.

Solution to Exercise 10.15

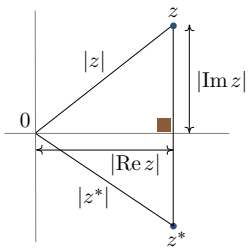
Let $z = x + iy$, where $x, y \in \mathbb{R}$. Then $(z^*)^* = (x - iy)^* = x - i(-y) = x + iy = z$.
Moreover, $zz^* = (x + iy)(x - iy) = x^2 + y^2 + i(-xy + xy) = x^2 + y^2 = |z|^2$, and
$\dfrac{z + z^*}{2} = \dfrac{x + iy + x - iy}{2} = x = \operatorname{Re} z$, and $\dfrac{z - z^*}{2i} = \dfrac{x + iy - x + iy}{2i} = \dfrac{2iy}{2i} = y = \operatorname{Im} z$.

Solution to Exercise 10.16

Let $z = x + iy$, where $x, y \in \mathbb{R}$. Then

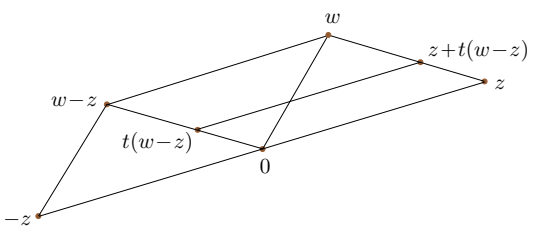
$$|z| = |x + iy| = \sqrt{x^2 + y^2} = \sqrt{x^2 + (-y)^2} = |x - iy| = |z^*|,$$
$$|\mathrm{Re}\, z| = |x| = \sqrt{x^2} = \sqrt{x^2 + 0} \leqslant \sqrt{x^2 + y^2} = |x + iy| = |z|,$$
$$|\mathrm{Im}\, z| = |y| = \sqrt{y^2} = \sqrt{0 + y^2} \leqslant \sqrt{x^2 + y^2} = |x + iy| = |z|.$$

Since z^* is the reflection of z in the real axis, and as $0 \in \mathbb{R}$, the distance of z to 0 is equal to the distance of z^* to 0, i.e., $|z| = |z^*|$. The inequalities $|\mathrm{Re}\, z| \leqslant |z|$ and $|\mathrm{Im}\, z| \leqslant |z|$ express the fact that the length of any side in a right-angled triangle is at most the length of the hypotenuse.



Solution to Exercise 10.17

In the picture below, $0, z, w, w - z$ form a parallelogram. So when we form the parallelogram determined by $0, z, t(w - z)$, the fourth vertex, namely $z + t(w - z)$, lies on the segment joining z and w (as the line joining 0 and $t(w - z)$ is parallel to the line joining z and w). As the t increases from 0 to 1, $z + t(w - z)$ moves from z to w along the segment joining them.



Solution to Exercise 10.18

As $p(w) = 0$, we get $(c_0 + c_1 w + \cdots + c_d w^d)^* = (p(w))^* = 0^* = 0$. We have

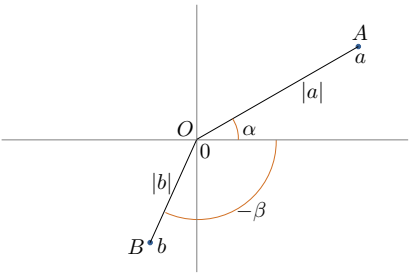
$$\begin{aligned} 0 &= (c_0 + c_1 w + \cdots + c_d w^d)^* = c_0^* + (c_1 w)^* + \cdots + (c_d w^d)^* \\ &= c_0^* + c_1^* w^* + \cdots + c_d^* (w^d)^* = c_0 + c_1 w^* + \cdots + c_d (w^*)^d \end{aligned}$$

(using the facts that $c_0, c_1, \cdots, c_d \in \mathbb{R}$, and $(w^k)^* = (w^*)^k$ for all $1 \leqslant k \leqslant d$). Hence $0 = c_0 + c_1 w^* + \cdots + c_d (w^*)^d = p(w^*)$.

Solution to Exercise 10.19

Let $a = |a|(\cos\alpha + i\sin\alpha)$, and $b = |b|(\cos\beta + i\sin\beta)$, where $\alpha, \beta \in (-\pi, \pi]$. Then $ab^* = |a|(\cos\alpha + i\sin\alpha)|b|(\cos\beta - i\sin\beta)$. So

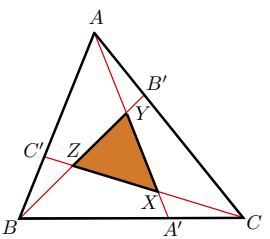
$$\mathrm{Im}(ab^*) = |a||b|((\cos\alpha)(-\sin\beta) + (\sin\alpha)(\cos\beta)) = |a||b|\sin(\alpha - \beta).$$



The area of the triangle OAB formed by $0, a, b$ (where $O \equiv 0, A \equiv a, B \equiv b$) is $\frac{1}{2}OA{\cdot}OB\sin\angle AOB = \frac{1}{2}|a||b||\sin(\alpha-\beta)| = \frac{1}{2}|\mathrm{Im}(ab^*)| = |\frac{\mathrm{Im}(ab^*)}{2}|$.

Solution to Exercise 10.20

In ΔABC, the points C', A', B' on AB, BC, CA, respectively, are such that $AC' : C'B = BA' : A'C = CB' : B'A = 2:1$. Join AA', BB', CC', and label the intersection points in the line segment pairs (CC', AA'), (AA', BB'), (BB', CC') as X, Y, Z, respectively. We need the ratio of the areas of the internal triangle ΔXYZ to that of ΔABC.



Suppose A, B, C correspond to the complex numbers $z, 0, w$, respectively. Then C', A', B' correspond to $\frac{z}{3}, \frac{2w}{3}, w + \frac{2}{3}(z - w) = \frac{2z + w}{3}$, respectively. As Y lies on the intersection of AA' and BB', for some $s, t \in [0, 1]$, we get

$$t\frac{2z + w}{3} = (1-s)z + s\frac{2w}{3},$$

i.e., $(2t - 3(1 - s))z = (2s - t)w$. Since $0, z, w$ are not collinear, it follows that $2t = 3(1 - s)$ and $t = 2s$. Solving, we get $s = \frac{3}{7}$. Hence Y corresponds to the complex number $\frac{4z + 2w}{7}$. Noticing the symmetry of the situation (i.e., by 'shifting the origin' to z and then to w), we can read off from the above that X, Z correspond to the complex numbers $\frac{4(w - z) + 2(-z)}{7} + z = \frac{4w + z}{2}$,

and $\frac{4(-w)+2(z-w)}{7}+w=\frac{w+2z}{7}$, respectively. Shifting the origin to say Z, we have that X, Y then correspond to $a=\frac{-z+3w}{7}$, $b=\frac{2z+w}{7}$, and so the area of the internal triangle, ΔXYZ, will be

$$\begin{aligned}|\tfrac{1}{2}\mathrm{Im}(\tfrac{(2z+w)}{7}\tfrac{(-z^*+3w^*)}{7})| &= |\tfrac{1}{2}\tfrac{1}{7(7)}\mathrm{Im}(-2|z|^2+3|w|^2-wz^*+6zw^*)| \\ &= |\tfrac{1}{2}\tfrac{1}{7(7)}\mathrm{Im}(-wz^*-w^*z+7zw^*)| \\ &= \tfrac{1}{7}\tfrac{|\mathrm{Im}(zw^*)|}{2}.\end{aligned}$$

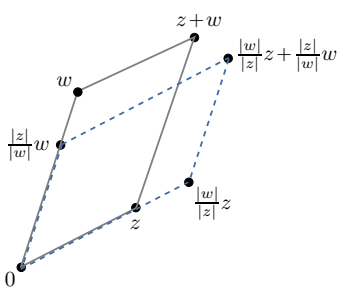
The area of ΔABC is $\frac{|\mathrm{Im}(zw^*)|}{2}$, and so the above shows that the area of ΔXYZ is one-seventh the area of ΔABC.

Solution to Exercise 10.21

The claim is a consequence of the following:

$$\begin{aligned}|\tfrac{|w|}{|z|}z+\tfrac{|z|}{|w|}w|^2 &= (\tfrac{|w|}{|z|}z+\tfrac{|z|}{|w|}w)(\tfrac{|w|}{|z|}z^*+\tfrac{|z|}{|w|}w^*) \\ &= \tfrac{|w|^2}{|z|^2}|z|^2+\tfrac{|z|^2}{|w|^2}|w|^2+\tfrac{|w|}{|z|}z\tfrac{|z|}{|w|}w^*+\tfrac{|z|}{|w|}w\tfrac{|w|}{|z|}z^* \\ &= |w|^2+|z|^2+zw^*+wz^* \\ &= (w+z)(w^*+z^*)=|w+z|^2.\end{aligned}$$

As $\frac{|w|}{|z|}z$ is a positive multiple of z, it lies on the ray joining 0 to z. The distance of $\frac{|w|}{|z|}z$ from the origin is $|w|$, which is one of the lengths of the sides of the parallelogram drawn with solid lines below. Similarly, $\frac{|z|}{|w|}w$ lies on the ray through 0 and w, and is at a distance of $|z|$ from 0, which is the other length of the side of the parallelogram drawn with solid lines. The included angle between $\frac{|w|}{|z|}z$ and $\frac{|z|}{|w|}w$ is the same as the angle between z and w. So the parallelogram drawn with solid lines is congruent to the parallelogram drawn with dashed lines, and in particular, the lengths of the corresponding diagonals are the same, giving $|z+w|=|\frac{|w|}{|z|}z+\frac{|z|}{|w|}w|$.



Solution to Exercise 10.22

If $z, w \in S^1$, then $|zw| = |z||w| = 1(1) = 1$, and so we have that $zw \in S^1$. Complex multiplication is associative and commutative. Also, $1 \in S^1$, and $z1 = z = 1z$ for all $z \in S^1$. If $z \in S^1$, then $z \neq 0$ and $\frac{1}{z} \in S^1$ too, because $|\frac{1}{z}| = \frac{1}{|z|} = \frac{1}{1} = 1$, and so every element in S^1 has a multiplicative inverse in S^1. Hence S^1 forms an Abelian group.

For $n \in \mathbb{N}$, let $\zeta = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$. If $m \in \mathbb{N}$ and $\zeta^m = 1$, then $\cos\frac{2\pi m}{n} = 1$, so that $(0 \neq)\frac{2\pi m}{n} = 2\pi k$ for some $k \in \mathbb{Z}\backslash\{0\}$, showing that n divides m. Thus $m \geqslant n$. So $\zeta, \cdots, \zeta^{n-1}$ are distinct and not 1. Also, $\zeta^n = 1$. Then $H_n = \{1, \zeta, \cdots, \zeta^{n-1}\}$ is a subgroup of S^1 with n distinct elements.

Let $H = \{\cos(\sqrt{2}\pi m) + i\sin(\sqrt{2}\pi m) : m \in \mathbb{Z}\}$. Then $1 = \cos 0 + i\sin 0 \in H$ and $H \subset S^1$. If $z, w \in H$, then there exist $m, n \in \mathbb{Z}$ such that

$$z = \cos(\sqrt{2}\pi m) + i\sin(\sqrt{2}\pi m) \quad \text{and} \quad w = \cos(\sqrt{2}\pi n) + i\sin(\sqrt{2}\pi n),$$

and so $zw = \cos(\sqrt{2}\pi(m+n)) + i\sin(\sqrt{2}\pi(m+n)) \in H$, as $m+n \in \mathbb{Z}$. Finally, if $z = \cos(\sqrt{2}\pi m) + i\sin(\sqrt{2}\pi m) \in H$, where $m \in \mathbb{Z}$, then we have that $z^{-1} = \cos(\sqrt{2}\pi(-m)) + i\sin(\sqrt{2}\pi(-m)) \in H$ too. The map $\varphi : \mathbb{Z} \to H$ given by $\mathbb{Z} \ni m \mapsto \cos(\sqrt{2}\pi m) + i\sin(\sqrt{2}\pi m) \in H$ is surjective, and injective (if for distinct $m, n \in \mathbb{Z}$, $\cos(\sqrt{2}\pi m) + i\sin(\sqrt{2}\pi m) = \cos(\sqrt{2}\pi n) + i\sin(\sqrt{2}\pi n)$, then $\sqrt{2}\pi m = \sqrt{2}\pi n + 2\pi k$ for some integer k, giving $\sqrt{2} = \frac{2k}{m-n} \in \mathbb{Q}$, a contradiction). Thus φ is a bijection. As $\mathbb{Z}$ is countable, so is H. (There are of course many other examples of infinite countable subgroups too.)

Solution to Exercise 10.23

Let $S = z_1 + \cdots + z_n$. If $\zeta = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$, then we have that the set $\{\zeta z_1, \cdots, \zeta z_n\} = \{z_1, \cdots, z_n\}$. Thus $S = z_1 + \cdots + z_n = \zeta z_1 + \cdots + \zeta z_n = \zeta S$, i.e., $(\zeta - 1)S = 0$. But as $n \geqslant 2$, $\zeta \neq 1$ (otherwise $\frac{2\pi}{n} = 2\pi k$ for some integer k, giving $1 = nk$ for integers n, k with $n \geqslant 2$, which is absurd). So $S = 0$.

Alternatively, by Exercise 10.11, p.115, $z^n - 1 = (z - z_1)\cdots(z - z_n)$, i.e., $z^n - 1 = z^n - (z_1 + \cdots + z_n)z^{n-1} + \cdots + (-1)^n z_1 \cdots z_n$, and so by 'comparing coefficients' of z^{n-1} (see Exercise 10.12, p.115), $z_1 + \cdots + z_n = 0$.

Solution to Exercise 10.24

For $z_1, z_2 \in \mathbb{C}$, $|z_1| = |z_1 - z_2 + z_2| \leqslant |z_1 - z_2| + |z_2|$. So $|z_1| - |z_2| \leqslant |z_1 - z_2|$. As this holds for *all* complex numbers z_1, z_2, by swapping z_1 and z_2, we also obtain $|z_2| - |z_1| \leqslant |z_2 - z_1| = |-(z_1 - z_2)| = |-1||z_1 - z_2| = |z_1 - z_2|$. Thus $||z_1| - |z_2|| \leqslant |z_1 - z_2|$.

Solution to Exercise 10.25

Let $C = \{z \in \mathbb{C} : |z|^2 - 2\mathrm{Re}(a^* z) + |a|^2 = r^2\}$. We have $z \in C$ if and only if $r^2 = |z|^2 - 2\mathrm{Re}(a^* z) + |a|^2 = zz^* - (a^* z + az^*) + aa^* = z(z^* - a^*) - a(z^* - a^*)$.

Thus $z \in C$ if and only if $r^2 = (z-a)((z-a)^*) = |z-a|^2$. So C is the set of all points $z \in \mathbb{C}$, whose distance $|z-a|$ to a is r. Thus C is a circle with centre a and radius r.

Solution to Exercise 10.26

We have

$$\begin{aligned}|z_1+z_2|^2+|z_1-z_2|^2 &= (z_1+z_2)(z_1^*+z_2^*) + (z_1-z_2)(z_1^*-z_2^*)\\ &= |z_1|^2 + z_1z_2^* + z_2z_1^* + |z_2|^2 + |z_1|^2 - z_1z_2^* - z_2z_1^* + |z_2|^2\\ &= 2(|z_1|^2+|z_2|^2).\end{aligned}$$

Consider the parallelogram P with vertices at $0, z_1, z_2, z_1+z_2$ in the complex plane. Then $|z_1+z_2|$ denotes the length of one diagonal of P, while $|z_1-z_2|$ is the length of the other diagonal of P. Also, $|z_1|, |z_2|$ are the lengths of the sides of P. So the above equality says:

In a parallelogram, the sum of the squares of the lengths of the diagonals equals the sum of the squares of the lengths of all the sides.

Solution to Exercise 10.27

Let z, w make angles α, β, respectively, with the positive x-axis in the counterclockwise direction. Then

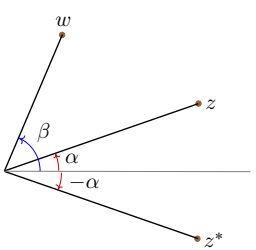
$$\begin{aligned}z^* &= (|z|(\cos\alpha + i\sin\alpha))^*\\ &= |z|(\cos\alpha - i\sin\alpha)\\ &= |z|(\cos(-\alpha) + i\sin(-\alpha)).\end{aligned}$$

So

$$\begin{aligned}z^*w &= |z|(\cos(-\alpha)+i\sin(-\alpha))|w|(\cos\beta + i\sin\beta)\\ &= |z||w|(\cos(\beta-\alpha)+i\sin(\beta-\alpha)).\end{aligned}$$

Thus $\mathrm{Re}(z^*w) = |z||w|\cos(\beta-\alpha) = (OA)(OB)\cos\angle AOB$. Hence

$$\begin{aligned}AB^2 &= |z-w|^2 = (z-w)(z^*-w^*)\\ &= |z|^2 + |w|^2 - z^*w - zw^*\\ &= |z|^2 + |w|^2 - 2\mathrm{Re}(z^*w)\\ &= OA^2 + OB^2 - 2(OA)(OB)\cos\angle AOB.\end{aligned}$$



Solution to Exercise 10.28

Let $(u,v)\in\mathbb{C}$. We seek a $\boldsymbol{p}=(x,y,z)\in S^2\backslash\{\boldsymbol{n}\}$ such that $\varphi(\boldsymbol{p})=(u,v)$, i.e., $(\frac{x}{1-z},\frac{y}{1-z})=(u,v)$. For such a triple (x,y,z), we have

$$u^2+v^2=\frac{x^2}{(1-z)^2}+\frac{y^2}{(1-z)^2}=\frac{x^2+y^2}{(1-z)^2}=\frac{1-z^2}{(1-z)^2}=\frac{1+z}{1-z},$$

and so $z=\dfrac{u^2+v^2-1}{u^2+v^2+1}$. Also,

$$x=u(1-z)=u(1-\frac{u^2+v^2-1}{u^2+v^2+1})=\frac{2u}{u^2+v^2+1}\quad\text{and}\quad y=v(1-z)=\frac{2v}{u^2+v^2+1}.$$

So we define $\boldsymbol{p}=(x,y,z)=(\dfrac{2u}{u^2+v^2+1},\dfrac{2v}{u^2+v^2+1},\dfrac{u^2+v^2-1}{u^2+v^2+1})$. Then

$$x^2+y^2+z^2=\frac{4u^2+4v^2+u^4+v^4+1+2u^2v^2-2u^2-2v^2}{(u^2+v^2+1)^2}=\frac{(u^2+v^2+1)^2}{(u^2+v^2+1)^2}=1.$$

Thus $\boldsymbol{p}\in S^2$. Also $\boldsymbol{p}\neq\boldsymbol{n}$, since otherwise

$$(\frac{2u}{u^2+v^2+1},\frac{2v}{u^2+v^2+1},\frac{u^2+v^2-1}{u^2+v^2+1})=(0,0,1),$$

and so from the first two entries, $u=0$ and $v=0$, implying

$$\frac{u^2+v^2-1}{u^2+v^2+1}=-1\neq 1,$$

a contradiction. So $\boldsymbol{p}\in S^2\backslash\{\boldsymbol{n}\}$. Finally,

$$\varphi(\boldsymbol{p})=\frac{1}{1-\frac{u^2+v^2-1}{u^2+v^2+1}}(\frac{2u}{u^2+v^2+1},\frac{2v}{u^2+v^2+1})=(u,v).$$

Hence $\varphi^{-1}(u,v)=(\dfrac{2u}{u^2+v^2+1},\dfrac{2v}{u^2+v^2+1},\dfrac{u^2+v^2-1}{u^2+v^2+1})$.

Let $w=(u,v)$ and $w'=(u',v')$. Let A,A',a,a' be the lengths in $\mathbb{R}^3$ of the segments joining $\boldsymbol{n}$ to $w,w',\varphi^{-1}w,\varphi^{-1}w'$, respectively. Then we have $A^2=u^2+v^2+1=|w|^2+1$ and $A'^2=u'^2+v'^2+1=|w'|^2+1$.

Considering the similar triangles shown on p.119, we obtain

$$\frac{a}{A}=\frac{1-z}{1},$$

where $z=\dfrac{u^2+v^2-1}{u^2+v^2+1}=\dfrac{|w|^2-1}{|w|^2+1}$ (z coordinate of $\varphi^{-1}w\in S^2\backslash\{\boldsymbol{n}\}$). Similarly,

$$\frac{a'}{A'}=\frac{1-z'}{1},$$

where $z'=\dfrac{u'^2+v'^2-1}{u'^2+v'^2+1}=\dfrac{|w'|^2-1}{|w'|^2+1}$. Thus $a=\dfrac{2}{\sqrt{1+|w|^2}}$ and $a'=\dfrac{2}{\sqrt{1+|w'|^2}}$.

The triangles formed by $\boldsymbol{n},\varphi^{-1}w,\varphi^{-1}w'$ and by $\boldsymbol{n},w,w'$ share the angle θ between the ray from $\boldsymbol{n}$ to w, and the ray from $\boldsymbol{n}$ to w'. The cosine formula gives $|w-w'|^2=A^2+A'^2-2AA'\cos\theta$ and $d_c(w,w')^2=a^2+a'^2-2aa'\cos\theta$.

Thus

$$\begin{aligned} d_c(w,w')^2 &= a^2 + a'^2 - 2aa'\frac{A^2 + A'^2 - |w-w'|^2}{2AA'} \\ &= \frac{4}{|w|^2+1} + \frac{4}{|w'|^2+1} - \frac{2}{\sqrt{|w|^2+1}}\,\frac{2}{\sqrt{|w'|^2+1}}\,\frac{|w|^2+1+|w'|^2+1-|w-w'|^2}{\sqrt{|w|^2+1}\sqrt{|w'|^2+1}} \\ &= \frac{4(|w|^2+1)+4(|w'|^2+1)-4(|w|^2+1+|w'|^2+1-|w-w'|^2)}{(1+|w|^2)(1+|w'|^2)} \\ &= \frac{4|w-w'|^2}{(1+|w|^2)(1+|w'|^2)}. \end{aligned}$$

Taking square roots, the result follows. As $d_c(w,w')$ is at most the diameter of the sphere, $d_c(w,w') \leqslant 2$. If $w, w' \in \mathbb{C}\backslash\{0\}$, then

$$\begin{aligned} d_c(\tfrac{1}{w},\tfrac{1}{w'}) &= \frac{2|\frac{1}{w}-\frac{1}{w'}|}{\sqrt{(1+\frac{1}{|w|^2})(1+\frac{1}{|w'|^2})}} = \frac{2\frac{|w'-w|}{|w||w'|}}{\frac{\sqrt{(|w|^2+1)(|w'|^2+1)}}{|w||w'|}} \\ &= \frac{2|w-w'|}{\sqrt{(1+|w|^2)(1+|w'|^2)}} = d_c(w,w'). \end{aligned}$$

Solution to Exercise 10.29

For $R \geqslant 1$, $k \in \{0,1,\cdots,d\}$, and all $z \in \mathbb{C}$ satisfying $|z| > R$, we have that $|z|^k = |z|^k 1 \leqslant |z|^k R^{d-k} \leqslant |z|^k |z|^{d-k} = |z|^d$. Hence for any $R \geqslant 1$ and all $z \in \mathbb{C}$ satisfying $|z| > R$, we have

$$\begin{aligned} |p(z)| = |c_0 + c_1 z + \cdots + c_d z^d| &\leqslant |c_0| + |c_1||z| + \cdots + |c_d||z^d| \\ &\leqslant |c_0||z|^d + |c_1||z|^d + \cdots + |c_d||z|^d = M|z|^d, \end{aligned}$$

where $M := |c_0| + |c_1| + \cdots + |c_d| > 0$.

For $z \neq 0$, we have $p(z) = z^d(c_d + \frac{c_{d-1}}{z} + \cdots + \frac{c_1}{z^{d-1}} + \frac{c_0}{z^d})$. As

$$\lim_{n\to\infty}(\frac{|c_{d-1}|}{n} + \cdots + \frac{|c_1|}{n^{d-1}} + \frac{|c_0|}{n^d}) = 0,$$

there exists an $N \in \mathbb{N}$ large enough so that

$$\frac{|c_{d-1}|}{N} + \cdots + \frac{|c_1|}{N^{d-1}} + \frac{|c_0|}{N^d} < \frac{|c_d|}{2}.$$

Hence for $|z| > R := N$ $(\geqslant 1)$, with $m := \frac{|c_d|}{2}$, we have

$$\begin{aligned} |p(z)| &= |z^d||c_d + \tfrac{c_{d-1}}{z} + \cdots + \tfrac{c_1}{z^{d-1}} + \tfrac{c_0}{z^d}| \\ &\geqslant |z|^d(|c_d| - |\tfrac{c_{d-1}}{z} + \cdots + \tfrac{c_1}{z^{d-1}} + \tfrac{c_0}{z^d}|) \\ &\geqslant |z|^d(|c_d| - (\tfrac{|c_{d-1}|}{|z|} + \cdots + \tfrac{|c_1|}{|z|^{d-1}} + \tfrac{|c_0|}{|z|^d})) \\ &\geqslant |z|^d(|c_d| - (\tfrac{|c_{d-1}|}{N} + \cdots + \tfrac{|c_1|}{N^{d-1}} + \tfrac{|c_0|}{N^d})) \\ &\geqslant |z|^d(|c_d| - \tfrac{|c_d|}{2}) = \tfrac{|c_d|}{2}|z|^d = m|z|^d. \end{aligned}$$

Solution to Exercise 10.30

(If:) Let $(\operatorname{Re} z_n)_{n\in\mathbb{N}}$, $(\operatorname{Im} z_n)_{n\in\mathbb{N}}$ be convergent in $\mathbb{R}$ to $\operatorname{Re} L$, $\operatorname{Im} L$ respectively. Given $\epsilon > 0$, let $N \in \mathbb{N}$ be such that for all $n \in \mathbb{N}$ satisfying $n > N$, we have $|\operatorname{Re} z_n - \operatorname{Re} L| < \frac{\epsilon}{\sqrt{2}}$ and $|\operatorname{Im} z_n - \operatorname{Im} L| < \frac{\epsilon}{\sqrt{2}}$. Then for all $n > N$, we have

$$|z_n - L| = \sqrt{(\operatorname{Re} z_n - \operatorname{Re} L)^2 + (\operatorname{Im} z_n - \operatorname{Im} L)^2} < \sqrt{\frac{\epsilon^2}{2} + \frac{\epsilon^2}{2}} = \epsilon.$$

So $(z_n)_{n\in\mathbb{N}}$ converges to L.

(Only if:) Let $(z_n)_{n\in\mathbb{N}}$ converge to L. Given $\epsilon > 0$, let $N \in \mathbb{N}$ be such that for all $n \in \mathbb{N}$ satisfying $n > N$, we have $|z_n - L| < \epsilon$. Then for $n > N$,

$$|\operatorname{Re} z_n - \operatorname{Re} L| = |\operatorname{Re}(z_n - L)| \leqslant |z_n - L| < \epsilon, \text{ and also}$$
$$|\operatorname{Im} z_n - \operatorname{Im} L| = |\operatorname{Im}(z_n - L)| \leqslant |z_n - L| < \epsilon.$$

So $(\operatorname{Re} z_n)_{n\in\mathbb{N}}$, $(\operatorname{Im} z_n)_{n\in\mathbb{N}}$ are convergent in $\mathbb{R}$ to $\operatorname{Re} L$, $\operatorname{Im} L$, respectively.

Solution to Exercise 10.31

Let $(z_n)_{n\in\mathbb{N}}$ be a Cauchy sequence in $\mathbb{C}$. Then the inequalities

$$|\operatorname{Re} z_n - \operatorname{Re} z_m| = |\operatorname{Re}(z_n - z_m)| \leqslant |z_n - z_m| \text{ and}$$
$$|\operatorname{Im} z_n - \operatorname{Im} z_m| = |\operatorname{Im}(z_n - z_m)| \leqslant |z_n - z_m|$$

show that the real sequences $(\operatorname{Re} z_n)_{n\in\mathbb{N}}$ and $(\operatorname{Im} z_n)_{n\in\mathbb{N}}$ are Cauchy sequences in $\mathbb{R}$. Since $\mathbb{R}$ is complete, they are convergent to, say, $a, b \in \mathbb{R}$, respectively. But then $(z_n)_{n\in\mathbb{N}}$ converges in $\mathbb{C}$ to $a + ib$. So $\mathbb{C}$ is complete.

Solution to Exercise 10.32

(If:) Suppose f is not continuous at z_0, that is[24],

$$\neg\big(\forall\epsilon > 0\ \exists\delta > 0 \text{ such that } \forall z \in S \text{ satisfying } |z - z_0| < \delta,\ |f(z) - f(z_0)| < \epsilon\big).$$

Thus there exists an $\epsilon_* > 0$ such that for all $\delta > 0$, there exists a $z \in S$ such that although $|z - z_0| < \delta$, we have $|f(z) - f(z_0)| \geqslant \epsilon_*$. For $n \in \mathbb{N}$, taking $\delta = \frac{1}{n} > 0$, there exists an element $z_n \in S$ such that $|z_n - z_0| < \delta = \frac{1}{n}$, but $|f(z_n) - f(z_0)| \geqslant \epsilon_*$. The sequence $(z_n)_{n\in\mathbb{N}}$ is contained in S and is convergent with limit z_0: indeed, given $\varepsilon > 0$, by the Archimedean property, there exists an $N \in \mathbb{N}$ such that $N > \varepsilon^{-1}$, and then for all $n > N$, we have $|z_n - z_0| < \frac{1}{n} < \frac{1}{N} < \varepsilon$. But the sequence $(f(z_n))_{n\in\mathbb{N}}$ does not converge to $f(z_0)$: otherwise e.g. for $\frac{\epsilon_*}{2} > 0$, if $N \in \mathbb{N}$ is such that for all $n > N$, we have $|f(z_n) - f(z_0)| < \frac{\epsilon_*}{2}$, then $\epsilon_* \leqslant |f(z_{N+1}) - f(z_0)| < \frac{\epsilon_*}{2}$, a contradiction.

[24] Here we use the notation '$\exists$' for the phrase 'there exists', and the notation '$\forall$' for the phrase 'for all/every/each'.

(Only if:) Let f be continuous at z_0, and $(z_n)_{n\in\mathbb{N}}$ be a sequence contained in S with limit z_0. As f is continuous at z_0, given $\epsilon>0$, there exists a $\delta>0$ such that for all $z\in S$ satisfying $|z-z_0|<\delta$, we have $|f(z)-f(z_0)|<\epsilon$. As $(z_n)_{n\in\mathbb{N}}$ is convergent with limit z_0, there exists an $N\in\mathbb{N}$ such that for all $n>N$, we have $|z_n-z_0|<\delta$. Consequently, for $n>N$, we have $|f(z_n)-f(z_0)|<\epsilon$. So $(f(z_n))_{n\in\mathbb{N}}$ is convergent with limit $f(z_0)$.

Solution to Exercise 10.33

Let $\epsilon>0$. Since f is continuous at z_0, it follows that there exists a $\delta>0$ such that whenever $z\in S$ satisfies $|z-z_0|<\delta$, we have $|f(z)-f(z_0)|<\epsilon$, and so also $||f(z)|-|f(z_0)||\leqslant|f(z)-f(z_0)|<\epsilon$. So $|f|$ is continuous at z_0.

Let $\epsilon>0$. As f is continuous at z_0, there exists a $\delta_1>0$ such that whenever $z\in S$ satisfies $|z-z_0|<\delta_1$, we have $|f(z)-f(z_0)|<\frac{\epsilon}{2}$. Also, since g is continuous at z_0, there exists a $\delta_2>0$ such that whenever $z\in S$ satisfies $|z-z_0|<\delta_2$, we have $|g(z)-g(z_0)|<\frac{\epsilon}{2}$. Thus with $\delta:=\min\{\delta_1,\delta_2\}>0$, we have for all $z\in S$ satisfying $|z-z_0|<\delta$, that

$$\begin{aligned}|(f+g)(z)-(f+g)(z_0)| &= |f(z)-f(z_0)+g(z)-g(z_0)|\\ &\leqslant |f(z)-f(z_0)|+|g(z)-g(z_0)|<\tfrac{\epsilon}{2}+\tfrac{\epsilon}{2}=\epsilon.\end{aligned}$$

This shows that $f+g$ is continuous at z_0.

Let $\epsilon>0$. As f is continuous at z_0, there exists a $\delta_1>0$ such that whenever $z\in S$ satisfies $|z-z_0|<\delta_1$, we have $|f(z)-f(z_0)|<\min\{1,\frac{\epsilon}{2(1+|g(z_0)|)}\}$. Also, since g is continuous at z_0, there exists a $\delta_2>0$ such that whenever $z\in S$ satisfies $|z-z_0|<\delta_2$, we have $|g(z)-g(z_0)|<\frac{\epsilon}{2(1+|f(z_0)|)}$.

So with $\delta:=\min\{\delta_1,\delta_2\}>0$, we have for all $z\in S$ satisfying $|z-z_0|<\delta$, that

$$\begin{aligned}&|(f\cdot g)(z)-(f\cdot g)(z_0)| = |f(z)g(z)-f(z_0)g(z_0)|\\ &\leqslant |f(z)g(z)-f(z)g(z_0)+f(z)g(z_0)-f(z_0)g(z_0)|\\ &\leqslant |f(z)||g(z)-g(z_0)|+|g(z_0)||f(z)-f(z_0)|\\ &\leqslant |f(z)-f(z_0)+f(z_0)||g(z)-g(z_0)|+|g(z_0)||f(z)-f(z_0)|\\ &\leqslant (|f(z)-f(z_0)|+|f(z_0)|)|g(z)-g(z_0)|+|g(z_0)||f(z)-f(z_0)|\\ &< (1+|f(z_0)|)\tfrac{\epsilon}{2(1+|f(z_0)|)}+|g(z_0)|\tfrac{\epsilon}{2(1+|g(z_0)|)}\\ &< \tfrac{\epsilon}{2}+\tfrac{\epsilon}{2}=\epsilon.\end{aligned}$$

This shows that $f\cdot g$ is continuous at z_0.

Solution to Exercise 10.34

(1) Given $R>0$, let $N\in\mathbb{N}$ be such that $N1>R$ (such an N exists by the Archimedean property). For $n>N$, we have $|z_n|=|ni^n|=n>N>R$. So $(ni^n)_{n\in\mathbb{N}}$ converges to ∞.

(2) Let $z_n = x_n + iy_n$, where $x_n, y_n \in \mathbb{R}$, $n \in \mathbb{N}$. From Exercise 10.28, p.119,

$$d_2(\varphi^{-1}z_n, \boldsymbol{n})^2 = (\frac{2x_n}{x_n^2+y_n^2+1})^2 + (\frac{2y_n}{x_n^2+y_n^2+1})^2 + (\frac{x_n^2+y_n^2-1}{x_n^2+y_n^2+1} - 1)^2$$
$$= \frac{4}{x_n^2+y_n^2+1} = \frac{4}{|z_n|^2+1}.$$

(Only if:) Given $\epsilon > 0$, let $R = \frac{2}{\epsilon} > 0$. As $(z_n)_{n\in\mathbb{N}}$ converges to ∞, there exists an $N \in \mathbb{N}$ such that for all $n \in \mathbb{N}$ satisfying $n > N$, we have $|z_n| > R$. So for all $n > N$, we have

$$|d_2(\varphi^{-1}z_n, \boldsymbol{n}) - 0| = \frac{2}{\sqrt{|z_n|^2+1}} < \frac{2}{|z_n|} < \frac{2}{R} = \epsilon.$$

So $\lim\limits_{n\to\infty} d_2(\varphi^{-1}z_n, \boldsymbol{n}) = 0$.

(If:) Let $R > 0$. Set $\epsilon = \frac{2}{\sqrt{1+R^2}} > 0$. As the sequence $(d_2(\varphi^{-1}z_n, \boldsymbol{n}))_{n\in\mathbb{N}}$ converges to 0, there exists an $N \in \mathbb{N}$ such that for all $n \in \mathbb{N}$ satisfying $n > N$, we have $\frac{2}{\sqrt{|z_n|^2+1}} = |d_2(\varphi^{-1}z_n, \boldsymbol{n}) - 0| < \frac{2}{\sqrt{1+R^2}}$, i.e., $|z_n| > R$. So $(z_n)_{n\in\mathbb{N}}$ converges to ∞.

Solution to Exercise 10.35

Let $z_0 \in \mathbb{C}$ and $\epsilon > 0$. Set $\delta = \epsilon > 0$. If $|z - z_0| < \delta$, then

$$|\mathrm{Re}\, z - \mathrm{Re}\, z_0| = |\mathrm{Re}(z - z_0)| \leqslant |z - z_0| < \delta = \epsilon,$$

and by the reverse triangle inequality (Exercise 10.24, p.118), we also have that $||z| - |z_0|| \leqslant |z - z_0| < \delta = \epsilon$. Hence $z \mapsto \mathrm{Re}\, z, |z|$ are continuous at z_0. As $z_0 \in \mathbb{C}$ was arbitrary, $z \mapsto \mathrm{Re}\, z, |z|$ are continuous on $\mathbb{C}$.

Solution to Exercise 10.36

(1) Suppose first that $c \neq 0$. We note that if $z \in \mathbb{C}$ and

$$\frac{az+b}{cz+d} = \frac{a}{c},$$

then we get $caz + bc = acz + ad$, giving $ad - bc = 0$, a contradiction. So, if $z \in \widehat{\mathbb{C}}$ and $f(z) = \frac{a}{c}$, then $z \notin \mathbb{C}$, that is, $z = \infty$.

1° If $z, z' \in \widehat{\mathbb{C}}$ are such that $f(z) = f(z') \in \mathbb{C}\backslash\{\frac{a}{c}\}$, then $z, z' \in \mathbb{C}$ and

$$\frac{az+b}{cz+d} = \frac{az'+b}{cz'+d}.$$

This yields upon simplifying that $(ad - bc)(z - z') = 0$, and so $z = z'$.

2° If $z, z' \in \widehat{\mathbb{C}}$ are such that $f(z) = f(z') = \frac{a}{c}$, then $z = z' = \infty$.

3° If $z, z' \in \widehat{\mathbb{C}}$ and $f(z) = f(z') = \infty$, then $z = z' = -\frac{d}{c}$.

Thus f is injective when $c \neq 0$. Next we show surjectivity.
(Solving for z in $w = \frac{az+b}{cz+d}$, we arrive at $z = \frac{-dw+b}{cw-a}$.)

For $w \in \mathbb{C}\backslash\{\frac{a}{c}\}$, we have

$$f(\tfrac{-dw+b}{cw-a}) = \frac{a(\frac{-dw+b}{cw-a})+b}{c(\frac{-dw+b}{cw-a})+d} = \frac{-adw+ab+bcw-ab}{-cdw+bc+cdw-ad} = \frac{-(ad-bc)w}{-(ad-bc)} = w.$$

Also, $f(\infty) = \frac{a}{c}$ and $f(-\frac{d}{c}) = \infty$. Hence every point of $\widehat{\mathbb{C}}$ is in the range of f, showing that f is surjective. To show the inverse is given by

$$g(w) = \frac{-dw+b}{cw-a} \quad \text{for} \quad w \in \mathbb{C}\backslash\{\tfrac{a}{c}\},$$

with the understanding that we have $g(\frac{a}{c}) = \infty$ and $g(\infty) = \frac{-d}{c}$, it suffices to show $f \circ g = \mathrm{id}$, where id is the identity map on $\widehat{\mathbb{C}}$ (because then $g = \mathrm{id} \circ g = (f^{-1} \circ f) \circ g = f^{-1} \circ (f \circ g) = f^{-1} \circ \mathrm{id} = f^{-1}$).

1° For $w \in \mathbb{C}\backslash\{\frac{a}{c}\}$, $\mathrm{id}(w) = w = f(\frac{-dw+b}{cw-a}) = f(g(w)) = (f \circ g)(w)$.

2° For $w = \frac{a}{c}$, $(f \circ g)(w) = f(g(\frac{a}{c})) = f(\infty) = \frac{a}{c} = w = \mathrm{id}(w)$.

3° For $w = \infty$, then $(f \circ g)(w) = f(g(\infty)) = f(\frac{-d}{c}) = \infty = w = \mathrm{id}(w)$.

This completes the proof when $c \neq 0$. Next we suppose that $c = 0$.

Then $0 \neq ad - bc = ad$ implies that $a \neq 0$ and $d \neq 0$. Thus $\alpha := \frac{a}{d} \neq 0$.

With $\beta := \frac{b}{d}$, we have $f(z) = \frac{a}{d}z + \frac{b}{d} = \alpha z + \beta \in \mathbb{C}$ for all $z \in \mathbb{C}$.

Also $f(\infty) = \infty$. If $z, z' \in \widehat{\mathbb{C}}$ are such that $f(z) = f(z') \in \mathbb{C}$, then we have $\alpha z + \beta = \alpha z' + \beta$, and so $\alpha(z - z') = 0$, giving $z = z'$ (as $\alpha \neq 0$). Also, if $z, z' \in \widehat{\mathbb{C}}$ and $f(z) = f(z') = \infty$, then $z = z' = \infty$. Thus f is injective.

For any $w \in \mathbb{C}$, $f(\frac{w-\beta}{\alpha}) = \alpha\frac{(w-\beta)}{\alpha} + \beta = w$. Moreover, $f(\infty) = \infty$. So every point of $\widehat{\mathbb{C}}$ belongs to the range of f, showing that f is surjective.

Finally, we will check that the inverse of f is given by the function g, where $g(w) := \frac{-dw+b}{0-a} = \frac{w}{\alpha} - \frac{\beta}{\alpha}$ for $w \in \mathbb{C}$, and $g(\infty) := \infty$.

1° For all $w \in \mathbb{C}$, $(f \circ g)(w) = f(\frac{w}{\alpha} - \frac{\beta}{\alpha}) = \alpha\frac{(w-\beta)}{\alpha} + \beta = w = \mathrm{id}(w)$.

2° Also, $(f \circ g)(\infty) = f(\infty) = \infty = \mathrm{id}(\infty)$.

(2) With the Möbius transformations

$$f(z) = \frac{az+b}{cz+d} \quad \text{and} \quad g(z) = \frac{a'z+b'}{c'z+d'},$$

we associate the matrices $F = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$ and $G = \left[\begin{smallmatrix} a' & b' \\ c' & d' \end{smallmatrix}\right]$.

Then the determinants of F and G are nonzero. For $z \in \mathbb{C}\backslash\{\frac{d'}{c'}\}$, we have

$$(f \circ g)(z) = f(\tfrac{a'z+b'}{c'z+d'}) = \frac{a(\frac{a'z+b'}{c'z+d'})+b}{c(\frac{a'z+b'}{c'z+d'})+d} = \frac{(aa'+bc')z+ab'+bd'}{(ca'+dc')z+(cb'+dd')} = \frac{Az+B}{Cz+D},$$

where $H := \left[\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right] := \left[\begin{smallmatrix} aa'+bc' & ab'+bd' \\ ca'+dc' & cb'+dd' \end{smallmatrix}\right] = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]\left[\begin{smallmatrix} a' & b' \\ c' & d' \end{smallmatrix}\right] = FG.$

To show $f \circ g$ is a Möbius transformation, we need $AD - BC \neq 0$. But $AD-BC = \det H = \det(FG) = (\det F)\det G = (ad-bc)(a'd'-b'c') \neq 0$.

(3) Let C be the circle $\{z\in\mathbb{C}:|z-z_0|=r\}$ centred at $z_0\in\mathbb{C}$ with radius $r>0$.

- For a translation $f(z)=z+b$, $f(C)=\{w\in\mathbb{C}:|w-(z_0+b)|=r\}$, which is a circle with centre z_0+b and radius $r>0$.
- For a dilation $f(z)=az$ $(a>0)$, $f(C)=\{w\in\mathbb{C}:|w-az_0|=ar\}$, which is a circle with centre az_0 and radius $ar>0$.
- For a rotation $f(z)=e^{i\theta}z$ $(\theta\in\mathbb{R})$, $f(C)=\{w\in\mathbb{C}:|w-e^{i\theta}z_0|=r\}$, which is a circle with centre $e^{i\theta}z_0$ and radius $r>0$.

So for translations, dilations, and rotations, the images of circles are circles.

For inversions, we'll now see that the image of a circle is either a circle or a straight line (which can be viewed as a circle with an infinite radius). If $f(z)=\frac{1}{z}$, then $f(C)\cap\mathbb{C}=\{w\in\mathbb{C}:|\frac{1}{w}-z_0|=r\}=\{w\in\mathbb{C}:|1-wz_0|=r|w|\}$. So:

$$\begin{aligned} w\in f(C)\cap\mathbb{C} &\Leftrightarrow 0=|1-wz_0|^2-r^2|w|^2 \\ &\Leftrightarrow 0=(1-wz_0)(1-w^*z_0^*)-r^2|w|^2 \\ &\Leftrightarrow (|z_0|^2-r^2)|w|^2-z_0w-z_0^*w^*+1=0. \end{aligned}$$

1° $|z_0|=r>0$. Then we have $w\in f(C)\cap\mathbb{C}$ if and only if $2\mathrm{Re}(z_0w)=1$. If $z_0=(x_0,y_0)\in\mathbb{R}^2$, we have with $w=(x,y)\in\mathbb{R}^2$ that $w\in f(C)$ if and only if $x_0x-y_0y=\frac{1}{2}$. Thus $f(C)=\{(x,y)\in\mathbb{R}^2:x_0x-y_0y=\frac{1}{2}\}\cup\{\infty\}$. $f(C)\cap\mathbb{C}=\{(x,y)\in\mathbb{R}^2:x_0x-y_0y=\frac{1}{2}\}$ is a line in the complex plane. (As $|z_0|=r>0$, $(x_0,y_0)\neq(0,0)$. Also, $0\in C$, and so $\infty\in f(C)$.)

2° $|z_0|>r$. Let $k>0$ be such that $k^2=|z_0|^2-r^2>0$. Also, let $a=\frac{z_0^*}{k^2}\in\mathbb{C}$. Then $w\in f(C)$ if and only if $|w|^2-2\mathrm{Re}(a^*w)+|a|^2=(\frac{r}{k^2})^2$. Thus (see e.g. Exercise 10.25, p.118) $f(C)=\{w\in\mathbb{C}:|w-a|=\frac{r}{k^2}\}$, which is a circle in the complex plane with centre a and radius $\frac{r}{k^2}$.

3° $|z_0|<r$. Let $k>0$ be such that $k^2=r^2-|z_0|^2>0$. Let $a=-\frac{z_0^*}{k^2}\in\mathbb{C}$. Then $w\in f(C)$ if and only if $|w|^2-2\mathrm{Re}(a^*w)+|a|^2=(\frac{r}{k^2})^2$. Hence $f(C)=\{w\in\mathbb{C}:|w-a|=\frac{r}{k^2}\}$, which is a circle with centre a and radius $\frac{r}{k^2}$.

Next we consider the case of lines. From Exercise 10.17, p.116, a line passing through z_0 and z_1, where $z_0\neq z_1$, is $L=\{tz_1+(1-t)z_0:t\in\mathbb{R}\}$. So if $d:=z_1-z_0\neq 0$, then $L=\{z_0+td:t\in\mathbb{R}\}$ is the line passing through z_0 and z_0+d $(\neq z_0)$.

- For a translation $f(z)=z+b$, $f(L)=\{z_0+b+td : t\in\mathbb{R}\}$, which is a line passing through z_0+b and z_0+b+d.
- For a dilation $f(z)=az$ $(a>0)$, $f(L)=\{az_0+tad : t\in\mathbb{R}\}$, which is a line passing through az_0 and az_0+ad.
- For a rotation $f(z)=e^{i\theta}z$ $(\theta\in\mathbb{R})$, $f(L)=\{e^{i\theta}z_0+te^{i\theta}d : t\in\mathbb{R}\}$, which is a line passing through $e^{i\theta}z_0$ and $e^{i\theta}z_0+e^{i\theta}d$.

Thus in each of the cases above, the images of lines are lines again.

With inversions $f(z)=\frac{1}{z}$, we will now see that the image of a line L is either a line (if L passes through the origin) or a circle (if L does not pass through the origin).

1° If $L=\{td : t\in\mathbb{R}\}$, then using $f(\infty)=0$ and $f(0)=\infty$, it can be seen that $f(L\cup\{\infty\})=\{s\frac{1}{d} : s\in\mathbb{R}\}\cup\{\infty\}$. The set $\{s\frac{1}{d} : s\in\mathbb{R}\}$ describes a line passing through 0 (obtained by reflecting L in the real axis).

2° If L doesn't pass through 0, then there is a nonzero complex number $z_0\in L$ closest to 0, so that the segment joining 0 to z_0 is perpendicular to L. So we may take $d=iz_0$, and $L=\{z_0+tiz_0 : t\in\mathbb{R}\}$.

For $w=\dfrac{1}{z_0(1+it)}$, we have

$$\begin{aligned}|w-\frac{1}{2z_0}| &= \frac{1}{2|z_0|}|\frac{2}{1+it}-1| = \frac{1}{2|z_0|}|\frac{2(1-it)}{1+t^2}-1|\\ &= \frac{1}{2|z_0|}|\frac{1-t^2}{1+t^2}-i\frac{2t}{1+t^2}| = \frac{1}{2|z_0|(1+t^2)}\sqrt{(1-t^2)^2+4t^2}\\ &= \frac{1}{2|z_0|(1+t^2)}(1+t^2) = \frac{1}{2|z_0|}.\end{aligned}$$

So $f(L\cup\{\infty\})=\{w\in\mathbb{C} : |w-\frac{1}{2z_0}|=\frac{1}{2|z_0|}\}$, which is a circle passing through 0.

Solution to Exercise 10.37

(1) We have

$$\begin{aligned}p(\boldsymbol{v}) &= (0^2-\boldsymbol{v}\cdot\boldsymbol{v}+0\boldsymbol{v}+0\boldsymbol{v}+\boldsymbol{v}\times\boldsymbol{v})+1\\ &= 0-1+\mathbf{0}+\mathbf{0}+\mathbf{0}+1=0.\end{aligned}$$

(2) We have $(q^*)^*=(q_0-\boldsymbol{q})^*=q_0-(-\boldsymbol{q})=q_0+\boldsymbol{q}=q$.
Also, $q+q^*=q_0+\boldsymbol{q}+q_0-\boldsymbol{q}=2q_0$. We have

$$\begin{aligned}q^*q &= (q_0-\boldsymbol{q})(q_0+\boldsymbol{q})\\ &= q_0^2-(-\boldsymbol{q})\cdot\boldsymbol{q}+q_0\boldsymbol{q}+q_0(-\boldsymbol{q})+(-\boldsymbol{q})\times\boldsymbol{q}\\ &= q_0^2+\boldsymbol{q}\cdot\boldsymbol{q}=q_0^2+\|\boldsymbol{q}\|_2^2\\ &= q_0^2+q_1^2+q_2^2+q_3^2.\end{aligned}$$

Hence $qq^*=(q^*)^*q^*=q_0^2+\|-\boldsymbol{q}\|_2^2=q_0^2+\|\boldsymbol{q}\|_2^2$.

Using $\boldsymbol{p}\times\boldsymbol{q}=-(\boldsymbol{q}\times\boldsymbol{p})=(-\boldsymbol{q})\times\boldsymbol{p}$ and $\boldsymbol{p}\cdot\boldsymbol{q}=\boldsymbol{q}\cdot\boldsymbol{p}=-(-\boldsymbol{q})\cdot\boldsymbol{p}$, we have

$$\begin{aligned}(pq)^* &= (p_0q_0 - \boldsymbol{p}\cdot\boldsymbol{q} + p_0\boldsymbol{q} + q_0\boldsymbol{p} + \boldsymbol{p}\times\boldsymbol{q})^*\\ &= p_0q_0 - \boldsymbol{p}\cdot\boldsymbol{q} - p_0\boldsymbol{q} - q_0\boldsymbol{p} - \boldsymbol{p}\times\boldsymbol{q}\\ &= q_0p_0 - (-\boldsymbol{p})\cdot(-\boldsymbol{q}) + p_0(-\boldsymbol{q}) + q_0(-\boldsymbol{p}) + \boldsymbol{q}\times\boldsymbol{p}\\ &= q_0p_0 - (-\boldsymbol{q})\cdot(-\boldsymbol{p}) + q_0(-\boldsymbol{p}) + p_0(-\boldsymbol{q}) + (-\boldsymbol{q})\times(-\boldsymbol{p}) = q^*p^*.\end{aligned}$$

We have $\lfloor pq\rfloor^2=(pq)^*pq=q^*p^*pq=q^*\lfloor p\rfloor^2q=\lfloor p\rfloor^2q^*q=\lfloor p\rfloor^2\lfloor q\rfloor^2$.

Finally, we have $q^{-1}q=\dfrac{q^*}{\lfloor q\rfloor^2}q=\dfrac{\lfloor q\rfloor^2}{\lfloor q\rfloor^2}=1=\dfrac{\lfloor q\rfloor^2}{\lfloor q\rfloor^2}=q\dfrac{q^*}{\lfloor q\rfloor^2}=qq^{-1}$.

(3) To check that $L_q(\boldsymbol{v})\in\mathbb{R}^3$, it suffices to check that the 'real part' of the quaternion $p:=q\boldsymbol{v}q^*$ vanishes. By the result from part (2), twice the real part of p is given by $p+p^*$, and so it is enough to check that the latter vanishes. We have

$$\begin{aligned}p+p^* &= q\boldsymbol{v}q^* + (q\boldsymbol{v}q^*)^* = q\boldsymbol{v}q^* + (q^*)^*\boldsymbol{v}^*q^*\\ &= q\boldsymbol{v}q^* + q(-\boldsymbol{v})q^* = q\boldsymbol{v}q^* - q\boldsymbol{v}q^* = 0.\end{aligned}$$

So $L_q:\mathbb{R}^3\to\mathbb{R}^3$ is well-defined. Next we check that L_q is a linear transformation. If $\boldsymbol{v}_1,\boldsymbol{v}_2\in\mathbb{R}^3$ and $c_1,c_2\in\mathbb{R}$, then we have (using the distributivity of multiplication over addition, and the commutativity of multiplication of real numbers with quaternions) that

$$\begin{aligned}L_q(c_1\boldsymbol{v}_1+c_2\boldsymbol{v}_2) &= q(c_1\boldsymbol{v}_1+c_2\boldsymbol{v}_2)q^* = q(c_1\boldsymbol{v}_1)q^* + q(c_2\boldsymbol{v}_2)q^*\\ &= c_1q\boldsymbol{v}_1q^* + c_2q\boldsymbol{v}_2q^* = c_1L_q(\boldsymbol{v}_1) + c_2L_q(\boldsymbol{v}_2).\end{aligned}$$

L_q is an isometry: Firstly, $\lfloor q^*\rfloor^2=q_0^2+\|-\boldsymbol{q}\|_2^2=q_0^2+\|\boldsymbol{q}\|_2^2=\lfloor q\rfloor^2=1$. For $\boldsymbol{v}\in\mathbb{R}^3$, we have

$$\begin{aligned}\|L_q(\boldsymbol{v})\|_2^2 &= \lfloor q\boldsymbol{v}q^*\rfloor^2 = (\lfloor q\rfloor\lfloor\boldsymbol{v}\rfloor\lfloor q^*\rfloor)^2 = \lfloor q\rfloor^2\lfloor\boldsymbol{v}\rfloor^2\lfloor q^*\rfloor^2\\ &= (1)\|\boldsymbol{v}\|_2^2(1) = \|\boldsymbol{v}\|_2^2.\end{aligned}$$

$\boldsymbol{q}$ is an eigenvector of L_q with eigenvalue 1: We have $\boldsymbol{q}\neq\boldsymbol{0}$ (given) and

$$\begin{aligned}L_q(\boldsymbol{q}) &= q\boldsymbol{q}q^* = (q_0+\boldsymbol{q})\boldsymbol{q}q^* = (-\boldsymbol{q}\cdot\boldsymbol{q} + q_0\boldsymbol{q} + \boldsymbol{q}\times\boldsymbol{q})q^*\\ &= (-\|\boldsymbol{q}\|_2^2 + q_0\boldsymbol{q} + \boldsymbol{0})(q_0-\boldsymbol{q})\\ &= -\|\boldsymbol{q}\|_2^2q_0 - q_0\boldsymbol{q}\cdot(-\boldsymbol{q}) - \|\boldsymbol{q}\|_2^2(-\boldsymbol{q}) + q_0(q_0\boldsymbol{q}) + (q_0\boldsymbol{q})\times(-\boldsymbol{q})\\ &= -q_0\|\boldsymbol{q}\|_2^2 + q_0\|\boldsymbol{q}\|_2^2 + \|\boldsymbol{q}\|_2^2\boldsymbol{q} + q_0^2\boldsymbol{q} + \boldsymbol{0}\\ &= (\|\boldsymbol{q}\|_2^2+q_0^2)\boldsymbol{q} = \lfloor q\rfloor^2\boldsymbol{q} = 1^2\boldsymbol{q} = 1\boldsymbol{q}.\end{aligned}$$

L_q describes a rotation about $\boldsymbol{q}$ by an angle of 2θ: Let $\boldsymbol{v}\in\mathbb{R}^3$. We can decompose $\boldsymbol{v}$ as $\boldsymbol{v}=c\boldsymbol{q}+\boldsymbol{n}$, where

$$c:=\frac{\boldsymbol{v}\cdot\boldsymbol{q}}{\boldsymbol{q}\cdot\boldsymbol{q}}\quad\text{and}\quad\boldsymbol{n}:=\boldsymbol{v}-\frac{\boldsymbol{v}\cdot\boldsymbol{q}}{\boldsymbol{q}\cdot\boldsymbol{q}}\boldsymbol{q}.$$

Then $\boldsymbol{n}$ lies in the plane $\boldsymbol{q}^\perp$ perpendicular to $\boldsymbol{q}$ because $\boldsymbol{n}\cdot\boldsymbol{q}=0$. As L_q is linear, we have $L_q(\boldsymbol{v})=L_q(c\boldsymbol{q}+\boldsymbol{n})=c\boldsymbol{q}+L_q\boldsymbol{n}$, it is enough to

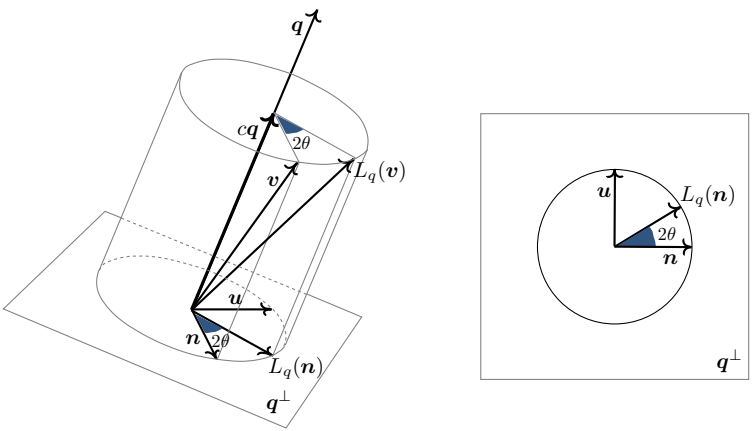
show that $L_q(\boldsymbol{n})$ is obtained by rotating $\boldsymbol{n}$ through an angle of 2θ in the plane $\boldsymbol{q}^\perp$, see the following picture. The vector $\boldsymbol{q}\times\boldsymbol{n}$ is perpendicular to $\boldsymbol{q}$, and hence lies in the plane $\boldsymbol{q}^\perp$. Moreover $\boldsymbol{q}\times\boldsymbol{n}$ is perpendicular to $\boldsymbol{n}$ within the plane $\boldsymbol{q}^\perp$. As $\|\boldsymbol{q}\times\boldsymbol{n}\|_2 = \|\boldsymbol{q}\|_2\|\boldsymbol{n}\|_2 \sin\frac{\pi}{2} = \|\boldsymbol{q}\|_2\|\boldsymbol{n}\|_2$, the vector

$$\boldsymbol{u} := \frac{\boldsymbol{q}\times\boldsymbol{n}}{\|\boldsymbol{q}\|_2}$$

has the same length as $\boldsymbol{n}$ and is perpendicular to $\boldsymbol{n}$. Furthermore, $(\boldsymbol{q}\times\boldsymbol{n})\times\boldsymbol{q} = \|\boldsymbol{q}\|_2^2\boldsymbol{n}$. So we have

$$\begin{aligned}
L_q(\boldsymbol{n}) &= q\boldsymbol{n}q^* = (q_0+\boldsymbol{q})\boldsymbol{n}q^* \\
&= (-\boldsymbol{q}\cdot\boldsymbol{n} + q_0\boldsymbol{n} + \boldsymbol{q}\times\boldsymbol{n})q^* = (q_0\boldsymbol{n} + \boldsymbol{q}\times\boldsymbol{n})(q_0-\boldsymbol{q}) \\
&= -(q_0\boldsymbol{n}+\boldsymbol{q}\times\boldsymbol{n})\cdot(-\boldsymbol{q}) + q_0(q_0\boldsymbol{n}+\boldsymbol{q}\times\boldsymbol{n}) + (q_0\boldsymbol{n}+\boldsymbol{q}\times\boldsymbol{n})\times(-\boldsymbol{q}) \\
&= q_0^2\boldsymbol{n} + q_0\boldsymbol{q}\times\boldsymbol{n} - q_0\boldsymbol{n}\times\boldsymbol{q} - (\boldsymbol{q}\times\boldsymbol{n})\times\boldsymbol{q} = q_0^2\boldsymbol{n} + 2q_0\boldsymbol{q}\times\boldsymbol{n} - \|\boldsymbol{q}\|_2^2\boldsymbol{n} \\
&= (q_0^2 - \|\boldsymbol{q}\|_2^2)\boldsymbol{n} + 2q_0\|\boldsymbol{q}\|_2\boldsymbol{u} \\
&= ((\cos\theta)^2 - (\sin\theta)^2)\boldsymbol{n} + 2(\cos\theta)(\sin\theta)\boldsymbol{u} \\
&= (\cos(2\theta))\boldsymbol{n} + (\sin(2\theta))\boldsymbol{u}.
\end{aligned}$$

The above shows that $L_q(\boldsymbol{n})$ lies in the plane $\boldsymbol{q}^\perp$, since $\boldsymbol{n},\boldsymbol{u}\in\boldsymbol{q}^\perp$. As $L_q(\boldsymbol{n})$ is an isometry, $\|L_q(\boldsymbol{n})\|_2 = \|\boldsymbol{n}\|_2$. Taking dot product with $\boldsymbol{n}$ in the above gives $L_q(\boldsymbol{n})\cdot\boldsymbol{n} = (\cos(2\theta))\|\boldsymbol{n}\|_2^2$, while taking dot product with $\boldsymbol{u}$ gives $L_q(\boldsymbol{n})\cdot\boldsymbol{u} = (\sin(2\theta))\|\boldsymbol{n}\|_2^2$. Thus $L_q(\boldsymbol{n})$ makes the angle 2θ with the vector $\boldsymbol{n}$, and the angle $\frac{\pi}{2}-2\theta$ with the vector $\boldsymbol{u}$. These facts show that $L_q(\boldsymbol{n})$ is obtained by rotating the vector $\boldsymbol{n}$ in the plane $\boldsymbol{q}^\perp$ through the angle 2θ.



The picture on the left shows the spatial picture in $\mathbb{R}^3$, while the one on the right shows the picture in the plane $\boldsymbol{q}^\perp$ (i.e., the view from above, along the axis containing $\boldsymbol{q}$, looking down at the plane $\boldsymbol{q}^\perp$).

Composition of rotations: Let $p = p_0 + \boldsymbol{p}$, $q = q_0 + \boldsymbol{q}$ be quaternions with $|p| = 1 = |q|$, and $\boldsymbol{p}, \boldsymbol{q} \in \mathbb{R}^3 \backslash \{\boldsymbol{0}\}$ not collinear. We claim that the vector $\widetilde{\boldsymbol{q}} := q_0\boldsymbol{p} + p_0\boldsymbol{q} + \boldsymbol{p} \times \boldsymbol{q}$ is nonzero. As $\boldsymbol{p} \times \boldsymbol{q}$ is perpendicular to the plane containing $\boldsymbol{p}$ and $\boldsymbol{q}$, $\widetilde{\boldsymbol{q}}$ will be nonzero whenever $\boldsymbol{p} \times \boldsymbol{q} \neq \boldsymbol{0}$. Since $\boldsymbol{p}, \boldsymbol{q}$ are both nonzero, the only way in which $\boldsymbol{p} \times \boldsymbol{q}$ will be the zero vector is if $\boldsymbol{p}$ and $\boldsymbol{q}$ are collinear, a contradiction to our assumption that $\boldsymbol{p}, \boldsymbol{q}$ are not collinear. (We remark that if $\boldsymbol{p}, \boldsymbol{q}$ were collinear, then the two rotations are along the same axis, and so composing them is trivial, as the final rotation is about the same axis, and the angle is obtained by just adding/subtracting the individual rotation angles.)
We have $|pq|^2 = |p|^2|q|^2 = 1^2 1^2 = 1$, and so $|pq| = 1$. Thus L_{pq} is well-defined. For all $\boldsymbol{v} \in \mathbb{R}^3$, we have

$$\begin{aligned}(L_p \circ L_q)(\boldsymbol{v}) &= L_p(L_q(\boldsymbol{v})) = L_p(q\boldsymbol{v}q^*) = p(q\boldsymbol{v}q^*)p^* \\ &= pq\boldsymbol{v}(q^*p^*) = pq\boldsymbol{v}(pq)^* = L_{pq}(\boldsymbol{v}).\end{aligned}$$

Hence $L_p \circ L_q = L_{pq}$.

Solution to Exercise 10.38

	$\boldsymbol{e}_1$	$\boldsymbol{e}_2$	$\boldsymbol{e}_3$	$\boldsymbol{e}_4$	$\boldsymbol{e}_5$	$\boldsymbol{e}_6$	$\boldsymbol{e}_7$
$\boldsymbol{e}_1$	-1	$\boldsymbol{e}_4$	$\boldsymbol{e}_7$	$-\boldsymbol{e}_2$	$\boldsymbol{e}_6$	$-\boldsymbol{e}_5$	$-\boldsymbol{e}_3$
$\boldsymbol{e}_2$	$-\boldsymbol{e}_4$	-1	$\boldsymbol{e}_5$	$\boldsymbol{e}_1$	$-\boldsymbol{e}_3$	$\boldsymbol{e}_7$	$-\boldsymbol{e}_6$
$\boldsymbol{e}_3$	$-\boldsymbol{e}_7$	$-\boldsymbol{e}_5$	-1	$\boldsymbol{e}_6$	$\boldsymbol{e}_2$	$-\boldsymbol{e}_4$	$\boldsymbol{e}_1$
$\boldsymbol{e}_4$	$\boldsymbol{e}_2$	$-\boldsymbol{e}_1$	$-\boldsymbol{e}_6$	-1	$\boldsymbol{e}_7$	$\boldsymbol{e}_3$	$-\boldsymbol{e}_5$
$\boldsymbol{e}_5$	$-\boldsymbol{e}_6$	$\boldsymbol{e}_3$	$-\boldsymbol{e}_2$	$-\boldsymbol{e}_7$	-1	$\boldsymbol{e}_1$	$\boldsymbol{e}_4$
$\boldsymbol{e}_6$	$\boldsymbol{e}_5$	$-\boldsymbol{e}_7$	$\boldsymbol{e}_4$	$-\boldsymbol{e}_3$	$-\boldsymbol{e}_1$	-1	$\boldsymbol{e}_2$
$\boldsymbol{e}_7$	$\boldsymbol{e}_3$	$\boldsymbol{e}_6$	$-\boldsymbol{e}_1$	$\boldsymbol{e}_5$	$-\boldsymbol{e}_4$	$-\boldsymbol{e}_2$	-1

We have $\boldsymbol{e}_1(\boldsymbol{e}_2\boldsymbol{e}_3) = \boldsymbol{e}_1\boldsymbol{e}_5 = \boldsymbol{e}_6 \neq -\boldsymbol{e}_6 = \boldsymbol{e}_4\boldsymbol{e}_3 = (\boldsymbol{e}_1\boldsymbol{e}_2)\boldsymbol{e}_3$.

Notes

Chapter 1: Exercise 1.5 is from [Artin (1991), Exercise 8, §5, Chapter 1].

Chapter 2: The proof of the recursion theorem is based on the one given in [Cohen and Ehrlich (1963), Theorem 1.2].

Chapter 3: The figure on p.26 is based on the one on [Ovchinnikov (2015), p.39].

Chapter 5: The illustrative analogy given for Theorem 5.7, p.57, stems from [Bryant (1990)]. The original proof seems to go back to [Newman (1951)], see also [Newman and Parsons (1988)].

Chapter 9: Section 9.3 is based on [Frame (1949)].

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Index

$\mathbb{C}[z]$, 115
$\mathbb{Z}[x]$, 102
n-tuple, 6
p-adic numbers, 107
mod , 75

Abelian group, 117
absolute value, 116
addition modulo n, 76
al-Khwarizmi, 102
Archimedean property, 54
arithmetic mean-geometric mean inequality, 49

Bezout equation, 66
bijective function, 7
binomial coefficients, 40
binomial theorem, 46
Bolzano-Weierstrass theorem, 59
Bombelli, 111
bounded above, 52
bounded below, 54
bounded sequence in $\mathbb{Q}$, 42
bounded sequence in $\mathbb{R}$, 56

Cantor set, 88
Cantor's theorem, 90
Cardano, 111
cardinal number, 22
Cartesian product, 6
Catalan number, 70
Cauchy sequence in $\mathbb{C}$, 120
Cauchy sequence in $\mathbb{Q}$, 42
Cauchy sequence in $\mathbb{R}$, 55
Cauchy-Schwarz inequality, 49
Chinese remainder theorem, 80
chordal metric, 119
compact set, 121
complete, 120
complex conjugate, 116
complex number, 109
complex plane, 112
composition of maps, 7
congruent modulo, 75
continued fraction, 97
continuous function, 120
convergent sequence, 121
convergent sequence in $\mathbb{C}$, 120
convergent sequence in $\mathbb{Q}$, 42
convergent sequence in $\mathbb{R}$, 55
coprime integers, 68
cosine formula, 119
countable set, 81

de Moivre's formula, 114
decreasing sequence, 56
degree of a polynomial, 102, 115
digit, 93
disjoint sets, 6
divides, 64
division algorithm, 63, 115
divisor, 64
domain, 7

equivalence class, 9
equivalence relation, 8
Euclid, 73
extended complex plane, 121
extended real number system, 105

factor, 64
Fermat prime, 74
Fibonacci sequence, 66
finite continued fraction, 97
finite set, 22
function, 7
fundamental theorem of algebra, 121
fundamental theorem of arithmetic, 71

Gaussian integer, 114
Gelfond-Schneider theorem, 104
golden ratio, 3
greatest common divisor, 65
greatest integer part, 54
greatest lower bound property, 54
group, 117

identity function/map, 7
image, 7
imaginary part, 109
increasing sequence, 56
infinite descent, 65
infinite set, 22
infinity, 105, 121
injective function, 7
integers, 26
integers modulo n, 76
intersection of sets, 6
inverse image, 7
inverse of a map, 7
ISBN number, 79

Kronecker's theorem, 60

least common multiple, 74
least upper bound, 52
length of a vector in $\mathbb{R}^3$, 125
Liouville number, 105
Liouville's approximation theorem, 104

Möbius transformation, 124
map, 7
maximum of a set, 52
metric space, 121
minimum of a set, 54
minus infinity, 105
modular arithmetic, 76
modulus, 116
monic polynomial, 102
monotone sequence, 56
multiple, 64
multiplication modulo n, 76

non-Archimedean absolute value, 107
norm, 126
normed space, 126
number line, 1

octonions, 127
one-point compactification, 121
one-to-one correspondence, 81
one-to-one function, 7
onto function, 7
open disc, 119
open set in $\mathbb{C}$, 119
order relation $<$ on $\mathbb{N}$, 20
order relation $<$ on $\mathbb{Q}$, 38
order relation $<$ on $\mathbb{R}$, 47
order relation $<$ on $\mathbb{Z}$, 32
ordered field, 110
ordered pair, 6
Ostrowski's theorem, 107

Peano axioms, 13
plus infinity, 105
postfix notation, 19
power set, 5
prime counting function, 73
prime number, 71
prime number theorem, 73
principle of induction, 17
product of real numbers, 43
product of complex numbers, 109

product of integers, 29
product of natural numbers, 17
product of rational numbers, 37

quaternions, 125

range, 7
rational approximation, 97
rational number, 36
rational zeroes theorem, 102
real part, 109
recursion theorem, 15
reflexive relation, 8
relation, 6, 8
relation on a set, 8
relatively prime integers, 68
restriction of a function, 7
Riemann sphere, 119

seaview property, 57
sequence, 10, 55
set, 4
singleton, 4
subsequence, 55
subset of a set, 5
successor, 13
successor function, 13
sum of rational numbers, 37
sum of real numbers, 43
sum of complex numbers, 109
sum of integers, 27
sum of natural numbers, 16
supremum of a set, 52
surds, 102
surjective function, 7
symmetric relation, 8

transitive relation, 8
triangle inequality in $\mathbb{C}$, 118
triangle inequality in $\mathbb{Q}$, 39
triangle inequality in $\mathbb{R}$, 49
triangle inequality in a metric space, 121
trichotomy law in $\mathbb{N}$, 20
trichotomy law in $\mathbb{Q}$, 39
trichotomy law in $\mathbb{R}$, 47
trichotomy law in $\mathbb{Z}$, 32
triple, 6
twin prime conjecture, 74
twin primes, 74

uncountable set, 81
union of sets, 6
upper bound, 52

well-ordering principle in $\mathbb{N}$, 22
well-ordering principle in $\mathbb{Z}$, 33
Wilson's theorem, 79

Zermelo-Frankel axioms, 5

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